Largan Precision

Core weakness outweighs progress in AI optics

Equities

Taiwan

Electric Components & Equipment

12-month rating	**Sell**
12m price target	**NT$5,000.00**
Price (08 Oct 2026)	**NT$6,660**

RIC: 3008.TW **BBG:** 3008 TT

Trading data and key metrics

52-wk range	NT$7,810.00-2,025.00
Market cap.	NT$875b/US$27.4b
Shares o/s	131m (ORD)
Free float	70%
Avg. daily volume ('000)	2,379
Avg. daily value (m)	NT$13,134.2
Common s/h equity (12/26E)	NT$199b
P/BV (12/26E)	4.4x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From	To	% ch	Cons.
12/26E	188.21	160.04	-15	184.91
12/27E	214.00	180.00	-16	204.02
12/28E	250.66	220.00	-12	238.78

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CPO progress is more tangible but substantial success is already priced in

We maintain Sell and our NT$5,000 target price as erosion in the core smartphone business outweighs CPO progress. The Q326 call confirmed that Largan's fiber array has passed customer qualification although its customer qualified another vendor's prism micro-lens assembly (PMLA) that Largan would now buy in. If Largan qualifies the full FAU plus carrier assembly for 2H27, content rises but with more pass-through cost and less margin upside. We roll our valuation to 2028E, the first full year of CPO production, though 25% below the NT$6,660 close on October 8.

Full FAU assembly would lift content, but with more pass-through

We believe Largan's FA customer is one of two module suppliers on NVIDIA's 2H27 program, which we estimate could reach 40% share. The program is still subject to successful yield ramp and could move to metalens in 2028, which would reopen supplier qualification. Assembling the full FAU could lift Largan's ASP to US$125-150 leveraging its precision alignment and automation but the PMLA, V-groove and carrier parts would be pass through that cuts into margin leverage. We expand our FAU TAM analysis (Figure 9) with FAU market in 2028/30 expanding to 12.3mn/35.6mn units, with Largan base case set at 25% of the TAM.

Core smartphone business stays soft into 2027

Q326 sales of NT$15.7bn, +15% QoQ/-11% YoY missedUBSe/Street +36%/+33% on weaker flagship smartphone demand. GMs fell 7.2ppts QoQ to 42.2%, lowest in 13 years on renewable energy costs and low variable-aperture yields, dropping OpM to 17 year lows at 30.3% and EPS to NT$39.6, below UBS/street NT$51.50/NT$49.69. We cut Q4 to NT$16.7bn +7% QoQ, 11% below consensus as its customer trimmed orders after a pricier iPhone 18 launch and possible lag to Genius which is picking up periscope share and outgrew in Q3 (+1% YoY vs. -11% YoY). We cut 2027 from NT$77bn to NT $67bn, before a recovery to NT$79bn in 2028 with the next periscope upgrade. We see GM recovering to 47.5%/48.9% in 2027-28 as yields improve and energy costs pass through in 2028 pricing.

Maintain Sell and our NT$5,000 target

We cut 2026E/27E EPS by 14%/16% to NT$160/NT$180 and introduce 2028E EPS of NT$220 for the core smartphone business. We value the lens business at NT$3,300 (15x 2028E ex-CPO EPS of NT$220, the middle of Largan's 13–18x historical range) and CPO option at NT$1,700 or US$7bn (49x our base-case 2028E CPO EPS of NT$35, near the top of optical peers' 14–54x). Our base case gives Largan 25% of a 12.3mn-unit, US $1.85bn FAU market at US$150 per unit and a 50% gross margin. That credits it with yields and margins it still needs to prove in a quarter when yields and energy costs cut 7.2ppts from its lens margin. At ~NT$6,660, the shares imply ~NT$69 of 2028E CPO EPS, about twice our base case.

Highlights (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	48,842	59,458	61,148	61,550	67,438	79,464	78,546	81,569
EBIT (UBS)	17,807	24,033	23,558	21,139	24,150	30,110	29,487	30,648
Net earnings (UBS)	18,047	22,480	23,613	20,953	23,567	28,805	28,397	29,473
EPS (UBS, diluted) (NT$)	135.22	168.43	176.92	160.04	180.00	220.00	216.89	225.11
DPS (net) (NT$)	85.50	67.50	97.50	80.00	81.97	90.34	110.41	108.85
Net (debt) / cash	111,655	117,781	122,309	109,131	120,267	127,842	144,348	160,697

Profitability/valuation	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
EBIT (UBS) margin %	36.5	40.4	38.5	34.3	35.8	37.9	37.5	37.6
ROIC (EBIT) %	39.3	46.2	40.9	30.3	30.1	35.1	33.1	35.3
EV/EBITDA (UBS core) x	7.7	7.1	6.0	24.7	21.7	18.1	17.7	16.6
P/E (UBS, diluted) x	16.2	15.1	13.3	41.6	37.0	30.3	30.7	29.6
Equity FCF (UBS) yield %	2.0	4.3	3.0	2.1	2.5	2.2	3.6	3.5
Dividend yield (net) %	3.9	2.7	4.1	1.2	1.2	1.4	1.7	1.6

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 6,660.00 on 12-Oct-2026 06:29:19 CST

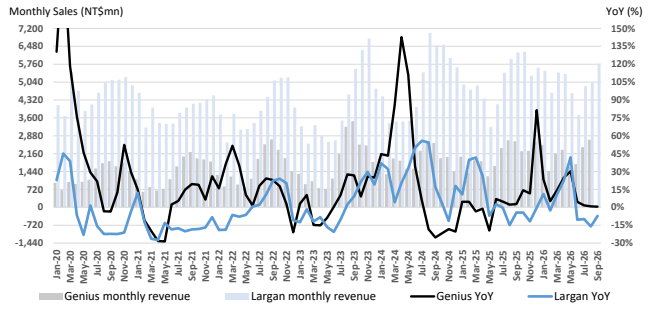
This report has been prepared by UBS Securities Pte. Ltd., Taipei Branch. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 13.**

Core smartphone lens seeing weakness with declining margins

Q326 sales was reported light at NT$15.7bn, +15% QoQ missing UBSe/Street +36%/ +33% amid weaker flagship smartphone demand. GMs fell 720bp to 42.2%, lowest in 13 years on renewable energy costs and low variable-aperture yields, dropping OpM to 17 year lows at 30.3% and EPS to NT$39.6, below UBS/street NT$51.50/NT$49.69. Renewable-energy costs are structural under Largan's long-term power commitments and rise with Apple's share of the mix; with 2027 pricing set, pass-through is more likely in 2028.

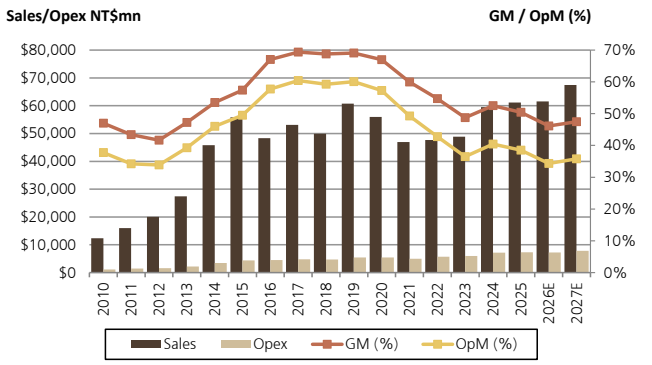
We cut Q4 to NT$16.7bn +7% QoQ, 11% below consensus as its customer trimmed orders after a pricier iPhone 18 launch and possible lag to Genius which is picking up periscope share and outgrew in Q3 (flat vs. -11% YoY). October revenue should be flat MoM and November lower as the key customer has shortened its forecast after cutting flagship orders in September. The spring-2027 model starts shipping in October but will not offset the flagship decline.

Figure 1: Genius outpacing Largan in recent months



Source: Company data

Figure 2: Largan margins dropping in recent years



Source: Company data, UBS estimates

We cut 2026E/27E revenue 9%/13% to NT$61.6bn/NT$67.4bn and EPS 14%/16% to NT$163/NT$180, and introduce 2028E revenue of NT$79.5bn, +18% YoY for the new smartphone lens upgrade, taking EPS to NT$220. Our estimates exclude CPO, which we value separately under base, bull and bear scenarios. Including our base case, 2008E EPS would be NT$255, 6% above the NT$240 consensus which likely includes some CPO. We see GM recovering to 47.5%/48.9% in 2027-28 as yields improve and energy costs pass-through in 2028 pricing. With a still a range of outcomes on 2028 CPO, we keep it in our valuation as a scenario for our upside/downside/base cases.

Figure 3: UBS vs. VA Consensus Q126A-Q426E and 2025A-28E

	Q126	Q226	Q326E			Q426E			2025	2026E			2027E			2028E		
(NT$m)	Actual	Actual	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA	Actual	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA
Revenue	**15,544**	**13,665**	**15,651**	**18,566**	**18,147**	**16,690**	**20,094**	**16,173**	**61,148**	**61,550**	**67,869**	**61,033**	**67,438**	**77,189**	**68,359**	**79,464**	**83,044**	**115,261**
- QoQ / YoY	-9.7%	-12.1%	14.5%	35.9%	32.8%	6.6%	28.4%	3.3%	2.8%	0.7%	11.0%	-0.2%	9.6%	13.7%	12.0%	17.8%	7.6%	68.6%
Gross profit	**7,680**	**6,753**	**6,607**	**9,242**	**8,968**	**7,346**	**10,044**	**6,980**	**30,837**	**28,386**	**33,718**	**28,019**	**32,008**	**38,405**	**30,319**	**38,823**	**42,004**	**53,520**
- Gross margin	49.4%	49.4%	42.2%	49.8%	49.4%	44.0%	50.0%	43.2%	50.4%	46.1%	49.7%	45.9%	47.5%	49.8%	44.4%	48.9%	50.6%	46.4%
Operating profit	**5,812**	**5,148**	**4,735**	**7,144**	**6,819**	**5,443**	**7,833**	**5,105**	**23,558**	**21,139**	**25,937**	**20,800**	**24,150**	**29,796**	**21,482**	**30,110**	**34,945**	**39,917**
- OP margin	37.4%	37.7%	30.3%	38.5%	37.6%	32.6%	39.0%	31.6%	38.5%	34.3%	38.2%	34.1%	35.8%	38.6%	31.4%	37.9%	42.1%	34.6%
Non-op profit	**1,476**	**951**	**1,357**	**1,027**	**1,071**	**1,064**	**1,064**	**1,477**	**2,342**	**4,847**	**4,518**	**5,261**	**4,467**	**4,467**	**4,734**	**4,595**	**4,595**	**4,937**
Pre-tax profit	**7,288**	**6,099**	**6,092**	**8,171**	**7,889**	**6,507**	**8,897**	**6,582**	**25,900**	**25,986**	**30,455**	**26,061**	**28,618**	**34,263**	**26,543**	**34,705**	**39,540**	**44,854**
Tax	**-1,112**	**-1,373**	**-853**	**-1,385**	**-1,250**	**-1,041**	**-1,499**	**-1,306**	**-4,340**	**-4,379**	**-5,369**	**-4,763**	**-5,050**	**-6,244**	**-5,150**	**-5,900**	**-6,722**	**-8,572**
- Tax rate	15.3%	22.5%	14.0%	17.0%	15.8%	16.0%	16.9%	19.8%	16.8%	16.9%	17.6%	18.3%	17.6%	18.2%	19.4%	17%	17%	19%
Net profit	**6,123**	**4,670**	**5,182**	**6,736**	**6,614**	**5,409**	**7,348**	**5,275**	**21,275**	**21,385**	**24,877**	**21,298**	**23,567**	**28,019**	**21,414**	**28,805**	**32,818**	**36,282**
EPS (NT$)	**46.63**	**35.70**	**39.62**	**51.50**	**49.69**	**41.36**	**56.18**	**40.04**	**159.41**	**163.33**	**190.00**	**161.25**	**180.00**	**214.00**	**161.64**	**220.00**	**250.66**	**275.28**

Source: Company data, Visible Alpha, UBS estimates. Note: EPS based on reported and may differ with cover page diluted EPS.

CPO fiber array qualified, but rising pass-through as it upgrades to FAU production

Largan's fiber array (FA) has passed qualification with a volume customer, and management expects mass-production capacity by mid-2027 in a newly completed fab dedicated to FA/FAU, using in-house equipment. The company's next step is to produce the full detachable FAU mounted on a carrier, a board with three alignment beams that sits on the optical engine's receptor. Largan can build FAUs now and says the customer

would prefer it to supply the full assembly, but qualification is pending with the customer preferring FAU and carrier sampled separately.

Even as its potential to supply the full FAU improve, Largan will buy in most of the content. The customer has qualified another PMLA vendor and prefers semiconductor-made optics; V-grooves and carrier beams and boards are outsourced and the customer does not want Largan to make MT ferrules. Management described its role as closer to a contractor and said margins hinge on yields, as the FA carries a high ASP but also a large bill of materials. Supplying the complete FAU could raise billed content, but a larger purchased bill of materials also dilutes the relationship between revenue and value added. Our full-FAU ASP assumption requires Largan to capture that invoice scope.

Next year's product is grating-coupled and works for both CPO and NPO while edge-coupled designs are being explored with the customer. The current product does not go directly to end customers, but management expects to work directly with end customers on next-generation designs (likely 2028) as it progresses toward multi-row FAs.

To ramp production, Largan has bought several land parcels and will build new fabs from scratch, with construction finishing around 2030. It may outsource some capacity in the interim and said it cannot yet be sure capacity will meet demand.

Annual FAU TAM reaches US$1.85bn in 2028 and US$4.5bn in 2030 in our base case

We update our FAU (fiber array unit) TAM factoring CPO/NPO penetration into AI networking switches, NVIDIA multi-rack systems, scale-up on compute and switch boards, and non-NVIDIA accelerators. We assume one passive FAU assembly connecting a fiber array to a photonic engine with one unit per optical engine.

We assume one FAU per optical engine and 32 per switch ASIC in our base case, matching NVIDIA's 32x3.2T engines on its 102.4T Spectrum-X package. Broadcom's 102.4Tb/s Davisson uses 16 engines at 6.4Tb/s, so our bear case assumes and even mix of these two designs (24 per ASIC). We note that higher switch bandwidth does not automatically increase FAU count if engine bandwidth rises at the same rate. Our 2030 assumption maintains the 32 FAUs per switch based on bandwidth rising to 204.8Tb/s ASICs using 6.4Tb/s optical engines.

Figure 4: UBS scenario based on 32 FAUs per switch although switches to date carry a range of specifications

	TH5 Bailly	TH6 Davisson	Quantum-X Q3450	Spectrum-X SN6810	Spectrum-X SN6800
Year of Introduction	**2023**	**H225**	**H225**	**H226**	**H226**
Switch ASIC	Tomahawk 5	Tomahawk 6	Quantum-3	Spectrum 6	Spectrum 6
Bandwidth per Switch box	51.2T	102.4T	115.2T	102.4T	409.6T
Switch ASICs per box	1	1	4	1	4
OE bandwidth (Tb/s each way)	6.4T	6.4T	1.6T	3.2T	3.2T
Optical engines per ASIC	8	16	18	32	32
FAU per Engine	1	1	1	1	1
FAU per Switch	**8**	**16**	**72**	**32**	**128**

Source: Company Data, UBS

Our base case for the FAU market sums up 4 opportunities for CPO/NPO attach:

1. Scale-out and scale-across switches

We estimate annual switch ASIC shipments x CPO + NPO penetration x optical engines per ASIC x FAUs per engine. In 2028, 25% CPO/NPO penetration or 1.0m 102.4T switch ASICs at 32 FAUs each produces 8.0m FAUs. In 2030, 55% of 1.1mn 204.8T switch ASICs at 32 FAUs drives a market size of 19.4mn FAUs.

Figure 5: Scale-out/scale-across could drive 8.0/19.4mn FAUs in 2028/2030

	2028			2030			Assumption
Scale out/Scale across	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
Switch ASIC shipments (k units)	800	1,000	1,050	880	1,100	1,155	
NPO penetration	3%	5%	7%	8%	10%	12%	Optics on the board beside the ASIC
CPO penetration	10%	20%	25%	35%	45%	55%	Optics inside the switch package.
NPO/CPO switch ASICs (k units)	**104**	**250**	**336**	**378**	**605**	**774**	
Bandwidth per Switch (Tbps)	102.4	102.4	102.4	204.8	204.8	204.8	
Optical engine (Tb)	3.2	3.2	3.2	6.4	6.4	6.4	
FAUs per switch ASIC	24	32	32	24	32	32	NVIDIA: 32 OE per ASIC, Broadcom 16
Scale out/Scale across # of FAUs (mn)	**2.5**	**8.0**	**10.8**	**9.1**	**19.4**	**24.8**	

Source: UBS

2. NVIDIA's multi-rack connections for Rubin Ultra and Feynman

NVIDIA at its 2026 GTC San Jose proposed a multi-rack option to network together 8 Oberon NVL72 racks for the Rubin Ultra generation and 8 Kyber NVL144 racks. Feedback at the time showed possible transceiver or NPO connections for Rubin Ultra and CPO connections for Feynman. Our 2028 base case uses 110,000 racks, 15% multi-rack adoption and 75% CPO/NPO penetration, giving approximately 12,400 connected racks.

For 2030, our rack count falls to 35,000 racks as we assume Kyber would have 4x the number of GPUs, equivalent to 140k Oberon equivalents. We assume the 35,000 racks at 55% multi-rack adoption x 80% CPO/NPO attach gives 15,400 connected racks. We assume about 216 and 576 inter-rack FAUs per rack, respectively, driving a market opportunity of 2.7mn FAUs in 2028 and 8.9mn in 2030.

Figure 6: NVIDIA rack connected pods could drive 2.7mn/8.9mn FAUs in 2028/2030

	2028			2030			Assumption
NVIDIA Pod Connected GPUs	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
NV platform	Oberon / Rubin Ultra			Kyber / Feynman			
NV racks shipments (k)	95	110	125	30	35.0	40	UBS rack assumptions
8-rack pods penetration (%)	10%	15%	20%	40%	55%	70%	
NPO/CPO penetration within pods (%)	50%	75%	80%	65%	80%	95%	Pluggable FAUs excluded from this model
NPO/CPO connected racks (k)	**4.8**	**12.4**	**20.0**	**7.8**	**15.4**	**26.6**	
GPUs per rack	72	72	72	144	144	144	
Switches per rack	72	72	72	144	144	144	72 ASICs in Oberon, 144 per Kyber
Optical engine (Tb) bandwidth	3.2	3.2	3.2	6.4	6.4	6.4	
FAUs attached per switch	2.5	3.0	4.0	2.5	4.0	4.5	3.2T OE in 2028; 6.4T in 2030 (each way)
NVLink bandwidth per NVIDIA GPU	14.4	14.4	14.4	28.8	28.8	28.8	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
FAUs per connected racks	**180**	**216**	**288**	**360**	**576**	**648**	
NV scale up # of FAUs (mn)	**0.9**	**2.7**	**5.8**	**2.8**	**8.9**	**17.2**	

Source: UBS. Switch count increases to connect to other racks

3. Initial CPO/NPO optical connections on compute and switch boards by 2030

We include by 2030 CPO native optical connections on compute and switch boards in 5% of racks each adding 1,152 FAUs across the GPUs and switches or 2.0mn FAUs by 2030.

Figure 7: NVIDIA compute board optics could drive 2.0mn FAUs in 2030

	2028			2030			Assumption
NVIDIA Compute Board Optics	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
NV racks shipments (k)	95	110	125	30	35.0	40	UBS rack assumptions
Optical GPU penetration	0%	0%	0%	3%	5%	15%	CPU-native GPUs
Compute tray FAUs #	0	0	0	144	144	144	36 trays x 16
GPUs per rack	72	72	72	144	144	144	
Optical engine (Tb)	3.2	3.2	3.2	6.4	6.4	6.4	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
NVLink bandwidth per NVIDIA GPU	14.4	14.4	14.4	28.8	28.8	28.8	
FAUs per connected racks	0	0	0	720	1,152	1,296	
NV optical GPU # of FAUs (mn)	**0.0**	**0.0**	**0.0**	**0.6**	**2.0**	**7.8**	

Source: UBS.

4. Non NVIDIA scale-up opportunity

Non-NVIDIA scale-up uses accelerator rack shipments x optical attach rate x FAUs per rack, producing about 0.5m units in 2028 and 3.7m in 2030.

Figure 8: ASIC market could drive additional compute board optics could drive 3.7mn FAUs in 2030

	2028			2030			Assumption
Non-NV scale up	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
ASIC rack shipments (k)	43.6	48.5	50.9	76.6	85.1	89.3	*Trainium 4/5/6, MI500, MI600, Jalapeno
Addressable NPO/CPO attach rate (%)	0%	5%	10%	5%	15%	25%	
NPO/CPO connected racks (k)	**0.0**	**2.4**	**5.1**	**3.8**	**12.8**	**22.3**	
Accelerators per rack	72	72	72	72	72	72	Average accelerators per rack
FAUs per accelerator	2.5	3.0	4.0	2.5	4.0	4.5	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
Scale-up per non-NVIDIA accelerator (Tb)	14.4	14.4	14.4	28.8	28.8	28.8	
FAUs per connected racks	180	216	288	180	288	324	
Non-NV # of FAUs (mn)	**0.0**	**0.5**	**1.5**	**0.7**	**3.7**	**7.2**	

Source: UBS

Overall market opportunity: 12.3mn/35.6mn generates a US$1.85bn/US $4.5bn TAM in 2028/2030

We sum four pools: scale-out/scale-across switches, NVIDIA multi-rack pods, CPO native NVIDIA GPUs and non-NVIDIA scale-up. We add 10%/5% for FAUs lost at downstream attach and test and for field spares, driving a market size of in our base case of 12.3mn FAUs in 2028 and 35.6mn in 2030. At US$150/US$125 per FAU, that is a US$1.85bn market in 2028 and US$4.5bn in 2030 (bear/bull US$0.5bn/US$3.5bn in 2028; US $1.4bn/US$9.0bn in 2030). The expansion to 2030 comes from wider optical adoption and greater content in scale-up systems, despite an assumed FAU ASP decline from US $150 to US$125.

Figure 9: Summary of FAU drivers: 12.3mn/35.6mn base case in 2028-2030

	2028			2030		
Summary of FAU drivers	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**
Scale out/Scale across # of FAUs (mn)	2.5	8.0	10.8	9.1	19.4	24.8
NV scale up # of FAUs (mn)	0.9	2.7	5.8	2.8	8.9	17.2
NV optical GPU # of FAUs (mn)	0.0	0.0	0.0	0.6	2.0	7.8
Non-NV # of FAUs (mn)	0.0	0.5	1.5	0.7	3.7	7.2
Downstream yield loss and spares	10%	10%	10%	5%	5%	5%
FAUs (mn units)	**3.7**	**12.3**	**19.8**	**13.9**	**35.6**	**59.9**
FAU ASP	$125	$150	$175	$100	$125	$150
Total FAU TAM (US$bn)	**$0.5**	**$1.8**	**$3.5**	**$1.4**	**$4.5**	**$9.0**

Source: UBS

Scenario analysis on Largan's CPO opportunity

Our base case gives Largan the benefit of the doubt for successfully ramping up a full FAU production and capturing 25% of the overall FAU TAM, yielding NT$35 of 2028E EPS, about 16% of our 2028E lens EPS. A 25% share means 3.1mn FAUs in its first full year of production: about 29% of NVIDIA-linked units and roughly three-quarters of the allocation our checks suggest its module partner holds. We book these at the full US $150 FAU price and 50% gross margin, allowing for an early-supply premium even though the optics, V-grooves and carrier parts are bought in. That gives 2028E CPO sales of US$462mn, rising to US$1.1bn and NT$84 of EPS in 2030 on a larger market.

Our bear case (15% share, 45% GM) gives NT$5 of CPO EPS in 2028E and NT$14 in 2030E; our bull case (35% share, 55% GM) gives NT$103 and NT$267. Each 5ppt of share moves 2028E CPO EPS by ~NT$6.9 (~NT$340/share at 49x), and each 5ppt of gross margin by ~NT$4.6 (~NT$225/share). If gross margin were 35% rather than 50%, holding operating expenses at 12% of sales, EPS would fall to approximately NT$21. A next-generation shift in lens or coupling design could also change content and alignment requirements including a potential metalens design which could invite in different suppliers.

Our 2030 base case implies ~8.9mn Largan FAUs a year (21mn in the upside case), though with risks around competition in a crowded field, yields and technology. A shift to semiconductor-made optics or metalens would loosen alignment tolerances and erode the precision premium behind Largan's pitch. The upside lies in direct work with cloud providers on multi-row FAs from 2028, where Largan's multi-layer alignment is most differentiated.

Figure 10: Largan downside/base/upside cases for 2028 and 2030

Largan CPO	2028			2030		
Scenario analysis	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**
FAUs (mn units)	3.7	12.3	19.8	13.9	35.6	59.9
FAU ASP	125	150	175	100	125	150
Total FAU TAM (US$mn)	**$461**	**$1,847**	**$3,461**	**$1,389**	**$4,452**	**$8,980**
Largan Sales (US$mn)	**$69.1**	**$461.9**	**$1,211.3**	**$208.3**	**$1,113.1**	**$3,142.8**
Largan share of the TAM (%)	15%	25%	35%	15%	25%	35%
GM %	45%	50%	55%	45%	50%	55%
Pretax %	33%	38%	43%	33%	38%	43%
Pretax income (US$mn)	**$22.8**	**$175.5**	**$520.8**	**$68.7**	**$423.0**	**$1,351.4**
Tax	18%	18%	18%	18%	18%	18%
Net income (US$mn)	**$18.7**	**$143.9**	**$427.1**	**$56.4**	**$346.8**	**$1,108.2**
Net income (NT$mn)	**$589**	**$4,533**	**$13,453**	**$1,776**	**$10,925**	**$34,907**
EPS (NT$)	**$5**	**$35**	**$103**	**$14**	**$84**	**$267**
Shares	130.8	130.8	130.8	130.8	130.8	130.8

Source: UBS

Valuation and investment implications

We roll our valuation forward to 2028E and value Largan on a sum of the parts. We value the lens business at NT$3,300 (15x 2028E EPS, the middle of Largan's 13–18x historical range) and CPO at NT$1,700 (49x our base-case 2028E CPO EPS of NT$35, near the top of optical peers' 14–54x). Our base case gives Largan 25% of a 12.3mn-unit, US$1.85bn FAU market at US$150 per unit and a 50% gross margin. The 49x multiple also assigns a substantial value to a business whose complete assembly still requires qualification and whose margins still need to demonstrate can reach similar levels to its mature smartphone business. At ~NT$6,660 recent share price level, the shares imply ~NT$69 of 2028E CPO EPS, about twice our base case.

Upside risks include faster full-FAU qualification, firm volume commitments from additional customers, stronger lens demand and better yields. Near-term catalysts are the customer's capacity forecast and its visit to the new fab in the coming month. A large forecast or faster FAU plus carrier qualification would be upside risks to our sell.

Figure 11: UBS SOTP valuation for Largan

(NT$)		UBSe			Prior UBS		Upside Case			Downside Case		
	2025	**2026**	**2027**	**2028**	**2026**	**2027**	**2026**	**2027**	**2028**	**2026**	**2027**	**2028**
Lens Business ex-CPO												
Sales	$61,148	$61,550	$67,438	$79,464	$67,869	$77,189	$62,166	$70,135	$82,642	$59,704	$64,066	$75,491
YoY Growth	2.8%	0.7%	9.6%	17.8%	11.0%	13.7%	1.7%	12.8%	17.8%	-2.4%	7.3%	17.8%
Gross Profit	**$30,837**	**$28,386**	**$32,008**	**$38,823**	**$33,718**	**$38,405**	**$28,981**	**$34,410**	**$41,699**	**$26,997**	**$29,446**	**$35,750**
GM%	**50.4%**	**46.1%**	**47.5%**	**48.9%**	**49.7%**	**49.8%**	**46.6%**	**49.1%**	**50.5%**	**45.2%**	**46.0%**	**47.4%**
Opex	$7,279	$7,247	$7,857	$8,714	$7,781	$8,609	$7,283	$8,015	$8,888	$7,138	$7,661	$8,496
Op Profit	**$23,558**	**$21,139**	**$24,150**	**$30,110**	**25,937**	**29,796**	**21,697**	**$26,396**	**$32,811**	**19,858**	**$21,785**	**$27,254**
OpM%	**38.5%**	**34.3%**	**35.8%**	**37.9%**	**38.2%**	**38.6%**	**34.9%**	**37.6%**	**39.7%**	**33.3%**	**34.0%**	**36.1%**
Non-Op	$2,342	$4,847	$4,467	$4,595	$4,518	$4,467	$4,847	$4,467	$4,595	$4,847	$4,467	$4,595
Pretax	$25,900	$25,986	$28,618	$34,705	$30,455	$34,263	$26,545	$30,863	$37,405	$24,706	$26,253	$31,849
Tax	($4,340)	($4,379)	($5,050)	($5,900)	($5,369)	($6,244)	($4,473)	($5,447)	($6,601)	($4,163)	($4,633)	($5,621)
Net income	**$21,275**	**$21,385**	**$23,567**	**$28,805**	**$24,877**	**$28,019**	**$21,849**	**$25,416**	**$30,804**	**$20,321**	**$21,620**	**$26,228**
EPS	**$159.41**	**$163.33**	**$180.00**	**$220.00**	**$190.00**	**$214.00**	**$166.88**	**$194.12**	**$235.28**	**$155.20**	**$165.13**	**$200.32**
Shares	133	131	131	131	131	131	131	131	131	131	131	131
EPS vs. base case							**-94.1%**	**7.8%**	**6.9%**	**-5.0%**	**-8.3%**	**-8.9%**
Current Price		**$6,660**										
P/E				15		15			16			14
Implied Valuation				**$3,300**		**$3,270**			**$3,670**			**$2,810**
CPO Scenario												
EPS				**$35**		**$35**			**$103**			**$5**
P/E				49		50			52			47
Implied Valuation				**$1,700**		**$1,730**			**$5,300**			**$210**
Total Largan												
Implied Share Price				**$5,000**		**$5,000**			**$8,970**			**$3,020**
Upside/Downside				**-25%**					**35%**			**-55%**

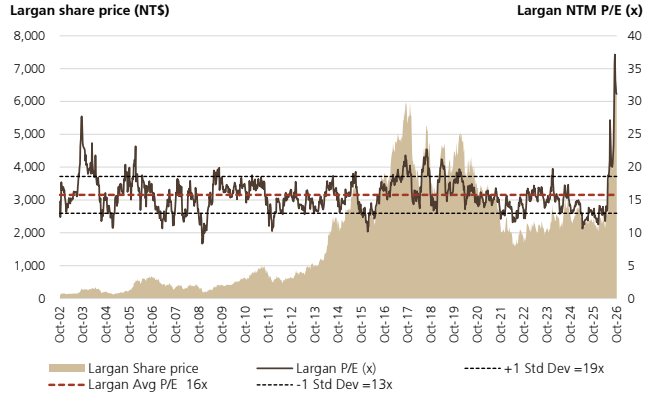
Source: Company data, UBS estimates

Figure 12: Optical component peers average 27x 2028 with a broad range of 12-75x 2028 P/E

Optical Company	Ticker	Market cap (US$m)	Price 11-Oct (LC$)	Price target (LC$)	P/E target (x)	Rating	EPS (LC$) 2025	EPS (LC$) 2026E	EPS (LC$) 2027E	EPS (LC$) 2028E	Earnings growth (%) 2026E	Earnings growth (%) 2027E	Earnings growth (%) 2028E	PE (x) 2026E	PE (x) 2027E	PE (x) 2028E	ROE (%) 2026E	ROE (%) 2027E	ROE (%) 2028E	Historical P/E (x) -1 Std Dev	Historical P/E (x) Avg.	Historical P/E (x) +1 Std Dev
Lens																						
Largan	3008.TW	27,378	6,660.0	5,000	23.4	Sell	176.92	188.21	214.00	250.66	4.4	13.7	17.1	35.4	31.1	26.6	12.7	13.7	14.8	13.1	15.9	18.8
Genius	3406.TW	3,741	1,060.0	1,250	16.9	Buy	36.28	38.72	41.47	47.75	6.7	7.1	15.1	27.4	25.6	22.2	15.8	15.6	16.4	1.1	2.3	3.5
Average											**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**
CPO Supply Chain																						
Lumentum	LITE.O	93,344	1,103.4	960	40.4	Neutral	2.12	10.52	23.78	35.81	434.4	160.4	51.7	52.0	56.0	37.9	27.1	36.1	37.9	8.5	19.4	30.3
Corning	GLW.N	137,095	156.7	201	42.1	Buy	2.57	3.34	4.78	7.12	30.6	43.5	47.3	47.7	33.4	22.4	22.9	31.8	41.1	-168.2	32.9	234.0
Coherent	COHR.N	123,129	312.6	-	-	-	4.49	7.42	11.74	16.82	86.0	64.0	42.0	42.1	26.6	18.6	13.5	19.7	23.8	14.1	19.6	25.1
Senko	9069.T	416,040	2,368.0	-	-	-	119.98	140.89	159.70	171.34	19.5	12.8	7.4	16.8	14.8	13.8	10.4	11.3	11.6	9.2	11.6	14.0
TFC	300394.SZ	256,388	235.0	-	-	-	2.02	3.32	5.36	6.88	65.5	60.7	28.5	70.8	43.9	34.2	47.0	47.4	42.1	21.2	33.9	46.5
FOCI	3363.TWO	75,798	667.0	-	-	-	0.85	0.22	13.89	43.01	(66.3)	6684.0	210.2	3,102.3	48.0	15.5	-1.6	39.2	74.6	51.6	115.2	178.8
EZconn	6442.TW	146,044	1,785.0	-	-	-	20.95	62.01	130.00	NULL	206.9	109.6	n.a.	28.8	13.7	n.a.	56.1	63.9	NULL	12.9	23.7	34.5
Himax	HIMX.N	2,320	13.3	-	-	-	0.32	0.52	1.01	1.96	62.3	88.2	93.5	25.5	13.2	6.8	10.5	14.5	26.1	0.6	21.5	42.3
Browave	3163.TWO	65,204	689.0	-	-	-	4.36	9.33	21.17	NULL	n.a.	n.a.	n.a.	73.8	32.5	n.a.	22.2	38.6	NULL	10.1	32.2	54.2
VisEra	6789.TW	153,102	481.0	-	-	-	3.60	6.13	7.45	8.95	70.9	21.5	20.2	78.4	64.6	53.7	10.7	13.3	12.2	31.3	50.4	69.6
Average											**101.1**	**805.0**	**62.6**	**353.8**	**34.7**	**25.4**	**21.9**	**31.6**	**33.7**	**-0.9**	**36.0**	**72.9**
Optical Average											**76.4**	**606.4**	**50.7**	**325.1**	**35.5**	**27.2**	**19.9**	**27.5**	**28.6**	**1.2**	**30.5**	**59.8**

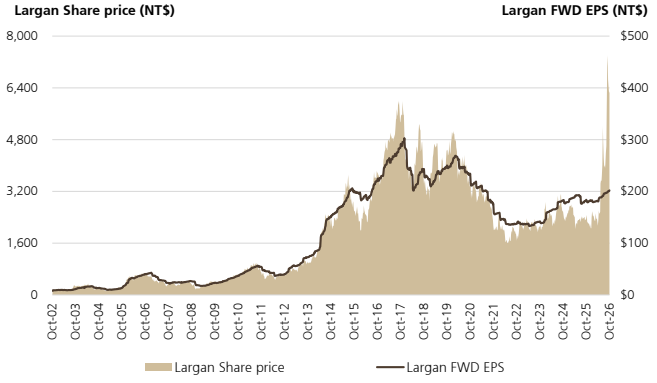
Source: UBSe, Eikon

Figure 13: Share price rally has re-rated Largan well above its historical range



Source: LSEG Workspace

Figure 14: Largan's share price has run well ahead of forward EPS as investors price in CPO



Source: LSEG Workspace

Figure 15: UBS P&L for Largan - mild growth at lower margins in recent years

Ticker		3008.TW	Current Price (NT$)			$6,660
Shares outstanding (mn)		130.8	Mkt cap (NT$ mn)			$871,115
Year	**2022**	**2023**	**2024**	**2025**	**2026E**	**2027E**
Revenues (NT$ mn)	**$47,675**	**$48,842**	**$59,458**	**$61,148**	**$61,550**	**$67,438**
YoY Growth (%)		**2.4%**	**21.7%**	**2.8%**	**0.7%**	**9.6%**
Gross profit (NT$mn)	$26,083	$23,794	$31,209	$30,837	$28,386	$32,008
GM (%)	**54.7%**	**48.7%**	**52.5%**	**50.4%**	**46.1%**	**47.5%**
Operating profit (NT$mn)	$20,384	$17,807	$24,033	$23,558	$21,139	$24,150
OPM (%)	**42.8%**	**36.5%**	**40.4%**	**38.5%**	**34.3%**	**35.8%**
Net profit (NT$ mn)	$22,625	$17,902	$25,915	$21,275	$21,385	$23,567
EPS (NT$)	**$169.52**	**$134.13**	**$194.17**	**$159.41**	**$163.33**	**$180.00**
EPS growth (%)		-21%	45%	-18%	2%	10%
P/E (x)	**39.3**	**49.7**	**34.3**	**41.8**	**40.8**	**37.0**
Dividend yield (%)	2%	1.1%	1.2%	1.3%	2.0%	1.2%
P/B (x)	**5.7**	**5.4**	**4.8**	**4.7**	**4.4**	**4.1**
ROE (%)	15%	11%	14%	11%	11%	12%

Quarters	**Q325**	**Q425**	**Q126**	**Q226**	**Q326E**	**Q426E**
Revenues (NT$ mn)	**$17,677**	**$17,219**	**$15,544**	**$13,665**	**$15,651**	**$16,690**
QoQ Growth (%)		**-2.6%**	**-9.7%**	**-12.1%**	**14.5%**	**6.6%**
Gross profit (NT$mn)	$8,352	$8,260	$7,680	$6,753	$6,607	$7,346
GM (%)	**47.2%**	**48.0%**	**49.4%**	**49.4%**	**42.2%**	**44.0%**
Operating profit (NT$mn)	$6,265	$6,329	$5,812	$5,148	$4,735	$5,443
OPM (%)	**35.4%**	**36.8%**	**37.4%**	**37.7%**	**30.3%**	**32.6%**
Net profit (NT$ mn)	$7,080	$6,720	$6,123	$4,670	$5,182	$5,409
EPS (NT$)	**$53.05**	**$50.35**	**$45.88**	**$34.99**	**$39.58**	**$41.32**

Source: Company data, UBS estimates. Note: 2026E EPS based on reported.

Figure 16: UBS Cash Flow Summary - staying positive to invest in CPO

Annual (NT$mn)	2021	2022	2023	2024	2025	2026E	2027E	21-25 Avg
Revenue	$46,962	$47,675	$48,842	$59,458	$61,148	$61,550	$67,438	$52,817
Capital spending	$5,850	$8,256	$8,080	$11,126	$12,338	$11,721	$11,135	$9,130
Capex/revenue (%)	**12%**	**17%**	**17%**	**19%**	**20%**	**19%**	**17%**	**17%**
Dep and amort	$4,650	$5,031	$5,299	$6,038	$7,511	$9,221	$10,460	$5,706
Depr/revenue (%)	**10%**	**11%**	**11%**	**10%**	**12%**	**15%**	**16%**	**11%**
Operating cash flow	$20,878	$44,207	$18,199	$31,579	$26,080	$20,614	$33,003	$28,188
Free cash flow	**$15,028**	**$35,951**	**$10,118**	**$20,453**	**$13,743**	**$8,893**	**$21,868**	**$19,058**

FCF and Dividends	2021	2022	2023	2024	2025	2026E	2027E	21-25 Avg
FCF / share (NT$)	$112	$269	$76	$153	$103	$68	$167	$143
FCF / EV Yield (%)	**2%**	**5%**	**1%**	**3%**	**2%**	**1%**	**3%**	**3%**
FCF Yield (%)	**2%**	**4%**	**1%**	**2%**	**2%**	**1%**	**3%**	**2%**
Dividend/share (NT$)	$91.51	$109.66	$72.50	$81.00	$85.50	$131.95	$81.89	$88.03
Payout rate (%)	**50%**	**79%**	**43%**	**60%**	**44%**	**83%**	**50%**	**55%**
Dividend amount (NT$)	$12,274	$14,636	$9,676	$10,811	$11,412	$17,276	$10,722	$11,762
Pre-Dividend close (NT$)	$3,040.00	$1,990.00	$2,310.00	$3,120.00	$2,260.00	$6,660.00	$6,660.00	
Dividend Yield (%)	**3.0%**	**5.5%**	**3.1%**	**2.6%**	**3.8%**	**2.0%**	**1.2%**	**3.6%**

Source: Company data, UBS estimates

Largan Precision (3008.TW)

Income Statement (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**48,842**	**59,458**	**61,148**	**61,550**	*0.7*	**67,438**	*9.6*	**79,464**	**78,546**	**81,569**
Gross profit	23,794	31,209	30,837	28,386	*-7.9*	32,008	*12.8*	38,823	38,398	39,903
EBITDA (UBS)	**23,106**	**30,070**	**31,069**	**30,360**	*-2.3*	**34,610**	*14.0*	**41,028**	**41,108**	**42,938**
Depreciation & amortisation	(5,299)	(6,038)	(7,511)	(9,221)	*-22.8*	(10,460)	*-13.4*	(10,918)	(11,621)	(12,290)
EBIT (UBS)	**17,807**	**24,033**	**23,558**	**21,139**	*-10.3*	**24,150**	*14.2*	**30,110**	**29,487**	**30,648**
Associates & investment income	212	0	3	12	*352.1*	11	*-7.1*	11	11	11
Other non-operating income	294	303	419	458	*9.5*	200	*-56.4*	200	200	200
Net interest	3,934	4,404	4,259	3,946	*-7.3*	4,256	*7.9*	4,384	4,516	4,651
Exceptionals (incl goodwill)	(145)	3,435	(2,338)	432	*-*	0	*-*	0	0	0
Pre-tax profit	**22,102**	**32,174**	**25,900**	**25,986**	*0.3*	**28,618**	*10.1*	**34,705**	**34,213**	**35,510**
Tax	(4,200)	(5,963)	(4,340)	(4,379)	*-0.9*	(5,050)	*-15.3*	(5,900)	(5,816)	(6,037)
Profit after tax	**17,902**	**26,211**	**21,560**	**21,607**	*0.2*	**23,567**	*9.1*	**28,805**	**28,397**	**29,473**
Preference dividends	0	0	0	0	*-*	0	*-*	0	0	0
Minorities	0	(296)	(285)	(222)	*22.1*	0	*-*	0	0	0
Extraordinary items	0	0	0	0	*-*	0	*-*	0	0	0
Net earnings (local GAAP)	**17,902**	**25,915**	**21,275**	**21,385**	*0.5*	**23,567**	*10.2*	**28,805**	**28,397**	**29,473**
Net earnings (UBS)	**18,047**	**22,480**	**23,613**	**20,953**	*-11.3*	**23,567**	*12.5*	**28,805**	**28,397**	**29,473**
Tax rate (%)	19.0	18.5	16.8	16.9	*0.6*	17.6	*4.7*	17.0	17.0	17.0

Per Share (NT$)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	135.22	168.43	176.92	160.04	*-9.5*	180.00	*12.5*	220.00	216.89	225.11
EPS (local GAAP, diluted)	134.13	194.17	159.41	163.33	*2.5*	180.00	*10.2*	220.00	216.89	225.11
EPS (UBS, basic)	135.22	168.43	176.92	160.04	*-9.5*	180.00	*12.5*	220.00	216.89	225.11
DPS (net) (NT$)	85.50	67.50	97.50	80.00	*-17.9*	81.97	*2.5*	90.34	110.41	108.85
Cash EPS (UBS, diluted) [1]	174.92	213.67	233.20	230.47	*-1.2*	259.89	*12.8*	303.39	305.65	318.97
Book value per share	1,234.64	1,382.91	1,430.07	1,518.08	*6.2*	1,616.11	*6.5*	1,745.87	1,849.64	1,963.23
Average shares (diluted)	133	133	133	131	*-1.9*	131	*0.0*	131	131	131

Balance Sheet (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	111,655	117,985	122,309	109,131	*-10.8*	120,267	*10.2*	127,842	144,348	160,697
Other current assets	22,666	27,779	31,977	42,218	*32.0*	45,034	*6.7*	56,368	55,712	57,851
Total current assets	**134,321**	**145,764**	**154,285**	**151,350**	*-1.9*	**165,301**	*9.2*	**184,210**	**200,060**	**218,547**
Net tangible fixed assets	41,135	46,936	51,472	54,428	*5.7*	55,103	*1.2*	54,763	53,191	50,448
Net intangible fixed assets	0	0	0	0	*-*	0	*-*	0	0	0
Investments / other assets	19,682	23,827	15,029	23,254	*54.7*	24,367	*4.8*	24,215	24,225	24,236
Total assets	**195,138**	**216,527**	**220,787**	**229,032**	*3.7*	**244,770**	*6.9*	**263,188**	**277,476**	**293,232**
Trade payables & other ST liabilities	29,517	30,375	29,749	30,078	*1.1*	32,982	*9.7*	34,573	35,276	36,158
Short term debt	0	203	0	0	*-*	0	*-*	0	0	0
Total current liabilities	**29,517**	**30,578**	**29,749**	**30,078**	*1.1*	**32,982**	*9.7*	**34,573**	**35,276**	**36,158**
Long term debt	0	0	0	0	*-*	0	*-*	0	0	0
Other long term liabilities	110	561	171	194	*13.7*	194	*0.0*	31	31	31
Preferred shares	0	0	0	0	*-*	0	*-*	0	0	0
Total liabilities (incl pref shares)	**29,627**	**31,139**	**29,919**	**30,272**	*1.2*	**33,176**	*9.6*	**34,604**	**35,307**	**36,189**
Common s/h equity	**165,510**	**185,388**	**190,868**	**198,760**	*4.1*	**211,594**	*6.5*	**228,584**	**242,170**	**257,042**
Minority interests	0	0	0	0	*-*	0	*-*	0	0	0
Total liabilities & equity	**195,138**	**216,527**	**220,787**	**229,032**	*3.7*	**244,770**	*6.9*	**263,188**	**277,476**	**293,232**

Cash Flow (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	17,902	25,915	21,275	21,385	*0.5*	23,567	*10.2*	28,805	28,397	29,473
Depreciation & amortisation	5,299	6,038	7,511	9,221	*22.8*	10,460	*13.4*	10,918	11,621	12,290
Net change in working capital	(8,990)	(6,633)	(7,328)	(1,082)	*85.2*	(6,064)	*NM*	(9,743)	1,359	(1,256)
Other operating	(212)	295	282	210	*-25.5*	5,040	*NM*	(11)	(11)	(11)
Operating cash flow	**13,999**	**25,615**	**21,740**	**29,735**	*36.8*	**33,003**	*11.0*	**29,968**	**41,366**	**40,496**
Tangible capital expenditure	(8,080)	(11,126)	(12,338)	(11,721)	*5.0*	(11,135)	*5.0*	(10,578)	(10,049)	(9,547)
Intangible capital expenditure	0	0	0	0	*-*	0	*-*	0	0	0
Net (acquisitions) & disposals	(2,111)	(13)	(1,200)	962	*-*	0	*-*	0	0	0
Other investing	(959)	(4,927)	4,302	(965)	*-*	0	*-*	0	0	0
Investing cash flow	**(11,151)**	**(16,066)**	**(9,236)**	**(11,724)**	*-26.9*	**(11,135)**	*5.0*	**(10,578)**	**(10,049)**	**(9,547)**
Equity dividends paid	(9,676)	(10,811)	(11,412)	(17,276)	*-51.4*	(10,732)	*37.9*	(11,816)	(14,811)	(14,601)
Share issues / (buybacks)	0	0	0	0	*-*	0	*-*	0	0	0
Other financing	(34)	1,255	(2,937)	(4,133)	*-40.7*	0	*-*	0	0	0
Change in debt & pref shares	(19)	212	(212)	0	*-*	0	*-*	0	0	0
Financing cash flow	**(9,729)**	**(9,344)**	**(14,561)**	**(21,408)**	*-47.0*	**(10,732)**	*49.9*	**(11,816)**	**(14,811)**	**(14,601)**
Cash flow inc/(dec) in cash	**(6,881)**	**205**	**(2,057)**	**(3,398)**	*-65.2*	**11,136**	*-*	**7,575**	**16,506**	**16,349**
FX / non cash items	6,823	6,125	6,380	(9,780)	*-*	0	*-*	0	0	0
Balance sheet inc/(dec) in cash	**(59)**	**6,330**	**4,324**	**(13,177)**	*-*	**11,136**	*-*	**7,575**	**16,506**	**16,349**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

Largan Precision (3008.TW)

Valuation (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	16.4	13.1	14.8	40.8	37.0	30.3	30.7	29.6
P/E (UBS, diluted)	16.2	15.1	13.3	41.6	37.0	30.3	30.7	29.6
P/CEPS	12.6	11.9	10.1	28.9	25.6	22.0	21.8	20.9
Equity FCF (UBS) yield %	2.0	4.3	3.0	2.1	2.5	2.2	3.6	3.5
Dividend yield (net) %	3.9	2.7	4.1	1.2	1.2	1.4	1.7	1.6
P/BV	1.8	1.8	1.6	4.4	4.1	3.8	3.6	3.4
EV/revenues (core)	3.6	3.6	3.1	NM	NM	9.3	9.3	8.7
EV/EBITDA (UBS core)	7.7	7.1	6.0	24.7	21.7	18.1	17.7	16.6
EV/EBIT (core)	10.0	8.9	8.0	35.4	31.1	24.6	24.7	23.2
EV/OpFCF (core)	8.1	7.4	6.3	25.1	22.0	18.0	17.7	16.5
EV/op. invested capital	3.9	4.1	3.3	10.7	9.3	8.6	8.2	8.2

Enterprise value (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	294,465	340,820	315,906	874,578	874,578	874,578	874,578	874,578
Net debt (cash)	(111,675)	(114,718)	(120,045)	(115,720)	(114,699)	(124,054)	(136,095)	(152,522)
Buy out of minorities	0	0	0	0	0	0	0	0
Pension provisions/other	0	0	0	0	0	0	0	0
Total enterprise value	**182,790**	**226,102**	**195,861**	**758,858**	**759,879**	**750,524**	**738,483**	**722,055**
Non core assets	(4,916)	(12,460)	(8,517)	(9,934)	(9,945)	(9,955)	(9,966)	(9,976)
Core enterprise value	**177,874**	**213,642**	**187,344**	**748,924**	**749,934**	**740,568**	**728,517**	**712,079**

Growth (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	2.4	21.7	2.8	0.7	9.6	17.8	(1.2)	3.8
EBITDA (UBS)	(9.1)	30.1	3.3	(2.3)	14.0	18.5	0.2	4.5
EBIT (UBS)	(12.6)	35.0	(2.0)	(10.3)	14.2	24.7	(2.1)	3.9
EPS (UBS, diluted)	3.8	24.6	5.0	(9.5)	12.5	22.2	(1.4)	3.8
Net DPS	21.9	(21.1)	44.4	(17.9)	2.5	10.2	22.2	(1.4)

Margins & Profitability (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	48.7	52.5	50.4	46.1	47.5	48.9	48.9	48.9
EBITDA margin	47.3	50.6	50.8	49.3	51.3	51.6	52.3	52.6
EBIT (UBS) margin	36.5	40.4	38.5	34.3	35.8	37.9	37.5	37.6
Net earnings (UBS) margin	37.0	37.8	38.6	34.0	34.9	36.2	36.2	36.1
ROIC (EBIT)	39.3	46.2	40.9	30.3	30.1	35.1	33.1	35.3
ROIC post tax	31.8	36.6	34.6	25.1	24.8	29.1	27.5	29.3
ROE (UBS)	11.3	12.8	12.6	10.8	11.5	13.1	12.1	11.8

Capital structure & Coverage (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	(4.8)	(3.9)	(3.9)	(3.6)	(3.5)	(3.1)	(3.5)	(3.7)
Net debt / total equity %	(67.5)	(63.5)	(64.1)	(54.9)	(56.8)	(55.9)	(59.6)	(62.5)
Net debt / (net debt + total equity) %	NM	NM	NM	NM	NM	NM	NM	NM
Net debt/EV %	(61.1)	(50.7)	(61.3)	(15.2)	(15.1)	(16.5)	(18.4)	(21.1)
Capex / depreciation %	152.5	184.3	164.3	127.1	106.5	96.9	86.5	77.7
Capex / revenue %	16.5	18.7	20.2	19.0	16.5	13.3	12.8	11.7
EBIT / net interest	-	-	-	-	-	-	-	-
Dividend cover (UBS)	1.6	2.5	1.8	2.0	2.2	2.4	2.0	2.1
Div. payout ratio (UBS) %	63.2	40.1	55.1	50.0	45.5	41.1	50.9	48.4

Revenues by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	48,842	59,458	61,148	61,550	67,438	79,464	78,546	81,569
Total	**48,842**	**59,458**	**61,148**	**61,550**	**67,438**	**79,464**	**78,546**	**81,569**

EBIT (UBS) by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	17,807	24,033	23,558	21,139	24,150	30,110	29,487	30,648
Total	**17,807**	**24,033**	**23,558**	**21,139**	**24,150**	**30,110**	**29,487**	**30,648**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	-24.9%
Forecast dividend yield	1.2%
Forecast stock return	-23.7%
Market return assumption	6.3%
Forecast excess return	-30.0%

Company Description

Founded in 1987, Largan Precision (Largan) is one of the world's leading manufacturers of lenses used in cameras and IT products. It specialises in plastic/hybrid lenses and, to a lesser extent, glass lenses and modules. In 2022, 20MP+ camera lenses were 10-20% of Largan's shipment mix, 10MP lenses were 50-60%, 8MP lenses were 0-10% and other lenses were 30-40%.

Valuation Method and Risk Statement

Our price target is based on SOTP valuation on the CPO opportunity and traditional lens business.
We think Largan's downside risks include: 1) ongoing pricing pressure from smartphone OEM clients and increasingly intense competition among lens suppliers; and 2) a slower-than-expected smartphone demand recovery in China, and 3) CPO commercial success below the market's elevated expectations.
We believe the key upside risk is a faster-than-expected CPO qualification process at Largan. In addition, if the company demonstrates exceptionally efficient and highly accurate automation capabilities, it could support greater market share gains within the industry than we currently anticipate.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Largan Precision

Question	Response
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	3
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	4
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	2
5. What's driving the difference?	We expect downside to consensus in 2026 due to slower smartphone sales through August. We indicate things getting better for the stock in the past 3-6 months on the smartphone traction.
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	2
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	Negative Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

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12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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UBS Securities Pte. Ltd., Taipei Branch: Irene Chang, Randy Abrams.

Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
Largan Precision[28]	3008.TW	Sell	NT$6,660	08 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.
28. UBS holds a long position of 0.5% or more of the listed shares of this company.

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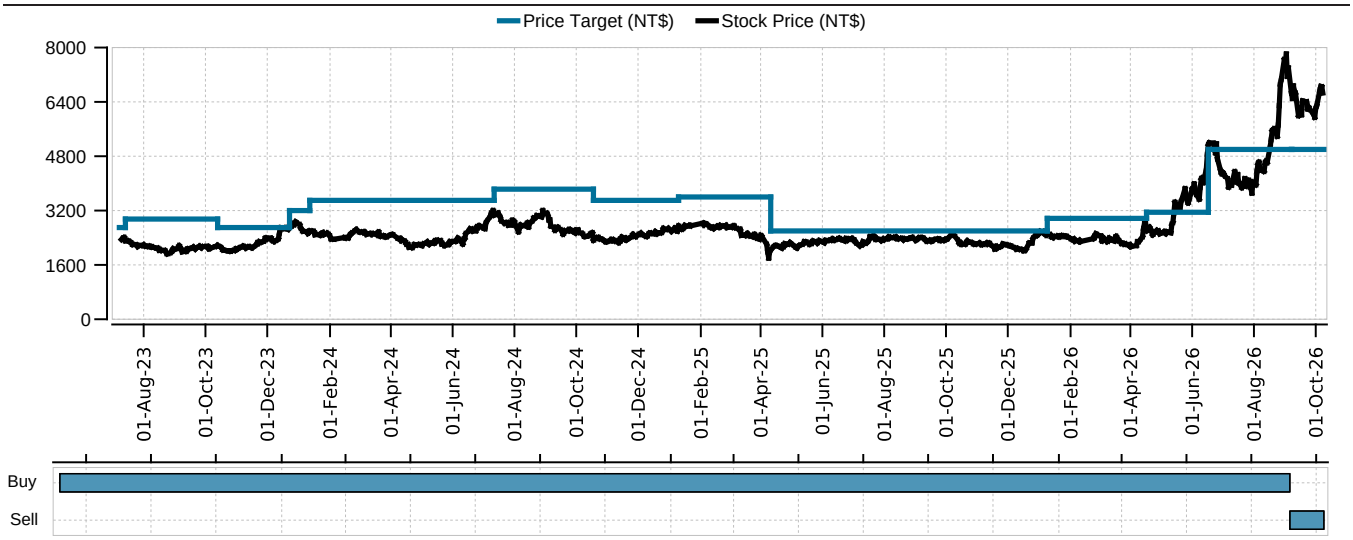
Largan Precision (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-07	2340.00	2700.00	Buy
2023-07-13	2405.00	2950.00	Buy
2023-10-12	2150.00	2700.00	Buy
2023-12-22	2650.00	3200.00	Buy
2024-01-11	2550.00	3500.00	Buy
2024-07-11	3200.00	3830.00	Buy
2024-10-17	2545.00	3500.00	Buy
2025-01-09	2595.00	3600.00	Buy
2025-04-10	1980.00	2600.00	Buy
2026-01-08	2475.00	2970.00	Buy
2026-04-16	2820.00	3150.00	Buy
2026-06-16	4860.00	5000.00	Buy
2026-09-06	7400.00	5000.00	Sell

Source: UBS Global Research; LSEG Eikon as of 08-Oct-2026. All prices as of local market close. Ratings as of date shown.

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Largan (3008.TW)

FAU to drive strong earnings growth and re-rating; TP up to NT$11,000, Buy

3008.TW	12m Price Target: **NT$11,000.00**	Price: **NT$6,660.00**	Upside: **65.2%**

BUY

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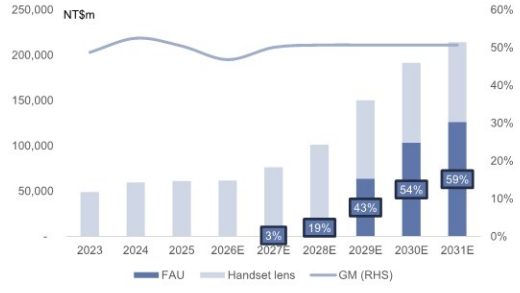
We are positive on Largan and expect its strong alignment technology to continue supporting its expansion from handset lens into FA and FAU for CPO switches. The company's FA has passed customer qualification with strong positive feedback, while customers prefer to purchase its integrated product (FA+FAU), reaffirming our positive view on Largan's business expansion. Management highlights that its strong alignment and automated production provide strong precision, fast production and testing, supporting customer penetration. We model FAU revenue contribution at 3% in 2027E and above 50% in 2030E, driving Largan's earnings growth, margin expansion, and valuation re-rating. Maintain Buy with TP up to NT$11,000 (implies 39.8x 2028E PE).

Exhibit 1: Largan revenues by products and GM trend



Source: Company data, Goldman Sachs Global Investment Research

3-month revenue preview: Largan delivered weak revenue growth in 3Q26 with revenue -11% YoY, which we attribute to a slow smartphone market amid high memory costs, fewer model launches of its major smartphone brand customer, and foldable phones released in 4Q26 (report link). The miss also reflects a slower handset lens pixel mix upgrade than we had previously estimated.

Key Data

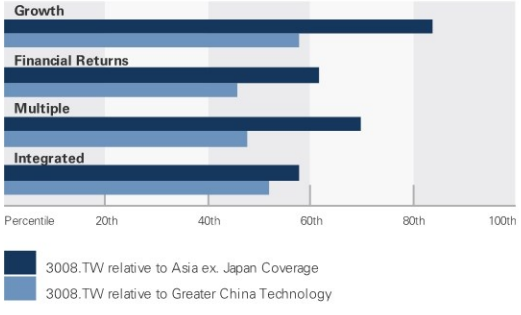
Market cap: NT$896.1bn / $28.1bn
Enterprise value: NT$782.7bn / $24.5bn
3m ADTV: NT$13.2bn / $413.8mn
Taiwan
Greater China Technology
M&A Rank: 3
Leases incl. in net debt & EV?: No

GS Forecast

	12/25	12/26E	12/27E	12/28E
Revenue (NT$ mn) New	**61,147.9**	**61,653.5**	**76,334.7**	**101,306.6**
Revenue (NT$ mn) Old	61,147.9	65,503.4	76,561.8	85,020.5
EBITDA (NT$ mn)	31,290.3	30,251.2	42,329.3	53,616.0
EPS (NT$) New	**159.41**	**162.60**	**207.41**	**278.57**
EPS (NT$) Old	159.41	187.34	217.11	239.29
P/E (X)	14.8	41.0	32.1	23.9
P/B (X)	1.7	4.5	4.2	3.9
Dividend yield (%)	3.3	1.2	1.5	2.1
CROCI (%)	11.2	11.5	14.4	16.2

	9/26	12/26E	3/27E	6/27E
EPS (NT$)	39.62	40.65	49.77	36.05

GS Factor Profile



Source: Company data, Goldman Sachs Research estimates. See disclosures for details.

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BUY

Largan (3008.TW)

Rating since Nov 12, 2024

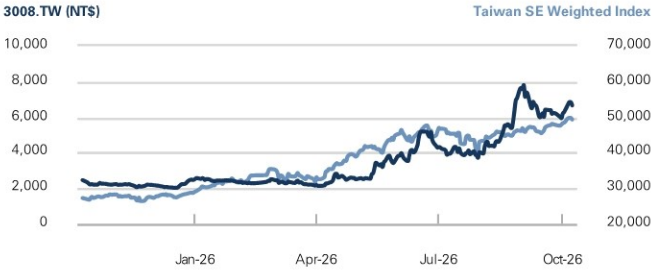
Ratios & Valuation

	12/25	12/26E	12/27E	12/28E
P/E (X)	14.8	41.0	32.1	23.9
P/B (X)	1.7	4.5	4.2	3.9
FCF yield (%)	4.3	1.1	3.7	3.3
EV/EBITDAR (X)	6.4	25.6	17.8	13.8
EV/EBITDA (excl. leases) (X)	6.4	25.6	17.8	13.8
CROCI (%)	11.2	11.5	14.4	16.2
ROE (%)	11.4	11.1	13.4	16.7
Net debt/equity (%)	(64.1)	(60.3)	(66.6)	(68.0)
Net debt/equity (excl. leases) (%)	(64.1)	(60.3)	(66.6)	(68.0)
Interest cover (X)	15,611.8	7,608.9	9,940.7	13,702.5
Days inventory outst, sales	37.1	47.9	40.9	40.4
Receivable days	62.1	62.0	50.0	50.0
Days payable outstanding	21.6	22.0	22.0	22.0
DuPont ROE (%)	11.1	10.7	12.9	15.9
Turnover (X)	0.3	0.3	0.3	0.4
Leverage (X)	1.2	1.2	1.2	1.2
Gross cash invested (ex cash) (NT$)	222,420.9	241,679.6	267,828.7	298,740.5
Average capital employed (NT$)	73,429.5	80,655.1	82,379.7	79,780.5
BVPS (NT$)	1,395.59	1,484.98	1,579.62	1,727.59

Growth & Margins (%)

	12/25	12/26E	12/27E	12/28E
Total revenue growth	2.8	0.8	23.8	32.7
EBITDA growth	3.4	(3.3)	39.9	26.7
EPS growth	(19.7)	2.0	27.6	34.3
DPS growth	(19.0)	2.4	27.0	35.0
EBIT margin	38.5	35.2	39.2	40.7
EBITDA margin	51.2	49.1	55.5	52.9
Net income margin	34.8	35.1	36.3	36.7

Price Performance



	3m	6m	12m
Absolute	68.6%	192.1%	168.0%
Rel. to the Taiwan SE Weighted Index	56.4%	105.9%	47.1%

Source: FactSet. Price as of 8 Oct 2026 close.

Income Statement (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Total revenue	61,147.9	61,653.5	76,334.7	101,306.6
Cost of goods sold	(30,310.8)	(32,786.7)	(38,112.7)	(49,958.8)
SG&A	(1,985.5)	(1,885.5)	(1,832.0)	(1,620.9)
R&D	(5,293.3)	(5,295.8)	(6,488.4)	(8,509.8)
Other operating inc./(exp.)	–	–	–	–
EBITDA	**31,290.3**	**30,251.2**	**42,329.3**	**53,616.0**
Depreciation & amortization	(7,732.0)	(8,565.7)	(12,427.8)	(12,398.9)
EBIT	**23,558.3**	**21,685.5**	**29,901.5**	**41,217.1**
Net interest inc./(exp.)	4,258.6	3,896.3	3,913.0	4,104.2
Income/(loss) from associates	87.0	83.3	–	–
Pre-tax profit	**25,900.2**	**26,446.2**	**33,814.6**	**45,321.3**
Provision for taxes	(4,340.0)	(4,539.3)	(5,883.7)	(7,891.5)
Minority interest	(284.9)	(241.0)	(248.7)	(250.2)
Preferred dividends	–	–	–	–
Net inc. (pre-exceptionals)	**21,275.4**	**21,665.9**	**27,682.2**	**37,179.6**
Post-tax exceptionals	–	–	–	–
Net inc. (post-exceptionals)	**21,275.4**	**21,665.9**	**27,682.2**	**37,179.6**
EPS (basic, pre-except) (NT$)	**159.41**	**162.60**	**207.41**	**278.57**
EPS (diluted, pre-except) (NT$)	**157.21**	**161.03**	**204.55**	**276.22**
EPS (basic, post-except) (NT$)	**159.41**	**162.60**	**207.41**	**278.57**
EPS (diluted, post-except) (NT$)	**157.21**	**161.03**	**204.55**	**276.22**
DPS (NT$)	77.87	79.76	101.32	136.83
Div. payout ratio (%)	48.9	49.1	48.9	49.1

Balance Sheet (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Cash & cash equivalents	115,941.7	115,177.0	136,906.6	152,609.6
Accounts receivable	10,461.7	10,483.6	10,430.0	17,325.2
Inventory	6,713.2	9,455.6	7,669.0	14,778.3
Other current assets	21,168.8	21,168.8	21,168.8	21,168.8
Total current assets	**154,285.5**	**156,285.0**	**176,174.5**	**205,881.9**
Net PP&E	51,472.4	60,978.8	57,943.2	52,866.6
Net intangibles	474.6	351.6	344.5	213.7
Total investments	26.5	26.5	26.5	26.5
Other long-term assets	14,528.1	14,528.1	14,528.1	14,528.1
Total assets	**220,787.0**	**232,169.9**	**249,016.8**	**273,516.7**
Accounts payable	1,727.9	2,224.5	2,369.9	3,652.5
Short-term debt	0.0	0.0	0.0	0.0
Short-term lease liabilities	–	–	–	–
Other current liabilities	28,020.6	28,214.1	31,194.1	35,898.7
Total current liabilities	**29,748.5**	**30,438.5**	**33,564.1**	**39,551.2**
Long-term debt	–	–	–	–
Long-term lease liabilities	–	–	–	–
Other long-term liabilities	170.6	170.6	170.6	170.6
Total long-term liabilities	**170.6**	**170.6**	**170.6**	**170.6**
Total liabilities	**29,919.1**	**30,609.1**	**33,734.6**	**39,721.7**
Preferred shares	–	–	–	–
Total common equity	188,867.2	199,801.1	213,771.1	232,534.0
Minority interest	**2,000.8**	**1,759.8**	**1,511.1**	**1,261.0**
Total liabilities & equity	**220,787.0**	**232,169.9**	**249,016.8**	**273,516.7**
Net debt, adjusted	(122,308.6)	(121,543.8)	(143,273.5)	(158,976.5)

Cash Flow (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Net income	25,900.2	21,665.9	27,682.2	37,179.6
D&A add-back	7,732.0	8,565.7	12,427.8	12,398.9
Minority interest add-back	(284.9)	(241.0)	(248.7)	(250.2)
Net (inc)/dec working capital	(1,208.2)	(2,267.7)	1,985.6	(12,721.8)
Other operating cash flow	(6,058.9)	–	–	–
Cash flow from operations	**26,080.2**	**27,723.0**	**41,846.9**	**36,606.6**
Capital expenditures	(12,340.0)	(17,849.1)	(9,160.2)	(7,091.5)
Acquisitions	(2,040.4)	–	–	–
Divestitures	(2.6)	(100.0)	(225.0)	(100.0)
Others	5,147.2	–	–	–
Cash flow from investing	**(9,235.7)**	**(17,949.1)**	**(9,385.2)**	**(7,191.5)**
Repayment of lease liabilities	–	–	–	–
Dividends paid (common & pref)	(13,013.1)	(10,538.6)	(10,732.1)	(13,712.2)
Inc/(dec) in debt	(203.4)	–	–	–
Other financing cash flows	(1,344.6)	0.0	0.0	0.0
Cash flow from financing	**(14,561.2)**	**(10,538.6)**	**(10,732.1)**	**(13,712.2)**
Total cash flow	**2,283.4**	**(764.7)**	**21,729.7**	**15,703.0**
Free cash flow	13,740.3	9,873.9	32,686.7	29,515.1

Source: Company data, Goldman Sachs Research estimates.

We expect Oct revenues to be at a similar level to Sep, and to sequentially decline in Nov and Dec, resulting in 4Q26 revenue growth of +7% QoQ.

Exhibit 2: Largan monthly revenues preview

	Jul 2026	Aug 2026	Sep 2026	Oct 2026E	Nov 2026E	Dec 2026E	2Q26	3Q26	4Q26E
Rev (NT$m)	4,877	5,012	5,779	5,895	5,600	5,299	13,665	15,651	16,794
YoY	-11%	-16%	-7%	-6%	6%	-6%	17%	-11%	-2%
MoM/QoQ	31%	3%	15%	2%	-5%	-5%	-12%	15%	7%
GS estimates (NT$m)	4,826	6,756	7,285				13,750	18,866	
Act. Vs. GS	1%	-26%	-21%				-1%	-17%	

Source: Company data, Goldman Sachs Global Investment Research

Earnings revision

Our net income estimates are revised by -13% / -4% / +16% in 2026-28E mainly on changes in revenues, GM, and Opex ratio, reflecting: **(1) handset lens:** we continue to expect Largan's lens pixel mix upgrade, but at a slower pace, reflecting the 3Q26 miss and slow smartphone market amid high memory costs, **(2) FAU:** we estimate revenue contribution will begin in 2H27, with 0.6m / 5m in 2027 / 28E, along with product mix upgrade toward higher bandwidth FAU that should support ASP and GM. We expect FAU revenue contribution of 3% in 2027E and exceed 50% in 2030E, **(3) GM:** we lower our GM estimates to reflect the slower pace of handset lens pixel mix upgrade and the company's higher costs on green energy. We expect GM to sustain at ~50% in 2026-28E, supported by continuous handset lens pixel mix upgrade and strong demand of FAU, **(4) Opex ratio:** we continue to expect rising opex, while the opex ratio would improve as strong revenue growth driven by FAU and FAU's higher ASP relative to handset lens enhance operating efficiency.

Exhibit 3: Largan earnings revision

	2026E			2027E			2028E		
NT m	Old	New	Chg	Old	New	Chg	Old	New	Chg
Revenue	65,503	61,654	-6%	76,562	76,335	0%	85,021	101,307	19%
GP	33,681	28,867	-14%	42,012	38,222	-9%	47,968	51,348	7%
OP	25,995	21,685	-17%	32,893	29,902	-9%	37,757	41,217	9%
Net income	24,904	21,666	-13%	28,976	27,682	-4%	31,937	37,180	16%
EPS (Diluted)	185.12	161.03	-13%	214.48	204.55	-5%	237.27	276.22	16%
Margins									
GM	51.4%	46.8%		54.9%	50.1%		56.4%	50.7%	
OPM	39.7%	35.2%		43.0%	39.2%		44.4%	40.7%	
NM	38.0%	35.1%		37.8%	36.3%		37.6%	36.7%	

Source: Goldman Sachs Global Investment Research

Exhibit 4: Largan revenues and GM revision

Revenues (NT$m)	2026E	2027E	2028E	GM	2026E	2027E	2028E
NEW	61,654	76,335	101,307	NEW	46.8%	50.1%	50.7%
FAU	-	2,074	19,191	FAU		46.0%	47.4%
Periscope lens	19,653	23,178	26,702	Periscope lens	38.4%	38.3%	38.2%
20MPx+ handset lens	11,911	26,562	34,714	20MPx+ handset lens	70.5%	70.4%	70.4%
Other handset lens	17,833	13,078	9,657	Other handset lens	57.1%	56.4%	55.6%
Others	12,256	11,443	11,042	Others	22.5%	20.3%	20.3%
OLD	65,503	76,562	85,021	OLD	51.4%	54.9%	56.4%
FAU	63	1,067	4,518	FAU	50.3%	50.4%	50.5%
Periscope lens	18,610	18,570	20,741	Periscope lens	38.9%	38.5%	38.5%
20MPx+ handset lens	16,705	31,766	38,447	20MPx+ handset lens	72.9%	73.0%	75.0%
Other handset lens	17,796	13,029	9,611	Other handset lens	62.7%	62.3%	62.0%
Others	12,330	12,130	11,705	Others	24.8%	24.8%	24.9%
CHG	-6%	0%	19%	CHG (NEW - OLD)	-5%	-5%	-6%
FAU	-100%	94%	325%	FAU	-50%	-4%	-3%
Periscope lens	6%	25%	29%	Periscope lens	-1%	0%	0%
20MPx+ handset lens	-29%	-16%	-10%	20MPx+ handset lens	-2%	-3%	-5%
Other handset lens	0%	0%	0%	Other handset lens	-6%	-6%	-6%
Others	-1%	-6%	-6%	Others	-2%	-5%	-5%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 5: Largan 3Q26 results snapshot

NT m	3Q25	2Q26	3Q26	QoQ	YoY	GS	Act / GS	Cons.	Act/Cons.
Revenue	17,677	13,665	15,651	15%	-11%	18,866	-17%	18,196	-14%
GP	8,352	6,753	6,607	-2%	-21%	9,339	-29%	9,120	-28%
OP	6,265	5,148	4,735	-8%	-24%	7,094	-33%	6,946	-32%
Net income	7,080	4,670	5,182	11%	-27%	6,699	-23%	6,723	-23%
Margins									
GM	47.2%	49.4%	42.2%			49.5%		50.1%	
OPM	35.4%	37.7%	30.3%			37.6%		38.2%	
NM	40.1%	34.2%	33.1%			35.5%		36.9%	

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Compared to Bloomberg consensus, we are 4% / 16% ahead in 2027 / 28E net income, mainly on higher revenues and lower opex ratio, reflecting our positive view on Largan's FAU, which is in strong demand with bandwidth upgrade.

Exhibit 6: GS vs. Bloomberg consensus

	2027E			2028E		
NT m	GS	BB	Diff %	GS	BB	Diff %
Revenue	76,335	72,074	6%	101,307	90,479	12%
GP	38,222	36,929	4%	51,348	45,410	13%
OP	29,902	28,454	5%	41,217	34,680	19%
Net income	27,682	26,732	4%	37,180	31,942	16%
Margins						
GM	50.1%	51.2%		50.7%	50.2%	
OPM	39.2%	39.5%		40.7%	38.3%	
NM	36.3%	37.1%		36.7%	35.3%	

Source: Goldman Sachs Global Investment Research, Bloomberg

Valuation

We change our valuation methodology from 2027E PE to discounted PE to better capture the company's long term opportunities in FAU. Our target PE multiple (29.8x) is derived from (1) peers' avg. ratio of trading PE to forward year fundamentals (NI YoY and OPM), which is at 0.5x, and (2) Largan's forward year fundamentals: 2030-31E NI YoY at

19% on avg. and OPM at 41% on avg. We apply the target PE multiple to Largan's 2030E EPS and discount it back to 2027E via a COE of 10.7%, which is based on beta at 1.1x, risk free at 3.5%, and market risk premium at 6.5%. Our new target price is NT$11,000 (vs. NT$6,263 previously). Our target price implies 39.8x 2028E PE, the first full year of FAU contribution, which is within Largan's historical trading range of 8.0x to 45.1x, reflecting our positive view on Largan's expansion from handset lens to FAU, which broadens its exposure from consumer electronics to AI data center that enjoys strong market growth, and could support a valuation re-rating. We note that Taiwan AI supply chain also enjoys re-rating when expanding from consumer electronics to AI data center. Maintain Buy.

Exhibit 7: Largan discounted PE

NT$m	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Key assumptions									
Revenues (NT$m)	**48,842**	**59,458**	**61,148**	**61,654**	**76,335**	**101,307**	**150,128**	**191,476**	**214,294**
FAU	-	-	-	-	2,074	19,191	63,908	103,530	126,348
Others	48,842	59,458	61,148	61,654	74,261	82,115	86,221	87,945	87,945
YoY	**2%**	**22%**	**3%**	**1%**	**24%**	**33%**	**48%**	**28%**	**12%**
FAU						825%	233%	62%	22%
Others	2%	22%	3%	1%	20%	11%	5%	2%	0%
FAU revenues contribution					*3%*	*19%*	*43%*	*54%*	*59%*
FAU shipment (k units)					592	4,789	14,368	21,551	25,862
YoY						*710%*	*200%*	*50%*	*20%*
FAU ASP (US$)					117	134	148	160	163
YoY						*14%*	*11%*	*8%*	*2%*
P&L									
Revenue	**48,842**	**59,458**	**61,148**	**61,654**	**76,335**	**101,307**	**150,128**	**191,476**	**214,294**
Gross profit	**23,808**	**31,209**	**30,837**	**28,867**	**38,222**	**51,348**	**76,093**	**97,050**	**108,616**
Gross margin	48.7%	52.5%	50.4%	46.8%	50.1%	50.7%	50.7%	50.7%	50.7%
OPEX	5,986	7,177	7,279	7,181	8,320	10,131	15,013	19,148	21,429
Opex ratio	12.3%	12.1%	11.9%	11.6%	10.9%	10.0%	10.0%	10.0%	10.0%
Operating profit	**17,821**	**24,033**	**23,558**	**21,685**	**29,902**	**41,217**	**61,081**	**77,903**	**87,186**
OPM	36.5%	40.4%	38.5%	35.2%	39.2%	40.7%	40.7%	40.7%	40.7%
Non-op	4,280	8,142	2,342	4,761	3,913	4,104	4,104	4,104	4,104
Pre-tax profit	**22,102**	**32,174**	**25,900**	**26,446**	**33,815**	**45,321**	**65,185**	**82,007**	**91,291**
Tax	4,200	5,963	4,340	4,539	5,884	7,891	11,350	14,279	15,896
Tax rate	19.0%	18.5%	16.8%	17.2%	17.4%	17.4%	17.4%	17.4%	17.4%
Minority	-	296	285	241	249	250	250	250	250
Net profit	**17,902**	**25,915**	**21,275**	**21,666**	**27,682**	**37,180**	**53,584**	**67,478**	**75,145**
EPS (NTD, diluted)	**132.62**	**191.44**	**157.21**	**161.03**	**204.55**	**276.22**	**398.10**	**501.32**	**558.28**
YoY	*-21%*	*44%*	*-18%*	*2%*	*27%*	*35%*	*44%*	*26%*	*11%*
TP implied P/E					53.8	39.8	27.6	21.9	19.7

2030E target P/E	29.8
Target multiple x EPS	14,939
Discounted back to 2027; TP (NTD)	**11,000**
Implied 2027 P/E	53.8

COE assumption	
Beta	1.1
Risk free	3.5%
Market risk premium	6.5%
COE	**10.7%**

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 8: Largan peer comparison

Company	Ticker	Rating	PE	NI YoY		OPM		Ratio
			2030E	2030E	2031E	2030E	2031E	
Largan	3008.TW	Buy	13.3	26%	11%	40.7%	40.7%	0.2
Largan (TP implied)	3008.TW	Buy	29.8	26%	11%	40.7%	40.7%	0.5
Company	**Ticker**	**Rating**	**PE**	**NI YoY**		**OPM**		**Ratio**
			2027E	**2027E**	**2028E**	**2027E**	**2028E**	
Peers								
AAC	2018.HK	Neutral	12.4	17%	12%	9.7%	10.0%	0.5
T&S	300570.SZ	NC	47.1	96%	62%	24.2%	25.4%	0.5
VPEC	2455.TW	Buy	48.9	67%	55%	35.2%	37.3%	0.5
Sumitomo	5802.JT	Buy	17.2	18%	25%	9.8%	11.2%	0.5
Average			**31.4**	**50%**	**38%**	**19.7%**	**21.0%**	**0.5**

NC: Not Covered; data from Bloomberg consensus. Price as of Oct 9, 2026.

Source: Goldman Sachs Global Investment Research, Bloomberg

Exhibit 9: Taiwan AI server supply chain enjoys strong re-rating when expanding from consumer electronics to AI data center

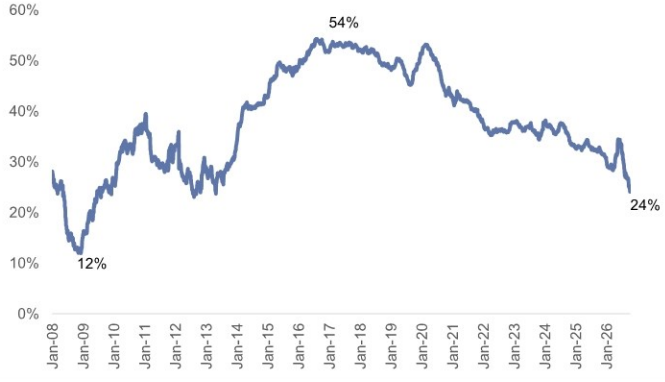
Company	Ticker	2020-2025 PE		
		Trough	Peak	Diff
Largan	3008.TW	10.6	28.0	163%
Taiwan AI server supply chain				
Landmark	3081.TWO	28.9	144.1	398%
VPEC	2455.TW	18.1	74.1	310%
Auras	3324.TWO	8.8	54.6	523%
Fositek	6805.TW	11.3	49.9	343%
AVC	3017.TW	4.7	32.8	604%
Inventec	2356.TW	9.9	31.0	213%
King Slide	2059.TW	9.0	30.2	236%
Gigabyte	2376.TW	3.8	29.6	685%
Mitac	3706.TW	2.5	28.7	1044%
Chenbro	8210.TW	6.7	25.6	282%
Wiwynn	6669.TW	6.1	23.4	287%
Wistron	3231.TW	6.0	21.8	262%
Quanta	2382.TW	6.8	20.4	201%
Hon Hai	2317.TW	7.7	17.5	129%
Avg.		**9.3**	**41.7**	**394%**

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 10: Largan 12M forward PE ratio

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 11: Largan QFII holdings



Source: TEJ

Financial tables

Exhibit 12: Largan P&L

NT$m	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2023	2024	2025	2026E	2027E	2028E
Revenues	15,544	13,665	15,651	16,794	18,095	13,322	19,655	25,263	48,842	59,458	61,148	61,654	76,335	101,307
GP	7,680	6,753	6,607	7,827	9,110	6,375	9,995	12,743	23,808	31,209	30,837	28,867	38,222	51,348
OP income	5,812	5,148	4,735	5,990	7,137	4,923	7,852	9,989	17,821	24,033	23,558	21,685	29,902	41,217
Net income	6,123	4,670	5,182	5,690	6,643	4,812	7,230	8,996	17,902	25,915	21,275	21,666	27,682	37,180
EPS, diluted (NT$)	46.11	35.61	39.52	39.79	49.09	35.56	53.43	66.48	132.62	191.44	157.21	161.03	204.55	276.22
YoY														
Revenues	7%	17%	-11%	-2%	16%	-3%	26%	50%	2%	22%	3%	1%	24%	33%
GP	-4%	8%	-21%	-5%	19%	-6%	51%	63%	-9%	31%	-1%	-6%	32%	34%
OP income	-5%	6%	-24%	-5%	23%	-4%	66%	67%	-13%	35%	-2%	-8%	38%	38%
Net income	-5%	352%	-27%	-15%	8%	3%	40%	58%	-21%	45%	-18%	2%	28%	34%
EPS, diluted (NT$)	-3%	362%	-25%	-19%	6%	0%	35%	67%	-21%	44%	-18%	2%	27%	35%
QoQ														
Revenues	-10%	-12%	15%	7%	8%	-26%	48%	29%						
GP	-7%	-12%	-2%	18%	16%	-30%	57%	27%						
OP income	-8%	-11%	-8%	27%	19%	-31%	60%	27%						
Net income	-9%	-24%	11%	10%	17%	-28%	50%	24%						
EPS, diluted (NT$)	-6%	-23%	11%	1%	23%	-28%	50%	24%						
Margins														
GM	49.4%	49.4%	42.2%	46.6%	50.3%	47.9%	50.9%	50.4%	48.7%	52.5%	50.4%	46.8%	50.1%	50.7%
OPM	37.4%	37.7%	30.3%	35.7%	39.4%	37.0%	40.0%	39.5%	36.5%	40.4%	38.5%	35.2%	39.2%	40.7%
NM	39.4%	34.2%	33.1%	33.9%	36.7%	36.1%	36.8%	35.6%	36.7%	43.6%	34.8%	35.1%	36.3%	36.7%
Ratios														
Opex ratio	12%	12%	12%	11%	11%	11%	11%	11%	12%	12%	12%	12%	11%	10%
Tax rate	15%	23%	14%	17%	17%	17%	17%	17%	19%	19%	17%	17%	17%	17%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 13: Largan balance sheet

NT$ mn	2023	2024	2025	2026E	2027E	2028E
Balance Sheet						
Cash and equivalents	107,490	113,658	115,942	115,177	136,907	152,610
Accounts receivable	10,091	10,360	10,462	10,484	10,430	17,325
Inventory	4,591	5,733	6,713	9,456	7,669	14,778
Other current assets	12,149	16,012	21,169	21,169	21,169	21,169
Current assets	**134,321**	**145,764**	**154,285**	**156,285**	**176,174**	**205,882**
Net PP&E/Fixed assets	41,135	46,936	51,472	60,979	57,943	52,867
Net intangibles	239	505	475	352	345	214
Other long-term assets	19,443	23,322	14,555	14,555	14,555	14,555
Non-current assets	**60,817**	**70,763**	**66,502**	**75,885**	**72,842**	**67,635**
Total assets	**195,138**	**216,527**	**220,787**	**232,170**	**249,017**	**273,517**
Accounts payable	1,731	1,855	1,728	2,224	2,370	3,653
Short-term debt	-	203	-	-	-	-
Other current liabilities	27,787	28,520	28,021	28,214	31,194	35,899
Current liabilities	**29,517**	**30,578**	**29,749**	**30,439**	**33,564**	**39,551**
Long-term debt	-	-	-	-	-	-
Other long-term liabilities	110	561	171	171	171	171
Non-current liabilities	**110**	**561**	**171**	**171**	**171**	**171**
Total liabilities	**29,627**	**31,139**	**29,919**	**30,609**	**33,735**	**39,722**
Common stock	1,335	1,335	1,335	1,335	1,335	1,335
Retained earnings	160,871	175,969	185,818	196,752	210,722	229,485
Other common equity	3,305	6,215	1,714	1,714	1,714	1,714
Total common equity	**165,510**	**183,519**	**188,867**	**199,801**	**213,771**	**232,534**
Minority interest	-	1,869	2,001	1,760	1,511	1,261
Total equity	**165,510**	**185,388**	**190,868**	**201,561**	**215,282**	**233,795**
BVPS	1,226.07	1,369.51	1,410.38	1,498.06	1,590.78	1,736.96
Cash conversion cycle						
Account receivable days	68	63	62	62	50	50
Inventory days	71	67	75	90	82	82
Net payable days	25	23	22	22	22	22
Cash conversion cycle	114	106	116	130	110	110
Ratios						
ROE (common equity)	11%	15%	11%	11%	13%	17%
ROA	9%	13%	10%	10%	12%	14%
Net debt to total equity	-67%	-64%	-64%	-60%	-67%	-68%
Total liabilities to total assets	15%	14%	14%	13%	14%	15%
Dupont analysis						
Asset turnover	0.3	0.3	0.3	0.3	0.3	0.4
Leverage (assets to equity)	1.2	1.2	1.2	1.2	1.2	1.2
Net margin	36.7%	43.6%	34.8%	35.1%	36.3%	36.7%
ROE (total equity)	11%	15%	11%	11%	13%	17%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 14: Largan cash flow

NT$ mn	2023	2024	2025	2026E	2027E	2028E
Cash flow statement						
Net income	22,102	32,174	25,900	21,666	27,682	37,180
Minority interest add-back	-	(296)	(285)	(241)	(249)	(250)
Depreciation and amortization add-back	5,422	6,230	7,732	8,566	12,428	12,399
(Increase)/decrease in working capital	(1,400)	(1,287)	(1,208)	(2,268)	1,986	(12,722)
Other operating cash flow items	(7,925)	(5,243)	(6,059)	-	-	-
Cash flow from operating	**18,199**	**31,579**	**26,080**	**27,723**	**41,847**	**36,607**
Capital expenditure	(8,237)	(11,133)	(12,340)	(17,849)	(9,160)	(7,091)
Other investment cash flow items	(2,915)	(4,933)	3,104	(100)	(225)	(100)
Cash flow from investing	**(11,151)**	**(16,066)**	**(9,236)**	**(17,949)**	**(9,385)**	**(7,191)**
Dividends paid	(11,412)	(9,009)	(13,013)	(10,539)	(10,732)	(13,712)
Change in common stock	-	-	-	-	-	-
Increase/(decrease) in short-term debt	(19)	203	(203)	-	-	-
Increase/(decrease) in long-term debt	-	-	-	-	-	-
Other financing cash flow items	1,680	(1,958)	(241)	-	-	-
Cash flow from financing	**(9,750)**	**(10,764)**	**(13,458)**	**(10,539)**	**(10,732)**	**(13,712)**
Net change in cash	(2,682)	6,168	2,283	(765)	21,730	15,703
FCF	9,962	20,446	13,740	9,874	32,687	29,515
Ratio						
Capex to revenue	17%	19%	20%	29%	12%	7%

Source: Company data, Goldman Sachs Global Investment Research

Price Target Risks and Methodology - Largan

Valuation: We are Buy rated on Largan and have a 12-month target price of NT$11,000, which is based on discounted PE. Our target P/E multiple of 29.8x is derived from (1) peers' avg. ratio of trading PE to forward year fundamentals (NI YoY and OPM), which is at 0.5x, and (2) Largan's forward year fundamentals: 2030-31E NI YoY at 19% on avg. and OPM at 41% on avg. We apply the target PE multiple to Largan's 2030E EPS and discount it back to 2027E via a COE of 10.7%, which is based on beta at 1.1x, risk free at 3.5%, and market risk premium at 6.5%. Our target price of NT$11,000 implies 39.8x 2028E PE, the first full year of FAU contribution, which is within Largan's historical trading range of 8.0x to 45.1x, reflecting our positive view on Largan's expansion from handset lens to FAU, which broadens its exposure from consumer electronics to AI data center that enjoys strong market growth, and could support a valuation re-rating. We note that Taiwan AI supply chain also enjoys re-rating when expanding from consumer electronics to AI data center. Maintain Buy.

Key risks: (1) slower-than-expected smartphone market growth, (2) fiercer competition in handset lenses, (3) slower-than-expected smartphone camera lens specification upgrades or Largan's handset lens pixel mix upgrade, (4) slower-than-expected FAU ramp up, and (5) fiercer competition in FAU.

Disclosure Appendix

Reg AC

We, Verena Jeng and Allen Chang, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

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Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Rating and pricing information

AAC (Neutral, HK$37.36), Sumitomo Electric Industries (Buy, ¥2,361) and VPEC (Buy, NT$558.00)

The rating(s) for Largan is/are relative to the other companies in its/their coverage universe: AAC, ACM Research, AMEC, ASMPT, AVC, AccoTest, Anji Micro, Asus, Auras, BOE, BYDE, Biren, CR Micro, CXMT, Cambricon, Chenbro, China Mobile (HK), China Telecom, China Tower Corp., China Unicom, Chinasoft Intl, Compal, Desay SV, E Ink, E-Town Semis, EHang, Empyrean, Eoptolink, FOCI, Fositek, Foxconn Industrial Internet, Gigabyte, Gigadevice, Glodon Co., HTC Corp., Hikvision, Hon Hai, Horizon Robotics, Hua Hong, Huace Navigation, Huaqin Co.(A), Huaqin Co.(H), Hwatsing, Iluvatar, InnoScience, Innolight (A), Innolight (H), Inspur, Insta360, Inventec, JCET, Kematek, King Slide, Kingdee, Kingsoft Office, LandMark, Largan, Lenovo, Lingyi, Luxshare (A), Luxshare (H), Maxscend, Meitu, MetaX, Mitac, Montage (A), Montage (H), NAURA, NSIG, Nexchip, OmniVision, Pegatron, Pony AI Inc. (ADR), Pony AI Inc. (H), Quanta, RoboTechnik, Ruijie Networks, SG Micro, SICC, SMIC (A), SMIC (H), SZS, Sangfor, SenseTime, Shengyi Tech, Shennan Circuits, StarPower, Sunny Optical, TFC Optical, Thundersoft, Tongyu Communication, Transsion, UMT, UNIS, VPEC, Vanchip, VeriSilicon, Victory Giant, WNC, WUS, WeRide, Wistron, Wiwynn, YJ Semitech, YOFC, Yonyou, ZTE (A), ZTE (H), iFlytek

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

There are no company-specific disclosures for: Largan (NT$6,660.00)

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	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	59%	43%

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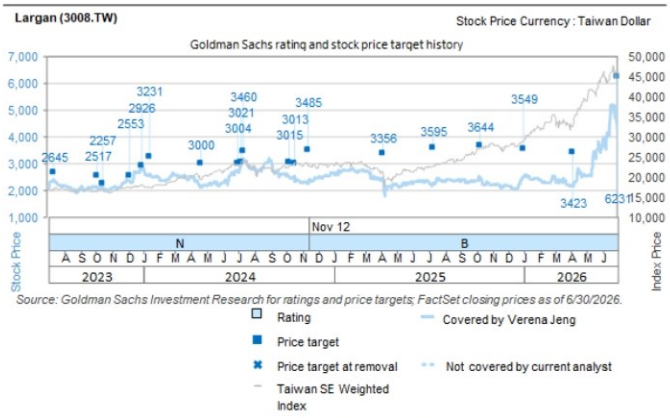
Price target and rating history chart(s)



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

Largan (3008.TW)

Date of report	Target price (NT$)	Closing price (NT$)
13-Jul-26	6,263.00	4,345.00
30-Jun-26	6,231.00	4,290.00
07-Apr-26	3,423.00	2,170.00
01-Jan-26	3,549.00	2,495.00
09-Oct-25	3,644.00	2,470.00
11-Jul-25	3,595.00	2,285.00
06-Apr-25	3,356.00	2,450.00
12-Nov-24	3,485.00	2,255.00
18-Oct-24	3,013.00	2,325.00
07-Oct-24	3,015.00	2,525.00
11-Jul-24	3,460.00	3,200.00
08-Jul-24	3,021.00	3,030.00
01-Jul-24	3,004.00	2,700.00
21-Apr-24	3,000.00	2,120.00
12-Jan-24	3,231.00	2,600.00
29-Dec-23	2,926.00	2,870.00
06-Dec-23	2,553.00	2,320.00
15-Oct-23	2,257.00	2,175.00

Price targets shown in table(s) are unadjusted for corporate actions.

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UPDATE

October 8, 2026 11:00 AM GMT

Largan Precision | Asia Pacific

3Q26 Results Miss on Lower GPM, but FAU/CPO Development Well on Track

AlphaSignals Earnings Reaction

Unchanged	**Meaningful shortfall**	**Modest revision lower**
Impact to our thesis	Financial results versus consensus	Direction of next 12-month consensus EPS

Source: Company data, Morgan Stanley Research

Key Takeaways

- 3Q26 gross margin is 42.2%, which is lower-than-expected and triggers a 31-36% GP and OP miss vs. our estimates.
- Revenue is likely to be flat MoM in October and decline sequentially in November.
- CPO-related (FAU) has seen good progress with new product verification passed.

Largan reported 3Q26 net income of NT$5.182bn, down 27% YoY.

The company reported a 11% YoY revenue decline, to NT$15.651bn; gross margin declined YoY and QoQ to 42.2% in 3Q26, lower than expected due to lower new-project yields and structural green electricity cost hikes.

Management expects shipments to be flat MoM in October and decline sequentially in November as order momentum is weaker-than-expected.

FAU/CPO development well on track: First FA project passed customer verification; expanding from FA into FAU, including coupling and potential FAU-to-carrier assembly, with a dedicated new facility expected to support mass production from around mid-2027. We maintain our long-term positive view on Largan's FAU/CPO opportunities.

Exhibit 1: 3Q26 results summary

NT$mn	3Q26	2Q26	QoQ	3Q25	YoY
Net sales	15,651	13,665	15%	17,677	-11%
Gross profit	6,607	6,753	-2%	8,352	-21%
% margin	42.2%	49.4%	-7%	47.2%	-5%
OPEX	-1,872	-1,605	17%	-2,088	-10%
% OPEX	12.0%	11.7%	0%	11.8%	0%
Operating income	4,735	5,148	-8%	6,265	-24%
% margin	30.3%	37.7%	-7%	35.4%	-5%
Net income	5,182	4,670	11%	7,080	-27%
% margin	33.1%	34.2%	-1%	40.1%	-7%
EPS (NT$)	39.62	35.73	11%	53.05	-25%

Source: Company data, Morgan Stanley Research

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Largan Precision (3008.TW, 3008 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Equal-weight**
Industry View	**In-Line**
Price target	NT$7,100.00
Up/downside to price target (%)	7
Shr price, close (Oct 8, 2026)	NT$6,660.00
52-Week Range	NT$7,995.00-2,020.00
Sh out, dil, curr (mn)	134
Mkt cap, curr (mn)	NT$893,374
EV, curr (mn)	NT$762,165
Avg daily trading value (mn)	NT$5,658

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	159.40	184.85	204.18	215.63
Prior EPS (NT$)**	-	-	-	-
Revenue, net (NT$ mn)	61,148	66,920	74,757	77,025
EBITDA (NT$ mn)	31,260	33,813	37,386	39,037
ModelWare net inc (NT $ mn)	21,275	24,179	26,707	28,204
P/E	15.7	36.0	32.6	30.9
P/BV	1.7	4.1	3.6	3.2
ROE (%)	11.5	12.7	12.4	11.7

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
e = Morgan Stanley Research estimates

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Exhibit 2: 3Q26 results vs. our estimates and consensus

NT$ mn	3Q26	MSe	Diff	Cons.	Diff
Net sales	15,651	19,255	-19%	18,403	-15%
Gross profit	6,607	9,627	-31%	9,131	-28%
% margin	42.2%	50.0%	-8%	49.6%	-7%
OPEX	-1,872	-2,274	-18%	-2,169	-14%
% OPEX	12.0%	11.8%	0%	11.8%	0%
Operating income	4,735	7,353	-36%	6,962	-32%
% margin	30.3%	38.2%	-8%	37.8%	-8%
Net income	5,182	6,988	-26%	6,743	-23%
% margin	33.1%	36.3%	-3%	36.6%	-4%
EPS (NT$)	39.62	53.43	-26%	51.35	-23%

Source: Company data, FactSet, Morgan Stanley Research estimates

Valuation Methodology and Risks

Largan Precision (3008.TW)

Base case: Our price target is based on an SOTP valuation, comprising the traditional business and potential FAU opportunity. We value the traditional business using a residual income model, consistent with our Greater China Hardware Tech coverage, assuming an 8.5% cost of equity, a 7% net profit CAGR over 2026-36E and a 3% terminal growth rate. We separately value the FAU business, assuming a 40% market share in 2028.

Risks to Upside

- Better-than-expected high-end smartphone demand
- Faster-than-expected penetration of periscopes
- Stronger-than-expected CPO volume growth

Risks to Downside

- Greater-than-expected competition, which would erode Largan's ASP and gross margin
- Weaker-than-expected smartphone demand, especially for high-end models
- Market share gains and more pricing pressure from Sunny
- Weaker-than-expected CPO volume growth

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(as of September 30, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
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Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
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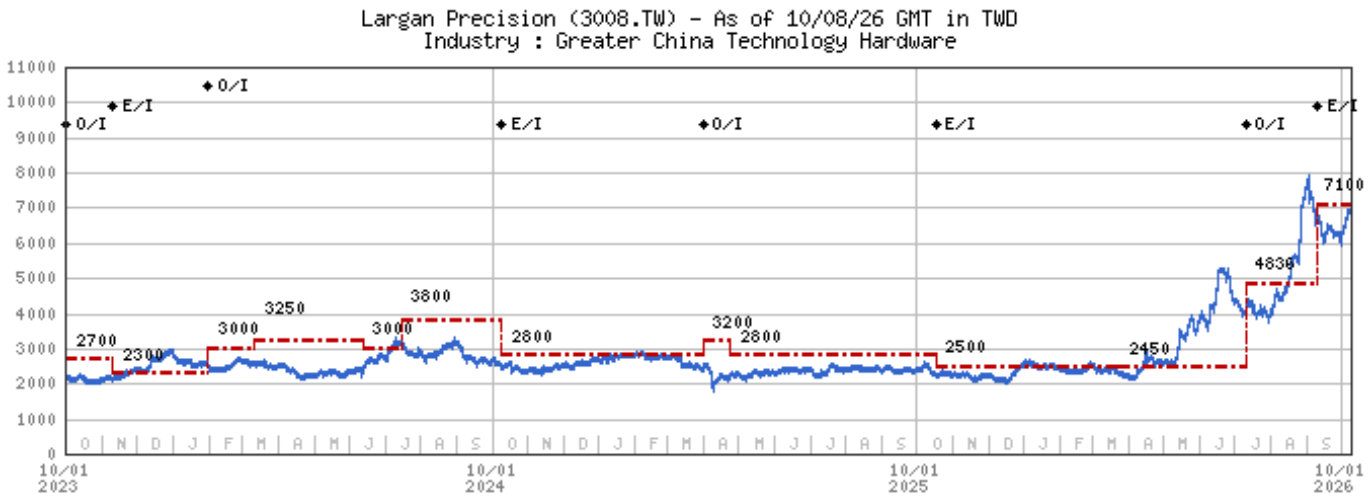
Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : E/I; 12/17/21 : O/I; 9/13/22 : E/I; 1/9/23 : U/I; 5/16/23 : O/I; 11/10/23 : E/I; 1/31/24 : O/I; 10/9/24 : E/I; 3/31/25 : O/I; 10/17/25 : E/I; 7/10/26 : O/I; 9/10/26 : E/I

Price Target History: 9/7/21 : 3050; 10/7/21 : 2300; 12/17/21 : 2600; 4/5/22 : 2300; 7/25/22 : 2400; 9/13/22 : 2200; 10/14/22 : 1900; 1/9/23 : 1700; 5/16/23 : 2700; 11/10/23 : 2300; 1/31/24 : 3000; 3/11/24 : 3250; 6/12/24 : 3000; 7/15/24 : 3800; 10/9/24 : 2800; 3/31/25 : 3200; 4/23/25 : 2800; 10/17/25 : 2500; 3/23/26 : 2450; 7/10/26 : 4836; 9/10/26 : 7100

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)

Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/08/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$37.24
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb152.01
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.14
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.18
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.29
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb378.50
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,060.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb21.70
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.10
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb33.00
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,660.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.83
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.42
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.18
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$50.80
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb48.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.39
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.84
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb68.49
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$35.06
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb46.10
Sunny Optical (2382.HK)	E (03/23/2026)	HK$60.05
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb235.07

Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.00
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$23.66
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb393.99
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$167.60
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb101.20
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb23.15
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb783.00
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$999.50
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$20.70
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb29.32
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,070.00
Advantech (2395.TW)	O (01/20/2021)	NT$727.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,205.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,440.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$36.60
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,520.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,360.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.50
Chenbro (8210.TW)	O (07/23/2025)	NT$874.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,330.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$1,965.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$144.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,370.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$332.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.00
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$10,615.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb30.04
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$297.50
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.63
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.18
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb175.19
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.30
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$996.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$195.50
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.50
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$6,000.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.67
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb55.69
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$363.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,285.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$249.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb64.56
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$50.15
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,960.00
Lenovo (0992.HK)	O (07/10/2026)	HK$32.32
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,915.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,475.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$89.70
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$325.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb131.86
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb368.02
Unimicron (3037.TW)	O (02/23/2026)	NT$1,270.00

Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$558.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$188.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,240.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$635.00
Zhen Ding (4958.TW)	++	NT$592.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$535.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,740.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

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Initiation of Coverage

Hermes Testing Solutions

Shifting-left, scaling-up: dual catalysts in chip probing and CPO

Equities

Taiwan
Semiconductors

12-month rating	**Buy**
	Prior : No Rating
12m price target	**NT$8,880.00**
	Prior :
Price (06 Oct 2026)	**NT$4,190**

RIC: 7856.TWO **BBG:** 7856 TT

Trading data and key metrics

52-wk range	NT$5,295.00-516.55
Market cap.	NT$114b/US$3.58b
Shares o/s	27.2m (ORD)
Free float	94%
Avg. daily volume ('000)	240
Avg. daily value (m)	NT$1,068.4
Common s/h equity (12/26E)	NT$15.2b
P/BV (12/26E)	9.1x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From		To	% ch	Cons.
12/26E	-		29.11	-	32.19
12/27E	-		153.28	-	89.82
12/28E	-		286.48	-	71.44

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Jack-of-all-trades for bespoke testing solutions in the semiconductor industry

We initiate coverage of Hermes Testing (HTSI) with a Buy rating and NT$8,880 PT. HTSI is uniquely positioned to benefit from two key semiconductor testing trends: shift-left testing and the emerging CPO testing market. Leveraging its long-standing relationship with TSMC, HTSI has established capabilities across electrical test engineering, customised test equipment, thermal control modules, probe cards, and CPO testing solutions. We forecast revenue growth of 127%/152%/78% in 2026E/27E/28E, driven by increasing adoption of thermal control modules for wafer-level test and initial deployment of CPO insertion 2 and insertion 3 test equipment through ficonTEC. Additional 2028E growth drivers include reflow tools, laser clean solutions, and MEMS probe cards. The stock trades at 27/15x 2027/28E PE, versus ~35x NTM PE for handler-equipment peers, offering re-rating potential as sales scale, new products are qualified, and its CPO presence expands. See our companion testing deep dive sector note.

Shift-left beneficiary with exceptional thermal control modules

Thermal control modules should be HTSI's largest growth driver over the next several years. We expect increasing chip power consumption, rising package complexity, and tighter probe pitches to drive adoption of advanced thermal control solutions used alongside leading-edge probers, in additional to more wafer level insertions and longer test time. We estimate HTSI will ship approximately 3,000 thermal-control modules in 2027E, up from around 1,200 units in 2026E, sales contribution rise from ~30% in Q226 to >50% of sales by 2027E. The launch of its second-generation platform in 2027E, which carries 25-50% higher ASPs, should further improve profitability. We forecast GM expansion from 55.5% in 2026E to 58.3%/60.0% in 2027E/28E.

Unique position to capture CPO upside

Through its partnership with German optical-automation leader ficonTEC, HTSI is positioned to participate in the initial ramp of CPO testing equipment. HTSI supports local manufacturing, installation, automation and after-sales service, making it a key enabler of volume production. We estimate HTSI captures approximately 30% of equipment value across the 40 insertion 2 and 70 insertion 3 systems we forecast for delivery in 2027E, with further potential upside in 2028E as adoption expands.

Valuation: initiate coverage with a Buy rating and price target NT$8,880

We forecast EPS of NT$29.1 in 2026E (including a one-off employee share purchase programme impact), rising to NT$153.3/286.5 in 2027E/28E. Our NT$8,880 price target is based on 31x average 2028E PE, versus handler-equipment peers forward PE at roughly 35x, reflecting HTSI's unique exposure to both shift-left testing, CPO, and our estimate of 32% LT earnings CAGR. We initiate with a Buy rating.

Highlights (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	1,598	2,425	5,505	13,879	24,713	29,218	32,810
EBIT (UBS)	54	297	1,076	6,154	11,604	14,294	16,170
Net earnings (UBS)	57	268	862	5,059	9,454	11,664	13,227
EPS (UBS, diluted) (NT$)	2.71	10.65	29.11	153.28	286.48	353.44	400.81
DPS (net) (NT$)	0.00	1.10	11.64	61.32	114.60	141.38	160.33
Net (debt) / cash	(469)	618	14,360	17,474	22,786	29,613	36,992
Profitability/valuation	**12/24**	**12/25**	**12/26E**	**12/27E**	**12/28E**	**12/29E**	**12/30E**
EBIT (UBS) margin %	3.4	12.2	19.5	44.3	47.0	48.9	49.3
ROIC (EBIT) %	-	41.7	147.5	381.2	334.1	282.5	261.7
EV/EBITDA (UBS core) x	-	35.1	80.1	15.1	7.7	5.8	4.7
P/E (UBS, diluted) x	-	68.6	NM	27.3	14.6	11.9	10.5
Equity FCF (UBS) yield %	-	2.5	0.6	3.0	6.4	9.3	10.6
Dividend yield (net) %	-	0.2	0.3	1.5	2.7	3.4	3.8

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 4,190.00 on 06-Oct-2026

This report has been prepared by UBS Securities Pte. Ltd., Taipei Branch. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 19.**

UBS Research THESIS MAP a guide to our thinking and what´s where in this report

PIVOTAL QUESTIONS

Q: Will HTSI be able to keep its moat within chip probing thermal control modules?
Yes. We expect HTSI's competitive position to strengthen as chip power consumption, package complexity, and probe density continue to increase. The company is developing next-generation thermal control solutions, including liquid cooling technologies, while also expanding into specialized chuck solutions designed to address wafer warpage. In addition, HTSI's close collaboration with TSMC, TEL, Advantest, and other OSATs should reinforce its position within advanced wafer-level testing.

Q: Will insertion 2 of CPO testing process be skipped?
No. We believe concerns surrounding insertion 2 adoption are overdone. While some industry participants favor bypassing insertion 2 due to throughput and ROI considerations, doing so increases the risk of defective optical engines being identified only after the components are delivered to OSATs, creating yield accountability challenges and higher packaging losses. Based on our industry studies, we expect insertion 2 testing to remain an important part of the CPO workflow over at least the next two years as the industry focuses on improving PIC yields and overall testing efficiency.

UBSVIEW

We initiate coverage of Hermes Testing (HTSI) with a Buy rating and NT$8,880 price target. HTSI offers unique exposure to two of the semiconductor testing industry's most attractive growth drivers: 1) shift-left testing and 2) CPO testing. The industry's shift-left testing strategy will have more test insertions move to the wafer level before packaging, in addition to longer CP time. Leveraging its longstanding relationship with TSMC, HTSI has established capabilities across thermal control modules, electrical-test engineering services, customized equipment, and emerging CPO testing solutions. We believe there could be re-rating potential as sales scale, new products are qualified, and its CPO presence expands.

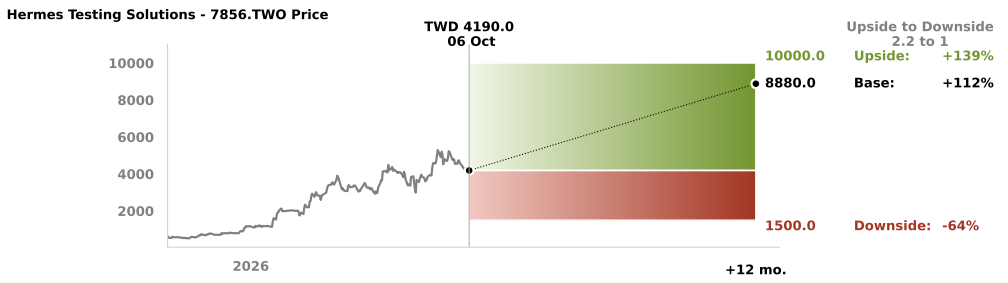
EVIDENCE

TSMC and OSATs have been expanding their CoWoS capacities aggressively, from 40kwpm in 2024 to around 90 kwpm in 2025, and is tracking to expand CoWoS capacity toward 160kwpm in 2026E. TSMC also ramped up expanding N3/N2 capacity, with multiple product launches from customers' pipelines—Nvidia's Rubin, AMD's MI400 series/Venice CPU and other cloud ASICs—translating into foreseeable, strong testing demand.

WHAT´S PRICED IN?

Despite a 264% YTD share-price increase, HTSI currently trades at just 15x of our 2028E EPS versus handler-equipment peers forward PE at roughly 35x. We believe market expectations for 2027 CPO testing equipment shipments remain conservative, particularly for insertion 2 equipment. Based on our industry analysis, we estimate HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, this should be higher than the market's expectation of no insertion 2 and limited insertion 3 shipment due to engineering bottleneck. We also see further upside in 2028E as CPO adoption expands. Our reverse DCF analysis indicates that the current market price implies only a 17% long-term sales CAGR vs our estimate of 27% and long-term GM level of 33% vs our estimate of 61%. We see scope for re-rating.

UPSIDE/DOWNSIDE SPECTRUM



Value drivers (2027/28E)	Thermal control module shipments	CPO insertion 2 & 3 shipments	GM
NT$10,000 upside	3,300/5,000	60&100 / 100&180	61.2%/63.6%
NT$8,880 base	3,000/4,100	40&70 / 80&160	58.3%/60.0%
NT$1,500 downside	1,300/2,000	5&20 / 10&50	53.3%/54.0%

Source: UBS estimates

COMPANY DESCRIPTION

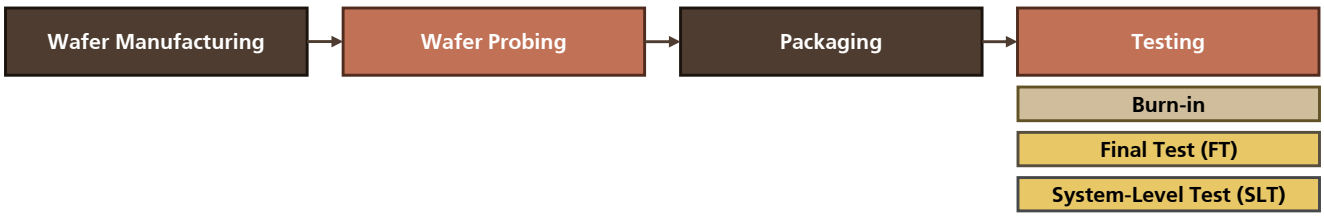
Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Testing TAM expanding with IC complexity: leaders likely to outperform industry growth

Our central view is that testing content per AI accelerator is growing faster than semiconductor unit volumes, creating a multi-year expansion in testing TAM. We believe semiconductor testing is entering a period of structural growth as AI-driven chip complexity increases testing intensity across wafer, package and system levels. The industry's shift-left strategy, which moves more testing upstream in the semiconductor manufacturing process, is resulting in longer and more frequent test insertions. As cloud AI GPUs and ASICs become increasingly complex, additional testing is required to ensure functionality, performance and yield, while reducing defective parts per million (DPPM). More robust testing should improve deployment efficiency and lower the cost of operating AI infrastructure.

Semiconductor testing generally comprises four stages: wafer/chip probing (CP), final test (FT), system-level test (SLT), and burn-in. CP is conducted at the wafer-level before packaging, while FT, SLT, and burn-in occur after packaging. The below Figure 3 illustrates the semiconductor manufacturing flow.

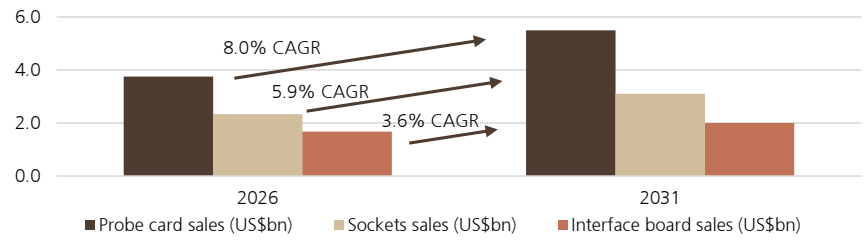
Figure 1: Semiconductor testing process: from wafer level to package level



Source: UBS

We estimate semiconductor testing historically accounts for ~1% of total semiconductor revenue. Within testing, consumables such as probe cards, sockets and interface boards represent ~10-15% of industry revenue. According to Yole, the probe card, socket and interface board markets should grow from US$3.8bn, US$2.3bn and US$1.7bn in 2026 to US$5.5bn, US$3.1bn and US$2.0bn by 2031, respectively.

Figure 2: Yole estimating testing consumables TAM to climb in the next 5 years



Source: Yole, UBS

Nonetheless, leaders within the testing sector will standout among competition, with greater sales CAGR comparing to the industry.

We see **HTSI**'s sales for the next few years to mainly be driven by 1) thermal control module shipment and 2) CPO equipment shipment. In the past, thermal control modules weren't necessary for chip probing, due to less TDP for the chips. However, as new iterations of AI accelerators now got substantial power draw, this unlocks a brand new TAM for HTSI to enter with their thermal control modules to pair with probers, in order to prevent probe card damages.

- **Robust thermal control module shipments.** We estimate HTSI to ship 3,000/4,100 thermal control modules in 2027E/28E, with continuous demand in future years, bringing their 2026-31E semiconductor equipment sales CAGR to 48%.

- **Thermal control module ASP expansion.** Gen 2 thermal control modules will start to ship in 2027E which will drive tool ASP up around 25-50%.
- **Demand supported by Advantest expansions.** We believe the 3,000 thermal control modules shipment forecast for 2027E is not far-fetched, as we read-across from Advantest's announced capacity expansion plan (testers will be paired with probers with a 1:1 ratio) with increasing capacity up to 5,000 testers within FY2027/3 and they also announced a 10,000-unit target within CY2028. Although the testers capacity expansion plan includes testers for final test, we do see a good portion to be allocated to chip probing given shift-left testing trend increasing both testing insertions and testing time before chip packaging.
- **CPO is another brand new TAM for semiconductor testing.** As insertion 2 and 3 testing equipment demand starts to rise in 2027 with scale-out CPO adoption, we see HTSI to ship 40/70 insertion 2 and 3 equipment in 2027E, with this number growing in 2028E to 80/160 and higher in future years as CPO proliferates, and driving 52% 2026-31E test equipment & engineering services sales CAGR.
- **CPO to see content increase.** Gen 2 insertion 3 equipment may start to ship in 2027E, with higher throughput and tool ASP.
- **Insertion 2 might scale down, but not happening anytime soon.** Insertion 2 equipment shipment might come down in 2029E and beyond, as foundries scale-up yield for photonic IC (PIC) in order to lessen the time consuming insertion 2 testing burden.

Figure 3: Hermes Testing is set to benefit with a 46% 2026-31E sales CAGR

Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Semiconductor Equipment & Customization products	**$2,815**	**$7,980**	**$13,324**	**$16,061**	**$18,315**	**$20,104**	**48%**
YoY		*183%*	*67%*	*21%*	*14%*	*10%*	
Test Equipment & Engineering services	**$1,480**	**$3,887**	**$7,901**	**$8,887**	**$9,898**	**$11,853**	**52%**
YoY		*163%*	*103%*	*12%*	*11%*	*20%*	
Probe cards and cleaning materials	**$1,210**	**$2,012**	**$3,488**	**$4,270**	**$4,597**	**$4,681**	**31%**
YoY		*66%*	*73%*	*22%*	*8%*	*2%*	
Key products shipment units, ASP, sales	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Thermal control module shipment	**1,200**	**3,000**	**4,100**	**4,500**	**4,700**	**4,900**	
Thermal control product blended ASP (NT$m)	$1.9	$2.4	$2.8	$3.0	$3.3	$3.5	
Total sales from thermal control module shipment	**$2,316**	**$7,168**	**$11,417**	**$13,505**	**$15,591**	**$17,067**	**49%**
% of total sales	*42%*	*52%*	*46%*	*46%*	*48%*	*47%*	
CPO Insertion 2 shipment	**8**	**40**	**80**	**60**	**20**	**20**	
Ins2 blended ASP (NT$m)	$13.8	$15.8	$17.4	$18.3	$19.2	$20.1	
Total sales from insertion 2	**$111**	**$633**	**$1,391**	**$1,095**	**$383**	**$402**	**29%**
% of total sales	*2%*	*5%*	*6%*	*4%*	*1%*	*1%*	
CPO insertion 3 shipment	**11**	**70**	**160**	**200**	**240**	**280**	
Ins3 blended ASP (NT$m)	$16.0	$17.2	$18.2	$19.1	$20.1	$21.1	
Total sales from insertion 3	**$176**	**$1,202**	**$2,917**	**$3,829**	**$4,824**	**$5,910**	**102%**
% of total sales	*3%*	*9%*	*12%*	*13%*	*15%*	*16%*	
Key product sales - total	**$2,602**	**$9,002**	**$15,725**	**$18,429**	**$20,798**	**$23,379**	**55%**
% of total sales	*47%*	*65%*	*64%*	*63%*	*63%*	*64%*	
Others - total	**$2,903**	**$4,877**	**$8,988**	**$10,789**	**$12,012**	**$13,260**	**36%**
% of total sales	*53%*	*35%*	*36%*	*37%*	*37%*	*36%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	**$5,505**	**$13,879**	**$24,713**	**$29,218**	**$32,810**	**$36,639**	**46%**
YoY		*152.1%*	*78.1%*	*18.2%*	*12.3%*	*11.7%*	
GM %	**55.5%**	**58.3%**	**60.0%**	**60.9%**	**61.3%**	**61.1%**	
OpM %	**19.5%**	**44.3%**	**47.0%**	**48.9%**	**49.3%**	**51.1%**	

Source: UBS estimates

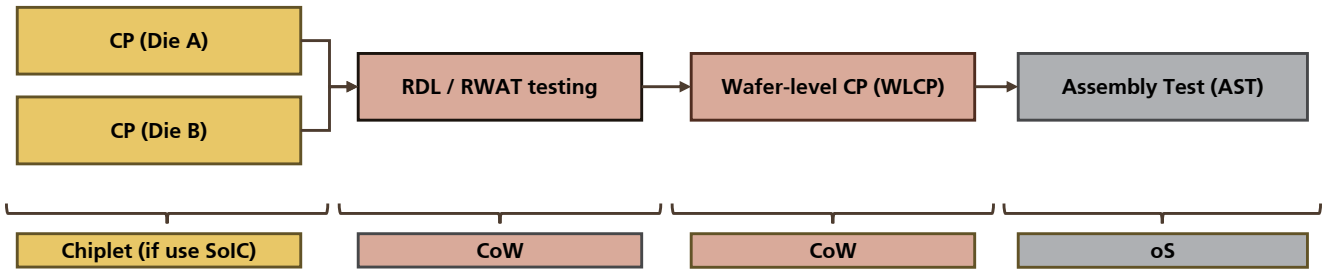
Robust growth drivers across semiconductor testing

We believe HTSI is set to benefit from several semiconductor testing growth drivers and capture the AI induced, expanding testing TAM. We see HTSI is uniquely positioned as the main beneficiary of two of the strongest growth drivers within semiconductor testing: testing shift-left and CPO.

Shift-left igniting secular growth for chip probing

The industry's adoption of shift-left testing should drive a multi-year structural growth opportunity for chip probing (CP) and its supply chain. As advanced packaging costs rise and AI accelerators become increasingly complex, semiconductor manufacturers are placing greater emphasis on identifying known-good dies (KGD) before packaging.

Figure 4: CP is set to grow substantially with shift-left, each packaging step set to be MORE costly than the prior step

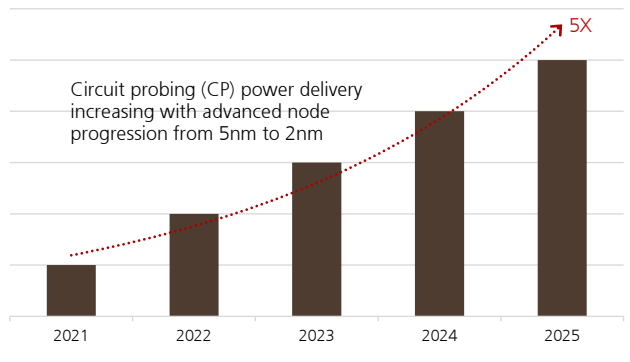


Source: UBS

The economics of advanced packaging reinforce this trend. Heterogeneous integration, chiplet architectures and 2.5D/3D packaging significantly increase package value, while defects identified late in the manufacturing process can lead to substantial scrap costs. To mitigate this risk, foundries and OSATs such as TSMC and ASE employ distributed testing flows incorporating KGD, known-good-stack (KGS) and assembly test screening, to catch defects before the components enter a multi-week packaging cycle (>2 weeks for CoW and 2 weeks for oS). The importance of these tests rises as AI clusters become increasingly dependent on component reliability and low defective-parts-per-million (DPPM) rates. Given the multiplicative yield degradation, a 300bps yield decline at just one manufacturing stage (e.g. dropping from 99.5% to 96.5%) reduces overall integrated four-stage yield from 98.0% to 95.0%; on a 500,000 unit ramp at US $10,000 per unit, this 3% scrap delta translates into around US$150m in unrecoverable losses given >6-month manufacturing lead times.

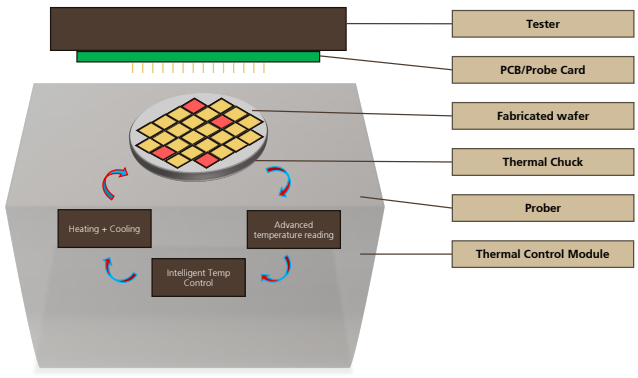
At the same time, rising chip power consumption is increasing the technical difficulty of wafer-level testing. As chip thermal design power scales beyond 1,000W and power density is expected by the industry to surge another 8x by 2030, traditional wafer-level chip probe (WLCP) encounters severe physical bottlenecks. Finer probe pitches and higher power densities increase the risk of overheating, probe needle burnout and yield degradation, creating demand for more advanced probing and thermal-control solutions.

Figure 5: CP power delivery trend rises with process tech evolution



Source: UBS

Figure 6: Thermal control module inside probers are essential to control wafer temperature



Source: UBS

Hermes Testing (HTSI) is a key beneficiary of this trend. Leveraging its long-standing relationship with TSMC and expertise in semiconductor test solutions, HTSI supplies advanced thermal control modules used alongside leading-edge probers (made by Tokyo Electron), particularly for AI applications. These thermal control modules are required for high TDP semiconductor wafer probe in order to maintain the temperature

of the chuck to avoid needle burns or probe card damages. We estimate HTSI will ship approximately 3,000 thermal-management modules in 2027E, up from 1,200 in 2026E, supporting sales growth of 127%/152% in 2026E/27E. HTSI is also the partner for prober installation, prober maintenance and services. In addition, HTSI provides high-end chuck processing services, an area that should benefit from increasing package sizes and higher wafer warpage associated with advanced packaging.

Beyond thermal control modules, HTSI also supplies consumables and customised solutions for wafer-level testing, including probe card cleaning sheets and laser-based probe head cleaning systems. As probing intensity increases under the shift-left testing model, demand for cleaning and maintenance solutions should rise alongside probe card utilisation. Currently, cleaning sheets is around 5% of sales, but we believe it could gain traction in 2027/28E and beyond. We see additional upside from future modular laser cleaning systems that could be integrated directly with probers, further expanding HTSI's content opportunity within advanced wafer-level testing.

CPO testing fuelling future growth for semiconductor testing

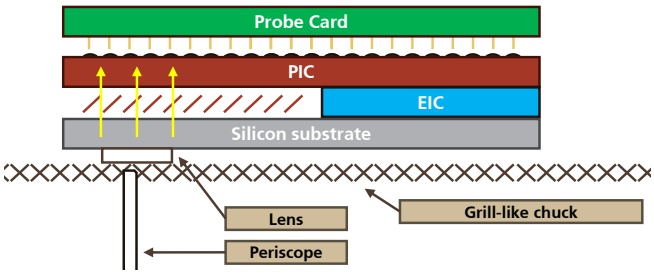
As interconnect bottlenecks increasingly shift from the memory wall to the copper wall, we believe co-packaged optics (CPO) adoption is inevitable over the longer term. Testing remains one of the key bottlenecks to CPO deployment, as optical validation differs fundamentally from the electrical testing processes that have dominated semiconductor manufacturing for decades. As a result, CPO is creating a new testing opportunity across the semiconductor equipment and consumables ecosystem. We have written extensively on CPO and opportunities that it brings to the semiconductor supply chain, see our CPO deep dive, our thoughts on CPO timing, 2026 SiPhIA conference takeaways, and SEMICON 2026 takeaways.

Various test equipment suppliers are scrambling to come up with ideal testing equipment across the four insertions of CPO testing, with solutions to ensure both optical and electrical signal integrity of the CPO solutions. The four insertions are:

- **Insertion 1: Electronic IC (EIC)/Photonic IC (PIC) test.** Single-side wafer test that has dedicated probe cards to test the fabricated EIC/PIC wafers.
- **Insertion 2: Optical/Electrical test.** Double sided wafer test, testing the hybrid bonded (e.g. via TSMC's COUPE) EIC and PIC wafers by an opto-electrical test that verifies the electrical signals on the top and optical signal on the bottom of the hybrid bonded wafers.
- **Insertion 3: Optical engine (OE) test.** The OE tests are done to ensure the quality of the OEs and with FAU (Fiber Array Unit) attached, to ensure the signal integrity of the OEs. The testing equipment needs to be able to handle 224G per channel data rates, requiring high unit per hour (UPH) throughput, meticulous dirt mitigation, and maintaining a near 100% contact rating on high-speed sockets.
- **Insertion 4: ASIC final test.** This final test step is established post the OEs are packaged together with the ASIC. The complete ASIC package (now with CPO) is now tested for opto-electrical functionalities.

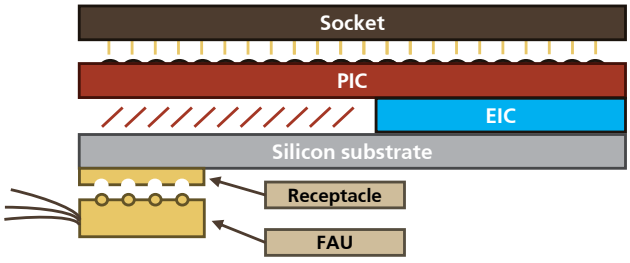
As it stands, insertion 2 and insertion 3 are the two areas where there are the most debates in the industry. Insertion 3 suppliers such as Chroma has noted the trend of "shift-to-middle" paradigm centered on insertion 3. In high-density scale-out switches deploying up to 32 optical engines (e.g. Nvidia's Spectrum-X architectures), any OE defect results in package scrap at the CoWoS level, making 100% known good die (KGD) validation at the singulated OE stage an absolute commercial requirement.

Figure 7: An illustration for insertion 2 test method



Source: UBS

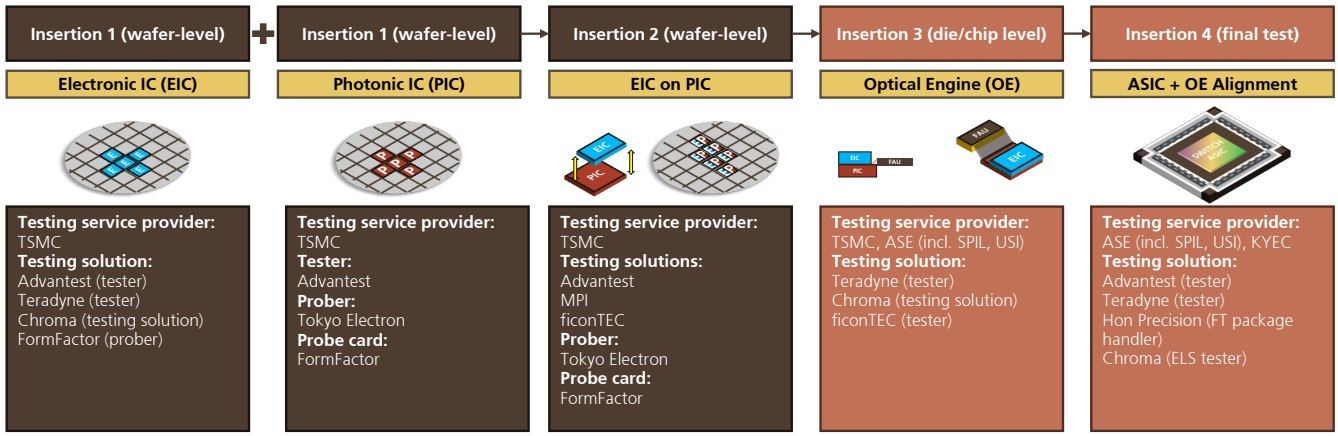
Figure 8: An illustration for insertion 3 test method



Source: UBS

As shown in the below figure, as there are no set standards for the testing equipment, throughput and stability are now the main target and area of competition for the major players in the field. Notable ones for each insertions are listed.

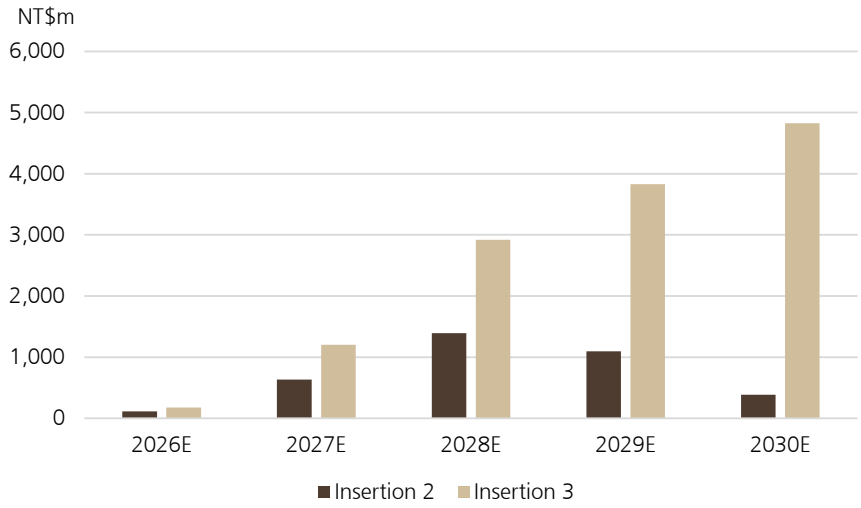
Figure 9: CPO testing flow chart with key suppliers



Source: UBS

With its 25 years of precision test equipment expertise, ficonTEC is an early mover for CPO test equipment and has key know-how for optical alignment, which is quintessential for optical test. **HTSI**, through its partnership with ficonTEC, is set to be a major beneficiary once CPO proliferates in 2027E-28E. As the partner of ficonTEC, Hermes Testing is the key enabler that can help ficonTEC achieve HVM for its CPO testing equipment. As of now, our view is that both insertion 2 and insertion 3 should be required within the next two years, despite the difficulties surrounding double-sided wafer probing (insertion 2), as the yield for OEs are still not high enough to avoid defective dies. Finding out known-good OEs will be crucial before the FAU coupling process during the latter stages of CPO packaging, to prevent impeding the final package with multiple OEs attached to the switch ASIC (e.g. 32 OEs on Nvidia's Spectrum-X). We estimate Hermes Testing will ship 40/70 units of insertion 2/3 testing equipment in 2027E, with potential upside to the numbers as we view Hermes Testing will likely assist with the ramp up for Gen 2 insertion 3 equipment in 2027E. Gen 2 insertion equipment will come with better throughput and ASP.

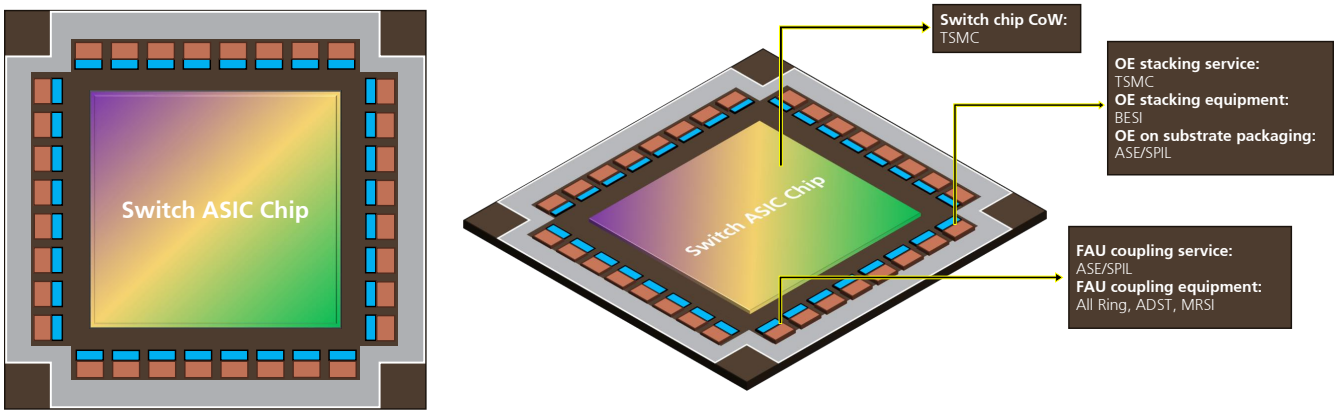
Figure 10: We believe HTSI CPO equipment sales will pick up in 2027/28E with 541%/135% YoY growth



Source: UBS estimates

Beyond insertions 2 and 3, HTSI is also positioned to participate in insertion 1 testing through customised prober solutions for photonic integrated circuit (PIC) testing. We believe these solutions could eventually be packaged as licensable intellectual property, creating an additional long-term growth opportunity as CPO adoption broadens. While near-term testing growth is driven by AI accelerators, CPO represents the next major expansion cycle for semiconductor testing. We believe HTSI is uniquely positioned to participate across multiple test insertions, providing exposure to a potentially significant source of incremental revenue beginning in 2027E.

Figure 11: A switch chip with 32 OEs and the CPO packaging supply chain



Source: UBS

HTSI's unique position for new TAM of thermal control modules and CPO test equipment

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan, as a spin-off from Hermes-Epitek. The company provides semiconductor testing and inspection solutions, engineering services, customized equipment, and test consumables. HTSI maintains a global service footprint across Taiwan (Hsinchu, Tainan), Mainland China (Suzhou), Japan (Tokyo, Kumamoto), Singapore, Malaysia, and the United States (Arizona), enabling close support for leading semiconductor customers.

Its main business segments are structured around three pillars:

- **Semiconductor equipment and customized products:** HTSI utilises a dual-track strategy of independent R&D and also equipment agency sales. They offers

highly customized solutions designed to address specific client pain points and bottlenecks. Key products include: thermal control modules that pair with a 1:1 ratio with wafer-level test probers, electron beam inspection equipment, white light interference optical inspection systems, laser probe cleaning machines, and the SYMPHONY fluxless reflow system.

- **Testing equipment engineering services:** The company operates the largest engineering service team in Taiwan, consisting of over 340 technical professionals. The engineering service team provides extensive equipment life-cycle support, including installation, relocations, modification, testing, routine maintenance, and repair services. CPO related equipment sales/services will be recognized within this segment, as CPO are not direct HTSI products, but rather sales recognition via its partnership with ficonTEC.
- **Probe card design, manufacturing, and cleaning materials:** HTSI delivers a comprehensive product portfolio that supports the entire probe card life-cycle. For probe cards, they supply membrane, cantilever, and vertical probe cards. As they are small in scale, HTSI targets niche probe card applications such as RF, 5G/6G, millimeter wave, LCD driver, low end logics, TIA (Transimpedance Amplifier), etc. In addition to the three existing probe technologies, HTSI also collaborates with MJC in Japan for MEMS probe development, targeting sales in 2028E for high-end, AI/HPC related applications. Cleaning materials is another important business within this segment, as HTSI source and process sand paper or other materials to offer cleaning sheets to clean the probe cards, removing oxidations, unwanted particles and substances that accumulated on the probe cards during the testing process.

The sales resiliency in the coming years mainly stems from its robust relationship with key industry players, especially as an engineering and service provider with TSMC for testing solutions. This sets up a solid foundation for its customized equipment business, as they know the specific testing bottleneck to tackle. For instance, as Hermes-Epitek manufacturers the probers for Tokyo Electron (TEL) and HTSI helps with installation, services, and maintenance of the probers, they understand the thermal bottlenecks that the wafer-level test industry has been facing and can effectively come up with module solutions to solve this.

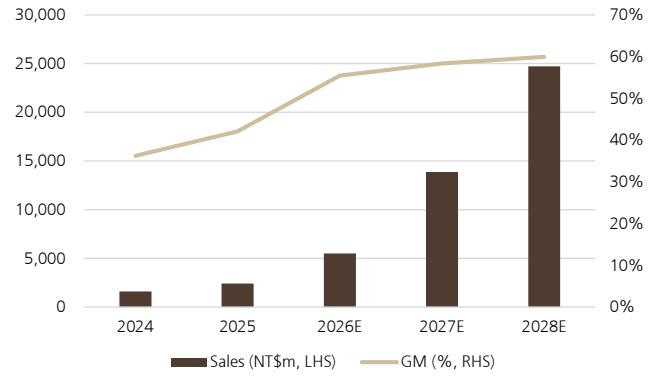
Margins set to expand with value-added products and services

The company has demonstrated strong execution in advanced wafer-level testing, with H126 revenue of NT$2.2bn already exceeding 90% of FY25 revenue (NT$2.4bn). Profitability has also improved significantly, with GM increasing from 42.1% in 2025 to 54.9% in H126.

With share of value-added products and services set to rise in 2027E in sales mix, HTSI is set to see GM and operating leverage expansion. As a niche engineering service provider that has robust ties to leading foundries and OSATs, we believe HTSI has substantial pricing power and estimate HTSI' GM for thermal control modules to be around 65%, with GM for CPO equipment of around 60% as HTSI provides an engineering-service-like business for it. We believe these factors will help the company to reach a 60% long-term GM range.

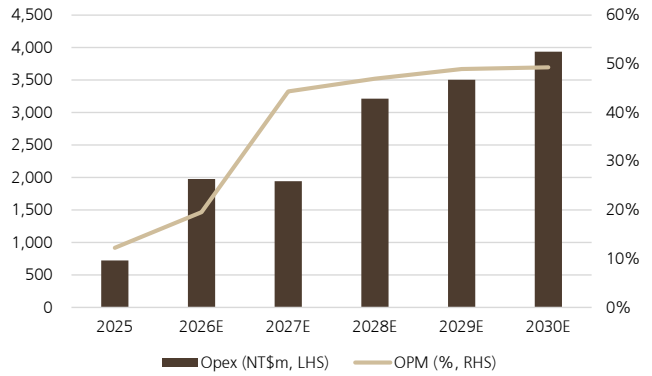
OPM is also set to expand with customized equipment sales rise in total sales mix. If we exclude the one-time expense to be recognized during Q326 for employee stock benefits (we estimate around NT$1bn), YoY opex growth will be around 99% for 2027E, with downside risk if the company conducts tight cost control measures. We forecast the company's OPM will expand from 19.5% in 2026E, to 44.3%/47.0% in 2027/28E.

Figure 12: HTSI sales set to continue to climb with 152% growth and GM at 58.3% in 2027E



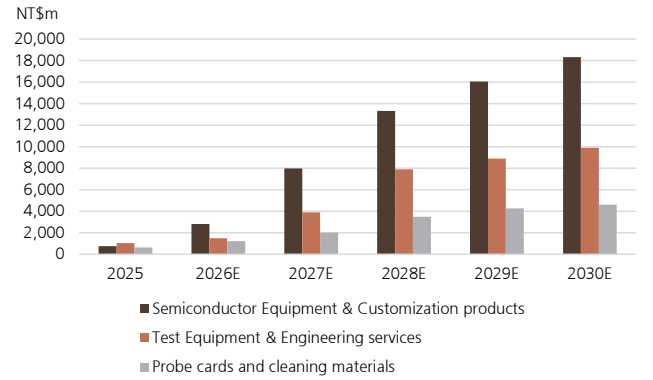
Source: UBS estimates

Figure 13: OPM set to grow to 44.3%/47.0% in 2027/28E



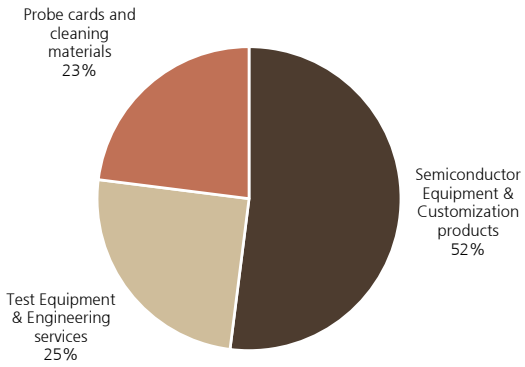
Source: UBS estimates. Note 2026E opex includes a one-time NT$1bn employee stock benefit.

Figure 14: We see customized equipment to continue to drive company growth



Source: Company data, UBS estimates

Figure 15: HTSI H126 sales mix breakdown



Source: Company data, UBS

Figure 16: UBS earnings forecast

(NT$m)	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E	2028E	2029E	2030E
Revenue	**2,425**	**985**	**1,200**	**1,513**	**1,807**	**5,505**	**2,634**	**3,197**	**3,760**	**4,289**	**13,879**	**24,713**	**29,218**	**32,810**
- YoY chg (%)	52%	123%	108%		28%	127%	167%	166%	149%	137%	152%	78%	18%	12%
- QoQ chg (%)		-30%	-51%	54%	51%		46%	-42%	43%	34%				
Gross profit	**1,021**	**545**	**655**	**843**	**1,012**	**3,054**	**1,534**	**1,856**	**2,184**	**2,523**	**8,097**	**14,817**	**17,801**	**20,107**
- Gross margin	42.1%	55.3%	54.5%	55.7%	56.0%	55.5%	58.3%	58.1%	58.1%	58.8%	58.3%	60.0%	60.9%	61.3%
Operating profit	**297**	**323**	**411**	**-399**	**741**	**1,076**	**1,166**	**1,408**	**1,657**	**1,923**	**6,154**	**11,604**	**14,294**	**16,170**
- Operating margin	12.2%	32.8%	34.3%	-26.4%	41.0%	19.5%	44.3%	44.1%	44.1%	44.8%	44.3%	47.0%	48.9%	49.3%
Pre-tax profit	**322**	**326**	**412**	**-396**	**743**	**1,085**	**1,205**	**1,449**	**1,700**	**1,968**	**6,323**	**11,818**	**14,580**	**16,534**
Net profit	**268**	**254**	**330**	**-317**	**594**	**862**	**964**	**1,160**	**1,360**	**1,574**	**5,059**	**9,454**	**11,664**	**13,227**
- YoY chg (%)	373%	586%	301%		299%	221%	280%	251%	-529%	165%	487%	87%	23%	13%
- QoQ chg (%)		70%	23%	-225%	80%		62%	35%	41%	36%				
Basic EPS (NT$)	**10.83**	**9.35**	**12.15**	**-10.20**	**18.01**	**29.11**	**29.22**	**35.14**	**41.22**	**47.70**	**153.29**	**286.49**	**353.45**	**400.82**
- YoY chg (%)	281%	492%	279%		200%	169%	213%	189%	-504%	165%	427%	87%	23%	13%
- QoQ chg (%)		55%	12%	-209%	48%		62%	21%	41%	36%				

Source: UBS estimates

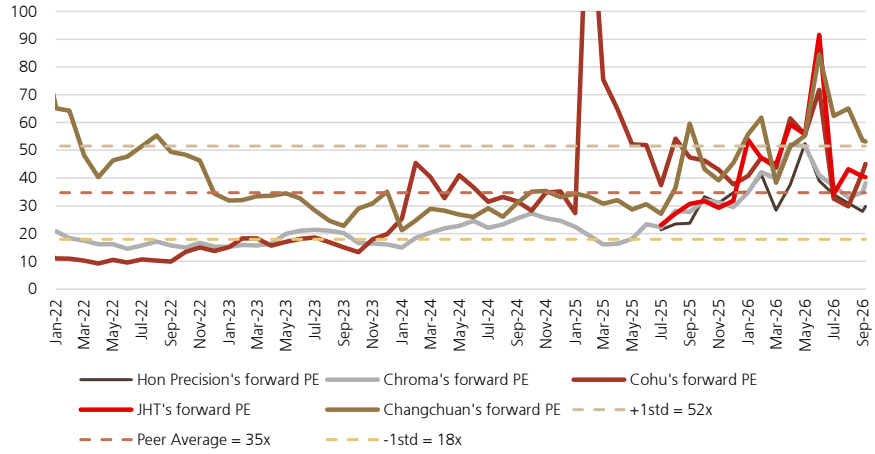
Figure 17: Testing supply chain valuation table

Company name	Ticker	Analyst	Market cap (US$m)	Price Target (LC)	Share price (LC)	Rating	EPS growth (%)		P/E (x)		P/BV (x)		ROE (%)	
							2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Testing service provider														
ASE Industrial	3711.TW	Sunny Lin	104,043	835	744	Buy	104	64	39.3	24.0	7.5	6.3	21	28
King Yuan Electronics	2449.TW	Sunny Lin	11,456	325	298	Buy	40	62	32.6	20.1	5.9	5.0	20	27
Amkor Technology	AMKR.O	Randy Abrams	13,114	80	53	Buy	73	8	20.6	19.1	2.6	2.4	14	13
JCET Group	600584.SS	Jimmy Yu	17,156	135	64	Buy	72	53	42.7	28.0	3.7	3.3	9	12
Tianshui Huatian Technology	002185.SZ	Jimmy Yu	7,928	13	16	Neutral	33	27	57.0	44.8	2.8	2.7	5	6
TongFu Microelectronics Co Ltd	002156.SZ	N/A	13,526	N/A	57	N/A	50	13	45.3	40.1	5.0	4.5	10	11
Shanghai V-Test Semiconductor Tech Co Ltd	688372.SS	N/A	2,517	N/A	100	N/A	36	106	40.6	19.7	3.8	3.2	11	18
Sector Average							**58**	**47**	**39.7**	**28.0**	**4.5**	**3.9**	**13**	**17**
Tester														
Advantest	6857.T	Atsuhiro Kinoshita	191,547	43,600	41,850	Buy	80	96	38.8	53.1	16.8	31.8	70	58
Teradyne Inc	TER.O	Timothy Arcuri	67,666	500	430	Buy	134	51	45.7	30.8	16.7	12.8	43	47
Sector Average							**107**	**74**	**42.2**	**42.0**	**16.7**	**22.3**	**57**	**52**
Prober/Prober solutions														
Hermes Testing Solutions	7856.TWO	Ryan Sun	3,580	8,880	4,190	Buy	221	487	143.9	27.3	9.1	7.0	10	29
Tokyo Electron	8035.T	Atsuhiro Kinoshita	184,031	18,720	12,805	Buy	-16	71	29.7	37.1	6.5	12.8	36	44
Sector Average							**102**	**279**	**86.8**	**32.2**	**7.8**	**9.9**	**23**	**36**
Handler														
Hon.Precision, Inc.	7769.TW	Sunny Lin	33,404	10,000	5,905	Buy	90	86	45.2	24.4	14.5	10.6	36	50
Chroma ATE	2360.TW	Jerry Su	31,434	2,960	2,350	Buy	64	38	51.8	37.7	28.6	20.8	58	64
Cohu, Inc.	COHU.O	N/A	3,460	N/A	73	N/A	n.a.	94	77.0	39.6	-	-	7	16
JHT Design Co Ltd	603061.SS	N/A	4,045	N/A	312	N/A	155	47	54.8	37.3	13.0	9.9	22	25
Hangzhou Changchuan Technology	300604.SZ	Yongwei Lai	22,690	310	240	Buy	60	53	76.1	49.8	23.4	16.5	36	39
Sector Average							**92**	**63**	**61.0**	**37.8**	**19.9**	**14.4**	**32**	**39**
Test Socket														
WinWay Technology	6515.TW	Ryan Sun	6,948	8,900	6,130	Buy	119	98	60.2	30.4	22.0	14.6	44	34
Leeno Industrial Inc	058470.KQ	N/A	4,936	N/A	86,900	N/A	33	17	35.4	30.3	8.0	6.8	24	24
ISC Co Ltd	095340.KQ	N/A	3,247	N/A	205,500	N/A	70	41	47.0	33.2	7.0	5.9	16	19
Yokowo Co Ltd	6800.T	N/A	744	N/A	4,925	N/A	28	93	38.3	19.8	-	1.8	-	-
Sector Average							**62**	**62**	**45.2**	**28.4**	**12.3**	**7.3**	**28**	**26**
Load Board														
Chunghwa Precision Test Tech	6510.TWO	Ryan Sun	3,149	5,000	3,055	Buy	112	99	47.5	23.8	9.2	7.3	21	34
Probe Card														
Technoprobe SpA	TPRO.MI	Harry Blaiklock, CFA	27,100	41	37	Buy	259	18	66.9	56.5	14.6	11.6	25	23
MPI Corp	6223.TWO	N/A	17,518	N/A	5,670	N/A	97	110	87.2	41.5	30.3	19.1	38	54
FormFactor Inc	FORM.O	N/A	11,038	N/A	141	N/A	159	24	46.2	37.1	9.3	8.9	19	23
MICRONICS JAPAN CO.,LTD.	6871.T	N/A	3,906	N/A	15,410	N/A	155	28	24.8	19.4	7.1	5.6	32	36
Japan Electronic Materials Corp	6855.T	N/A	804	N/A	8,650	N/A	46	43	24.9	17.4	-	-	-	-
Sector Average							**138**	**54**	**49.6**	**32.6**	**14.1**	**10.5**	**27**	**34**

Source: LSEG, UBS estimates. Priced as of October 6 (HKT).

WHAT´S PRICED IN?

HTSI is trading at NT$4,190 as of 6 Oct 2026



Source: LSEG, UBS

Despite a 264% YTD share-price increase, HTSI still trades at just 15x 2028E P/E. We believe market expectations for 2027E CPO testing equipment shipments remain conservative, particularly for insertion 2 equipment. Based on our industry analysis, we estimate HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, this should be higher than the market's expectation of no insertion 2 and limited insertion 3 shipment due to engineering bottleneck. We also see further upside in 2028E as CPO adoption expands. Our reverse DCF analysis indicates that the current market price implies only a 17% long-term sales CAGR vs our estimate of 27% and long-term GM level of 33% vs our estimate of 61%. We see scope for re-rating.

We believe the market is not fully reflecting three key growth drivers:

1) Shift-left testing is still underappreciated.

As advanced packaging costs rise and AI accelerators become increasingly complex, more thorough testing is done at the wafer-level. This should drive sustained demand for HTSI's thermal control modules, probe card consumables, and other wafer-level testing solutions.

2) CPO revenue potential remains underestimated.

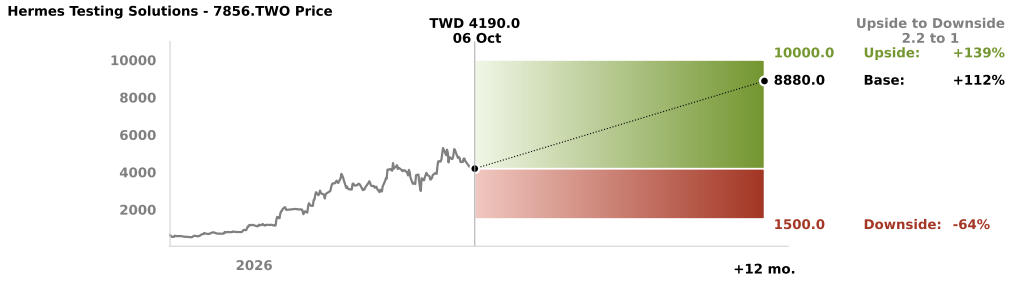
Investor expectations for 2027 CPO equipment shipments, particularly insertion 2 testing systems, remain conservative. Based on our industry studies, we forecast HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, with further potential upside in 2028E as CPO adoption broadens.

3) New products provide additional optionality.

Beyond its core businesses, HTSI is expanding into higher-value solutions including mass-reflow tools, MEMS probe cards and laser cleaning systems, creating incremental growth opportunities that we believe are largely absent from current expectations.

We value HTSI using 31x average 2028E PE, underpinned by a 32% 2027-31E earnings CAGR. We use 2028E PE as our view is that 2028 is the year for thermal control modules and CPO adoption grow from the initial ramping up stage to mass adoption stage. We believe using handlers as a peer group is justifiable for a unique company like HTSI, with thermal control modules for chip probing sharing similar characteristics as active thermal control used for final test, in addition to Hon Precision and Chroma having exposure to CPO test equipment. For equipment vendors, the peer group's average NTM PE post 2022 is around 35x, and the group is currently trading at around 41x NTM PE (see What's Priced In chart).

UPSIDE/DOWNSIDE SPECTRUM



HTSI is trading at NT $4,190 as of 6 Oct 2026

Value drivers (2027/28E)	Thermal control module shipments	CPO insertion 2 & 3 shipments	GM
NT$10,000 upside	3,300/5,000	60&100 / 100&175	61.2%/63.6%
NT$8,880 base	3,000/4,100	40&70 / 80&160	58.3%/60.0%
NT$1,500 downside	1,300/2,000	5&20 / 10&50	53.3%/54.0%

Source: UBS estimates

Risk to the current share price is heavily skewed (2.2:1) to the upside.

UPSIDE (NT$10,000): Our upside scenario reflects stronger than expected thermal module and CPO equipment shipment in 2027-28E, with additional upside from better traction of probe cards and mass reflow tools, forming 157%/91% YoY sales growth in 2027/28E and even higher operating leverage with OPM of 46.3%/50.0% in 2027/28E. This scenario implies a 34x 2028E PE, a upside driving by more equipment demand stemming for robust shift-left test induced equipment bottleneck, alongside CPO proliferation with scale-up CPO equipment pull-in in 2028E.

BASE (NT$8,880): Our price target of NT$8,880 is based on 31x 2028E PE. We expect an earnings CAGR of 32% in 2027-31E. With shift-left trend for more thermal module requirement during testing, alongside 40/70 CPO insertion 2/3 shipment. We see these growth drivers driving YoY sales growth from a high base of 127% in 2026E to 152%/78% YoY in 2027/2028E, with OPM of 44.3%/47.0% in 2027/28E.

DOWNSIDE (NT$1,500): Our downside scenario reflects lower shipment of thermal control modules due to less advanced prober demand with less insertion 2 and 3 equipment shipment due to bottlenecks with CPO production, alongside limited additional growth drivers to be picked up in 2028 such as MEMS probe card and mass reflow tools. This leads to weaker YoY sales growth of 67%/50% in 2027/28E and operating margin of 29.3%/32.0% in 2027/28E as equipment shipment fails to scale. This scenario implies a 14x 2028E PE, as valuation de-rate with dimmed outlook without CPO and limited upside from thermal control module shipments.

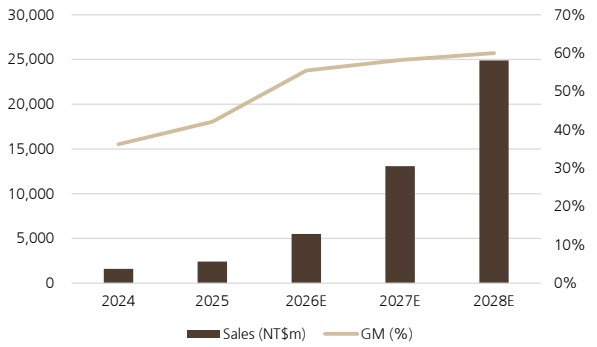
COMPANY DESCRIPTION

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Industry Outlook

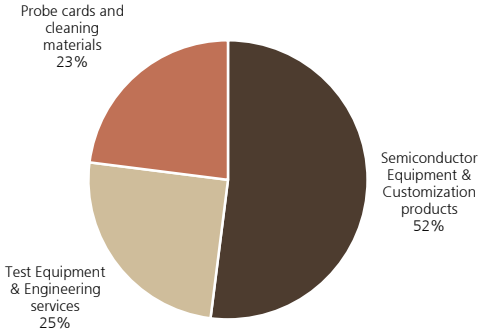
New iterations of semiconductors, especially for AI / HPC applications, are set to increase in both front-end wafer fabrication advancements and back-end packaging complexities with each iterations. With the industry now moves to utilize "shift left", chip probing and its relevant supply chain will be key beneficiaries of this trend which we believe will unlock multi-year, secular growth trends.

HTSI sales and GM trend



Source: Company data, UBS estimates

HTSI H126 sales mix breakdown



Source: Company data, UBS

Hermes Testing Solutions (7856.TWO)

Income Statement (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**1,598**	**2,425**	**5,505**	*127.0*	**13,879**	*152.1*	**24,713**	**29,218**	**32,810**
Gross profit	579	1,021	3,054	*199.2*	8,097	*165.1*	14,817	17,801	20,107
EBITDA (UBS)	**280**	**556**	**1,329**	*139.0*	**6,512**	*389.8*	**12,138**	**15,053**	**17,182**
Depreciation & amortisation	(226)	(260)	(254)	*2.3*	(358)	*-41.0*	(534)	(759)	(1,012)
EBIT (UBS)	**54**	**297**	**1,076**	*262.7*	**6,154**	*NM*	**11,604**	**14,294**	**16,170**
Associates & investment income	0	0	0	-	0	-	0	0	0
Other non-operating income	13	35	1	*-96.8*	0	-	0	0	0
Net interest	(10)	(9)	8	-	169	*NM*	214	286	364
Exceptionals (incl goodwill)	0	0	0	-	0	-	0	0	0
Pre-tax profit	**57**	**322**	**1,085**	*236.7*	**6,323**	*NM*	**11,818**	**14,580**	**16,534**
Tax	0	(54)	(223)	*-314.2*	(1,265)	*NM*	(2,364)	(2,916)	(3,307)
Profit after tax	**57**	**268**	**862**	*221.2*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Preference dividends	0	0	0	-	0	-	0	0	0
Minorities	0	0	0	*-83.3*	0	-	0	0	0
Extraordinary items	0	0	0	-	0	-	0	0	0
Net earnings (local GAAP)	**57**	**268**	**862**	*221.1*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Net earnings (UBS)	**57**	**268**	**862**	*221.1*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Tax rate (%)	0.0	16.7	20.6	*23.0*	20.0	*-2.7*	20.0	20.0	20.0

Per Share (NT$)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	2.71	10.65	29.11	*173.4*	153.28	*NM*	286.48	353.44	400.81
EPS (local GAAP, diluted)	2.71	10.65	29.11	*173.4*	153.28	*NM*	286.48	353.44	400.81
EPS (UBS, basic)	2.84	10.83	29.11	*168.8*	153.29	*NM*	286.49	353.45	400.82
DPS (net) (NT$)	0.00	1.10	11.64	*NM*	61.32	*NM*	114.60	141.38	160.33
Cash EPS (UBS, diluted) [1]	13.49	20.94	37.67	*79.9*	164.12	*335.6*	302.66	376.44	431.48
Book value per share	12.85	46.45	459.82	*NM*	602.66	*31.1*	827.84	1,066.69	1,326.13
Average shares (diluted)	21	25	30	*17.5*	33	*11.5*	33	33	33

Balance Sheet (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	182	709	14,360	*NM*	17,474	*21.7*	22,786	29,613	36,992
Other current assets	589	806	1,732	*114.7*	3,913	*125.9*	6,302	6,856	7,786
Total current assets	**771**	**1,515**	**16,092**	*NM*	**21,386**	*32.9*	**29,088**	**36,469**	**44,777**
Net tangible fixed assets	509	441	429	*-2.7*	765	*78.5*	1,467	2,169	2,797
Net intangible fixed assets	0	0	0	-	0	-	0	0	0
Investments / other assets	122	111	224	*101.6*	224	*0.0*	224	224	224
Total assets	**1,402**	**2,067**	**16,745**	*NM*	**22,376**	*33.6*	**30,780**	**38,863**	**47,799**
Trade payables & other ST liabilities	384	694	1,485	*113.8*	2,401	*61.8*	3,374	3,575	3,950
Short term debt	525	0	0	-	0	-	0	0	0
Total current liabilities	**909**	**694**	**1,485**	*113.8*	**2,401**	*61.8*	**3,374**	**3,575**	**3,950**
Long term debt	125	90	0	-	0	-	0	0	0
Other long term liabilities	28	19	86	*349.5*	86	*0.0*	86	86	86
Preferred shares	0	0	0	-	0	-	0	0	0
Total liabilities (incl pref shares)	**1,062**	**804**	**1,571**	*95.4*	**2,488**	*58.4*	**3,461**	**3,662**	**4,037**
Common s/h equity	**339**	**1,263**	**15,174**	*NM*	**19,888**	*31.1*	**27,319**	**35,201**	**43,762**
Minority interests	0	0	0	*-1.1*	0	*0.0*	0	0	0
Total liabilities & equity	**1,402**	**2,067**	**16,745**	*NM*	**22,376**	*33.6*	**30,780**	**38,863**	**47,799**

Cash Flow (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	57	268	862	*221.1*	5,059	*NM*	9,454	11,664	13,227
Depreciation & amortisation	226	260	254	*-2.3*	358	*41.0*	534	759	1,012
Net change in working capital	(254)	(65)	(374)	*NM*	(1,264)	*-238.1*	(1,416)	(354)	(554)
Other operating	88	139	256	*84.3*	0	-	0	0	0
Operating cash flow	**117**	**602**	**998**	*65.9*	**4,152**	*316.1*	**8,572**	**12,069**	**13,685**
Tangible capital expenditure	(157)	(122)	(275)	*-126.4*	(694)	*-152.1*	(1,236)	(1,461)	(1,641)
Intangible capital expenditure	0	0	0	-	0	-	0	0	0
Net (acquisitions) & disposals	(29)	17	13	*-24.0*	0	-	0	0	0
Other investing	(19)	(20)	(2)	*89.5*	0	-	0	0	0
Investing cash flow	**(206)**	**(125)**	**(265)**	*-111.9*	**(694)**	*-162.2*	**(1,236)**	**(1,461)**	**(1,641)**
Equity dividends paid	0	0	(27)	-	(345)	*NM*	(2,023)	(3,782)	(4,666)
Share issues / (buybacks)	23	626	13,102	*NM*	0	-	0	0	0
Other financing	(46)	(54)	(33)	*38.9*	0	-	0	0	0
Change in debt & pref shares	141	(525)	(125)	*76.2*	0	-	0	0	0
Financing cash flow	**117**	**46**	**12,916**	*NM*	**(345)**	-	**(2,023)**	**(3,782)**	**(4,666)**
Cash flow inc/(dec) in cash	**29**	**523**	**13,649**	*NM*	**3,113**	*-77.2*	**5,313**	**6,827**	**7,379**
FX / non cash items	-	4	2	*-42.4*	0	*-100.0*	0	0	0
Balance sheet inc/(dec) in cash	**-**	**527**	**13,652**	*NM*	**3,113**	*-77.2*	**5,313**	**6,827**	**7,379**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

Hermes Testing Solutions (7856.TWO)

Valuation (x)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	-	68.6	NM	27.3	14.6	11.9	10.5
P/E (UBS, diluted)	-	68.6	NM	27.3	14.6	11.9	10.5
P/CEPS	-	34.3	NM	25.5	13.8	11.1	9.7
Equity FCF (UBS) yield %	-	2.5	0.6	3.0	6.4	9.3	10.6
Dividend yield (net) %	-	0.2	0.3	1.5	2.7	3.4	3.8
P/BV	-	15.7	9.1	7.0	5.1	3.9	3.2
EV/revenues (core)	-	8.1	NM	7.1	3.8	3.0	2.5
EV/EBITDA (UBS core)	-	35.1	80.1	15.1	7.7	5.8	4.7
EV/EBIT (core)	-	65.8	99.0	15.9	8.1	6.1	5.0
EV/OpFCF (core)	-	65.8	99.0	15.9	8.1	6.1	5.0
EV/op. invested capital	-	27.4	>100	60.7	27.0	17.3	13.1

Enterprise value (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	-	19,581	113,884	113,884	113,884	113,884	113,884
Net debt (cash)	-	(75)	(7,489)	(15,917)	(20,130)	(26,200)	(33,302)
Buy out of minorities	28	19	86	86	86	86	86
Pension provisions/other	0	0	0	0	0	0	0
Total enterprise value	**-**	**19,526**	**106,481**	**98,054**	**93,841**	**87,771**	**80,668**
Non core assets	(29)	0	0	0	0	0	0
Core enterprise value	**-**	**19,526**	**106,481**	**98,054**	**93,841**	**87,771**	**80,668**

Growth (%)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	-	51.8	127.0	152.1	78.1	18.2	12.3
EBITDA (UBS)	-	98.9	139.0	NM	86.4	24.0	14.1
EBIT (UBS)	-	NM	NM	NM	88.6	23.2	13.1
EPS (UBS, diluted)	-	NM	173.4	NM	86.9	23.4	13.4
Net DPS	-	-	NM	NM	86.9	23.4	13.4

Margins & Profitability (%)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	36.2	42.1	55.5	58.3	60.0	60.9	61.3
EBITDA margin	17.5	22.9	24.1	46.9	49.1	51.5	52.4
EBIT (UBS) margin	3.4	12.2	19.5	44.3	47.0	48.9	49.3
Net earnings (UBS) margin	3.6	11.1	15.7	36.4	38.3	39.9	40.3
ROIC (EBIT)	-	41.7	NM	NM	NM	NM	NM
ROIC post tax	-	34.7	NM	NM	NM	NM	NM
ROE (UBS)	-	33.5	10.5	28.9	40.1	37.3	33.5

Capital structure & Coverage (x)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	1.7	(1.1)	(10.8)	(2.7)	(1.9)	(2.0)	(2.2)
Net debt / total equity %	138.1	(49.0)	(94.6)	(87.9)	(83.4)	(84.1)	(84.5)
Net debt / (net debt + total equity) %	58.0	(95.9)	NM	NM	NM	NM	NM
Net debt/EV %	-	(0.4)	(7.0)	(16.2)	(21.5)	(29.8)	(41.3)
Capex / depreciation %	69.5	46.8	108.5	194.1	NM	192.5	162.1
Capex / revenue %	9.8	5.0	5.0	5.0	5.0	5.0	5.0
EBIT / net interest	5.3	31.3	-	-	-	-	-
Dividend cover (UBS)	-	9.9	2.5	2.5	2.5	2.5	2.5
Div. payout ratio (UBS) %	-	10.1	40.0	40.0	40.0	40.0	40.0

Revenues by division (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	1,598	2,425	5,505	13,879	24,713	29,218	32,810
Total	**1,598**	**2,425**	**5,505**	**13,879**	**24,713**	**29,218**	**32,810**

EBIT (UBS) by division (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	54	297	1,076	6,154	11,604	14,294	16,170
Total	**54**	**297**	**1,076**	**6,154**	**11,604**	**14,294**	**16,170**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	111.9%
Forecast dividend yield	0.3%
Forecast stock return	112.2%
Market return assumption	6.3%
Forecast excess return	105.9%

Company Description

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Valuation Method and Risk Statement

We value Hermes Testing based on PE methodology.
Risks for the semis industry:
- Geopolitical uncertainties: Geopolitical uncertainties including tariffs and more export control could negatively impact semis cycle, and a downturn would lead to slower demand in the overall market and moderating capacity expansion.
- Meaningful deceleration of new technology development: Slower demand and less investment are expected If there is meaningful deceleration of new technology development such as Cloud AI, autonomous driving, etc.
- Rising utility cost: Larger-than-expected utility cost increases would further pressure profitability. Rising utility costs in Taiwan have negatively impacted margins for Taiwan tech companies.
Company risks:
- Faster- or slower-than-expected front-end/CoWoS capacity expansion: This could result in Hermes Testing gaining/losing equipment orders from major foundries and OSATs.
- Competitors achieving technological advancement: Potential risk of competition (from either local or foreign companies) achieving technological advancements could result in Hermes Testing losing market share from major foundries and OSATs.
- Better- or worse-than-expected AI growth: Better- or worse-than-expected AI growth could lead to a ramp-up or a slowdown of equipment orders from major foundries and OSATs to Hermes Testing, affecting long-term revenue growth potential.
- Slowdown or delay of new technological advancement: A potential slowdown or delay of the rollout for technological advancements for technologies such as SoIC, CoPoS, and CPO, would limit future equipment orders.
- Increase of raw material costs: Rising raw material costs may directly impact Hermes Testing's margins for equipment shipments.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Hermes Testing Solutions

Question	Response
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	4
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	4
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	3
5. What's driving the difference?	
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	3
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	No Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

Required Disclosures

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UBS Global Research: Global Equity Rating Definitions

12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

KEY DEFINITIONS:Forecast Stock Return (FSR) is defined as expected percentage price appreciation plus gross dividend yield over the next 12 months. In some cases, this yield may be based on accrued dividends. **Market Return Assumption (MRA)** is defined as the one-year local market interest rate plus 5% (a proxy for, and not a forecast of, the equity risk premium). **Under Review (UR)** Stocks may be flagged as UR by the analyst, indicating that the stock's price target and/or rating are subject to possible change in the near term, usually in response to an event that may affect the investment case or valuation. **Equity Price Targets** have an investment horizon of 12 months.

EXCEPTIONS AND SPECIAL CASES:UK and European Investment Fund ratings and definitions are: Buy: Positive on factors such as structure, management, performance record, discount; **Neutral:** Neutral on factors such as structure, management, performance record, discount; **Sell:** Negative on factors such as structure, management, performance record, discount. **Core Banding Exceptions (CBE):** Exceptions to the standard +/-6% bands may be granted by the Investment Review Consultation (IRC). Factors considered by the IRC include the stock's volatility and the credit spread of the respective company's debt. As a result, stocks deemed to be very high or low risk may be subject to higher or lower bands as they relate to the rating. When such exceptions apply, they will be identified in the Company Disclosures table in the relevant research piece.

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UBS Securities Pte. Ltd., Taipei Branch: Ryan Sun, Sunny Lin.

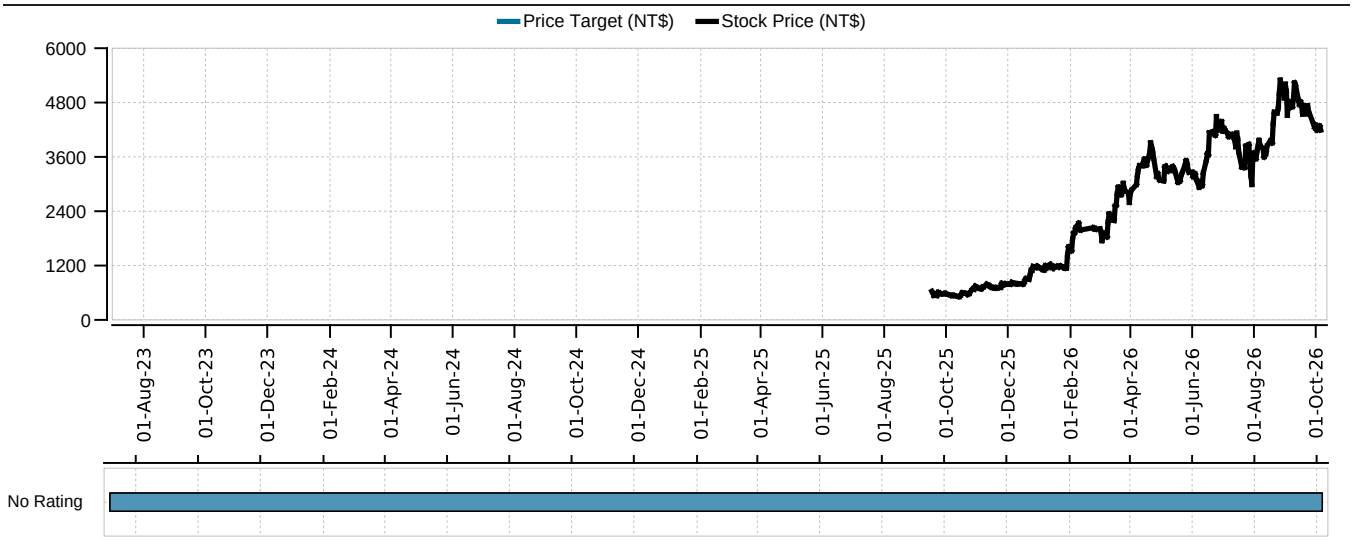
Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
Hermes Testing Solutions	7856.TWO	No Rating	NT$4,190	06 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

Unless otherwise indicated, please refer to the Valuation and Risk sections within the body of this report. For a complete set of disclosure statements associated with the companies discussed in this report, including information on valuation and risk, please contact UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research.

Hermes Testing Solutions (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-06	NaN	-	No Rating

Source: UBS Global Research; LSEG Eikon as of 06-Oct-2026. All prices as of local market close. Ratings as of date shown.

Additional Prices: Hangzhou Changchuan Technology, Rmb239.79 (30 Sep 2026); ASE Industrial, NT$744.00 (06 Oct 2026); Taiwan Semiconductor Manufacturing, NT$2585.00 (06 Oct 2026); Chroma ATE, NT$2350.00 (06 Oct 2026); Teradyne Inc, US$430.32 (06 Oct 2026); Advantest, ¥41850 (06 Oct 2026); Hon.Precision, Inc., NT$5905.00 (06 Oct 2026); Source: UBS. All prices as of local market close.

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Initiation of Coverage

Hermes Testing Solutions

Shifting-left, scaling-up: dual catalysts in chip probing and CPO

Equities

Taiwan
Semiconductors

12-month rating	**Buy**
	Prior : No Rating
12m price target	**NT$8,880.00**
	Prior :
Price (06 Oct 2026)	**NT$4,190**

RIC: 7856.TWO **BBG:** 7856 TT

Trading data and key metrics

52-wk range	NT$5,295.00-516.55
Market cap.	NT$114b/US$3.58b
Shares o/s	27.2m (ORD)
Free float	94%
Avg. daily volume ('000)	240
Avg. daily value (m)	NT$1,068.4
Common s/h equity (12/26E)	NT$15.2b
P/BV (12/26E)	9.1x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From	To	% ch	Cons.
12/26E	-	29.11	-	32.19
12/27E	-	153.28	-	89.82
12/28E	-	286.48	-	71.44

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Jack-of-all-trades for bespoke testing solutions in the semiconductor industry

We initiate coverage of Hermes Testing (HTSI) with a Buy rating and NT$8,880 PT. HTSI is uniquely positioned to benefit from two key semiconductor testing trends: shift-left testing and the emerging CPO testing market. Leveraging its long-standing relationship with TSMC, HTSI has established capabilities across electrical test engineering, customised test equipment, thermal control modules, probe cards, and CPO testing solutions. We forecast revenue growth of 127%/152%/78% in 2026E/27E/28E, driven by increasing adoption of thermal control modules for wafer-level test and initial deployment of CPO insertion 2 and insertion 3 test equipment through ficonTEC. Additional 2028E growth drivers include reflow tools, laser clean solutions, and MEMS probe cards. The stock trades at 27/15x 2027/28E PE, versus ~35x NTM PE for handler-equipment peers, offering re-rating potential as sales scale, new products are qualified, and its CPO presence expands. See our companion testing deep dive sector note.

Shift-left beneficiary with exceptional thermal control modules

Thermal control modules should be HTSI's largest growth driver over the next several years. We expect increasing chip power consumption, rising package complexity, and tighter probe pitches to drive adoption of advanced thermal control solutions used alongside leading-edge probers, in additional to more wafer level insertions and longer test time. We estimate HTSI will ship approximately 3,000 thermal-control modules in 2027E, up from around 1,200 units in 2026E, sales contribution rise from ~30% in Q226 to >50% of sales by 2027E. The launch of its second-generation platform in 2027E, which carries 25-50% higher ASPs, should further improve profitability. We forecast GM expansion from 55.5% in 2026E to 58.3%/60.0% in 2027E/28E.

Unique position to capture CPO upside

Through its partnership with German optical-automation leader ficonTEC, HTSI is positioned to participate in the initial ramp of CPO testing equipment. HTSI supports local manufacturing, installation, automation and after-sales service, making it a key enabler of volume production. We estimate HTSI captures approximately 30% of equipment value across the 40 insertion 2 and 70 insertion 3 systems we forecast for delivery in 2027E, with further potential upside in 2028E as adoption expands.

Valuation: initiate coverage with a Buy rating and price target NT$8,880

We forecast EPS of NT$29.1 in 2026E (including a one-off employee share purchase programme impact), rising to NT$153.3/286.5 in 2027E/28E. Our NT$8,880 price target is based on 31x average 2028E PE, versus handler-equipment peers forward PE at roughly 35x, reflecting HTSI's unique exposure to both shift-left testing, CPO, and our estimate of 32% LT earnings CAGR. We initiate with a Buy rating.

Highlights (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	1,598	2,425	5,505	13,879	24,713	29,218	32,810
EBIT (UBS)	54	297	1,076	6,154	11,604	14,294	16,170
Net earnings (UBS)	57	268	862	5,059	9,454	11,664	13,227
EPS (UBS, diluted) (NT$)	2.71	10.65	29.11	153.28	286.48	353.44	400.81
DPS (net) (NT$)	0.00	1.10	11.64	61.32	114.60	141.38	160.33
Net (debt) / cash	(469)	618	14,360	17,474	22,786	29,613	36,992
Profitability/valuation	**12/24**	**12/25**	**12/26E**	**12/27E**	**12/28E**	**12/29E**	**12/30E**
EBIT (UBS) margin %	3.4	12.2	19.5	44.3	47.0	48.9	49.3
ROIC (EBIT) %	-	41.7	147.5	381.2	334.1	282.5	261.7
EV/EBITDA (UBS core) x	-	35.1	80.1	15.1	7.7	5.8	4.7
P/E (UBS, diluted) x	-	68.6	NM	27.3	14.6	11.9	10.5
Equity FCF (UBS) yield %	-	2.5	0.6	3.0	6.4	9.3	10.6
Dividend yield (net) %	-	0.2	0.3	1.5	2.7	3.4	3.8

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 4,190.00 on 06-Oct-2026

UBS Research THESIS MAP a guide to our thinking and what´s where in this report

PIVOTAL QUESTIONS

Q: Will HTSI be able to keep its moat within chip probing thermal control modules?
Yes. We expect HTSI's competitive position to strengthen as chip power consumption, package complexity, and probe density continue to increase. The company is developing next-generation thermal control solutions, including liquid cooling technologies, while also expanding into specialized chuck solutions designed to address wafer warpage. In addition, HTSI's close collaboration with TSMC, TEL, Advantest, and other OSATs should reinforce its position within advanced wafer-level testing.

Q: Will insertion 2 of CPO testing process be skipped?
No. We believe concerns surrounding insertion 2 adoption are overdone. While some industry participants favor bypassing insertion 2 due to throughput and ROI considerations, doing so increases the risk of defective optical engines being identified only after the components are delivered to OSATs, creating yield accountability challenges and higher packaging losses. Based on our industry studies, we expect insertion 2 testing to remain an important part of the CPO workflow over at least the next two years as the industry focuses on improving PIC yields and overall testing efficiency.

UBSVIEW

We initiate coverage of Hermes Testing (HTSI) with a Buy rating and NT$8,880 price target. HTSI offers unique exposure to two of the semiconductor testing industry's most attractive growth drivers: 1) shift-left testing and 2) CPO testing. The industry's shift-left testing strategy will have more test insertions move to the wafer level before packaging, in addition to longer CP time. Leveraging its longstanding relationship with TSMC, HTSI has established capabilities across thermal control modules, electrical-test engineering services, customized equipment, and emerging CPO testing solutions. We believe there could be re-rating potential as sales scale, new products are qualified, and its CPO presence expands.

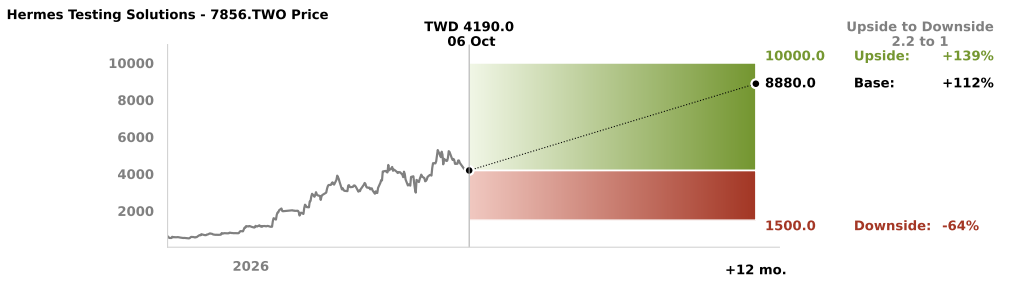
EVIDENCE

TSMC and OSATs have been expanding their CoWoS capacities aggressively, from 40kwpm in 2024 to around 90 kwpm in 2025, and is tracking to expand CoWoS capacity toward 160kwpm in 2026E. TSMC also ramped up expanding N3/N2 capacity, with multiple product launches from customers' pipelines—Nvidia's Rubin, AMD's MI400 series/Venice CPU and other cloud ASICs—translating into foreseeable, strong testing demand.

WHAT´S PRICED IN?

Despite a 264% YTD share-price increase, HTSI currently trades at just 15x of our 2028E EPS versus handler-equipment peers forward PE at roughly 35x. We believe market expectations for 2027 CPO testing equipment shipments remain conservative, particularly for insertion 2 equipment. Based on our industry analysis, we estimate HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, this should be higher than the market's expectation of no insertion 2 and limited insertion 3 shipment due to engineering bottleneck. We also see further upside in 2028E as CPO adoption expands. Our reverse DCF analysis indicates that the current market price implies only a 17% long-term sales CAGR vs our estimate of 27% and long-term GM level of 33% vs our estimate of 61%. We see scope for re-rating.

UPSIDE/DOWNSIDE SPECTRUM



Value drivers (2027/28E)	Thermal control module shipments	CPO insertion 2 & 3 shipments	GM
NT$10,000 upside	3,300/5,000	60&100 / 100&180	61.2%/63.6%
NT$8,880 base	3,000/4,100	40&70 / 80&160	58.3%/60.0%
NT$1,500 downside	1,300/2,000	5&20 / 10&50	53.3%/54.0%

Source: UBS estimates

COMPANY DESCRIPTION

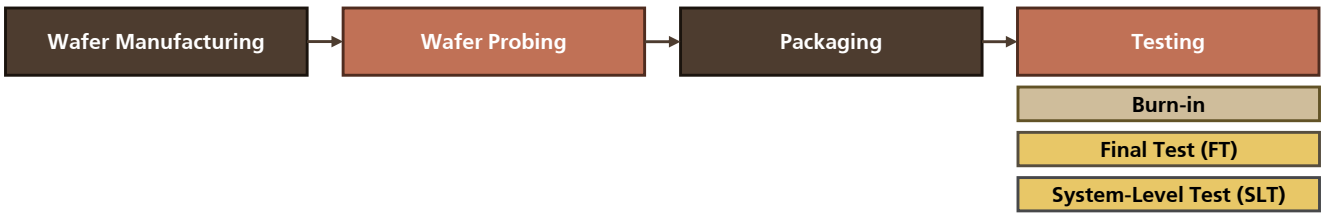
Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Testing TAM expanding with IC complexity: leaders likely to outperform industry growth

Our central view is that testing content per AI accelerator is growing faster than semiconductor unit volumes, creating a multi-year expansion in testing TAM. We believe semiconductor testing is entering a period of structural growth as AI-driven chip complexity increases testing intensity across wafer, package and system levels. The industry's shift-left strategy, which moves more testing upstream in the semiconductor manufacturing process, is resulting in longer and more frequent test insertions. As cloud AI GPUs and ASICs become increasingly complex, additional testing is required to ensure functionality, performance and yield, while reducing defective parts per million (DPPM). More robust testing should improve deployment efficiency and lower the cost of operating AI infrastructure.

Semiconductor testing generally comprises four stages: wafer/chip probing (CP), final test (FT), system-level test (SLT), and burn-in. CP is conducted at the wafer-level before packaging, while FT, SLT, and burn-in occur after packaging. The below Figure 3 illustrates the semiconductor manufacturing flow.

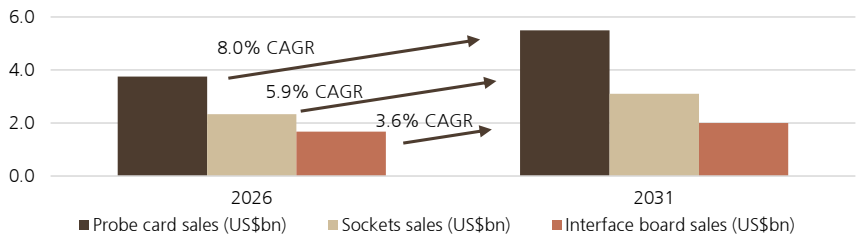
Figure 1: Semiconductor testing process: from wafer level to package level



Source: UBS

We estimate semiconductor testing historically accounts for ~1% of total semiconductor revenue. Within testing, consumables such as probe cards, sockets and interface boards represent ~10-15% of industry revenue. According to Yole, the probe card, socket and interface board markets should grow from US$3.8bn, US$2.3bn and US$1.7bn in 2026 to US$5.5bn, US$3.1bn and US$2.0bn by 2031, respectively.

Figure 2: Yole estimating testing consumables TAM to climb in the next 5 years



Source: Yole, UBS

Nonetheless, leaders within the testing sector will standout among competition, with greater sales CAGR comparing to the industry.

We see **HTSI**'s sales for the next few years to mainly be driven by 1) thermal control module shipment and 2) CPO equipment shipment. In the past, thermal control modules weren't necessary for chip probing, due to less TDP for the chips. However, as new iterations of AI accelerators now got substantial power draw, this unlocks a brand new TAM for HTSI to enter with their thermal control modules to pair with probers, in order to prevent probe card damages.

- **Robust thermal control module shipments.** We estimate HTSI to ship 3,000/4,100 thermal control modules in 2027E/28E, with continuous demand in future years, bringing their 2026-31E semiconductor equipment sales CAGR to 48%.

- **Thermal control module ASP expansion.** Gen 2 thermal control modules will start to ship in 2027E which will drive tool ASP up around 25-50%.
- **Demand supported by Advantest expansions.** We believe the 3,000 thermal control modules shipment forecast for 2027E is not far-fetched, as we read-across from Advantest's announced capacity expansion plan (testers will be paired with probers with a 1:1 ratio) with increasing capacity up to 5,000 testers within FY2027/3 and they also announced a 10,000-unit target within CY2028. Although the testers capacity expansion plan includes testers for final test, we do see a good portion to be allocated to chip probing given shift-left testing trend increasing both testing insertions and testing time before chip packaging.
- **CPO is another brand new TAM for semiconductor testing.** As insertion 2 and 3 testing equipment demand starts to rise in 2027 with scale-out CPO adoption, we see HTSI to ship 40/70 insertion 2 and 3 equipment in 2027E, with this number growing in 2028E to 80/160 and higher in future years as CPO proliferates, and driving 52% 2026-31E test equipment & engineering services sales CAGR.
- **CPO to see content increase.** Gen 2 insertion 3 equipment may start to ship in 2027E, with higher throughput and tool ASP.
- **Insertion 2 might scale down, but not happening anytime soon.** Insertion 2 equipment shipment might come down in 2029E and beyond, as foundries scale-up yield for photonic IC (PIC) in order to lessen the time consuming insertion 2 testing burden.

Figure 3: Hermes Testing is set to benefit with a 46% 2026-31E sales CAGR

Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Semiconductor Equipment & Customization products	**$2,815**	**$7,980**	**$13,324**	**$16,061**	**$18,315**	**$20,104**	**48%**
YoY		*183%*	*67%*	*21%*	*14%*	*10%*	
Test Equipment & Engineering services	**$1,480**	**$3,887**	**$7,901**	**$8,887**	**$9,898**	**$11,853**	**52%**
YoY		*163%*	*103%*	*12%*	*11%*	*20%*	
Probe cards and cleaning materials	**$1,210**	**$2,012**	**$3,488**	**$4,270**	**$4,597**	**$4,681**	**31%**
YoY		*66%*	*73%*	*22%*	*8%*	*2%*	
Key products shipment units, ASP, sales	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Thermal control module shipment	**1,200**	**3,000**	**4,100**	**4,500**	**4,700**	**4,900**	
Thermal control product blended ASP (NT$m)	$1.9	$2.4	$2.8	$3.0	$3.3	$3.5	
Total sales from thermal control module shipment	**$2,316**	**$7,168**	**$11,417**	**$13,505**	**$15,591**	**$17,067**	**49%**
% of total sales	*42%*	*52%*	*46%*	*46%*	*48%*	*47%*	
CPO Insertion 2 shipment	**8**	**40**	**80**	**60**	**20**	**20**	
Ins2 blended ASP (NT$m)	$13.8	$15.8	$17.4	$18.3	$19.2	$20.1	
Total sales from insertion 2	**$111**	**$633**	**$1,391**	**$1,095**	**$383**	**$402**	**29%**
% of total sales	*2%*	*5%*	*6%*	*4%*	*1%*	*1%*	
CPO insertion 3 shipment	**11**	**70**	**160**	**200**	**240**	**280**	
Ins3 blended ASP (NT$m)	$16.0	$17.2	$18.2	$19.1	$20.1	$21.1	
Total sales from insertion 3	**$176**	**$1,202**	**$2,917**	**$3,829**	**$4,824**	**$5,910**	**102%**
% of total sales	*3%*	*9%*	*12%*	*13%*	*15%*	*16%*	
Key product sales - total	**$2,602**	**$9,002**	**$15,725**	**$18,429**	**$20,798**	**$23,379**	**55%**
% of total sales	*47%*	*65%*	*64%*	*63%*	*63%*	*64%*	
Others - total	**$2,903**	**$4,877**	**$8,988**	**$10,789**	**$12,012**	**$13,260**	**36%**
% of total sales	*53%*	*35%*	*36%*	*37%*	*37%*	*36%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	**$5,505**	**$13,879**	**$24,713**	**$29,218**	**$32,810**	**$36,639**	**46%**
YoY		*152.1%*	*78.1%*	*18.2%*	*12.3%*	*11.7%*	
GM %	**55.5%**	**58.3%**	**60.0%**	**60.9%**	**61.3%**	**61.1%**	
OpM %	**19.5%**	**44.3%**	**47.0%**	**48.9%**	**49.3%**	**51.1%**	

Source: UBS estimates

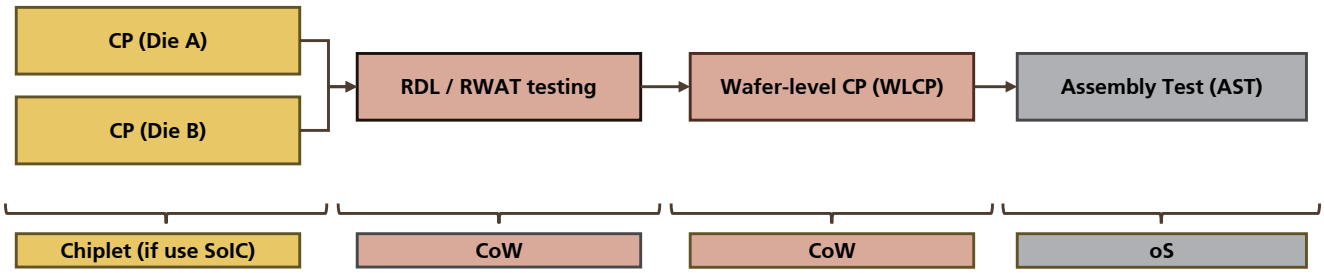
Robust growth drivers across semiconductor testing

We believe HTSI is set to benefit from several semiconductor testing growth drivers and capture the AI induced, expanding testing TAM. We see HTSI is uniquely positioned as the main beneficiary of two of the strongest growth drivers within semiconductor testing: testing shift-left and CPO.

Shift-left igniting secular growth for chip probing

The industry's adoption of shift-left testing should drive a multi-year structural growth opportunity for chip probing (CP) and its supply chain. As advanced packaging costs rise and AI accelerators become increasingly complex, semiconductor manufacturers are placing greater emphasis on identifying known-good dies (KGD) before packaging.

Figure 4: CP is set to grow substantially with shift-left, each packaging step set to be MORE costly than the prior step

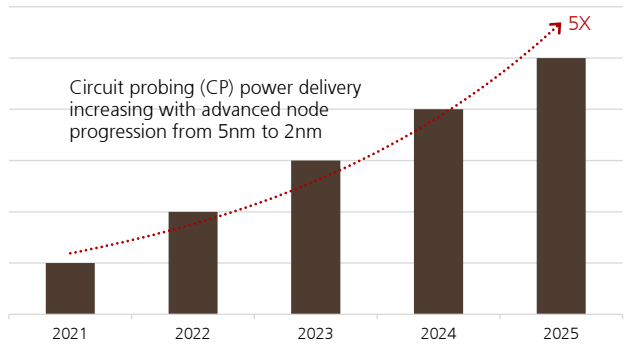


Source: UBS

The economics of advanced packaging reinforce this trend. Heterogeneous integration, chiplet architectures and 2.5D/3D packaging significantly increase package value, while defects identified late in the manufacturing process can lead to substantial scrap costs. To mitigate this risk, foundries and OSATs such as TSMC and ASE employ distributed testing flows incorporating KGD, known-good-stack (KGS) and assembly test screening, to catch defects before the components enter a multi-week packaging cycle (>2 weeks for CoW and 2 weeks for oS). The importance of these tests rises as AI clusters become increasingly dependent on component reliability and low defective-parts-per-million (DPPM) rates. Given the multiplicative yield degradation, a 300bps yield decline at just one manufacturing stage (e.g. dropping from 99.5% to 96.5%) reduces overall integrated four-stage yield from 98.0% to 95.0%; on a 500,000 unit ramp at US $10,000 per unit, this 3% scrap delta translates into around US$150m in unrecoverable losses given >6-month manufacturing lead times.

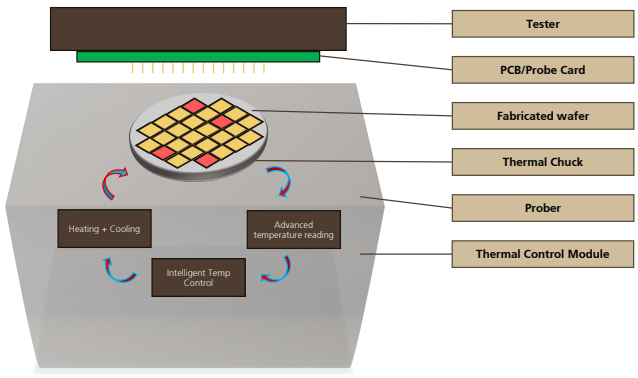
At the same time, rising chip power consumption is increasing the technical difficulty of wafer-level testing. As chip thermal design power scales beyond 1,000W and power density is expected by the industry to surge another 8x by 2030, traditional wafer-level chip probe (WLCP) encounters severe physical bottlenecks. Finer probe pitches and higher power densities increase the risk of overheating, probe needle burnout and yield degradation, creating demand for more advanced probing and thermal-control solutions.

Figure 5: CP power delivery trend rises with process tech evolution



Source: UBS

Figure 6: Thermal control module inside probers are essential to control wafer temperature



Source: UBS

Hermes Testing (HTSI) is a key beneficiary of this trend. Leveraging its long-standing relationship with TSMC and expertise in semiconductor test solutions, HTSI supplies advanced thermal control modules used alongside leading-edge probers (made by Tokyo Electron), particularly for AI applications. These thermal control modules are required for high TDP semiconductor wafer probe in order to maintain the temperature

of the chuck to avoid needle burns or probe card damages. We estimate HTSI will ship approximately 3,000 thermal-management modules in 2027E, up from 1,200 in 2026E, supporting sales growth of 127%/152% in 2026E/27E. HTSI is also the partner for prober installation, prober maintenance and services. In addition, HTSI provides high-end chuck processing services, an area that should benefit from increasing package sizes and higher wafer warpage associated with advanced packaging.

Beyond thermal control modules, HTSI also supplies consumables and customised solutions for wafer-level testing, including probe card cleaning sheets and laser-based probe head cleaning systems. As probing intensity increases under the shift-left testing model, demand for cleaning and maintenance solutions should rise alongside probe card utilisation. Currently, cleaning sheets is around 5% of sales, but we believe it could gain traction in 2027/28E and beyond. We see additional upside from future modular laser cleaning systems that could be integrated directly with probers, further expanding HTSI's content opportunity within advanced wafer-level testing.

CPO testing fuelling future growth for semiconductor testing

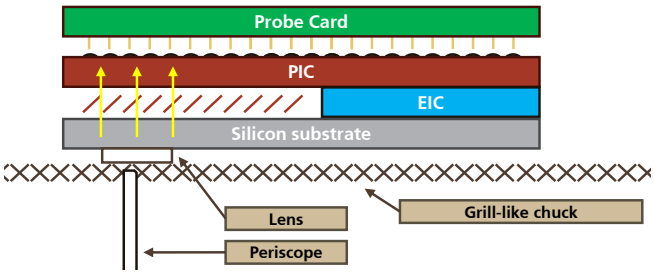
As interconnect bottlenecks increasingly shift from the memory wall to the copper wall, we believe co-packaged optics (CPO) adoption is inevitable over the longer term. Testing remains one of the key bottlenecks to CPO deployment, as optical validation differs fundamentally from the electrical testing processes that have dominated semiconductor manufacturing for decades. As a result, CPO is creating a new testing opportunity across the semiconductor equipment and consumables ecosystem. We have written extensively on CPO and opportunities that it brings to the semiconductor supply chain, see our CPO deep dive, our thoughts on CPO timing, 2026 SiPhIA conference takeaways, and SEMICON 2026 takeaways.

Various test equipment suppliers are scrambling to come up with ideal testing equipment across the four insertions of CPO testing, with solutions to ensure both optical and electrical signal integrity of the CPO solutions. The four insertions are:

- **Insertion 1: Electronic IC (EIC)/Photonic IC (PIC) test.** Single-side wafer test that has dedicated probe cards to test the fabricated EIC/PIC wafers.
- **Insertion 2: Optical/Electrical test.** Double sided wafer test, testing the hybrid bonded (e.g. via TSMC's COUPE) EIC and PIC wafers by an opto-electrical test that verifies the electrical signals on the top and optical signal on the bottom of the hybrid bonded wafers.
- **Insertion 3: Optical engine (OE) test.** The OE tests are done to ensure the quality of the OEs and with FAU (Fiber Array Unit) attached, to ensure the signal integrity of the OEs. The testing equipment needs to be able to handle 224G per channel data rates, requiring high unit per hour (UPH) throughput, meticulous dirt mitigation, and maintaining a near 100% contact rating on high-speed sockets.
- **Insertion 4: ASIC final test.** This final test step is established post the OEs are packaged together with the ASIC. The complete ASIC package (now with CPO) is now tested for opto-electrical functionalities.

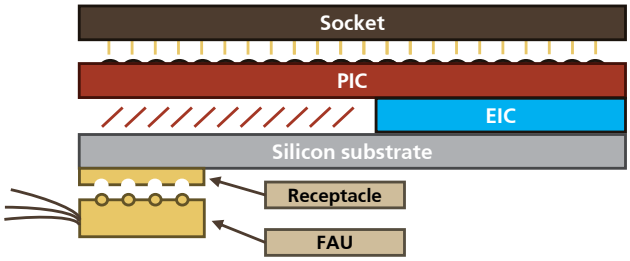
As it stands, insertion 2 and insertion 3 are the two areas where there are the most debates in the industry. Insertion 3 suppliers such as Chroma has noted the trend of "shift-to-middle" paradigm centered on insertion 3. In high-density scale-out switches deploying up to 32 optical engines (e.g. Nvidia's Spectrum-X architectures), any OE defect results in package scrap at the CoWoS level, making 100% known good die (KGD) validation at the singulated OE stage an absolute commercial requirement.

Figure 7: An illustration for insertion 2 test method



Source: UBS

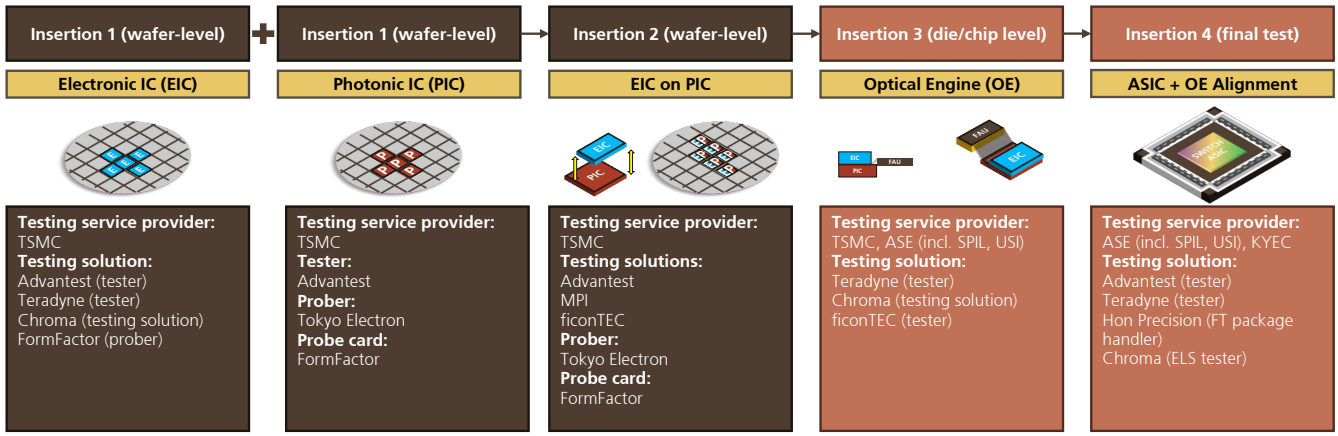
Figure 8: An illustration for insertion 3 test method



Source: UBS

As shown in the below figure, as there are no set standards for the testing equipment, throughput and stability are now the main target and area of competition for the major players in the field. Notable ones for each insertions are listed.

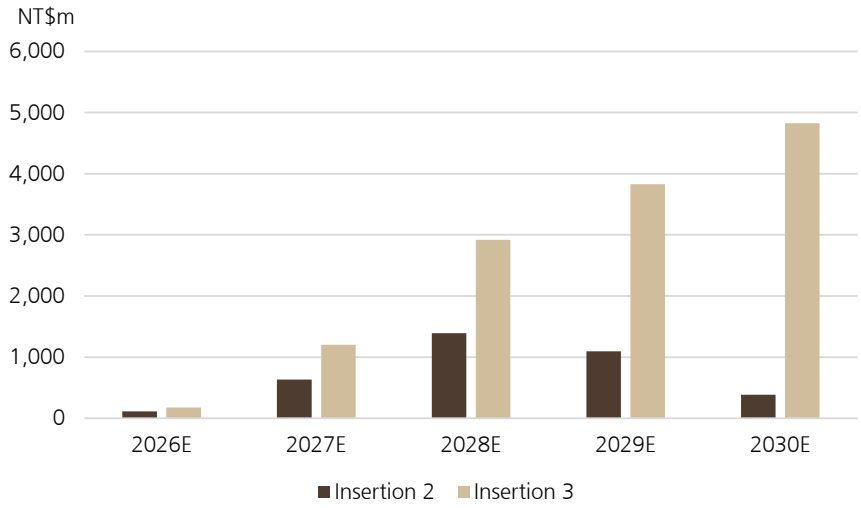
Figure 9: CPO testing flow chart with key suppliers



Source: UBS

With its 25 years of precision test equipment expertise, ficonTEC is an early mover for CPO test equipment and has key know-how for optical alignment, which is quintessential for optical test. **HTSI**, through its partnership with ficonTEC, is set to be a major beneficiary once CPO proliferates in 2027E-28E. As the partner of ficonTEC, Hermes Testing is the key enabler that can help ficonTEC achieve HVM for its CPO testing equipment. As of now, our view is that both insertion 2 and insertion 3 should be required within the next two years, despite the difficulties surrounding double-sided wafer probing (insertion 2), as the yield for OEs are still not high enough to avoid defective dies. Finding out known-good OEs will be crucial before the FAU coupling process during the latter stages of CPO packaging, to prevent impeding the final package with multiple OEs attached to the switch ASIC (e.g. 32 OEs on Nvidia's Spectrum-X). We estimate Hermes Testing will ship 40/70 units of insertion 2/3 testing equipment in 2027E, with potential upside to the numbers as we view Hermes Testing will likely assist with the ramp up for Gen 2 insertion 3 equipment in 2027E. Gen 2 insertion equipment will come with better throughput and ASP.

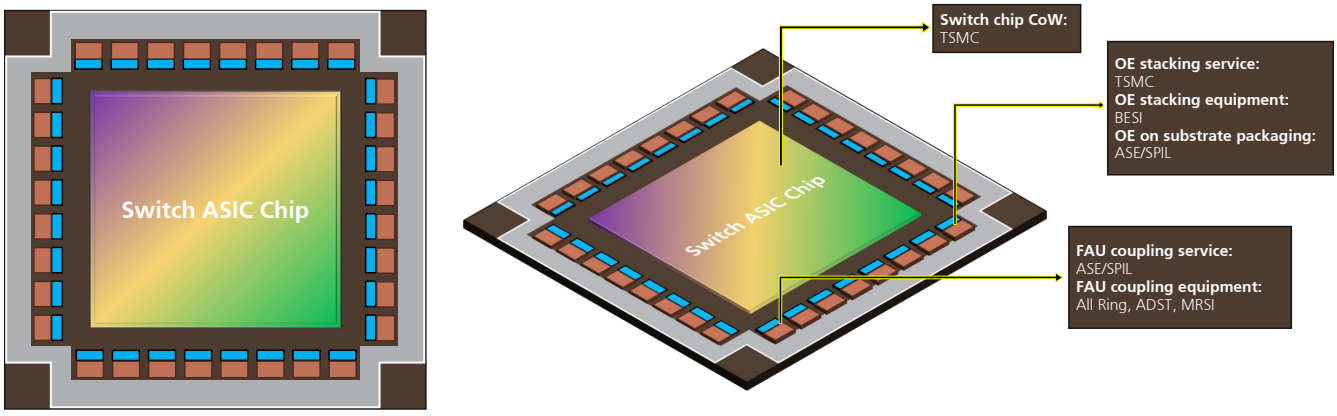
Figure 10: We believe HTSI CPO equipment sales will pick up in 2027/28E with 541%/135% YoY growth



Source: UBS estimates

Beyond insertions 2 and 3, HTSI is also positioned to participate in insertion 1 testing through customised prober solutions for photonic integrated circuit (PIC) testing. We believe these solutions could eventually be packaged as licensable intellectual property, creating an additional long-term growth opportunity as CPO adoption broadens. While near-term testing growth is driven by AI accelerators, CPO represents the next major expansion cycle for semiconductor testing. We believe HTSI is uniquely positioned to participate across multiple test insertions, providing exposure to a potentially significant source of incremental revenue beginning in 2027E.

Figure 11: A switch chip with 32 OEs and the CPO packaging supply chain



Source: UBS

HTSI's unique position for new TAM of thermal control modules and CPO test equipment

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan, as a spin-off from Hermes-Epitek. The company provides semiconductor testing and inspection solutions, engineering services, customized equipment, and test consumables. HTSI maintains a global service footprint across Taiwan (Hsinchu, Tainan), Mainland China (Suzhou), Japan (Tokyo, Kumamoto), Singapore, Malaysia, and the United States (Arizona), enabling close support for leading semiconductor customers.

Its main business segments are structured around three pillars:

- **Semiconductor equipment and customized products:** HTSI utilises a dual-track strategy of independent R&D and also equipment agency sales. They offers

highly customized solutions designed to address specific client pain points and bottlenecks. Key products include: thermal control modules that pair with a 1:1 ratio with wafer-level test probers, electron beam inspection equipment, white light interference optical inspection systems, laser probe cleaning machines, and the SYMPHONY fluxless reflow system.

- **Testing equipment engineering services:** The company operates the largest engineering service team in Taiwan, consisting of over 340 technical professionals. The engineering service team provides extensive equipment life-cycle support, including installation, relocations, modification, testing, routine maintenance, and repair services. CPO related equipment sales/services will be recognized within this segment, as CPO are not direct HTSI products, but rather sales recognition via its partnership with ficonTEC.
- **Probe card design, manufacturing, and cleaning materials:** HTSI delivers a comprehensive product portfolio that supports the entire probe card life-cycle. For probe cards, they supply membrane, cantilever, and vertical probe cards. As they are small in scale, HTSI targets niche probe card applications such as RF, 5G/6G, millimeter wave, LCD driver, low end logics, TIA (Transimpedance Amplifier), etc. In addition to the three existing probe technologies, HTSI also collaborates with MJC in Japan for MEMS probe development, targeting sales in 2028E for high-end, AI/HPC related applications. Cleaning materials is another important business within this segment, as HTSI source and process sand paper or other materials to offer cleaning sheets to clean the probe cards, removing oxidations, unwanted particles and substances that accumulated on the probe cards during the testing process.

The sales resiliency in the coming years mainly stems from its robust relationship with key industry players, especially as an engineering and service provider with TSMC for testing solutions. This sets up a solid foundation for its customized equipment business, as they know the specific testing bottleneck to tackle. For instance, as Hermes-Epitek manufacturers the probers for Tokyo Electron (TEL) and HTSI helps with installation, services, and maintenance of the probers, they understand the thermal bottlenecks that the wafer-level test industry has been facing and can effectively come up with module solutions to solve this.

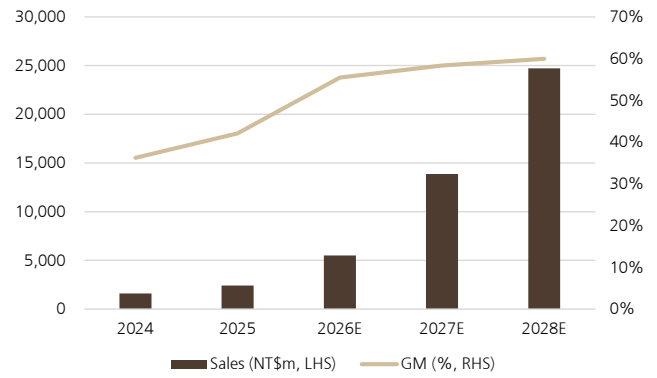
Margins set to expand with value-added products and services

The company has demonstrated strong execution in advanced wafer-level testing, with H126 revenue of NT$2.2bn already exceeding 90% of FY25 revenue (NT$2.4bn). Profitability has also improved significantly, with GM increasing from 42.1% in 2025 to 54.9% in H126.

With share of value-added products and services set to rise in 2027E in sales mix, HTSI is set to see GM and operating leverage expansion. As a niche engineering service provider that has robust ties to leading foundries and OSATs, we believe HTSI has substantial pricing power and estimate HTSI' GM for thermal control modules to be around 65%, with GM for CPO equipment of around 60% as HTSI provides an engineering-service-like business for it. We believe these factors will help the company to reach a 60% long-term GM range.

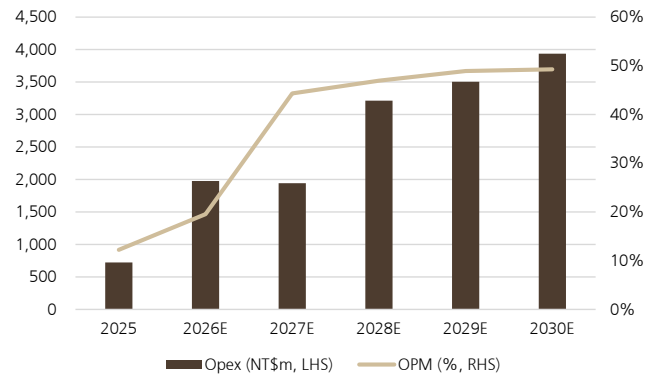
OPM is also set to expand with customized equipment sales rise in total sales mix. If we exclude the one-time expense to be recognized during Q326 for employee stock benefits (we estimate around NT$1bn), YoY opex growth will be around 99% for 2027E, with downside risk if the company conducts tight cost control measures. We forecast the company's OPM will expand from 19.5% in 2026E, to 44.3%/47.0% in 2027/28E.

Figure 12: HTSI sales set to continue to climb with 152% growth and GM at 58.3% in 2027E



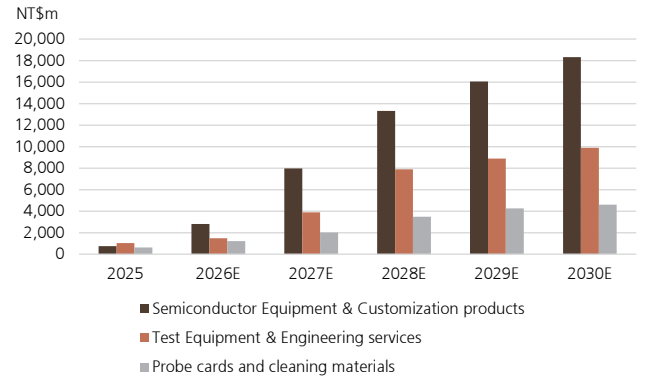
Source: UBS estimates

Figure 13: OPM set to grow to 44.3%/47.0% in 2027/28E



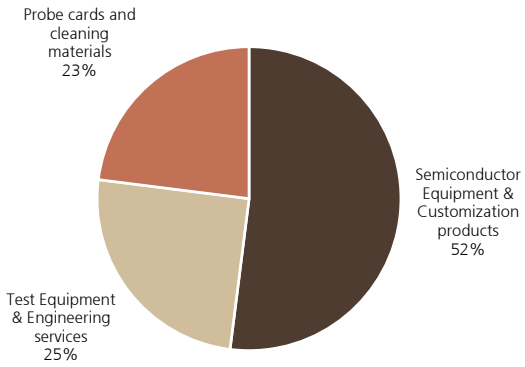
Source: UBS estimates. Note 2026E opex includes a one-time NT$1bn employee stock benefit.

Figure 14: We see customized equipment to continue to drive company growth



Source: Company data, UBS estimates

Figure 15: HTSI H126 sales mix breakdown



Source: Company data, UBS

Figure 16: UBS earnings forecast

(NT$m)	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E	2028E	2029E	2030E
Revenue	**2,425**	**985**	**1,200**	**1,513**	**1,807**	**5,505**	**2,634**	**3,197**	**3,760**	**4,289**	**13,879**	**24,713**	**29,218**	**32,810**
- YoY chg (%)	52%	123%	108%		28%	127%	167%	166%	149%	137%	152%	78%	18%	12%
- QoQ chg (%)		-30%	-51%	54%	51%		46%	-42%	43%	34%				
Gross profit	**1,021**	**545**	**655**	**843**	**1,012**	**3,054**	**1,534**	**1,856**	**2,184**	**2,523**	**8,097**	**14,817**	**17,801**	**20,107**
- Gross margin	42.1%	55.3%	54.5%	55.7%	56.0%	55.5%	58.3%	58.1%	58.1%	58.8%	58.3%	60.0%	60.9%	61.3%
Operating profit	**297**	**323**	**411**	**-399**	**741**	**1,076**	**1,166**	**1,408**	**1,657**	**1,923**	**6,154**	**11,604**	**14,294**	**16,170**
- Operating margin	12.2%	32.8%	34.3%	-26.4%	41.0%	19.5%	44.3%	44.1%	44.1%	44.8%	44.3%	47.0%	48.9%	49.3%
Pre-tax profit	**322**	**326**	**412**	**-396**	**743**	**1,085**	**1,205**	**1,449**	**1,700**	**1,968**	**6,323**	**11,818**	**14,580**	**16,534**
Net profit	**268**	**254**	**330**	**-317**	**594**	**862**	**964**	**1,160**	**1,360**	**1,574**	**5,059**	**9,454**	**11,664**	**13,227**
- YoY chg (%)	373%	586%	301%		299%	221%	280%	251%	-529%	165%	487%	87%	23%	13%
- QoQ chg (%)		70%	23%	-225%	80%		62%	35%	41%	36%				
Basic EPS (NT$)	**10.83**	**9.35**	**12.15**	**-10.20**	**18.01**	**29.11**	**29.22**	**35.14**	**41.22**	**47.70**	**153.29**	**286.49**	**353.45**	**400.82**
- YoY chg (%)	281%	492%	279%		200%	169%	213%	189%	-504%	165%	427%	87%	23%	13%
- QoQ chg (%)		55%	12%	-209%	48%		62%	21%	41%	36%				

Source: UBS estimates

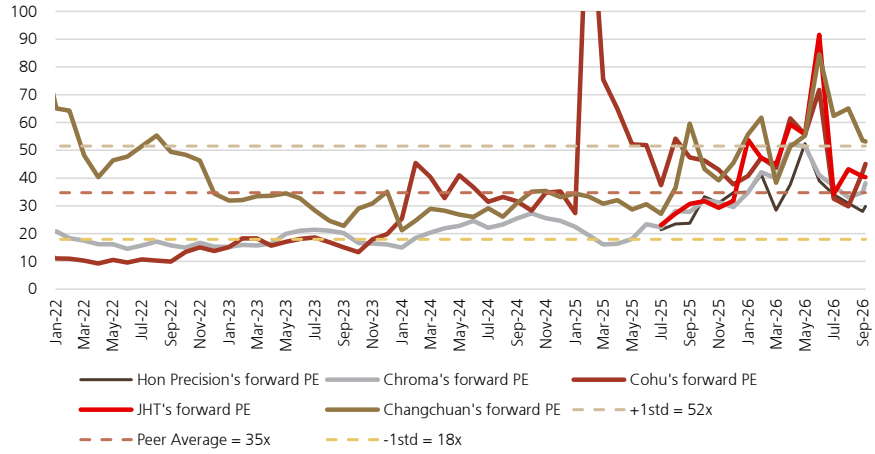
Figure 17: Testing supply chain valuation table

Company name	Ticker	Analyst	Market cap (US$m)	Price Target (LC)	Share price (LC)	Rating	EPS growth (%)		P/E (x)		P/BV (x)		ROE (%)	
							2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Testing service provider														
ASE Industrial	3711.TW	Sunny Lin	104,043	835	744	Buy	104	64	39.3	24.0	7.5	6.3	21	28
King Yuan Electronics	2449.TW	Sunny Lin	11,456	325	298	Buy	40	62	32.6	20.1	5.9	5.0	20	27
Amkor Technology	AMKR.O	Randy Abrams	13,114	80	53	Buy	73	8	20.6	19.1	2.6	2.4	14	13
JCET Group	600584.SS	Jimmy Yu	17,156	135	64	Buy	72	53	42.7	28.0	3.7	3.3	9	12
Tianshui Huatian Technology	002185.SZ	Jimmy Yu	7,928	13	16	Neutral	33	27	57.0	44.8	2.8	2.7	5	6
TongFu Microelectronics Co Ltd	002156.SZ	N/A	13,526	N/A	57	N/A	50	13	45.3	40.1	5.0	4.5	10	11
Shanghai V-Test Semiconductor Tech Co Ltd	688372.SS	N/A	2,517	N/A	100	N/A	36	106	40.6	19.7	3.8	3.2	11	18
Sector Average							**58**	**47**	**39.7**	**28.0**	**4.5**	**3.9**	**13**	**17**
Tester														
Advantest	6857.T	Atsuhiro Kinoshita	191,547	43,600	41,850	Buy	80	96	38.8	53.1	16.8	31.8	70	58
Teradyne Inc	TER.O	Timothy Arcuri	67,666	500	430	Buy	134	51	45.7	30.8	16.7	12.8	43	47
Sector Average							**107**	**74**	**42.2**	**42.0**	**16.7**	**22.3**	**57**	**52**
Prober/Prober solutions														
Hermes Testing Solutions	7856.TWO	Ryan Sun	3,580	8,880	4,190	Buy	221	487	143.9	27.3	9.1	7.0	10	29
Tokyo Electron	8035.T	Atsuhiro Kinoshita	184,031	18,720	12,805	Buy	-16	71	29.7	37.1	6.5	12.8	36	44
Sector Average							**102**	**279**	**86.8**	**32.2**	**7.8**	**9.9**	**23**	**36**
Handler														
Hon.Precision, Inc.	7769.TW	Sunny Lin	33,404	10,000	5,905	Buy	90	86	45.2	24.4	14.5	10.6	36	50
Chroma ATE	2360.TW	Jerry Su	31,434	2,960	2,350	Buy	64	38	51.8	37.7	28.6	20.8	58	64
Cohu, Inc.	COHU.O	N/A	3,460	N/A	73	N/A	n.a.	94	77.0	39.6	-	-	7	16
JHT Design Co Ltd	603061.SS	N/A	4,045	N/A	312	N/A	155	47	54.8	37.3	13.0	9.9	22	25
Hangzhou Changchuan Technology	300604.SZ	Yongwei Lai	22,690	310	240	Buy	60	53	76.1	49.8	23.4	16.5	36	39
Sector Average							**92**	**63**	**61.0**	**37.8**	**19.9**	**14.4**	**32**	**39**
Test Socket														
WinWay Technology	6515.TW	Ryan Sun	6,948	8,900	6,130	Buy	119	98	60.2	30.4	22.0	14.6	44	34
Leeno Industrial Inc	058470.KQ	N/A	4,936	N/A	86,900	N/A	33	17	35.4	30.3	8.0	6.8	24	24
ISC Co Ltd	095340.KQ	N/A	3,247	N/A	205,500	N/A	70	41	47.0	33.2	7.0	5.9	16	19
Yokowo Co Ltd	6800.T	N/A	744	N/A	4,925	N/A	28	93	38.3	19.8	-	1.8	-	-
Sector Average							**62**	**62**	**45.2**	**28.4**	**12.3**	**7.3**	**28**	**26**
Load Board														
Chunghwa Precision Test Tech	6510.TWO	Ryan Sun	3,149	5,000	3,055	Buy	112	99	47.5	23.8	9.2	7.3	21	34
Probe Card														
Technoprobe SpA	TPRO.MI	Harry Blaiklock, CFA	27,100	41	37	Buy	259	18	66.9	56.5	14.6	11.6	25	23
MPI Corp	6223.TWO	N/A	17,518	N/A	5,670	N/A	97	110	87.2	41.5	30.3	19.1	38	54
FormFactor Inc	FORM.O	N/A	11,038	N/A	141	N/A	159	24	46.2	37.1	9.3	8.9	19	23
MICRONICS JAPAN CO.,LTD.	6871.T	N/A	3,906	N/A	15,410	N/A	155	28	24.8	19.4	7.1	5.6	32	36
Japan Electronic Materials Corp	6855.T	N/A	804	N/A	8,650	N/A	46	43	24.9	17.4	-	-	-	-
Sector Average							**138**	**54**	**49.6**	**32.6**	**14.1**	**10.5**	**27**	**34**

Source: LSEG, UBS estimates. Priced as of October 6 (HKT).

WHAT´S PRICED IN?

HTSI is trading at NT$4,190 as of 6 Oct 2026



Source: LSEG, UBS

Despite a 264% YTD share-price increase, HTSI still trades at just 15x 2028E P/E. We believe market expectations for 2027E CPO testing equipment shipments remain conservative, particularly for insertion 2 equipment. Based on our industry analysis, we estimate HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, this should be higher than the market's expectation of no insertion 2 and limited insertion 3 shipment due to engineering bottleneck. We also see further upside in 2028E as CPO adoption expands. Our reverse DCF analysis indicates that the current market price implies only a 17% long-term sales CAGR vs our estimate of 27% and long-term GM level of 33% vs our estimate of 61%. We see scope for re-rating.

We believe the market is not fully reflecting three key growth drivers:

1) Shift-left testing is still underappreciated.

As advanced packaging costs rise and AI accelerators become increasingly complex, more thorough testing is done at the wafer-level. This should drive sustained demand for HTSI's thermal control modules, probe card consumables, and other wafer-level testing solutions.

2) CPO revenue potential remains underestimated.

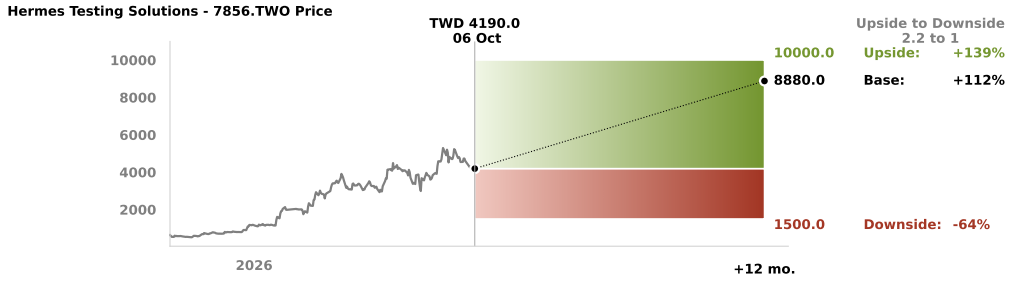
Investor expectations for 2027 CPO equipment shipments, particularly insertion 2 testing systems, remain conservative. Based on our industry studies, we forecast HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, with further potential upside in 2028E as CPO adoption broadens.

3) New products provide additional optionality.

Beyond its core businesses, HTSI is expanding into higher-value solutions including mass-reflow tools, MEMS probe cards and laser cleaning systems, creating incremental growth opportunities that we believe are largely absent from current expectations.

We value HTSI using 31x average 2028E PE, underpinned by a 32% 2027-31E earnings CAGR. We use 2028E PE as our view is that 2028 is the year for thermal control modules and CPO adoption grow from the initial ramping up stage to mass adoption stage. We believe using handlers as a peer group is justifiable for a unique company like HTSI, with thermal control modules for chip probing sharing similar characteristics as active thermal control used for final test, in addition to Hon Precision and Chroma having exposure to CPO test equipment. For equipment vendors, the peer group's average NTM PE post 2022 is around 35x, and the group is currently trading at around 41x NTM PE (see What's Priced In chart).

UPSIDE/DOWNSIDE SPECTRUM



HTSI is trading at NT $4,190 as of 6 Oct 2026

Value drivers (2027/28E)	Thermal control module shipments	CPO insertion 2 & 3 shipments	GM
NT$10,000 upside	3,300/5,000	60&100 / 100&175	61.2%/63.6%
NT$8,880 base	3,000/4,100	40&70 / 80&160	58.3%/60.0%
NT$1,500 downside	1,300/2,000	5&20 / 10&50	53.3%/54.0%

Source: UBS estimates

Risk to the current share price is heavily skewed (2.2:1) to the upside.

UPSIDE (NT$10,000): Our upside scenario reflects stronger than expected thermal module and CPO equipment shipment in 2027-28E, with additional upside from better traction of probe cards and mass reflow tools, forming 157%/91% YoY sales growth in 2027/28E and even higher operating leverage with OPM of 46.3%/50.0% in 2027/28E. This scenario implies a 34x 2028E PE, a upside driving by more equipment demand stemming for robust shift-left test induced equipment bottleneck, alongside CPO proliferation with scale-up CPO equipment pull-in in 2028E.

BASE (NT$8,880): Our price target of NT$8,880 is based on 31x 2028E PE. We expect an earnings CAGR of 32% in 2027-31E. With shift-left trend for more thermal module requirement during testing, alongside 40/70 CPO insertion 2/3 shipment. We see these growth drivers driving YoY sales growth from a high base of 127% in 2026E to 152%/78% YoY in 2027/2028E, with OPM of 44.3%/47.0% in 2027/28E.

DOWNSIDE (NT$1,500): Our downside scenario reflects lower shipment of thermal control modules due to less advanced prober demand with less insertion 2 and 3 equipment shipment due to bottlenecks with CPO production, alongside limited additional growth drivers to be picked up in 2028 such as MEMS probe card and mass reflow tools. This leads to weaker YoY sales growth of 67%/50% in 2027/28E and operating margin of 29.3%/32.0% in 2027/28E as equipment shipment fails to scale. This scenario implies a 14x 2028E PE, as valuation de-rate with dimmed outlook without CPO and limited upside from thermal control module shipments.

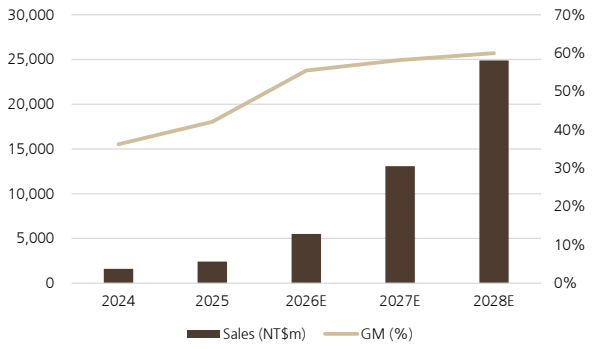
COMPANY DESCRIPTION

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Industry Outlook

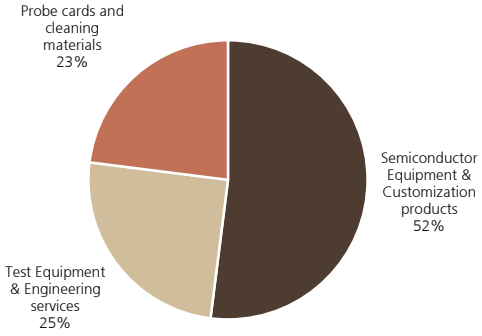
New iterations of semiconductors, especially for AI / HPC applications, are set to increase in both front-end wafer fabrication advancements and back-end packaging complexities with each iterations. With the industry now moves to utilize "shift left", chip probing and its relevant supply chain will be key beneficiaries of this trend which we believe will unlock multi-year, secular growth trends.

HTSI sales and GM trend



Source: Company data, UBS estimates

HTSI H126 sales mix breakdown



Source: Company data, UBS

Hermes Testing Solutions (7856.TWO)

Income Statement (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**1,598**	**2,425**	**5,505**	*127.0*	**13,879**	*152.1*	**24,713**	**29,218**	**32,810**
Gross profit	579	1,021	3,054	*199.2*	8,097	*165.1*	14,817	17,801	20,107
EBITDA (UBS)	**280**	**556**	**1,329**	*139.0*	**6,512**	*389.8*	**12,138**	**15,053**	**17,182**
Depreciation & amortisation	(226)	(260)	(254)	*2.3*	(358)	*-41.0*	(534)	(759)	(1,012)
EBIT (UBS)	**54**	**297**	**1,076**	*262.7*	**6,154**	*NM*	**11,604**	**14,294**	**16,170**
Associates & investment income	0	0	0	-	0	-	0	0	0
Other non-operating income	13	35	1	*-96.8*	0	-	0	0	0
Net interest	(10)	(9)	8	-	169	*NM*	214	286	364
Exceptionals (incl goodwill)	0	0	0	-	0	-	0	0	0
Pre-tax profit	**57**	**322**	**1,085**	*236.7*	**6,323**	*NM*	**11,818**	**14,580**	**16,534**
Tax	0	(54)	(223)	*-314.2*	(1,265)	*NM*	(2,364)	(2,916)	(3,307)
Profit after tax	**57**	**268**	**862**	*221.2*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Preference dividends	0	0	0	-	0	-	0	0	0
Minorities	0	0	0	*-83.3*	0	-	0	0	0
Extraordinary items	0	0	0	-	0	-	0	0	0
Net earnings (local GAAP)	**57**	**268**	**862**	*221.1*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Net earnings (UBS)	**57**	**268**	**862**	*221.1*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Tax rate (%)	0.0	16.7	20.6	*23.0*	20.0	*-2.7*	20.0	20.0	20.0

Per Share (NT$)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	2.71	10.65	29.11	*173.4*	153.28	*NM*	286.48	353.44	400.81
EPS (local GAAP, diluted)	2.71	10.65	29.11	*173.4*	153.28	*NM*	286.48	353.44	400.81
EPS (UBS, basic)	2.84	10.83	29.11	*168.8*	153.29	*NM*	286.49	353.45	400.82
DPS (net) (NT$)	0.00	1.10	11.64	*NM*	61.32	*NM*	114.60	141.38	160.33
Cash EPS (UBS, diluted) [1]	13.49	20.94	37.67	*79.9*	164.12	*335.6*	302.66	376.44	431.48
Book value per share	12.85	46.45	459.82	*NM*	602.66	*31.1*	827.84	1,066.69	1,326.13
Average shares (diluted)	21	25	30	*17.5*	33	*11.5*	33	33	33

Balance Sheet (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	182	709	14,360	*NM*	17,474	*21.7*	22,786	29,613	36,992
Other current assets	589	806	1,732	*114.7*	3,913	*125.9*	6,302	6,856	7,786
Total current assets	**771**	**1,515**	**16,092**	*NM*	**21,386**	*32.9*	**29,088**	**36,469**	**44,777**
Net tangible fixed assets	509	441	429	*-2.7*	765	*78.5*	1,467	2,169	2,797
Net intangible fixed assets	0	0	0	-	0	-	0	0	0
Investments / other assets	122	111	224	*101.6*	224	*0.0*	224	224	224
Total assets	**1,402**	**2,067**	**16,745**	*NM*	**22,376**	*33.6*	**30,780**	**38,863**	**47,799**
Trade payables & other ST liabilities	384	694	1,485	*113.8*	2,401	*61.8*	3,374	3,575	3,950
Short term debt	525	0	0	-	0	-	0	0	0
Total current liabilities	**909**	**694**	**1,485**	*113.8*	**2,401**	*61.8*	**3,374**	**3,575**	**3,950**
Long term debt	125	90	0	-	0	-	0	0	0
Other long term liabilities	28	19	86	*349.5*	86	*0.0*	86	86	86
Preferred shares	0	0	0	-	0	-	0	0	0
Total liabilities (incl pref shares)	**1,062**	**804**	**1,571**	*95.4*	**2,488**	*58.4*	**3,461**	**3,662**	**4,037**
Common s/h equity	**339**	**1,263**	**15,174**	*NM*	**19,888**	*31.1*	**27,319**	**35,201**	**43,762**
Minority interests	0	0	0	*-1.1*	0	*0.0*	0	0	0
Total liabilities & equity	**1,402**	**2,067**	**16,745**	*NM*	**22,376**	*33.6*	**30,780**	**38,863**	**47,799**

Cash Flow (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	57	268	862	*221.1*	5,059	*NM*	9,454	11,664	13,227
Depreciation & amortisation	226	260	254	*-2.3*	358	*41.0*	534	759	1,012
Net change in working capital	(254)	(65)	(374)	*NM*	(1,264)	*-238.1*	(1,416)	(354)	(554)
Other operating	88	139	256	*84.3*	0	-	0	0	0
Operating cash flow	**117**	**602**	**998**	*65.9*	**4,152**	*316.1*	**8,572**	**12,069**	**13,685**
Tangible capital expenditure	(157)	(122)	(275)	*-126.4*	(694)	*-152.1*	(1,236)	(1,461)	(1,641)
Intangible capital expenditure	0	0	0	-	0	-	0	0	0
Net (acquisitions) & disposals	(29)	17	13	*-24.0*	0	-	0	0	0
Other investing	(19)	(20)	(2)	*89.5*	0	-	0	0	0
Investing cash flow	**(206)**	**(125)**	**(265)**	*-111.9*	**(694)**	*-162.2*	**(1,236)**	**(1,461)**	**(1,641)**
Equity dividends paid	0	0	(27)	-	(345)	*NM*	(2,023)	(3,782)	(4,666)
Share issues / (buybacks)	23	626	13,102	*NM*	0	-	0	0	0
Other financing	(46)	(54)	(33)	*38.9*	0	-	0	0	0
Change in debt & pref shares	141	(525)	(125)	*76.2*	0	-	0	0	0
Financing cash flow	**117**	**46**	**12,916**	*NM*	**(345)**	-	**(2,023)**	**(3,782)**	**(4,666)**
Cash flow inc/(dec) in cash	**29**	**523**	**13,649**	*NM*	**3,113**	*-77.2*	**5,313**	**6,827**	**7,379**
FX / non cash items	-	4	2	*-42.4*	0	*-100.0*	0	0	0
Balance sheet inc/(dec) in cash	**-**	**527**	**13,652**	*NM*	**3,113**	*-77.2*	**5,313**	**6,827**	**7,379**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

Hermes Testing Solutions (7856.TWO)

Valuation (x)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	-	68.6	NM	27.3	14.6	11.9	10.5
P/E (UBS, diluted)	-	68.6	NM	27.3	14.6	11.9	10.5
P/CEPS	-	34.3	NM	25.5	13.8	11.1	9.7
Equity FCF (UBS) yield %	-	2.5	0.6	3.0	6.4	9.3	10.6
Dividend yield (net) %	-	0.2	0.3	1.5	2.7	3.4	3.8
P/BV	-	15.7	9.1	7.0	5.1	3.9	3.2
EV/revenues (core)	-	8.1	NM	7.1	3.8	3.0	2.5
EV/EBITDA (UBS core)	-	35.1	80.1	15.1	7.7	5.8	4.7
EV/EBIT (core)	-	65.8	99.0	15.9	8.1	6.1	5.0
EV/OpFCF (core)	-	65.8	99.0	15.9	8.1	6.1	5.0
EV/op. invested capital	-	27.4	>100	60.7	27.0	17.3	13.1

Enterprise value (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	-	19,581	113,884	113,884	113,884	113,884	113,884
Net debt (cash)	-	(75)	(7,489)	(15,917)	(20,130)	(26,200)	(33,302)
Buy out of minorities	28	19	86	86	86	86	86
Pension provisions/other	0	0	0	0	0	0	0
Total enterprise value	**-**	**19,526**	**106,481**	**98,054**	**93,841**	**87,771**	**80,668**
Non core assets	(29)	0	0	0	0	0	0
Core enterprise value	**-**	**19,526**	**106,481**	**98,054**	**93,841**	**87,771**	**80,668**

Growth (%)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	-	51.8	127.0	152.1	78.1	18.2	12.3
EBITDA (UBS)	-	98.9	139.0	NM	86.4	24.0	14.1
EBIT (UBS)	-	NM	NM	NM	88.6	23.2	13.1
EPS (UBS, diluted)	-	NM	173.4	NM	86.9	23.4	13.4
Net DPS	-	-	NM	NM	86.9	23.4	13.4

Margins & Profitability (%)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	36.2	42.1	55.5	58.3	60.0	60.9	61.3
EBITDA margin	17.5	22.9	24.1	46.9	49.1	51.5	52.4
EBIT (UBS) margin	3.4	12.2	19.5	44.3	47.0	48.9	49.3
Net earnings (UBS) margin	3.6	11.1	15.7	36.4	38.3	39.9	40.3
ROIC (EBIT)	-	41.7	NM	NM	NM	NM	NM
ROIC post tax	-	34.7	NM	NM	NM	NM	NM
ROE (UBS)	-	33.5	10.5	28.9	40.1	37.3	33.5

Capital structure & Coverage (x)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	1.7	(1.1)	(10.8)	(2.7)	(1.9)	(2.0)	(2.2)
Net debt / total equity %	138.1	(49.0)	(94.6)	(87.9)	(83.4)	(84.1)	(84.5)
Net debt / (net debt + total equity) %	58.0	(95.9)	NM	NM	NM	NM	NM
Net debt/EV %	-	(0.4)	(7.0)	(16.2)	(21.5)	(29.8)	(41.3)
Capex / depreciation %	69.5	46.8	108.5	194.1	NM	192.5	162.1
Capex / revenue %	9.8	5.0	5.0	5.0	5.0	5.0	5.0
EBIT / net interest	5.3	31.3	-	-	-	-	-
Dividend cover (UBS)	-	9.9	2.5	2.5	2.5	2.5	2.5
Div. payout ratio (UBS) %	-	10.1	40.0	40.0	40.0	40.0	40.0

Revenues by division (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	1,598	2,425	5,505	13,879	24,713	29,218	32,810
Total	**1,598**	**2,425**	**5,505**	**13,879**	**24,713**	**29,218**	**32,810**

EBIT (UBS) by division (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	54	297	1,076	6,154	11,604	14,294	16,170
Total	**54**	**297**	**1,076**	**6,154**	**11,604**	**14,294**	**16,170**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	111.9%
Forecast dividend yield	0.3%
Forecast stock return	112.2%
Market return assumption	6.3%
Forecast excess return	105.9%

Company Description

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Valuation Method and Risk Statement

We value Hermes Testing based on PE methodology.
Risks for the semis industry:
- Geopolitical uncertainties: Geopolitical uncertainties including tariffs and more export control could negatively impact semis cycle, and a downturn would lead to slower demand in the overall market and moderating capacity expansion.
- Meaningful deceleration of new technology development: Slower demand and less investment are expected If there is meaningful deceleration of new technology development such as Cloud AI, autonomous driving, etc.
- Rising utility cost: Larger-than-expected utility cost increases would further pressure profitability. Rising utility costs in Taiwan have negatively impacted margins for Taiwan tech companies.
Company risks:
- Faster- or slower-than-expected front-end/CoWoS capacity expansion: This could result in Hermes Testing gaining/losing equipment orders from major foundries and OSATs.
- Competitors achieving technological advancement: Potential risk of competition (from either local or foreign companies) achieving technological advancements could result in Hermes Testing losing market share from major foundries and OSATs.
- Better- or worse-than-expected AI growth: Better- or worse-than-expected AI growth could lead to a ramp-up or a slowdown of equipment orders from major foundries and OSATs to Hermes Testing, affecting long-term revenue growth potential.
- Slowdown or delay of new technological advancement: A potential slowdown or delay of the rollout for technological advancements for technologies such as SoIC, CoPoS, and CPO, would limit future equipment orders.
- Increase of raw material costs: Rising raw material costs may directly impact Hermes Testing's margins for equipment shipments.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Hermes Testing Solutions

Question	Response
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	4
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	4
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	3
5. What's driving the difference?	
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	3
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	No Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

Required Disclosures

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Analyst Certification:
Each research analyst primarily responsible for the content of this research report, in whole or in part, certifies that with respect to each security or issuer that the analyst covered in this report: (1) all of the views expressed accurately reflect his or her personal views about those securities or issuers and were prepared in an independent manner, including with respect to UBS, and (2) no part of his or her compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in the research report.

UBS Global Research: Global Equity Rating Definitions

12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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Company Name	Reuters	12-month rating	Price	Price date
Hermes Testing Solutions	7856.TWO	No Rating	NT$4,190	06 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

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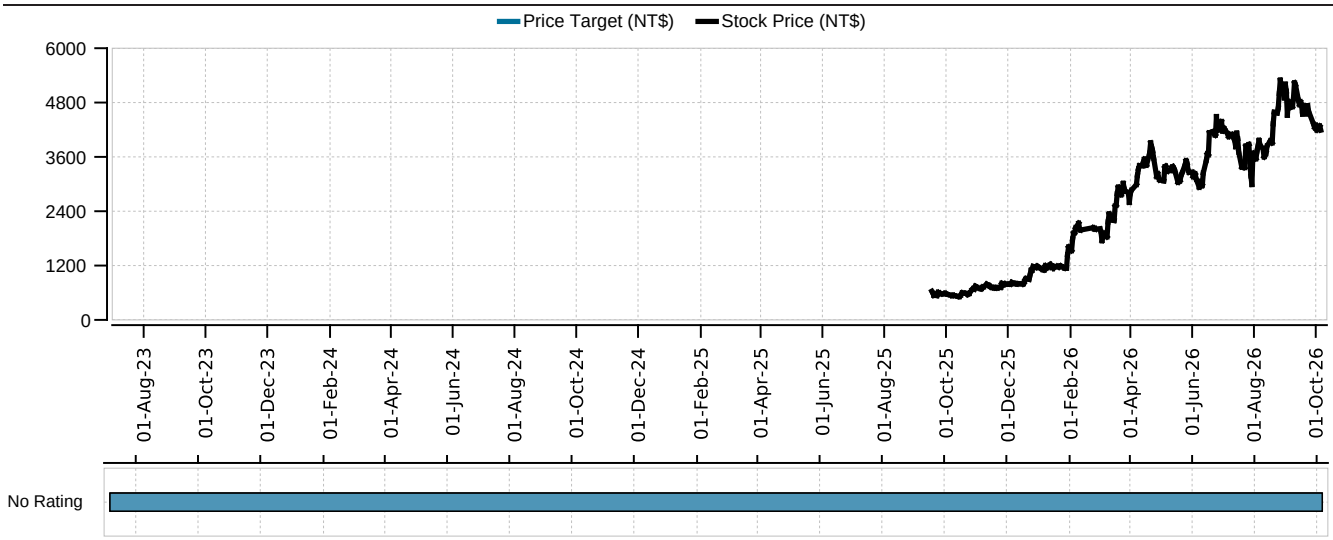
Hermes Testing Solutions (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-06	NaN	-	No Rating

Source: UBS Global Research; LSEG Eikon as of 06-Oct-2026. All prices as of local market close. Ratings as of date shown.

Additional Prices: Hangzhou Changchuan Technology, Rmb239.79 (30 Sep 2026); ASE Industrial, NT$744.00 (06 Oct 2026); Taiwan Semiconductor Manufacturing, NT$2585.00 (06 Oct 2026); Chroma ATE, NT$2350.00 (06 Oct 2026); Teradyne Inc, US$430.32 (06 Oct 2026); Advantest, ¥41850 (06 Oct 2026); Hon.Precision, Inc., NT$5905.00 (06 Oct 2026); Source: UBS. All prices as of local market close.

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Asia Semiconductor Insight

AI put to the test: the new era of chip testing

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Time to revisit semiconductor testing as content growth accelerates

Our core thesis is that AI-driven chip complexity is drastically increasing testing content across wafer, package, and system levels, creating a multi-year growth opportunity that the market has yet to fully price in. Testing companies in Taiwan have underperformed TAIEX in the past three months (indexing at -9% vs TAIEX +7%), yet to recover from the July market deleveraging. Investor hesitation currently stems from 1) fear of test content growth expectations already priced-in, 2) lingering concerns over shorter test times, and 3) near-term capacity constraints among major suppliers of probe cards, load boards, socket pins, testers, and handlers, caused by heightened testing demand. Ultimately, these concerns overshadow the structural increase in test insertions required by AI accelerators, chiplets, and advanced packaging. As advanced packaging scales, testing intensity must rise in tandem, driving growth through higher content per device rather than semiconductor unit growth alone.

Expanding testing opportunities with shift-left for CP and increasing FT content

Chip probing (CP) supply chain should be the key beneficiary of the industry's shift-left testing trend, as more test insertions move to the wafer level before packaging, in addition to longer CP time. We believe investors underestimate this trend, which should accelerate with broader adoption of 2.5D/3D packaging, chiplets and larger AI packages, driving demand for high pin count probe cards, probers, and thermal-control equipment. We expect strong growth for **Hermes Testing (HTSI)** as thermal-management requirements rise with increasing chip power and complexity, with prober thermal-control module shipments reaching 3,000/4,100 units in 2027E/28E, up from 1,200 in 2026E. We forecast **Chunghwa Precision Test (CHPT)**'s sales to exceed consensus by 7%/15% in 2027E/28E, supported by structural AI/HPC testing demand and rising value-added content. Importantly, shift-left testing does not diminish the role of final test (FT) and mission-mode testing. Test content and test time should continue to increase as validation becomes increasingly critical for complex, large-form-factor AI accelerators. We expect larger package sizes and rising system-level testing requirements to drive demand for **WinWay**'s high-pin-count sockets, supporting 74%/71% YoY sales growth in 2027E/28E.

Future of testing: co-packaged optics (CPO) will be a major opportunity

CPO testing represents a significant new market opportunity, though we expect the ecosystem to remain fragmented rather than consolidating around a single winner. As AI scaling shifts the industry bottleneck from memory to interconnect, CPO adoption is becoming increasingly important, creating new testing challenges that will require collaboration across equipment, optics, and test solution providers. Leveraging ficonTEC's optical alignment expertise and **HTSI**'s probe, testing, and manufacturing capabilities, we estimate the HTSI will deliver 40 units of insertion 2 equipment and 70 units of insertion 3 testing equipment in 2027, with further upside in 2028 as CPO deployment expands from scale-out towards scale-up.

Initiate Hermes Testing, Chunghwa Precision Test, WinWay with Buy ratings

We initiate coverage on semiconductor testing solution provider **Hermes Testing with a Buy rating, NT$8,880 price target** based on 31x 2028E PE, for thermal control/CPO with the most attractive risk/reward in our testing coverage; test interface PCB supplier **CHPT with a Buy rating, NT$5,000 price target** based on 31x avg 2027-28E PE, for CPU/TPU programs and test board content gain; and semiconductor test socket leader **WinWay with a Buy rating, NT$8,900 price target** based on 30x avg 2027-28E PE, for socket content growth and margin recovery. Among our testing coverage, we also like Hon Precision and Chroma ATE.

This report has been prepared by UBS Securities Pte. Ltd., Taipei Branch. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 22.**

UBS Research THESIS MAP	MOST FAVORED	LEAST FAVORED
	HTSI, CHPT, WinWay, Hon Precision, Chroma ATE	N/A

PIVOTAL QUESTIONS

Q: Is chip probing becoming increasingly important in the testing supply chain?

Yes. As package costs rise, foundries must improve known-good-die (KGD) screening before packaging, increasing demand for wafer-level testing. This should drive demand for probe cards, probers and advanced thermal control solutions capable of managing higher power consumption, wafer warpage and tighter test tolerances. We forecast HTSI/CHPT/WinWay' probe card business to see a 31%/35%/33% sales CAGR for 2026-2031E, as they are positioned to grow strongly given their smaller scale and robust growth drivers from shift left, CPU, AI accelerators. We forecast HTSI to ship 1,200/3,000/4,100 thermal control modules in 2026E/27E/28E, as advanced wafer-level test for AI accelerators will require thermal control solutions in order to prevent probe card needle burnt.

Q: Will FT and SLT content continue to grow in the coming years?

Yes. Despite concerns that final-test (FT) time could flatten or decline with Nvidia's Rubin GPUs, we expect overall FT content to continue growing as advanced packaging complexity and chip sizes increase. We also expect broader adoption of system-level testing (SLT)/mission-mode testing, as AI accelerators require increasingly stringent validation to ensure optimal system and rack-level performance. We forecast WinWay's socket business to see a 33% sales CAGR for 2026-2031E.

Q: Will CPO testing be a revenue generator in 2027-28 to supply chain players?

Yes. Testing remains a key bottleneck, creating a new equipment opportunity beginning in 2027E. We expect initial CPO volumes in 2027, followed by broader adoption in 2028. Testing remains one of the key bottlenecks to commercialization, driving demand for new equipment and test solutions. We believe early leaders such as ficonTEC (partnership with HTSI) are well positioned to benefit from the initial ramp, alongside suppliers including Chroma ATE and Hon Precision.

WHAT´S PRICED IN?

1) We believe upside from agentic AI CPU demand is not reflected in CHPT and WinWay shares. We estimate CPU-related demand could contribute 20-30% of 2027E sales for CHPT and WinWay, respectively, with further upside in 2028E. **2)** We believe market expectations for 2027 CPO testing equipment shipments remain conservative, particularly for insertion 2 equipment. Based on our industry analysis, we estimate HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, this should be higher than the market's expectation of no insertion 2 and limited insertion 3 shipment due to engineering bottleneck. We also see further upside in 2028E as CPO adoption expands.

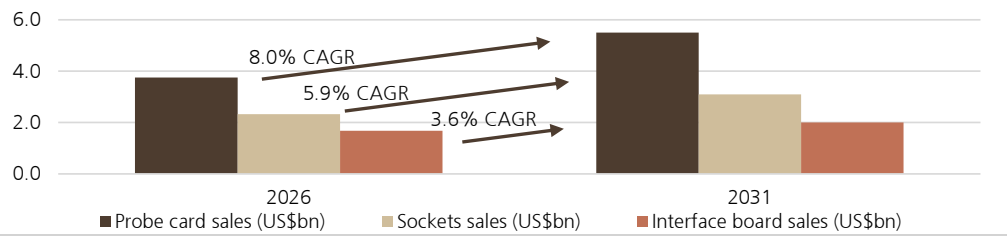
UBSVIEW

Testing is becoming one of the largest incremental beneficiaries of AI-driven semiconductor complexity. As more test insertions are required across wafer, package and system levels, testing content per device should increase materially, expanding the industry's addressable market and supporting multi-year growth across the supply chain.

EVIDENCE

TSMC and OSATs have been expanding their CoWoS capacities aggressively, from 40kwpm in 2024 to around 90 kwpm in 2025, and tracking to expand CoWoS capacity toward 160kwpm in 2026E. TSMC also ramped up expanding N3/N2 capacity, with multiple product launches from customers' pipelines—Nvidia's Rubin, AMD's MI400 series/Venice CPU and other cloud ASICs—translating into foreseeable, strong testing demand.

Figure 1: Testing consumables TAM to continue to climb in the next 5 years



Source: Yole, UBS

Executive Summary

Figure 2: Summary of our investment thesis for HTSI, CHPT, and WinWay

Company	Value Chain Positioning	Primary Product Focus	Investment Thesis	Price Target	Upside
Hermes Testing Solutions (HTSI)	Wafer-level Test	Semiconductor testing related installation and services, customized equipment (e.g. thermal control modules for probers), engineering services (CPO test equipment), probe cards.	A key semiconductor testing solution provider and a shift-left beneficiary, with thermal management capabilities that positions the company in the center of AI/HPC chip probing. Uniquely positioned with CPO exposure through its engineering expertise and partnership with ficonTEC.	**NT$8,880**	
Chunghwa Precision Test (CHPT)	Wafer-level Test, Final Test, System-level Test, Burn-in Test	Semiconductor test PCB, final test load board (interface board), probe cards, burn-in board.	A key semiconductor test PCB maker with unique vertical integration capabilities for fast product cadence to enable design-wins. Probe card business is benefiting from the shift-left trend, driving robust growth with design-wins from projects such as CPU and AI ASICs.	**NT$5,000**	
WinWay Technology	Wafer-level Test, Final Test, System-level Test, Burn-in Test	Final test sockets, system-level test sockets, probe cards.	Leading semiconductor test socket provider, beneficiary of final test and mission-level test content expansion with advanced packaging and chiplet proliferation. With its partnership with Technoprobe, WinWay's probe card strength is adding another leg of growth.	**NT$8,900**	

Source: UBS estimates

Shift-left igniting secular growth for chip probing

With the industry increasingly adopting shift-left testing, chip probing (CP) and its supply chain should be key beneficiaries of a multi-year structural growth trend. This underpins our above-consensus forecasts for **Chunghwa Precision Test (CHPT)**, with 2027E/28E sales 7%/15% above consensus driven by stronger AI/HPC demand, capacity expansion, and rising value-added content. **WinWay** should also benefit through its partnership with Technoprobe, with probe card revenue growth remaining robust at 94%/72% in 2026E/27E following 161%/260% growth in 2024/25.

Hermes Testing (HTSI) should be another major beneficiary as increasing chip power and complexity drive demand for advanced thermal-management modules used alongside high-end probing systems. With TEL's leading AI-oriented probers typically paired with advanced thermal-control solutions, we expect HTSI's thermal-module shipments to reach 3,000 units in 2027E from 1,200 in 2026E. Combined with testing, installation and service revenue, this should drive HTSI sales growth of 127%/152% in 2026E/27E.

Accelerators still in focus with TPU as another significant growth driver

Nvidia remains a key growth driver for both CHPT and WinWay. We expect both companies to gain share with Rubin, with WinWay's socket share improving from roughly 50% historically to 60% with Rubin, while CHPT is positioned to supply around 70% of probe card and final test PCB content. Nvidia currently accounts for roughly 50% of WinWay's revenue and 30% of CHPT's.

Beyond Nvidia, we view Google's TPU roadmap as a meaningful incremental opportunity. Following the successful ramp of TPU v8t with MediaTek as design partner, **CHPT** is positioned to supply probe card PCBs, final test load boards, system-level test boards and burn-in boards. **WinWay** will supply sockets for both TPU v8t and v9, with the larger v9 package providing another growth leg. We estimate production of 4mn TPU v8t units in 2027E and 2.3mn TPU v9 packages (9.2mn dies) alongside 2m v8t units in 2028E.

CPU business as a supplemental catalyst for test interface in an agentic world

With agentic AI proliferation, CPU demand is increasing significantly and this directly correlates to increasing demand for testing consumables. WinWay and CHPT should both benefit from the strong ramp up of Nvidia's ARM based Vera CPU and Google's ARM based Axion CPU in 2026E and 2027E. WinWay will be supplying sockets and probe cards for Vera and Axion, and CHPT will be the probe card and PCB supplier for both Vera and Axion. WinWay will have the additional upside from AMD's x86 Venice CPU, and CHPT could capture some share through AMD's equipment vendor procurement. We estimate **CHPT**'s CPU related sales to be low to mid-teens % of total sales in 2026E and growing to 20-30% in 2027E with CHPT's share expansions with turnkey probe card solutions for Google's Axion. For **WinWay**, we believe CPU should account for around 30% of sales in 2026E/27E.

CPO testing fuelling future growth for semiconductor testing

ficonTEC's 25 years of precision optical-test expertise makes it an early leader in CPO test equipment, particularly in optical alignment, a critical capability for high-volume manufacturing. Through its partnership with ficonTEC, **HTSI** is well positioned to benefit from the initial CPO ramp in 2027E-28E by supporting manufacturing scale-up.

We expect both insertion 2 and insertion 3 testing to be required over the next two years. Despite the technical challenges of double-sided wafer probing, insertion 2 testing remains important given current optical-engine yields and the need to identify known-good optical engines before fibre-array-unit (FAU) coupling. This becomes increasingly critical as switch designs integrate larger numbers of optical engines.

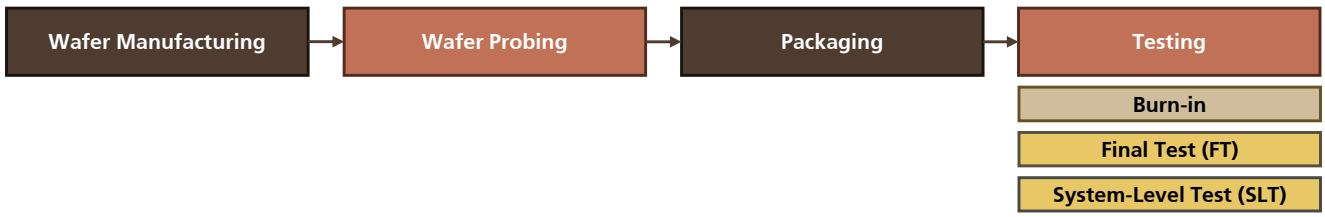
We estimate HTSI will ship 40 insertion 2 and 70 insertion 3 systems in 2027E, with further upside in 2028E as CPO adoption expands and next-generation platforms move into volume production.

Testing TAM expanding with IC complexity: leaders likely to outperform industry growth

Our central view is that testing content per AI accelerator is growing faster than semiconductor unit volumes, creating a multi-year expansion in testing TAM. We believe semiconductor testing is entering a period of structural growth as AI-driven chip complexity increases testing intensity across wafer, package and system levels. The industry's shift-left strategy, which moves more testing upstream in the semiconductor manufacturing process, is resulting in longer and more frequent test insertions. As cloud AI GPUs and ASICs become increasingly complex, additional testing is required to ensure functionality, performance and yield, while reducing defective parts per million (DPPM). More robust testing should improve deployment efficiency and lower the cost of operating AI infrastructure.

Semiconductor testing generally comprises four stages: wafer/chip probing (CP), final test (FT), system-level test (SLT), and burn-in. CP is conducted at the wafer-level before packaging, while FT, SLT, and burn-in occur after packaging. The below Figure 3 illustrates the semiconductor manufacturing flow.

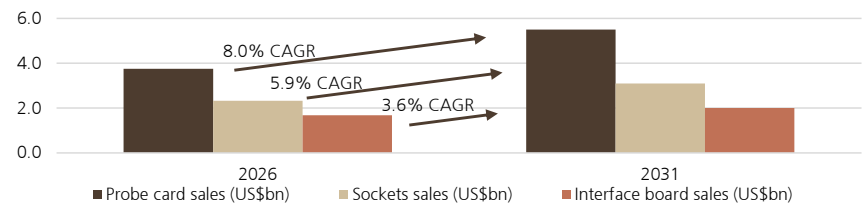
Figure 3: Semiconductor testing process: from wafer level to package level



Source: UBS

We estimate semiconductor testing historically accounts for ~1% of total semiconductor revenue. Within testing, consumables such as probe cards, sockets and interface boards represent ~10-15% of industry revenue. According to Yole, the probe card, socket and interface board markets should grow from US$3.8bn, US$2.3bn and US$1.7bn in 2026 to US$5.5bn, US$3.1bn and US$2.0bn by 2031, respectively.

Figure 4: Yole estimating testing consumables TAM to climb in the next 5 years



Source: Yole, UBS

Nonetheless, leaders within the testing sector are likely to stand out among competition with greater sales CAGR comparing to the industry. Below we highlight **Hermes Testing (HTSI)**, **Chunghwa Precision Test (CHPT)**, and **WinWay**'s impressive growth drivers and how this compares with broader industry sales CAGR.

We see **HTSI**'s sales for the next few years to mainly be driven by 1) thermal control module shipment and 2) CPO equipment shipment. In the past, thermal control modules weren't necessary for chip probing, due to less TDP for the chips. However, as new iterations of AI accelerators now have substantial power draw, this unlocks a brand new TAM for HTSI to enter with their thermal control modules to pair with probers, in order to prevent probe card damage.

- **Robust thermal control module shipments.** We estimate HTSI to ship 3,000/4,100 thermal control modules in 2027E/28E, with continuous demand in future years, bringing their 2026-31E semiconductor equipment sales CAGR to 48%.
- **Thermal control module ASP expansion.** Gen 2 thermal control modules should start to ship in 2027E which will drive tool ASP up around 25-50%.
- **Demand supported by Advantest expansions.** We believe the 3,000 thermal control modules shipment forecast for 2027E is not far-fetched, as we read-across from Advantest's announced capacity expansion plan (testers will be paired with probers with a 1:1 ratio) with increasing capacity up to 5,000 testers within FY2027/3 and they also announced a 10,000-unit target within CY2028. Although the testers capacity expansion plan includes testers for final test, we do see a good portion to be allocated to chip probing given the shift-left testing trend increasing both testing insertions and testing time before chip packaging.
- **CPO is another brand new TAM for semiconductor testing.** As insertion 2 and 3 testing equipment demand starts to rise in 2027E with scale-out CPO adoption, we expect HTSI to ship 40/70 insertion 2 and 3 equipment in 2027E, with this number growing in 2028E to 80/160 and higher in future years as CPO proliferates, and driving 52% 2026-31E test equipment & engineering services sales CAGR.
- **CPO to see content increase.** Gen 2 insertion 3 equipment may start to ship in 2027E, with higher throughput and tool ASP.
- **Insertion 2 might scale down, but not likely any time soon.** Insertion 2 equipment shipment might come down in 2029E and beyond, as foundries scale-up yield for photonic IC (PIC) in order to lessen the time consuming insertion 2 testing burden.

Figure 5: Hermes Testing is set to benefit with a 46% 2026-31E sales CAGR

Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Semiconductor Equipment & Customization products	**$2,815**	**$7,980**	**$13,324**	**$16,061**	**$18,315**	**$20,104**	**48%**
YoY		*183%*	*67%*	*21%*	*14%*	*10%*	
Test Equipment & Engineering services	**$1,480**	**$3,887**	**$7,901**	**$8,887**	**$9,898**	**$11,853**	**52%**
YoY		*163%*	*103%*	*12%*	*11%*	*20%*	
Probe cards and cleaning materials	**$1,210**	**$2,012**	**$3,488**	**$4,270**	**$4,597**	**$4,681**	**31%**
YoY		*66%*	*73%*	*22%*	*8%*	*2%*	
Key products shipment units, ASP, sales	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Thermal control module shipment	**1,200**	**3,000**	**4,100**	**4,500**	**4,700**	**4,900**	
Thermal control product blended ASP (NT$m)	$1.9	$2.4	$2.8	$3.0	$3.3	$3.5	
Total sales from thermal control module shipment	**$2,316**	**$7,168**	**$11,417**	**$13,505**	**$15,591**	**$17,067**	**49%**
% of total sales	*42%*	*52%*	*46%*	*46%*	*48%*	*47%*	
CPO Insertion 2 shipment	**8**	**40**	**80**	**60**	**20**	**20**	
Ins2 blended ASP (NT$m)	$13.8	$15.8	$17.4	$18.3	$19.2	$20.1	
Total sales from insertion 2	**$111**	**$633**	**$1,391**	**$1,095**	**$383**	**$402**	**29%**
% of total sales	*2%*	*5%*	*6%*	*4%*	*1%*	*1%*	
CPO insertion 3 shipment	**11**	**70**	**160**	**200**	**240**	**280**	
Ins3 blended ASP (NT$m)	$16.0	$17.2	$18.2	$19.1	$20.1	$21.1	
Total sales from insertion 3	**$176**	**$1,202**	**$2,917**	**$3,829**	**$4,824**	**$5,910**	**102%**
% of total sales	*3%*	*9%*	*12%*	*13%*	*15%*	*16%*	
Key product sales - total	**$2,602**	**$9,002**	**$15,725**	**$18,429**	**$20,798**	**$23,379**	**55%**
% of total sales	*47%*	*65%*	*64%*	*63%*	*63%*	*64%*	
Others - total	**$2,903**	**$4,877**	**$8,988**	**$10,789**	**$12,012**	**$13,260**	**36%**
% of total sales	*53%*	*35%*	*36%*	*37%*	*37%*	*36%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	**$5,505**	**$13,879**	**$24,713**	**$29,218**	**$32,810**	**$36,639**	**46%**
YoY		*152.1%*	*78.1%*	*18.2%*	*12.3%*	*11.7%*	
GM %	**55.5%**	**58.3%**	**60.0%**	**60.9%**	**61.3%**	**61.1%**	
OpM %	**19.5%**	**44.3%**	**47.0%**	**48.9%**	**49.3%**	**51.1%**	

Source: UBS estimates

Uniquely positioned as a company with vertical integration capabilities with MEMS probes, multi-layer PCB capabilities, and automation equipment all-in-house, **CHPT** is set to capture the growing test shift-left trend in the coming years, increasing both testing insertions and testing time before chip packaging.

- **CP to scale from a lower base.** As a smaller probe card vendor, CHPT can benefit from the growing demand for AI probe cards, with 2026-31E sales CAGR for turnkey probe card robust at 41%.
- **Also a FT/SLT/burn-in beneficiary.** The company will also see final test, system-level test, burn-in content increase to play a role in their sales growth from their IC Test Board sales CAGR of 39% in 2026-31E.
- **Capacity expansion supporting growth.** As CHPT is currently capacity constrained, the robust projected sales growth is enabled by CHPT's rapid capacity expansion strategy with strong demand for its products, with one production expansion scheduled for August 2026 and a second in March 2027, doubling and tripling year-end 2025 capacity, respectively. Its Fab 3 will come online in Q428, doubling existing capacity when fully ramped.
- **MEMS capability to support share gain.** CHPT has MEMS probe capabilities stemming from its history as a smartphone application processor (AP) probe card supplier. It is in a good position to capture the growing demand for MEMS probe cards as AI/HPC are shifting to MEMS probes.
- **Probe card content expect to rise.** We also expect probe card content per device to continue rising as AI accelerators grow in die size, pin count, power consumption and packaging complexity. This should drive secular growth for both probe card suppliers and upstream component providers, making chip probing one of the most attractive areas within the semiconductor testing value chain.

Figure 6: CHPT is set to benefit with a 35% 2026-31E sales CAGR

Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Probe Card	$1,753	$2,903	$4,535	$7,194	$8,361	$9,711	41%
YoY		*66%*	*56%*	*59%*	*16%*	*16%*	
Board Only	$3,574	$6,420	$8,836	$12,650	$13,899	$14,528	32%
YoY		*80%*	*38%*	*43%*	*10%*	*5%*	
Chip Probing	$5,327	$9,324	$13,371	$19,844	$22,261	$24,239	35%
YoY		*75%*	*43%*	*48%*	*12%*	*9%*	
IC Test Board	$1,315	$2,520	$3,579	$5,005	$5,996	$6,825	39%
YoY		*92%*	*42%*	*40%*	*20%*	*14%*	
Service and Others	$502	$640	$918	$1,137	$1,308	$1,504	25%
YoY		*28%*	*43%*	*24%*	*15%*	*15%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	$7,144	$12,484	$17,868	$25,985	$29,565	$32,567	35%
YoY		*74.7%*	*43.1%*	*45.4%*	*13.8%*	*10.2%*	
GM %	56.4%	56.1%	56.2%	56.4%	56.3%	56.3%	
OpM %	35.3%	41.5%	44.0%	45.2%	47.1%	47.3%	

Source: UBS estimates

As the leader in AI/HPC sockets, **WinWay** is set to deliver 2026-31E socket sales CAGR of 33%, underpinned by robust demand for final test and system-level test; with test content and test times rising which each iteration of new AI accelerators/CPU, more sockets and equipment will be needed in order to increase the throughput.

- **Strong accelerator/CPU roadmap to support growth.** Robust testing demand can also be inferred from AI accelerator/CPU roadmaps of Nvidia, AMD, Google, alongside tester capacity expansions by Advantest.
- **Capacity expansion underpinning future strength.** WinWay's socket pin capacity expansion will support this exceptional sales growth. from 3.5mn pins per month at year-end 2025 to 6mn currently, reaching 9mn by year-end 2026 and 14mn by year-end 2027 through expansions in Renwu, Kaohsiung. By 2027, approximately 60% of pogo-pin requirements should be internally sourced, strengthening both supply security and margins.
- **Probe card business to rise from smaller base.** Its probe card business is also set to benefit from the shift-left trend for high-end AI/HPC probe cards. WinWay leverages its existing customers' demand in addition to their MEMS partnership with Technoprobe, as we forecast WinWay's probe card sales to rise from a low base, with a 33% 2026-31E sales CAGR.

Figure 7: WinWay is set to benefit with a 33% 2026-31E sales CAGR

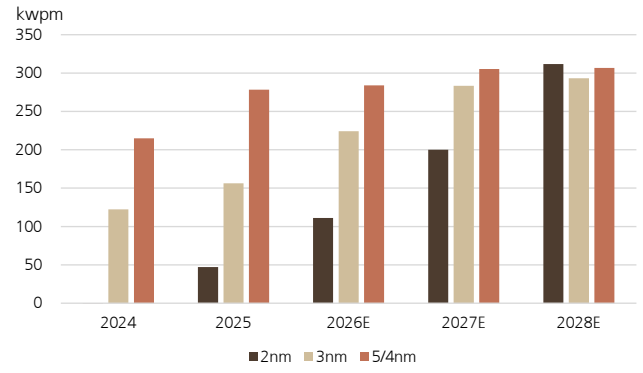
Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Coaxial Socket (HyperSocket)	$7,661	$13,705	$23,103	$25,872	$30,603	$34,726	35%
YoY		*79%*	*69%*	*12%*	*18%*	*13%*	
RF & Plastic Socket	$2,311	$3,835	$6,465	$6,960	$7,259	$7,252	26%
YoY		*66%*	*69%*	*8%*	*4%*	*0%*	
Burn-in Socket	$253	$306	$516	$715	$903	$1,075	34%
YoY		*21%*	*69%*	*39%*	*26%*	*19%*	
Test Socket	$10,224	$17,845	$30,083	$33,547	$38,764	$43,053	33%
YoY		*75%*	*69%*	*12%*	*16%*	*11%*	
Probe Card	$4,478	$7,700	$13,690	$15,507	$16,095	$18,403	33%
YoY		*72%*	*78%*	*13%*	*4%*	*14%*	
Contact Element	$937	$1,596	$2,691	$3,014	$3,465	$3,899	33%
YoY		*70%*	*69%*	*12%*	*15%*	*13%*	
Others	$444	$811	$1,367	$1,671	$2,133	$2,654	43%
YoY		*83%*	*69%*	*22%*	*28%*	*24%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	$16,084	$27,952	$47,831	$53,739	$60,457	$68,010	33%
YoY		*73.8%*	*71.1%*	*12.4%*	*12.5%*	*12.5%*	
GM %	41.1%	43.1%	44.7%	44.7%	44.9%	45.1%	
OpM %	27.5%	32.1%	36.7%	36.7%	36.9%	37.1%	

Source: UBS estimates

The AI era has materially increased the value of semiconductor testing. Compared with traditional mobile and automotive chips, AI accelerators incorporate larger compute dies, more advanced nodes, chiplet architectures, advanced packaging technologies, multiple I/O dies, and increasing HBM content. These factors drive higher-value test consumables, including advanced MEMS probe cards, multilayer interface boards, and next-generation sockets designed to support higher pin counts, thermal loading, and power delivery requirements.

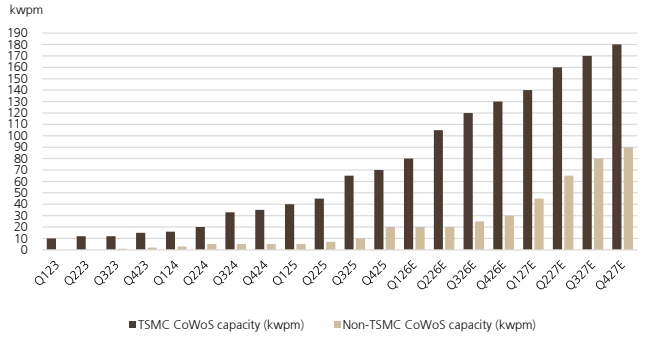
New iterations of semiconductors, especially for AI / HPC applications, are set to increase in both front-end wafer fabrication advancements and back-end packaging complexities with each iterations (i.e. Nvidia's roadmap from Blackwell to Rubin to Feynman or Google's TPU roadmap from v7 to v8 to v9). This trend is evident with TSMC's front-end and TSMC, Intel, and OSAT's back-end advanced packaging capacity expansions. For front-end, we see industry N3 capacity expanding from 156kwpm in year end 2025 to 224/283/293kpwm in year end 2026E/27E/28E, and N2 capacity from 47kwpm in year-end 2025 to 111/200/312kpwm in year end 2026E/27E/28E. For back-end, we are seeing industry CoWoS capacity rising from 90kwpm in 2025 to 160kwpm in 2026E and further expanding to 270kwpm in 2027E.

Figure 8: Industry's leading edge capacity



Source: UBS estimates

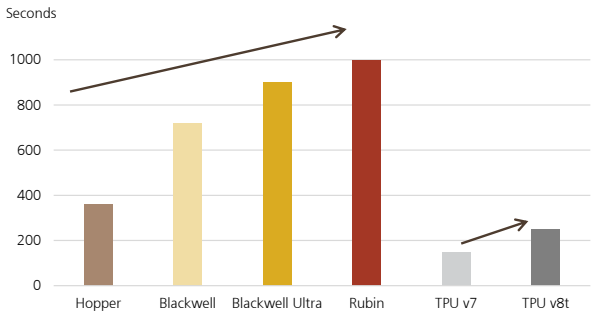
Figure 9: Industry CoWoS capacity



Source: UBS estimates

Final-test, system-level test, and burn-in content should continue to increase as AI accelerators become larger, more power-hungry and more complex. Rising thermal design power and packaging complexity increase validation requirements, driving longer test times and higher demand for test equipment to prevent testing from becoming a production bottleneck.

Figure 10: Our FT test time estimates



Source: UBS estimates

While architectural changes can affect test requirements for individual product generations, we expect the long-term trend of rising testing content to remain intact. For example, Nvidia's Rubin Ultra adopts a simpler package structure than previously expected, from a 4-die structure to a 2-die structure and Rubin Ultra HBM de-spec from HBM4E 12-Hi to initially use HBM4 8-Hi and upgrade to HBM4E 8-Hi at the latter stages, which may moderate near-term test growth. However, future products such as Nvidia's Feynman are expected to reintroduce greater package complexity, with a four-GPU per package structure. More importantly, any reduction in test time can be offset by higher production volumes, making throughput requirements equally important for testing demand.

A rising test trend that lifts all test interface vendors

Figure 11: Key vendors for each stage of the semiconductor testing industry

	Chip Probing	Final Test	System-Level Test
Testing service providers	TSMC, ASE, KYEC, Amkor, Intel	ASE, Amkor, KYEC, JCET, Tianshui Huatian, Tongfu	ASE, Amkor, KYEC, JCET, Tianshui Huatian, Tongfu
Testers	Advantest, Teradyne	Advantest, Teradyne	N / A
Handlers (Probers)	TEL, Tokyo Seimitsu, TSK	Advantest, Chroma, Cohu, Hangzhou Changchuan Technology, Hon Precision, JHT, Teradyne	Hon Precision, Chroma
Probe Cards	FormFactor, Technoprobe, MPI, MJC, JEM, WinWay, CHPT	N / A	N / A
Socket	N / A	WinWay, Cohu, Leeno, Yokowo, Yamaichi, Smiths	WinWay
Load board	N / A	Yamaichi, Advantest, CHPT	Yamaichi, Advantest, CHPT

Source: UBS

Below we summarize key technical items of probe cards, sockets, and load boards/PCB used for semiconductor testing:

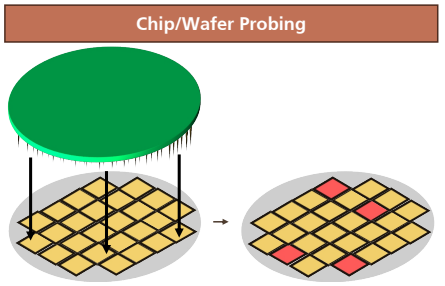
Probe cards and chip probing (CP)

Probe cards are used during chip probing (CP), the wafer-level stage of semiconductor testing that identifies known-good dies (KGD) before packaging. As advanced packaging technologies such as CoWoS, 3D stacking and hybrid bonding become more prevalent, CP is becoming increasingly important because a single defective die or HBM stack can compromise the entire package. This is driving the industry's shift-left testing strategy, with more testing conducted earlier in the manufacturing process to identify defects before costly packaging steps.

Probe cards consist of contact elements (probes) mounted on a PCB and are used together with probers and automated test equipment (ATE). During testing, wafers are placed on the prober chuck while the probe card makes electrical contact with individual dies. The tester then performs electrical, parametric, and functional tests to determine whether each die meets performance requirements. Dies that fail testing are removed from the production flow prior to dicing and packaging.

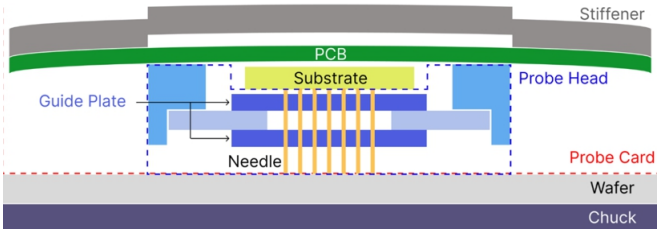
Furthermore, probe cards are highly customised products designed for specific chip architectures and testing platforms. As AI accelerators adopt larger dies, higher pin counts and more advanced packaging technologies, probe card complexity and content value continue to increase.

Figure 12: Chip probing utilise probe cards to find known-good dies (KGD) and eliminate faulty dies



Source: UBS

Figure 13: CP test illustration with probe card structure

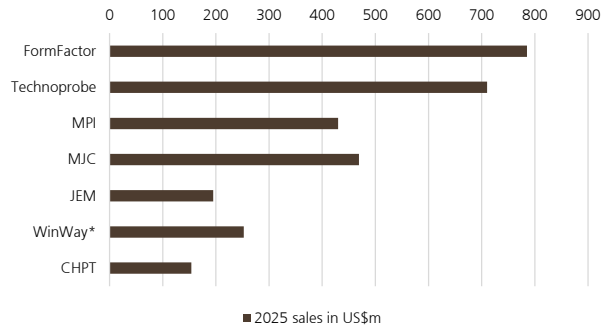


Source: Company data (CHPT)

The global probe card market remains concentrated among a handful of suppliers:

- **FormFactor (FORM.O)**: World's largest probe card company (including both logic and memory probe cards) by revenue as of year end 2025. Key supplier for HBM probe cards.
- **Technoprobe (TPRO.MI)**: World's largest logic probe card company by revenue as of year end 2025. Key supplier for AI accelerators MEMS probe cards, especially for GPU.
- **MJC (6871.T)**: Japanese company, mainly specializing in memory probe cards, 3rd largest probe card company by revenue as of year end 2025.
- **MPI (6223.TWO)**: Traditionally specializes in cantilever probe cards, now expanding its MEMS probe capacity. 4th largest probe card company by revenue as of year end 2025. A probe card supplier of AI ASICs.
- **JEM (6855.T)**: Mainly specialize in memory probe cards, 5th largest probe card company by revenue as of year end 2025.
- **WinWay (6515.TWO)**: One of the fastest-growing logic probe card suppliers. Following its collaboration with Technoprobe, WinWay rose from the 15th-largest probe card supplier in 2024 to the fifth-largest logic probe card supplier in 2025. Probe card revenue increased from 11% of sales in 2024 to 29% in 2025.
- **Chunghwa Precision Test Tech (6510.TW)**: A key supplier for smartphone application processor (AP) probe cards, and has been migrating to HPC related applications in recent years. Although its share within the probe card market is minor, it is a major supplier for probe card PCB, especially for high end probe cards with advanced PCB that has high layer counts (>40 layers, and over 80 layers for AI applications). Its key competitive advantage is its vertical integration as it manufacturers the MEMS probe and PCB in-house.

Figure 14: 2025 sales of major probe card suppliers



Source: Company data, UBS. Note: Only 29% of WinWay's 2025 sales are from probe cards.

Probe cards generally employ either **vertical** or **cantilever** architectures. **Vertical** probe cards use probes arranged perpendicular to the substrate, enabling higher pin density

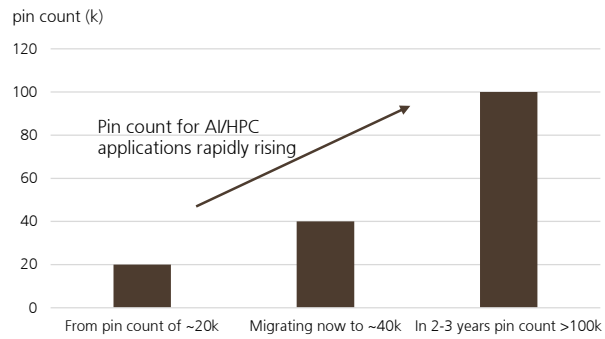
and finer pitch capability. They have become the standard for AI, HPC and other advanced semiconductor applications. **Cantilever** probe cards use angled probes extending from the PCB and are typically used in analog, driver IC and selected memory applications where pin-count and pitch requirements are less demanding. The latest generation of AI accelerators increasingly relies on **MEMS** (Micro-Electro-Mechanical Systems) based vertical probe cards, which support ultra-fine pitches, higher pin counts and improved probing precision. Technoprobe currently leads this segment with pitch capability reaching approximately 40 micrometres.

For leading-edge AI applications, probe cards must increasingly support:

1. Higher current-carrying capability (CCC).
2. Stable contact resistance.
3. Lower parasitic inductance for signal integrity.
4. Greater durability under varying thermal conditions.

For **CHPT**, with their MEMS probe capabilities stemming from their smartphone application processor probe card supply, they are in a good position to capture the growing demand for MEMS probe cards as AI/HPC are shifting to MEMS probes. We also expect probe card content per device to continue rising as AI accelerators grow in die size, pin count, power consumption and packaging complexity. This should drive secular growth for both probe card suppliers and upstream component providers, making chip probing one of the most attractive areas within the semiconductor testing value chain.

Figure 15: Probe card pin count growth trajectory

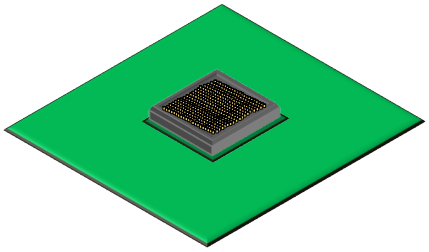


Source: UBS estimates

Sockets

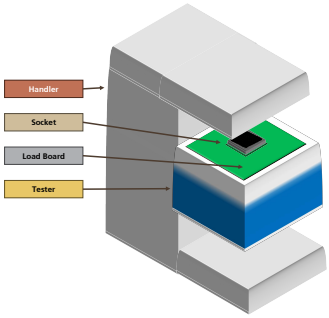
Semiconductor test sockets are electro-mechanical interfaces that are used post chip probing and chip packaging process to test the functionality of the chips. Sockets are placed on to load boards on top of testers during the final test/system level test processes. Handlers place the chips onto the sockets, and pins inside the sockets enable the testers to read the signals from the device under test (DUT, i.e. the packaged chips), with tester signals fanning out from beneath the PCB load board.

Figure 16: Socket illustration



Source: UBS

Figure 17: FT equipment/ consumables



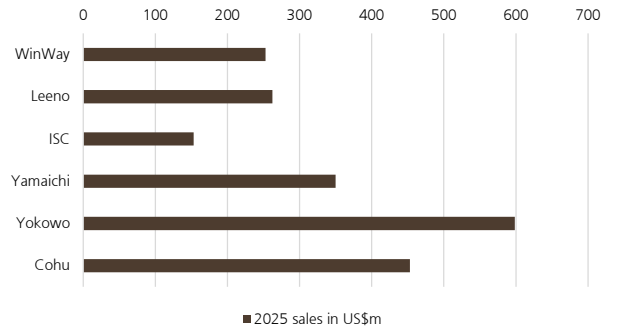
Source: UBS

As AI accelerators become larger and more complex, socket requirements continue to increase. High-end sockets typically comprise a socket housing, alignment plate, integrated thermal-control components and contact elements that connect the DUT to the tester. Spring probes (pogo pins) are the most common choice for HPC sockets, with high durability, individual probe replacement flexibility, and consistent contact force. FT and SLT sockets account for more than 70% of the total socket market, with burn-in sockets representing the remainder.

Compared with the probe card market, the socket industry is more fragmented. Key suppliers include:

- **WinWay (6515.TW)**: A major share gainer in AI/HPC sockets, with strong exposure to Nvidia and AMD. The company specializes in high-performance coaxial sockets and is developing its next-generation HyperSocket platform. It is also expanding into probe cards through its partnership with Technoprobe.
- **Leeno (058470.KQ)**: Vertically integrated supplier focused on smartphone AP and memory sockets.
- **ISC (095340.KQ)**: Leading memory socket supplier with elastomer expertise.
- **Smiths Interconnect (Acquired by Molex)**: A key player in AI/HPC socket industry, with its DaVinci coaxial sockets able to support high frequency connectivity testing.
- **Yamaichi (6941.T)**: Leading Japanese socket supplier with strong automotive exposure.
- **Yokowo (6800.T)**: Major provider of contact probes and socket pins.
- **Cohu (COHU.O)**: Broad test-solutions provider spanning ATE, handlers and sockets, with greater auto/industrial/consumer exposure.

Figure 18: 2025 sales of major socket suppliers



Source: Company data, UBS. Note: Most socket suppliers are companies with multiple business segments, i.e. Yokowo and Cohu.

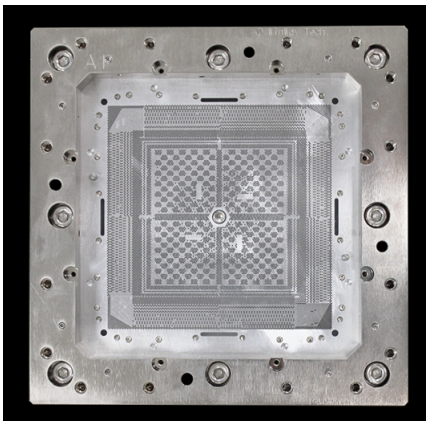
AI accelerators are pushing socket requirements beyond 10,000 pins, making **coaxial sockets** the industry standard for leading-edge AI/HPC devices. These sockets must support transmission speeds of up to 224G while handling significantly higher thermal design power (TDP) during testing.

Elastomer technology is increasingly being adopted alongside spring-probe (pogo-pin) architectures to reduce chip damage and improve contact reliability. WinWay's next-generation **HyperSocket** integrates elastomer materials between the DUT and spring probes, improving probe lifetime, reducing cleaning requirements and enhancing production efficiency. We believe HyperSocket could become an important differentiator as next-generation AI accelerators demand higher pin counts, bandwidth and thermal performance. We estimate HyperSocket could contribute approximately 5% of WinWay's revenue in 2027E while carrying a 20-30% ASP premium versus conventional coaxial sockets.

The transition toward larger AI packages, higher pin counts and faster data-transfer speeds should continue to increase socket content per device. We expect this trend to support long-term growth in the socket market, with technology leaders such as WinWay well positioned to gain share through next-generation solutions including

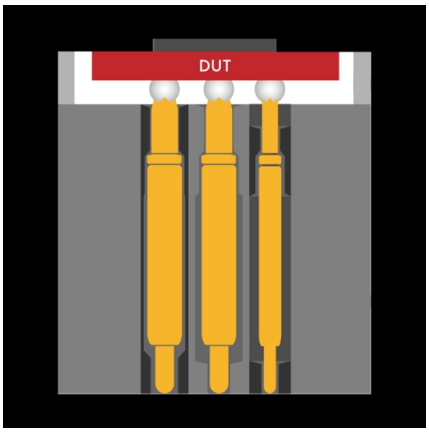
HyperSocket.

Figure 19: Coaxial socket



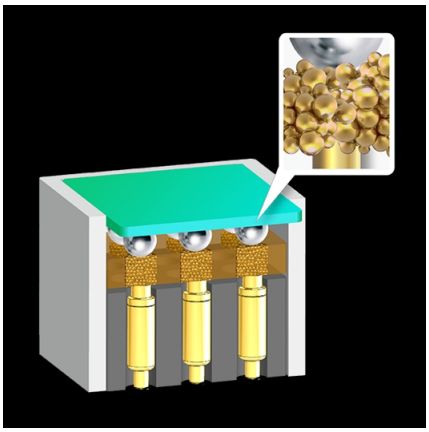
Source: Company data (WinWay)

Figure 20: DUT socket contact



Source: Company data (WinWay)

Figure 21: HyperSocket



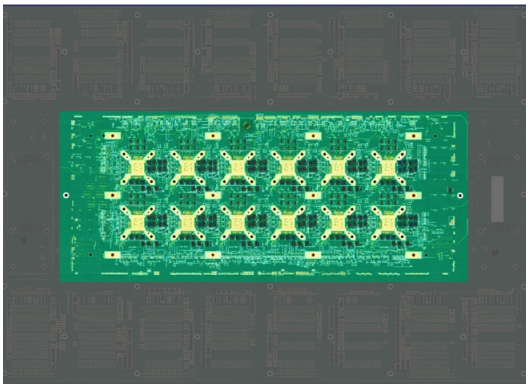
Source: Company data (WinWay)

Load Board/Interface board

Load boards are specialised printed circuit boards (PCBs) that connect the tester to the device under test (DUT) during final test (FT) and system-level test (SLT). Acting as the electrical interface between the packaged chip and automated test equipment (ATE), load boards are responsible for power delivery and high-speed signal transmission throughout the testing process. As AI and HPC devices become increasingly complex, load board requirements continue to rise. **Multilayer construction** is crucial as modern load boards often require more than 50 layers for smartphone application processors and over 80 layers for leading-edge AI accelerators to support higher pin counts, dense signal routing and greater power delivery demands. At the same time, **impedance control** is an important factor as maintaining signal integrity is becoming increasingly important as interconnect bandwidth rises and high-speed SerDes blocks become more prevalent. These trends are driving greater technology content and value per load board.

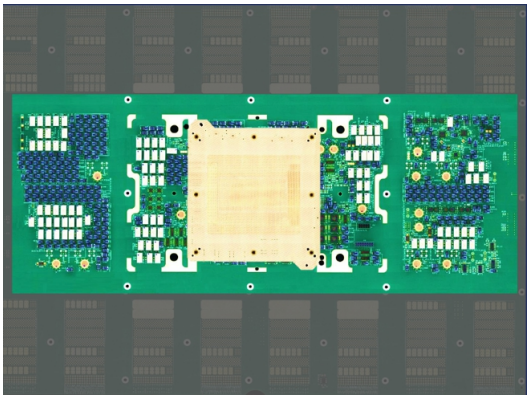
We expect load board content to increase alongside AI chip complexity, larger package sizes and higher interface speeds, benefiting suppliers with advanced multilayer PCB capabilities such as CHPT.

Figure 22: CHPT's AP load board



Source: Company data (CHPT)

Figure 23: CHPT's AI load board



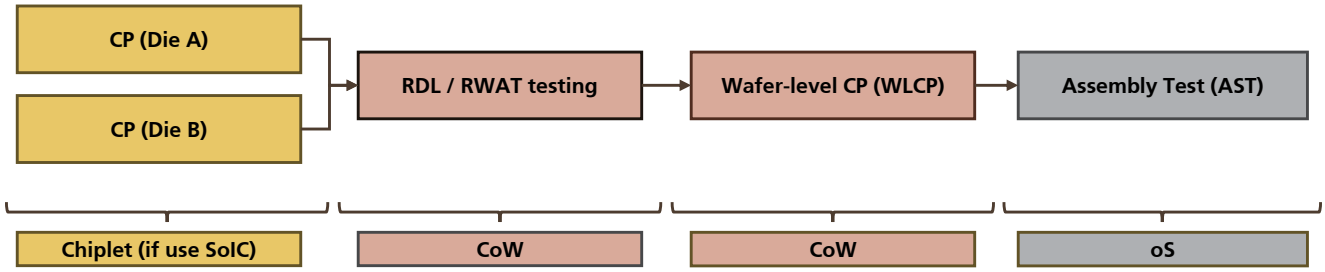
Source: Company data (CHPT)

Shift-left igniting secular growth for chip probing

The industry's adoption of shift-left testing should drive a multi-year structural growth opportunity for chip probing (CP) and its supply chain. As advanced packaging costs rise and AI accelerators become increasingly complex, semiconductor manufacturers are placing greater emphasis on identifying known-good dies (KGD) before packaging. This

underpins our above-consensus forecasts for **Chunghwa Precision Test (CHPT)**, with 2027E/28E sales 7%/15% above consensus driven by structural AI/HPC probe card and test consumables demand, capacity expansion, and increasing test interface content as test insertions and test time increase. **WinWay** should also benefit through its growing probe card business alongside partner Technoprobe, with probe card sales expected to grow 94%/72% YoY in 2026E/27E following 161%/260% growth in 2024/25.

Figure 24: CP is set to grow substantially with shift-left, each packaging step set to be MORE costly than the prior step

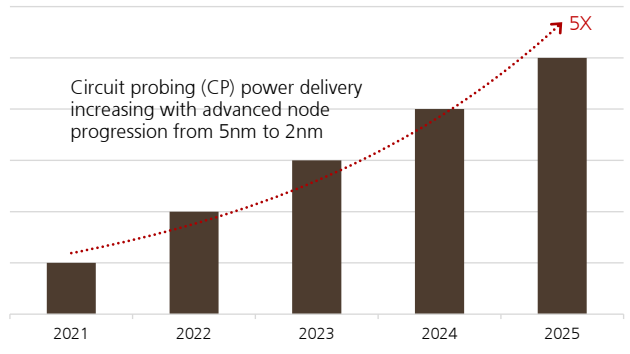


Source: UBS

The economics of advanced packaging reinforce this trend. Heterogeneous integration, chiplet architectures and 2.5D/3D packaging significantly increase package value, while defects identified late in the manufacturing process can lead to substantial scrap costs. To mitigate this risk, foundries and OSATs such as TSMC and ASE employ distributed testing flows incorporating KGD, known-good-stack (KGS) and assembly test screening, to catch defects before the components enter a multi-week packaging cycle (>2 weeks for CoW and 2 weeks for oS). The importance of these tests rises as AI clusters become increasingly dependent on component reliability and low defective-parts-per-million (DPPM) rates. Given the multiplicative yield degradation, a 300bps yield decline at just one manufacturing stage (e.g. dropping from 99.5% to 96.5%) reduces overall integrated four-stage yield from 98.0% to 95.0%; on a 500,000 unit ramp at US $10,000 per unit, this 3% scrap delta translates into around US$150m in unrecoverable losses given >6-month manufacturing lead times.

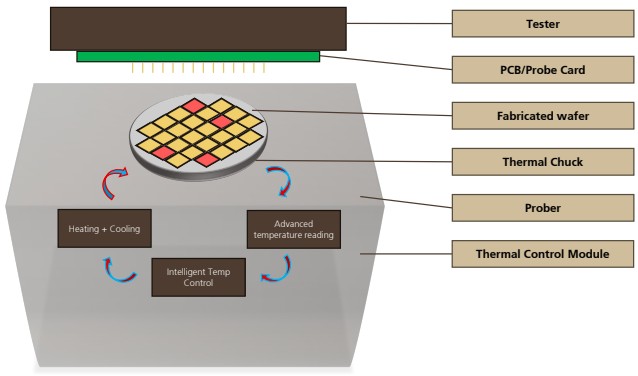
At the same time, rising chip power consumption is increasing the technical difficulty of wafer-level testing. As chip thermal design power scales beyond 1,000W and power density is expected to surge another 8x by 2030, traditional wafer-level chip probe (WLCP) encounters severe physical bottlenecks. Finer probe pitches and higher power densities increase the risk of overheating, probe needle burnout and yield degradation, creating demand for more advanced probing and thermal control solutions.

Figure 25: CP power delivery trend rises with process tech evolution



Source: UBS

Figure 26: Thermal control module inside probers are essential to control wafer temperature



Source: UBS

Hermes Testing (HTSI) is a key beneficiary of this trend. Leveraging its long-standing relationship with TSMC and expertise in semiconductor test solutions, HTSI supplies advanced thermal control modules used alongside leading-edge probers (made by Tokyo Electron), particularly for AI applications. These thermal control modules are

required for high TDP semiconductor wafer probe in order to maintain the temperature of the chuck to avoid needle burns or probe card damages. We estimate HTSI will ship approximately 3,000 thermal-management modules in 2027E, up from 1,200 in 2026E, supporting sales growth of 127%/152% in 2026E/27E. HTSI is also the partner for prober installation, prober maintenance and services. In addition, HTSI provides high-end chuck processing services, an area that should benefit from increasing package sizes and higher wafer warpage associated with advanced packaging.

Beyond thermal control modules, HTSI also supplies consumables and customised solutions for wafer-level testing, including probe card cleaning sheets and laser-based probe head cleaning systems. As probing intensity increases under the shift-left testing model, demand for cleaning and maintenance solutions should rise alongside probe card utilisation. Currently, cleaning sheets is around 5% of sales, but we believe it could gain traction in 2027/28E and beyond. We see additional upside from future modular laser cleaning systems that could be integrated directly with probers, further expanding HTSI's content opportunity within advanced wafer-level testing.

Rapid capacity expansions accelerating growth

Industry test interface suppliers are expanding capacity aggressively to meet rising AI and HPC demand. As advanced semiconductors require higher-performance probe cards, sockets and interface boards, capacity additions are becoming increasingly important to support growth across both wafer-level and package-level testing.

We believe **CHPT** and **WinWay** are well positioned to gain share within the high-end AI/HPC test interface market. Taiwanese suppliers benefit from their proximity to leading fabless customers and the growing preference for localised supply chains, particularly for advanced testing applications where responsiveness and engineering support are critical. Against a backdrop of tight industry supply, both companies have announced significant capacity expansion plans over the next two to three years.

CHPT: capacity expansion supporting AI/HPC share gains

CHPT's existing capacity is effectively fully utilised and the company is pursuing both near-term and long-term expansion. Near-term growth will come from leased facilities, with one production expansion scheduled for August 2026 and a second in March 2027, doubling and tripling year-end 2025 capacity, respectively. Longer term, CHPT's new Fab 3 facility in Pingzhen will significantly expand in-house PCB manufacturing and wet-processing capacity, with mass production expected around Q428, expected to double existing capacity when fully ramped.

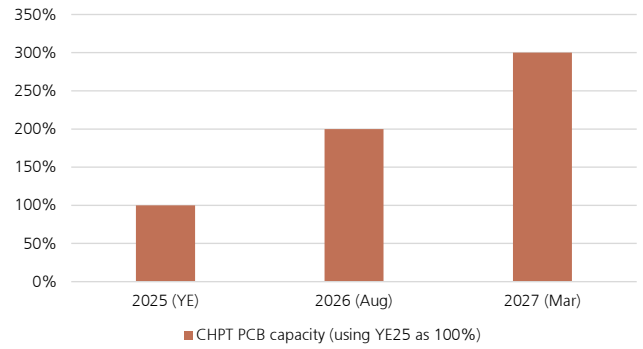
Combined with increasing exposure to higher-value AI/HPC products, including turnkey CPU probe cards and TPU FT/SLT/burn-in boards, we expect CHPT to deliver revenue growth of 49%/75%/43% in 2026E/27E/28E.

WinWay: scaling for the next phase of AI demand

WinWay's capacity is expected to increase from 3.5mn pins per month at year-end 2025 to 6mn currently, reaching 9mn by year-end 2026 and 14mn by year-end 2027 through expansions in Renwu, Kaohsiung. By 2027, approximately 60% of pogo-pin requirements should be internally sourced, strengthening both supply security and margins.

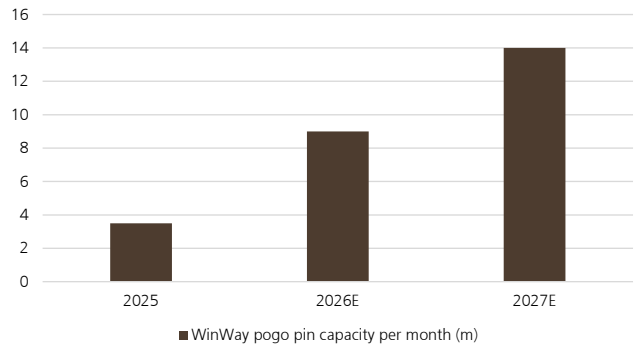
We estimate these capacity additions will support revenue growth of 105%/73%/71% in 2026E/27E/28E.

Figure 27: Announced capacity expansions for CHPT



Source: Company data, UBS

Figure 28: Announced capacity expansions for WinWay



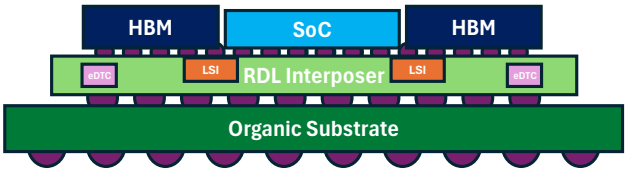
Source: Company data, UBS

Advanced packaging growth remains the key demand driver

Capacity expansion by test interface vendors is ultimately supported by continued investment throughout the advanced packaging ecosystem, from providers such as TSMC, ASE, Intel, and Amkor. During the Q226 earnings calls, TSMC raised its recent 2026 capex from US$52-56bn to US$60-64bn and ASE raised its 2026 capex from US $8.5bn to US$10.5bn, with both companies citing robust demand supporting the capex raise. We believe capex for the next two years will be even higher than 2026, implying continuous growth in future years. With advanced packaging technologies getting significantly more complex and with larger form factor, illustrated by the upcoming ramp of both TSMC's CoPoS (chip-on-panel-on-substrate) and Intel's EMIB-T (embedded multi-die interconnect bridge) packaging technologies, test interface solution providers will benefit with more value added consumables (see our CoPoS vs EMIB-T deep dive).

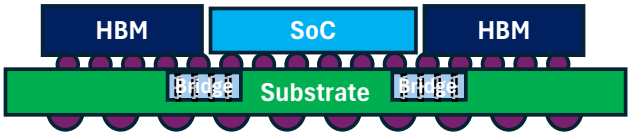
We expect test interface content growth to outpace unit growth. For instance, test socket's ASP directly correlates with its pin counts - with the addition of 20-30% more pins inside the sockets, socket ASP can grow 20-30% or more as the socket might require significant cooling capabilities as the TDP of the DUT rises with its larger form factor. Unlike prior semiconductor cycles, much of the current demand is driven by rising test interface content per device rather than unit growth alone.

Figure 29: TSMC's CoPoS



Source: UBS

Figure 30: Intel's EMIB-T



Source: UBS

Accelerators still in focus with TPU as another significant growth driver

Nvidia has been the primary driver of testing content growth over the past several years, benefiting the entire semiconductor testing supply chain. We expect this trend to continue as GPU architectures evolve from Blackwell to Rubin and then to Feynman, with increasing process node complexity, larger package sizes and more advanced packaging technologies driving higher testing value per device.

Nvidia should remain a major contributor to both CHPT and WinWay, with GPUs continuing to serve as their core growth engine. Both companies are positioned to gain share with Rubin. We estimate WinWay's socket share could improve from a 50/50 split with Smiths Interconnect in prior generations to 60/40 with Rubin, while CHPT is gaining

share in test interface PCBs and is positioned to supply around 70% of Rubin probe card and final test PCB content. Nvidia currently accounts for approximately 50% of WinWay's sales and 30% of CHPT's sales. Both Nvidia GPUs and Google TPUs increase testing demand through higher content per device rather than unit growth alone.

Beyond Nvidia, Google's TPU roadmap represents a significant incremental growth opportunity. Following the successful ramp of TPU v8t with MediaTek as design partner, **CHPT** is positioned to supply probe card PCBs, final test load boards, system-level test boards and burn-in boards. For v9, although the final test interface vendors are yet to be decided, we believe as the current supplier, CHPT will still be supplying PCB for final test, system level test, and burn-in, with potential to continue to supply PCB for probe cards. **WinWay** should also benefit as the socket supplier for both TPU v8t and the next-generation TPU v9 platform. We believe MediaTek's increasing share gain for Google TPUs will also benefit the local Taiwanese test interface supply chain.

We view TPU v9 as a particularly attractive opportunity given its large four-die package architecture based on Intel's EMIB-T packaging technology, which should drive higher testing content. We estimate production of 4mn TPU v8t units in 2027E, followed by 2.3mn TPU v9 packages (9.2mn dies) and an additional 2mn TPU v8t units in 2028E. Importantly, while EMIB-T packaging is expected to be performed in Malaysia, we believe the majority of advanced testing value should remain within the Taiwanese testing ecosystem. Leading testing providers, including ASE and KYEC, maintain substantial testing operations across Taiwan and Southeast Asia, supporting continued participation by Taiwanese test interface suppliers. See our latest MediaTek report.

CPU business as a supplemental catalyst for test interface in an agentic world

We believe semiconductor test interfaces are an underappreciated beneficiary of the emerging agentic AI ecosystem. UBS estimates the CPU market will expand to US $170bn by 2030E from US$31bn in 2025 and US$71bn in 2027E, driven by rising deployment of AI infrastructure. As CPU-to-GPU ratios increasingly approach 1:1 in AI clusters, demand for testing interfaces should expand alongside CPU volumes. TSMC is the main CP test provider in the industry, with probe card suppliers aiding its massive demand by supplying probes, such as TSMC's GPU probe cards supported by Technoprobe's probes. However, with capacity now becoming scarce with strong front and back-end demand, TSMC is outsourcing its CP business to ASE and KYEC, further broadening the testing supply chain. This is where new probe card vendors come in—smaller probe card players in the past, such as **WinWay** and **CHPT**, now have the opportunity to gain significant, non-GPU share from TSMC, as they have MEMS probe capabilities (WinWay's MEMS probes are from its strategic partnership with Technoprobe) and has a long established relationship with the fabless companies such as Nvidia via supplying non-probe card consumables such as sockets and PCB. Of which, CPU projects have become the main area for CHPT and WinWay to expand their footprint.

We expect both companies to benefit from the ramp of Nvidia's ARM-based Vera CPU and Google's ARM-based Axion CPU in 2026E-27E. WinWay is positioned to supply sockets and probe cards for Vera and sockets for Axion, while CHPT should provide probe cards and test interface PCBs for both Vera and Axion, with particularly strong exposure to Axion. Additional upside could come from AMD's Venice CPU, where WinWay is well positioned and CHPT could gain indirect exposure through testing-equipment procurement channels. We estimate CPU-related revenue will represent low to mid-teens % of CHPT sales in 2026E, increasing to 20-30% in 2027E as the company expands its turnkey probe card solutions for Axion. For WinWay, CPU-related demand could account for approximately 30% of sales in 2026E-27E. Importantly, CPU programmes diversify exposure beyond AI GPUs and provide a second structural growth driver for test interface suppliers.

CPO testing fuelling future growth for semiconductor testing

As interconnect bottlenecks increasingly shift from the memory wall to the copper wall, we believe co-packaged optics (CPO) adoption is inevitable over the longer term. Testing remains one of the key bottlenecks to CPO deployment, as optical validation differs

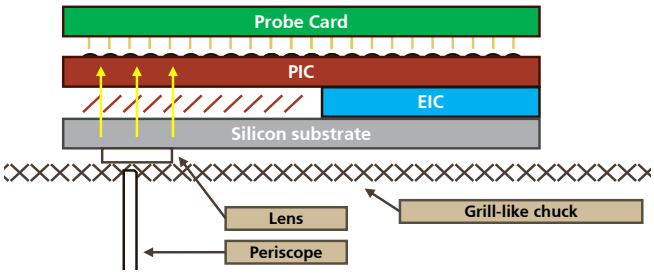
fundamentally from the electrical testing processes that have dominated semiconductor manufacturing for decades. As a result, CPO is creating a new testing opportunity across the semiconductor equipment and consumables ecosystem. We have written extensively on CPO and the opportunities that it brings to the semiconductor supply chain, see our CPO deep dive, our thoughts on CPO timing, 2026 SiPhIA conference takeaways, and SEMICON 2026 takeaways.

Various test equipment suppliers are scrambling to come up with ideal testing equipment across the four insertions of CPO testing, with solutions to ensure both optical and electrical signal integrity of the CPO solutions. The four insertions are:

- **Insertion 1: Electronic IC (EIC)/Photonic IC (PIC) test.** Single-side wafer test that has dedicated probe cards to test the fabricated EIC/PIC wafers.
- **Insertion 2: Optical/Electrical test.** Double sided wafer test, testing the hybrid bonded (e.g. via TSMC's COUPE) EIC and PIC wafers by an opto-electrical test that verifies the electrical signals on the top and optical signal on the bottom of the hybrid bonded wafers.
- **Insertion 3: Optical engine (OE) test.** The OE tests are done to ensure the quality of the OEs and with FAU (Fiber Array Unit) attached, to ensure the signal integrity of the OEs. The testing equipment needs to be able to handle 224G per channel data rates, requiring high unit per hour (UPH) throughput, meticulous dirt mitigation, and maintaining a near 100% contact rating on high-speed sockets.
- **Insertion 4: ASIC final test.** This final test step is established post the OEs are packaged together with the ASIC. The complete ASIC package (now with CPO) is now tested for opto-electrical functionalities.

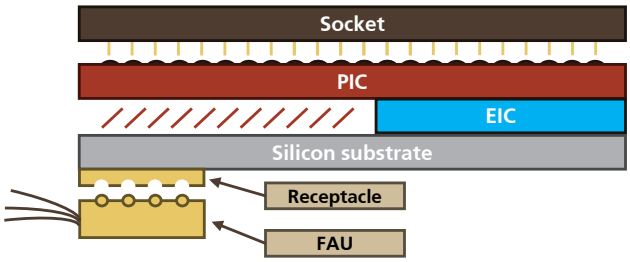
As it stands, insertion 2 and insertion 3 are the two areas where there are the most debates in the industry. Insertion 3 suppliers such as Chroma have noted the trend of "shift-to-middle" paradigm centered on insertion 3. In high-density scale-out switches deploying up to 32 optical engines (e.g. Nvidia's Spectrum-X architectures), any OE defect results in package scrap at the CoWoS level, making 100% known-good die (KGD) validation at the singulated OE stage an absolute commercial requirement.

Figure 31: An illustration for insertion 2 test method



Source: UBS

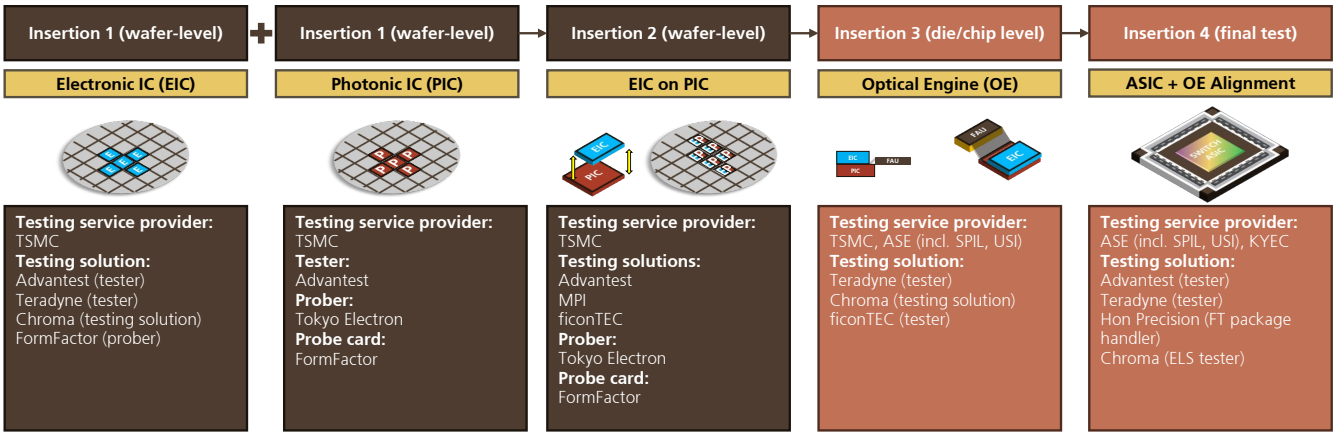
Figure 32: An illustration for insertion 3 test method



Source: UBS

As shown in the below figure, as there are no set standards for the testing equipment, throughput and stability are now the main targets and areas of competition for the major players in the field. Notable ones for each insertions are listed.

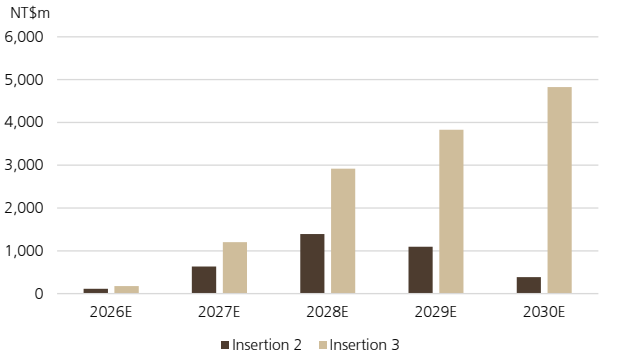
Figure 33: CPO testing flow chart with key suppliers



Source: UBS

With its 25 years of precision test equipment expertise, ficonTEC is an early mover for CPO test equipment and has key know-how for optical alignment, which is quintessential for optical test. **Hermes Testing**, through its partnership with ficonTEC, is set to be a major beneficiary once CPO proliferates in 2027E-28E. As the partner of ficonTEC, Hermes Testing is the key enabler that can help ficonTEC achieve HVM for its CPO testing equipment. As of now, our view is that both insertion 2 and insertion 3 should be required within the next two years, despite the difficulties surrounding double-sided wafer probing (insertion 2), as the yield for OEs is still not high enough to avoid defective dies. Finding out known-good OEs will be crucial before the FAU coupling process during the latter stages of CPO packaging, to prevent impeding the final package with multiple OEs attached to the switch ASIC (e.g. 32 OEs on Nvidia's Spectrum-X). We estimate Hermes Testing will ship 40/70 units of insertion 2/3 testing equipment in 2027E, with upside to these numbers as we view Hermes Testing will likely assist with the ramp up for Gen 2 insertion 3 equipment in 2027E. Gen 2 insertion equipment will come with better throughput and ASP.

Figure 34: HTSI CPO equipment sales to pick up in 2027/28E with 541%/135% YoY growth



Source: UBS estimates

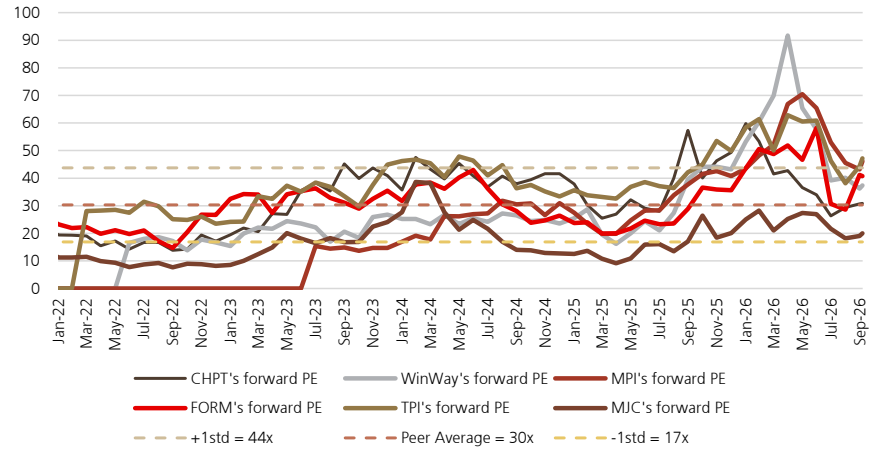
Beyond insertions 2 and 3, HTSI is also positioned to participate in insertion 1 testing through customised prober solutions for photonic integrated circuit (PIC) testing. We believe these solutions could eventually be packaged as licensable intellectual property, creating an additional long-term growth opportunity as CPO adoption broadens. While near-term testing growth is driven by AI accelerators, CPO represents the next major expansion cycle for semiconductor testing. We believe HTSI is uniquely positioned to participate across multiple test insertions, providing exposure to a potentially significant source of incremental revenue beginning in 2027E.

Valuation comparison

We set up three peer groups for valuation comparison: probe card, sockets, and equipment vendors.

For probe card vendors, the peer group's average NTM PE post 2022 is around 30x, and the group is currently trading at around 39x NTM PE. We value CHPT using 31x average 2027-28E PE, in-line with its historical average of 31x and underpinned by a 31% 2027-31E earnings CAGR.

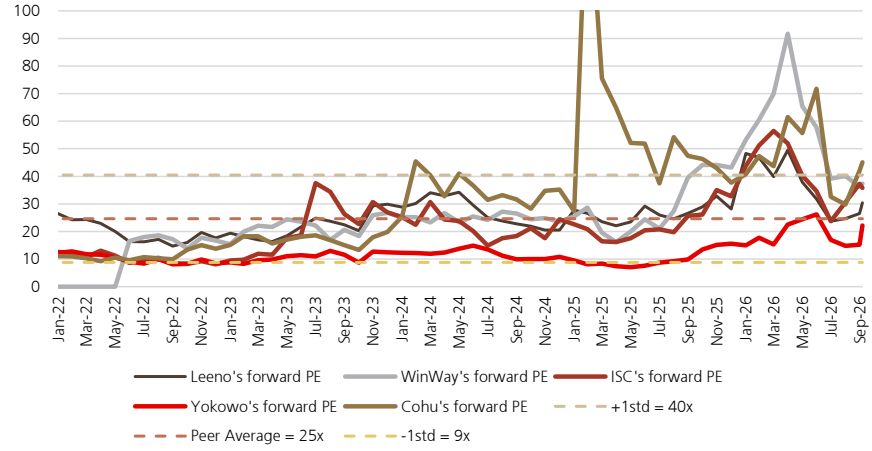
Figure 35: Key probe card vendors' 12-month forward PE band (x)



Source: LSEG, UBS

For socket vendors, the peer group's average NTM PE post 2022 is around 25x, and the group is currently trading at around 39x NTM PE. We value WinWay using 30x average 2027-28E PE, in-line with its historical average of 30x and underpinned by a 30% 2027-31E earnings CAGR.

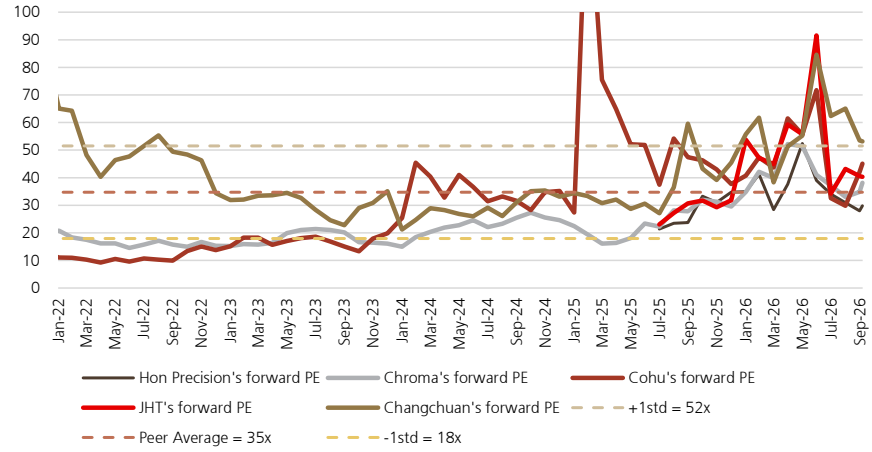
Figure 36: Key socket vendors' 12-month forward PE band (x)



Source: LSEG, UBS

For equipment vendors, the peer group's average NTM PE post 2022 is around 35x, and the group is currently trading at around 41x NTM PE. We value HTSI using 31x average 2028E PE, underpinned by a 32% 2027-31E earnings CAGR. We use 2028 PE as our view is that 2028 is the year thermal control modules and CPO adoption grow from the initial ramping up stage to mass adoption stage. We believe that using handlers as a peer group is justifiable for a unique company like HTSI, with thermal control modules for chip probing sharing similar characteristics to active thermal control used for final test, in addition to Hon Precision and Chroma having exposure to CPO test equipment.

Figure 37: Key equipment vendors' 12-month forward PE band (x)



Source: LSEG, UBS

Figure 38: Testing supply chain valuation table

Company name	Ticker	Analyst	Market cap (US$m)	Price Target (LC)	Share price (LC)	Rating	EPS growth (%)		P/E (x)		P/BV (x)		ROE (%)	
							2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Testing service provider														
ASE Industrial	3711.TW	Sunny Lin	104,043	835	744	Buy	104	64	39.3	24.0	7.5	6.3	21	28
King Yuan Electronics	2449.TW	Sunny Lin	11,456	325	298	Buy	40	62	32.6	20.1	5.9	5.0	20	27
Amkor Technology	AMKR.O	Randy Abrams	13,114	80	53	Buy	73	8	20.6	19.1	2.6	2.4	14	13
JCET Group	600584.SS	Jimmy Yu	17,156	135	64	Buy	72	53	42.7	28.0	3.7	3.3	9	12
Tianshui Huatian Technology	002185.SZ	Jimmy Yu	7,928	13	16	Neutral	33	27	57.0	44.8	2.8	2.7	5	6
TongFu Microelectronics Co Ltd	002156.SZ	N/A	13,526	N/A	57	N/A	50	13	45.3	40.1	5.0	4.5	10	11
Shanghai V-Test Semiconductor Tech Co Ltd	688372.SS	N/A	2,517	N/A	100	N/A	36	106	40.6	19.7	3.8	3.2	11	18
Sector Average							**58**	**47**	**39.7**	**28.0**	**4.5**	**3.9**	**13**	**17**
Tester														
Advantest	6857.T	Atsuhiro Kinoshita	191,547	43,600	41,850	Buy	80	96	38.8	53.1	16.8	31.8	70	58
Teradyne Inc	TER.O	Timothy Arcuri	67,666	500	430	Buy	134	51	45.7	30.8	16.7	12.8	43	47
Sector Average							**107**	**74**	**42.2**	**42.0**	**16.7**	**22.3**	**57**	**52**
Prober/Prober solutions														
Hermes Testing Solutions	7856.TWO	Ryan Sun	3,580	8,880	4,190	Buy	221	487	143.9	27.3	9.1	7.0	10	29
Tokyo Electron	8035.T	Atsuhiro Kinoshita	184,031	18,720	12,805	Buy	-16	71	29.7	37.1	6.5	12.8	36	44
Sector Average							**102**	**279**	**86.8**	**32.2**	**7.8**	**9.9**	**23**	**36**
Handler														
Hon.Precision, Inc.	7769.TW	Sunny Lin	33,404	10,000	5,905	Buy	90	86	45.2	24.4	14.5	10.6	36	50
Chroma ATE	2360.TW	Jerry Su	31,434	2,960	2,350	Buy	64	38	51.8	37.7	28.6	20.8	58	64
Cohu, Inc.	COHU.O	N/A	3,460	N/A	73	N/A	n.a.	94	77.0	39.6	-	-	7	16
JHT Design Co Ltd	603061.SS	N/A	4,045	N/A	312	N/A	155	47	54.8	37.3	13.0	9.9	22	25
Hangzhou Changchuan Technology	300604.SZ	Yongwei Lai	22,690	310	240	Buy	60	53	76.1	49.8	23.4	16.5	36	39
Sector Average							**92**	**63**	**61.0**	**37.8**	**19.9**	**14.4**	**32**	**39**
Test Socket														
WinWay Technology	6515.TW	Ryan Sun	6,948	8,900	6,130	Buy	119	98	60.2	30.4	22.0	14.6	44	34
Leeno Industrial Inc	058470.KQ	N/A	4,936	N/A	86,900	N/A	33	17	35.4	30.3	8.0	6.8	24	24
ISC Co Ltd	095340.KQ	N/A	3,247	N/A	205,500	N/A	70	41	47.0	33.2	7.0	5.9	16	19
Yokowo Co Ltd	6800.T	N/A	744	N/A	4,925	N/A	28	93	38.3	19.8	-	1.8	-	-
Sector Average							**62**	**62**	**45.2**	**28.4**	**12.3**	**7.3**	**28**	**26**
Load Board														
Chunghwa Precision Test Tech	6510.TWO	Ryan Sun	3,149	5,000	3,055	Buy	112	99	47.5	23.8	9.2	7.3	21	34
Probe Card														
Technoprobe SpA	TPRO.MI	Harry Blaiklock, CFA	27,100	41	37	Buy	259	18	66.9	56.5	14.6	11.6	25	23
MPI Corp	6223.TWO	N/A	17,518	N/A	5,670	N/A	97	110	87.2	41.5	30.3	19.1	38	54
FormFactor Inc	FORM.O	N/A	11,038	N/A	141	N/A	159	24	46.2	37.1	9.3	8.9	19	23
MICRONICS JAPAN CO.,LTD.	6871.T	N/A	3,906	N/A	15,410	N/A	155	28	24.8	19.4	7.1	5.6	32	36
Japan Electronic Materials Corp	6855.T	N/A	804	N/A	8,650	N/A	46	43	24.9	17.4	-	-	-	-
Sector Average							**138**	**54**	**49.6**	**32.6**	**14.1**	**10.5**	**27**	**34**

Source: LSEG, UBS estimates. Priced as of October 6 (HKT).

Valuation Method and Risk Statement

Tech investing involves risk. It is difficult for the investment community, UBS included, to project the financial results of tech companies, as their operating models are highly volatile and unpredictable, and they compete in a highly dynamic marketplace. In addition to their low predictability, valuing tech stocks can be challenging because neither traditional nor non-traditional valuation measures have provided much insight into how tech stocks trade.

We value Chroma ATE using a PE methodology.
Downside risks include weaker-than-expected demand for high-performance computing dampening its semiconductor testing business and slower-than-expected recovery in EV investments into battery capacity installations.

We value Chunghwa Precision Test, Hermes Testing and WinWay based on PE methodology.
Risks: Faster- or slower-than-expected front-end/CoWoS capacity expansion could result in CHPT, HTSI and WinWay gaining/losing probe card/board orders from major foundries and OSATs. Potential risk of competition (from either local or foreign companies) achieving technological advancements could result in these companies losing market share from end customers, major foundries, and OSATs. Better- or worse-than-expected AI growth could lead to a ramp-up or a slowdown of product orders from customers, major foundries, and OSATs to CHPT, HTSI and WinWay, affecting long-term revenue growth potential. A potential slowdown or delay of the rollout for technological advancements for technologies such as SoIC, CoPoS, and even CPO, would limit future growth potential. Rising raw material costs may directly impact their margins for equipment shipments.

We value Hon Precision based on PE multiple.
Faster- or slower-than-expected CoWoS capacity expansion could indirectly result in Hon Precision gaining or losing equipment orders from major customers; the risk of competition from either local or foreign companies achieving technological advancements could result in Hon Precision losing market share from its major customers; better- or worse-than-expected AI growth could lead to a ramp-up or a slowdown of equipment orders from customers to Hon Precision, affecting long-term revenue growth potential; a potential slowdown or delay in the rollout of technological advancements for technologies, such as MCL and CPO, would limit future equipment orders; rising raw material costs may directly impact Hon Precision's margins for equipment shipments.

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12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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UBS Securities Pte. Ltd., Taipei Branch: Ryan Sun, Sunny Lin.

Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
Chroma ATE[28]	2360.TW	Buy	NT$2,400	07 Oct 2026
Chunghwa Precision Test Tech	6510.TWO	No Rating	NT$3,055	06 Oct 2026
Hermes Testing Solutions	7856.TWO	No Rating	NT$4,190	06 Oct 2026
Hon.Precision, Inc.	7769.TW	Buy	NT$5,885	07 Oct 2026
WinWay Technology	6515.TW	No Rating	NT$5,520	07 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.
28. UBS holds a long position of 0.5% or more of the listed shares of this company.

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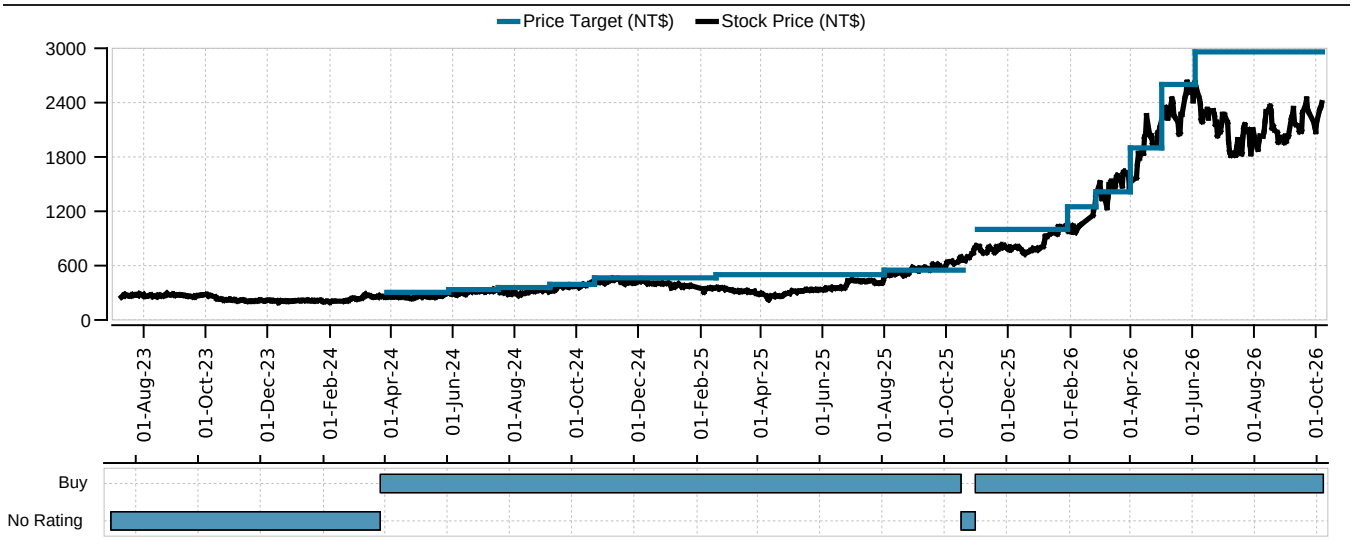
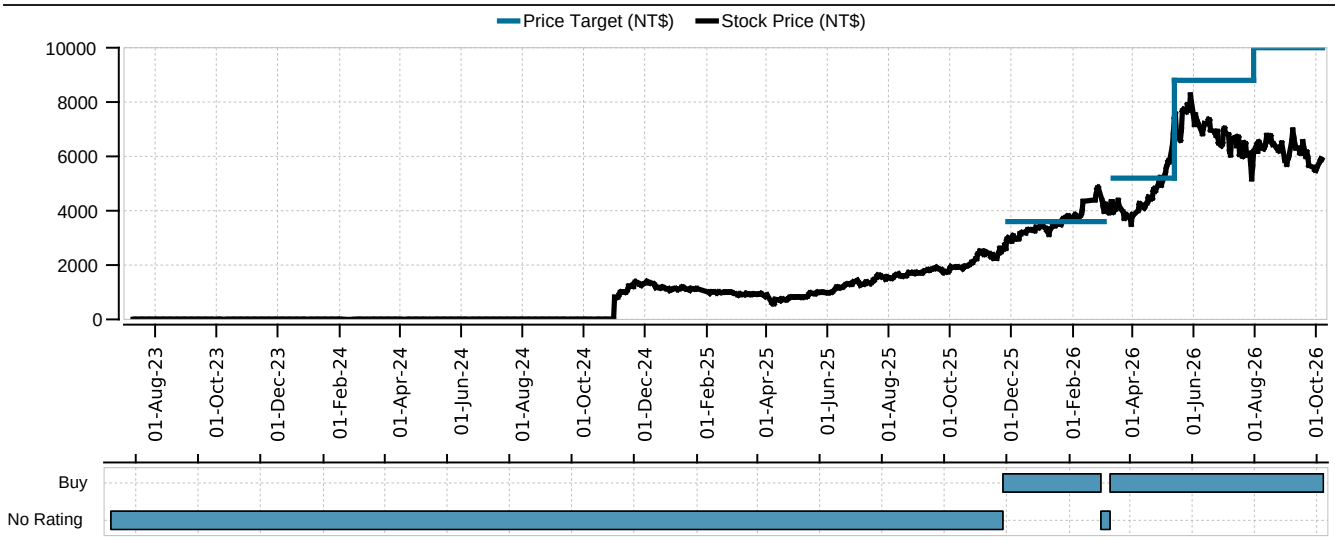
Chroma ATE (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-07	256.50	-	No Rating
2024-03-27	256.00	305.00	Buy
2024-05-27	293.50	335.00	Buy
2024-07-15	335.00	358.00	Buy
2024-09-04	311.00	393.00	Buy
2024-10-18	416.00	465.00	Buy
2025-02-15	350.00	500.00	Buy
2025-07-31	430.50	550.00	Buy
2025-10-17	662.00	-	No Rating
2025-10-31	819.00	1000.00	Buy
2026-01-28	1035.00	1250.00	Buy
2026-02-25	1255.00	1415.00	Buy
2026-03-31	1465.00	1900.00	Buy
2026-05-01	2120.00	2600.00	Buy
2026-06-03	2545.00	2960.00	Buy

Source: UBS Global Research; LSEG Eikon as of 07-Oct-2026. All prices as of local market close. Ratings as of date shown.

Hon.Precision, Inc. (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-07	0.00	-	No Rating
2025-11-27	2940.00	3600.00	Buy
2026-03-03	4150.00	-	No Rating
2026-03-12	4325.00	5200.00	Buy
2026-05-12	6870.00	8800.00	Buy
2026-07-30	5620.00	10000.00	Buy

Source: UBS Global Research; LSEG Eikon as of 07-Oct-2026. All prices as of local market close. Ratings as of date shown.

Additional Prices: Hangzhou Changchuan Technology, Rmb239.79 (30 Sep 2026); ASE Industrial, NT$744.00 (06 Oct 2026); Technoprobe SpA, €36.84 (06 Oct 2026); Alphabet Inc. - Class A, US$347.68 (06 Oct 2026); King Yuan Electronics, NT$298.00 (06 Oct 2026); Taiwan Semiconductor Manufacturing, NT$2585.00 (06 Oct 2026); Tokyo Electron, ¥12805 (06 Oct 2026); MediaTek Inc., NT $4920.00 (06 Oct 2026); NVIDIA Corp, US$239.24 (06 Oct 2026); Teradyne Inc, US$430.32 (06 Oct 2026); Advantest, ¥41850 (06 Oct 2026); Advanced Micro Devices Inc, US$649.42 (06 Oct 2026); Source: UBS. All prices as of local market close.

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Initiation of Coverage

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Final point of contact: securing AI yield through advanced testing socket

Equities

Taiwan
Semiconductors

12-month rating	**Buy**
	Prior : No Rating
12m price target	**NT$8,900.00**
	Prior :
Price (06 Oct 2026)	**NT$6,130**

RIC: 6515.TW **BBG:** 6515 TT

Trading data and key metrics

52-wk range	NT$10,795.00-2,125.00
Market cap.	NT$221b/US$6.95b
Shares o/s	36.0m (ORD)
Free float	69%
Avg. daily volume ('000)	496
Avg. daily value (m)	NT$3,249.8
Common s/h equity (12/26E)	NT$10.0b
P/BV (12/26E)	22.0x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From	To	% ch	Cons.
12/26E	-	99.30	-	97.71
12/27E	-	196.30	-	187.07
12/28E	-	383.04	-	348.32

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Solid growth ahead with robust capacity expansions

We initiate coverage of AI/HPC test socket leader WinWay with a Buy rating and NT $8,900 price target. WinWay's share price has corrected 43% from its April 2026 peak due to broader market de-rating and margin pressure associated with its Technoprobe partnership and capacity expansion. We believe the recent correction presents an opportunity to re-engage with WinWay's long-term growth trajectory, supported by three key drivers: 1) increasing content from AI GPUs, CPUs, and TPUs, 2) substantial capacity expansion, and 3) broader adoption of system-level testing (SLT). Together, these drivers should support revenue growth of 105%/74%/71% in 2026E/27E/28E. Importantly, WinWay's growth is increasingly being driven by higher socket content per device rather than semiconductor unit growth alone and the stock currently trades at 16x of our 2028E P/E vs historical average of 30x, potential to re-rate with GM recovery in the coming quarters. See our companion testing deep dive sector note.

Rising yields and utilisation to offset probe card dilution

Gross margin declined to 38.3% in Q226 from 45.3% in FY25, pressured by outsourced MEMS probes and expansion-related costs. While probe-card margins are unlikely to return to H125 levels of 48.9% given the Technoprobe partnership structure, we expect margin pressure to ease as utilization and yields improve, supporting GM recovery to 43.1% in 2027E and 44.7% in 2028E.

CPU and TPU as fundamental growth drivers

Beyond its core GPU socket franchise, WinWay is well positioned to benefit from the expansion of AI-oriented CPU and TPU platforms. We expect the company to supply sockets for AMD's Venice CPU, Nvidia's Vera CPU, and Google's TPU v8t and v9, and also Vera CPU probe cards, providing an additional growth engine beyond GPUs. WinWay is well positioned in this segment due to its historical collaboration with end customers and its proximity to the local Taiwanese test supply chain. We view AI CPUs and TPUs as an increasingly important contributor to test interface demand as compute architectures diversify in the agentic AI era, driving WinWay's socket and probe card sales growth of 132%/75%/69% and 94%/72%/78% in 2026E/27E/28E, respectively. We estimate CPU/TPU to contribute around 30%/20% of WinWay's 2027E sales.

Valuation: Initiate with a Buy rating, with price target of NT$8,900

We forecast EPS of NT$101.8 in 2026E, rising to NT$201.4/393.0 in 2027E/28E. Our NT $8,900 price target is based on 30x average 2027E-28E P/E, in line with the company's historical average. We believe this valuation is supported by WinWay's exposure to structural growth in AI/HPC testing, expanding socket content, and our long-term earnings CAGR estimate of 30%. We initiate with a Buy rating.

Highlights (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	3,682	5,798	7,857	16,084	27,952	47,831	53,739	60,457
EBIT (UBS)	557	1,368	2,070	4,428	8,986	17,560	19,699	22,336
Net earnings (UBS)	464	1,186	1,673	3,656	7,241	14,130	15,870	18,029
EPS (UBS, diluted) (NT$)	13.34	33.02	46.71	99.30	196.30	383.04	430.19	488.73
DPS (net) (NT$)	10.87	25.75	25.07	61.10	120.85	235.82	264.85	300.89
Net (debt) / cash	302	977	2,594	1,155	2,200	7,340	10,417	14,723
Profitability/valuation	**12/23**	**12/24**	**12/25**	**12/26E**	**12/27E**	**12/28E**	**12/29E**	**12/30E**
EBIT (UBS) margin %	15.1	23.6	26.3	27.5	32.1	36.7	36.7	36.9
ROIC (EBIT) %	20.7	35.3	49.2	70.5	83.8	116.7	100.9	93.9
EV/EBITDA (UBS core) x	33.4	21.5	22.9	46.1	23.0	11.7	10.1	8.6
P/E (UBS, diluted) x	51.2	30.5	32.9	61.7	31.2	16.0	14.2	12.5
Equity FCF (UBS) yield %	(1.1)	2.1	2.8	0.4	1.5	4.3	5.2	6.3
Dividend yield (net) %	1.6	2.6	1.6	1.0	2.0	3.8	4.3	4.9

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 6,130.00 on 06-Oct-2026

This report has been prepared by UBS Securities Pte. Ltd., Taipei Branch. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 24.**

UBS Research THESIS MAP a guide to our thinking and what´s where in this report

PIVOTAL QUESTIONS

Q: Will capacity constraints limit WinWay's growth in the next two to three years?
Near term, yes. Beyond 2026E, no. WinWay's current capacity remains constrained, but ongoing expansions should significantly increase output over the next two years. With new capacity coming online and the company shifting toward higher-value AI/HPC applications, including CPU probe cards, >10k-pin coaxial sockets and next-generation HyperSockets, we forecast revenue growth of 105%/74%/71% in 2026E/27E/28E, supported by capacity additions, rising AI/HPC content and expanding socket demand. We also expect operating leverage to improve as capacity scales and utilization increases, with OPM rising from 27.5% in 2026E to 32.1%/36.7% in 2027E/28E.

Q: Will WinWay's GM recover from Q226's 38.3%?
Yes. We view recent margin pressure as largely temporary. GM has been impacted by MEMS probe sourcing through Technoprobe and start-up costs associated with capacity expansion. Although margin dilution from probe card business is unlikely to unravel due to the MEMS partnership terms with Technoprobe, as new facilities ramp and yields improve, we expect profitability to recover gradually from the Q226 trough, with gross margin improving to 43.1% in 2027E and 44.7% in 2028E.

UBSVIEW

We initiate coverage of WinWay with a Buy rating and NT$8,900 price target. We believe the market is overly focused on near-term margin pressure and underappreciating WinWay's long-term growth outlook. Key drivers include 1) expanding content from AI GPUs, CPUs, and TPUs, 2) substantial capacity expansion, and 3) growing adoption of system-level testing (SLT). Together, these drivers should support revenue growth of 105%/74%/71% in 2026E/27E/28E.

EVIDENCE

TSMC and OSATs have been expanding their CoWoS capacities aggressively, from 40kwpm in 2024 to around 90 kwpm in 2025, and is tracking to expand CoWoS capacity toward 160kwpm in 2026E. TSMC also ramped up expanding N3/N2 capacity, with multiple product launches from customers' pipelines—Nvidia's Rubin, AMD's MI400 series/Venice CPU and other cloud ASICs—translating into foreseeable, strong testing demand as the majority are WinWay's customers. AI accelerator test times have increased meaningfully for new platforms. We expect continued front-end and advanced packaging investments to drive higher testing intensity and support long-term demand for advanced sockets and test interfaces.

WHAT´S PRICED IN?

WinWay's share price has corrected 43% from its April 2026 peak and currently trades at 16x of our 2028E P/E vs historical average of 30x. Our reverse DCF analysis indicates that the current market price implies only a 19% long-term sales CAGR vs our estimate of 25%, alongside long-term GM level of 32% vs our estimate of 45%. We see scope for re-rating.

UPSIDE/DOWNSIDE SPECTRUM

Value drivers (2027/28E)	Test socket sales growth	Probe card sales growth	GM %
NT$10,000 upside	90%/87%	87%/93%	45.1%/49.7%
NT$8,900 base	75%/69%	72%/78%	43.1%/44.7%
NT$4,000 downside	35%/34%	32%/38%	38.1%/39.7%

Source: UBS estimates

COMPANY DESCRIPTION

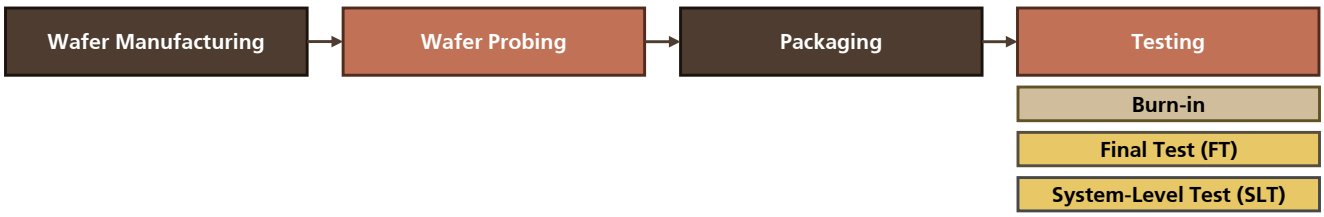
WinWay was founded in 2001 in Kaoshiung, Taiwan. It is a semiconductor testing interface solutions provider, with final test socket as its main business segment. The company has high-end solutions for AI/HPC testing, providing final test/system-level test and burn-in test sockets and probe cards through its own made probes and partnership with Technoprobe.

Testing TAM expanding with IC complexity: leaders likely to outperform industry growth

Our central view is that testing content per AI accelerator is growing faster than semiconductor unit volumes, creating a multi-year expansion in testing TAM. We believe semiconductor testing is entering a period of structural growth as AI-driven chip complexity increases testing intensity across wafer, package and system levels. The industry's shift-left strategy, which moves more testing upstream in the semiconductor manufacturing process, is resulting in longer and more frequent test insertions. As cloud AI GPUs and ASICs become increasingly complex, additional testing is required to ensure functionality, performance and yield, while reducing defective parts per million (DPPM). More robust testing should improve deployment efficiency and lower the cost of operating AI infrastructure.

Semiconductor testing generally comprises four stages: wafer/chip probing (CP), final test (FT), system-level test (SLT), and burn-in. CP is conducted at the wafer-level before packaging, while FT, SLT, and burn-in occur after packaging. The below Figure 1 illustrates the semiconductor manufacturing flow.

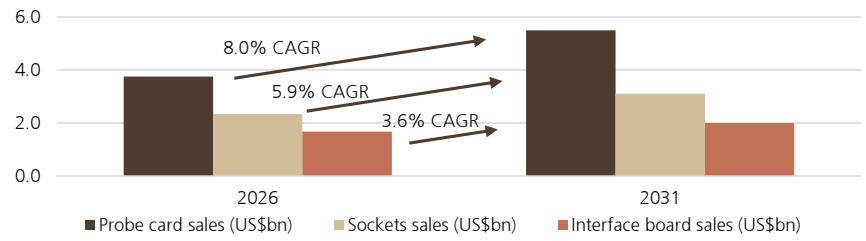
Figure 1: Semiconductor testing process: from wafer level to package level



Source: UBS

We estimate semiconductor testing historically accounts for ~1% of total semiconductor revenue. Within testing, consumables such as probe cards, sockets and interface boards represent ~10-15% of industry revenue. According to Yole, the probe card, socket and interface board markets should grow from US$3.8bn, US$2.3bn and US$1.7bn in 2026 to US$5.5bn, US$3.1bn and US$2.0bn by 2031, respectively.

Figure 2: Yole estimating testing consumables TAM to climb in the next 5 years



Source: Yole, UBS

Nonetheless, leaders within the testing sector will standout among competition, with greater sales CAGR comparing to the industry.

As the leader of AI/HPC sockets, **WinWay** is set to deliver 2026-31E socket sales CAGR of 33%, underpinned by robust demand for final test and system-level test, with test content and test times rising which each iteration of new AI accelerators/CPU, more sockets and equipment will be needed in order to increase the throughput.

- **Strong accelerator/CPU roadmap to support growth.** Robust testing demand can also be inferred from AI accelerator/CPU roadmaps of Nvidia, AMD, Google, alongside tester capacity expansions by Advantest.
- **Capacity expansion underpinning future strength.** WinWay's socket pin capacity expansion is expected by the company to support this exceptional sales growth from 3.5mn pins per month at year-end 2025 to 6mn currently, reaching

9mn by year-end 2026 and 14mn by year-end 2027 through expansions in Renwu, Kaohsiung. By 2027, approximately 60% of pogo-pin requirements should be internally sourced, strengthening both supply security and margins.

- **Probe card business to rise from smaller base.** Its probe card business is also set to benefit from the shift-left trend for high-end AI/HPC probe cards. WinWay leverages its existing customers' demand in addition to their MEMS partnership with Technoprobe, as we forecast WinWay's probe card sales to rise from a low base, with a 33% 2026-31E sales CAGR.

Figure 3: WinWay is set to benefit with a 33% 2026-31E sales CAGR

Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Coaxial Socket (HyperSocket)	$7,661	$13,705	$23,103	$25,872	$30,603	$34,726	35%
YoY		*79%*	*69%*	*12%*	*18%*	*13%*	
RF & Plastic Socket	$2,311	$3,835	$6,465	$6,960	$7,259	$7,252	26%
YoY		*66%*	*69%*	*8%*	*4%*	*0%*	
Burn-in Socket	$253	$306	$516	$715	$903	$1,075	34%
YoY		*21%*	*69%*	*39%*	*26%*	*19%*	
Test Socket	$10,224	$17,845	$30,083	$33,547	$38,764	$43,053	33%
YoY		*75%*	*69%*	*12%*	*16%*	*11%*	
Probe Card	$4,478	$7,700	$13,690	$15,507	$16,095	$18,403	33%
YoY		*72%*	*78%*	*13%*	*4%*	*14%*	
Contact Element	$937	$1,596	$2,691	$3,014	$3,465	$3,899	33%
YoY		*70%*	*69%*	*12%*	*15%*	*13%*	
Others	$444	$811	$1,367	$1,671	$2,133	$2,654	43%
YoY		*83%*	*69%*	*22%*	*28%*	*24%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	$16,084	$27,952	$47,831	$53,739	$60,457	$68,010	33%
YoY		*73.8%*	*71.1%*	*12.4%*	*12.5%*	*12.5%*	
GM %	41.1%	43.1%	44.7%	44.7%	44.9%	45.1%	
OpM %	27.5%	32.1%	36.7%	36.7%	36.9%	37.1%	

Source: UBS estimates

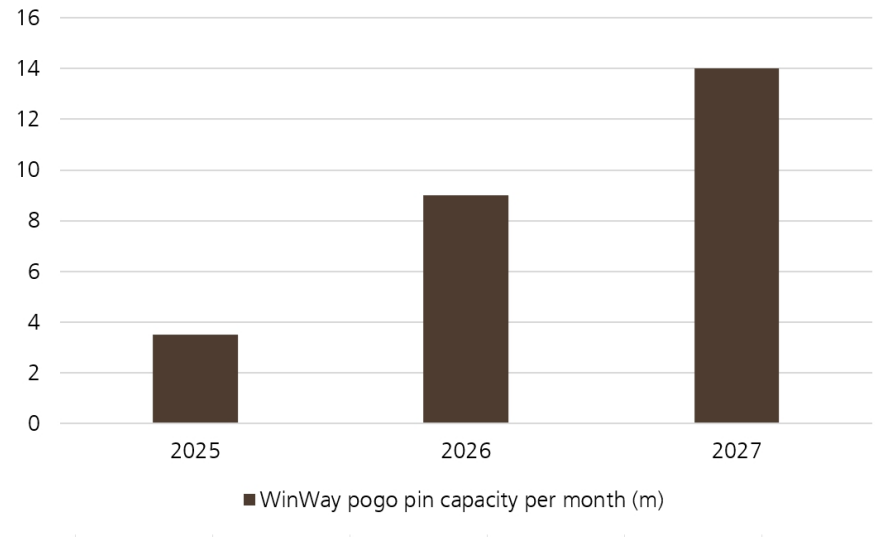
WinWay is expanding capacity rapidly to capture this growing TAM. Industry test interface suppliers are expanding capacity aggressively to meet rising AI and HPC demand. As advanced semiconductors require higher-performance probe cards, sockets and interface boards, capacity additions are becoming increasingly important to support growth across both wafer-level and package-level testing.

We believe WinWay is well positioned to gain share within the high-end AI/HPC test interface market. Taiwanese suppliers benefit from their proximity to leading fabless customers and the growing preference for localised supply chains, particularly for advanced testing applications where responsiveness and engineering support are critical. Against a backdrop of tight industry supply, WinWay announced significant capacity expansion plans over the next two to three years.

WinWay's capacity is expected by the company to increase from 3.5mn pins per month at year-end 2025 to 6mn currently, reaching 9mn by year-end 2026 and 14mn by year-end 2027 through expansions in Renwu, Kaohsiung. By 2027, approximately 60% of pogo-pin requirements should be internally sourced, strengthening both supply security and margins.

We estimate these capacity additions will support revenue growth of 105%/74%/71% in 2026E/27E/28E.

Figure 4: Announced capacity expansions for WinWay



Source: Company data

For WinWay, this growth opportunity is substantial, as the capacity expansions for its socket pins will satisfy around 60% of its own pogo pin demand in 2027E, we estimate socket sales will likely contribute for around NT$163 of EPS in 2027E and to further expand to NT$307 in 2028E. This paves the way for substantial share gaining opportunity for WinWay for AI/HPC sockets as we believe there are advantages to have footprint within Taiwan, where there's already an established testing ecosystem with testing providers such as ASE/SPIL/KYEC, and handler providers such as Hon Precision, all positioned locally in Taiwan to provide around the clock service and support for AI/HPC customers.

We see industry capacity tightness to support socket pin ASP growth, as we forecast pin ASP to grow from US$2.0 in 2026E to US$2.5 in 2027E and US$3.0 in 2028E. We believe WinWay's capacity expansion creates a significant growth opportunity with a better margin profile. By 2027E, approximately 60% of pogo pin demand should be internally sourced, improving both supply security and profitability. We estimate sockets could contribute around NT$123 of EPS in 2027E, rising to NT$261 in 2028E. Combined with its expanding capacity, strong customer relationships and proximity to Taiwan's semiconductor testing ecosystem, including ASE/SPIL, KYEC and major handler suppliers, WinWay is well positioned to gain share in the AI/HPC socket market.

Figure 5: We see socket pin expansions to drive significant growth in the next 2-3 years

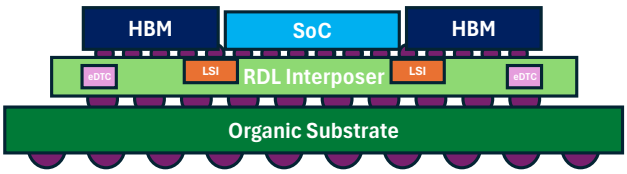
Socket Pin Analysis	2025	2026E	2027E	2028E
Pin Capacity (m/month)	3.2	6.7	11.2	18.3
Self sufficiency	*56%*	*50%*	*60%*	*70%*
Pin demand (m/month)	5.63	13.40	18.59	26.11
ASP (US$)	$2.0	$2.0	$2.5	$3.0
Sales contribution (US$m/month)	**$11**	**$27**	**$46**	**$78**
Sales contribution (US$m/year)	**$135**	**$322**	**$558**	**$940**
Socket sales (NT$m)	**$4,214**	**$10,224**	**$17,845**	**$30,083**
GM (%)	*50%*	*42%*	*47%*	*50%*
Gross Profit (NT$m)	**$2,107**	**$4,294**	**$8,387**	**$15,041**
OPEX (%)	*19%*	*17%*	*16%*	*11%*
Operating Profit (NT$m)	**$1,306**	**$2,556**	**$5,532**	**$11,732**
OPM (%)	*31%*	*25%*	*31%*	*39%*
Tax rate (%)	*19%*	*20%*	*20%*	*20%*
Net profit (NT$m)	**$1,060**	**$2,041**	**$4,426**	**$9,386**
EPS (NT$m)	**$29.7**	**$56.9**	**$123.1**	**$261.1**

Source: UBS estimates

The AI era has materially increased the value of semiconductor testing. Compared with traditional mobile and automotive chips, AI accelerators incorporate larger compute dies, more advanced nodes, chiplet architectures, advanced packaging technologies, multiple I/O dies, and increasing HBM content. These factors drive higher-value test consumables, including advanced MEMS probe cards, multilayer interface boards, and next-generation sockets designed to support higher pin counts, thermal loading, and power delivery requirements.

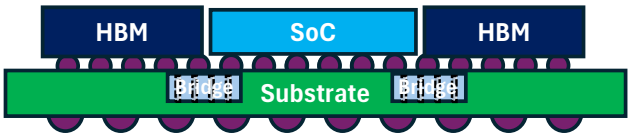
Capacity expansion by test interface vendors is ultimately supported by continued investment throughout the advanced packaging ecosystem, from providers such as TSMC, ASE, Intel, and Amkor. During the Q226 earnings calls, TSMC raised its recent 2026 capex from US$52-56bn to US$60-64bn and ASE raised its 2026 capex from US $8.5bn to US$10.5bn, with both companies citing robust demand supporting capex raise, we believe capex for the next two years will be even higher than 2026, indicating continuous growth in future years. With advanced packaging technologies getting significantly more complex and with larger form factor, illustrated by the upcoming ramp of both TSMC's CoPoS (chip-on-panel-on-substrate) and Intel's EMIB-T (embedded multi-die interconnect bridge) packaging technologies, test interface solution providers will benefit with more value added consumables (see our CoPoS vs EMIB-T deep dive).

Figure 6: TSMC's CoPoS



Source: UBS

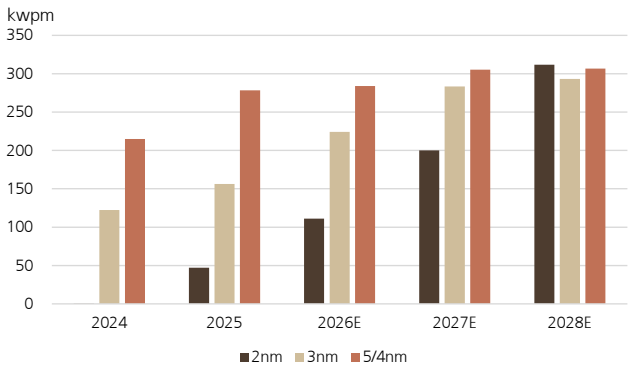
Figure 7: Intel's EMIB-T



Source: UBS

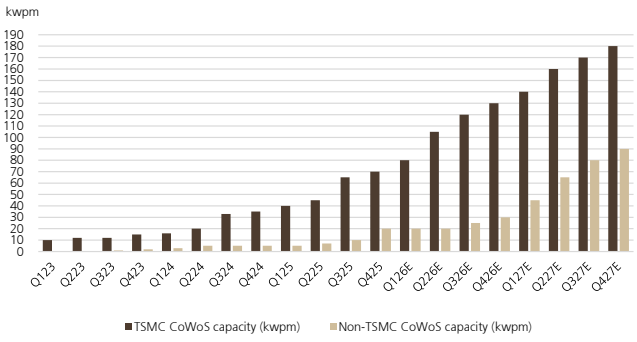
New iterations of semiconductors, especially for AI / HPC applications, are set to increase in both front-end wafer fabrication advancements and back-end packaging complexities with each iterations (i.e. Nvidia's roadmap from Blackwell to Rubin to Feynman or Google's TPU roadmap from v7 to v8 to v9). This trend is evident with TSMC's front-end and TSMC, Intel, and OSAT's back-end advanced packaging capacity expansions. For front-end, we see industry N3 capacity expanding from 156kwpm in year end 2025 to 224/283/293kpwm in year end 2026E/27E/28E, and N2 capacity from 47kwpm in year-end 2025 to 111/200/312kpwm in year end 2026E/27E/28E. For back-end, we are seeing industry CoWoS capacity to rise from 90kwpm in 2025 to 160kwpm in 2026E and further expand to 270kwpm in 2027E.

Figure 8: Industry's leading edge capacity



Source: UBS estimates

Figure 9: Industry CoWoS capacity

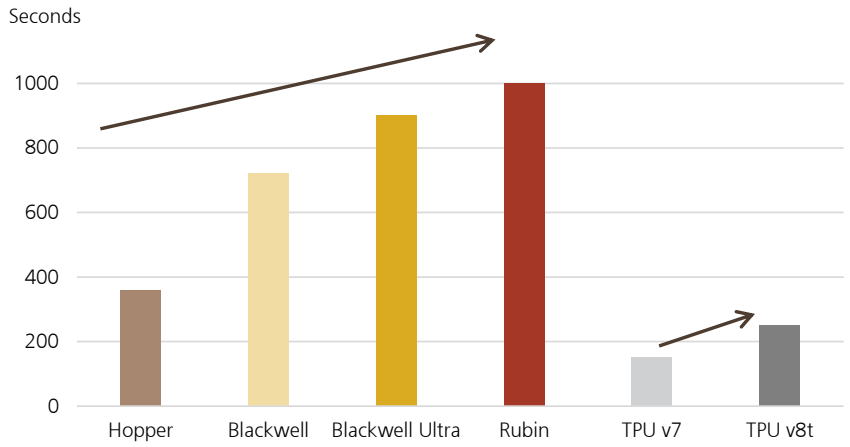


Source: UBS estimates

Final-test, system-level test, and burn-in content should continue to increase as AI accelerators become larger, more power-hungry and more complex. Rising thermal design power and packaging complexity increase validation requirements, driving

longer test times and higher demand for test equipment to prevent testing from becoming a production bottleneck. This is where **WinWay**'s sockets comes into play—as high-end chips with 2.5D/3D/3.5D packaging requires intricate FT sockets to handle the high thermal design power (TDP) and high data rates of up to 224G during the FT process for reliable test results.

Figure 10: Our FT test time estimates



Source: UBS estimates

While architectural changes can affect test requirements for individual product generations, we expect the long-term trend of rising testing content to remain intact. For example, Nvidia's Rubin Ultra adopts a simpler package structure than previously expected, from a 4-die structure to a 2-die structure and Rubin Ultra HBM de-spec from HBM4E 12-Hi to initially use HBM4 8-Hi and upgrade to HBM4E 8-Hi at the latter stages, which may moderate near-term test growth. However, future products such as Nvidia's Feynman are expected to reintroduce greater package complexity, with a four-GPU per package structure. More importantly, any reduction in test time can be offset by higher production volumes, making throughput requirements equally important for testing demand.

We expect test interface content growth to outpace unit growth, For instance, test socket's ASP directly correlates with its pin counts, with the addition of 20-30% more pins inside the sockets, socket ASP can grow 20-30% or more as the socket might require significant cooling capabilities as the TDP of the DUT rises with its larger form factor. Unlike prior semiconductor cycles, much of the current demand is driven by rising test interface content per device rather than unit growth alone.

Robust growth drivers across semiconductor testing

We believe WinWay is set to benefit from several semiconductor testing growth drivers and capture the AI induced, expanding testing TAM. We see three main growth drivers for WinWay: 1) AI accelerators (GPU + TPU), 2) agentic AI driven CPU demand, and 3) shift-left testing trend. Below we list the three main growth drivers in detail.

Accelerators still in focus with TPU as another significant growth driver

Nvidia has been the primary driver of testing content growth over the past several years, benefiting the entire semiconductor testing supply chain. We expect this trend to continue as GPU architectures evolve from Blackwell to Rubin and then to Feynman, with increasing process node complexity, larger package sizes and more advanced packaging technologies driving higher testing value per device.

We believe Nvidia should remain a major contributor to WinWay, with GPUs continuing to serve as their core growth engine. We estimate WinWay's socket share could improve from a roughly 50/50 split with Smiths Interconnect in prior generations to 60/40 with Rubin. Nvidia currently accounts for approximately 50% of WinWay's revenue. Both Nvidia GPUs and Google TPUs increase testing demand through higher content per device rather than unit growth alone.

Beyond Nvidia, Google's TPU roadmap represents a significant incremental growth opportunity. In our view, WinWay should benefit as the socket supplier for both TPU v8t and the next-generation TPU v9 platform. We believe MediaTek's increasing share gain for Google TPUs will also benefit the local Taiwanese test interface supply chain. We estimate TPU to contribute to around 20% of WinWay 2027E sales.

We view TPU v9 as a particularly attractive opportunity given its large four-die package architecture based on Intel's EMIB-T packaging technology, which should drive higher testing content. We estimate production of 4mn TPU v8t units in 2027E, followed by 2.3mn TPU v9 packages (9.2mn dies) and an additional 2mn TPU v8t units in 2028E. Importantly, while EMIB-T packaging is expected to be performed in Malaysia, we believe the majority of advanced testing value should remain within the Taiwanese testing ecosystem. Leading testing providers, including ASE and KYEC, maintain substantial testing operations across Taiwan and Southeast Asia, supporting continued participation by Taiwanese test interface suppliers. See our latest MediaTek report and EMIB-T deep dive.

CPU business as a supplemental catalyst for test interface companies in the agentic world

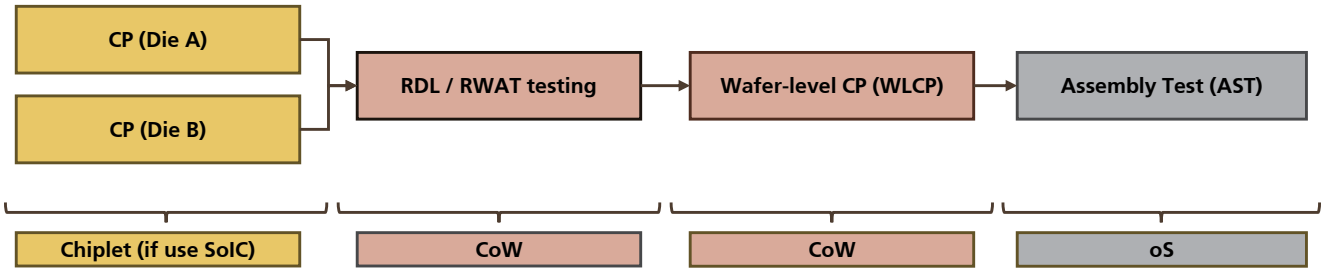
We believe semiconductor test interface are an underappreciated beneficiary of the emerging agentic AI ecosystem. UBS estimates the CPU market will expand to US $170bn by 2030E from US$31bn in 2025 and US$71bn in 2027E, driven by rising deployment of AI infrastructure. As CPU-to-GPU ratios increasingly approach 1:1 in AI clusters, demand for testing interfaces should expand alongside CPU volumes. TSMC is the main CP test provider in the industry, with probe card suppliers aiding its massive demand by supplying probes, such as TSMC's GPU probe cards supported by Technoprobe's probes. However, with capacity now becoming scarce with strong front and back-end demand, TSMC is outsourcing its CP business to ASE and KYEC, further broadening the testing supply chain. This is where new probe card vendors come in—smaller probe card players in the past, such as WinWay, now has the opportunity to gain significant, non-GPU share from TSMC, as they have MEMS probe capabilities (WinWay's MEMS probe are from its strategic partnership with Technoprobe) and has a long established relationship with the fabless companies such as Nvidia via supplying non-probe card consumables such as sockets. Of which, CPU projects has become the main area for WinWay to expand their footprint.

We expect WinWay to benefit from the ramp of Nvidia's ARM-based Vera CPU and Google's ARM-based Axion CPU in 2026E-27E. WinWay is positioned to supply sockets and probe cards for Vera and sockets for Axion. Additional upside could come from AMD's Venice CPU, where WinWay is well positioned. For WinWay, CPU-related demand could account for approximately 30% of sales in 2026E-27E. Importantly, CPU programmes diversify exposure beyond AI GPUs and provide a second structural growth driver for test interface suppliers.

Shift-left igniting secular growth for chip probing

The industry's adoption of shift-left testing should drive a multi-year structural growth opportunity for chip probing (CP) and its supply chain. As advanced packaging costs rise and AI accelerators become increasingly complex, semiconductor manufacturers are placing greater emphasis on identifying known-good dies (KGD) before packaging. The test shift-left trend will drive structural AI/HPC probe card and test consumables demand, with increasing test interface content as test insertions and test time increase. WinWay should also benefit through its partnership with Technoprobe, with probe card revenue growth remaining robust at 94%/72% in 2026E/27E following 161%/260% growth in 2024/25.

Figure 11: CP is set to grow substantially with shift-left, each packaging step set to be MORE costly than the prior step

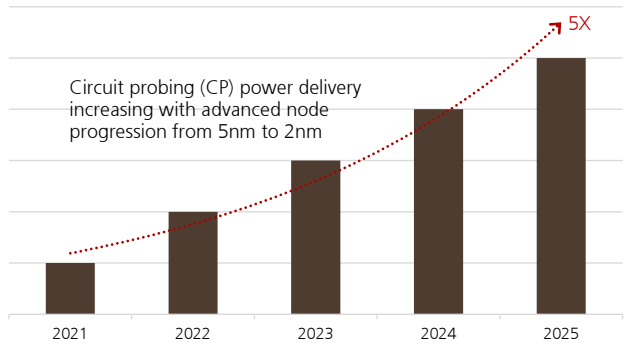


Source: UBS

The economics of advanced packaging reinforce this trend. Heterogeneous integration, chiplet architectures and 2.5D/3D packaging significantly increase package value, while defects identified late in the manufacturing process can lead to substantial scrap costs. To mitigate this risk, foundries and OSATs such as TSMC and ASE employ distributed testing flows incorporating KGD, known-good-stack (KGS) and assembly test screening, to catch defects before the components enter a multi-week packaging cycle (>2 weeks for CoW and 2 weeks for oS). The importance of these tests rises as AI clusters become increasingly dependent on component reliability and low defective-parts-per-million (DPPM) rates. Given the multiplicative yield degradation, a 300bps yield decline at just one manufacturing stage (e.g. dropping from 99.5% to 96.5%) reduces overall integrated four-stage yield from 98.0% to 95.0%; on a 500,000 unit ramp at US $10,000 per unit, this 3% scrap delta translates into around US$150m in unrecoverable losses given >6-month manufacturing lead times.

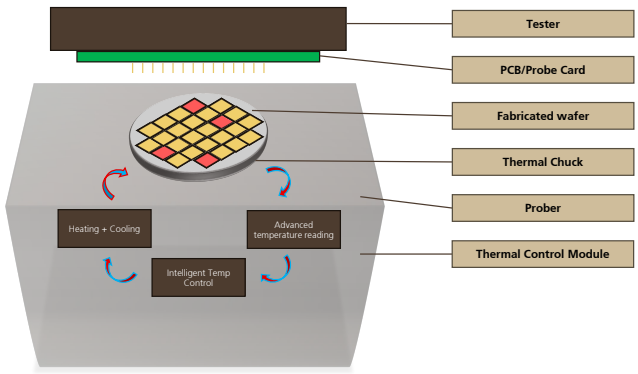
At the same time, rising chip power consumption is increasing the technical difficulty of wafer-level testing. As chip thermal design power scales beyond 1,000W and power density is expected by the industry to surge another 8x by 2030, traditional wafer-level chip probe (WLCP) encounters severe physical bottlenecks. Finer probe pitches and higher power densities increase the risk of overheating, probe needle burnout and yield degradation, creating demand for more advanced probing and thermal-control solutions.

Figure 12: CP power delivery trend rises with process tech evolution



Source: UBS

Figure 13: Thermal control module inside probers are essential to control wafer temperature



Source: UBS

WinWay set to benefit from the AI trend with its leading position within advanced test socket market

Figure 14: Key vendors for each stage of the semiconductor testing industry

	Chip Probing	Final Test	System-Level Test
Testing service providers	TSMC, ASE, KYEC, Amkor, Intel	ASE, Amkor, KYEC, JCET, Tianshui Huatian, Tongfu	ASE, Amkor, KYEC, JCET, Tianshui Huatian, Tongfu
Testers	Advantest, Teradyne	Advantest, Teradyne	N / A
Handlers (Probers)	TEL, Tokyo Seimitsu, TSK	Advantest, Chroma, Cohu, Hangzhou Changchuan Technology, Hon Precision, JHT, Teradyne	Hon Precision, Chroma
Probe Cards	FormFactor, Technoprobe, MPI, MJC, JEM, WinWay, CHPT	N / A	N / A
Socket	N / A	WinWay, Cohu, Leeno, Yokowo, Yamaichi, Smiths	WinWay
Load board	N / A	Yamaichi, Advantest, CHPT	Yamaichi, Advantest, CHPT

Source: UBS

WinWay was founded in 2001 in Kaohsiung, Taiwan, and is a leading provider of semiconductor testing interface solutions, with test sockets representing its core business. The company focuses on advanced AI/HPC applications, supplying sockets for final test (FT), system-level test (SLT) and burn-in, as well as cobra probe cards and partnership with Technoprobe for MEMS probe cards.

WinWay's business is primarily divided into two segments:

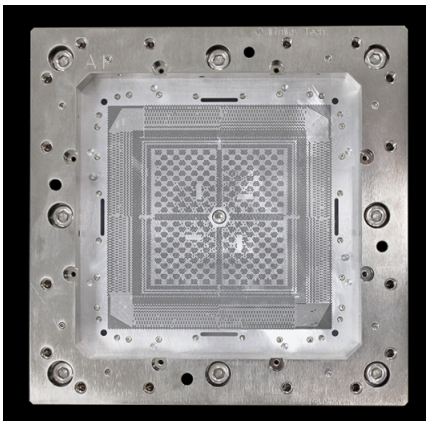
- **Test socket:** Sockets are WinWay's largest revenue contributor. The company supplies a broad range of testing sockets, including coaxial, PoP/plastic and burn-in sockets, serving applications ranging from consumer electronics to high-performance computing.
    - **Coaxial socket:** utilised as the mainstream option for nearly all AI/HPC chips, as traditional sockets are near its physical limits. Coaxial sockets need to handle data rates of up to 224G for high speed transmission and high chip TDP during the FT process.
    - **HyperSocket:** a rubber-silicone like layer called **elastomer** are placed on top of the pogo pins, these elastomer pins causes zero damage to the chips being tested under high contact forces. WinWay's next generation HyperSocket technology utilises elastomers to provide a softer contact between the DUT and the pogo pins, avoiding damage to solder ball. This structure significantly enhances spring probe's lifetime, reduces cleaning frequency, and achieves maximum production efficiency. We estimate it will contribute around 5% of sales in 2027E with a 20-30% ASP premium as compared to standard coaxial socket.
- **Probe cards:** WinWay historically supplies cobra probe cards (vertical probe cards), but with the company announcing their five-year probe supply deal with Technoprobe for MEMS probe in December 2025, they are now one of Nvida Vera CPU's main probe card suppliers.

AI accelerators are pushing socket requirements beyond 10,000 pins, making **coaxial sockets** the industry standard for leading-edge AI/HPC devices. These sockets must support transmission speeds of up to 224G while handling significantly higher thermal design power (TDP) during testing.

Elastomer technology is increasingly being adopted alongside spring-probe (pogo-pin) architectures to reduce chip damage and improve contact reliability. WinWay's next-generation **HyperSocket** integrates elastomer materials between the DUT and spring probes, improving probe lifetime, reducing cleaning requirements and enhancing production efficiency. We believe HyperSocket could become an important differentiator as next-generation AI accelerators demand higher pin counts, bandwidth and thermal performance. We estimate HyperSocket could contribute approximately 5% of WinWay's revenue in 2027E while carrying a 20-30% ASP premium versus conventional coaxial sockets. The transition toward larger AI packages, higher pin counts and faster data-transfer speeds should continue to increase socket content per device. We expect this trend to support long-term growth in the socket market, with technology leaders such as WinWay well positioned to gain share through next-generation solutions including HyperSocket.

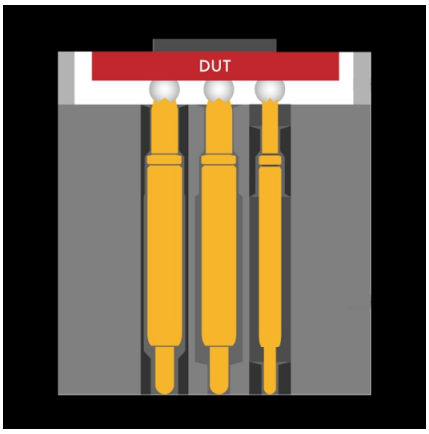
Beyond its existing product portfolio, WinWay is developing next-generation technologies including CPO electrical test sockets and liquid-cooled sockets for future high TDP AI accelerators, reinforcing its position within the AI/HPC testing ecosystem.

Figure 15: Coaxial socket



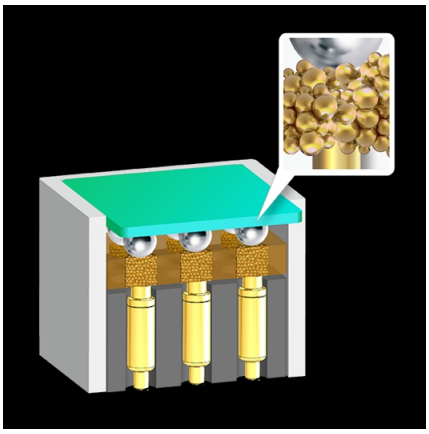
Source: Company data (WinWay)

Figure 16: DUT socket contact



Source: Company data (WinWay)

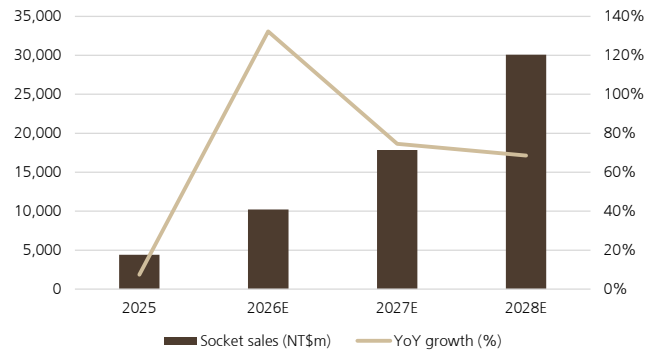
Figure 17: HyperSocket



Source: Company data (WinWay)

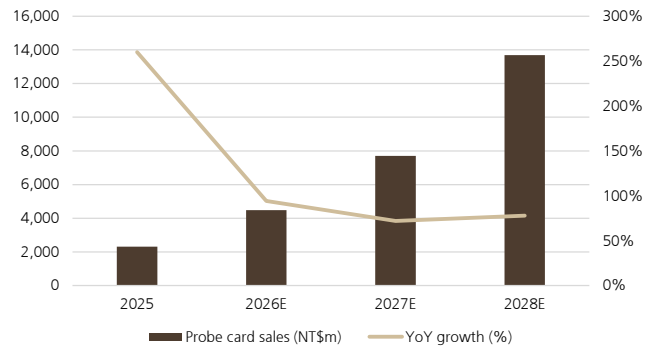
WinWay's competitive advantages are its 1) technology leadership with coaxial socket and HyperSocket, 2) proximity to the semiconductor testing value chain in Taiwan, and 3) historical relationship with key AI companies such as Nvidia and AMD. We believe these advantages may allow WinWay to capture significant upside in the growing semiconductor testing TAM driven by cloud AI proliferation, and gain share across its competition such as Smiths Interconnect for AI and HPC applications.

Figure 18: WinWay's socket sales growth



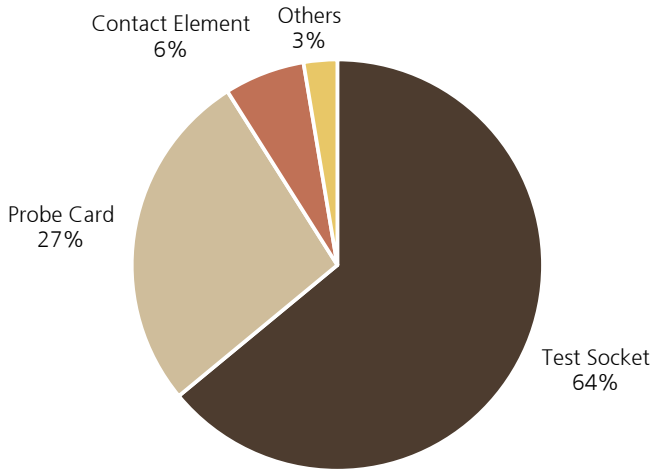
Source: Company data, UBS estimates

Figure 19: WinWay' probe card sales growth



Source: Company data, UBS estimates

Figure 20: WinWay's H126 product mix

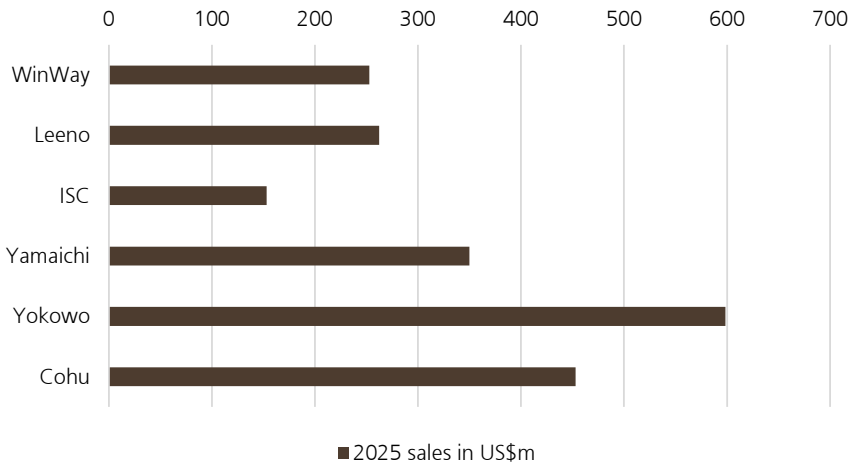


Source: Company data, UBS

Compared with the probe card market, the socket industry is more fragmented. Competition include:

- **WinWay (6515.TW)**: A major share gainer in AI/HPC sockets, with strong exposure to Nvidia and AMD. The company specializes in high-performance coaxial sockets and is developing its next-generation HyperSocket platform. It is also expanding into probe cards through its partnership with Technoprobe.
- **Leeno (058470.KQ)**: Vertically integrated supplier focused on smartphone AP and memory sockets.
- **ISC (095340.KQ)**: Memory socket supplier with elastomer expertise.
- **Smiths Interconnect (Acquired by Molex)**: A key player in AI/HPC socket industry, with its DaVinci coaxial sockets able to support high frequency connectivity testing.
- **Yamaichi (6941.T)**: Japanese socket supplier with substantial automotive exposure.
- **Yokowo (6800.T)**: Major provider of contact probes and socket pins.
- **Cohu (COHU.O)**: Broad test-solutions provider spanning ATE, handlers and sockets, with high exposure to automotive, industrial and consumer applications.

Figure 21: 2025 sales of major socket suppliers

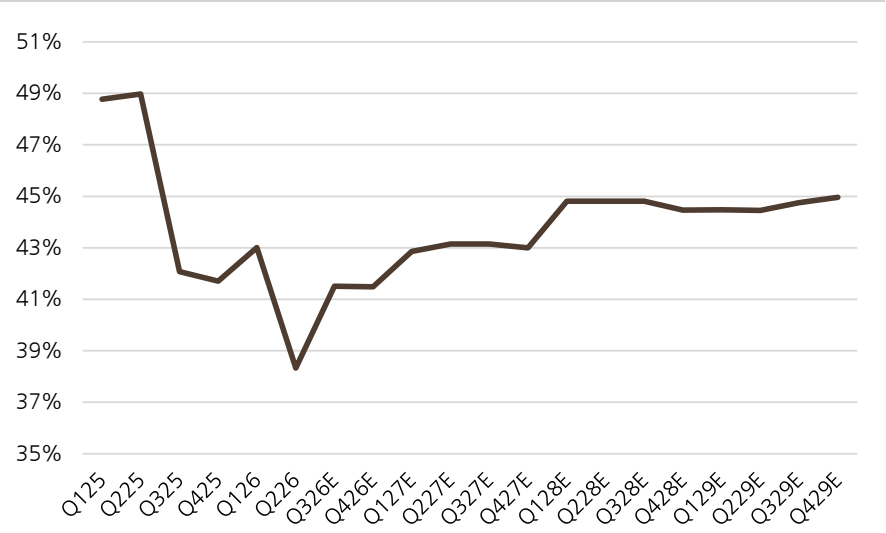


Source: Company data, UBS. Note: Most socket suppliers are companies with multiple business segments, i.e. Yokowo and Cohu.

WinWay's margin is set to recover in 2027E/28E

Company GM hit a trough in Q226, at 38.3%. We view recent margin pressure as largely temporary. GM has been impacted by MEMS probe sourcing through Technoprobe (probe card GM now below 40% and dilutive to >50% coaxial socket GM) and start-up costs associated with WinWay's rapid capacity expansion. During the early part of 2025, WinWay's GM was high at 48.9% in H125, with more cobra probe cards (60% GM) in mix with solid pogo pin and socket yield. With MEMS probe card now accounting for more than 70% of probe card revenue and GM at around low to mid 30%, margin dilution from probe card business is unlikely to unravel for GM to return to above 45% in the near term due to the MEMS partnership terms with Technoprobe. However, we expect that as new facilities ramp and yields improve, profitability to recover gradually from the Q226 trough. We see GM improving to 43.1% in 2027E and 44.7% in 2028E.

Figure 22: We believe WinWay' GM will gradually recover in the next two-three years



Source: UBS estimates

Figure 23: UBS earnings forecast

(NT$m)	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E	2028E	2029E	2030E
Revenue	**7,857**	**2,980**	**3,523**	**4,556**	**5,024**	**16,084**	**5,488**	**6,978**	**7,327**	**8,159**	**27,952**	**47,831**	**53,739**	**60,457**
- YoY chg (%)	36%	30%	131%	153%	125%	105%	84%	98%	61%	62%	74%	71%	12%	13%
- QoQ chg (%)		33%	-55%	53%	43%		9%	-57%	33%	17%				
Gross profit	**3,556**	**1,282**	**1,350**	**1,891**	**2,084**	**6,608**	**2,353**	**3,011**	**3,162**	**3,508**	**12,034**	**21,387**	**23,998**	**27,173**
- Gross margin	45.3%	43.0%	38.3%	41.5%	41.5%	41.1%	42.9%	43.2%	43.2%	43.0%	43.1%	44.7%	44.7%	44.9%
Operating expense	1,487	505	525	547	603	2,180	659	768	806	816	3,048	3,827	4,299	4,837
Operating profit	**2,070**	**777**	**825**	**1,344**	**1,482**	**4,428**	**1,694**	**2,244**	**2,356**	**2,692**	**8,986**	**17,560**	**19,699**	**22,336**
- Operating margin	26.3%	26.1%	23.4%	29.5%	29.5%	27.5%	30.9%	32.2%	32.2%	33.0%	32.1%	36.7%	36.7%	36.9%
Non-op profit	-8	97	22	17	14	150	15	17	18	15	66	103	138	200
Pre-tax profit	**2,061**	**874**	**848**	**1,361**	**1,495**	**4,578**	**1,710**	**2,261**	**2,374**	**2,707**	**9,052**	**17,663**	**19,837**	**22,537**
Tax	388	175	175	272	299	922	342	452	475	541	1,810	3,533	3,967	4,507
- Effective tax rate	18.8%	20.1%	20.7%	20.0%	20.0%	20.1%	20.0%	20.0%	20.0%	20.0%	20.0%	20.0%	20.0%	20.0%
Outstanding shares (m)	36	36	36	36	36	36	36	36	36	36	36	36	36	36
Net profit	**1,673**	**699**	**672**	**1,089**	**1,196**	**3,656**	**1,368**	**1,809**	**1,899**	**2,166**	**7,241**	**14,130**	**15,870**	**18,029**
- YoY chg (%)	41%	14%	228%	193%	147%	119%	96%	169%	74%	81%	98%	95%	12%	14%
- QoQ chg (%)		45%	-60%	56%	78%		14%	-51%	39%	20%				
Basic EPS (NT$)	**46.93**	**19.54**	**18.70**	**30.28**	**33.27**	**101.83**	**38.04**	**50.31**	**52.83**	**60.24**	**201.42**	**393.03**	**441.42**	**501.48**
- YoY chg (%)	37%	14%	225%	190%	145%	117%	95%	169%	74%	81%	98%	95%	12%	14%
- QoQ chg (%)		44%	-60%	55%	78%		14%	-51%	39%	20%				

Source: UBS estimates

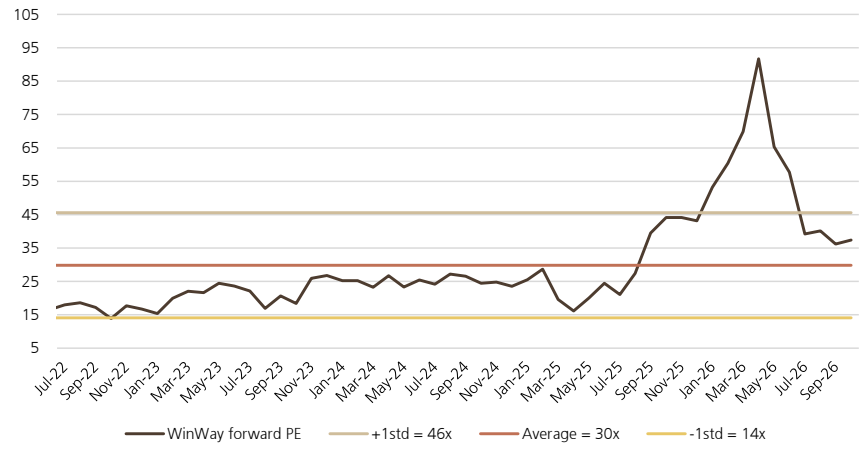
Figure 24: UBS estimates vs consensus

	UBS			Consensus			Difference		
(NT$m)	**2026E**	**2027E**	**2028E**	**2026F**	**2027F**	**2028F**	**2026E**	**2027E**	**2028E**
Revenue	**16,084**	**27,952**	**47,831**	**16,322**	**26,839**	**45,477**	**-1%**	**4%**	**5%**
- YoY chg (%)	105%	74%	71%	108%	64%	69%			
Gross profit	**6,608**	**12,034**	**21,387**	**6,825**	**12,136**	**20,622**	**-3%**	**-1%**	**4%**
- Gross margin	41.1%	43.1%	44.7%	41.8%	45.2%	45.3%			
Operating profit	**4,428**	**8,986**	**17,560**	**4,438**	**8,983**	**16,451**	**0%**	**0%**	**7%**
- Operating margin	27.5%	32.1%	36.7%	27.2%	33.5%	36.2%			
Pretax profit	**4,578**	**9,052**	**17,663**	**4,305**	**8,652**	**15,545**	**6%**	**5%**	**14%**
Net profit	**3,656**	**7,241**	**14,130**	**3,477**	**6,992**	**12,642**	**5%**	**4%**	**12%**
- Net margin	22.7%	25.9%	29.5%	21.3%	26.1%	27.8%			
Basic EPS (NT$)	**101.83**	**201.42**	**393.03**	**97.71**	**187.07**	**348.32**	**4%**	**8%**	**13%**
- YoY chg (%)	117%	98%	95%	108%	91%	86%			

Source: UBS estimates, LSEG

Our NT$8,900 price target is based on our valuation for WinWay at 30x average 2027E-28E P/E, in-line with its historical average of 30x and underpinned by a 30% 2027-31E earnings CAGR.

Figure 25: WinWay's 12-month forward PE (x)



Source: LSEG, UBS

Figure 26: Testing supply chain valuation table

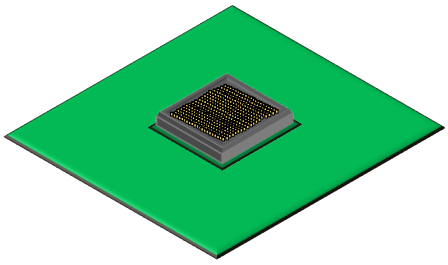
Company name	Ticker	Analyst	Market cap (US$m)	Price Target (LC)	Share price (LC)	Rating	EPS growth (%)		P/E (x)		P/BV (x)		ROE (%)	
							2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Testing service provider														
ASE Industrial	3711.TW	Sunny Lin	104,043	835	744	Buy	104	64	39.3	24.0	7.5	6.3	21	28
King Yuan Electronics	2449.TW	Sunny Lin	11,456	325	298	Buy	40	62	32.6	20.1	5.9	5.0	20	27
Amkor Technology	AMKR.O	Randy Abrams	13,114	80	53	Buy	73	8	20.6	19.1	2.6	2.4	14	13
JCET Group	600584.SS	Jimmy Yu	17,156	135	64	Buy	72	53	42.7	28.0	3.7	3.3	9	12
Tianshui Huatian Technology	002185.SZ	Jimmy Yu	7,928	13	16	Neutral	33	27	57.0	44.8	2.8	2.7	5	6
TongFu Microelectronics Co Ltd	002156.SZ	N/A	13,526	N/A	57	N/A	50	13	45.3	40.1	5.0	4.5	10	11
Shanghai V-Test Semiconductor Tech Co Ltd	688372.SS	N/A	2,517	N/A	100	N/A	36	106	40.6	19.7	3.8	3.2	11	18
Sector Average							**58**	**47**	**39.7**	**28.0**	**4.5**	**3.9**	**13**	**17**
Tester														
Advantest	6857.T	Atsuhiro Kinoshita	191,547	43,600	41,850	Buy	80	96	38.8	53.1	16.8	31.8	70	58
Teradyne Inc	TER.O	Timothy Arcuri	67,666	500	430	Buy	134	51	45.7	30.8	16.7	12.8	43	47
Sector Average							**107**	**74**	**42.2**	**42.0**	**16.7**	**22.3**	**57**	**52**
Prober/Prober solutions														
Hermes Testing Solutions	7856.TWO	Ryan Sun	3,580	8,880	4,190	Buy	221	487	143.9	27.3	9.1	7.0	10	29
Tokyo Electron	8035.T	Atsuhiro Kinoshita	184,031	18,720	12,805	Buy	-16	71	29.7	37.1	6.5	12.8	36	44
Sector Average							**102**	**279**	**86.8**	**32.2**	**7.8**	**9.9**	**23**	**36**
Handler														
Hon.Precision, Inc.	7769.TW	Sunny Lin	33,404	10,000	5,905	Buy	90	86	45.2	24.4	14.5	10.6	36	50
Chroma ATE	2360.TW	Jerry Su	31,434	2,960	2,350	Buy	64	38	51.8	37.7	28.6	20.8	58	64
Cohu, Inc.	COHU.O	N/A	3,460	N/A	73	N/A	n.a.	94	77.0	39.6	-	-	7	16
JHT Design Co Ltd	603061.SS	N/A	4,045	N/A	312	N/A	155	47	54.8	37.3	13.0	9.9	22	25
Hangzhou Changchuan Technology	300604.SZ	Yongwei Lai	22,690	310	240	Buy	60	53	76.1	49.8	23.4	16.5	36	39
Sector Average							**92**	**63**	**61.0**	**37.8**	**19.9**	**14.4**	**32**	**39**
Test Socket														
WinWay Technology	6515.TW	Ryan Sun	6,948	8,900	6,130	Buy	119	98	60.2	30.4	22.0	14.6	44	34
Leeno Industrial Inc	058470.KQ	N/A	4,936	N/A	86,900	N/A	33	17	35.4	30.3	8.0	6.8	24	24
ISC Co Ltd	095340.KQ	N/A	3,247	N/A	205,500	N/A	70	41	47.0	33.2	7.0	5.9	16	19
Yokowo Co Ltd	6800.T	N/A	744	N/A	4,925	N/A	28	93	38.3	19.8	-	1.8	-	-
Sector Average							**62**	**62**	**45.2**	**28.4**	**12.3**	**7.3**	**28**	**26**
Load Board														
Chunghwa Precision Test Tech	6510.TWO	Ryan Sun	3,149	5,000	3,055	Buy	112	99	47.5	23.8	9.2	7.3	21	34
Probe Card														
Technoprobe SpA	TPRO.MI	Harry Blaiklock, CFA	27,100	41	37	Buy	259	18	66.9	56.5	14.6	11.6	25	23
MPI Corp	6223.TWO	N/A	17,518	N/A	5,670	N/A	97	110	87.2	41.5	30.3	19.1	38	54
FormFactor Inc	FORM.O	N/A	11,038	N/A	141	N/A	159	24	46.2	37.1	9.3	8.9	19	23
MICRONICS JAPAN CO.,LTD.	6871.T	N/A	3,906	N/A	15,410	N/A	155	28	24.8	19.4	7.1	5.6	32	36
Japan Electronic Materials Corp	6855.T	N/A	804	N/A	8,650	N/A	46	43	24.9	17.4	-	-	-	-
Sector Average							**138**	**54**	**49.6**	**32.6**	**14.1**	**10.5**	**27**	**34**

Source: LSEG, UBS estimates. Priced as of October 6 (HKT).

Appendix: What are semiconductor test sockets?

Semiconductor test socket are electro-mechanical interface that are used post chip probing and chip packaging process to test the functionality of the chips. Sockets are placed onto load boards on top of testers during the final test/system level test processes. Handlers will place the chips onto the sockets, and pins inside the sockets will enable the testers to read the signals from the device under test (DUT, i.e. the packaged chips), with tester signals fan out from beneath the PCB load board.

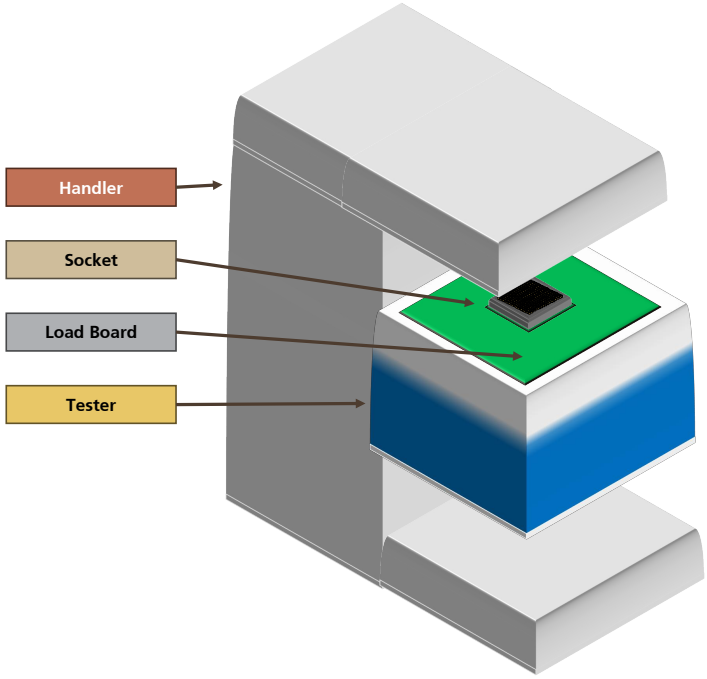
Figure 27: Socket illustration (with load board underneath)



Source: UBS

As AI accelerators become larger and more complex, socket requirements continue to increase. High-end sockets typically comprise a socket housing, alignment plate, integrated thermal-control components and contact elements that connect the DUT to the tester. Spring probes (pogo pins) are the most common choice for HPC sockets, with high durability, individual probe replacement flexibility, and consistent contact force. FT and SLT sockets account for more than 70% of the total socket market, with burn-in sockets representing the remainder.

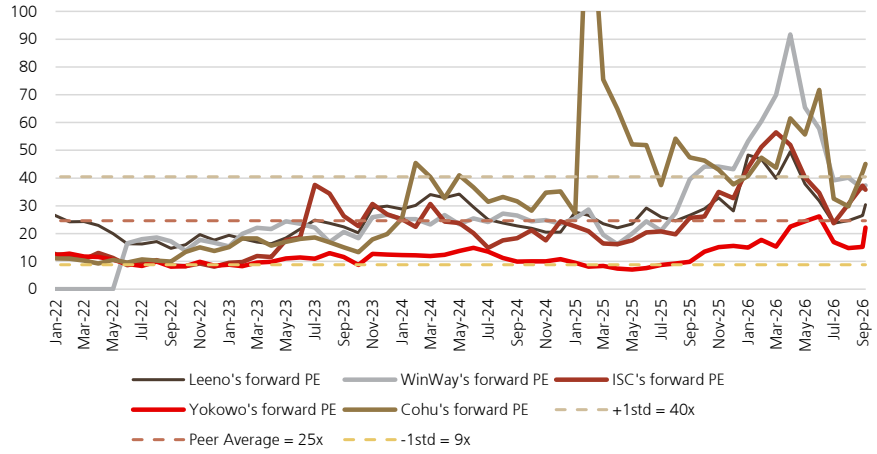
Figure 28: Final test equipment and consumables (simplified for illustration purposes, handlers should come with ATC)



Source: UBS

WHAT´S PRICED IN?

WinWay is trading at NT$6,130 as of 6 Oct 2026



Source: LSEG, UBS estimates

Despite a 120% YTD share-price increase, WinWay remains 43% below its April 2026 peak and trades at only 16x our 2028E P/E. We believe the market remains overly focused on near-term GM pressure while underestimating the company's multi-year growth opportunity. Our reverse DCF analysis indicates that the current market price implies only a 19% long-term sales CAGR vs our estimate of 25%, alongside long-term GM level of 32% vs our estimate of 45%.

We believe investors are not fully reflecting three key drivers:

1) CPU and TPU opportunities are becoming increasingly meaningful.

Beyond its core GPU socket franchise, WinWay is positioned to benefit from Nvidia's Vera CPU, AMD's Venice CPU, and Google's TPU. These programmes should broaden revenue sources and create an additional demand driver for high-end sockets and probe cards.

2) Larger AI packages are driving higher socket content per device.

As AI accelerators adopt larger form factors, higher pin counts and greater power consumption, socket complexity and ASPs continue to rise. We expect socket content growth to outpace semiconductor unit growth, creating a structural tailwind for WinWay.

3) Capacity ramp and margin recovery are not fully reflected.

Following substantial capacity expansion, we expect revenue growth of 105%/74%/71% in 2026E/27E/28E. At the same time, we forecast gross margin recovery from 38.3% in Q226 to 43.1%/44.7% in 2027E/28E as utilization and yields improve.

Our NT$8,900 price target is based on our valuation for WinWay at 30x average 2027E-28E P/E, in-line with its historical average of 30x and underpinned by a 30% 2027-31E earnings CAGR. Comparing with historical NTM peer average PE of roughly 25x (currently trading at 39x NTM PE), we believe WinWay deserves to trade at a premium given its leadership in AI/HPC sockets, expanding CPU and TPU exposure, and our long-term earnings CAGR estimate.

UPSIDE/DOWNSIDE SPECTRUM

WinWay is trading at NT $6,130 as of 6 Oct 2026

Value drivers (2027/28E)	Test socket sales growth	Probe card sales growth	GM %
NT$10,000 upside	90%/87%	87%/93%	45.1%/49.7%
NT$8,900 base	75%/69%	72%/78%	43.1%/44.7%
NT$4,000 downside	35%/34%	32%/38%	38.1%/39.7%

Source: UBS estimates

Risk to the current share price is skewed (1.8:1) to the upside

UPSIDE (NT$10,000): Our upside scenario reflects stronger than expected market share gain among Nvidia's Rubin sockets and Vera CPU probe card strength, with strong TPU v9 shipment in 2028E, forming 90%/87% YoY test socket sales growth and 87%/93% probe card sales growth in 2027/28E, translating to higher sales growth of 82%/75% in 2027/28E and steeper GM recovery in 2027/28E.

BASE (NT$8,900): Our price target of NT$8,900 is based on 30x average 2027-28E PE. We expect an earnings CAGR of 30% in 2027-31E. With extending FT time and larger form factor coaxial sockets, we see these growth drivers driving YoY sales growth from a high base of 105% in 2026E to 74%/71% YoY in 2027/2028E, with OPM of 32.1%/36.7% in 2027/28E.

DOWNSIDE (NT$4,000): Our downside scenario reflects slow GM recovery of 38.1%/39.7% in 2027/28E, with weaker growth for both socket and probe card businesses for slower capacity ramp up, leading to YoY sales growth of 22.3%/34.8% in 2027/28E, respectively.

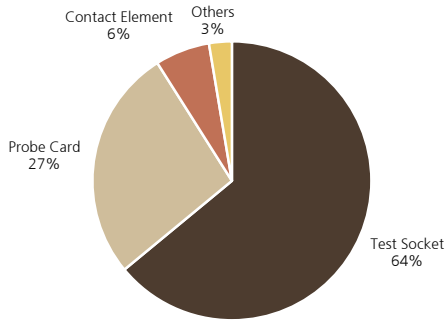
COMPANY DESCRIPTION

WinWay was founded in 2001 in Kaoshiung, Taiwan. It is a semiconductor testing interface solution provider, with final test sockets as its main business segment. The company has high-end solutions for AI/HPC testing, providing final test/system-level test and burn-in test sockets and probe cards through its own-made probes and a partnership with Technoprobe.

Industry Outlook

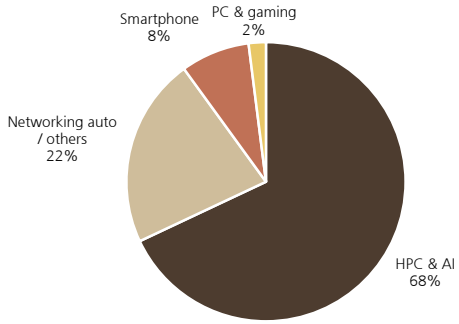
New iterations of semiconductors, especially for AI / HPC applications, are set to increase in both front-end wafer fabrication advancements and back-end packaging complexities with each iterations. With the industry now moves to utilize "shift left", chip probing and its relevant supply chain will be key beneficiaries of this trend which we believe will unlock multi-year, secular growth trends.

WinWay's H126 product mix



Source: Company data, UBS

WinWay's 4M26 revenue by end market



Source: Company data, UBS

WinWay Technology (6515.TW)

Income Statement (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**3,682**	**5,798**	**7,857**	**16,084**	*104.7*	**27,952**	*73.8*	**47,831**	**53,739**	**60,457**
Gross profit	1,365	2,534	3,556	6,608	*85.8*	12,034	*82.1*	21,387	23,998	27,173
EBITDA (UBS)	**684**	**1,605**	**2,328**	**4,753**	*104.2*	**9,531**	*100.5*	**18,478**	**21,047**	**24,147**
Depreciation & amortisation	(127)	(237)	(258)	(325)	*-25.9*	(546)	*-68.0*	(918)	(1,348)	(1,811)
EBIT (UBS)	**557**	**1,368**	**2,070**	**4,428**	*113.9*	**8,986**	*102.9*	**17,560**	**19,699**	**22,336**
Associates & investment income	0	0	0	0	-	0	-	0	0	0
Other non-operating income	(7)	57	(53)	98	-	0	-	0	0	0
Net interest	11	10	45	52	*14.6*	66	*27.6*	103	138	200
Exceptionals (incl goodwill)	0	0	0	0	-	0	-	0	0	0
Pre-tax profit	**561**	**1,435**	**2,061**	**4,578**	*122.1*	**9,052**	*97.7*	**17,663**	**19,837**	**22,537**
Tax	(97)	(249)	(388)	(922)	*-137.5*	(1,810)	*-96.3*	(3,533)	(3,967)	(4,507)
Profit after tax	**464**	**1,186**	**1,673**	**3,656**	*118.5*	**7,241**	*98.1*	**14,130**	**15,870**	**18,029**
Preference dividends	0	0	0	0	-	0	-	0	0	0
Minorities	0	0	0	0	-	0	-	0	0	0
Extraordinary items	0	0	0	0	-	0	-	0	0	0
Net earnings (local GAAP)	**464**	**1,186**	**1,673**	**3,656**	*118.5*	**7,241**	*98.1*	**14,130**	**15,870**	**18,029**
Net earnings (UBS)	**464**	**1,186**	**1,673**	**3,656**	*118.5*	**7,241**	*98.1*	**14,130**	**15,870**	**18,029**
Tax rate (%)	17.3	17.3	18.8	20.1	*6.9*	20.0	*-0.7*	20.0	20.0	20.0

Per Share (NT$)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	13.34	33.02	46.71	99.30	*112.6*	196.30	*97.7*	383.04	430.19	488.73
EPS (local GAAP, diluted)	13.34	33.02	46.71	99.30	*112.6*	196.30	*97.7*	383.04	430.19	488.73
EPS (UBS, basic)	13.52	34.31	46.93	101.83	*117.0*	201.42	*97.8*	393.03	441.42	501.48
DPS (net) (NT$)	10.87	25.75	25.07	61.10	*143.7*	120.85	*97.8*	235.82	264.85	300.89
Cash EPS (UBS, diluted) [1]	17.00	39.61	53.91	108.13	*100.6*	211.09	*95.2*	407.93	466.73	537.81
Book value per share	100.33	158.19	178.79	278.86	*56.0*	418.88	*50.2*	691.05	896.65	1,133.28
Average shares (diluted)	35	36	36	37	*2.8*	37	*0.2*	37	37	37

Balance Sheet (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	639	1,124	2,594	2,632	*1.5*	3,677	*39.7*	8,818	11,894	16,201
Other current assets	1,737	3,921	3,832	11,091	*189.4*	14,824	*33.7*	18,720	20,653	23,217
Total current assets	**2,377**	**5,045**	**6,426**	**13,724**	*113.6*	**18,501**	*34.8*	**27,537**	**32,547**	**39,418**
Net tangible fixed assets	2,109	2,256	2,206	2,991	*35.6*	4,401	*47.2*	6,831	9,245	11,667
Net intangible fixed assets	0	0	0	0	-	0	-	0	0	0
Investments / other assets	390	278	412	837	*103.2*	837	*0.0*	837	837	837
Total assets	**4,876**	**7,580**	**9,044**	**17,551**	*94.1*	**23,740**	*35.3*	**35,205**	**42,629**	**51,921**
Trade payables & other ST liabilities	979	1,812	2,524	5,968	*136.4*	7,109	*19.1*	8,790	8,822	9,606
Short term debt	0	0	0	0	-	0	-	0	0	0
Total current liabilities	**979**	**1,812**	**2,524**	**5,968**	*136.4*	**7,109**	*19.1*	**8,790**	**8,822**	**9,606**
Long term debt	337	147	0	1,477	-	1,477	*0.0*	1,477	1,477	1,477
Other long term liabilities	69	81	75	93	*25.2*	93	*0.0*	93	93	93
Preferred shares	0	0	0	0	-	0	-	0	0	0
Total liabilities (incl pref shares)	**1,386**	**2,040**	**2,599**	**7,539**	*190.1*	**8,680**	*15.1*	**10,361**	**10,393**	**11,177**
Common s/h equity	**3,489**	**5,540**	**6,445**	**10,011**	*55.3*	**15,059**	*50.4*	**24,845**	**32,236**	**40,744**
Minority interests	0	0	0	0	-	0	-	0	0	0
Total liabilities & equity	**4,876**	**7,580**	**9,044**	**17,551**	*94.1*	**23,740**	*35.3*	**35,205**	**42,629**	**51,921**

Cash Flow (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	464	1,186	1,673	3,656	*118.5*	7,241	*98.1*	14,130	15,870	18,029
Depreciation & amortisation	127	237	258	325	*25.9*	546	*68.0*	918	1,348	1,811
Net change in working capital	327	(784)	(44)	(2,212)	*NM*	(2,592)	*-17.2*	(2,215)	(1,901)	(1,780)
Other operating	(318)	411	(84)	443	-	0	-	0	0	0
Operating cash flow	**600**	**1,049**	**1,803**	**2,212**	*22.7*	**5,195**	*134.9*	**12,834**	**15,316**	**18,060**
Tangible capital expenditure	(856)	(298)	(277)	(1,287)	*-364.4*	(1,957)	*-52.1*	(3,348)	(3,762)	(4,232)
Intangible capital expenditure	0	0	0	0	-	0	-	0	0	0
Net (acquisitions) & disposals	(230)	(834)	1,009	(3,887)	-	0	-	0	0	0
Other investing	(9)	48	(10)	(5)	*53.3*	0	-	0	0	0
Investing cash flow	**(1,094)**	**(1,084)**	**722**	**(5,178)**	-	**(1,957)**	*62.2*	**(3,348)**	**(3,762)**	**(4,232)**
Equity dividends paid	(754)	(378)	(890)	(894)	*-0.4*	(2,193)	*-145.4*	(4,345)	(8,478)	(9,522)
Share issues / (buybacks)	0	0	0	0	-	0	-	0	0	0
Other financing	(14)	(15)	(17)	(11)	*33.4*	0	-	0	0	0
Change in debt & pref shares	400	909	(151)	3,896	-	0	-	0	0	0
Financing cash flow	**(369)**	**516**	**(1,057)**	**2,991**	-	**(2,193)**	-	**(4,345)**	**(8,478)**	**(9,522)**
Cash flow inc/(dec) in cash	**(863)**	**481**	**1,468**	**25**	*-98.3*	**1,045**	*NM*	**5,141**	**3,076**	**4,306**
FX / non cash items	(1)	4	2	14	*NM*	0	*-100.0*	0	0	0
Balance sheet inc/(dec) in cash	**(865)**	**485**	**1,470**	**39**	*-97.4*	**1,045**	*NM*	**5,141**	**3,076**	**4,306**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

WinWay Technology (6515.TW)

Valuation (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	51.2	30.5	32.9	61.7	31.2	16.0	14.2	12.5
P/E (UBS, diluted)	51.2	30.5	32.9	61.7	31.2	16.0	14.2	12.5
P/CEPS	39.6	24.4	28.4	55.3	28.3	14.6	12.8	11.1
Equity FCF (UBS) yield %	(1.1)	2.1	2.8	0.4	1.5	4.3	5.2	6.3
Dividend yield (net) %	1.6	2.6	1.6	1.0	2.0	3.8	4.3	4.9
P/BV	6.8	6.4	8.6	22.0	14.6	8.9	6.8	5.4
EV/revenues (core)	6.2	5.9	6.8	NM	7.8	4.5	3.9	3.4
EV/EBITDA (UBS core)	33.4	21.5	22.9	46.1	23.0	11.7	10.1	8.6
EV/EBIT (core)	41.0	25.2	25.8	49.5	24.4	12.3	10.8	9.3
EV/OpFCF (core)	41.0	25.2	25.8	49.5	24.4	12.3	10.8	9.3
EV/op. invested capital	8.5	8.9	12.7	34.9	20.5	14.4	10.9	8.8

Enterprise value (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	23,660	35,016	55,053	220,986	220,986	220,986	220,986	220,986
Net debt (cash)	(903)	(640)	(1,785)	(1,874)	(1,677)	(4,770)	(8,879)	(12,570)
Buy out of minorities	69	81	75	93	93	93	93	93
Pension provisions/other	0	0	0	0	0	0	0	0
Total enterprise value	**22,827**	**34,457**	**53,343**	**219,206**	**219,403**	**216,310**	**212,201**	**208,510**
Non core assets	0	0	(5)	(138)	(138)	(138)	(138)	(138)
Core enterprise value	**22,827**	**34,457**	**53,338**	**219,068**	**219,265**	**216,172**	**212,063**	**208,372**

Growth (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	(28.1)	57.5	35.5	104.7	73.8	71.1	12.4	12.5
EBITDA (UBS)	(53.3)	134.6	45.0	104.2	100.5	93.9	13.9	14.7
EBIT (UBS)	(58.5)	145.7	51.3	113.9	102.9	95.4	12.2	13.4
EPS (UBS, diluted)	(58.1)	147.5	41.5	112.6	97.7	95.1	12.3	13.6
Net DPS	(50.2)	136.9	(2.6)	143.7	97.8	95.1	12.3	13.6

Margins & Profitability (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	37.1	43.7	45.3	41.1	43.1	44.7	44.7	44.9
EBITDA margin	18.6	27.7	29.6	29.5	34.1	38.6	39.2	39.9
EBIT (UBS) margin	15.1	23.6	26.3	27.5	32.1	36.7	36.7	36.9
Net earnings (UBS) margin	12.6	20.5	21.3	22.7	25.9	29.5	29.5	29.8
ROIC (EBIT)	20.7	35.3	49.2	70.5	NM	NM	NM	NM
ROIC post tax	17.1	29.2	39.9	56.3	67.1	NM	NM	NM
ROE (UBS)	12.9	26.3	27.9	44.4	57.8	70.8	55.6	49.4

Capital structure & Coverage (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	(0.4)	(0.6)	(1.1)	(0.2)	(0.2)	(0.4)	(0.5)	(0.6)
Net debt / total equity %	(8.7)	(17.6)	(40.2)	(11.5)	(14.6)	(29.5)	(32.3)	(36.1)
Net debt / (net debt + total equity) %	(9.5)	(21.4)	(67.3)	(13.0)	(17.1)	(41.9)	(47.7)	(56.6)
Net debt/EV %	(4.0)	(1.9)	(3.3)	(0.9)	(0.8)	(2.2)	(4.2)	(6.0)
Capex / depreciation %	NM	126.1	107.4	NM	NM	NM	NM	NM
Capex / revenue %	23.2	5.1	3.5	8.0	7.0	7.0	7.0	7.0
EBIT / net interest	-	-	-	-	-	-	-	-
Dividend cover (UBS)	1.2	1.3	1.9	1.7	1.7	1.7	1.7	1.7
Div. payout ratio (UBS) %	80.4	75.1	53.4	60.0	60.0	60.0	60.0	60.0

Revenues by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	3,682	5,798	7,857	16,084	27,952	47,831	53,739	60,457
Total	**3,682**	**5,798**	**7,857**	**16,084**	**27,952**	**47,831**	**53,739**	**60,457**

EBIT (UBS) by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	557	1,368	2,070	4,428	8,986	17,560	19,699	22,336
Total	**557**	**1,368**	**2,070**	**4,428**	**8,986**	**17,560**	**19,699**	**22,336**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	45.2%
Forecast dividend yield	1.0%
Forecast stock return	46.2%
Market return assumption	6.3%
Forecast excess return	39.9%

Company Description

WinWay was founded in 2001 in Kaoshiung, Taiwan. It is a semiconductor testing interface solution provider, with final test sockets as its main business segment. The company has high-end solutions for AI/HPC testing, providing final test/system-level test and burn-in test sockets and probe cards through its own-made probes and a partnership with Technoprobe.

Valuation Method and Risk Statement

We value WinWay based on PE methodology.
Risks for the semis industry:
- Geopolitical uncertainties: Geopolitical uncertainties including tariffs and more export control could negatively impact semis cycle, and a downturn would lead to slower demand in the overall market and moderating capacity expansion.
- Meaningful deceleration of new technology development: Slower demand and less investment are expected If there is meaningful deceleration of new technology development such as Cloud AI, autonomous driving, etc.
- Rising utility cost: Larger-than-expected utility cost increases would further pressure profitability. Rising utility costs in Taiwan have negatively impacted margins for Taiwan tech companies.
Company risks:
- Faster- or slower-than-expected CoWoS capacity expansion: This could result in WinWay gaining/losing socket/probe card orders from major end customers, foundries, and OSATs.
- Competitors achieving technological advancement: Potential risk of competition (from either local or foreign companies) achieving technological advancements could result in WinWay losing market share from major end customers, foundries and OSATs.
- Better- or worse-than-expected AI growth: Better- or worse-than-expected AI growth could lead to a ramp-up or a slowdown of equipment orders from major foundries and OSATs to WinWay, affecting long-term revenue growth potential.
- Slowdown or delay of new technological advancement: A potential slowdown or delay of the rollout for technological advancements for technologies such as SoIC, CoPoS, and CPO, would limit future socket/probe card orders.
- Increase of raw material costs: Rising raw material costs may directly impact WinWay's margins for socket/probe card shipments.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

WinWay Technology

Question	Response
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	4
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	3
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	3
5. What's driving the difference?	
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	3
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	No Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

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12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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UBS Securities Pte. Ltd., Taipei Branch: Ryan Sun, Sunny Lin.

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Company Name	Reuters	12-month rating	Price	Price date
WinWay Technology	6515.TW	No Rating	NT$5,520	07 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

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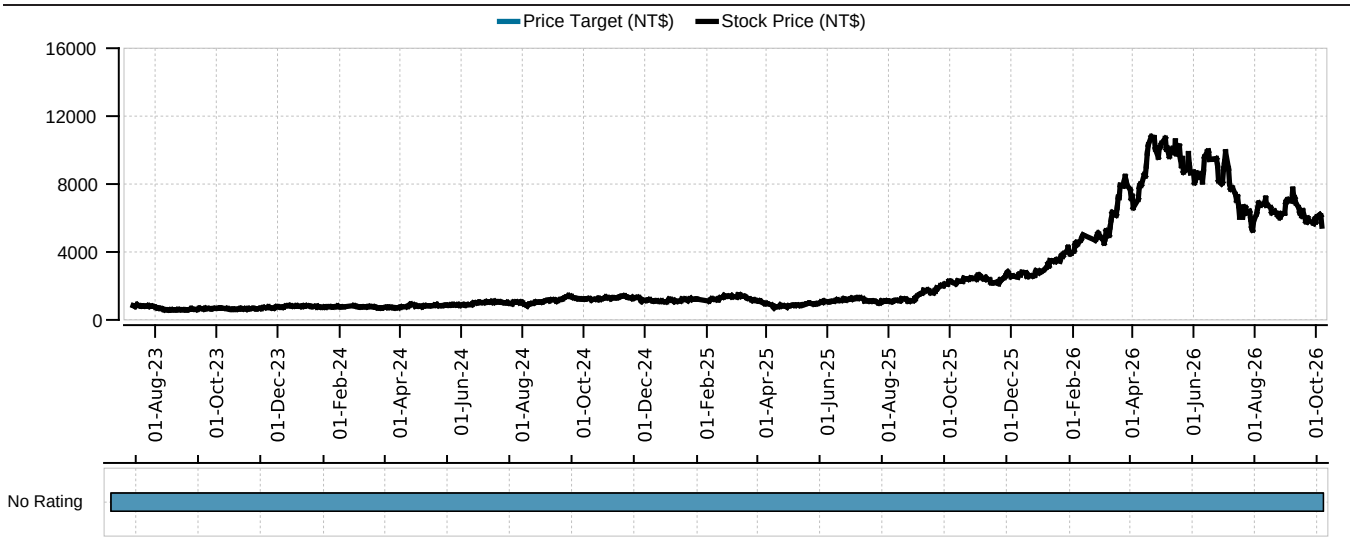
WinWay Technology (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-07	824.00	-	No Rating

Source: UBS Global Research; LSEG Eikon as of 07-Oct-2026. All prices as of local market close. Ratings as of date shown.

Additional Prices: ASE Industrial, NT$732.00 (07 Oct 2026); Alphabet Inc. - Class A, US$347.68 (06 Oct 2026); Taiwan Semiconductor Manufacturing, NT$2585.00 (07 Oct 2026); NVIDIA Corp, US$239.24 (06 Oct 2026); Advanced Micro Devices Inc, US$649.42 (06 Oct 2026); Source: UBS. All prices as of local market close.

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Daiwa Capital Markets Hong Kong Limited

Acter (5536 TT)

Target price: **TWD1,415.00**

Share price (7 Oct): **TWD1,045.00** | Up/downside: **+35.4%**

5 4 3 2 **1**

★ **Buy** (initiation)

Initiation: acting on the demand for cleanrooms

- Key beneficiary of the cleanroom shortage in the high-tech industry
- Fast-growing order backlog likely to be realised over 2026-28E
- Initiating with a Buy (1) rating and a 12-month TP of TWD1,415

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Investment case: We initiate coverage of Acter, an established multinational cleanroom engineering, procurement, construction and management (EPCM) turnkey service provider, with a Buy (1) rating. Its core business includes cleanroom electromechanical engineering, water/gas/chemical delivery systems, and other cleanroom services. We believe now is a good time to revisit the stock as it underperformed the TAIEX Index by 32.7% in 3Q26, mainly on weaker-than-expected 2Q26 results and soft sector sentiment. Given its stronger revenue momentum over recent months and upcoming TSMC results on 15 October, which may drive sentiment, we suggest investors to buy the stock. We forecast a 24.6% revenue CAGR and 36.9% net profit CAGR over 2026-28E.

Intact global demand for cleanroom technology, the edge AI market, and rapid cleanroom expansion in Asia further support Acter's order backlog. Its backlog reached TWD66bn in July 2026 vs. TWD46bn in 2025 and TWD30bn in 2022 (15.3% 2022-25 CAGR), which largely outpaced the global cleanroom technology market's forecast CAGR over 2023-2032E, according to Precedence Research.

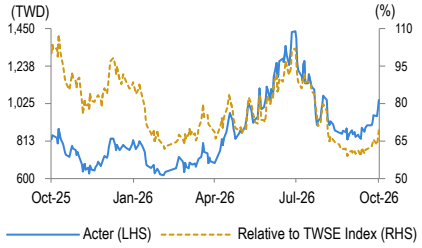
Strong competitive edge supports order share gains across multiple segments. Acter's competitive edge lies in 1) intelligent management enabled by digital twins and AI, using 3D building information modelling (BIM) and AI-driven digital twin management; 2) intellectualisation of green engineering, which comprises green design, green procurement, and green techniques; 3) experience across multiple industries (ie, data centres and electromechanical engineering for the consumer industry) which supports better integration; and 4) a complete network, enabling on-the-spot support. Acter believes it has the highest market share among the operators of three major US data centre players due largely to its capabilities in liquid cooling and its expertise in electromechanical and cleanroom engineering.

Catalysts: Near-term: project wins from key clients, supporting stronger monthly revenue momentum; mid- to long-term: an improving operating-margin trend and a resilient order backlog.

Valuation: We are the first non-domestic broker to cover Acter. We initiate coverage with a 12-month TP of TWD1,415, based on a target PER of 20x (equivalent to a 0.54x PEG over 2026-28E). We believe our valuation is fair, given Acter's strong earnings visibility, which we believe warrants a rerating. Our TP is supported by a DCF valuation of TWD1,454.

Risks: 1) weaker-than-expected expansion in the semiconductor industry; and 2) project delays or cancellations.

Share price performance



12-month range	618.00-1,435.00
Market cap (USDbn)	4.08
3m avg daily turnover (USDm)	31.93
Shares outstanding (m)	124
Major shareholder	Xiang-Hui Development (4.4%)

Financial summary (TWD)

Year to 31 Dec	26E	27E	28E
Revenue (m)	53,844	66,983	83,593
Operating profit (m)	9,853	13,598	18,014
Net profit (m)	6,325	8,797	11,851
Core EPS (fully-diluted)	50.788	70.635	95.161
EPS change (%)	79.3	39.1	34.7
Daiwa vs Cons. EPS (%)	(6.4)	(5.4)	n.a.
PER (x)	20.6	14.8	11.0
Dividend yield (%)	3.4	4.7	6.4
DPS	35.7	49.6	66.9
PBR (x)	9.3	7.2	5.5
EV/EBITDA (x)	11.6	8.3	6.1
ROE (%)	46.8	54.9	56.8

Source: FactSet, Daiwa forecasts

See important disclosures, including any required research certifications, beginning on page 25

Daiwa Capital Markets Hong Kong Limited

5 4 3 2 **1**

Buy (initiation)

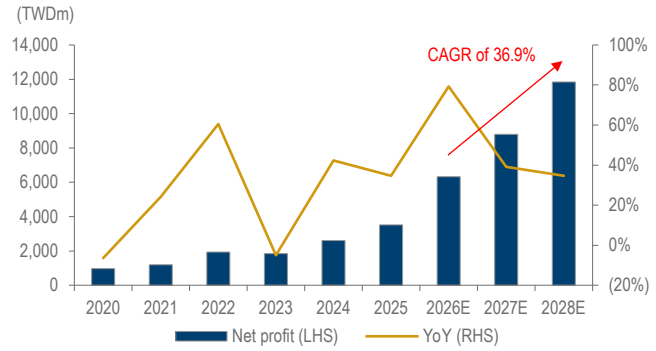
How do we justify our view?

Growth outlook	Valuation	Earnings revisions
✔ ✔ ✔ ✔ ✔	✔ ✔ ✔ ✔ ✔	✔ ✔ ✔ ✔ ✔

Growth outlook

We forecast Acter to record a 36.9% net profit CAGR over 2026-28E, on the back of a 24.6% revenue CAGR over the same period, driven by 1) larger project scale, as strong demand for cleanroom EPCM services from the high-tech sector is likely to support continued growth in Acter's order backlog and 2) improving gross margins and operating efficiency. We expect operating margin to expand by 2.0pp in 2027E and 1.3pp in 2028E, driven by favourable gross margins on new projects and operating leverage.

Acter: net profit and net-profit growth



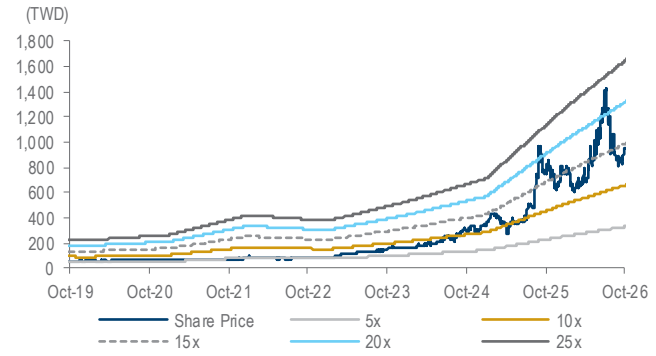
Source: Company, Daiwa forecasts

Valuation

Our 12-month TP of TWD1,415 is based on a target PER of 20x, applied to our 1-year-forward EPS forecast. Our target multiple of 20x is higher than the stock's past-3-year 1-year-forward PER (12.2x) and past-5-year 1-year-forward PER (9.6x) and implies a PEG of 0.54x on our 36.9% net profit CAGR forecast over 2026-28E. We believe the stock deserves a rerating in the next 12 months, largely supported by resilient cleanroom demand globally.

Our TP is also supported by an alternative DCF-derived valuation of TWD1,454/share.

Acter: 1-year-forward PER bands

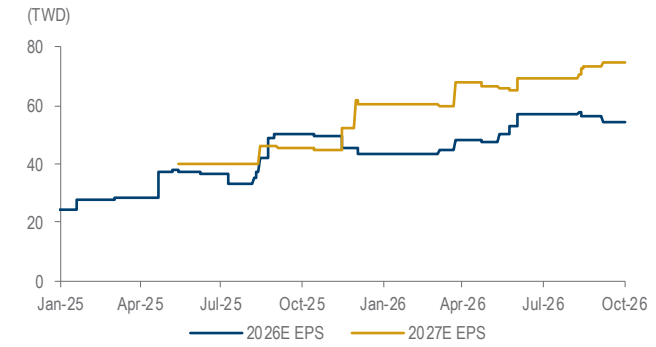


Source: Bloomberg, Daiwa forecasts

Earnings revisions

Bloomberg consensus EPS estimates for 2026E and 2027E have risen 24.8% and 24.3% YTD, respectively. This is likely due to upward revisions to order backlog and earnings forecasts following better-than-expected industry outlook. We are the first foreign broker to cover Acter.

Acter: Bloomberg consensus EPS forecast revisions



Source: Bloomberg

Financial summary

Key assumptions

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cleanroom eletromechanical revenue growth (%)	36.9	44.1	(19.0)	30.6	70.7	18.0	15.1	15.8
Water, gas and chemical delivery systems revenue growth (%)	58.5	35.0	11.0	14.8	(16.8)	67.7	46.7	40.4
Consumer industry electromechanical revenue growth (%)	55.2	32.4	(7.3)	5.7	(4.7)	78.7	38.8	33.1
Biomedical revenue growth (%)	(10.8)	83.7	(27.6)	(38.5)	68.0	52.5	8.3	33.1
High-tech equipment and materials sales and services revenue growth (%)	143.9	2.5	(54.0)	14.7	98.0	(44.9)	(5.5)	0.0

Profit and loss (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cleanroom electromechanical	11,279	16,256	13,173	17,200	29,359	34,641	39,859	46,141
Water, gas and chemical delivery systems	5,915	7,985	8,861	10,174	8,467	14,198	20,829	29,253
Other Revenue	3,024	4,021	3,026	2,879	3,656	5,005	6,296	8,198
Total Revenue	**20,217**	**28,262**	**25,061**	**30,254**	**41,482**	**53,844**	**66,983**	**83,593**
Other income	0	0	0	0	0	0	0	0
COGS	(16,934)	(23,484)	(20,143)	(23,676)	(33,634)	(41,190)	(50,238)	(61,859)
SG&A	(892)	(1,121)	(1,292)	(1,616)	(1,718)	(2,369)	(2,646)	(3,093)
Other op.expenses	(252)	(335)	(362)	(555)	(427)	(431)	(502)	(627)
Operating profit	**2,139**	**3,323**	**3,264**	**4,407**	**5,703**	**9,853**	**13,598**	**18,014**
Net-interest inc./(exp.)	9	7	95	160	164	227	50	50
Assoc/forex/extraord./others	30	90	157	290	195	92	42	42
Pre-tax profit	**2,179**	**3,420**	**3,516**	**4,857**	**6,062**	**10,172**	**13,690**	**18,106**
Tax	(625)	(921)	(916)	(1,269)	(1,458)	(2,447)	(3,293)	(4,355)
Min. int./pref. div./others	(349)	(566)	(761)	(971)	(1,077)	(1,400)	(1,600)	(1,900)
Net profit (reported)	**1,204**	**1,933**	**1,838**	**2,617**	**3,527**	**6,325**	**8,797**	**11,851**
Net profit (adjusted)	**1,204**	**1,933**	**1,838**	**2,617**	**3,527**	**6,325**	**8,797**	**11,851**
EPS (reported)(TWD)	**21.079**	**16.838**	**15.027**	**21.093**	**28.424**	**50.976**	**70.897**	**95.513**
EPS (adjusted)(TWD)	**21.079**	**16.838**	**15.027**	**21.093**	**28.424**	**50.976**	**70.897**	**95.513**
EPS (adjusted fully-diluted)(TWD)	**19.548**	**15.477**	**14.585**	**20.982**	**28.325**	**50.788**	**70.635**	**95.161**
DPS (TWD)	**15.034**	**8.823**	**6.087**	**10.000**	**14.500**	**35.683**	**49.628**	**66.859**
EBIT	**2,139**	**3,323**	**3,264**	**4,407**	**5,703**	**9,853**	**13,598**	**18,014**
EBITDA	**2,292**	**3,556**	**3,595**	**4,893**	**6,097**	**10,482**	**14,182**	**18,610**

Cash flow (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Profit before tax	2,179	3,420	3,516	4,857	6,062	10,172	13,690	18,106
Depreciation and amortisation	123	144	174	196	199	537	542	554
Tax paid	(490)	(638)	(943)	(804)	(1,249)	(2,447)	(3,293)	(4,355)
Change in working capital	(1,809)	(1,544)	2,386	(2,536)	421	(2,122)	(1,006)	(1,946)
Other operational CF items	(82)	304	(775)	530	368	(227)	(50)	(50)
Cash flow from operations	**(80)**	**1,686**	**4,358**	**2,244**	**5,801**	**5,913**	**9,883**	**12,309**
Capex	(24)	(147)	(137)	(151)	(128)	(150)	(180)	(180)
Net (acquisitions)/disposals	(116)	(413)	(285)	(194)	(120)	(122)	(122)	(122)
Other investing CF items	(36)	22	4	21	35	0	0	0
Cash flow from investing	**(176)**	**(537)**	**(418)**	**(324)**	**(213)**	**(272)**	**(302)**	**(302)**
Change in debt	(231)	470	197	(80)	115	0	0	0
Net share issues/(repurchases)	0	0	0	0	0	0	0	0
Dividends paid	(686)	(859)	(1,416)	(1,303)	(1,861)	(1,799)	(4,428)	(6,158)
Other financing CF items	(211)	1,883	(55)	127	(721)	0	0	0
Cash flow from financing	**(1,128)**	**1,494**	**(1,274)**	**(1,256)**	**(2,467)**	**(1,799)**	**(4,428)**	**(6,158)**
Forex effect/others	0	0	0	0	0	0	0	0
Change in cash	**(1,384)**	**2,643**	**2,665**	**664**	**3,120**	**3,841**	**5,153**	**5,849**
Free cash flow	(31)	1,228	4,900	1,403	5,141	5,763	9,703	12,129

Source: FactSet, Daiwa forecasts

Financial summary continued …

Balance sheet (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & short-term investment	3,699	6,411	8,951	9,899	12,996	16,838	21,991	27,839
Inventory	1,560	2,904	2,615	2,302	1,862	2,907	2,848	3,612
Accounts receivable	6,027	8,416	6,619	9,150	11,517	15,574	18,958	23,919
Other current assets	5,971	7,405	6,419	8,614	9,972	1,833	1,833	1,833
Total current assets	**17,258**	**25,136**	**24,604**	**29,964**	**36,346**	**37,153**	**45,630**	**57,204**
Fixed assets	573	690	768	833	890	4,464	4,101	3,728
Goodwill & intangibles	108	92	75	81	71	71	71	71
Other non-current assets	765	1,199	2,139	3,199	4,012	4,012	4,012	4,012
Total assets	**18,704**	**27,117**	**27,587**	**34,078**	**41,320**	**45,700**	**53,814**	**65,015**
Short-term debt	651	896	541	524	591	591	591	591
Accounts payable	6,568	8,708	7,727	9,454	11,676	14,657	16,975	20,754
Other current liabilities	2,554	4,893	4,977	6,160	8,199	7,403	7,403	7,403
Total current liabilities	**9,773**	**14,497**	**13,245**	**16,138**	**20,467**	**22,652**	**24,970**	**28,749**
Long-term debt	0	0	0	3	3	3	3	3
Other non-current liabilities	1,635	1,820	1,257	1,211	1,433	1,433	1,433	1,433
Total liabilities	**11,408**	**16,318**	**14,502**	**17,351**	**21,902**	**24,088**	**26,405**	**30,184**
Share capital	572	586	620	620	620	1,241	1,241	1,241
Reserves/R.E./others	5,063	6,998	8,191	10,350	12,499	12,673	16,870	22,391
Shareholders' equity	**5,635**	**7,585**	**8,812**	**10,970**	**13,119**	**13,913**	**18,110**	**23,632**
Minority interests	1,661	3,214	4,273	5,756	6,299	7,699	9,299	11,199
Total equity & liabilities	**18,704**	**27,117**	**27,587**	**34,078**	**41,320**	**45,700**	**53,814**	**65,015**
EV	128,278	127,364	125,527	126,048	123,561	121,120	117,566	113,618
Net debt/(cash)	**(3,048)**	**(5,514)**	**(8,410)**	**(9,373)**	**(12,402)**	**(16,244)**	**(21,397)**	**(27,245)**
BVPS (TWD)	**98.613**	**66.063**	**72.041**	**88.411**	**105.729**	**112.132**	**145.957**	**190.453**

Key ratios (%)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Sales (YoY)	44.6	39.8	(11.3)	20.7	37.1	29.8	24.4	24.8
EBITDA (YoY)	30.5	55.2	1.1	36.1	24.6	71.9	35.3	31.2
Operating profit (YoY)	25.8	55.3	(1.8)	35.0	29.4	72.8	38.0	32.5
Net profit (YoY)	24.2	60.5	(4.9)	42.4	34.8	79.3	39.1	34.7
Core EPS (fully-diluted) (YoY)	18.1	(20.8)	(5.8)	43.9	35.0	79.3	39.1	34.7
Gross-profit margin	16.2	16.9	19.6	21.7	18.9	23.5	25.0	26.0
EBITDA margin	11.3	12.6	14.3	16.2	14.7	19.5	21.2	22.3
Operating-profit margin	10.6	11.8	13.0	14.6	13.7	18.3	20.3	21.6
Net profit margin	6.0	6.8	7.3	8.7	8.5	11.7	13.1	14.2
ROAE	22.3	29.2	22.4	26.5	29.3	46.8	54.9	56.8
ROAA	7.1	8.4	6.7	8.5	9.4	14.5	17.7	19.9
ROCE	28.7	33.8	25.8	28.5	30.6	46.7	54.2	56.8
ROIC	50.5	50.9	48.5	54.1	60.3	n.a	n.a	n.a
Net debt to equity	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Effective tax rate	28.7	26.9	26.1	26.1	24.1	24.1	24.1	24.1
Accounts receivable (days)	91.7	93.3	109.5	95.1	90.9	91.8	94.1	93.6
Current ratio (x)	1.8	1.7	1.9	1.9	1.8	1.6	1.8	2.0
Net interest cover (x)	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Net dividend payout	71.3	52.4	40.5	47.4	51.0	70.0	70.0	70.0
Free cash flow yield	n.a.	0.9	3.8	1.1	4.0	4.4	7.5	9.4

Source: FactSet, Daiwa forecasts

Company profile

Established in 1979, Acter Group (Acter) is a leading engineering services provider specialising in cleanroom, electromechanical, and high-tech facility integration solutions for semiconductor, optoelectronics, biotech, green energy, and advanced manufacturing clients across Taiwan, China and the rest of the world. As of 2025, its revenue comprised cleanroom electromechanical (71%), water, gas and chemical delivery systems (20%), consumer industry electromechanical (4%), biomedical (2%) and high-tech equipment & materials sales & services (3%).

Acting on the demand for cleanrooms

Why are cleanrooms a bottleneck in the high-tech industry?

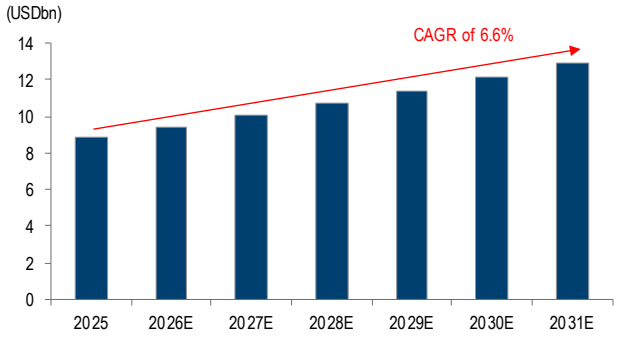
In recent years, cleanrooms have become a critical bottleneck for high-tech industries, especially semiconductor manufacturing, OSAT, AI hardware, and PCB, mainly because of its indispensable nature, severe supply-demand imbalance, labour shortages, and high technical and regulatory barriers.

Cleanrooms are a bottleneck in the high-tech industry, including semiconductor foundry, memory, and electronic assembling

Cleanrooms are an indispensable core infrastructure as they determine product yield and quality stability. Advanced manufacturing nodes in semiconductors, optoelectronics, and biopharmaceuticals require highly particle-free, constant-temperature, and constant-humidity environments to ensure reliable production. As process technologies continue to advance, such as 2nm node (or even newer nodes) and advanced packaging, existing cleanrooms must be rebuilt, and the complexity of building new ones thus increases. This keeps overall demand for cleanrooms at high levels.

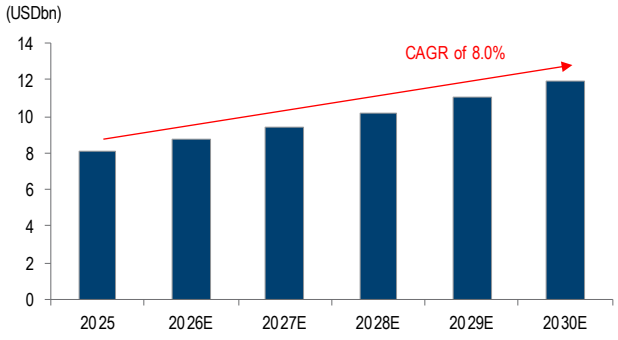
A severe global supply-demand imbalance with demand for cleanroom construction outpacing supply. The surge in AI servers, high-performance computing (HPC), and high-bandwidth memory (HBM) has triggered a global expansion for front-end wafer fabs, packaging and testing facilities, and PCB manufacturing. Order visibility remains extremely strong across the industry. According to Markets and Markets, the global cleanroom market size is set to expand at a CAGR of 6.6% over 2025-31E. At the same time, the global semiconductor cleanroom market size is set to grow at a CAGR of 8.0% over 2025-30E. For instance, Acter's order backlog exceeded TWD66bn at the end of July 2026, up from TWD46bn at the end of 2025, showing the urgency of demand for cleanroom construction.

Global: cleanroom market size and growth



Source: Markets and Markets

Global: semiconductor cleanroom market size and growth



Source: Markets and Markets

Human resources are an additional limitation to capacity growth. Cleanroom engineering is highly technical and labour-intensive, relying on specialised on-site installation and supervision that cannot be replaced by robots. At the same time, the industry faces severe labour shortages and an aging workforce, and younger generations are reluctant to join field operations. Cross-border staff deployment is also less cost-efficient than local hiring.

Cleanrooms involve complex system integration, creating high barriers to entry. A cleanroom construction includes 4 sequential phases, where each phase usually requires totally different areas of expertise. Therefore, the competitive landscape in each area differs. The process begins with **civil & structural (C&S) works** to establish the physical plant, followed by **mechanical, electrical, and plumbing (MEP) works** for essential utility generation, where Acter is one of the leading global experts. After MEP is finished, chemicals/gas/water are then distributed via **main systems** through primary piping network and finally delivered to production equipment through **hook-up systems**. As

global supply chains restructure under N+1 (domestic market + 1) strategies, companies building overseas facilities in places like the US face strict and complicated licensing processes, particularly on fire safety and hazardous chemicals. Without experience dealing with such engineering and safety issues, construction timelines can be severely delayed.

As a result, even with advanced equipment and chip designs, a high-tech company cannot begin production without adequate cleanroom space, skilled engineers, and system-integration capabilities. Therefore, cleanroom engineering contractors hold significant bargaining power and capacity allocation leverage in today's tech supply chain.

IC foundry

In the foundry sector, where the cleanroom supply-demand shortage is the tightest, project activity remains centred on advanced process capacity expansion in Taiwan and Arizona. Simultaneously, domestic advanced node development remains intense, with multi-phase fab builds advancing across Hsinchu, Taichung, and Kaohsiung. Local cleanroom and MEP integration demand is further boosted by advanced packaging facilities in Taiwan (such as Chiayi and Tainan) and upcoming projects in the US are set to contribute revenue for the cleanroom providers in the following years.

Memory

The memory segment is exhibiting robust momentum as well, mainly catalysed by HBM capacity additions. Projects such as the plant in Japan, transitioned to contribution for the cleanroom EPCM providers starting from 2026. In Taiwan, turnkey engineering contracts in Miaoli are also actively underway. This trend is reinforced by stronger-than-expected expansion activity from memory producers in China.

OSAT, PCB and ODM/OEM

For OSAT companies, demand is also surging due to the spill-over effect of advanced packaging protocols and N+1 supply chain restructuring. The OSAT names are expanding their footprint both in Southern Taiwan and in overseas locations including in the US and Southeast Asia. Major engineering contracts signed between OSAT clients and Acter show stable long-term workload for strong cleanroom demand over our forecast horizon. Similarly, overseas expansion for PCB manufacturing is peaking in Southeast Asia, especially in Thailand. We expect Chinese PCB players to also increase their capacity, just like memory companies in the past year. Except for front-end and back-end chip manufacturing, ODM/OEM and data centre infrastructure is also a key growth engine for the EPCM industry. Major Taiwanese ODMs are building production facilities in Texas and Mexico to meet local manufacturing needs. Cleanroom contractors, such as Acter, providing MEP and system integration in Texas are directly benefiting from these projects. The shift toward liquid cooling systems has also increased engineering complexity, favouring few cleanroom companies with the expertise to deliver complex integration for CSPs in both North America and Southeast Asia.

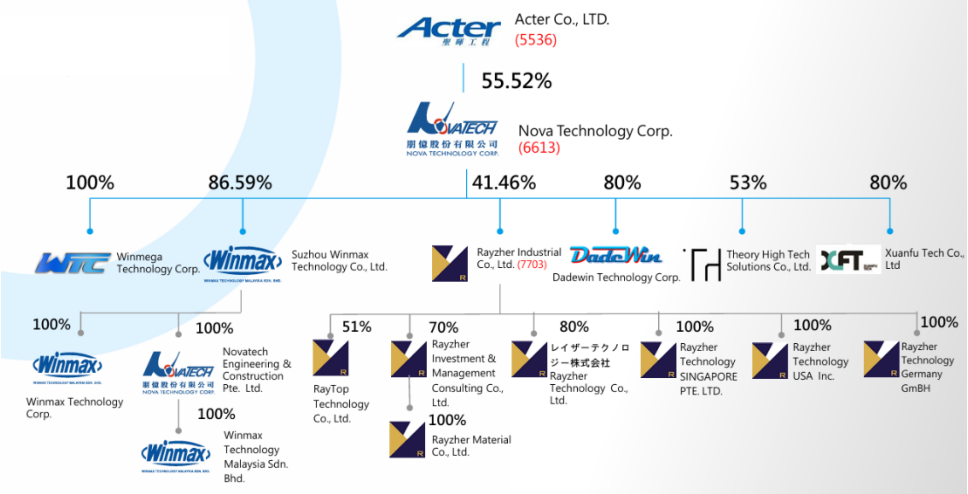
Acter: an integrated cleanroom solutions provider

Acter provides integrated cleanroom solutions via various subsidiaries

Acter Group (Acter), an integrated cleanroom solutions provider, maintains its competitive positioning within the high-tech cleanroom and MEP engineering sector, driven by its group synergy and global footprint. Acter operates as a comprehensive vendor through a group business model. The parent company, Acter, supervises overall cleanroom and MEP services, while subsidiary Nova Technology (Novatech, 6613 TT, not rated) mainly provides chemical primary system and hook-up, and Novatech's subsidiary Rayzher (7703 TT, not rated) provides gas primary system, hook-up and equipment. In China, Acter operates through Acter Technology Integration Group (603163 CH, not rated) and Suzhou Winmax (not listed) for Chinese EPCM (engineering, procurement, construction, and management) and equipment business. Except for C&S (civil and structural) works, where the margin is low (7-8%) and the contract duration is long, Acter's services span nearly the entire cleanroom construction process. Its one-stop solutions model supports customer retention

and improves bid-win rates, particularly for overseas expansion, where customers have few reliable providers to choose from.

Acter: organisational chart



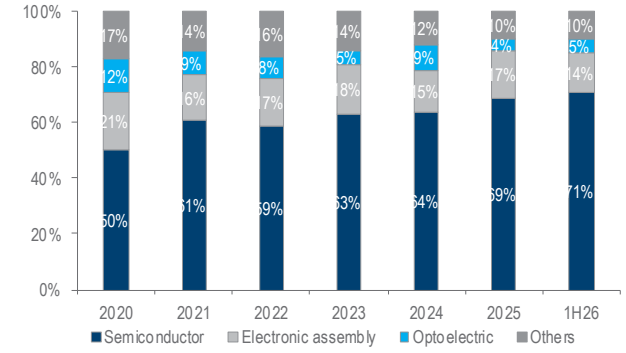
Source: Company, Daiwa

To avoid the risk of over-concentration, Acter diversifies its client portfolio across semiconductor foundries, memory, PCB, OSAT, ODM hardware assembly, CSP data centres and the consumer industry. By sector, semiconductor projects represent the largest share of Acter's portfolio, accounting for 66% of total revenue in 1H26, vs. 69% in 2025. Foundry manufacturing is the largest source of demand for cleanrooms in Taiwan, especially as advanced-node capacity expands at sites in Hsinchu, Taichung and Kaohsiung. In the memory segment, the company executes turnkey engineering builds to expand its capacity aggressively, creating cleanroom opportunities across Taiwan, Japan and Singapore. This multi-sector coverage decreases the risk of cyclical downturns in any individual segment, in our view.

Acter provides cleanroom services for numerous industries, across multiple regions

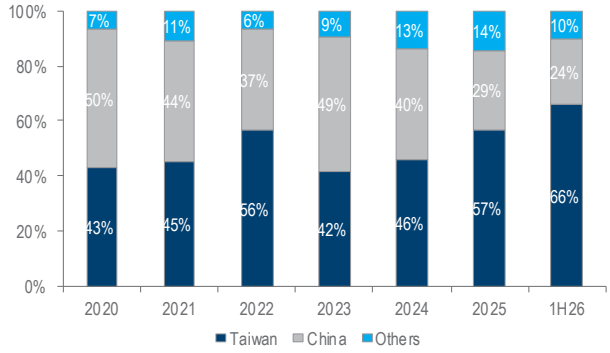
By establishing 33 operating entities across 11 international regions, the company effectively captures capex migration driven by global N+1 supply chain diversification. This allows Acter to support Taiwanese and international clients as they expand manufacturing capacity from Taiwan into Southeast Asia, Japan and the US. Due to the complexity of advanced-node IC manufacturing, packaging and testing, it is especially important for these Taiwanese semiconductor companies to work with familiar cleanroom builders when expanding in the US. This is one of the reasons why Taiwanese cleanroom companies are receiving overseas orders and starting to set up global bases, in our view. Despite higher management, material, and manpower costs in the US, most cleanroom companies are benefitting from the trend by improving margins, given the higher complexity and risks to operate overseas. Across overseas hubs in Texas and Phoenix (expected to begin from 2027), Acter prioritises local hiring and partners with regional civil contractors to shorten the learning curve and to avoid the high overhead. To further address the potential labour shortages in the future, the company has established cross-border academic and vocational partnerships to bring Southeast Asian talent to Taiwan for technical training before deploying them back to their home markets, securing future human capacity to support global expansion projects.

Acter: revenue by clients' industry



Source: Company, Daiwa

Acter: revenue by geography

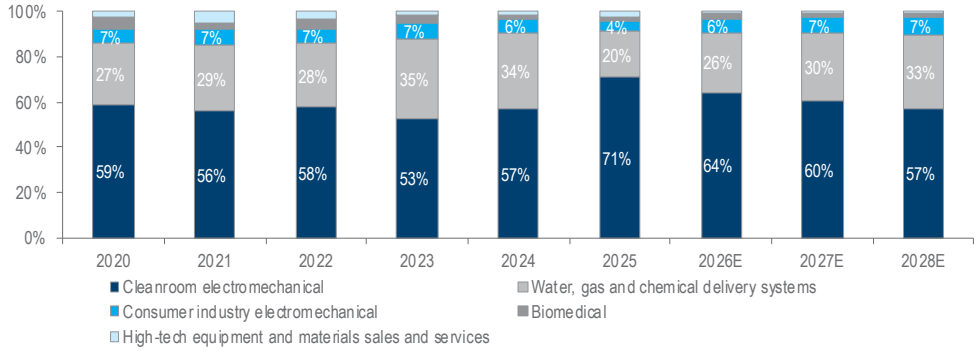


Source: Company, Daiwa

Semiconductor-related business in Taiwan is the major revenue stream for Acter

By product, cleanroom electromechanical services remain Acter's largest source of revenue (accounting for 66% of 1H26 revenue, per the company). These services include cleanroom architectural and MEP engineering, which powers airflow, filtration, environmental regulation and utility controls. Water, gas and chemical delivery systems are its second-largest source of revenue (accounting for 24% of 1H26 revenue, per the company). Contributed mainly by Novatech (6613 TT, not rated) and Rayzher (7703 TT, not rated), these services cover the main systems and hook-up systems, for ultra-pure water (UPW), specialty gases, chemical piping, etc., all of which require highly precise build-up work. Except for cleanroom electromechanical and water, gas and chemical delivery systems, the rest of Acter's revenue comes from the consumer industry services electromechanical (5%), biomedical (3%), and high-tech equipment and materials sales and services (1%).

Acter: revenue by product



Source: Company, Daiwa

Technically, Acter integrates digital transformation and green engineering into its business model, helping it differentiate from peers. Acter deploys digital twin technology and 1:1 3D BIM modelling to detect piping interferences and optimise spatial layout before installation, minimising schedule delays. This makes Acter a more reliable option, in our view, and thus being a favourable supplier, especially for the semiconductor clients that have extremely high quality and swiftness criterion.

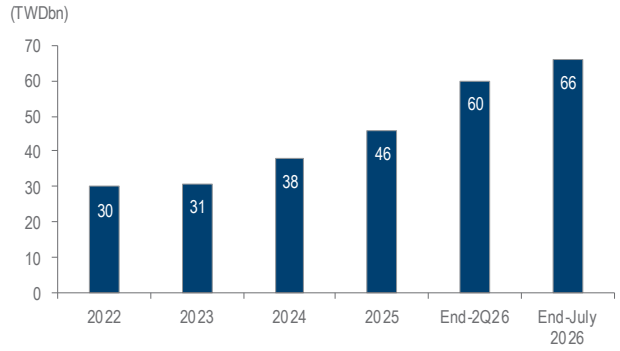
Bright outlook: strong order visibility

Rising orders on hand due to supply-demand imbalance for cleanrooms

Acter's order backlog has increased from TWD55bn in 1Q26 to TWD60bn in 2Q26 and TWD66bn in July 2026. We expect this figure to further increase in the following quarters, given the supply-demand imbalance in the cleanroom sector. This is composed of semiconductor (63%), electronic assembly (11%), consumer (9%) and others (17%), per the company. In 2H25, the company accelerated the digestion of small-scale projects below TWD1bn, and, as a result, starting from 1H26, it successfully shifted its project

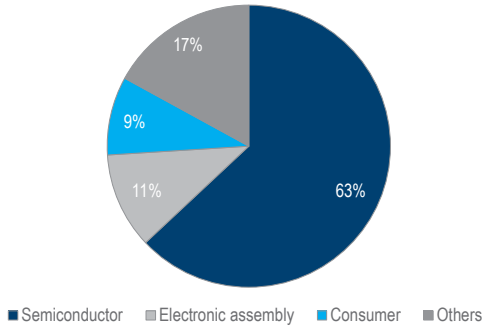
selection strategy to focus on large-scale projects valued at TWD1bn or above while reducing smaller contracts. This drives the growth in top-line.

Acter: order backlog



Source: Company, Daiwa

Acter: order backlog by clients' industry



Source: Company, Daiwa

OSAT customers have large order backlogs with Acter, which should drive future growth for the company

Back-end packaging, cloud infrastructure and hardware assembly are additional growth drivers. Acter serves as a contractor for both Taiwanese and overseas OSAT providers, including TWD21.7bn contracts YTD, awarded by a major OSAT client vs. TWD16.6bn contract obtained from the same client in 2025. In cloud infrastructure, Acter maintains qualified supplier status with US CSPs, providing mechanical, electrical and plumbing (MEP) integration, especially for those data centres that implement liquid-cooling systems. The company also wins factory construction projects for Taiwanese ODMs across the US and Southeast Asia, as well as PCB facility projects in Thailand and China. Non-tech sectors, including biotechnology, commercial buildings, smart logistics hubs and medical facilities, comprise 9% of the total backlog, per management.

Acter: contracts signed with one major OSAT client

Date of announcement	Type	Constructor	Amount (TWDm)
2026/1/7	EPCM	Acter	596
2026/1/16	EPCM	Acter	969
2026/1/22	EPCM	Acter	880
2026/2/23	EPCM	Acter	659
2026/4/15	EPCM	Acter	1,166
2026/4/27	EPCM	Acter	1,075
2026/5/6	EPCM & equipment	Acter	365
2026/5/19	EPCM	Acter	535
2026/5/27	EPCM	Novatech	947
2026/6/1	EPCM	Acter	550
2026/6/22	EPCM & equipment	Acter	349
2026/6/25	EPCM	Novatech	508
2026/7/2	EPCM	Acter	662
2026/7/8	EPCM	Acter	679
2026/7/13	EPCM	Acter	1,990
2026/7/20	EPCM	Acter	1,074
2026/8/5	EPCM	Acter	2,046
2026/8/5	EPCM	Novatech	714
2026/8/10	EPCM	Acter	841
2026/8/12	EPCM	Acter	582
2026/8/17	EPCM & equipment	Acter	389
2026/9/2	EPC	Acter	542
2026/9/17	EPC & equipment	Acter	462
2026/9/30	EPC	Acter	2,037
2026/10/1	EPC	Acter	1,055
Total for 2026 YTD			**21,675**
Total for 2025			**16,554**

Source: Companies, Daiwa

Competitive advantages

High entry barriers to the segment

Comprehensive product portfolio, empowered by subsidiaries

The supply of cleanroom services is limited due to high entry barriers in the segment

Acter provides a one-stop EPCM solution, leveraging its subsidiaries to deliver complete solutions. Within the group, the parent company Acter focuses on overall cleanroom and MEP system integration; subsidiary Novatech provides primary chemical systems and hook-up systems; and sub-subsidiary Rayzer specialises in primary specialty-gas systems and hook-up systems. In China, Suzhou Winmax, also a subsidiary of Novatech, focuses on the supply of wet processing and chemical handling equipment. This integrated model covers cleanroom engineering integration, water, power, chemical and gas-utility systems, as well as post-completion validation and maintenance, ensuring cross-system compatibility while mitigating risks associated with multi-contractor hiring.

Under global supply chain diversification trends, this unified group strategy strengthens order-winning capabilities in overseas markets. Clients expanding internationally face regulatory, linguistic and supply-chain complexities, favouring partners that offer one-stop services with a proven cross-border track record.

A diverse client base helps Acter weather economic cycles

Acter's main competitive advantage is its diversification of clients by industry and region. This guides the company through the peaks and troughs of the economy, which is especially crucial given the strong cyclicality of the semiconductor/high-tech industry. Acter serves a broad client base across semiconductors, OSAT, PCB, electronics assembly (ODM), CSP data centres, biopharmaceuticals and even the consumer industry (including commercial real estate and logistics), whereas many peers rely on a few large foundry clients. When semiconductor demand slows, capacity expansion in the PCB or data centre sectors offset the decline. Additionally, Acter avoids concentration risk and margin compression from major foundry accounts, as projects from leading foundries yield lower margins; by serving niche markets like US-based CSPs, the company achieves higher gross margins.

Geographically, Acter operates across Taiwan, China, Southeast Asia, the US and other countries. The company has OSAT and foundry customers in Taiwan and China, PCB clients expanding into Thailand, and ODM and data centre clients establishing operations in the US. Across different countries, local hiring and local supply chain implementation can further reduce costs and increase efficiency relative to peers reliant on a cross-border labour force.

Labour shortages

Amid global labour shortages, Acter minimises the operational impact and turns workforce constraints into a moat through localised talent development. Since engineering labour cannot easily be automated, Acter has built a cross-border talent pipeline via industry-academia partnerships. The company recruits Southeast Asian students for specialised M&E engineering training in Taiwan, deploying them to global project sites after their graduation to solve labour bottlenecks. This structure maintains an engineering team with an average tenure of more than 7 years, creating an operational barrier in an industry with a high employee turnover rate.

Compared with peers relying on high-cost personnel sent from Taiwan, we note that Acter's local hiring strategy creates cost and efficiency advantages. Prioritising compliant local hiring across overseas markets reduces travel expenses while accelerating integration with local supply chains and regulations. Localised teams also handle post-completion services and technical support directly on-site, building long-term client trust in overseas markets.

Numerous competitors, each with different specialisation

Some competitors, unlike Acter, lack a diversified client base

Acter faces different competitors across its main product lines. In the EPCM segment, including cleanroom and MEP works, key competitors include UIS (2404 TT, not rated), L&K (6139 TT, not rated), Yankey (6691 TT, not rated) and Jiuhan (6903 TT, not rated). Despite having business models similar to Acter, UIS, Yankey and Jiuhan focus more on major semiconductor players in both Taiwan and the US, and L&K concentrates on ASEAN semiconductor projects. In the equipment/material sales and agency space, Acter mainly competes with Marketech (6196 TT, not rated). In the water, gas and chemical delivery systems area, through Novatech and Rayzher, Acter group competes with Marketech, Mega Union (6944 TT, not rated), Taiwan Puritic (6826 TT, not rated), Trusval (6667 TT, not rated) and Cica-Huntek (6725 TT, not rated).

In our view, despite the competitive environment, Acter maintains a formidable competitive moat built on several key advantages. We attribute this to its more than 45 year experience in the industry, enabling it to develop specialised expertise that is hard for competitors to replicate.

Acter: peer comparison (cleanroom related)

Company	Ticker	Market Cap (USDm)	Market	Main thermal product	2025 Product mix	2025 Segment breakdown	2025 Market breakdown
Acter	**5536 TT**	**4,070**	**Taiwan**	**EPCM turnkey solutions**	**Cleanroom electromechanical (71%), water, gas and chemical delivery systems (20%), consumer industry electromechanical (4%), biomedical (2%) and high-tech equipment & materials sales & services (3%).**	**Semiconductor (69%), electronics assembly (17%), optoelectronics (10%), others (4%)**	**Taiwan (57%), China (29%), others (14%)**
Novatech	6613 TT	708	Taiwan	Gas and chemical delivery systems and wet processing equipment	Equipment (46%), engineering (47%), eco-integration (7%)	Semiconductor (85%), optoelectronics (7%), others (8%)	China (50%), Taiwan (45%), others (5%)
Rayzher	7703 TT	225	Taiwan	Gas delivery systems	Hook-up systems (52%), primary system (28%), small-scale works (18%), equipment & material sales (3%)	Semiconductor, equipment and material providers, gas system providers, optoelectronics	Taiwan (99%), China & Japan (1%)
Acter Technology Integration	603163 CH	1,102	China	EPCM turnkey solutions	Cleanroom electromechanical (65%), other electromechanical (32%), equipment & material sales (2%)	Semiconductor (60%), precision manufacturing (34%), optoelectronics (2%), others (4%)	China (58%), others (42%)
UIS	2404 TT	7,957	Taiwan	EPCM turnkey solutions	System integration (100%)	Foundries, memory	Taiwan (67%), U.S. (25%), China (3%), Singapore (3%), Japan (1%)
L&K	6139 TT	5,820	Taiwan	EPCM turnkey solutions	Turnkey solutions (63%), electromechanical (25%), cleanroom builds (8%), delivery systems (4%)	Semiconductor (74%), railway/highway/airport (10%), MRT (7%), commercial (4%), energy (2%), expropriation (2%), others (1%)	ASEAN (59%), Taiwan (34%), China (7%)
Marketech	6196 TT	4,634	Taiwan	Gas and chemical delivery systems, Cleanroom equipment manufacturing/representative, EPCM turnkey solutions	Gas and chemical delivery systems (39%), representative for equipment (25%), equipment manufacturing (18%), turnkey solutions (18%)	Semiconductor (81%), optoelectronics (6%), others (13%)	Taiwan (45%), China (18%), U.S. (17%), others (20%)
Yankey	6691 TT	3,253	Taiwan	EPCM turnkey solutions	CSA (civil, structural & architectural), MEP (mechanical, electrical & plumbing), CR (cleanroom)	Semiconductor (58%), PCB (15%), optoelectronics (11%), data centres (6%), office (6%), electronic components (2%), others (1%)	Taiwan (89%), China (6%), others (5%)
Jiuhan	6903 TT	684	Taiwan	EPCM turnkey solutions	EPCM turnkey solutions (100%)	AI-related (83%), public works (17%)	Taiwan (100%)
Mega Union	6944 TT	2,842	Taiwan	UWS (ultrapure water systems), wastewater recycling, water treatment engineering	Water treatment engineering (80%), O&M/chemicals/consumables (20%)	Semiconductor and other high-tech industry	Taiwan (69%), U.S. (19%), Singapore (10%), China (2%)
Taiwan Puritic	6826 TT	1,223	Taiwan	Gas delivery systems	Gas primary system (83%), gas equipment (17%)	High-tech industry (100%)	Taiwan (73%), Americas (18%), Asia (9%)
Trusval	6667 TT	468	Taiwan	Water, gas and chemical delivery systems	Primary and hook-up systems (92%), eco-integration (4%)		Taiwan (96%), others (6%)
Cica-Huntek	6725 TT	253	Taiwan	Chemical delivery systems	Primary and hook-up systems (73%), system integration engineering (20%), system maintenance (7%)		Taiwan (87%), Asia (13%)

Source: Companies, Bloomberg, Daiwa

Financial analysis

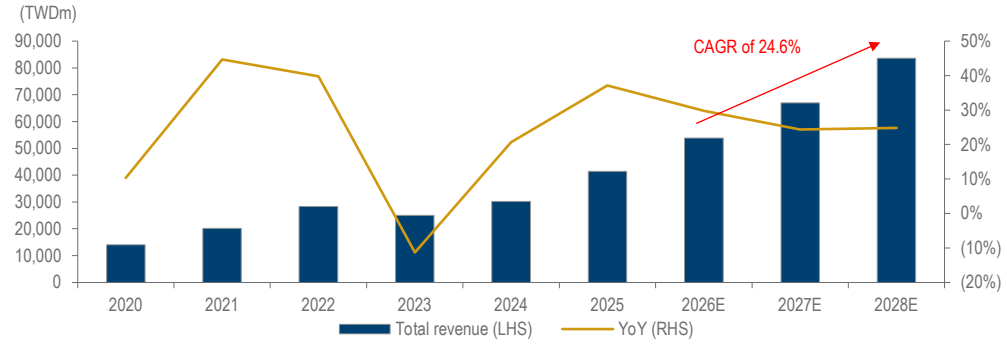
We forecast the company's net profit to rise at a CAGR of 36.9% over 2026-28E, driven by a revenue CAGR of 24.6% due to surging demand from the high-tech industry (an increase in the order backlog to TWD66bn at the end of July 2026 vs. TWD46bn at end-2025) and a 3.3pp expansion in operating margin over 2026-28E, driven by the gross margin improvement from more favourable project wins and operating leverage.

Strong order visibility, with revenue expected to grow across all sectors over 2026-28E

We forecast Acter's revenue to record a CAGR of 24.6% over 2026-28E

We forecast Acter's revenue to increase at a CAGR of 24.6% over 2026-28E, driven by scale expansion and a growing demand from various industries, including semiconductor, electronic assembly, optoelectronics and others.

Acter: revenue growth trend



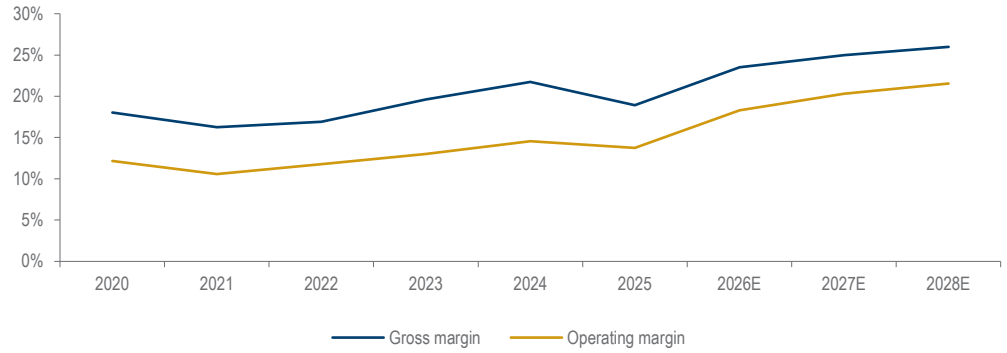
Source: Company, Daiwa forecasts

Gross- and operating-margin expansion trend

We forecast an upward operating-margin trend over 2026-28E

Over 2026-28E, we forecast Acter's gross and operating margins to expand on rising revenue scale and a higher complexity of cleanrooms for advanced manufacturing. We note that the gross margin from high-tech cleanrooms is higher than that from consumer industries, such as commercial buildings and biopharmaceuticals. Thus, we estimate the company's gross margin to reach 26.0% in 2028E from 18.9% in 2025 and its operating margin to reach 21.6% in 2028E from 13.7% in 2025.

Acter: gross- and operating-margin trend



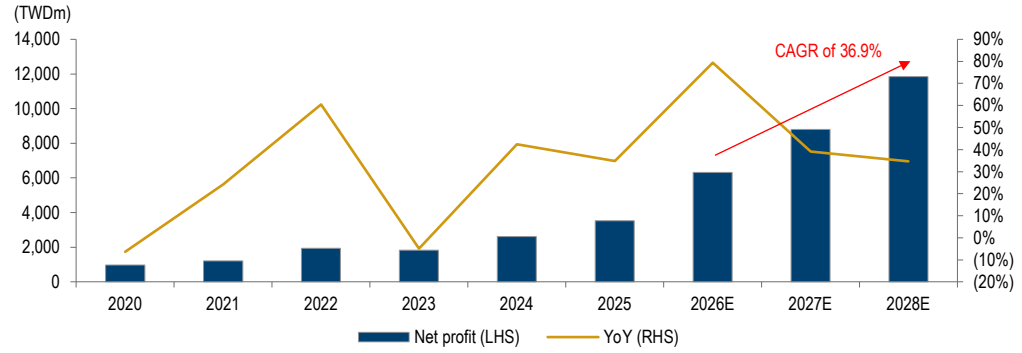
Source: Company, Daiwa forecasts

Net margin likely to rise over 2026-28E

We forecast a net profit CAGR of 36.9% for Acter over 2026-28E

We forecast Acter's net profit to rise to TWD8,797m in 2027E and TWD11,851m in 2028E (from TWD6,325m in 2026E). We expect its net margin to come in at 13.1% in 2027E and 14.2% in 2028E (vs. 11.7% in 2026E) and for its net profit to grow at a CAGR of 36.9% over 2026-28E.

Acter: net profit trend



Source: Company, Daiwa forecasts

Quarterly P&L

Acter: quarterly P&L

(TWDm)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E
Revenue	11,603	13,381	13,959	14,900	14,900	16,128	17,242	18,713
Gross profit	2,945	2,762	3,211	3,736	3,427	4,032	4,310	4,976
Operating profit	2300	2126	2499	2930	2697	3258	3517	4125
Profit before tax	2371	2232	2569	3000	2720	3281	3540	4148
Net profit	1583	1361	1547	1835	1786	2092	2269	2651
Basic EPS (TWD)	12.76	10.97	12.46	14.79	14.39	16.86	18.28	21.36
Margin								
Gross margin	25.4%	20.6%	23.0%	25.1%	23.0%	25.0%	25.0%	26.6%
Operating margin	19.8%	15.9%	17.9%	19.7%	18.1%	20.2%	20.4%	22.0%
Pre-tax margin	20.4%	16.7%	18.4%	20.1%	18.3%	20.3%	20.5%	22.2%
Net margin	13.6%	10.2%	11.1%	12.3%	12.0%	13.0%	13.2%	14.2%
YoY								
Revenue	35.6%	22.7%	26.4%	35.8%	28.4%	20.5%	23.5%	25.6%
Gross profit	91.7%	31.5%	63.1%	66.6%	16.4%	46.0%	34.3%	33.2%
Operating profit	124.6%	35.5%	58.3%	91.1%	17.3%	53.3%	40.8%	40.8%
Profit before tax	109.3%	45.8%	46.2%	82.8%	14.7%	47.0%	37.8%	38.3%
Net profit	146.3%	57.9%	51.6%	83.1%	12.8%	53.7%	46.7%	44.5%
QoQ								
Revenue	5.7%	15.3%	4.3%	6.7%	0.0%	8.2%	6.9%	8.5%
Gross profit	31.3%	-6.2%	16.3%	16.4%	-8.3%	17.7%	6.9%	15.4%
Operating profit	50.0%	-7.6%	17.5%	17.2%	-7.9%	20.8%	8.0%	17.3%
Profit before tax	44.5%	-5.9%	15.1%	16.8%	-9.3%	20.6%	7.9%	17.2%
Net profit	58.0%	-14.1%	13.7%	18.6%	-2.7%	17.1%	8.5%	16.8%

Source: Company, Daiwa forecasts

Intact demand to support revenue growth over our forecast horizon; ROE expansion underway

We expect stronger cash balances and FCF over 2026-28E due to better profitability

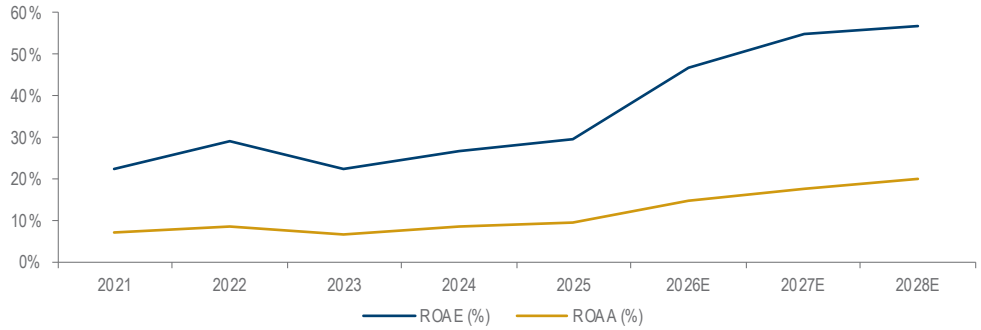
We forecast Acter's total cash and equivalents to reach TWD16,838m (36.8% of its total assets) in 2026E and increase to TWD27,839m in 2028E (42.8% of total assets) with net cash likely to come in at TWD16,244m in 2026E and TWD27,245m in 2028E.

After factoring in our 2027 and 2028 capex assumptions of TWD180m, we expect Acter's free cash flow (FCF) to improve to TWD12,129m in 2028E, from TWD5,763m in 2026E. We believe the capacity expansion cost will initially be low and that the company will continue to benefit from the better profitability.

Strong ROE outlook over 2026-28E

We expect Acter's ROE to improve to a record-high of 56.8% in 2028E from 29.3% in 2025 (following steady annual increases to 46.8% in 2026E and 54.9% in 2027E), which we attribute to margin expansion from the revenue size increase and gross margin expansion led by increases in cleanroom complexity.

Acter: ROAE and ROAA trend



Source: Company, Daiwa forecasts

Valuation and risks

Undemanding valuation, alongside its leading global positions in the OSAT and memory segments and an optimistic industry outlook

12-month TP of TWD1,415, based on a target PER of 20x

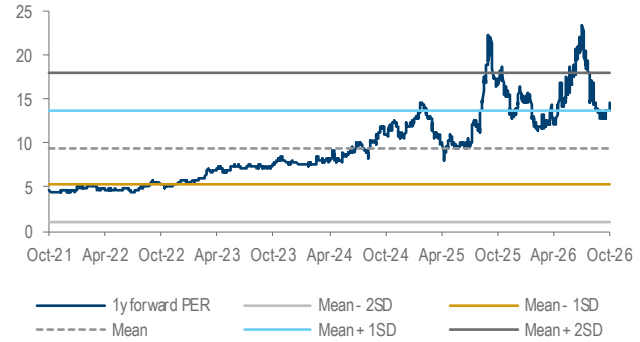
We initiate coverage of Acter with a Buy (1) rating and 12-month TP of TWD1,415, based on a target PER of 20x applied to our one-year-forward EPS forecast. This target multiple is higher than the average of its past-3-year 1-year-forward PER of 12.2x (ranged from 7.4x to 23.5x) and the average of its past-5-year 1-year-forward PER of 9.6x (ranged from 4.4x to 23.5x), which we view as undemanding as we forecast a 36.9% net profit CAGR over 2026-28E, implying a 0.54x PEG. This is supported by its business expansion in the next 2-3 years. We see potential for a rerating of the stock, given Acter's global leadership in cleanroom EPCM, particularly in the OSAT and memory industries.

The stock is trading currently at a 15.8x 1-year-forward PER vs. its earnings CAGR of 36.9% over 2026-28E, translating to a 0.43x PEG ratio. Its share price has underperformed TAIEX Index by 32.7% in 3Q26, mainly due to its weaker-than-expected 2Q26 results. However, given the potential improvement in sector sentiment following TSMC's results on 15 October 2026, along with the likelihood that Acter's revenue momentum will strengthen in the coming months, we recommend that investors buy the stock.

Acter: past-3-year-forward PER (average: 12.0x)

Source: Bloomberg, Daiwa estimates

Acter: past-5-year-forward PER (average: 9.5x)



Source: Bloomberg, Daiwa estimates

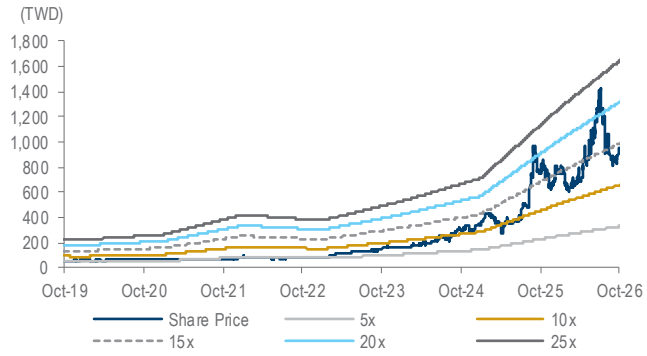
We compare Acter with other leading cleanroom EPCM companies such as UIS (2404 TT, TWD1,330.0, not rated), L&K (6139 TT, TWD787.0, not rated) and Yankey (6691 TT, TWD776.0, not rated). UIS, L&K and Yankey are trading at a PER of 14.7x (earnings CAGR of 40.7% over 2025-27E), 9.5x (earnings CAGR of 36.9% over 2026-28E) and 15.1x (earnings CAGR of 53.6% over 2025-27E), on their 1-year-forward EPS, respectively.

Stronger earnings growth outlook and more room to grow vs. peers

Acter's peers are trading at 18.7x for 2026E and 14.2x for 2027E. We forecast Acter to deliver an EPS CAGR of 36.9% over 2026-28E vs. 31.5% CAGR for the segment over the same period.

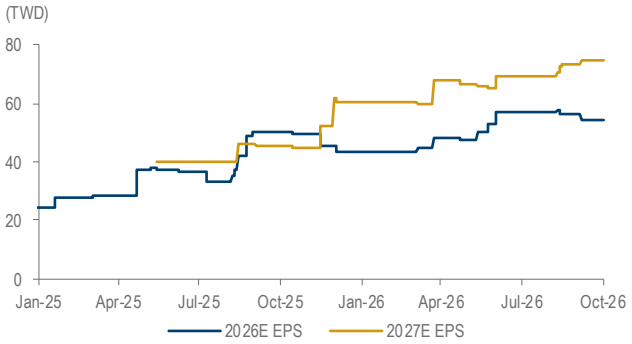
We believe our target one-year-forward PER of 20x (equivalent to a 0.54x PEG using 2026-28E EPS CAGR) is undemanding as: **1) Acter is ramping up capacity to meet growing demand from high-tech customers and is in an expansion cycle;** and **2) its leading position among OSAT and memory clients warrants a valuation premium, compared to its global peers**.

Acter: 1-year-forward PER bands



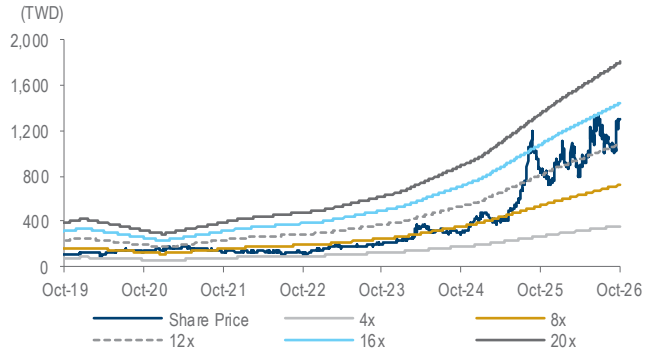
Source: Bloomberg, Daiwa forecasts

Acter: earnings revision trend



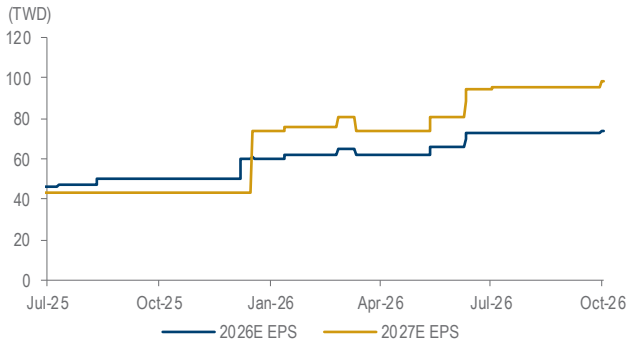
Source: Bloomberg, Daiwa

UIS: 1-year-forward PER bands



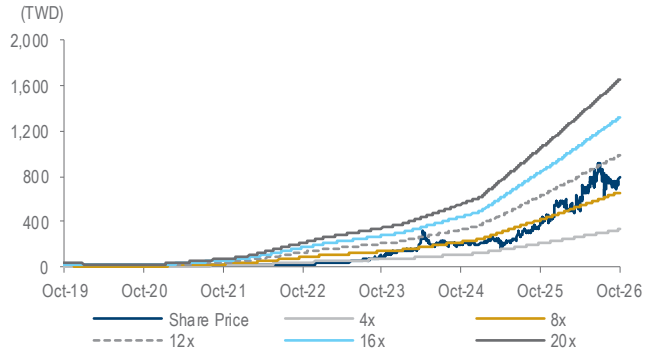
Source: Bloomberg, Daiwa forecasts

UIS: earnings revision trend



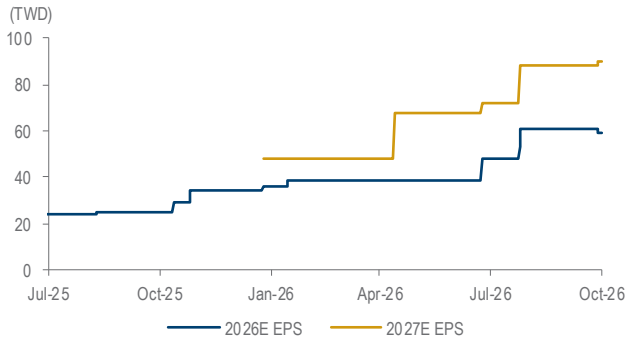
Source: Bloomberg, Daiwa

L&K: 1-year-forward PER bands



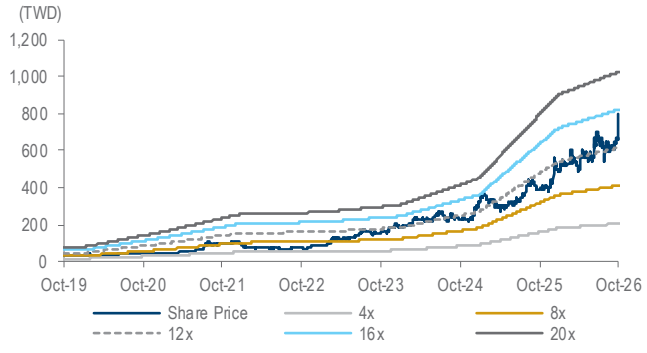
Source: Bloomberg, Daiwa forecasts

L&K: earnings revision trend



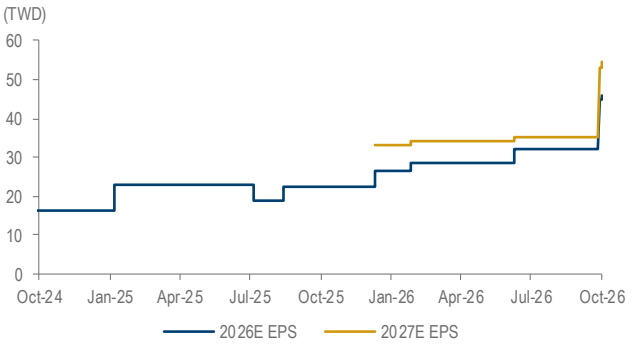
Source: Bloomberg, Daiwa

Yankey: 1-year-forward PER bands



Source: Bloomberg, Daiwa forecasts

Yankey: earnings revision trend



Source: Bloomberg, Daiwa

Acter: peers valuation comparison

Companies	Bloomberg code	Price (lc) 7/10/26	Daiwa rating	Market cap (USDm)	D. turnover (USDm)	PER (x) 2026E	PER (x) 2027E	PER (x) 2028E	EPS CAGR 2026-28E	ROE (%) 2026E	ROE (%) 2027E	ROE (%) 2028E	PBR (x) 2026E	PBR (x) 2027E	PBR (x) 2028E	Dividend yield (%) 2026E	Dividend yield (%) 2027E	Dividend yield (%) 2028E
Acter*	***5536 TT***	***1,045.0***	***Buy***	***4,070***	***33.4***	***20.6***	***14.8***	***11.0***	***36.9***	***46.8***	***54.9***	***56.8***	***9.3***	***7.2***	***5.5***	***3.4***	***4.7***	***6.4***
Novatech	6613 TT	290.0	Non rated	708	3.8	15.5	11.6	8.1	38.4	27.9	34.3	n.a.	n.a.	n.a.	n.a.	5.1	6.8	n.a.
Rayzher	7703 TT	206.0	Non rated	225	0.6	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Acter Technology Integration	603163 CH	73.9	Non rated	1,102	9.6	33.8	22.3	17.6	38.5	17.0	22.4	24.9	5.7	4.9	4.2	1.6	2.4	3.7
UIS	2404 TT	1,330.0	Non rated	7,957	68.0	17.9	13.5	11.0	27.7	56.0	48.0	47.3	n.a.	n.a.	n.a.	4.6	5.5	n.a.
L&K	6139 TT	787.0	Non rated	5,820	81.6	13.3	8.8	7.1	36.9	64.0	64.1	n.a.	n.a.	n.a.	n.a.	5.2	8.3	n.a.
Marketech	6196 TT	670.0	Non rated	4,634	27.8	23.5	20.5	19.3	10.4	34.5	32.3	30.2	7.2	6.1	5.4	2.3	2.6	2.6
Yankey	6691 TT	776.0	Non rated	3,253	16.8	16.9	14.2	n.a.	n.a.	67.2	62.1	n.a.	8.6	5.5	n.a.	4.9	5.8	n.a.
Jiuhan	6903 TT	327.0	Non rated	684	4.7	9.8	7.4	n.a.	n.a.	46.9	42.7	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Mega Union	6944 TT	908.0	Non rated	2,842	12.2	22.9	18.2	n.a.	n.a.	44.1	46.0	n.a.	6.6	8.2	n.a.	2.7	3.2	n.a.
Taiwan Puritic	6826 TT	455.5	Non rated	1,223	1.6	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Trusval	6667 TT	278.5	Non rated	468	2.4	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Cica-Huntek	6725 TT	217.0	Non rated	253	0.8	12.5	10.5	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	Industry average					***18.7***	***14.2***	***12.4***	***31.5***	***44.9***	***45.2***	***39.8***	***7.5***	***6.4***	***5.0***	***3.7***	***4.9***	***4.2***

*Source: Bloomberg; *Daiwa forecasts*

Our 12-month TP of TWD1,415 is also supported by our DCF valuation

We also reference a DCF methodology (see charts below), which indicates a DCF-derived fair value estimate of TWD1,454/share, which supports our 12-month TP of TWD1,415.

In our DCF calculation, we apply a WACC of 19.3% to Acter's FCF over 2026-35E and a terminal growth rate of 7.0% thereafter. We expect the company to continue to record steadily increasing FCF over 2026-35E.

Acter: DCF calculation

Target gearing (debt/capital)	**0.7%**
Market risk premium (%)	**18.0%**
Risk-free rate (%)	**5.0%**
Cost of debt (%)	**5.0%**
Cost of equity (%)	**19.4%**
WACC (%)	**19.3%**
Terminal growth rate	7.0%
Terminal WACC	19.3%
Enterprise value (TWDm)	167,994
Less: net debt (as of end-2025) (TWDm)	-12,402
Equity value (TWDm)	180,396
No. shares (m)	124
Per share equity value (TWD)	**TWD1,454**

Source: Company, Daiwa estimates

Acter: DCF sensitivity

		WACC (%) 18.9	19.1	19.3	19.5	19.7
Terminal growth rate (%)	8.0	1,571	1,545	1,520	1,496	1,473
	7.5	1,533	1,509	1,486	1,463	1,442
	7.0	1,498	1,476	1,454	1,433	1,413
	6.5	1,466	1,445	1,425	1,405	1,386
	6.0	1,437	1,417	1,398	1,379	1,361

Source: Daiwa estimates

Acter: free cash flow forecasts

(TWDm)	2026E	2027E	2028E	2029E	...	2035E
EBITDA	**10,482**	**14,182**	**18,610**	**22,332**	...	**66,683**
Less: taxation (EBIT*tax rate)	-2,447	-3,293	-4,355	-5,226	...	-15,605
Plus: decrease in working capital	-2,122	-1,006	-1,946	-1,946	...	-1,946
Less: capital expenditure	-150	-180	-180	-180	...	-180
Free cash flow (TWDm)	**5,763**	**9,703**	**12,129**	**14,980**	...	**48,952**
Discount rate	19.3%	19.3%	19.3%	19.3%	...	19.3%
Discount factor	0.96	0.80	0.67	0.56	...	0.20
NPV of free cash flow	**5,510**	**7,776**	**8,145**	**8,432**	...	**9,558**

Source: Daiwa estimates and forecasts

Risks to our call

Primary/key risks:

Weaker-than-expected expansion in semiconductor industry

The biggest downside risk to our call for Acter is weaker-than-expected expansion from semiconductor clients, especially in advanced-node foundries, advanced packaging and OSAT. If global demand for AI servers, HPC or HBM slows, or if clients delay fab build-ups and overseas expansion plans, it will directly pressure order inflows and backlog conversion for Acter.

Project execution delays or cancellations

Acter's revenue recognition is tied to large, complex EPCM and integrated engineering projects, which are often subject to customer-driven delays. Any postponement in project

schedule, construction bottlenecks or cancellation of large projects could defer revenue recognition and hurt overall profit. This risk is more obvious in overseas projects, where licensing, fire-safety approvals and hazardous-chemical regulations are more complicated.

Key risks to our call: weaker-than-expected expansion in semiconductor industry, project delays or cancellations, and higher-than-expected raw-material price hikes

Higher-than-expected raw-material price hikes

Raw-material costs (especially copper and oil-related costs) account for 50-60% of COGS and thus the price volatility makes costs difficult to manage. Acter needs to maintain supplier relations, negotiate appropriate price-adjustment mechanisms with clients, and invest in R&D to identify composite material substitutes, thereby reducing reliance on any single component.

Secondary risks:

Labour issues or larger-than-expected increases in labour costs

Cleanroom engineering is highly labour-intensive and relies on on-site installation and supervision. Although Acter has built a local hiring and talent-development model, a larger-than-expected shortage of engineers or rising wage pressure could still raise costs and constrain capacity expansion. Any inability to secure enough skilled labour may also limit the company's ability to convert orders into revenue in time.

Margin pressure from lower-margin projects or mix shift

Acter's business mix includes a few low-margin C&S services as well as larger EPCM and system integration contracts. If the company takes on a higher proportion of lower-margin projects, or if competition intensifies in certain geographies or customer segments, gross margin could come under pressure. In addition, an unfavourable product mix shift towards less profitable work could offset benefits from backlog growth.

Overseas expansion and regulatory risks

Acter is expanding across markets in the US, Southeast Asia, Japan and China, which exposes it to FX movements, local compliance requirements, tax complexity and regulatory uncertainty. Delays in licensing, changes in safety standards or local-content requirements could increase project costs and reduce execution efficiency. Any misstep in managing overseas operations could damage profitability and client retention.

Appendix

Company background

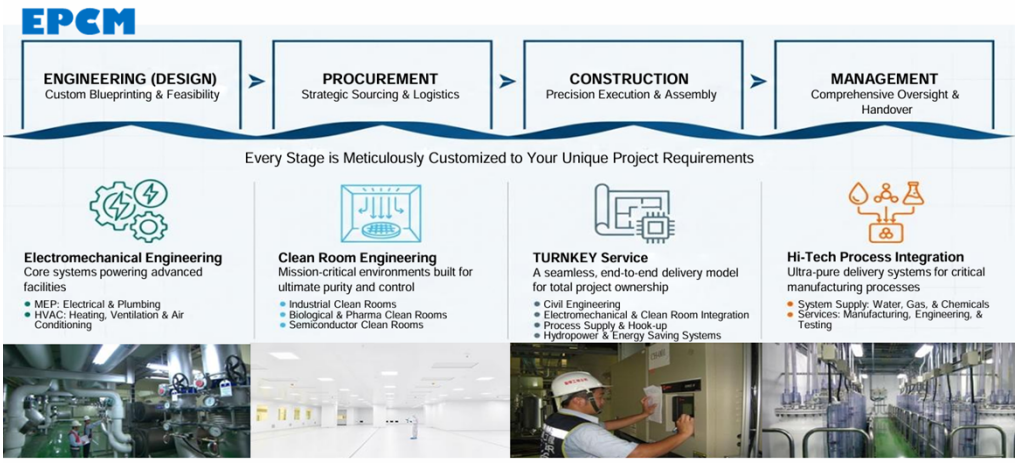
History

Comprehensive cleanroom EPCM provider in Taiwan; dominates both high-tech market and consumer industry

Acter (5536 TT), founded in 1979 and headquartered in Taichung, Taiwan, is a cleanroom EPCM provider with over four decades of experience. Starting with early mechanical and electrical (M&E) engineering through vertical and horizontal integration, Acter has now developed into an engineering technology group with cross-border execution capabilities. A cleanroom building process begins with **civil & structural (C&S) works** to establish the physical plant, followed by **mechanical, electrical, and plumbing (MEP) works** for essential utility generation, where Acter is one of the experts. After MEP is finished, chemicals/gas/water are then distributed via main systems **through primary piping networks**, and finally delivered to production equipment through **hook-up systems**. The company's core competency is providing one-stop integrated total solutions, spanning services, primary piping networks and hook-up systems, leaving C&S services to other engineering peers, given the nature of the latter segment's low margin and long timespan.

In terms of business model, Acter delivers EPCM (Engineering, Procurement, Construction, Management) capabilities. From initial customised planning and feasibility assessments to strategic procurement, installation, and post-completion maintenance, Acter builds production environments tailored to client process requirements.

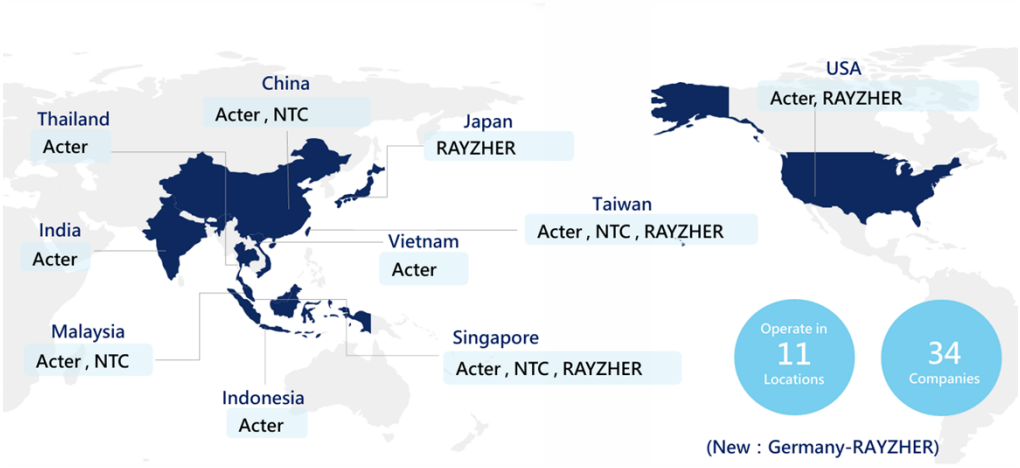
Acter: EPCM solutions offered



Source: Company, Daiwa

Regarding global expansion, Acter actively responds to the global supply chain's "N+1" strategy, operating across 11 regions with 33 companies globally. Its service network spans Taiwan, China, Southeast Asia (including Thailand, Malaysia, Vietnam, Indonesia, and Singapore), Japan, India, and the US. The company leverages a group synergy strategy by integrating resources from key subsidiaries – such as Novatech (6613 TT, not rated; provides chemical primary and hook-up systems), Rayzher (7703 TT, not rated; provides special gas primary and hook-up supply systems), and Acter Group (603163 CH, not rated; provides cleanroom engineering in China) – to form a complete technical ecosystem, ensuring cross-border capability and quality.

Acter: global footprint



Source: Company, Daiwa

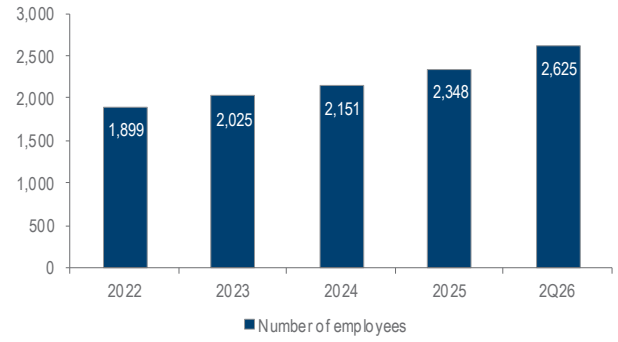
In 2Q26, Acter classified revenue as: cleanroom electromechanical (63%), water, gas and chemical delivery systems (28%), consumer industry electromechanical (6%), biomedical (2%) and high-tech equipment & materials sales & services (1%). In 2025, the product mix was: cleanroom electromechanical (71%), water, gas and chemical delivery systems (20%), consumer industry electromechanical (4%), biomedical (2%) and high-tech equipment & materials sales & services (3%).

Young, capable human resource is a crucial asset for Acter

Acter serves a diversified client base divided into high-tech and consumer divisions. The high-tech division focuses on advanced sectors including semiconductors, OSAT, PCB, AI data centres (for CSPs), and electronics assembly. The consumer division extends into healthcare, premium commercial properties, logistics facilities, and green engineering. This multi-industry strategy allows the company to capitalise on tech sector upgrades while maintaining resilient operations across technology (especially semiconductor) cycles.

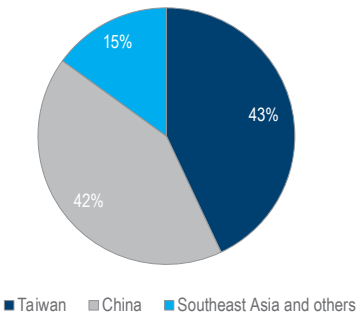
Acter adopts a management strategy involving local hiring in overseas markets, prioritising collaboration with local supply chains and compliance with local regulations to reduce cross-border deployment costs and thus build strong local teams. Acter has more than 2,600 employees worldwide (average age: 42 years; average length of service: 7 years). With more than 45 years of experience in EPCM, the company plays a reliable role for both high-tech and consumer industry.

Acter: number of employees



Source: Company, Daiwa

Acter: location of employees (as of June 2026)



Source: Company, Daiwa

Furthermore, Acter integrates ESG sustainability into its core corporate identity and technological innovation. As a green engineering provider, the company develops energy-saving construction methods, digital twins, ice-storage energy conservation, wastewater recycling, and AI smart facility management systems to help clients achieve carbon neutrality while expanding capacity. Backed by solid corporate governance, Acter

Acter integrates ESG capabilities into its core corporate value

consistently ranks among the top performers in Taiwan's corporate governance evaluations, maintaining a stable dividend payout policy and strong ROE as one of the leading facility engineering vendors.

Acter: management team (according to 2025 annual report)

Management	Position	Background and experience
Liang, Chin-Li	Chairman/CEO	Director since founding the company in 2008. Liang possesses working experience related to refrigeration and air-conditioning; graduated from EMBA, National Chiao Tung University, and Department of Electrical Engineering – Refrigerating and Air-conditioning, Taipei Tech
Hu, Tai-Tsen	Director	Director since founding the company in 2009. Hu possesses working experience related to refrigeration and air-conditioning; graduated from EMBA, Tunghai University, and Department of Electrical Engineering – Refrigerating and Air-conditioning, Taipei Tech
Lai, Ming-Kun	President	Graduated from EMBA, National Taiwan University of Science and Technology, and Department of Electrical Engineering – Refrigerating and Air-conditioning, Taipei Tech
Wang, Chun-Sheng	President	Graduated from EMBA, Feng Chia University, and Department of Electrical Engineering – Refrigerating and Air-conditioning, Taipei Tech

Source: Company

Acter: major shareholders (as of August 2026)

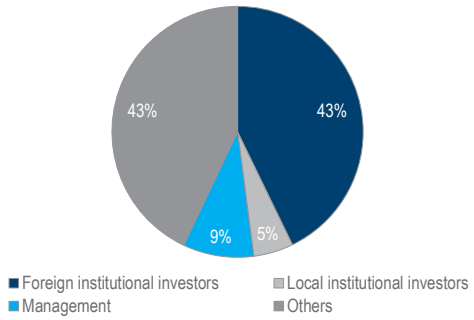
Holder	Share (%)
Xiang-Hui Development Co., Ltd.	4.35%
Chiu-Chang Investment Co., Ltd.	3.96%
Liang, Chin-Li	3.72%
JFE Engineering Co., Ltd.	2.23%

Source: TEJ

Acter: QFII holdings

Source: TEJ

Acter: shareholding structure (as of September 2026)



Source: TEJ

Glossary of terms

EPCM (engineering, procurement, construction, and management): Cleanroom construction services that cover the whole process including project design, purchasing, construction and management.

BIM (building information modelling): A digital modelling system of a building and its key information used in cleanroom construction.

C&S (civil & structural): Site infrastructure and the design and construction of load-bearing elements.

MEP (mechanical, electrical, and plumbing): Building essential systems for environment control, power, lighting, water, and drainage.

ESG analysis

ESG risks

Risks		Management	Analyst comments
G	Executive/board quality	1	Acter maintains a structured governance framework that supports disciplined management. The chairman is Chin-Li Liang. The company operates with defined responsibility across the board, management, cyber security oversight, operations, and audit functions. It has established an organisation function led by the President, a Chief Cyber Security Officer, a Response Team, an Operations and Policy Enforcement Team, and an Audit Team to supervise policy execution and review security implementation.
	Capital management	1	Acter manages capital in a stable and shareholder-friendly manner. Over the past several years, Acter has typically maintained a dividend payout ratio of 65-80%. It also maintains a strong financing network across multiple banks, supporting flexibility for project execution and expansion while preserving financial stability.
	Related party & transaction	1	Acter's business is supported by a transparent network of suppliers, project partners, and financial counterparties, including companies with equity ties and reciprocal business relationships that are common in its operating ecosystem. These relationships are managed through formal internal procedures for fund lending, endorsements and guarantees, and asset acquisition and disposal, all of which are approved through shareholder mechanisms and company rules. Such a structure helps keep inter-company and supply-chain transactions organised and traceable, reducing the risk of unclear dealings.
E	Product quality & safety	1	Acter executes its projects through a complete, disciplined process covering bidding assessment, project planning, design, procurement, construction management, verification and testing, completion and acceptance, closing review, and warranty service. Its cleanroom and electromechanical engineering business is built around maintaining temperature, humidity, and dust-free environments that support high-precision manufacturing. The company strengthens quality control through BIM, Revit, simulation software, ISO drawing output tools, and automated modeling, which improve design accuracy, construction efficiency, and project consistency.
E	Ecological impacts	1	Acter has built a concrete sustainability programme with measurable environmental outcomes. The company completed Scope 1, 2, and 3 greenhouse gas emissions calculation, obtained third-party verification under ISO 14064-1:2018, and publicly disclosed energy consumption and emissions data. It also implemented a strong waste reduction programme, achieving a 95% wastepaper recycling rate at headquarters in 2025 and 100% paper recycling at the 19F and 20F floors after establishing paper segregation and reuse procedures. The company also promoted sustainability measures through emissions inventory management, environmental disclosures, and resource-efficiency practices. These actions show active environmental management and a clear commitment to lowering operational impact.
	Health & safety		

Note: Management score represents a company's ability to manage/benefit from certain ESG topics. The scores range from 1 to 3, with 1 being the strongest.

Update Date: 7 Oct 2026

Source: Daiwa, Company

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Rating	Percentage of total
Buy*	73.96%
Hold**	19.58%
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Source: Daiwa

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- Before engaging in any trading, please thoroughly confirm accounting and tax treatments regarding your trading in financial instruments with such experts as certified public accountants.

* The amount of the trading commission cannot be stated here in advance because it will be determined between our company and you based on current market conditions and the content of each transaction etc.

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國泰期貨 國泰證券子公司

產業更新 – 國防軍工產業

無人機、無人艇兵分二路，無人載具供應鏈蓄勢

What's New – 國防預算持續提高，六年編列 2,400 億元

2026 年國防總支出規劃達 9,495 億元，2027 年行政院總預算案進一步提高至 1 兆 1,225 億元，年增約 18%，後續待立法院審議。《強化國防自主暨無人載具產業發展條例》規劃六年編列 2,400 億元，每年以 400 億元為原則，支持可信任供應鏈、研發及測試能力建置。此外，行政院提出 1,457 億元國防追加預算，其中濱海監偵型、濱海攻擊型無人機及小型自殺無人艇合計 559 億元，盼於預算通過後加快採購，於 2027 年形成初始戰力。隨政策需求逐步落實，國內軍工供應鏈將由樣機開發與測試，進一步朝產線建置、零組件穩定供貨及批量交付發展。

Trends/Outlook – 無人載具供應鏈蓄勢，持續性發展可期

台灣無人機需求將沿低成本大量部署與高階情監偵兩條路線發展。前者重視採購成本、生產速度與快速補充能力，帶動模組化設計及共用零組件需求；後者強調長航時、感測品質與穩定資料鏈，商機涵蓋機體、光電酬載、導控系統及後勤服務。反無人機系統則由單一干擾設備走向多感測器、指揮控制與反制手段整合，跨品牌相容性、場域驗證及持續更新能力將成為競爭重點。無人艇方面，產業價值由船體製造延伸至自主航行、通信及任務酬載，台海風浪環境亦提高海試與可靠度驗證的重要性。研究部認為，隨部署規模擴大，具備系統整合、穩定交付及長期維保能力的業者，有機會建立持續性收入來源。

Investment Summary

1. 長榮航太(2645 TT)以軍用商規艦載監偵無人機切入市場，延伸航太製造與品質管理優勢；漢翔(2634 TT)與 Shield AI 合作，以 V-BAT 垂直起降無人機取得海巡案，後續可望拓展在地整合、維保及新增採購需求。創泓科技(7714 TT)布局 C2 無人機防禦系統與 Sentrycs 反無人機方案，商機涵蓋設備供應、偵測反制及指揮控制整合，後續關注標案導入與服務收入成長。
2. 台船(2208 TT)的「奮進魔鬼魚」採三體船與模組化酬載設計，並透過與 Havoc 合作布局自主水面載具；龍德造船(6753 TT)的「黑潮」以高速操縱與匿蹤設計為特色，結合既有軍用小型船建造經驗，爭取量產與後勤需求。研究部對台灣軍工產業中期發展持正面看法，後續隨預算、採購及產品驗證逐步落實，相關業者有望將技術與製造能力轉化為實質營運成長。

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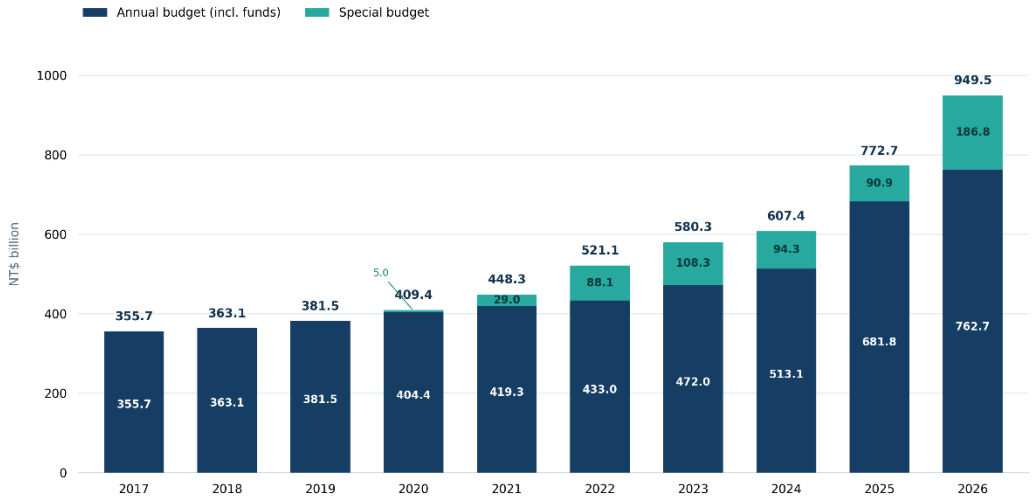
個股	代碼	預估每股盈餘(當地幣別)			每股盈餘成長率(%)			預估本益比/本淨比			目標價(當地幣別)	最新收盤價(當地幣別)	評等
		2025	2026E	2027E	2025	2026E	2027E	2025	2026E	2027E			
長榮航太	2645 TT	5.46	8.63	9.68	11.4	58.0	12.2	31.6	20	17.8	220	172.5	買進
漢翔	2634 TT	0.78	2.58	2.45	-66.2	230.8	-6.2	82.2	24.8	26.2	73.5	64.1	中立
創泓科技	7714 TT	6.33	-	-	24.9	-	-	26.5	-	-	-	168.0	未評等
台船	2208 TT	-1.7	-	-	-	-	-	N/A	-	-	-	18.45	未評等
龍德造船	6753 TT	5.98	7.73	5.28	-4.3	29.3	-31.7	21.9	16.9	24.8	-	131.0	未評等

國泰期貨 國泰證券子公司

從預算增加到產線擴張，台灣軍工迎來新一輪成長

過去十年，台灣國防支出的成長，逐步為軍工產業累積更大的發展空間。如今隨著無人機、無人艇及反制系統成為國防重點，這股需求正從大型軍備採購，延伸至電子、航太與造船供應鏈。以廣義國防總支出觀察，2026 年原規劃規模達 9,495 億元，若將其分為年度預算與特別預算兩類，年度預算含基金約 7,627 億元，特別預算約 1,868 億元。年度預算支撐人員、裝備維持與持續性軍事投資，特別預算則加快重點裝備籌獲，兩者共同擴大產業需求。行政院先前也通過的 2027 年總預算案，進一步將廣義國防總支出推升至 1 兆 1,225 億元，較 2026 年原規劃增加約 18%，目前送至立法院審議，仍待後續觀察。

表一：Taiwan defense budget from 2017 to 2026



資料來源：行政院主計處、國防部、國泰證期研究部整理

六年 2,400 億元，行政院再提無人載具追加預算

如果說國防預算擴張為產業打開市場，無人載具專法則讓企業看見更長期的投入方向。9 月行政院副署的《強化國防自主暨無人載具產業發展條例》，要求循年度預算程序，連續六年編列 2,400 億元，每年以 400 億元為原則，必要時可依需求增編。政策範圍涵蓋可信任供應鏈、研發與測試場域，也將資源引導至具實質技術及製造能力的業者，讓國防採購同時成為產業升級的推力。此外，行政院也提出的 1,457 億元國防追加預算，其中「濱海監偵型」、「濱海攻擊型無人機」及「小型自殺無人艇」三項合計 559 億元。國防部已完成招商說明，希望在預算通過後加快採購，於 2027 年形成初始戰力。換句話說，產業正走到從樣機、測試走向批量交付的關鍵階段：無人機廠需要準備產線，零組件供應商需要建立穩定供貨能力，系統整合商則必須完成通信、控制與任務酬載的驗證。後續追加案等待立法院三讀。

國泰期貨 國泰證券子公司

國內無人機需求轉向大量部署及偵察視野延伸

烏克蘭讓產業看見小型無人機也能改變軍事裝備的需求結構。當無人機從少數專業單位的裝備，成為前線頻繁使用、需要持續補充的工具，產業競爭也開始從單架性能，延伸至生產速度、成本控制與快速改版。台灣正沿著這條路徑前進：政府已透過前幾年軍用商規無人機採購累積量產經驗，9 月經濟部亦公布中光電相關平台已完成軍用商規量產交付，顯示國內供應鏈逐步跨過展示與驗證階段。未來市場將形成兩條並行的成長路線。

1. **低成本、大量部署型無人機**，重點在可負擔的採購成本、穩定補充能力與持續更新；烏克蘭常見的 FPV(第一人稱視角)機型，即展現這類需求的規模潛力。台灣的濱海任務還需要兼顧飛行距離、海島環境與可信任供應鏈，因此，共用零組件、模組化設計及多家供應，將是逐步降低成本的重要方向。

2. **高階情監偵無人機**，透過更長滯空時間、更高品質的感測器與穩定資料鏈，擴大海域觀測範圍。如漢翔已確認與 Shield AI 合作取得海巡署 V-BAT 艦載無人機案，該平台具垂直起降與超過 12 小時續航能力，有利在有限甲板空間下執行長時間監偵。海軍公開規劃亦包括 2026~2029 年以約 167.9 億元籌購 32 架銳鳶二及 8 套導控站，另以 360 億元籌購 280 架艦載智慧垂直起降無人機及 140 套導控系統。這類跨年度計畫帶來的不僅是機體製造，還涵蓋光電酬載、地面站、訓練與後勤服務；後續採購與在地分工逐步落實後，具備整合及長期維保能力的廠商，有機會取得更高附加價值。

圖一：Ukrainian FPV Drone - General Cherry

資料來源：烏克蘭國防部、國泰證期研究部整理

當前線需要更多「看得見」的能力，後方的基地、機場與重要設施，也需要更完整的低空防護能力。烏克蘭國防部先前就曾指出，Mavic、Autel、Matrice 等多旋翼系統仍持續大量採購，用於前線觀測，並同步測試替代產品，也說明小型監偵與大型長航時平台各有用途。對台灣而言，這種任務分層將帶動不同等級的飛控、射頻、光學與電池管理需求，也讓資安、韌體更新及檢測認證成為供應鏈的重要環節。

另一方面，反無人機系統正從單一干擾器，走向涵蓋偵測、辨識、指揮與處置的完整架構：雷達與射頻感測發現目標，光電及熱影像協助確認，指揮控制平台整合資訊，再搭配適合的反制手段。北約 2026 年公布的反無人機框架採購，同樣採取多感測器與統一指管架構，顯示市場對整合能力的重視。國內創泓科技已推出 C2 無人機防禦系統，整合訊號偵測、光學辨識及反制模組，9 月進一步介紹 Sentrycs 代理及在地合作布局，商業機會可由設備供應延伸至系統整合、技術支援與持續更新。隨光纖操控與自主飛行等技術發展，單靠切斷無線通信的反制方式，也需要更多感測器與處

置方案配合。研究部認為，反無人機市場的競爭核心，將逐步轉向跨品牌整合、現場驗證及軟體更新能力；能在複雜環境中降低誤警、穩定辨識目標並持續維護系統的業者，更有機會建立長期客戶關係。後續觀察國防及關鍵基礎設施標案的導入進度。

圖二：Ukrainian Small ISR Drone - Mavic 3 Pro

資料來源：烏克蘭國防部、國泰證期研究部整理

無人艇從船體製造延伸至自主航行與海洋感測

當船艇能夠在無人乘載的情況下完成巡邏、觀測與運補，造船產業的競爭範圍，也從船體與推進系統延伸至自主航行、通信與感測器整合。水面無人艇(USV)可執行海域監偵、通信中繼、搜救、設施巡檢及軍事任務；水下無人載具(UUV)則將觀測範圍帶到海面之下，應用於海床測繪、海纜與管線檢查、反水雷及水下環境偵測。台灣既有的軍艦、商船與高速小型船經驗，也為這個新市場提供製造基礎。台船「奮進魔鬼魚」船長 8.6 公尺、寬 3.7 公尺，最高航速超過 35 節，採三體船與模組化酬載設計，並於 2025 年取得 ABS 原則性設計認可(AIP)，為後續產品驗證建立基礎。9 月台船進一步與 Havoc 宣布合作，擬推動自主水面載具授權共同生產，展現國內造船能力與國際自主控制技術結合的方向。龍德造船則以「黑潮」切入市場，船長 9 公尺、寬 2 公尺、最高航速超過 40 節，在 25 節巡航條件下航程約 250 公里，並具自主航行及目標識別功能。隨產品逐步成形，後續量產採購與任務驗證將成為主要觀察重點。水面平台有望先帶動船體、航控、資料鏈與任務模組需求，水下市場則進一步考驗聲學設備、耐壓密封、能源及回收能力。

圖三：台船-奮進魔鬼魚(Endeavor Manta USV)

資料來源：台船、國泰證期研究部整理

另一方面，台海海象則是無人載具商業化不可忽略的限制。交通部觀測數據顯示，臺中港測站冬季平均波高約 2.1 公尺，比夏季高約 1.3 公尺，足以說明不同季節的差異性。對小型無人艇，波浪會增加船體拍擊與姿態變化，影響感測器穩定、推進效率、航速及回收；水下航行雖可減少直接受表面波浪影響，出入水、母船收放及浮出通信仍受海況約束。對無人機而言，浪高主要透過甲板運動

與海面飛濺影響艦載起降，飛行本身則更直接受到陣風、降雨及鹽霧影響。因此，廠商需要驗證的不只是最高航速，還包括不同波高、週期、浪向及酬載下的任務完成率、穩定通信、自主避碰、故障處置與回收成功率。研究部認為，能累積跨季節海試資料、維修經驗及可重複交付能力的船廠將更具競爭優勢。

從政策走向可持續交付，軍工供應鏈蓄勢

國防預算擴張、無人載具產業政策及非紅供應鏈需求，有利支持業者持續投入研發與量產。無人系統的產業效益亦將由載具製造延伸至感測器、通信、任務整合與反制設備，並隨部署規模增加，帶動備件、維修、軟體更新及訓練等後續需求，為供應鏈建立持續性收入來源。

無人機方面，長榮航太(2645)則以軍用商規艦載監偵無人機切入市場，具備航太製造與品質管理能力的延伸優勢。漢翔(2634)透過與 Shield AI 合作的 V-BAT 垂直起降無人機取得海巡案，後續成長機會包括在地整合、維保及新增採購；反無人機方面，創泓科技(7714)布局 C2 無人機防禦系統與 Sentrycs 反無人機方案，可望拓展偵測、辨識、反制及指揮控制整合業務。

表二：無人機規格與任務定位比較表

類型	代表機型 / 製造商	機體規模	續航/滯空時間	主要用途與產業重點
FPV 無人機	General Cherry/烏克蘭	多種尺寸機架	17~30 分鐘	前線任務；低成本、量產與快速補充
小型多旋翼監偵機	Mavic 3 Pro/DJI	展開尺寸約 35cm	43 分鐘	近距離監偵；便攜、影像品質與快速部署
艦載垂直起降監偵機	V-BAT/Shield AI	翼展 3.8m	12 小時以上	艦載監偵；垂直起降、酬載整合與維保
中型固定翼海洋監偵機	銳鳶二型/中科院	翼展約 12m	16 小時以上	海域監視與預警；資料鏈、感測及指管整合
大型中高度長航時無人機	MQ-9B SkyGuardian/ General Atomics	翼展約 24m	40 小時以上	長航時廣域監偵；地面站、通信與後勤支援

資料來源：國泰證期研究部整理

無人艇方面，台船(2208)的「奮進魔鬼魚」採三體船設計與模組化酬載，具備拓展多元海上任務的潛力；龍德造船(6753)的「黑潮」則以高速操縱與匿蹤設計為特色，結合既有軍用小型船建造經驗，爭取量產與後勤需求。研究部認為，後續隨產品驗證與採購逐步落實，具備系統整合、穩定交付及長期維保能力的業者，將有機會將政策需求轉化為實質營運成長。

國泰期貨 國泰證券子公司

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國泰期貨 國泰證券子公司

投資評等定義

英文評等	中文評等	潛在報酬率
STRONG BUY	強力買進	> +30%
BUY	買進	介於+15% ~ +30%
NEUTRAL	中立	介於-15% ~ +15%
SELL	賣出	< -15%

註:潛在報酬率=(6個月內目標價/本次報告評等最近一天收盤價－1) 100

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國泰期貨 國泰證券子公司

聯友金屬 7610 TT (綠能環保)

供應跟不上需求擴大

投資與評價

1. 3Q26 受惠國際鎢供應持續吃緊、鎢價維持高檔，加上屏東二廠新增 30%產能陸續開出，自售比重持續提升，預估營收季增 47.5%，毛利率 66.5%，EPS 27.32 元創單季新高。
2. 2027 年成長動能由鎢價轉向擴產與產品高值化，屏南三廠、屏科廠陸續開出，產品由鎢酸鈉延伸至 APT、氧化鎢、鎢粉及碳化鎢，自售比重有機會挑戰 60%，預估 2026/27 EPS 83.77/109.27 元。
3. 目前股價評價位於近兩年本益比 10～24 倍區間中段，考量非中鎢供應鏈稀缺性、自售比重提升及 2027 年新產能接棒，給予區間中間偏上的 22 倍 FY27E PER，給予強力買進評等，目標價 2,400 元(22x FY27E EPS)。

重點更新

3Q26 鎢價與自售業務推升獲利續增

預估 3Q26 營收 28.43 億元 (+47.5% QoQ)，主要受惠國際鎢價維持高檔、鎢酸鈉自售業務成長，以及屏東二廠新增 30%產能陸續開出；毛利 18.90 億元 (+44.8% QoQ)，毛利率 66.5% (-1.3ppts QoQ)，營收與毛利仍維持高速成長。營業費用 1.88 億元(+55.6% QoQ)，稅後淨利 13.27 億元(+40.3% QoQ)，EPS 27.32 元創單季新高。預估 4Q26 隨新產能進一步量產、自售比重過半，營收 29.59 億元 (+4.1% QoQ)，惟費用投入增加，EPS 降至 26.17 元。

2027 年產能與高值化產品接棒

預估 2027 年營收 132.79 億元 (+52.1% YoY)，成長動能由鎢價逐步轉向產能擴張及產品高值化，屏南三廠與屏科廠新產能陸續開出，產品由鎢酸鈉延伸至 APT、氧化鎢、鎢粉及碳化鎢，自售比重有機會挑戰 60%；毛利 87.22 億元 (+48.9% YoY)，毛利率 65.7% (-1.4ppts YoY)。惟新廠與新工序投入增加，營業費用 20.62 億元(+202.8% YoY)，使營業利益 66.60 億元(+28.7% YoY)、營益率 50.2%(-9.1ppts YoY)；稅後淨利 53.09 億元(+30.4% YoY)，EPS 109.27 元再創年度新高。後續營運重點將轉向新產能利用率，以及高值化產品的提升速率。

強力買進 – 初次評等

目標價：2,400 元 +34%
收盤價：1,790 元

報告日期	評等	目標價
2026/10/07	買進	2,400 元

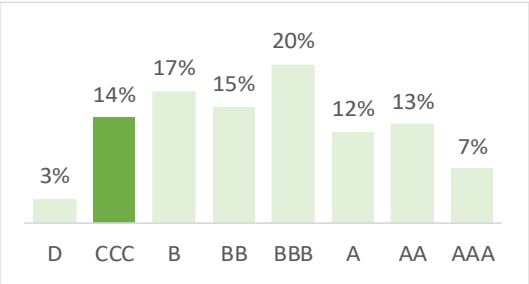
	國泰	市場	差異
目標價	2,400	--	--
2026 EPS	83.77	77.60	+7.9%
2027 EPS	109.27	107.51	+1.6%
現在 P/E	16.38	--	--

公司資訊

市值(百萬元)	105,430
自由流通股數比例(free float ; %)	72.93
20 日平均成交量(百萬元)	1,770
每股淨值(元,2027E)	219.48
股價淨值比(倍)	7.68
現金股利率(%)	0.56
股東權益報酬率(%,2026E)	50.98
外資持股比率(%)	18.99

聯友金屬 ESG 評等

ESG 評等	社會(S)	經濟(E)	環境(E)	揭露(D)
CCC	40%	39%	16%	33%



呂理舜/ 綠色能源、電動車
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項目(百萬元)	2Q'26	3Q'26(E)	4Q'26(E)	1Q'27(E)	2025	2026(E)	2027(E)
營業收入	1,928	2,843	2,959	3,010	1,406	8,731	13,279
稅前盈餘	1,188	1,702	1,620	1,599	303	5,176	6,655
稅後盈餘(母公司業主)	946	1,327	1,271	1,263	241	4,070	5,309
營收成長率(%)(QoQ/YoY)	92.42	47.50	4.07	1.72	27.90	521.20	52.08
毛利率(%)	67.73	66.47	64.94	64.10	30.78	67.08	65.68
營益率(%)	61.47	59.87	54.81	53.22	22.00	59.28	50.15
稅後盈餘成長率(%)(QoQ/YoY)	80.22	40.25	-4.2	-0.64	311	1587.07	30.44
每股盈餘 EPS(元)	19.48	27.32	26.17	26.00	4.97	83.77	109.27

EPS 以調整後股數 48.59 百萬股計算

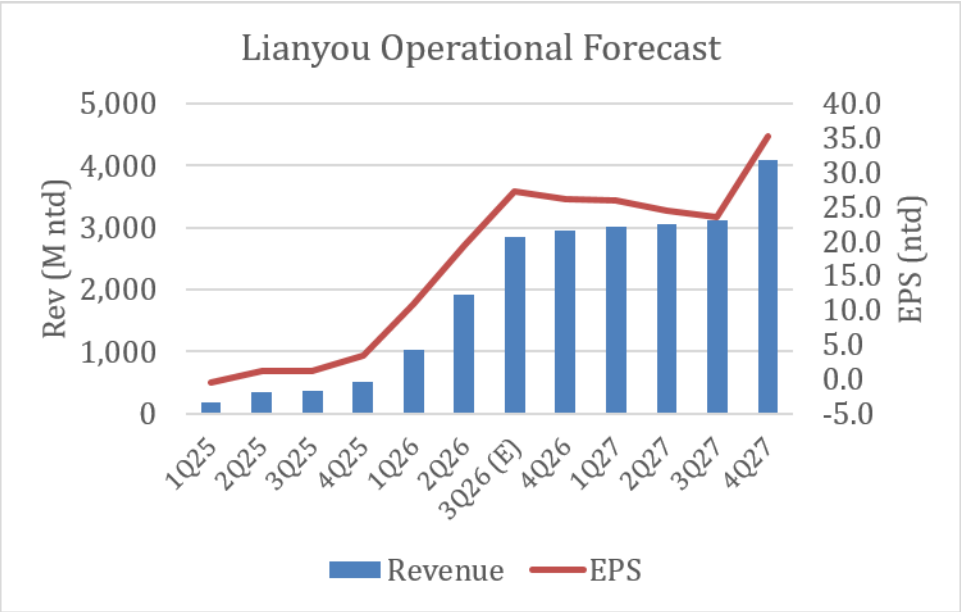
聯友金屬 7610 TT

訪談內容

3Q26 鎢價與自售業務推升獲利續增

預估 3Q26 營收 28.43 億元（+47.5% QoQ），主要受惠國際鎢供應持續吃緊、鎢價維持高檔，加上鎢酸鈉自售業務成長，以及屏東二廠增加的 30%新產能部分陸續開出；國際市場中國持續限制鎢相關產品出口，尤其對美國與日本出口限制趨嚴，使國際客戶加速建立非中供應鏈，聯友以二次硬質合金回收生產鎢產品，在競爭者有限的情況下，可望維持穩定的需求與產品售價。預估毛利 18.90 億元（+44.8% QoQ），毛利率 66.5%（-1.3ppts QoQ），雖自售產品比例提高，但毛利率較 2Q26 高基期略降；營業費用 1.88 億元（+55.6% QoQ），營業利益 17.02 億元（+43.7% QoQ），營益率 59.9%（-1.6ppts QoQ）。預估稅後淨利 13.27 億元（+40.3% QoQ），EPS 27.32 元創歷史新高。除鎢產品外，公司亦持續推進循環再製鈷粉客戶認證與出貨，作為後續產品高值化方向。

圖 1：聯友金屬獲利預估



資料來源：國泰證期研究部整理

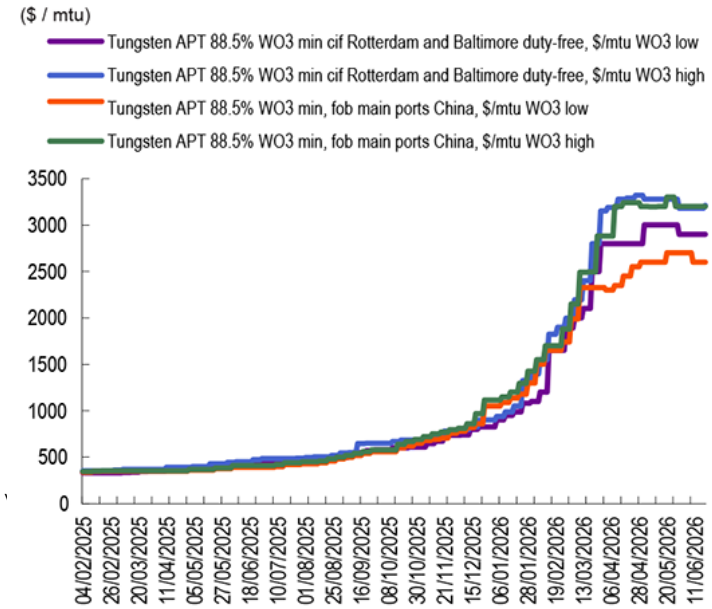
4Q26 營收續創高峰

預估 4Q26 營收 29.59 億元（+4.1% QoQ），從上季大幅成長後仍維持季增，主要受惠鎢產品自售業務持續成長，以及預期國際鎢供需仍偏緊，且公司新產能進一步進入量產階段，使單季自售比重過半。中國掌握全球主要原生鎢供應，出口限制使海外供應鏈短期難以快速補足缺口，軍工、半導體及 AI 相關精密加工需求則持續支撐國際鎢價。預估毛利 19.21 億元（+1.7% QoQ），毛利率 64.9%（-1.5ppts QoQ）；惟營業費用增至 3.00 億元（+59.6% QoQ），在於新的工序流程投資持續投入，使營業利益 16.22 億元（-4.7% QoQ），營益率 54.8%（-5.1ppts QoQ）。預估稅後淨利 12.71 億元（-4.2% QoQ），EPS 26.17 元；整體而言 4Q26 營收與毛利仍向上，但費用增加使獲利較 3Q26 高峰略為回落。

2026 年鎢價大漲與自售比例增加

預估 2026 年營收 87.31 億元(+521.2% YoY)，受惠國際鎢價大幅上漲、鎢酸鈉自售業務放量，以及非中鎢供應鏈需求快速提升。中國鎢出口限制使國際市場供給趨緊，聯友具備廢硬質合金回收、濕法冶金及既有海外客戶基礎，使鎢價上漲透過自售產品，直接轉化為獲利成長。預估毛利 58.57 億元(+1,254.0% YoY)，毛利率由 2025 年的 30.8%大幅提升至 67.1%(+36.3ppts YoY)，主因仍在產品國際報價及自售業務成長明顯；營業費用 6.81 億元 (+452.3% YoY)，營業利益 51.76 億元 (+1,573.7% YoY)，營益率 59.3% (+37.3ppts YoY)。預估稅後淨利 40.70 億元 (+1,587.1% YoY)，EPS 83.77 元創歷史新高，2026 年為公司材料處理量由加工大幅轉向自售產品後，獲利結構明顯轉變。

圖 2：鎢金屬價格走勢



資料來源：Fastmarkets, 國泰證期研究部整理

2027 年產能與高值化產品接棒

預估 2027 年營收 132.79 億元 (+52.1% YoY)，在 2026 年鎢價與自售業務快速成長後，2027 年成長動能逐步由價格轉向產能擴張及產品高值化。公司規劃由既有鎢酸鈉進一步向 APT、氧化鎢、鎢粉及碳化鎢延伸，屏南三廠與屏科廠相關新產能預計陸續開出，鎢相關加工產能有機會擴增倍翻，若產品認證與量產順利，可望提高公司在鎢產業鏈的垂直整合程度。預估毛利 87.22 億元 (+48.9% YoY)，毛利率 65.7% (-1.4ppts YoY)，在營收快速擴張下仍維持六成以上高毛利率；惟配合新廠、新產品及營運規模擴張，營業費用增至 20.62 億元 (+202.8% YoY)，使營業利益 66.60 億元 (+28.7% YoY)，營益率降至 50.2% (-9.1ppts YoY)。預估稅後淨利 53.09 億元 (+30.4% YoY)，EPS 109.27 元再創年度新高；2027 年營收與毛利仍維持高速成長，但費用投入增加使獲利增速低於營收，後續營運重點將轉向新產能利用率，及 APT、氧化鎢、鎢粉、碳化鎢等高值化產品的提升速率。

非中鎢供應鏈稀缺性提升

中國鎢出口限制使國際供應持續偏緊，聯友藉二次硬質合金回收與濕法冶金技術，受惠國際客戶加速建立非中供應鏈，2026 年鎢價上漲、自售比重提升及屏東二廠新增產能開出，帶動獲利明顯成長；2027 年成長動能則由格進一步轉向產能擴張與產品高值化，隨屏南三廠及屏科廠新產能陸續開出，產品由鎢酸鈉延伸至 APT、氧化鎢、鎢粉及碳化鎢，自售部分有機會在 2027 年挑戰 60%。預估 2026/27 EPS 為 83.77/109.27 元。3. 目前股價評價位於近兩年本益比 10～24 倍區間中段，考量非中鎢供應鏈稀缺性、自售比重提升及 2027 年新產能接棒，給予區間中間偏上的 22 倍 FY27E PER，給予強力買進評等，目標價 2,400 元(22x FY27E EPS)。

產業概況

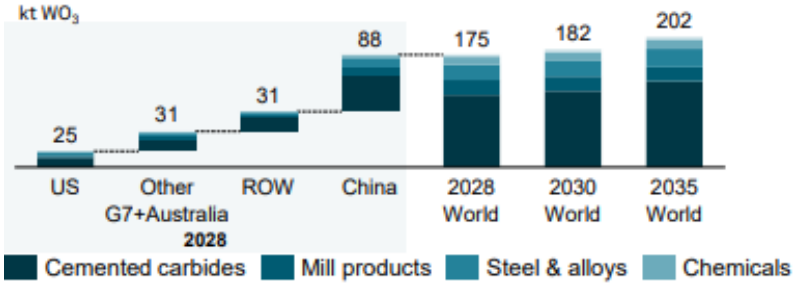
鎢金屬需求持續由工業與國防驅動

2026 年中國鎢總供給約 8.51 萬噸 W、YoY +12.4%，其中原生礦約 4.36 萬噸、幾乎零成長，回收鎢則由高價刺激增加約 50%至約 3.0 萬噸，使回收料占供給由 2025 年的 25–26%升至約 35%；換言之 2026 幾乎所有可快速增加的供給彈性都來自回收處理，使得非新礦山供應的競爭也明顯升高。顯示高鎢價已開始抑制價格敏感度較高的傳統用途，但切削工具、航太與國防等剛性需求仍維持成長。全球鎢需求主要仍來自硬質合金，應用於切削工具、礦山鑽頭、汽車與精密加工；其中礦業與營建約占 2025 年全球 WC 市場 38.5%。

國防需求的重要性在 2026 年進一步提高，美國與以色列對伊朗軍事行動、俄烏戰爭持續消耗飛彈與穿甲彈用鎢，這類用途與硬質合金不同，使用後無法回收，使其對原生鎢形成永久性消耗。因全球每年鎢需求約 10～11 萬噸，其中原生礦約占 65%、回收料約 35%，現有需求須依賴二次資源補足。短期新礦與冶煉產能建置時間長，因此即使高價刺激新增供給，供需緊張仍不容易快速逆轉。

各應用來看硬質合金(Cemented carbide)佔比最大，主要用於切削刀具、採礦鑽頭、耐磨零件與國防彈藥等。加工材/材加工品 (Mill products): 如鎢絲、鎢棒、鎢板等。鋼鐵與合金 (Steel & alloys) 用於高速鋼、耐熱合金鋼等。化學品 (Chemicals): 催化劑、顏料及化學試劑。其中 PCB 鑽針需求在近年也快速增加，據估計因 AI server PCB 變得更厚、更多層、更難鑽，高速低損耗材料使 WC 鑽針磨耗大幅提高。AI server PCB 已朝 28–46 層甚至更高層數發展，而一般鑽針原本約可加 工 3,000 孔，在高階材料下可能降到約 600 孔，在相同鑽孔量可能需要接近 4–5 倍鑽針消耗。半導體需求在 4Q26 仍然偏強，其中第一類是半導體製程本身使用的高純度鎢/WF_6 等材料，另一類是晶圓、封裝、設備與零組件加工所使用的 WC 超硬工具。整體而言隨著 AI GPU/ASIC、HBM 與先進封裝擴產，使高純度 W 與精密 WC 工具需求維持穩定成長。

圖 3：鎢金屬主要應用



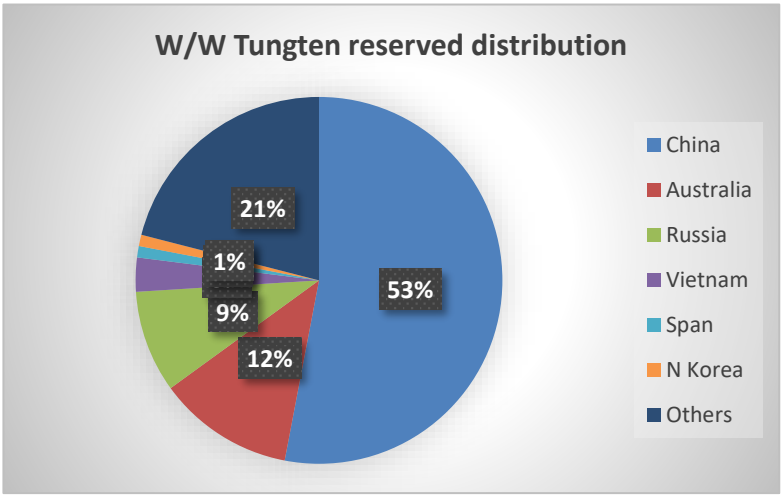
資料來源：S&P Global, 國泰證期研究部整理

中國七成產能限制鎢出口影響大

全球鎢供應高度集中，在 2025 年全球鎢精礦開採產量約 8 萬多噸，中國占約 75%，且中國同時掌握從礦山、APT、鎢粉到碳化鎢的完整加工鏈，鎢的供應風險除了礦源集中，更包含中游精煉能力集中。2H25 開始中國開始收緊鎢礦開採配額並實施出口管制後，非中國市場供給明顯轉緊，價格也顯著提升。

全球鎢市場主要依賴中國供應，因此中國價格大致主導全球定價；但出口限制與供應鏈安全政策出現後，國際買家開始建立中國以外的礦源、冶煉及回收體系，使得國際報價持續走高，鎢回收目前約占全球供給三成，雖然產地來源風險較低，但是既有供應鏈的產能瓶頸，使得回收流程取得鎢的價格持續走高。雖陸續有些新礦場宣告將投入開發，但仍需兩年以上時間準備，舊有礦場的重啟也面臨到回收率與經濟性的挑戰，且對於環境衝擊高未必適合歐美國家，目前最為環保與的作法仍是從回收鎢開始加工。透過廢鎢回收陸續轉化為鎢酸鈉、APT、氧化鎢、鎢粉、碳化鎢，將有利於賣方市場的精鍊供應商，邁向更高值化的末端產品發展。

圖 4：鎢礦蘊藏分布



資料來源：USGS, 國泰證期研究部整理

美國積極多元化鎢礦來源

3Q26 美國政府承諾向 Elmet Group 投入 US$450mn，並搭配 DLA 最高 US$2bn 鎢戰略庫存採購，政策目標已從單純取得礦源轉向礦山+APT/精煉+零組件的垂直整合。US$450mn 投資將用於擴張美國鎢製造能力、取得長期原料、投資美國及盟國礦山與建立新的 refining/conversion 能力；其中初期資金約 US$200mn。另一方面 DLA 與 Elmet 簽訂最高 US$2bn IDIQ 合約，最低保證採購 US$150mn，品項直接涵蓋鎢礦、鎢精礦及鎢酸鈉；Elmet 待新增礦山、offtake 與 processing capacity 形成後才向戰略庫存交貨，以避免政府囤貨擠壓既有民間供應。EQ Resources 規劃向美國供應 4,000 噸鎢精礦；但 Nevada APT processing facility 全面恢復營運預計要到 2H28，因此 4Q26 政策首先增加的是戰略採購與長約需求，而非立即增加 APT 現貨供給。這代表短期甚至可能強化非中國的原料競爭，但預計要數年後後才有機會逐步改善美國中游加工缺口。

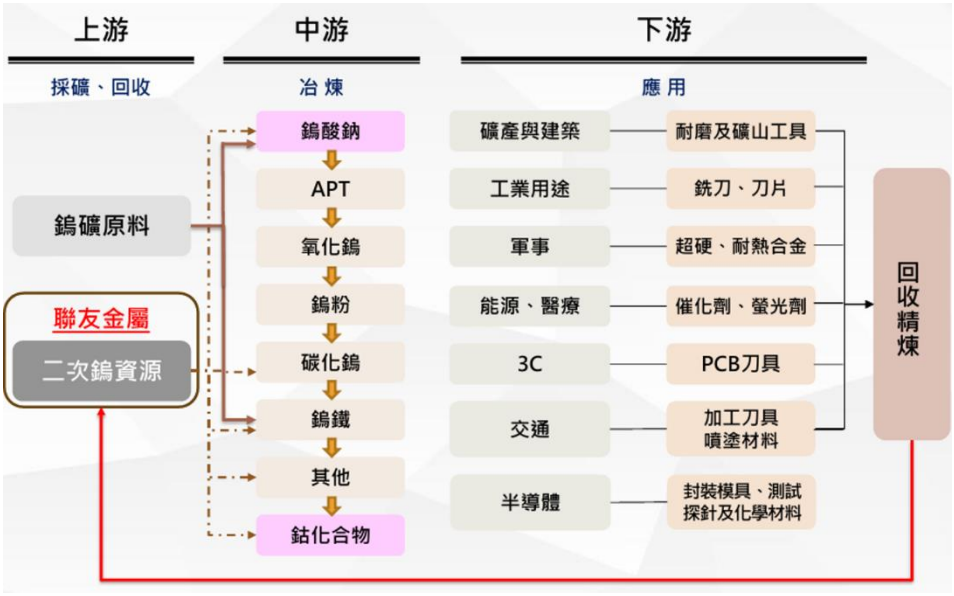
鈷金屬供給仍有短缺風險

2025 年全球鈷礦產量約 31 萬噸，其中剛果民主共和國約 23 萬噸、占七成比重，全球約 50%儲量亦集中於剛果，而中國又掌握全球七成以上鈷精煉能力，因此上游礦源與中游精煉均高度集中。需求方面 2025 年全球鈷消費動力電池占比超過 65%，EV 仍是最大需求來源；雖電池高鎳化、低鈷化以及 LFP 滲透率提升，會限制單位電池鈷用量成長，但 IT 電池的使用仍以氧化鈷為主要材料。而長期更值得注意的是航空發動機、燃氣渦輪、國防設備所使用的高溫合金，以及硬質合金與高性能磁鐵需求。高溫合金目前約占鈷需求 15～20%，工業工具、硬質合金與催化劑約一成，使鈷逐漸從單一新能源金屬轉變成兼具新能源與戰略材料屬性的關鍵金屬；在供應集中及可追溯要求提高下，循環再生鈷、高純度鈷粉與非中國精煉能力的戰略價值亦將持續提高。

公司簡介

聯友金屬不依賴原生鎢礦，而是以廢硬質合金、鎢鋼及研磨屑等二次料作為鎢來源，經濕式冶金回收後製成鎢酸鈉（Sodium Tungstate），同時回收鈷，形成 WC/鎢鋼廢料、鎢酸鈉、APT、氧化鎢的城市礦山供應鏈。公司目前鎢酸鈉年產能約 6,000 噸，2025 年鎢酸鈉出貨約 4,728 噸、YoY +8%，其中代工約占 52%、自售約 48%，相較 2024 年代工 42%、自售 58%，可看出自有設備利用率已處相對高檔；公司規劃透過新廠將鎢酸鈉產能進一步提高至約 12,000 噸/年、較現有產能增加 100%，若以鎢酸鈉 $Na_2WO_4{\cdot}2H_2O$ 含鎢重量約 55.7%換算，滿產相當於一年可處理約 6,700 噸 W contained。公司並規劃進一步向下游 APT 與氧化鎢延伸，使產品由目前以鎢酸鈉為主，逐步建立回收料、鎢酸鈉、APT、WO_3的垂直整合能力。

圖 5：聯友金屬產業地位



資料來源：聯友金屬，國泰證期研究部整理

ESG 政策

公司在社會面評分上屬中上，然而需在經濟與環境面與揭露面強化:
經濟面: 股東權益方面，公司優點為董事長親自出席股東常會；股東會結束後公開揭露逐項議案之詳細表決資訊。公司可加強處為當年度未召開 2 次以上法人說明會。
法規遵循方面，公司於題項評估期間未有不合理或受主管機關處分之情事。獨立董事方面，公司優點為至少 2 名獨立董事之連續任期不超過 9 年。公司可加強處為獨立董事席次未達董事席次二分之一以上。董事會永續治理方面，公司優點為有設置永續發展相關委員會；有制訂環境、社會及治理之中長期發展策略；當年度永續報告書經董事會通過。公司可加強處為負責永續發展之最終負責人非董事長或總經理。董事會績效與薪酬方面，公司優點為將董事績效評估與薪資報酬連結。公司可加強處為企業永續績效尚未與高階主管及董事薪酬進行連結。
董事會多元性方面，公司優點為揭露董事會多元化政策並揭露落實情形；董事會成員具備不同專業背景。公司可加強處為任一性別董事席次未達 1/3。永續創新方面，公司優點為取得綠色認證。公司可加強處為未揭露綠色相關部門設置情形與職責；未揭露綠色/永續採購之政策或金額。
此外，產業創新方面，公司優點為對其專利之智慧財產權進行管理與保護，並揭露量化績效。
法規遵循方面，公司近 3 年並未受公平交易委員會懲罰。

環境面: 環境系統與治理方面，公司優點為積極將綠色概念融入營運過程，打造綠色價值鏈；榮獲 ISO 14001 環境管理認證；將環境管理作為與聯合國永續發展目標(SDGs)連結。空氣管理方面，公司優點為制定空氣管理政策、揭露空氣汙染排放數據；公司可加強處為未揭露量化空氣汙染減量目標。法規遵循方面，評估期間未違反空氣汙染相關法規或已針對違反情形提出具體改善措施。氣候變遷管理方面，公司可加強處為未說明董事會或管理階層對氣候變遷議題之監督頻率或事項；未揭露使用具公信力的 1.5 °C 以下及 3 °C 以上溫度情境評估氣候風險；未揭露與供應鏈或合作夥伴共同推動因應氣候變遷相關策略；未揭露氣候變遷對公司所產生的機會項目。
溫室氣體管理方面，公司優點為揭露至少三年的溫室氣體範疇一與二排放量，並揭露涵蓋範圍或佔營收比率。公司可加強處為未揭露溫室氣體管理專責單位及相關管理政策；未揭露範疇三排放量；未揭露量化溫室氣體減量目標；未揭露是否通過碳盤查認證。節能管理方面，公司優點為揭露用電或節能數據。公司可加強處為未揭露能源管理專責單位及相關管理政策；未揭露節能量化目標；未揭露能源風險對營運的影響評估結果。再生能源管理方面，公司可加強處為未揭露再生能源管理專責單位及相關推動政策；未揭露再生能源使用量或發電量；未揭露量化再生能源發展目標。廢水管理方面，公司優點為制定廢水管理政策、揭露廢污水排放量；公司可加強處為未揭露量化廢汙水減量目標。
揭路面: ESG 揭露方面，公司優點為依據 GRI 準則編寫永續報告書；於永續報告書清楚說明揭露範疇；建置英文網站，並包含財務、業務及公司治理相關資訊；於永續報告書揭露重大性分析細節。公司可加強處為永續報告書未依據 SASB 進行對應揭露；未上傳英文版永續報告書於公開資訊觀測站；永續報告書尚未經過第三方機構確信或保證；未於官網揭露利害關係人問卷，供利害關係人回饋；未說明公司營運對應 SDGs 目標之具體作為；未揭露使用雙重重大性分析重大性議題。

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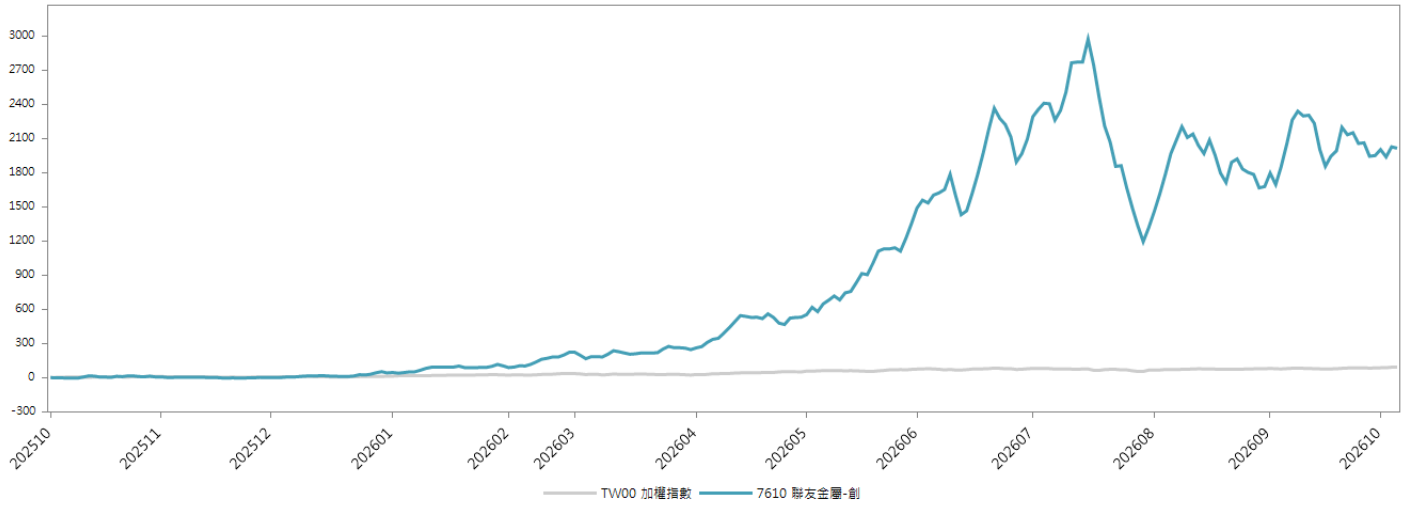
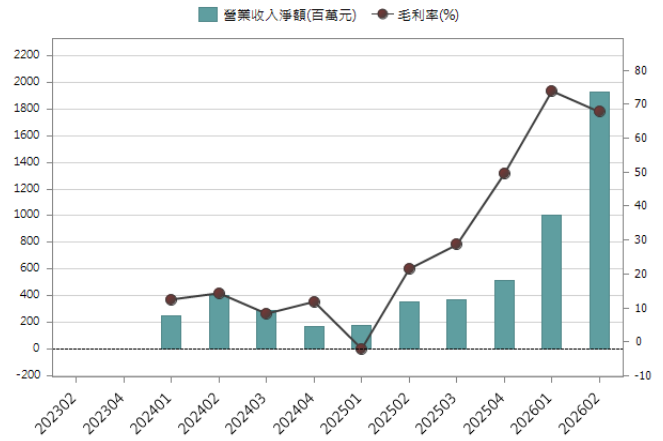
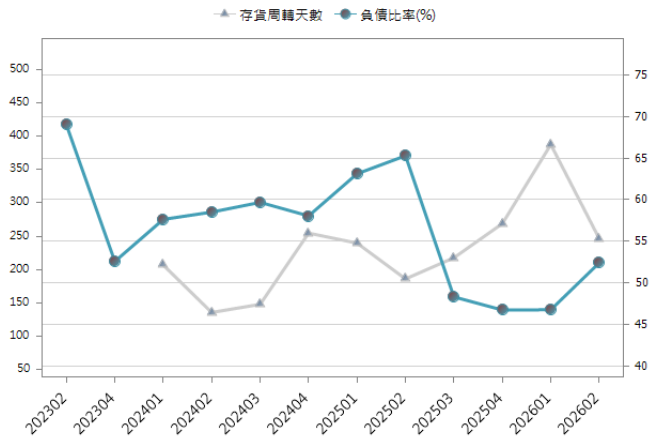
股價相對大盤走勢圖



近三年單季營收與毛利率走勢圖

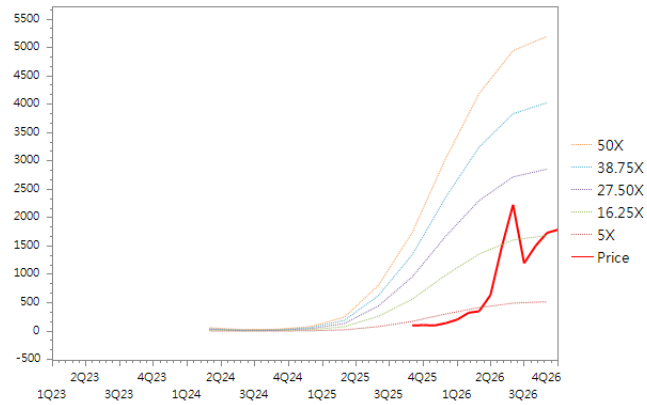


近三年負債比率與存貨週轉天數



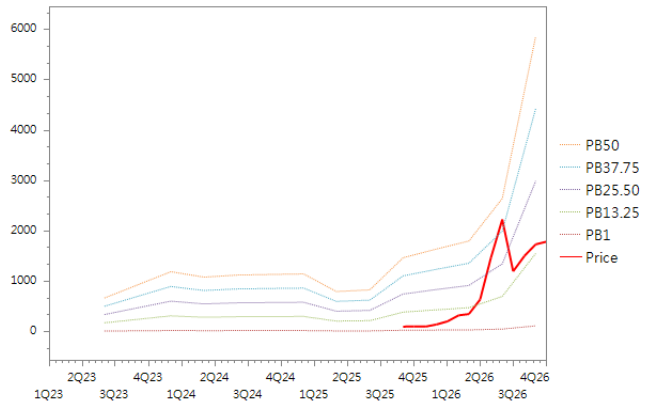
未來 12 個月 P/E(本益比)區間圖

PE band



未來 12 個月 P/B(股價淨值比)區間圖

PB band



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損益表

至 12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
營業收入	916	1,099	1,406	8,731	13,279
營業成本	835	967	973	2,874	4,557
營業毛利	82	132	433	5,857	8,722
營業費用	66	69	123	681	2,062
營業利益	16	63	309	5,176	6,660
營業外收入與支出	-14	3	-6	0	-5
其他收入	2	2	2	5	4
其他利益與損失	0	11	13	25	23
財務成本	17	16	28	40	41
權益法認列之投資損益	0	0	0	0	0
其他項目	1	6	7	10	9
稅前純益	1	66	303	5,176	6,655
所得稅費用	-3	7	62	1,105	1,345
稅後純益	4	59	241	4,071	5,310
母公司業主- 稅後純益	4	59	241	4,070	5,309
非控制權益- 稅後純益	0	0	0	1	1
稅前息前折舊攤銷前淨利	61	114	387	5,272	6,660
每股盈餘 (元)	0.08	1.21	4.97	83.77	109.27

資料來源:公司資料及國泰預估

資產負債表

12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
資產總額	1,554	1,894	3,025	11,001	21,003
流動資產	747	761	1,731	8,904	18,017
現金及約當現金	48	85	261	4,279	12,526
短期投資	14	95	49	66	66
存貨	435	431	923	2,758	3,302
應收帳款及票據	141	42	167	845	1,167
其他流動資產	108	109	331	957	957
非流動資產	807	1,133	1,294	2,097	2,985
長期投資	0	0	3	11	4
固定資產	460	980	1,228	1,965	2,861
其他非流動資產	347	153	63	121	121
負債總額	817	1,100	1,413	5,287	10,323
流動負債	709	599	881	3,473	8,011
應付帳款及票據	188	132	207	667	793
短期借款	199	438	591	2,352	6,764
其他流動負債	322	29	82	454	454
長期負債	108	500	533	1,813	2,313
長期借款	105	496	511	1,735	2,235
其他非流動負債	3	4	22	78	78
股東權益總額	737	794	1,611	5,714	10,679
股本	309	347	486	486	486
資本公積	399	370	843	916	916
保留盈餘	31	81	268	4,201	9,165
非控制權益	0	0	15	15	15
其他權益	-3	-3	-1	97	97

資料來源:公司資料及國泰預估

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現金流量表

至 12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
營業活動之現金流量	-157	152	-361	1,821	1,881
稅前純益	1	66	303	5,176	6,655
折舊及攤提	45	51	78	96	188
本期營運資金變動	-144	34	-548	-1,916	-2,101
其他營業活動現金流量	-60	1	-194	-1,536	-2,861
投資活動之現金流量	-148	-742	-186	-863	-1,496
資本支出淨額	-128	-236	-220	-829	-1,642
本期長期投資變動	2	0	-3	6	14
其他投資活動現金流量	-21	-506	37	-40	132
籌資活動之現金流量	341	627	723	3,059	7,426
現金增資	289	0	553	0	0
本期借款變動	65	631	168	4,086	8,824
發放現金股利	-7	-3	-7	-24	-486
其他籌資活動現金流量	-6	-1	9	-1,018	-1,038
匯率調整數	0	0	0	0	0
自由現金流量	-285	-84	-581	992	239
本期產生現金流量	36	37	176	4,017	7,810

資料來源:公司資料及國泰預估

資產負債表主要財務報表分析

至 12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
年成長率(%)					
營業收入	53.0	19.9	27.9	521.2	52.1
營業利益	-85.72	299.64	391.8	1573.66	28.67
稅前息前折舊攤銷前淨利	-57.17	86.58	240.41	1261.88	26.32
稅後純益	-94.89	1364.86	311	1587.07	30.44
每股盈餘	-94.89	1364.86	311	1587.07	30.44
獲利能力分析(%)					
營業毛利率	8.9	12.0	30.8	67.1	65.7
營業利益率	1.7	5.7	22.0	59.3	50.2
稅前息前折舊攤銷前淨利率	6.7	10.3	27.5	60.4	50.2
稅後純益率	0.4	5.3	17.2	46.6	40.0
資產報酬率	0.3	3.4	9.8	58.0	25.3
股東權益報酬率	0.7	7.7	20.2	111.1	49.7
穩定 / 償債能力分析					
負債/資產（%）	52.6	58.1	46.7	48.1	49.2
淨負債比率（%）	32.7	95.0	49.6	-4.5	-33.6
利息保障倍數（X）	1.1	5.1	11.8	129.3	161.9
利息及短期債保障倍數	0.1	0.2	0.6	2.2	1.0
現金流量 / 利息保障倍數(X)	-9.0	9.4	-12.9	45.1	45.5
現金流量 / 利息,短債保障倍數(X)	-0.9	0.4	-0.7	0.8	0.3
流動比率(%)	105.4	127.0	196.6	256.3	224.9
速動比率(%)	32.0	40.3	57.7	149.4	171.8
淨負債(百萬元)	241	755	792	-257	-3,593
每股資料分析(元)					
每股盈餘	0.1	1.2	5.0	83.8	109.3
每股營業現金流量	-3.2	3.1	-7.4	37.5	38.7
每股淨值	15.2	16.4	32.8	117.3	219.5
每股營收	18.9	22.6	28.9	179.7	273.3
每股稅前息前折舊攤銷前淨利	1.3	2.3	8.0	108.5	137.1
每股現金股利	0.1	0.2	0.5	10.0	13.0

資料來源:公司資料及國泰預估

國泰期貨 國泰證券子公司

聯友金屬 7610 TT

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投資評等定義

英文評等	中文評等	潛在報酬率
STRONG BUY	強力買進	> +30%
BUY	買進	介於+15% ~ +30%
NEUTRAL	中立	介於-15% ~ +15%
SELL	賣出	< -15%

註:潛在報酬率=(6 個月內目標價/本次報告評等最近一天收盤價 – 1) * 100

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本報告係根據本公司所取得資訊加以彙集及研究分析， 本公司雖盡力使用可靠且廣泛的資訊， 但本公司並不保證各項資訊之完整性及正確性。本報告中所提出之意見係為本報告出版當時的意見，邇後相關資訊或意見若有變更， 本公司將不會另行通知。本公司亦無義務持續更新本報告之內容或追蹤研究本報告所涵蓋之主題。本報告中提及的標的物價格、價值及收益隨時可能因各種本公司無法控制之政治、經濟、市場等因素而產生變化。本報告中之各項預測， 如營收及目標價格， 均係基於對目前所得資訊作合理假設下所完成， 所以並不必然實現。從事特定交易， 如涉及期貨、選擇權及衍生性金融商品， 因具有高度風險並不適合每一個投資人。本報告不得視為買賣有價證券或其他金融商品的要約或要約之引誘。

國泰金融控股公司所屬子公司暨關係企業(以下稱 “ 國泰金融集團”) 從事廣泛金融業務，包括但不限於銀行、保險、證券經紀、自有資金投資與交易、資產管理及證券投資信託等。國泰金融集團對於本報告所涵蓋之標的公司可能有投資或其他的業務往來關係。國泰金融集團及其所屬員工， 可能會投資本報告所涵蓋之標的公司， 或以該等公司為標的的衍生性金融商品， 而其交易的方向與本報告中所提及的交易方向可能不一致。除此之外， 國泰金融集團於法令許可的範圍內，亦有可能以本報告所涵蓋之標的公司作為發行衍生性

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2026 年 10 月 6 日

研究員：王美珍 iris.wang@capital.com.tw

前日收盤價	2,005.00 元
目標價	
3 個月	2,680.00 元
12 個月	2,680.00 元

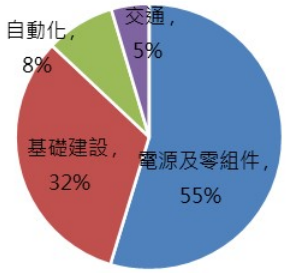
近期報告日期、評等及前日股價

07/31/2026	Buy	1,530 元
06/26/2026	Trading Buy	1,980 元
05/04/2026	Buy	2,165 元
03/02/2026	Buy	1,430 元
01/07/2026	Buy	1,050 元

公司基本資訊

目前股本(百萬元)	25,975
市值(億元)	52,080
目前每股淨值(元)	113.07
外資持股比(%)	61.71
投信持股比(%)	4.42
董監持股比(%)	7.32
融資餘額(張)	9,854
現金股息配發率(%)	50.13

產品組合



股價相對大盤走勢

台達電(2308 TT) Buy

產品組合較佳，獲利持續提升。

投資建議：受惠於 AI、Data Center 領域需求成長，營收與獲利維持成長趨勢，投資評等為 Buy，目標價為 2,680 元。

預估 3Q26 營收 QoQ+7.28%：受缺料影響，預估 3Q26 營收為 1,965.94 億元、QoQ+7.28%；在無較大業外收支下，稅後淨利為 257.42 億元、QoQ+2.41%，稅後 EPS 為 9.91 元。受惠於新一代 AI 伺服器放量與缺料緩解，預估 4Q26 營收為 2,218.52 億元、QoQ+12.85%；稅後淨利為 309.71 億元、QoQ+20.32%，稅後 EPS 為 11.92 元。

產品規格提升，電源類產品展望佳：台達電為 AI 伺服器電源解決方案主要供應商，受惠於 AI 需求提升，帶動 AC/DC 電源大幅成長外，CSP 客戶在新一代 ASIC 伺服器平台採用 DC/DC 電源模組，亦成為另一成長動能。800 VDC 與±400VDC 預計於 4Q26 出貨，2026 年營收比重約 1%，預期 2H27 營收貢獻較顯著。

水冷散熱 2026 年營收維持成長趨勢：水冷散熱產品有水冷板、Manifold、CDU、Sidecar。目前出貨以系統類的 Sidecar 為主，歸類於基礎建設；CDU 則歸類於電源類與零組件。水冷散熱 2025 年營收比重為 9%，台達電 Sidecar 產品 2025 年市佔率極高，雖有競爭者加入，但整體市場需求持續增長，2026 年 Sidecar 營收仍將維持成長趨勢，預估 2026 年水冷散熱營收比重為 12%。

固態變壓器為長期成長動能：SST(固態變壓器)目前已少量出貨，客戶為中系 CSP、800V 高壓直流電動車充電站，亦出貨給美系客戶測試。SST 導入需從資料中心設計初期即納入規劃，預期 2~3 年後會產生顯著的營收貢獻，仍持續改良產品規格中。

(百萬元)	2026F	2027F	2028F	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F
營業收入淨額	761,055	1,032,458	1,312,055	150,318	161,613	159,353	183,256	196,594	221,852	218,056	252,669	281,284
營業毛利淨額	274,229	364,987	466,513	52,416	55,903	58,961	65,308	70,186	79,775	76,350	89,480	99,787
營業利益	132,954	182,531	298,210	24,809	26,418	28,417	30,570	33,684	40,282	39,550	51,103	33,684
稅後純益	102,404	160,037	230,476	18,606	17,324	20,556	25,136	25,742	30,971	30,878	38,884	45,411
稅後 EPS(元)	39.42	61.61	88.73	7.16	6.67	7.91	9.68	9.91	11.92	11.89	14.97	17.48
毛利率(%)	36.03%	35.35%	35.56%	34.87%	34.59%	37.00%	35.64%	35.70%	35.96%	35.01%	35.41%	35.48%
營業利益率(%)	17.47%	17.68%	22.73%	16.50%	16.35%	17.83%	16.68%	17.13%	18.16%	18.14%	20.23%	11.98%
稅後純益率(%)	13.46%	15.50%	17.57%	12.38%	10.72%	12.90%	13.72%	13.09%	13.96%	14.16%	15.39%	16.14%
營業收入 YoY/QoQ(%)	37.16%	35.66%	27.08%	21.19%	7.51%	-1.40%	15.00%	7.28%	12.85%	-1.71%	15.87%	11.33%
稅後純益 YoY/QoQ(%)	70.37%	56.28%	44.01%	33.39%	-6.89%	18.66%	22.28%	2.41%	20.32%	-0.30%	25.93%	16.78%

註：稅後純益係指本期淨利歸屬於母公司業主；稅後 EPS 以股本 259.75 億元計算。

ESG：

台達電超過一半的董事會成員實際參與台達永續委員會、節能管理、節水減廢管理，及再生電力的重要會議；永續長每季向董事會報告氣候變遷的管理進展，並且由董事會針對重要議題做決策。在進行重要併購或是新建廠房等決策時，亦會考量氣候變遷的風險與機會。

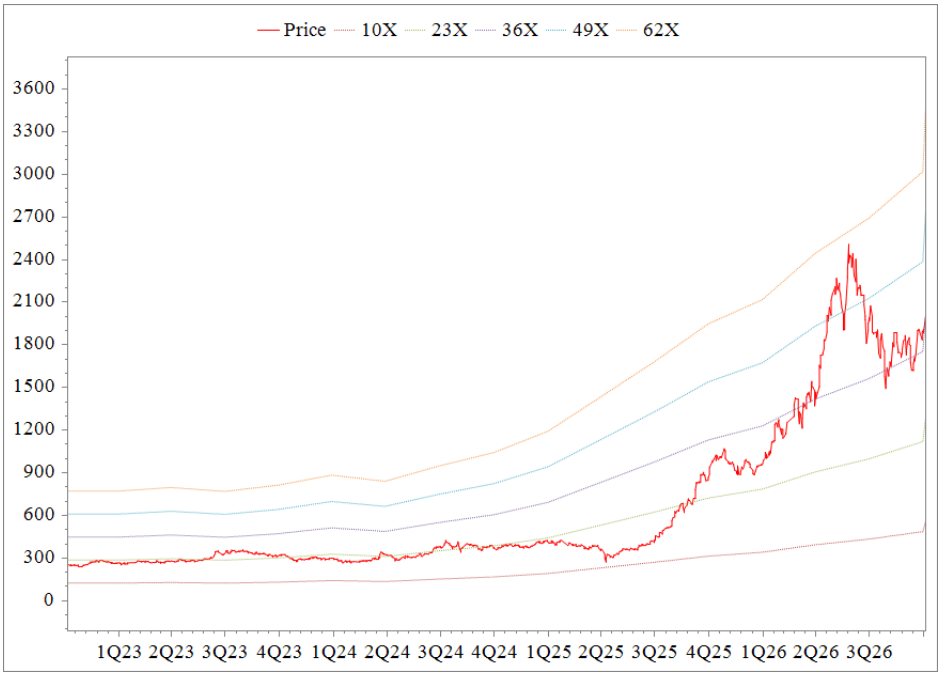
ESG 委員會為內部最高層級氣候相關推動組織。管理職責回歸包含企業永續發展部、節能服務部、財務、採購、人資、事業群等單位策略。依企業策略、氣候風險大數據、各類研究報告，以及外部評比指標鑑別風險與機會項目。

在治理層面，本報告新增「產品安全」與「品質管理」專章，藉由制定符合國際標準的相關政策與指引，強化控管機制，確保產品皆經嚴謹把關。此外，台達確立風險組織架構，由風管推動委員會辨別集團短期至長期（新興）風險，並完善風險管理流程。在供應鏈管理上，台達則首次舉辦「供應商 ESG 溝通大會」，明確傳達 ESG 與低碳供應鏈管理策略。

為推動價值鏈減碳，台達成立兩項子委員會，分別針對 C1(採購的產品與服務)與 C11(售出產品的使用)開展減碳行動。透過跨部門合作，盤點原物料採購與產品使用階段之碳排數據，並推動包括產品能效提升、低碳材料設計、以及與供應商與客戶議合，致力於實現範疇三減碳目標。

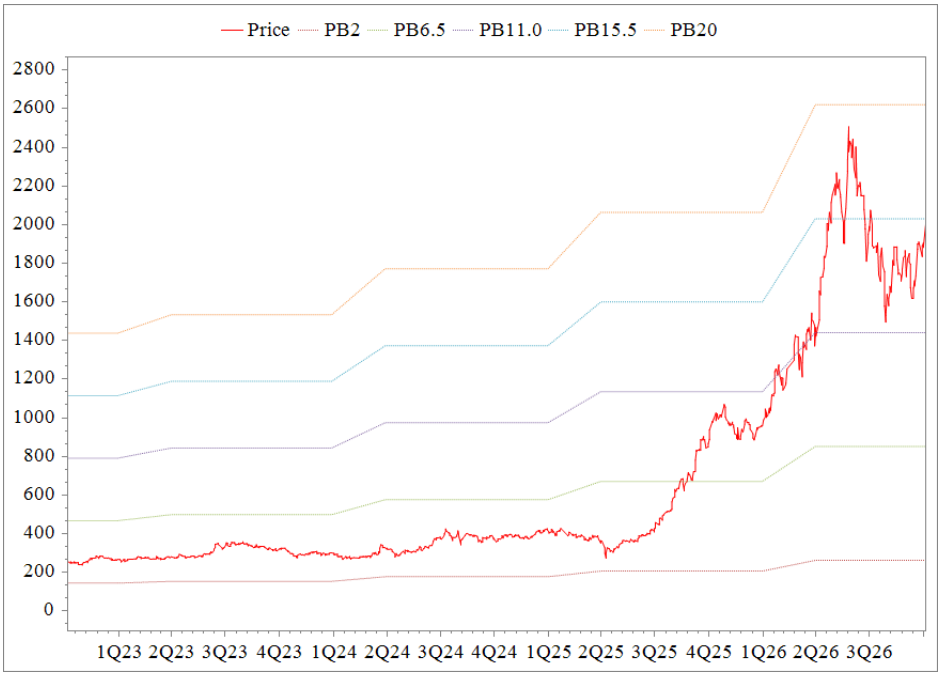
在自然資本保育方面，台達目標於 2050 年實現淨值正面效益(NPI)。2024 年成為 TNFD Early Adopters，並首次發表《氣候與自然生物多樣性報告書》，依循 TCFD、TNFD 與 IFRS S2 等國際準則，提升揭露品質與透明度。同年，參與聯合國生物多樣性 COP16 分享自身實踐，並連署 Business for Nature 發起之「Renewed policy ambition on nature」企業聲明，倡議將自然納入政策決策。

Forward PE Band



資料來源：CMoney，群益預估彙整

Forward PB Band



資料來源：CMoney，群益預估彙整

資產負債表

(百萬元)	2024	2025	2026F	2027F	2028F
資產總計	531,898	639,619	781,606	1,024,618	1,372,413
流動資產	310,925	395,781	577,622	772,878	946,158
現金及約當現金	117,459	151,172	219,564	294,364	360,057
應收帳款與票據	89,921	121,717	172,363	235,415	280,420
存貨	83,856	101,478	146,309	179,292	258,536
採權益法之投資	234	380	224	397	1,042
不動產、廠房設備	115,710	142,040	143,865	145,462	147,082
負債總計	**252,332**	**314,230**	**373,776**	**487,043**	**651,229**
流動負債	152,283	209,665	287,089	381,198	466,547
應付帳款及票據	69,223	94,452	117,644	156,149	211,488
非流動負債	100,049	104,565	63,494	44,148	67,646
權益總計	279,565	325,389	407,830	537,575	721,184
普通股股本	**25,975**	**25,975**	**25,975**	**25,975**	**25,975**
保留盈餘	141,468	183,526	265,967	395,713	579,321
母公司業主權益	230,087	268,058	340,330	458,806	632,137
負債及權益總計	**531,898**	**639,619**	**781,606**	**1,024,618**	**1,372,413**

現金流量表

(百萬元)	2024	2025	2026F	2027F	2028F
營業活動現金	72,895	98,474	139,167	220,752	314,346
稅前純益	51,316	87,866	144,118	215,680	304,946
折舊及攤銷	25,016	27,839	31,075	31,420	31,637
營運資金變動	-2,486	-24,190	-72,285	-57,530	-68,911
其他營運現金	-951	6,959	36,259	31,182	46,674
投資活動現金	-40,360	-53,214	-8,476	-90,987	-217,249
資本支出淨額	-33,380	-45,580	-50,000	-45,000	-45,000
長期投資變動	-547	-2,349	0	0	0
其他投資現金	-6,433	-5,285	41,524	-45,987	-172,249
籌資活動現金	-11,118	-9,401	-62,299	-54,967	-31,404
長借/公司債變動	8,552	11,283	-41,070	-19,346	23,497
現金增資	0	0	0	0	0
發放現金股利	-16,702	-18,183	-30,132	-41,561	-57,146
其他籌資現金	-2,967	-2,501	8,903	5,941	2,245
淨現金流量	30,003	33,713	68,392	74,799	65,693
期初現金	87,456	117,459	151,172	219,564	294,364
期末現金	117,459	151,172	219,564	294,364	360,057

損益表

(百萬元)	2024	2025	2026F	2027F	2028F
營業收入淨額	421,148	554,885	761,055	1,032,458	1,312,055
營業成本	284,567	364,729	486,826	626,157	845,541
營業毛利淨額	136,580	190,157	274,229	364,987	466,513
營業費用	88,928	106,224	141,276	155,495	168,304
營業利益	47,652	83,932	132,954	182,531	298,210
EBITDA	74,472	114,204	154,768	203,660	210,360
業外收入及支出	3,664	3,934	11,164	11,154	7,872
稅前純益	51,316	87,866	144,118	215,680	304,946
所得稅	10,925	19,930	31,546	44,374	64,191
稅後純益	35,229	60,108	102,404	160,037	230,476
稅後 EPS(元)	13.56	23.14	39.42	61.61	88.73
完全稀釋 EPS**	13.56	23.14	39.42	61.61	88.73

註 1：稅後純益係指本期淨利歸屬於母公司業主。

註 2：稅後 EPS 以股本 259.75【最新股本】億元計算

註 3：完全稀釋 EPS 以股本 259.75 億元計算

比率分析

(百萬元)	2024	2025	2026F	2027F	2028F
成長力分析(%)					
營業收入淨額	4.97%	31.76%	37.16%	35.66%	27.08%
營業毛利淨額	16.52%	39.23%	44.21%	33.10%	27.82%
營業利益	16.37%	76.14%	58.41%	37.29%	63.38%
稅後純益	5.50%	70.62%	70.37%	56.28%	44.01%
獲利能力分析(%)					
毛利率	32.43%	34.27%	36.03%	35.35%	35.56%
EBITDA(%)	17.68%	20.58%	20.34%	19.73%	16.03%
營益率	11.31%	15.13%	17.47%	17.68%	22.73%
稅後純益率	8.36%	10.83%	13.46%	15.50%	17.57%
總資產報酬率	6.62%	9.40%	13.10%	15.62%	16.79%
股東權益報酬率	12.60%	18.47%	25.11%	29.77%	31.96%
償債能力檢視					
負債比率(%)	47.44%	49.13%	47.82%	47.53%	47.45%
負債/淨值比(%)	90.26%	96.57%	91.65%	90.60%	90.30%
流動比率(%)	204.18%	188.77%	201.20%	202.75%	202.80%
其他比率分析					
存貨天數	102.67	92.74	92.89	94.90	94.50
應收帳款天數	73.36	69.61	70.52	72.08	71.75

資料來源：CMoney、群益

季度損益表

(百萬元)	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	1Q28F
營業收入淨額	124,035	150,318	161,613	159,353	183,256	196,594	221,852	218,056	252,669	281,284	280,449	276,346
營業成本	79,986	97,902	105,710	100,392	117,948	126,409	142,077	100,392	163,188	181,498	181,078	178,964
營業毛利淨額	44,049	52,416	55,903	58,961	65,308	70,186	79,775	76,350	89,480	99,787	99,371	97,382
營業費用	25,380	27,607	29,485	30,544	34,738	36,502	39,493	36,800	38,377	40,825	39,493	39,300
營業利益	18,669	24,809	26,418	28,417	30,570	33,684	40,282	39,550	51,103	33,684	58,194	58,082
業外收入及支出	902	2,174	-770	2,676	4,430	2,080	1,980	2,676	4,507	2,035	1,935	1,915
稅前純益	19,571	26,983	25,649	31,093	35,000	35,764	42,262	41,466	53,088	60,997	60,129	59,997
所得稅	4,237	6,062	6,011	7,258	7,831	7,582	8,875	7,258	11,679	12,809	12,627	11,999
稅後純益	13,948	18,606	17,324	20,556	25,136	25,742	30,971	30,878	38,884	45,411	44,864	45,492
最新股本	25,975	25,975	25,975	25,975	25,975	25,975	25,975	25,975	25,975	25,975	25,975	25,975
稅後EPS(元)	5.37	7.16	6.67	7.91	9.68	9.91	11.92	11.89	14.97	17.48	17.27	17.51
獲利能力(%)												
毛利率(%)	35.51%	34.87%	34.59%	37.00%	35.64%	35.70%	35.96%	35.01%	35.41%	35.48%	35.43%	35.24%
營業利益率(%)	15.05%	16.50%	16.35%	17.83%	16.68%	17.13%	18.16%	18.14%	20.23%	11.98%	20.75%	21.02%
稅後純益率(%)	11.25%	12.38%	10.72%	12.90%	13.72%	13.09%	13.96%	14.16%	15.39%	16.14%	16.00%	16.46%
QoQ(%)												
營業收入淨額	4.30%	21.19%	7.51%	-1.40%	15.00%	7.28%	12.85%	-1.71%	15.87%	11.33%	-0.30%	-1.46%
營業利益	33.01%	32.89%	6.49%	7.57%	7.58%	10.19%	19.59%	-1.82%	29.21%	-34.09%	72.76%	-0.19%
稅前純益	24.95%	37.87%	-4.95%	21.23%	12.56%	2.18%	18.17%	-1.88%	28.03%	14.90%	-1.42%	-0.22%
稅後純益	36.33%	33.39%	-6.89%	18.66%	22.28%	2.41%	20.32%	-0.30%	25.93%	16.78%	-1.20%	1.40%
YoY(%)												
營業收入淨額	19.91%	33.97%	41.51%	34.00%	47.75%	30.79%	37.27%	36.84%	37.88%	43.08%	26.41%	26.73%
營業利益	42.16%	51.06%	146.87%	102.46%	63.75%	35.77%	52.48%	39.18%	67.17%	0.00%	44.47%	46.86%
稅前純益	30.16%	52.67%	160.67%	98.51%	78.83%	32.54%	64.77%	33.36%	51.68%	70.56%	42.28%	44.69%
稅後純益	40.21%	50.79%	141.36%	100.92%	80.21%	38.35%	78.78%	50.22%	54.70%	76.41%	44.86%	47.33%

註1：稅後EPS以股本259.75億元計算

註2：稅後純益係指本期淨利歸屬於母公司業主。

【投資評等說明】

評等	定義
強力買進(Strong Buy)	首次評等潛在上漲空間≥35%
買進(Buy)	15%≤首次評等潛在上漲空間<35%
區間操作(Trading Buy)	5%≤首次評等潛在上漲空間<15%
中立(Neutral)	無法由基本面給予投資評等 預期近期股價將處於盤整 建議降低持股

【投資評等調整機制】

投資評等係「首次」給予特定個股投資評等時，其前一日收盤價相較 3 個月目標價之潛在上漲空間計算而得。個股投資評等分為四個等級，定義如上。爾後的投資評等係依循「首次評等」，直到停止推薦。

「強力買進」、「買進」及「區間操作」均有 upside 目標價。差別僅在於，首次評等時潛在報酬率不同。「中立」則無目標價。

一旦我們給予特定個股「強力買進」、「買進」或「區間操作」之投資評等，就是責任的開始，爾後將透過各式研究報告作定期性、持續性追蹤基本面及股價變化，直到停止推薦。

停止推薦情境：

1. 達目標價。
2. 雖未達原訂目標價，但檢視基本面、訊息面、籌碼面等多方訊息，研判股價上漲空間已然有限，將適時出具降評報告。
3. 推薦後股價不漲反跌。

【免責聲明】

本研究報告僅提供予特定人之客戶作為參考資料「非經同意不得轉載」。我們並不確保此資訊的完整性與正確性，投資人應了解，報告中有關未來預測之陳述可能不會實現，因而不應被依賴。而且此報告並非根據特定投資目的或依預定對象之財務狀況所撰寫出來的，因此，此研究報告的目的，既非對投資人於買賣證券、選擇權、期貨或其他證券相關之衍生性商品提供詢價服務，亦非作為進行交易的要約。投資人應注意到相關證券之價值及收益，可能會有無預警地上升或下降，產生投資回報金額可能比原始投資來得少的情形。

個股報告

2026 年 10 月 6 日

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前日收盤價	817.00 元
目標價	
3 個月	1,080.00 元
12 個月	1,080.00 元

近期報告日期、評等及前日股價

禾伸堂(3026 TT) Buy

NP0 MLCC 受惠 AI SERVER 電源架構升級

投資建議：受惠 AI SERVER 電源架構升級，禾伸堂在 NP0 具領先同業優勢，營收及毛利率將同步攀升，投資建議為 Buy，目標價 1,080 元。

高階 MLCC 占比持續攀升，2Q26 財報表現優異：

禾伸堂 2Q26 營收 40.97 億元(QOQ+13.2%/YOY+25.2%)，優於市場預期，主因受惠於高階 MLCC 產出與銷售增加，帶動毛利率達 26.18% (QOQ+1.94ppts/YOY+6.37ppts)，費用率 10.89%與 2Q26 相當，本業淨利 6.27 億元(QOQ+29.6%/YOY+111.2%)，稅後純益 6.42 億元(QOQ+35.2%/YOY+179.1%)，較去年同期倍增，2Q26 稅後 EPS 3.87 元，獲利呈現大幅增長，優於市場預期。

展望未來，NP0 MLCC 市佔率有望持續攀升，獲利將同步上揚：

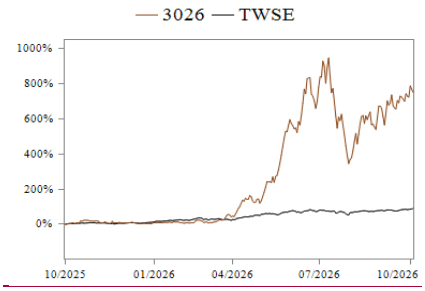
以 NP0 與 X7R/X8R 競爭態勢來看，目前電源類產品的驗證需要經過直接客戶、終端客戶以及安規認證，整體認證時間至少需要 2~3 年，以 VR 機種來看，台廠目前通過認證家數不到 3 家，當前 NP0 以禾伸堂為主要份額。日韓競爭方面，因高階小尺寸 MLCC 產能持續緊缺，日韓等主要大廠傾向優先轉向生產 0201、0402、0603 等小尺寸規格，且擴產意願不高，故客戶有望轉向要求禾伸堂增加產能與供給，2027 年 NP0 市佔率有望進一步提升。禾伸堂 2Q26 AI 應用占比約為 32%，考量 Vera Ribin 將於 4Q26 開始小量生產，於 2027 年正式放量，加上 TPU 與 ASIC 平台需求快速成長，皆有望帶動高階 MLCC 用量需求持續提升，群益預估禾伸堂 2026 年底 AI 應用占比有望達 45%，2027 年底將提升至 60%。綜上所述，預估 2026 年營收為 171.22 億元(YOY+27.5%)、2027 年營收為 245.70 億元(YOY+43.5%)，毛利率受惠 AI 比重持續拉升分別為 26.8%及 34%，2026 年、2027 年稅後 EPS 為 16.43 元及 31.59 元，獲利呈現大幅增長。

公司基本資訊

目前股本(百萬元)	1,659
市值(億元)	1,355
目前每股淨值(元)	62.28
外資持股比(%)	11.02
投信持股比(%)	16.10
董監持股比(%)	9.55
融資餘額(張)	13,652
現金股息配發率(%)	88.15

產品組合

股價相對大盤走勢



(百萬元)	2026F	2027F	2028F	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F
營業收入淨額	17,122	24,570	31,986	3,433	3,320	3,618	4,097	4,494	4,912	5,296	5,887	6,687
營業毛利淨額	4,584	8,365	12,818	625	585	877	1,073	1,231	1,404	1,625	1,930	2,332
營業利益	2,715	5,674	9,309	268	193	483	627	753	853	1,040	1,284	1,610
稅後純益	2,725	5,240	8,304	280	303	475	642	746	863	979	1,197	1,479
稅後 EPS(元)	16.43	31.59	50.06	1.69	1.83	2.86	3.87	4.49	5.20	5.90	7.21	8.91
毛利率(%)	26.77%	34.05%	40.07%	18.19%	17.62%	24.24%	26.18%	27.38%	28.58%	30.68%	32.78%	34.88%
營業利益率(%)	15.86%	23.09%	29.10%	7.81%	5.81%	13.36%	15.29%	16.74%	17.36%	19.63%	21.81%	24.08%
稅後純益率(%)	15.91%	21.33%	25.96%	8.15%	9.14%	13.12%	15.66%	16.59%	17.57%	18.49%	20.33%	22.11%
營業收入 YoY/QoQ(%)	27.52%	43.49%	30.18%	4.92%	-3.30%	9.00%	13.23%	9.69%	9.29%	7.82%	11.15%	13.59%
稅後純益 YoY/QoQ(%)	149.60%	92.31%	58.47%	21.71%	8.37%	56.45%	35.24%	16.17%	15.73%	13.48%	22.21%	23.56%

註：稅後純益係指本期淨利歸屬於母公司業主；稅後 EPS 以股本 16.59 億元計算。

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ESG(Environment, Social, Governance)：

禾伸堂（3026 TT）致力於永續發展，在不斷精進技術研發及調整產品組合之下，同時將保護環境及節能減廢納入重要管理項目。在溫室氣體管理的部分，2025 年臺灣地區溫室氣體排放已進行盤查，並於 2026 年完成第三方查驗，2025 年也依據 ISO 14067:2018 標準完成積層陶瓷電容產品碳足跡查證，透過了解產品碳排放量，以制定有效的減碳計畫；此外，為了有效減少溫室氣體排放，透過自發自用太陽能光電 2025 年已取得國家再生能源憑證中心發放之綠電憑證 T-REC 628 張（626,064 度），推估減少溫室氣體排放約 309.28 公噸。在能源與資源管理方面，落實廢棄物分類管理，並進行製程原物料與廢溶劑回收再利用的「轉廢為能」計畫，與供應商合作將廢塑料作為固體再生燃料（SRF），2025 年共計回收 117.05 公噸，大幅減少廢棄物直接焚化的處理量；另外，2025 年回收溶劑量約 194.70 公噸，有效減少溶劑（原物料）採購及廢棄物處理費用，並降低對環境的衝擊。

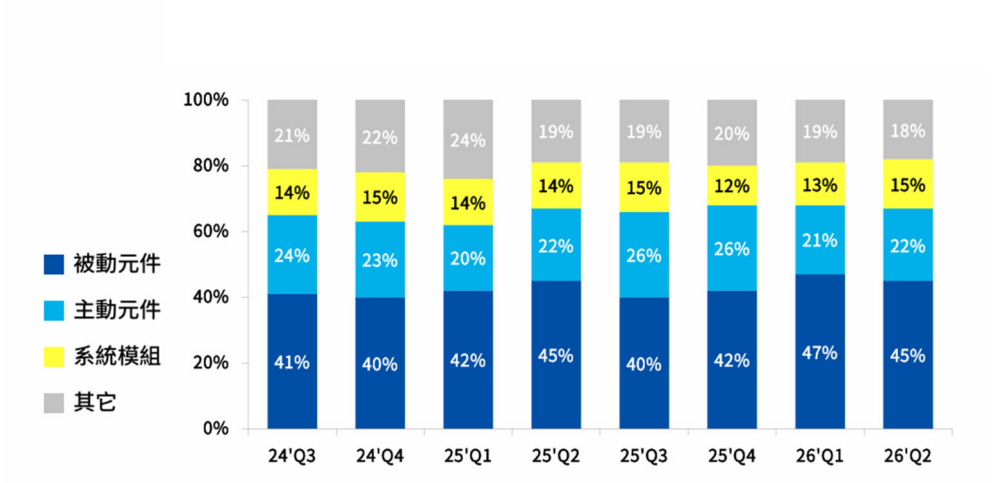
公司簡介：

禾伸堂(3026 TT)成立於 1981 年。公司總部位於台灣台北，生產基地分布於桃園龍潭、宜蘭利澤以及日本北海道。早期以被動元件代理商起家，後期切入利基型積層陶瓷電容器(MLCC)及固態電容研發製造。主要產品包含：積層陶瓷電容(MLCC)、主動元件、系統模組與其他代理產品。公司代理產品以銷售通訊、車用及消費性電子市場為主。2Q26 產品營收比重:被動元件 45%、主動元件 22%、系統模組 15%、其他 18%。

圖一、禾伸堂 MLCC 產品圖:

資料來源：禾伸堂、群益投顧彙整

圖二、禾伸堂產品營收比重:



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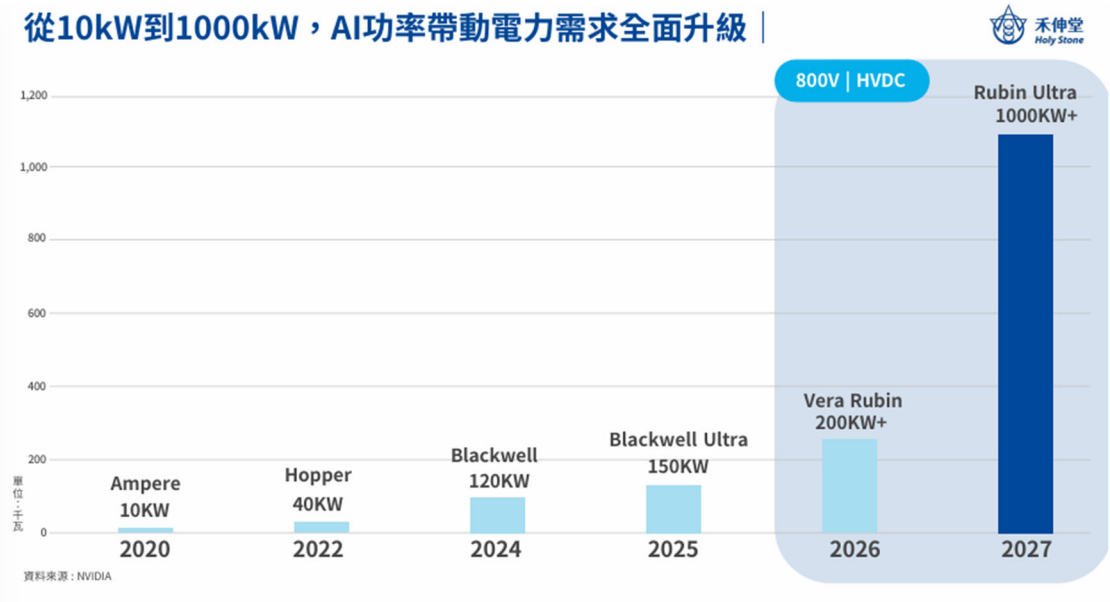
因 AI 伺服器電源架構升級，帶動 NP0 MLCC 用量顯著大增：

禾伸堂目前產品主要應用於電源類，尤其是 AI 伺服器隨著電源供應器瓦數增加，電源從 IT Rack 移出獨立成 Power Rack(Sidecar Rack)，新一代 Rubin 平台架構也將朝向更高功率密度的獨立電源櫃與多元電力備援系統發展。這項架構轉型大幅推升了高電壓、高效能與高穩定性被動元件的需求，其中 NP0 材質之積層陶瓷電容（MLCC）因具備極佳的溫度穩定性、低介質損耗與高可靠度，成為高功率電源供應器與電壓調節模組（VRM）中不可或缺的關鍵零組件，因過去多採用單相 PSU(如 3.3kW、4.2kW、4.3kW 等)，而下一世代為了在高瓦數下增加效率，Power Rack 已轉成三相 AC/DC PSU，此外，內部仍需要進行 DC/DC 轉換(400V 降至 48V/50V，或 800V 降至 48V)，群益預估單一機櫃所需的 NP0 MLCC 數量可提升 3~5 倍。

圖三、AI 功率持續提升，電力需求全面升級:



資料來源：禾伸堂、群益投顧彙整

圖四、AI Server Power、BBU 使用客製化、低損耗、高壓、高容 NP0:

年份	NVIDIA GPU 型號	伺服器電源供應器瓦數	機櫃電壓	客製化低損耗 (NP0) 高壓、高容需求規格	單一機櫃使用數量	AI Server Power 供應商	全球主要CSP
2020	A100	1600W~2200W	208~240 VAC	1206 NP0 10nF 630V	800~1000	DELTA LITEON flex 欧陆通 HONOTO	NVIDIA aws Microsoft Google Meta
2022	H100	3000W~3200W	240~415 VAC	1210 NP0 22nF 1000V	1800~2000		
2024	H200	3300W~3600W	415~480 VAC	1210 NP0 33nF 630V	1800~2000		
2025	GB200	4200W~5500W	480 VAC	1210 NP0 33nF 1000V	3000~5000		
2026	GB300	5500W~18KW	480 VAC	1210 NP0 33nF 1000V	3000~5000		
	Vera Rubin	5500W~18KW	480 VAC / 800V	1210 NP0 33nF 1000V	3000~5000		
2027	Rubin Ultra	>30KW	800V | HVDC	1210 NP0 47nF 630V	N/A		
				1210 NP0 33nF 1250V	N/A		

資料來源：禾伸堂、群益投顧彙整

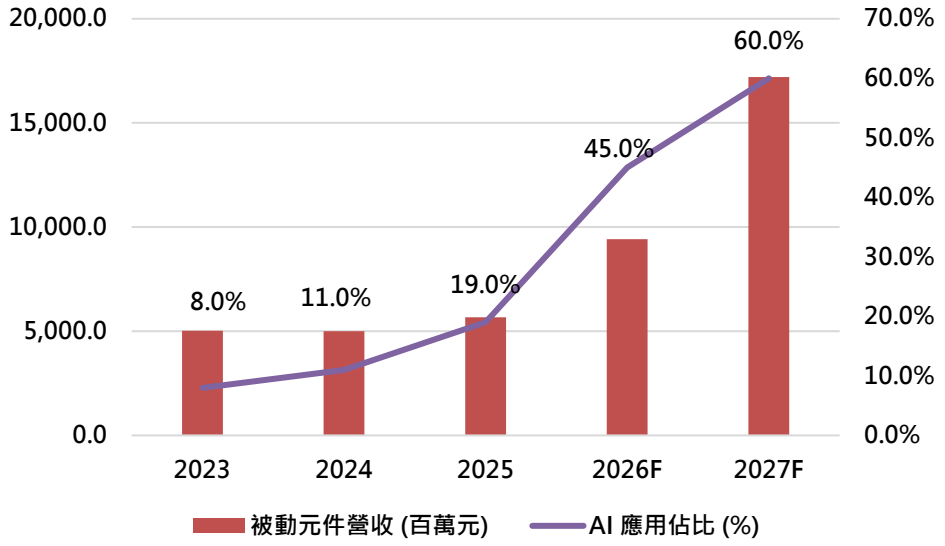
<u>展望未來，NP0 MLCC 市佔率有望持續攀升，獲利將同步上揚</u>：

以 NP0 與 X7R/X8R 競爭態勢來看，目前電源類產品的驗證需要經過直接客戶、終端客戶以及安規認證，整體認證時間至少需要 2~3 年，以 VR 機種來看，台廠目前通過認證家數不到 3 家，當前 NP0 以禾伸堂為主要份額。日韓競爭方面，因高階小尺寸 MLCC 產能持續緊缺，日韓等主要大廠傾向優先轉向生產 0201、0402、0603 等小尺寸規格，且擴產意願不高，故客戶有

望轉向要求禾伸堂增加產能與供給，2027 年 NP0 市佔率有望進一步提升。

禾伸堂 2Q26 AI 應用占比約為 32%，考量 Vera Ribin 將於 4Q26 開始小量生產，於 2027 年正式放量，加上 TPU 與 ASIC 平台需求快速成長，皆有望帶動高階 MLCC 用量需求持續提升，群益預估禾伸堂 2026 年底 AI 應用占比有望達 45%，2027 年底將提升至 60%。綜上所述，預估 2026 年營收為 171.22 億元(YOY+27.5%)、2027 年營收為 245.70 億元(YOY+43.5%)，毛利率受惠 AI 比重持續拉升分別為 26.8%及 34%，2026 年、2027 年稅後 EPS 為 16.43 元及 31.59 元，獲利呈現大幅增長。

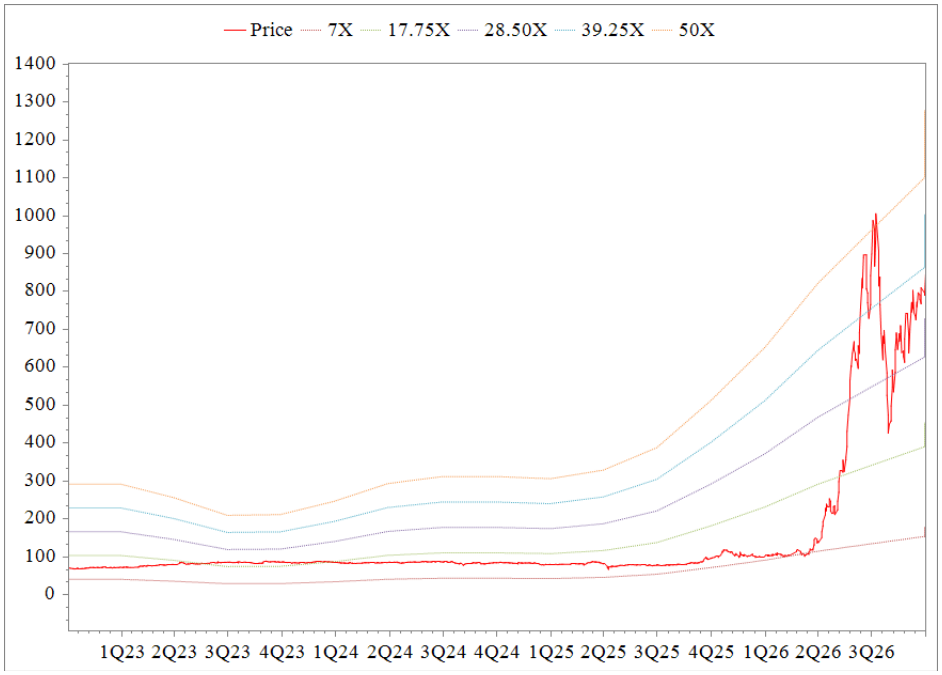
圖五、禾伸堂 AI 占比 2027 年預估達 60%:



資料來源：群益投顧預估

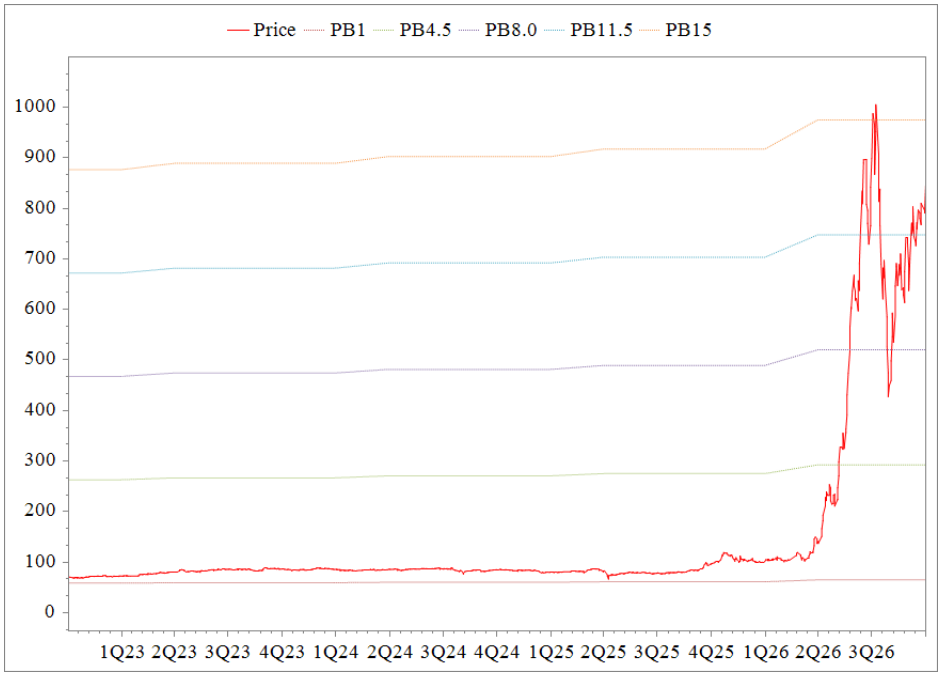
投資建議：受惠 AI SERVER 電源架構升級，禾伸堂在 NP0 具領先同業優勢，營收及毛利率將同步攀升，投資建議為 Buy，目標價 1,080 元。

Forward PE Band



資料來源：CMoney，群益預估彙整

Forward PB Band



資料來源：CMoney，群益預估彙整

資產負債表

(百萬元)	2024	2025	2026F	2027F	2028F
資產總計	15,826	16,188	18,568	24,253	32,784
流動資產	10,912	11,724	13,996	19,777	28,405
現金及約當現金	4,502	4,619	6,135	11,838	20,464
應收帳款與票據	3,087	3,784	3,937	4,380	3,941
存貨	2,575	2,156	2,255	2,646	2,277
採權益法之投資	443	444	581	581	581
不動產、廠房設備	4,231	3,764	3,603	3,365	3,117
負債總計	**5,446**	**5,789**	**7,327**	**10,411**	**14,784**
流動負債	3,791	4,932	5,804	6,733	5,415
應付帳款及票據	989	946	1,024	1,047	1,049
非流動負債	1,655	857	1,523	3,678	9,369
權益總計	10,379	10,399	11,038	13,639	17,797
普通股股本	**1,659**	**1,659**	**1,659**	**1,659**	**1,659**
保留盈餘	5,259	5,442	6,401	9,021	13,173
母公司業主權益	9,983	10,148	10,786	13,387	17,545
負債及權益總計	**15,826**	**16,188**	**18,365**	**24,050**	**32,581**

現金流量表

(百萬元)	2024	2025	2026F	2027F	2028F
營業活動現金	1,611	1,080	2,156	4,696	7,065
稅前純益	1,103	1,191	3,149	5,545	8,607
折舊及攤銷	724	734	698	687	687
營運資金變動	393	-322	-810	208	172
其他營運現金	-610	-522	-881	-1,743	-2,401
投資活動現金	146	-253	-677	-584	-587
資本支出淨額	138	-266	-486	-448	-438
長期投資變動	1	0	0	0	0
其他投資現金	6	13	-190	-137	-150
籌資活動現金	-379	-702	4	1,077	1,154
長借/公司債變動	160	-680	608	1,077	1,153
現金增資	0	0	0	0	0
發放現金股利	-829	-912	-962	-2,402	-4,193
其他籌資現金	290	890	1,567	1,402	3,183
淨現金流量	1,422	117	1,516	5,220	7,658
期初現金	3,080	4,502	4,619	6,135	11,355
期末現金	4,502	4,619	6,135	11,355	19,012

資料來源：CMoney、群益

損益表

(百萬元)	2024	2025	2026F	2027F	2028F
營業收入淨額	12,786	13,427	17,122	24,570	31,986
營業成本	10,697	10,907	12,538	16,205	19,168
營業毛利淨額	2,089	2,520	4,584	8,365	12,818
營業費用	1,468	1,442	1,869	2,691	3,509
營業利益	621	1,078	2,715	5,674	9,309
EBITDA	1,855	1,985	3,414	6,360	9,996
業外收入及支出	482	113	434	447	449
稅前純益	1,103	1,191	3,149	6,121	9,758
所得稅	177	165	524	984	1,557
稅後純益	973	1,092	2,725	5,240	8,304
稅後 EPS(元)	5.87	6.58	16.43	31.59	50.06
完全稀釋 EPS**	5.87	6.58	16.43	31.59	50.06

註 1：稅後純益係指本期淨利歸屬於母公司業主。

註 2：稅後 EPS 以股本 16.59【最新股本】億元計算

註 3：完全稀釋 EPS 以股本 16.59 億元計算

比率分析

(百萬元)	2024	2025	2026F	2027F	2028F
成長力分析(%)					
營業收入淨額	-3.43%	5.01%	27.52%	43.49%	30.18%
營業毛利淨額	-8.84%	20.65%	81.90%	82.46%	53.23%
營業利益	-12.76%	73.70%	151.83%	108.95%	64.07%
稅後純益	14.44%	12.17%	149.60%	92.31%	58.47%
獲利能力分析(%)					
毛利率	16.34%	18.77%	26.77%	34.05%	40.07%
EBITDA(%)	14.50%	14.78%	19.94%	25.89%	31.25%
營益率	4.85%	8.03%	15.86%	23.09%	29.10%
稅後純益率	7.61%	8.13%	15.91%	21.33%	25.96%
總資產報酬率	6.15%	6.74%	14.68%	21.61%	25.33%
股東權益報酬率	9.38%	10.50%	24.69%	38.42%	46.66%
償債能力檢視					
負債比率(%)	34.41%	35.76%	39.46%	42.93%	45.10%
負債/淨值比(%)	52.47%	55.67%	66.38%	76.33%	83.07%
流動比率(%)	287.84%	237.73%	241.14%	293.74%	524.58%
其他比率分析					
存貨天數	97.29	79.15	64.20	55.19	46.87
應收帳款天數	83.61	93.40	82.30	61.78	47.48

季度損益表

(百萬元)	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	1Q28F
營業收入淨額	3,272	3,433	3,320	3,618	4,097	4,494	4,912	5,296	5,887	6,687	6,700	7,219
營業成本	2,624	2,808	2,735	2,741	3,025	3,264	3,508	3,671	3,957	4,354	4,222	4,463
營業毛利淨額	648	625	585	877	1,073	1,231	1,404	1,625	1,930	2,332	2,478	2,756
營業費用	352	356	392	394	446	478	551	585	646	722	738	796
營業利益	297	268	193	483	627	753	853	1,040	1,284	1,610	1,740	1,960
業外收入及支出	-1	68	74	85	136	98	115	111	116	111	109	111
稅前純益	295	336	267	569	762	850	968	1,150	1,400	1,721	1,849	2,072
所得稅	58	71	0	113	150	129	132	197	229	268	290	335
稅後純益	230	280	303	475	642	746	863	979	1,197	1,479	1,585	1,762
最新股本	1,659	1,659	1,659	1,659	1,659	1,659	1,659	1,659	1,659	1,659	1,659	1,659
稅後EPS(元)	1.39	1.69	1.83	2.86	3.87	4.49	5.20	5.90	7.21	8.91	9.56	10.62
獲利能力(%)												
毛利率(%)	19.81%	18.19%	17.62%	24.24%	26.18%	27.38%	28.58%	30.68%	32.78%	34.88%	36.98%	38.18%
營業利益率(%)	9.06%	7.81%	5.81%	13.36%	15.29%	16.74%	17.36%	19.63%	21.81%	24.08%	25.97%	27.15%
稅後純益率(%)	7.03%	8.15%	9.14%	13.12%	15.66%	16.59%	17.57%	18.49%	20.33%	22.11%	23.67%	24.41%
QoQ(%)												
營業收入淨額	-3.82%	4.92%	-3.30%	9.00%	13.23%	9.69%	9.29%	7.82%	11.15%	13.59%	0.19%	7.76%
營業利益	-7.46%	-9.60%	-28.01%	150.44%	29.61%	20.12%	13.32%	21.92%	23.49%	25.39%	8.06%	12.66%
稅前純益	1.04%	13.87%	-20.44%	112.66%	34.08%	11.50%	13.88%	18.82%	21.71%	22.91%	7.47%	12.02%
稅後純益	-17.42%	21.71%	8.37%	56.45%	35.24%	16.17%	15.73%	13.48%	22.21%	23.56%	7.22%	11.15%
YoY(%)												
營業收入淨額	1.01%	0.64%	3.88%	6.36%	25.22%	30.92%	47.97%	46.37%	43.68%	48.78%	36.39%	36.30%
營業利益	95.35%	37.77%	70.37%	50.81%	111.23%	180.69%	341.88%	115.11%	104.95%	113.94%	104.01%	88.51%
稅前純益	19.12%	-17.26%	46.71%	94.65%	158.31%	152.94%	262.08%	102.30%	83.63%	102.41%	91.01%	80.07%
稅後純益	0.41%	-6.23%	33.03%	70.41%	179.09%	166.38%	184.47%	106.34%	86.47%	98.33%	83.73%	79.96%

註1：稅後EPS以股本16.59億元計算

註2：稅後純益係指本期淨利歸屬於母公司業主。

【投資評等說明】

評等	定義
強力買進(Strong Buy)	首次評等潛在上漲空間≥35%
買進(Buy)	15%≤首次評等潛在上漲空間<35%
區間操作(Trading Buy)	5%≤首次評等潛在上漲空間<15%
中立(Neutral)	無法由基本面給予投資評等 預期近期股價將處於盤整 建議降低持股

【投資評等調整機制】

投資評等係「首次」給予特定個股投資評等時，其前一日收盤價相較 3 個月目標價之潛在上漲空間計算而得。個股投資評等分為四個等級，定義如上。爾後的投資評等係依循「首次評等」，直到停止推薦。

「強力買進」、「買進」及「區間操作」均有 upside 目標價。差別僅在於，首次評等時潛在報酬率不同。「中立」則無目標價。

一旦我們給予特定個股「強力買進」、「買進」或「區間操作」之投資評等，就是責任的開始，爾後將透過各式研究報告作定期性、持續性追蹤基本面及股價變化，直到停止推薦。

停止推薦情境：

1. 達目標價。
2. 雖未達原訂目標價，但檢視基本面、訊息面、籌碼面等多方訊息，研判股價上漲空間已然有限，將適時出具降評報告。
3. 推薦後股價不漲反跌。

【免責聲明】

本研究報告僅提供予特定人之客戶作為參考資料「非經同意不得轉載」。我們並不確保此資訊的完整性與正確性，投資人應了解，報告中有關未來預測之陳述可能不會實現，因而不應被依賴。而且此報告並非根據特定投資目的或依預定對象之財務狀況所撰寫出來的，因此，此研究報告的目的，既非對投資人於買賣證券、選擇權、期貨或其他證券相關之衍生性商品提供詢價服務，亦非作為進行交易的要約。投資人應注意到相關證券之價值及收益，可能會有無預警地上升或下降，產生投資回報金額可能比原始投資來得少的情形。

2026/10/07

紡織

聚陽 (1477 TT)

維持評等

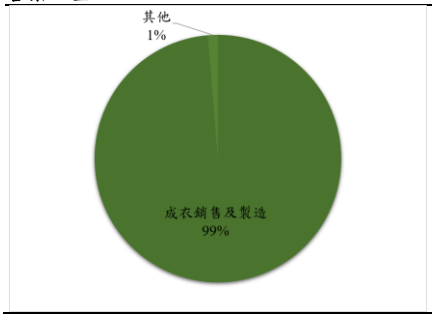
增加持股(Overweight)

潛在漲幅	18.9%
前日收盤價	203.50 元
目標價	242.00 元
前次目標價	242.00 元
前次評等	增加持股

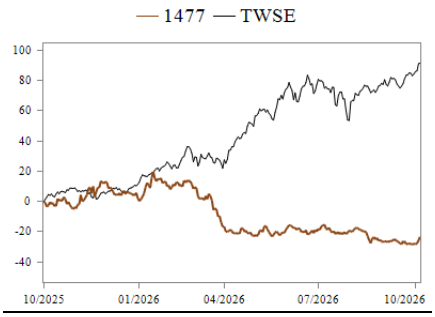
公司基本資訊

目前股本(百萬元)	2,467
市值(億元)	502
目前每股淨值(元)	55.89
外資持股比(%)	15.32
董監持股比(%)	9.39

營業比重



個股與大盤走勢(%)



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4Q26 營運展望正向；訂單篩選策略啟動

- **3Q26 成長動能延續；管理層審慎樂觀展望 4Q26**
- **下單模式轉變；客戶與訂單篩選支撐獲利能力**
- **2027 年復甦幅度仍取決於客戶補庫存時點**

營運維持溫和復甦，4Q26 展望轉趨正向：聚陽 3Q26 營收 94.8 億元、YoY+6.7%；前 9 月累計營收由負轉正為年增 0.2%，整體出貨大致符合公司預期。9 月營收 29.3 億元、YoY+24.1%，因去年同期低基期而有較高成長，但適逢聖誕節與感恩節出貨旺季導致船班擁擠，約 8,000 萬元營收遞延至次月。3Q26 出貨量估維持個位數年增，ASP 大致持平，預期毛利率仍可維持 24%以上水準。展望 4Q26，管理層看法轉趨正向，並啟動訂單篩選，包含挑選客戶、產品類別及提高短單比重，優先承接獲利能力較佳的業務，預期將有助於提升獲利表現。

短單占比提升，客戶與訂單篩選支撐獲利能力：受中東戰爭、油價及終端需求不確定性影響，客戶下單模式轉變，初期訂單通常未一次下滿，後續再以追加單補足，此趨勢預計延續至 4Q26。客戶下單模式轉向短單，使訂單能見度維持在約 4 個月，但聚陽具備快速反應與急短單生產優勢，而短單報價相對較佳，反而有利毛利率表現。公司將篩選價格及獲利條件較佳的訂單，減少關稅與原物料成本壓力較大的品項；4Q26 成長動能主要來自百貨公司、美系品牌及通路商回補庫存需求，在訂單結構及產品組合優化帶動下，獲利能力有望持續改善。

東南亞持續擴產，2027 年復甦幅度仍取決於客戶補庫存時點：產能佈局方面，由於菲律賓廠關閉抵銷部分新增產能，2026 年整體產能增幅有限，未來將持續以印尼與孟加拉作為主要擴產據點。展望 2027 年，客戶庫存已處低檔，管理層認為待美國期中選舉結束、政策與外部環境明朗後，客戶下單週期有望逐步恢復正常，將對具規模的大型供應商相對有利。不過，目前終端需求仍屬溫和，回補庫存啟動時點及力道尚未明確，因此短期營運可望延續緩和改善趨勢。公司表示內部營運體質已完成調整，後續較明顯的成長仍須等待外部需求回溫。

4Q26 營運正向，2027 年也有望持續復甦，惟時程與幅度仍須觀察。因此維持審慎的 14 倍 PER，以未來四季預估 EPS 計算，目標價 242 元。**風險因子：**外部環境變化、終端需求、客戶回補庫存速度。

(百萬元)	2022	2023	2024	2025	2026F	2027F
營業收入淨額	32,083	32,459	35,524	34,428	35,190	37,741
營業毛利淨額	8,256	8,366	8,991	8,564	8,899	9,626
營業利益	4,432	4,840	5,300	4,681	5,053	5,531
稅後純益	3,515	3,991	4,115	3,614	3,901	4,282
公告基本 EPS(元)	14.53	16.18	16.68	14.65	--	--
調整後 EPS(元)	14.53	16.50	16.68	14.65	15.81	17.36
營業收入 YoY(%)	10.90%	1.17%	9.44%	-3.08%	2.21%	7.25%
稅後純益 YoY(%)	32.38%	13.56%	3.10%	-12.17%	7.94%	9.77%
本益比	12.07	17.58	17.16	18.51	12.87	11.72
本淨比	2.76	4.35	4.82	4.76	3.40	3.35
股息(元)	14.50	16.40	17.10	15.00	16.00	17.20
殖利率(%)	6.16%	4.13%	4.98%	5.60%	7.86%	8.45%
ROE(%)	24.18%	25.60%	27.02%	25.20%	27.08%	28.77%

圖表一：季度財報預估差異

損益表 (單位：百萬元)	2Q26					3Q26(F)				
	實際數	中信預估	差異	市場共識	差異	調整後	調整前	差異	市場共識	差異
營業收入	8,166	8,166	0.0%	8,413	-2.9%	9,484	9,303	1.9%	9,501	-0.2%
營業毛利	1,990	2,012	-1.1%	2,057	-3.3%	2,329	2,316	0.6%	2,326	0.1%
營業利益	1,086	1,191	-8.8%	1,179	-7.9%	1,354	1,442	-6.1%	1,338	1.3%
稅前純益	1,072	1,177	-8.9%	1,178	-9.0%	1,331	1,387	-4.1%	1,316	1.1%
稅後淨利	814	904	-9.9%	904	-9.9%	1,037	1,081	-4.1%	1,029	0.7%
EPS(元)	3.30	3.66	-10.0%	3.61	-8.5%	4.20	4.38	-4.1%	4.15	1.3%
財務比率(%/百分點)										
毛利率	24.4%	24.6%	-0.3 ppts	24.5%	-0.1 ppts	24.6%	24.9%	-0.3 ppts	24.5%	0.1 ppts
營益率	13.3%	14.6%	-1.3 ppts	14.0%	-0.7 ppts	14.3%	15.5%	-1.2 ppts	14.1%	0.2 ppts
稅後淨利率	10.0%	11.1%	-1.1 ppts	10.7%	-0.8 ppts	10.9%	11.6%	-0.7 ppts	10.8%	0.1 ppts

資料來源：Bloomberg、中信投顧預估

圖表二：年度財報預估差異

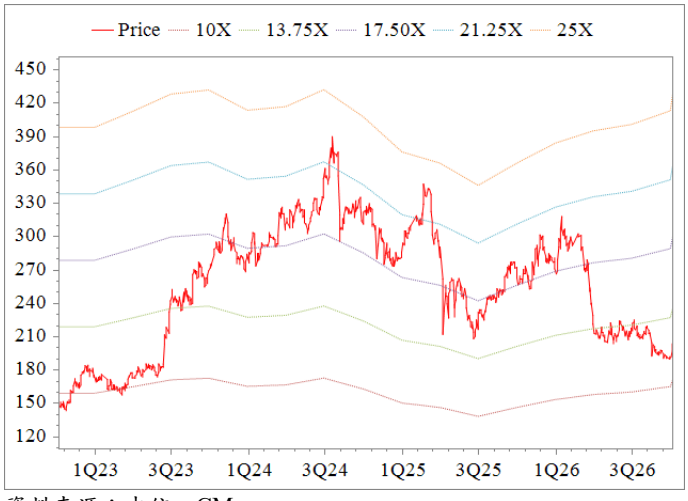
損益表 (單位：百萬元)	2026(F)					2027(F)				
	調整後	調整前	差異	市場共識	差異	調整後	調整前	差異	市場共識	差異
營業收入	35,190	34,759	1.2%	35,276	-0.2%	37,741	37,584	0.4%	37,758	0.0%
營業毛利	8,899	8,807	1.1%	8,888	0.1%	9,626	9,632	-0.1%	9,627	0.0%
營業利益	5,053	5,173	-2.3%	4,973	1.6%	5,531	5,724	-3.4%	5,451	1.5%
稅前純益	4,981	5,068	-1.7%	4,891	1.8%	5,439	5,602	-2.9%	5,382	1.1%
稅後淨利	3,901	3,976	-1.9%	3,810	2.4%	4,282	4,455	-3.9%	4,189	2.2%
EPS(元)	15.81	16.12	-1.9%	15.51	1.9%	17.36	18.06	-3.9%	17.07	1.7%
財務比率(%/百分點)										
毛利率	25.3%	25.3%	0.0 ppts	25.2%	0.1 ppts	25.5%	25.6%	-0.1 ppts	25.5%	0.0 ppts
營益率	14.4%	14.9%	-0.5 ppts	14.1%	0.3 ppts	14.7%	15.2%	-0.6 ppts	14.4%	0.2 ppts
稅後淨利率	11.1%	11.4%	-0.4 ppts	10.8%	0.3 ppts	11.3%	11.9%	-0.5 ppts	11.1%	0.3 ppts

資料來源：Bloomberg、中信投顧預估

2026/10/07

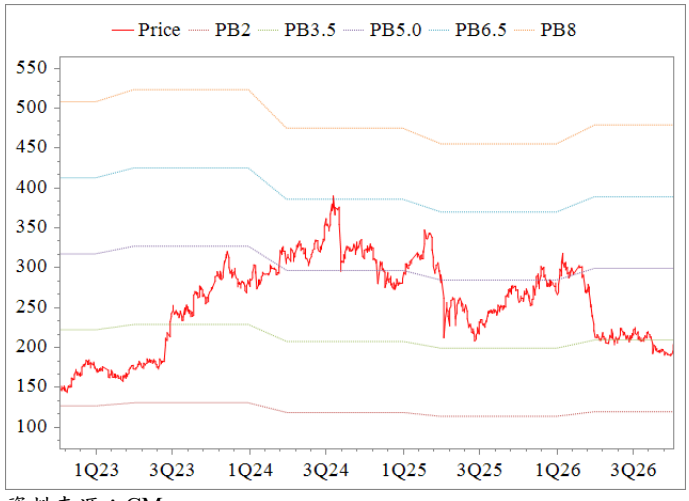
(百萬元)	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F
營業收入	8,897	8,166	9,484	8,643	9,451	8,706	10,334	9,250
營業成本	6,571	6,176	7,155	6,389	6,968	6,549	7,748	6,851
營業毛利	2,326	1,990	2,329	2,255	2,484	2,157	2,586	2,399
營業費用	949	903	975	1,020	1,034	906	1,059	1,096
營業利益	1,377	1,086	1,354	1,235	1,449	1,251	1,527	1,303
營業外收支淨額	-20	-15	-24	-14	-15	-35	-25	-16
稅前純益	1,357	1,072	1,331	1,221	1,434	1,216	1,502	1,287
所得稅	279	236	283	234	298	261	301	247
稅後純益	1,073	814	1,037	977	1,130	934	1,188	1,030
調整後 EPS(元)	4.35	3.30	4.20	3.96	4.58	3.79	4.82	4.17
毛利率(%)	26.14%	24.37%	24.56%	26.08%	26.28%	24.78%	25.03%	25.94%
營業利益率(%)	15.48%	13.30%	14.28%	14.29%	15.33%	14.37%	14.78%	14.09%
稅後純益率(%)	12.06%	9.97%	10.93%	11.30%	11.96%	10.73%	11.50%	11.13%
營業收入								
QoQ(%)	12.25%	-8.22%	16.14%	-8.86%	9.35%	-7.88%	18.70%	-10.49%
YoY(%)	-10.35%	6.20%	6.69%	9.05%	6.23%	6.62%	8.97%	7.02%
營業利益								
QoQ(%)	29.11%	-21.13%	24.68%	-8.82%	17.34%	-13.65%	22.07%	-14.69%
YoY(%)	-12.84%	15.37%	24.04%	15.76%	5.21%	15.18%	12.77%	5.51%
稅後純益								
QoQ(%)	23.45%	-24.11%	27.32%	-5.81%	15.72%	-17.34%	27.22%	-13.36%
YoY(%)	-15.54%	32.19%	20.85%	12.34%	5.31%	14.71%	14.62%	5.43%

PER Band



資料來源：中信、CMoney

PBR Band



資料來源：CMoney

2026/10/07

資產負債表

(百萬元)	2022	2023	2024	2025	2Q26
資產總計	24,429	24,876	23,198	23,904	23,967
流動資產	17,963	18,419	16,093	15,905	16,021
現金及約當現金	2,760	3,014	2,900	2,624	2,255
應收帳款與票據	1,199	1,527	829	899	930
存貨	5,241	4,843	5,702	5,034	5,362
採權益法之投資	289	364	315	346	323
不動產、廠房設備	5,227	5,264	5,815	5,978	5,960
負債總計	8,984	8,949	8,429	9,730	10,077
流動負債	8,511	8,597	8,019	8,565	8,894
應付帳款及票據	1,741	2,052	2,115	2,070	2,474
非流動負債	474	352	411	1,166	1,183
權益總計	15,445	15,927	14,769	14,174	13,890
普通股股本	2,419	2,419	2,467	2,467	2,467
保留盈餘	6,725	7,209	5,706	5,350	5,017
母公司業主權益	15,357	15,819	14,644	14,045	13,789
負債及權益總計	24,429	24,876	23,198	23,904	23,967

損益表

(百萬元)	2023	2024	2025	2026F	2027F
營業收入淨額	32,459	35,524	34,428	35,190	37,741
營業成本	24,092	26,532	25,864	26,291	28,115
營業毛利淨額	8,366	8,991	8,564	8,899	9,626
營業費用	3,527	3,691	3,883	3,846	4,096
營業利益	4,840	5,300	4,681	5,053	5,531
EBITDA	5,764	6,080	5,442	5,880	6,391
業外收入及支出	188	-57	-89	-73	-92
稅前純益	5,027	5,243	4,592	4,981	5,439
所得稅	992	1,071	922	1,033	1,107
稅後純益	3,991	4,115	3,614	3,901	4,282
公告基本 EPS(元)	16.18	16.68	14.65	- -	- -
調整後 EPS(元)	16.50	16.68	14.65	15.81	17.36

註：稅後純益係指本期淨利歸屬於母公司業主。

現金流量表

(百萬元)	2022	2023	2024	2025	2Q26
營業活動現金	3,715	5,785	4,731	4,057	957
稅前純益	4,589	5,027	5,243	4,592	1,072
折舊及攤銷	562	548	565	608	167
營運資金變動	-223	397	-72	486	-625
其他營運現金	-1,213	-188	-1,004	-1,629	342
投資活動現金	-1,066	-1,478	1,177	-150	568
資本支出淨額	-390	-437	-791	-692	-109
長期投資變動	-17	28	28	20	-7
其他投資現金	-659	-1,069	1,941	522	684
籌資活動現金	-2,640	-4,037	-6,169	-4,084	-2,283
長借/公司債變動	-81	-164	0	0	0
現金增資	0	0	0	0	0
發放現金股利	-2,298	-3,507	-5,659	-3,972	0
其他籌資現金	-261	-366	-510	-113	-2,283
淨現金流量	118	254	-115	-276	-764
期初現金	2,643	2,760	3,014	2,900	3,019
期末現金	2,760	3,014	2,900	2,624	2,255

資料來源：中信、CMoney

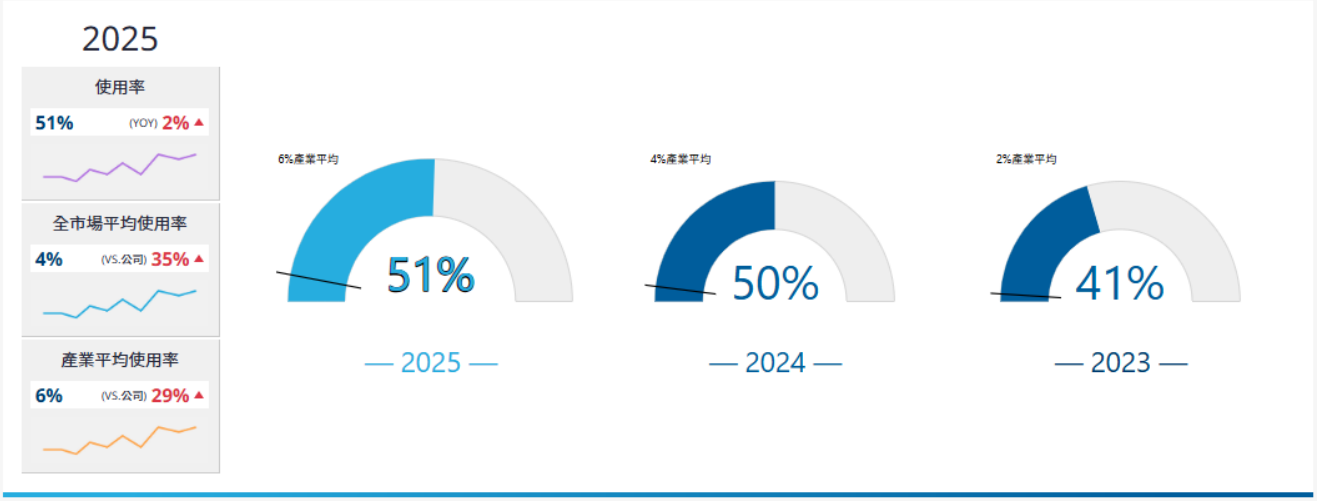
比率分析

(百萬元)	2023	2024	2025	2026F	2027F
成長力分析(%)					
營業收入淨額	1.17%	9.44%	-3.08%	2.21%	7.25%
營業毛利淨額	1.34%	7.47%	-4.75%	3.92%	8.17%
營業利益	9.20%	9.50%	-11.68%	7.96%	9.45%
稅後純益	13.98%	3.39%	-12.04%	7.94%	9.77%
獲利能力分析(%)					
毛利率	25.78%	25.31%	24.87%	25.29%	25.51%
EBITDA(%)	17.76%	17.12%	15.81%	16.71%	16.93%
營益率	14.91%	14.92%	13.60%	14.36%	14.65%
稅後純益率	12.43%	11.74%	10.66%	11.09%	11.35%
ROE(%)	25.60%	27.02%	25.20%	27.08%	28.77%

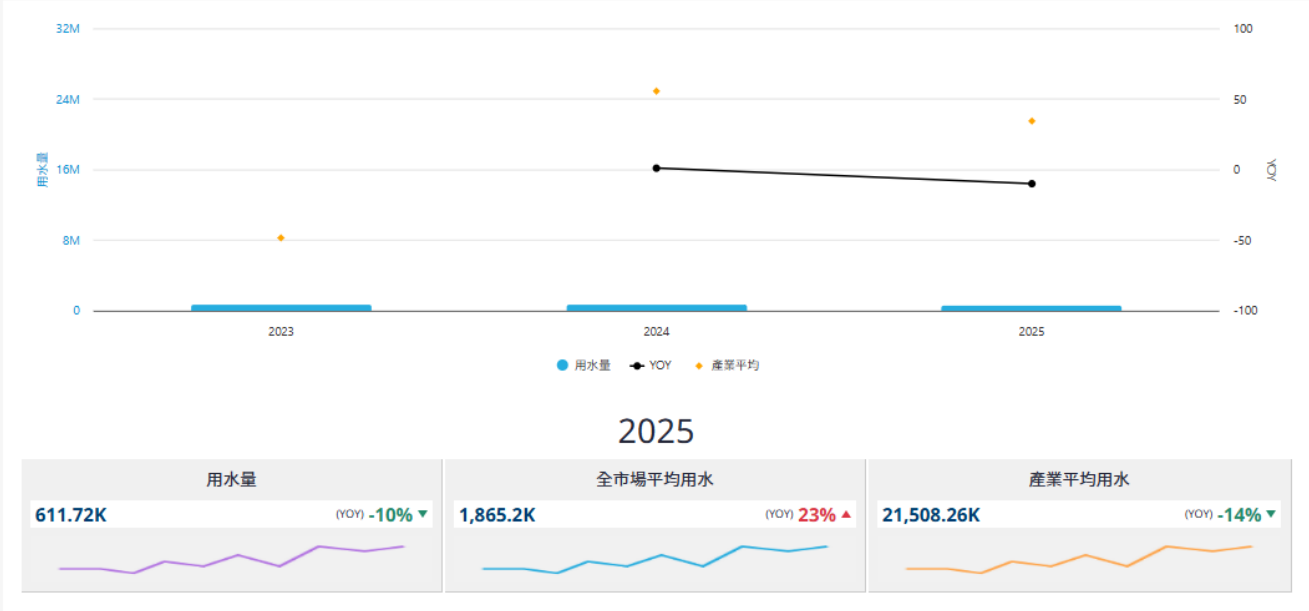
ESG 資訊-環境保護(Environmental)

溫室氣體(範疇一/二)排放量

再生能源使用率

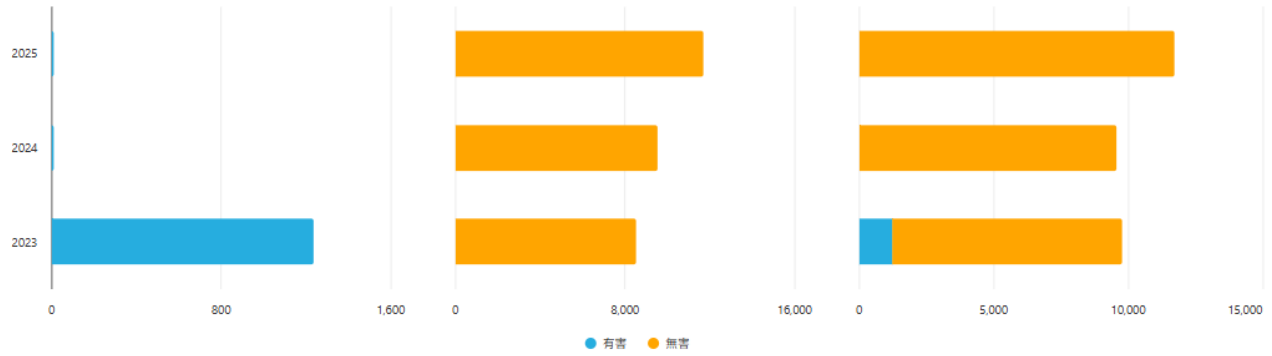


用水量



2026/10/07

廢棄物量



2025

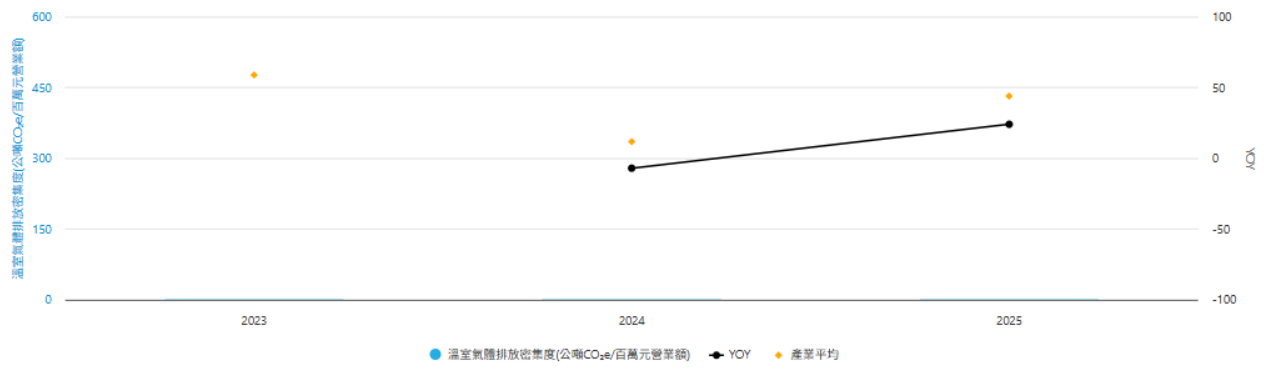
有害	無害	總量
0.01K (YOY) -13% ▼	11.7K (YOY) 23% ▲	11.71K (YOY) 23% ▲

密集度

2025

溫室氣體排放密集度	用水密集度	廢棄物密集度
1.081	17.768	0.34

溫室氣體排放密集度(公噸CO_2e/百萬元營業額)

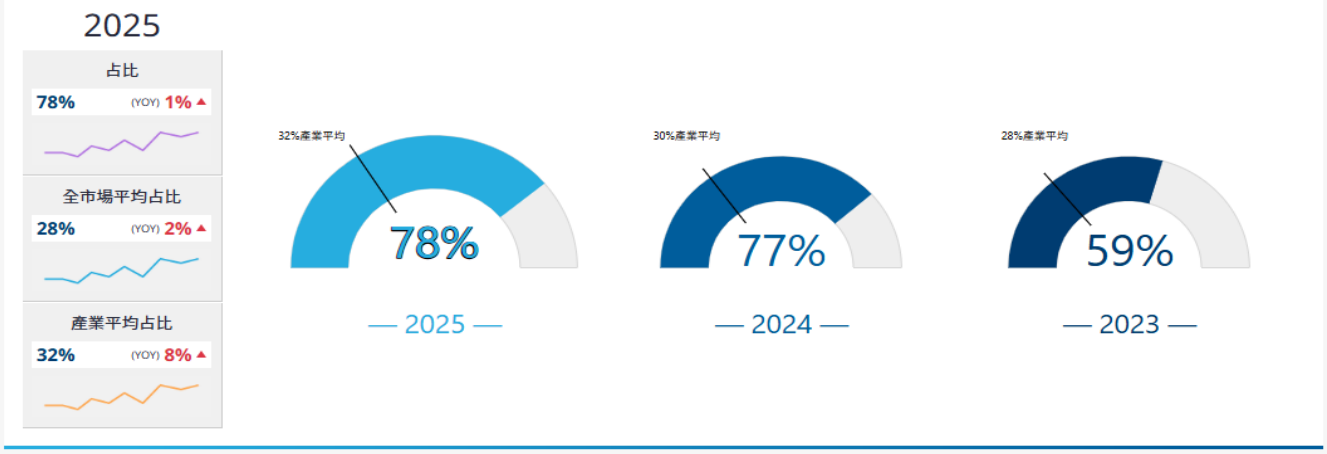


2025

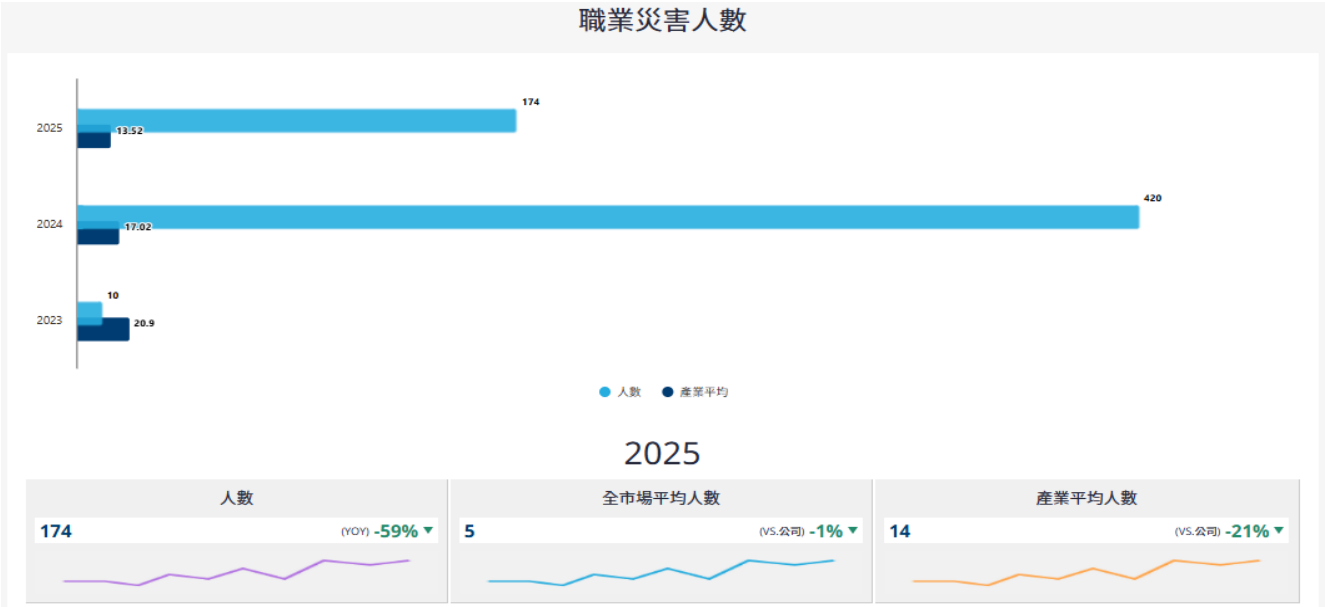
溫室氣體排放密集度(公噸CO_2e/百萬元營業額)	全市場平均溫室氣體排放密集度(公噸CO_2e/百萬元營業額)	產業平均溫室氣體排放密集度(公噸CO_2e/百萬元營業額)
0K (YOY) 24% ▲	0.03K (YOY) -24% ▼	0.43K (YOY) 29% ▲

ESG 資訊-社會責任(Social)

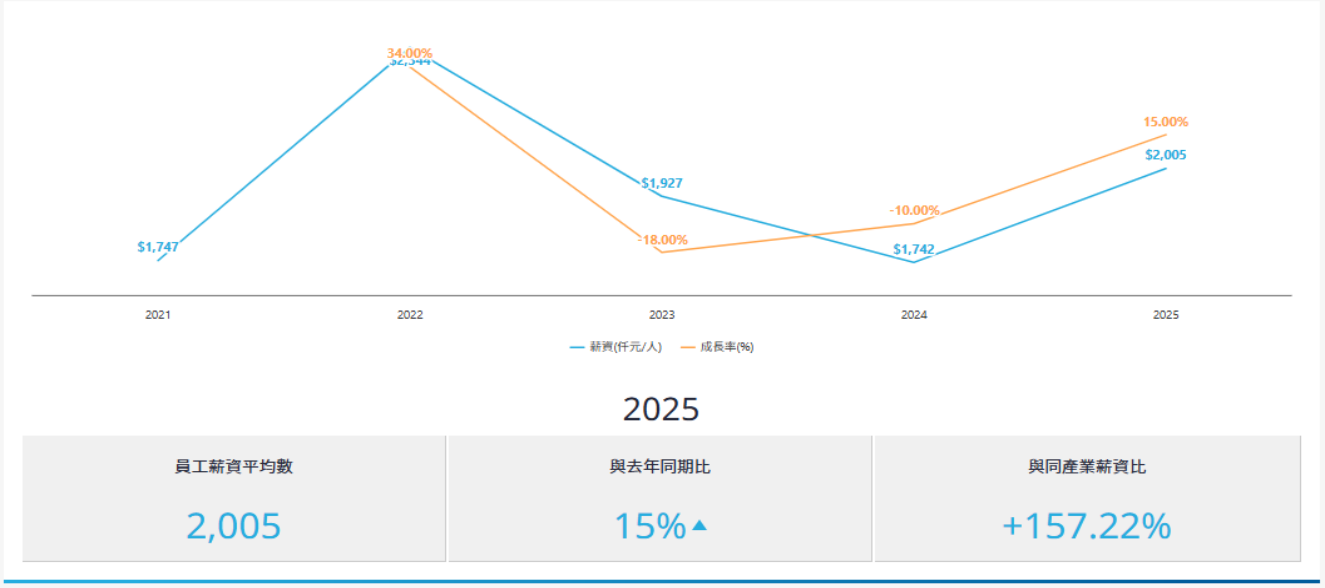
管理職女性主管占比



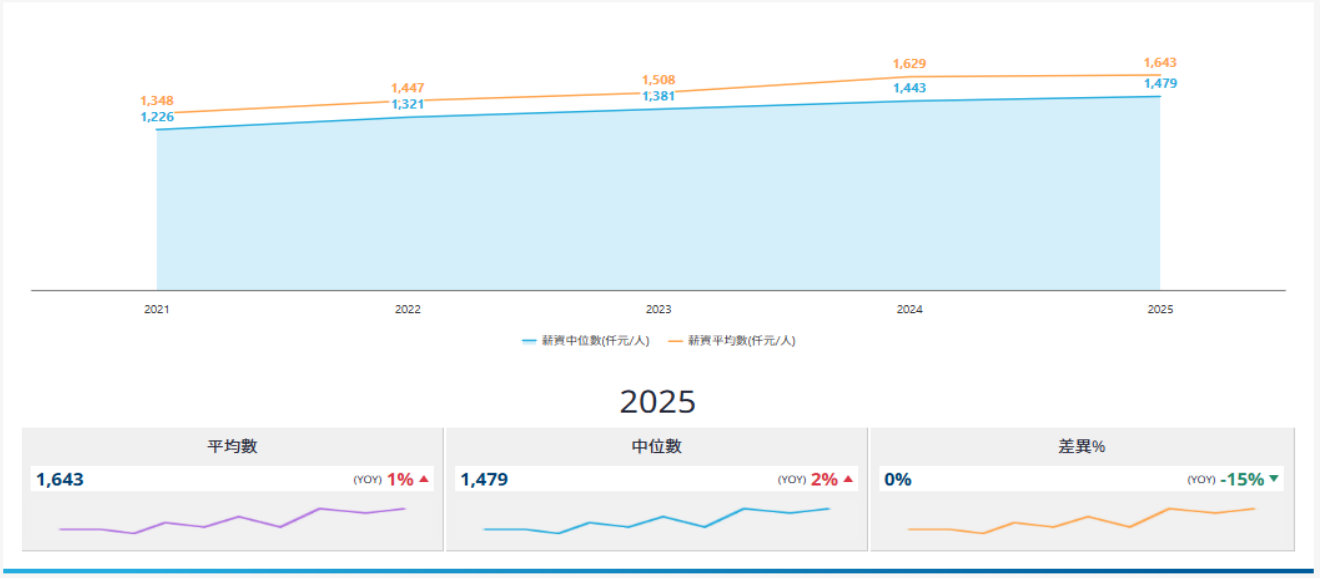
職業災害人數



員工薪資年度趨勢



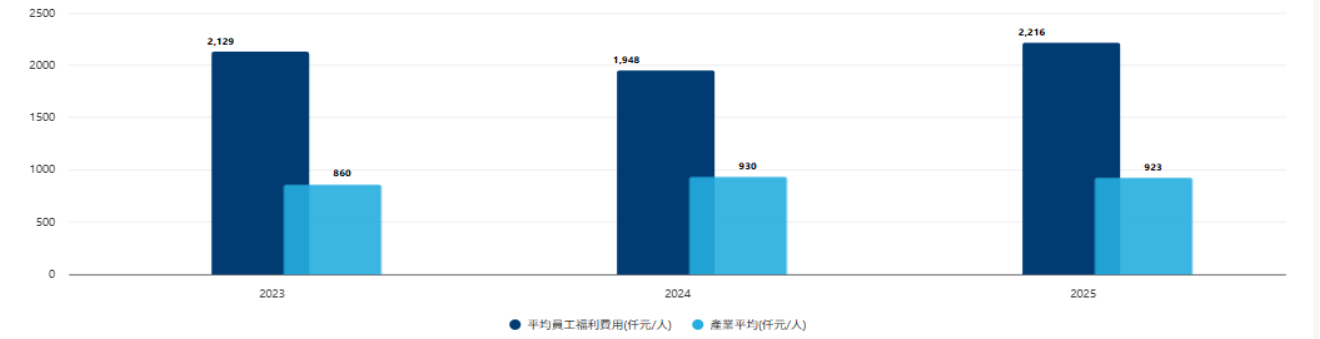
非擔任主管職務之全時員工薪資



2025

平均數	中位數	差異%
1,643 (YOY) 1% ▲	1,479 (YOY) 2% ▲	0% (YOY) -15% ▼

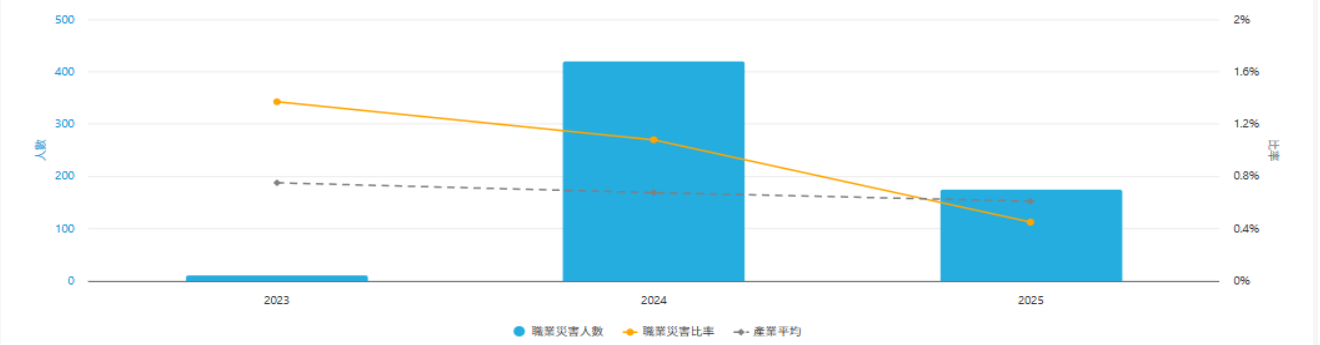
平均員工福利費用



2025

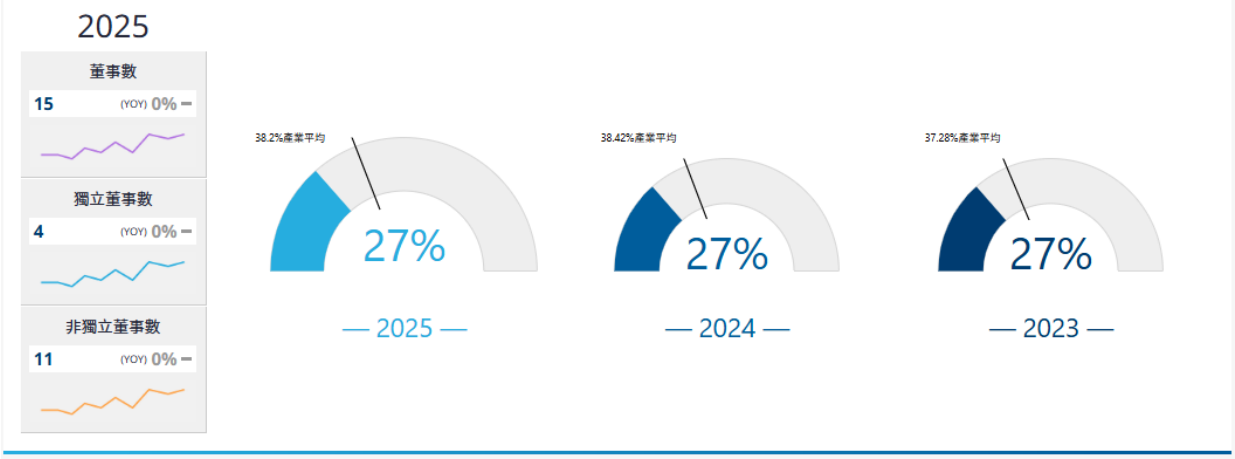
平均福利	全市場平均福利	產業平均福利
2,216 (YOY) 14% ▲	1,350 (YOY) 3% ▲	923 (YOY) -1% ▼

職業災害人數與比率

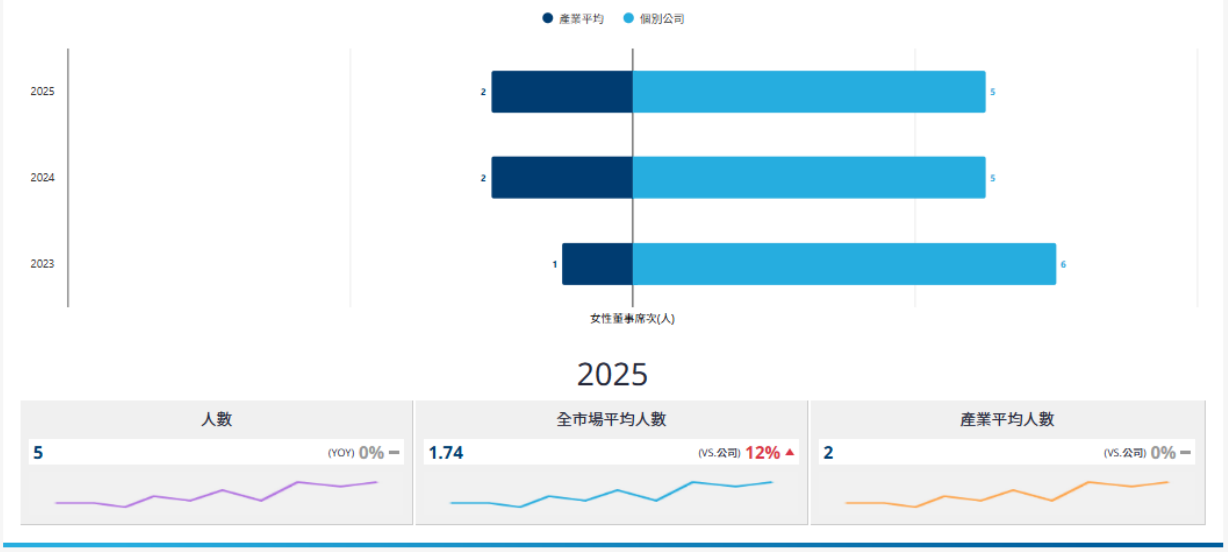


ESG 資訊-公司治理(Governance)

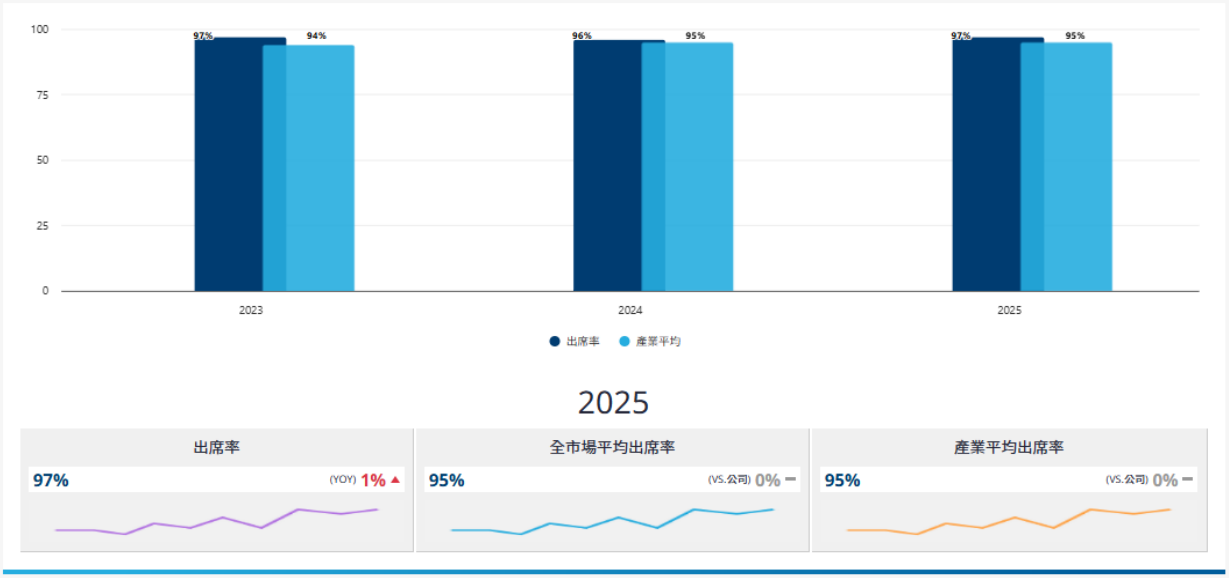
獨立董事vs.董事席次占比



女性董事席次



董事之董事會出席率



2026/10/07

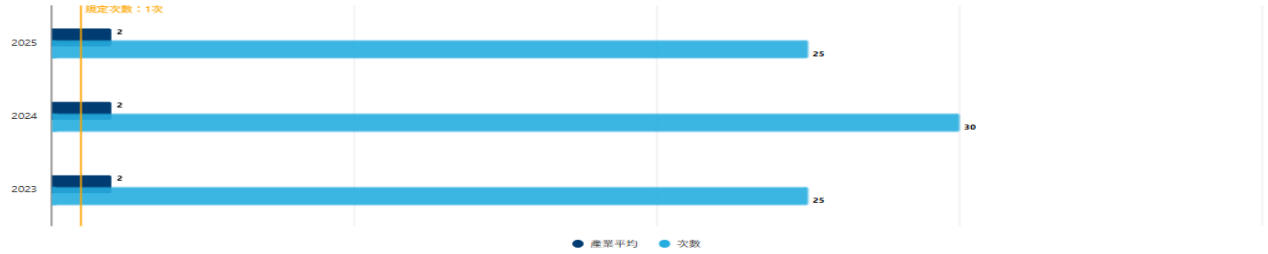
董事進修符合進修要點比率

內圈:個別公司　外圈:產業平均

2025

進修率	全市場平均進修率	產業平均進修率
100% (YOY) 0%	94% (VS.公司) 1%	93% (VS.公司) 3%

法說會次數



2025

次數	全市場平均次數	產業平均次數
25 (YOY) -17%	2 (VS.公司) 0%	2 (VS.公司) 0%

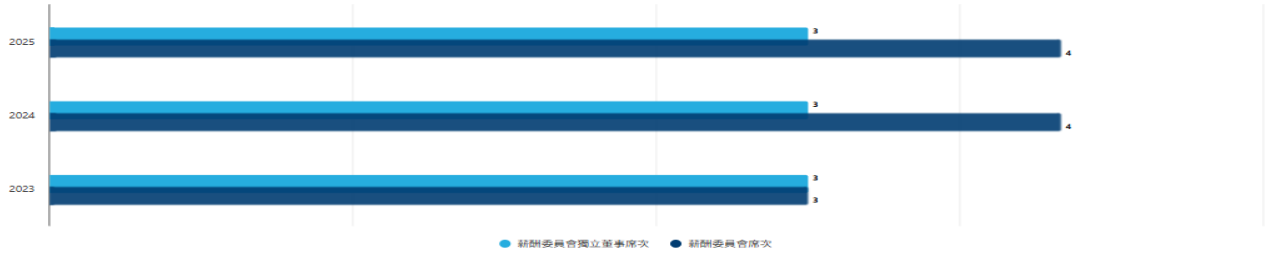
薪酬委員會出席率

內圈:個別公司出席率　外圈:產業平均

2025

進修率	全市場平均進修率	產業平均進修率
92% (YOY) 0%	97% (VS.公司) 1%	97% (VS.公司) 3%

薪酬委員會席次



資料來源：證交所

2026/10/07

投資評等說明

個股評等	Rating	定義
買進	Buy (B)	相較於市場指數之回報潛力極具吸引力
增加持股	Overweight (OW)	相較於市場指數之回報潛力略有吸引力
中立	Neutral (N)	相較於市場指數之回報潛力相似
降低持股	Underweight (UW)	相較於市場指數之回報潛力稍低
賣出	Sell (S)	相較於市場指數之回報潛力較低
未評等	Not Rated (NR)	

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台北市南港區經貿二路 188 號 5 樓及 18 樓 / (02) 5569-1988
營業執照字號：114 金管投顧新字第 014 號

06 Oct 2026 05:09:42 ET | 11 pages

Chroma ATE (2360.TW)

3Q26 Another Strong Beat; September Revenue 3x YoY

CITI'S TAKE

Chroma ATE reported Sep revenue that tripled YoY to NT$6,860mn (+22% MoM), concluding 3Q26 with revenue of NT$17,926mn (+180% YoY/+32% QoQ), another quarterly record and beating Street/Citi estimates by 23%/25%, respectively. We believe the robust revenue was mainly driven by datacenter power testers (HDVC, China's ESS build) and photonics (CPO ramp-up), and the 3Q revenue beat suggests upside to Street estimates for FY26 and FY27. Looking ahead, we expect robust AI-driven revenue growth to continue for Chroma across segments, including power tester (HVDC), SLT (GPU & ASIC), metrology (CoWoS, panel), and photonics. CPO shipment remains at an early stage and could expand several times next year. New products in metrology, FT handler, burn-in system could see more meaningful contribution starting 2027. We maintain our positive view and Buy rating for Chroma.

Buy	
Price (06 Oct 26 13:30)	NT$2,350.00
Target price	NT$2,500.00
Expected share price return	6.4%
Expected dividend yield	0.8%
Expected total return	7.2%
Market Cap	NT$999,327M
	US$31,497M

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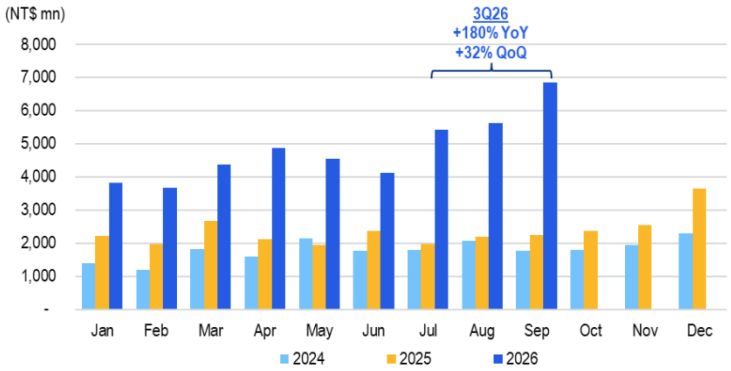
Figure 1. Chroma ATE – Monthly Revenue Trend



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Source: Company Reports, Citi Research

See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations

Figure 2. Chroma ATE – 3Q26 Preliminary Results vs. Estimates

(NT$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26 Prelim	3Q26 BBG	3Q26 Diff	3Q26 Citi est	3Q26 Diff
Revenue	**6,865**	**6,455**	**6,410**	**8,580**	**11,859**	**13,529**	**17,926**	**14,618**	**23%**	**14,314**	**25%**
YoY %	*55%*	*17%*	*14%*	*42%*	*73%*	*110%*	*180%*	*128%*		*123%*	
Gross Profit	4,148	4,224	3,835	5,217	7,424	8,185		9,032		8,755	
GPM	**60.4%**	**65.4%**	**59.8%**	**60.8%**	**62.6%**	**60.5%**		**61.8%**		**61.2%**	
Opex	(1,984)	(2,027)	(1,987)	(2,229)	(2,627)	(2,943)		(3,036)		(2,816)	
Opex %	28.9%	31.4%	31.0%	26.0%	22.2%	21.8%		20.8%		19.7%	
EBIT	**2,164**	**2,197**	**1,848**	**2,988**	**4,797**	**5,242**		**5,997**		**5,938**	
EBIT margin	31.5%	34.0%	28.8%	34.8%	40.5%	38.7%		41.0%		41.5%	
Pre-Tax Profit- Adj.	2,597	2,438	2,288	3,408	5,094	6,514		6,268		6,183	
Net Profit - Adj.	**2,123**	**1,953**	**1,880**	**2,549**	**3,864**	**5,123**		**5,091**		**4,923**	
YoY %	*122%*	*39%*	*32%*	*73%*	*82%*	*162%*		*171%*		*162%*	
EPS (NT$) - Adj.	**4.98**	**4.60**	**4.42**	**5.99**	**9.07**	**12.10**		**11.85**		**11.63**	

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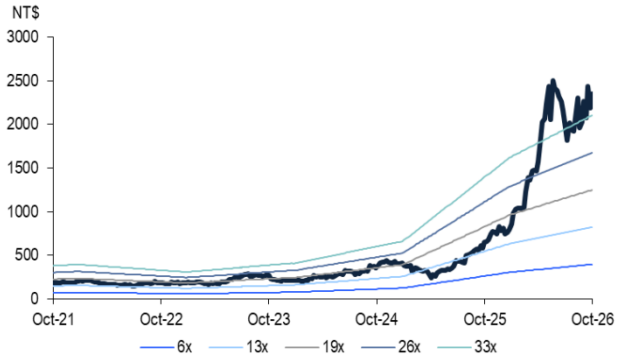
Source: Company Reports, Bloomberg, Citi Research Estimates

Figure 3. Chroma ATE – 12M Fwd. P/E Band



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Source: dataCentral, Citi Research

Figure 4. Chroma ATE – 12M Fwd. P/E Chart

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Source: dataCentral, Citi Research

RELATED:

Chroma ATE (2360.TW): August Revenue +156% YoY, Another Record High

Chroma ATE (2360.TW): Chroma Plans 13.6mn Share GDR Issuance; Dilution Manageable with Long-Term Upside Potential

Chroma ATE (2360.TW): 2Q Beat on Non-Op, Positive Outlook Intact

Chroma ATE

Valuation

Our target price of NT$2,500 is based on 40x 2027E P/E, at close to 1x PEG. Chroma shares have re-rated above its historical range as its growth outlook is now strongly tied to AI infrastructure build out. We estimate 60%+ of its revenue is now AI-related, with strong growth across product segments including power testers, SLT, metrology, and photonics. Chroma has a commanding market share in most of these segments and should be well positioned to capitalize on the global AI expansion.

Risks

Downside risks to achieving our target price include: 1) softer AI/CPO growth expectation due to changes in capex outlook or production bottleneck; 2) fewer SLT needed as a result of shortened cycle time as GPU yield rate improves; 3) disappointing metrology shipment due to changes at ramp-up schedule; 4) testing process change leading to loss in market share; 5) FX risks (50-60% of Chroma's sales are in US$, 10-20% in Rmb).

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Appendix A-1

ANALYST CERTIFICATION

The research analysts primarily responsible for the preparation and content of this research report are either (i) designated by "AC" in the author block or (ii) listed in bold alongside content which is attributable to that analyst. If multiple AC analysts are designated in the author block, each analyst is certifying with respect to the entire research report other than (a) content attributable to another AC certifying analyst listed in bold alongside the content and (b) views expressed solely with respect to a specific issuer which are attributable to another AC certifying analyst identified in the price charts or rating history tables for that issuer shown below. Each of these analysts certify, with respect to the sections of the report for which they are responsible: (1) that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc. and its affiliates; and (2) no part of the research analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in this report.

IMPORTANT DISCLOSURES

Chroma ATE (2360.TW)

Ratings and Target Price History
Fundamental Research

Analyst: Kevin Chen

	Date	Rating	Target Price	Closing Price
1	11-Oct-23 14:29:57	1	*330.00	253.00
2	21-Feb-24 11:50:58	1	*280.00	227.50
3	30-Apr-24 13:48:08	1	*320.00	266.50
4	31-Jul-24 15:19:03	1	*360.00	301.00
5	04-Nov-24 17:53:01	1	*500.00	437.00
6	30-Apr-25 14:29:54	1	*425.00	292.50
7	29-Jun-25 19:17:54	1	*500.00	429.50
8	06-Aug-25 09:55:28	1	*550.00	494.50
9	30-Oct-25 18:40:38	1	*900.00	804.00
10	22-Feb-26 19:26:56	1	*1,200.00	1,050.00
11	11-Mar-26 02:42:34	1	*1,650.00	1,495.00
12	03-May-26 18:43:42	1	*2,500.00	2,120.00

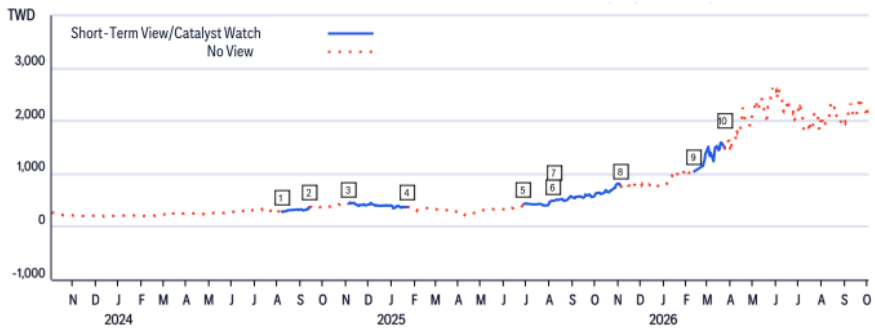
*Indicates Change

Rating/target price changes above reflect Eastern Time

Chroma ATE (2360.TW)

Short-Term View/Catalyst Watch Research

Analyst: Kevin Chen



	Date	Action	Expected Direction	Duration	Closing Price
1	07-Aug-24 18:28:11	Add CW	Upside	90 Days	292.00
2	13-Sep-24 00:34:47	Remove CW	Upside	90 Days	380.00
3	04-Nov-24 12:53:01	Add STV	Upside	90 Days	437.00
4	02-Feb-25 19:44:53	Remove STV	Upside	90 Days	379.50
5	29-Jun-25 15:17:54	Add STV	Upside	90 Days	429.50
6	06-Aug-25 05:55:28	Remove STV	Upside	90 Days	494.50
7	06-Aug-25 05:55:28	Add CW	Upside	90 Days	494.50
8	04-Nov-25 19:35:17	Remove CW	Upside	90 Days	776.00
9	22-Feb-26 14:26:56	Add CW	Upside	30 Days	1,050.00
10	24-Mar-26 21:33:21	Remove CW	Upside	30 Days	1,480.00

CW - Catalyst Watch , STV - Short-Term View

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	12 Month Rating			Catalyst Watch		
Data current as of 01 Oct 2026	**Buy**	**Hold**	**Sell**	**Buy**	**Hold**	**Sell**
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% of companies in each rating category that are investment banking clients	39%	43%	30%	42%	38%	36%

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06 Oct 2026 07:28:09 ET | 11 pages

WinWay Technology (6515.TW)

Sep-26 Sales Dip on Probe Card, Mix Shift to Support Margins

CITI'S TAKE

WinWay reported Sep-26 revenues of NT$1.25bn (-27% MoM/+88% YoY). QTD 3Q26 revenues of NT$4.56bn (+29% QoQ/+153% YoY) beat Citi/cons forecasts by 3%/7%, on robust AI/HPC test interface demand. The MoM decline was mainly driven by lower probe card revenues as major project shipments have temporarily wrapped up, and we expect probe card contribution to stay light in 4Q26E. While this may weigh on the top line near term, we see margins could recover gradually as product mix improves alongside the continued AI GPU and ASIC socket ramp. In the long term, we continue to favor WinWay as a core beneficiary of rising testing complexity for advanced chips across diverse platforms. Reiterate Buy/High Risk.

Buy / High Risk	
Price (06 Oct 26 13:30)	NT$6,130.00
Target price	NT$11,500.00
Expected share price return	87.6%
Expected dividend yield	1.1%
Expected total return	88.7%
Market Cap	NT$223,407M
	US$7,041M

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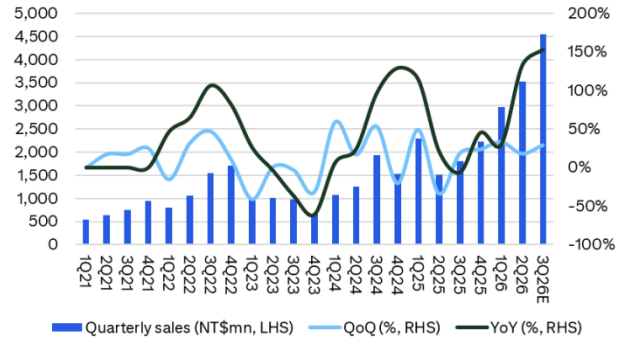
Figure 1. WinWay - Quarterly sales trend



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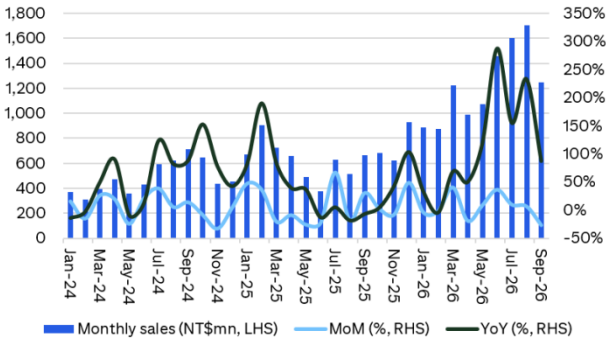
Source: Company Reports, Citi Research

Figure 2. WinWay - Monthly sales trend



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WinWay Technology

Valuation

We value WinWay using a PE-based approach that factors in earnings momentum and likely expansion of its addressable market. Our target price of NT$11,500 is derived from applying c.41x to 2H27E-1H28E EPS, higher than the upper end of its long-term PE range, which we believe is justified given the company's entrenched position in the high-end AI GPU testing supply chain, rising contribution from coaxial socket and SLT solutions, and the increasing value of test content per module. With these structural drivers in place, and as we believe the market is starting to value it as an AI/HPC play (rather than a consumer electronics stock), we expect the multiple to hold at a premium relative to its historical averages.

Risks

Our quant rating system rates the WinWay stock High Risk based on its historical volatility.

Key downside risks that could impede the stock from reaching our target price and our earnings forecasts include: 1) client and sector concentration – specifically any delays in US AI GPU product roadmap could impact the business outlook for WinWay; 2) intensifying competition in coaxial and hypersockets from global peers; 3) faster-than-expected testing algorithm optimization, which could shorten testing durations and reduce the anticipated hardware demand multiplier; and 4) delay of MEMS Phase 2 mass production.

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Appendix A-1

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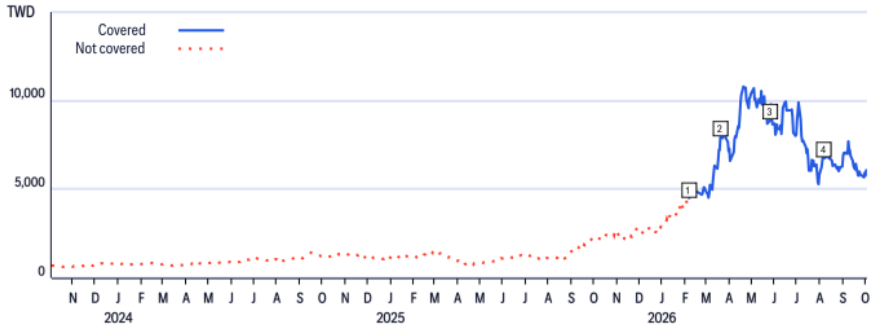
WinWay Technology (6515.TW)
Ratings and Target Price History
Fundamental Research

Analyst: Michael Hung



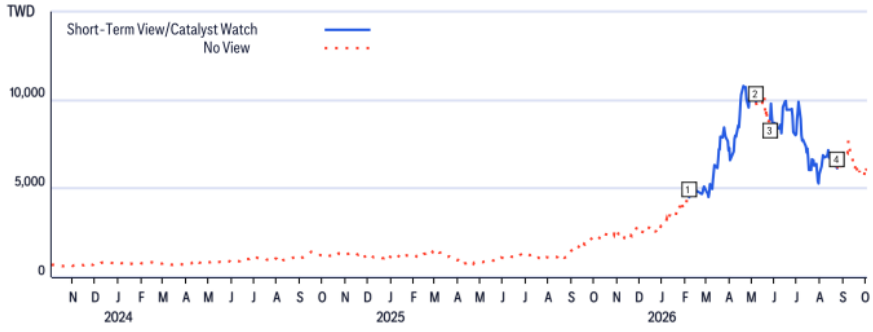
	Date	Rating	Target Price	Closing Price
1	06-Feb-26 05:59:37	*1H	*6,000.00	4,510.00
2	20-Mar-26 02:14:37	1H	*10,000.00	7,930.00
3	26-May-26 19:30:25	1H	*13,000.00	8,905.00
4	06-Aug-26 12:43:24	1H	*11,500.00	6,775.00

*Indicates Change

Rating/target price changes above reflect Eastern Time

WinWay Technology (6515.TW)
Short-Term View/Catalyst Watch Research

Analyst: Michael Hung



	Date	Action	Expected Direction	Duration	Closing Price
1	06-Feb-26 00:59:37	Add CW	Upside	90 Days	4,510.00
2	07-May-26 21:43:12	Remove CW	Upside	90 Days	9,860.00
3	26-May-26 15:30:25	Add CW	Upside	90 Days	8,905.00
4	24-Aug-26 21:35:40	Remove CW	Upside	90 Days	6,145.00

CW - Catalyst Watch , STV - Short-Term View

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	12 Month Rating			Catalyst Watch		
Data current as of 01 Oct 2026	**Buy**	**Hold**	**Sell**	**Buy**	**Hold**	**Sell**
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% of companies in each rating category that are investment banking clients	39%	43%	30%	42%	38%	36%

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Kinsus Interconnect Technology (3189 TT)

Target price: **TWD1,580.00** (from TWD1,068.00)

Share price (6 Oct): **TWD1,070.00** | Up/downside: **+47.7%**

5 4 3 2 **1**

Buy (unchanged)

3Q26 results preview: several favourable trends ahead

- Rising high-end ABF substrate likely to drive blended ASP increase
- Multicore and optics likely to be the next growth drivers
- Reaffirming our Buy (1) call; raising 12M TP to TWD1,580

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What's new: We believe an improving product mix, driven by rising Nvidia's CPU contribution, to be a key near-term share price catalyst.

What's the impact: **Near-term update.** After Kinsus released its August preliminary results on 23 September, we note that we had underestimated the 3Q26E gross margin expansion on product optimisation and price hike scale. The results imply a GM of 33% in August (vs. 26% in 2Q26). We therefore lift our 3Q-4Q26E gross margin to 33.2-35.1% (from 28.9-30.3%) to reflect an improving product mix, including greater CPU substrate contribution. We believe the gross margin of high-end chipset substrate may be c.50%, supporting margin expansion. We also expect a c.10% QoQ price hike for both BT and ABF substrate in 3Q26, above Kinsus' guidance of a high-single-digit% QoQ, making these substrates the other growth driver. We see further gross margin expansion in 2027-28 on the ongoing price hike trend and increasing high-end ABF substrate product mix.

Business outlook. We believe several favourable trends lie ahead. First, we see a rising contribution from high-end GPU and CPU ABF substrates. We believe Nvidia's CPU currently accounts for 5-10% of total revenue, with Kinsus holding a 70-80% market share. We also expect Kinsus to acquire a teens% market share in AMD's Venice, which should gradually ramp up from 4Q26. These will likely become key growth drivers of the ABF substrate segment in 2027, in our view. We believe the ramp-up progress for GPU substrate for these two clients is slightly slower than that for CPU. Second, we believe Kinsus will benefit from the rising BT substrate demand for optical transceivers. In our view, migration to on-board optics (OBO) and NPO/CPO will require ABF substrate beneath the optics engine (OE) in OBO; the OE and IC will need a bigger ABF substrate as we move to the NPO/CPO generation. As such, we expect the total ABF substrate demand to increase with rising adoption of optics in AI servers. Lastly, we expect the adoption of multiple-core ABF from 2028 to improve the rigidity and transmission performance before the official launch of ceramic and glass-core substrate, which we expect to increase the ABF TAM as well.

What we recommend: We raise our 2026-28E earnings by 19-29% to reflect the better-than-expected gross margin in August. We reaffirm our Buy (1) call and raise our 12M TP to TWD1,580, based on an unchanged target PER of 36x (high end of 3-year-trading range of 10-36x), applied to the 1-year-forward EPS forecast. Key downside risks: weaker-than-expected demand for memory, PCs, gaming GPUs or AI servers.

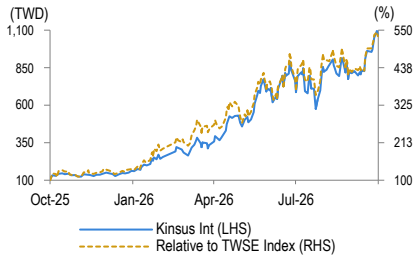
How we differ: Our 2026-28E EPS are 27-57% above the Bloomberg consensus, likely on our more bullish view on its gross margin expansion.

Forecast revisions (%)

Year to 31 Dec	26E	27E	28E
Revenue change	1.3	-	2.2
Net profit change	29.1	28.7	18.8
Core EPS (FD) change	18.6	18.3	9.2

Source: Daiwa forecasts

Share price performance



12-month range	111.18-1,095.00
Market cap (USDbn)	17.18
3m avg daily turnover (USDm)	428.04
Shares outstanding (m)	510
Major shareholder	Asus Investment Co Ltd (13.2%)

Financial summary (TWD)

Year to 31 Dec	26E	27E	28E
Revenue (m)	54,062	88,448	125,676
Operating profit (m)	9,142	24,773	43,198
Net profit (m)	7,558	22,325	39,673
Core EPS (fully-diluted)	14.824	43.787	77.810
EPS change (%)	322.9	195.4	77.7
Daiwa vs Cons. EPS (%)	29.1	57.1	46.7
PER (x)	72.2	24.4	13.8
Dividend yield (%)	0.3	0.4	0.4
DPS	3.5	4.5	4.5
PBR (x)	11.6	8.1	5.3
EV/EBITDA (x)	30.7	15.1	8.8
ROE (%)	15.2	32.4	40.0

Source: FactSet, Daiwa forecasts

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Flash

Kinsus: revenue and earnings forecast revisions

	2026E			2027E			2028E		
(TWDm)	Previous	New	Consensus	Previous	New	Consensus	Previous	New	Consensus
Revenue	53,385	54,062	53,912	88,484	88,448	80,794	122,920	125,676	120,243
Diff (%)		1.3%	0.3%		0.0%	9.5%		2.2%	4.5%
Gross Margin (%)	27.0%	29.65%	27.3%	35.6%	40.5%	33.7%	43.2%	46.37%	38.5%
Operating profit	7,578	9,142	7,907	20,449	24,773	18,170	38,330	43,198	32,345
Op Margin (%)	14.2%	16.9%	14.7%	23.1%	28.0%	22.5%	31.2%	34.4%	26.9%
Net profit	5,855	7,558	5,857	17,344	22,325	14,212	33,395	39,673	27,051
EPS	12.50	14.82	11.49	37.02	43.79	27.87	71.29	77.81	53.06
Diff (%)		18.6%	29.1%		18.3%	57.1%		9.2%	46.7%

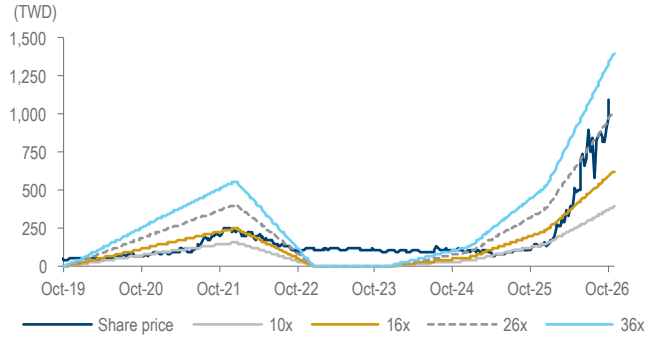
Source: Bloomberg, Daiwa forecasts

Kinsus: quarterly and annual P&L statement

	2026E				2027E				2025	2026E	2027E	2028E
(TWDm)	1Q	2Q	3QE	4QE	1QE	2QE	3QE	4QE				
Net revenue	**11,105**	**12,496**	**14,539**	**15,922**	**18,046**	**21,099**	**23,941**	**25,361**	**39,351**	**54,062**	**88,448**	**125,676**
COGS	(8,755)	(9,235)	(9,716)	(10,326)	(11,304)	(12,750)	(14,067)	(14,480)	(31,037)	(38,032)	(52,601)	(67,397)
Gross profit	**2,349**	**3,261**	**4,824**	**5,596**	**6,743**	**8,349**	**9,874**	**10,881**	**8,314**	**16,030**	**35,847**	**58,279**
Operating expenses	1,459	1,582	1,841	2,006	2,274	2,637	2,993	3,170	5,645	6,888	11,074	15,081
Operating profit	**891**	**1,679**	**2,983**	**3,590**	**4,469**	**5,712**	**6,881**	**7,711**	**2,670**	**9,142**	**24,773**	**43,198**
Non-operating profit	74	106	88	90	89	84	90	92	417	358	354	359
Pre-tax profit	**964**	**1,784**	**3,071**	**3,680**	**4,558**	**5,796**	**6,971**	**7,803**	**3,087**	**9,499**	**25,127**	**43,557**
Income taxes	(89)	(101)	(174)	(208)	(258)	(328)	(394)	(441)	(370)	(571)	(1,420)	(2,462)
Net profit	**549**	**1,311**	**2,586**	**3,112**	**3,986**	**5,132**	**6,230**	**6,978**	**1,596**	**7,558**	**22,325**	**39,673**
Net EPS (TWD)	**1.17**	**2.57**	**5.07**	**6.10**	**7.82**	**10.06**	**12.22**	**13.69**	**3.5**	**14.8**	**43.8**	**77.8**
Operating Ratios												
Gross margin	21.2%	26.1%	33.2%	35.1%	37.4%	39.6%	41.2%	42.9%	21.1%	29.7%	40.5%	46.4%
Operating margin	8.0%	13.4%	20.5%	22.5%	24.8%	27.1%	28.7%	30.4%	6.8%	16.9%	28.0%	34.4%
Pre-tax margin	8.7%	14.3%	21.1%	23.1%	25.3%	27.5%	29.1%	30.8%	7.8%	17.6%	28.4%	34.7%
Net margin	4.9%	10.5%	17.8%	19.5%	22.1%	24.3%	26.0%	27.5%	4.1%	14.0%	25.2%	31.6%
YoY (%)												
Net revenue	29%	31%	40%	47%	63%	69%	65%	59%	29%	37%	64%	42%
Gross profit	21%	62%	146%	134%	187%	156%	105%	94%	-4%	93%	124%	63%
Operating profit	78%	171%	375%	290%	402%	240%	131%	115%	144%	242%	171%	74%
Pre-tax profit	38%	188%	331%	249%	373%	225%	127%	112%	93%	208%	165%	73%
Net profit	99%	288%	662%	384%	626%	292%	141%	124%	3164%	374%	195%	78%
QoQ (%)												
Net revenue	3%	13%	16%	10%	13%	17%	13%	6%				
Gross profit	-2%	39%	48%	16%	20%	24%	18%	10%				
Operating profit	-3%	88%	78%	20%	24%	28%	20%	12%				
Pre-tax profit	-9%	85%	72%	20%	24%	27%	20%	12%				
Net profit	-14%	139%	97%	20%	28%	29%	21%	12%				

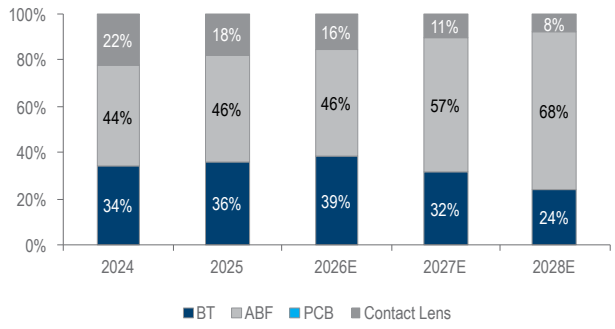
Source: Company, Daiwa forecasts

Kinsus: 1-year forward PER bands



Source: TEJ, Daiwa forecasts

Kinsus: revenue breakdown by product



Source: Company, Daiwa forecasts

Financial summary

Key assumptions

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Global PC shipment (mn)	361	301	260	263	287	244	244	244
Regular server shipmnet (mn)	14	15	12	14	16	18	22	26
Global smartphone shipment (mn)	1,655	1,437	1,380	1,437	1,429	1,249	1,255	1,300

Profit and loss (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
ABF	10,191	16,526	11,978	13,319	17,998	24,558	48,820	82,562
BT	17,565	18,796	8,065	10,398	14,314	21,234	30,530	33,105
Other Revenue	5,580	6,336	6,790	6,817	7,039	8,271	9,098	10,008
Total Revenue	**33,336**	**41,658**	**26,832**	**30,535**	**39,351**	**54,062**	**88,448**	**125,676**
Other income	0	0	0	0	0	0	0	0
COGS	(22,583)	(25,771)	(20,075)	(21,867)	(31,037)	(38,032)	(52,601)	(67,397)
SG&A	(2,879)	(3,532)	(3,301)	(5,010)	(3,433)	(4,241)	(6,960)	(9,312)
Other op.expenses	(2,413)	(2,484)	(2,397)	(2,612)	(2,212)	(2,647)	(4,114)	(5,769)
Operating profit	**5,456**	**9,851**	**1,042**	**1,095**	**2,670**	**9,142**	**24,773**	**43,198**
Net-interest inc./(exp.)	(68)	(172)	(332)	(394)	(391)	(336)	(356)	(350)
Assoc/forex/extraord./others	253	607	716	902	809	694	710	709
Pre-tax profit	**5,641**	**10,287**	**1,426**	**1,603**	**3,087**	**9,499**	**25,127**	**43,557**
Tax	(672)	(2,158)	(256)	(272)	(370)	(571)	(1,420)	(2,462)
Min. int./pref. div./others	(633)	(957)	(1,123)	(1,282)	(1,121)	(1,370)	(1,381)	(1,422)
Net profit (reported)	**4,336**	**7,172**	**48**	**49**	**1,596**	**7,558**	**22,325**	**39,673**
Net profit (adjusted)	**4,336**	**7,172**	**48**	**49**	**1,596**	**7,558**	**22,325**	**39,673**
EPS (reported)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
EPS (adjusted)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
EPS (adjusted fully-diluted)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
DPS (TWD)	**4.500**	**6.476**	**1.000**	**1.000**	**1.518**	**3.500**	**4.500**	**4.500**
EBIT	**5,456**	**9,851**	**1,042**	**1,095**	**2,670**	**9,142**	**24,773**	**43,198**
EBITDA	**9,833**	**14,958**	**6,394**	**7,228**	**9,914**	**17,313**	**33,761**	**53,085**

Cash flow (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Profit before tax	5,641	10,287	1,426	1,603	3,087	9,499	25,127	43,557
Depreciation and amortisation	4,377	5,107	5,351	6,133	7,244	8,171	8,988	9,887
Tax paid	(672)	(2,158)	(256)	(272)	(370)	(571)	(1,420)	(2,462)
Change in working capital	(1,347)	(176)	1,528	(906)	(1,959)	(3,452)	(2,982)	(3,707)
Other operational CF items	2,575	2,901	(1,793)	1,147	91	(634)	0	0
Cash flow from operations	**10,574**	**15,961**	**6,256**	**7,705**	**8,093**	**13,013**	**29,713**	**47,274**
Capex	(13,384)	(17,094)	(10,136)	(10,408)	(6,117)	(7,832)	(5,483)	(3,838)
Net (acquisitions)/disposals	0	0	0	0	0	0	0	0
Other investing CF items	489	559	(2,351)	1,301	(1,752)	(2,374)	(44)	(85)
Cash flow from investing	**(12,895)**	**(16,535)**	**(12,486)**	**(9,107)**	**(7,869)**	**(10,206)**	**(5,526)**	**(3,922)**
Change in debt	6,544	1,903	6,043	1,624	(1,684)	(8,891)	0	(45)
Net share issues/(repurchases)	0	0	0	0	0	0	0	0
Dividends paid	(451)	(2,443)	(3,438)	(454)	(457)	(774)	(1,785)	(2,294)
Other financing CF items	(94)	2,411	2,730	(173)	(1,130)	8,625	0	0
Cash flow from financing	**6,000**	**1,870**	**5,335**	**996**	**(3,270)**	**(1,041)**	**(1,785)**	**(2,340)**
Forex effect/others	(12)	56	(88)	71	(39)	51	39	35
Change in cash	**3,667**	**1,352**	**(983)**	**(335)**	**(3,084)**	**1,817**	**22,441**	**41,047**
Free cash flow	(2,809)	(1,133)	(3,880)	(2,703)	1,976	5,181	24,230	43,436

Source: FactSet, Daiwa forecasts

Financial summary continued …

Balance sheet (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & short-term investment	15,968	17,923	21,013	16,676	16,336	21,188	43,628	84,676
Inventory	3,466	3,481	2,612	2,981	4,828	6,237	8,747	10,924
Accounts receivable	5,679	5,050	4,299	5,284	6,804	10,300	16,405	22,824
Other current assets	1,312	1,647	1,445	4,735	970	1,080	1,080	1,080
Total current assets	**26,424**	**28,100**	**29,369**	**29,677**	**28,938**	**38,804**	**69,860**	**119,503**
Fixed assets	31,586	44,101	48,209	49,658	50,897	51,643	48,138	42,089
Goodwill & intangibles	0	0	0	0	0	0	0	0
Other non-current assets	376	432	489	100	138	434	478	562
Total assets	**58,386**	**72,633**	**78,067**	**79,435**	**79,973**	**90,882**	**118,476**	**162,154**
Short-term debt	1,100	377	1,409	2,452	3,053	915	915	869
Accounts payable	10,150	10,798	8,457	7,521	10,030	14,004	19,638	24,526
Other current liabilities	2,733	4,199	4,393	7,018	7,023	5,571	5,571	5,571
Total current liabilities	**13,983**	**15,374**	**14,259**	**16,990**	**20,106**	**20,490**	**26,124**	**30,966**
Long-term debt	9,387	10,770	15,280	13,779	11,197	6,219	6,219	6,219
Other non-current liabilities	2,027	7,674	9,369	8,788	7,297	6,318	6,318	6,318
Total liabilities	**25,397**	**33,818**	**38,908**	**39,557**	**38,600**	**33,027**	**38,661**	**43,503**
Share capital	4,508	4,528	4,544	4,566	4,568	5,269	5,269	5,269
Reserves/R.E./others	28,481	34,288	34,615	35,312	36,805	52,585	74,546	113,382
Shareholders' equity	**32,989**	**38,816**	**39,159**	**39,878**	**41,373**	**57,854**	**79,815**	**118,651**
Minority interests	0	0	0	0	0	0	0	0
Total equity & liabilities	**58,386**	**72,633**	**78,067**	**79,435**	**79,973**	**90,882**	**118,476**	**162,154**
EV	540,075	538,779	541,231	545,110	543,470	531,501	509,061	467,968
Net debt/(cash)	**(5,481)**	**(6,776)**	**(4,325)**	**(445)**	**(2,085)**	**(14,054)**	**(36,495)**	**(77,588)**
BVPS (TWD)	**64.478**	**75.804**	**69.748**	**69.314**	**71.656**	**92.357**	**131.414**	**202.420**

Key ratios (%)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Sales (YoY)	23.0	25.0	(35.6)	13.8	28.9	37.4	63.6	42.1
EBITDA (YoY)	70.8	52.1	(57.3)	13.1	37.1	74.6	95.0	57.2
Operating profit (YoY)	307.0	80.6	(89.4)	5.1	143.7	242.4	171.0	74.4
Net profit (YoY)	700.2	65.4	(99.3)	2.9	3,164.4	373.6	195.4	77.7
Core EPS (fully-diluted) (YoY)	698.1	65.3	(99.3)	2.5	3,153.4	322.9	195.4	77.7
Gross-profit margin	32.3	38.1	25.2	28.4	21.1	29.7	40.5	46.4
EBITDA margin	29.5	35.9	23.8	23.7	25.2	32.0	38.2	42.2
Operating-profit margin	16.4	23.6	3.9	3.6	6.8	16.9	28.0	34.4
Net profit margin	13.0	17.2	0.2	0.2	4.1	14.0	25.2	31.6
ROAE	13.9	20.0	0.1	0.1	3.9	15.2	32.4	40.0
ROAA	8.6	10.9	0.1	0.1	2.0	8.8	21.3	28.3
ROCE	14.0	21.1	2.0	2.0	4.8	15.2	32.6	40.6
ROIC	19.9	26.1	2.6	2.4	6.0	20.7	53.7	96.6
Net debt to equity	net cash	net cash	2.5	4.6	4.8	net cash	net cash	net cash
Effective tax rate	11.9	21.0	17.9	17.0	12.0	6.0	5.7	5.7
Accounts receivable (days)	55.2	47.0	63.6	57.3	56.1	57.7	55.1	57.0
Current ratio (x)	1.9	1.8	2.1	1.7	1.4	1.9	2.7	3.9
Net interest cover (x)	84.0	61.0	5.3	5.1	8.9	29.3	71.6	125.5
Net dividend payout	52.6	41.9	909.2	909.2	43.2	23.5	10.3	5.8
Free cash flow yield	n.a.	n.a.	n.a.	n.a.	0.4	0.9	4.4	8.0

Source: FactSet, Daiwa forecasts

Company profile

Kinsus was founded in 2000 as a subsidiary of Asustek. The company mainly focus on niche ball grid array (BGA) substrates to avoid price competition among mainstream products like graphics card and chipsets, and it has become top-3 flip chip manufacturer in global. Besides, its manufacturing sites are based in Taiwan and mainland China.

ESG analysis

ESG risks

Risks		Management	Analyst comments
G	Executive/board quality	2	Kinsus's leadership team comprises experienced professionals dedicated to steering the company towards sustainable growth. This diverse board composition ensures a comprehensive range of expertise and perspectives, contributing to Kinsus's strategic decision-making and corporate governamce.
	Capital management	2	Kinsus maintains a robust financial framework, ensuring the availability of resources necessary for operational efficiency and strategic investments. The company's financial health is regularly disclosed through annual reports, providing transparency to stakeholders.
	Related party & transaction	1	The company adheres to stringent corporate governance policies to manage related party transactions, ensuring transparency and compliance with regulatory standards. The company discloses such transactions in its financial reports, maintaining accountability and stakeholder trust. For instance, the company's annual general meeting documents provide details on related party transactions, ensuring transparency and accountability.
S	Supply chain management	1	Kinsus is committed to sustainable supply chain practices, emphasizing environmental, social, and governance considerations. The company has established comprehensive ESG policies, including commitments to labor standards, environmental health and safety, ethics, and conflict-free minerals. These policies guide the company's interactions with suppliers and partners, promoting sustainability throughout the supply chain.
E	Data security	3	The company recognizes the importance of information security in its operations. As part of its corporate governance framework, Kinsus is likely to implement standard data protection practices to safeguard sensitive information and ensure compliance with relevant regulations. Stakeholders are encouraged to consult the company's official communications or contact the company directly for detailed information on data security policies.

Note: Management score represents a company's ability to manage/benefit from certain ESG topics. The scores range from 1 to 3, with 1 being the strongest.

Update Date: 31 Jan 2026

Source: Daiwa, Company

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Rating	Percentage of total
Buy*	73.96%
Hold**	19.58%
Sell***	6.46%

Source: Daiwa

Notes: data is for single-branded Daiwa research in Asia (ex Japan) and correct as of 30 September 2026.
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* The amount of the trading commission cannot be stated here in advance because it will be determined between our company and you based on current market conditions and the content of each transaction etc.

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Kinsus Interconnect Technology (3189 TT)

Target price: **TWD1,580.00** (from TWD1,068.00)

Share price (6 Oct): **TWD1,070.00** | Up/downside: **+47.7%**

5 4 3 2 **1**

Buy (unchanged)

3Q26 results preview: several favourable trends ahead

- Rising high-end ABF substrate likely to drive blended ASP increase
- Multicore and optics likely to be the next growth drivers
- Reaffirming our Buy (1) call; raising 12M TP to TWD1,580

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What's new: We believe an improving product mix, driven by rising Nvidia's CPU contribution, to be a key near-term share price catalyst.

What's the impact: **Near-term update.** After Kinsus released its August preliminary results on 23 September, we note that we had underestimated the 3Q26E gross margin expansion on product optimisation and price hike scale. The results imply a GM of 33% in August (vs. 26% in 2Q26). We therefore lift our 3Q-4Q26E gross margin to 33.2-35.1% (from 28.9-30.3%) to reflect an improving product mix, including greater CPU substrate contribution. We believe the gross margin of high-end chipset substrate may be c.50%, supporting margin expansion. We also expect a c.10% QoQ price hike for both BT and ABF substrate in 3Q26, above Kinsus' guidance of a high-single-digit% QoQ, making these substrates the other growth driver. We see further gross margin expansion in 2027-28 on the ongoing price hike trend and increasing high-end ABF substrate product mix.

Business outlook. We believe several favourable trends lie ahead. First, we see a rising contribution from high-end GPU and CPU ABF substrates. We believe Nvidia's CPU currently accounts for 5-10% of total revenue, with Kinsus holding a 70-80% market share. We also expect Kinsus to acquire a teens% market share in AMD's Venice, which should gradually ramp up from 4Q26. These will likely become key growth drivers of the ABF substrate segment in 2027, in our view. We believe the ramp-up progress for GPU substrate for these two clients is slightly slower than that for CPU. Second, we believe Kinsus will benefit from the rising BT substrate demand for optical transceivers. In our view, migration to on-board optics (OBO) and NPO/CPO will require ABF substrate beneath the optics engine (OE) in OBO; the OE and IC will need a bigger ABF substrate as we move to the NPO/CPO generation. As such, we expect the total ABF substrate demand to increase with rising adoption of optics in AI servers. Lastly, we expect the adoption of multiple-core ABF from 2028 to improve the rigidity and transmission performance before the official launch of ceramic and glass-core substrate, which we expect to increase the ABF TAM as well.

What we recommend: We raise our 2026-28E earnings by 19-29% to reflect the better-than-expected gross margin in August. We reaffirm our Buy (1) call and raise our 12M TP to TWD1,580, based on an unchanged target PER of 36x (high end of 3-year-trading range of 10-36x), applied to the 1-year-forward EPS forecast. Key downside risks: weaker-than-expected demand for memory, PCs, gaming GPUs or AI servers.

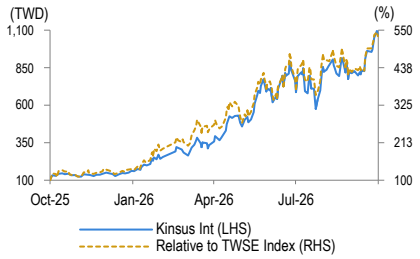
How we differ: Our 2026-28E EPS are 27-57% above the Bloomberg consensus, likely on our more bullish view on its gross margin expansion.

Forecast revisions (%)

Year to 31 Dec	26E	27E	28E
Revenue change	1.3	-	2.2
Net profit change	29.1	28.7	18.8
Core EPS (FD) change	18.6	18.3	9.2

Source: Daiwa forecasts

Share price performance



12-month range	111.18-1,095.00
Market cap (USDbn)	17.18
3m avg daily turnover (USDm)	428.04
Shares outstanding (m)	510
Major shareholder	Asus Investment Co Ltd (13.2%)

Financial summary (TWD)

Year to 31 Dec	26E	27E	28E
Revenue (m)	54,062	88,448	125,676
Operating profit (m)	9,142	24,773	43,198
Net profit (m)	7,558	22,325	39,673
Core EPS (fully-diluted)	14.824	43.787	77.810
EPS change (%)	322.9	195.4	77.7
Daiwa vs Cons. EPS (%)	29.1	57.1	46.7
PER (x)	72.2	24.4	13.8
Dividend yield (%)	0.3	0.4	0.4
DPS	3.5	4.5	4.5
PBR (x)	11.6	8.1	5.3
EV/EBITDA (x)	30.7	15.1	8.8
ROE (%)	15.2	32.4	40.0

Source: FactSet, Daiwa forecasts

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Flash

Kinsus: revenue and earnings forecast revisions

	2026E			2027E			2028E		
(TWDm)	Previous	New	Consensus	Previous	New	Consensus	Previous	New	Consensus
Revenue	53,385	54,062	53,912	88,484	88,448	80,794	122,920	125,676	120,243
Diff (%)		1.3%	0.3%		0.0%	9.5%		2.2%	4.5%
Gross Margin (%)	27.0%	29.65%	27.3%	35.6%	40.5%	33.7%	43.2%	46.37%	38.5%
Operating profit	7,578	9,142	7,907	20,449	24,773	18,170	38,330	43,198	32,345
Op Margin (%)	14.2%	16.9%	14.7%	23.1%	28.0%	22.5%	31.2%	34.4%	26.9%
Net profit	5,855	7,558	5,857	17,344	22,325	14,212	33,395	39,673	27,051
EPS	12.50	14.82	11.49	37.02	43.79	27.87	71.29	77.81	53.06
Diff (%)		18.6%	29.1%		18.3%	57.1%		9.2%	46.7%

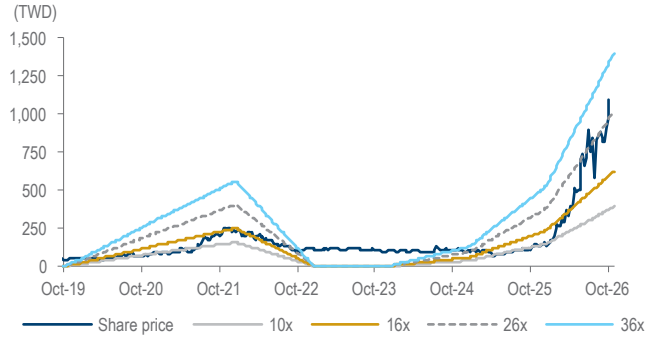
Source: Bloomberg, Daiwa forecasts

Kinsus: quarterly and annual P&L statement

	2026E				2027E				2025	2026E	2027E	2028E
(TWDm)	1Q	2Q	3QE	4QE	1QE	2QE	3QE	4QE				
Net revenue	**11,105**	**12,496**	**14,539**	**15,922**	**18,046**	**21,099**	**23,941**	**25,361**	**39,351**	**54,062**	**88,448**	**125,676**
COGS	(8,755)	(9,235)	(9,716)	(10,326)	(11,304)	(12,750)	(14,067)	(14,480)	(31,037)	(38,032)	(52,601)	(67,397)
Gross profit	**2,349**	**3,261**	**4,824**	**5,596**	**6,743**	**8,349**	**9,874**	**10,881**	**8,314**	**16,030**	**35,847**	**58,279**
Operating expenses	1,459	1,582	1,841	2,006	2,274	2,637	2,993	3,170	5,645	6,888	11,074	15,081
Operating profit	**891**	**1,679**	**2,983**	**3,590**	**4,469**	**5,712**	**6,881**	**7,711**	**2,670**	**9,142**	**24,773**	**43,198**
Non-operating profit	74	106	88	90	89	84	90	92	417	358	354	359
Pre-tax profit	**964**	**1,784**	**3,071**	**3,680**	**4,558**	**5,796**	**6,971**	**7,803**	**3,087**	**9,499**	**25,127**	**43,557**
Income taxes	(89)	(101)	(174)	(208)	(258)	(328)	(394)	(441)	(370)	(571)	(1,420)	(2,462)
Net profit	**549**	**1,311**	**2,586**	**3,112**	**3,986**	**5,132**	**6,230**	**6,978**	**1,596**	**7,558**	**22,325**	**39,673**
Net EPS (TWD)	**1.17**	**2.57**	**5.07**	**6.10**	**7.82**	**10.06**	**12.22**	**13.69**	**3.5**	**14.8**	**43.8**	**77.8**
Operating Ratios												
Gross margin	21.2%	26.1%	33.2%	35.1%	37.4%	39.6%	41.2%	42.9%	21.1%	29.7%	40.5%	46.4%
Operating margin	8.0%	13.4%	20.5%	22.5%	24.8%	27.1%	28.7%	30.4%	6.8%	16.9%	28.0%	34.4%
Pre-tax margin	8.7%	14.3%	21.1%	23.1%	25.3%	27.5%	29.1%	30.8%	7.8%	17.6%	28.4%	34.7%
Net margin	4.9%	10.5%	17.8%	19.5%	22.1%	24.3%	26.0%	27.5%	4.1%	14.0%	25.2%	31.6%
YoY (%)												
Net revenue	29%	31%	40%	47%	63%	69%	65%	59%	29%	37%	64%	42%
Gross profit	21%	62%	146%	134%	187%	156%	105%	94%	-4%	93%	124%	63%
Operating profit	78%	171%	375%	290%	402%	240%	131%	115%	144%	242%	171%	74%
Pre-tax profit	38%	188%	331%	249%	373%	225%	127%	112%	93%	208%	165%	73%
Net profit	99%	288%	662%	384%	626%	292%	141%	124%	3164%	374%	195%	78%
QoQ (%)												
Net revenue	3%	13%	16%	10%	13%	17%	13%	6%				
Gross profit	-2%	39%	48%	16%	20%	24%	18%	10%				
Operating profit	-3%	88%	78%	20%	24%	28%	20%	12%				
Pre-tax profit	-9%	85%	72%	20%	24%	27%	20%	12%				
Net profit	-14%	139%	97%	20%	28%	29%	21%	12%				

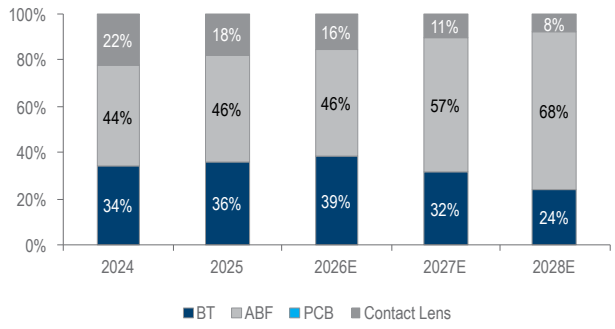
Source: Company, Daiwa forecasts

Kinsus: 1-year forward PER bands



Source: TEJ, Daiwa forecasts

Kinsus: revenue breakdown by product



Source: Company, Daiwa forecasts

Financial summary

Key assumptions

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Global PC shipment (mn)	361	301	260	263	287	244	244	244
Regular server shipmnet (mn)	14	15	12	14	16	18	22	26
Global smartphone shipment (mn)	1,655	1,437	1,380	1,437	1,429	1,249	1,255	1,300

Profit and loss (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
ABF	10,191	16,526	11,978	13,319	17,998	24,558	48,820	82,562
BT	17,565	18,796	8,065	10,398	14,314	21,234	30,530	33,105
Other Revenue	5,580	6,336	6,790	6,817	7,039	8,271	9,098	10,008
Total Revenue	**33,336**	**41,658**	**26,832**	**30,535**	**39,351**	**54,062**	**88,448**	**125,676**
Other income	0	0	0	0	0	0	0	0
COGS	(22,583)	(25,771)	(20,075)	(21,867)	(31,037)	(38,032)	(52,601)	(67,397)
SG&A	(2,879)	(3,532)	(3,301)	(5,010)	(3,433)	(4,241)	(6,960)	(9,312)
Other op.expenses	(2,413)	(2,484)	(2,397)	(2,612)	(2,212)	(2,647)	(4,114)	(5,769)
Operating profit	**5,456**	**9,851**	**1,042**	**1,095**	**2,670**	**9,142**	**24,773**	**43,198**
Net-interest inc./(exp.)	(68)	(172)	(332)	(394)	(391)	(336)	(356)	(350)
Assoc/forex/extraord./others	253	607	716	902	809	694	710	709
Pre-tax profit	**5,641**	**10,287**	**1,426**	**1,603**	**3,087**	**9,499**	**25,127**	**43,557**
Tax	(672)	(2,158)	(256)	(272)	(370)	(571)	(1,420)	(2,462)
Min. int./pref. div./others	(633)	(957)	(1,123)	(1,282)	(1,121)	(1,370)	(1,381)	(1,422)
Net profit (reported)	**4,336**	**7,172**	**48**	**49**	**1,596**	**7,558**	**22,325**	**39,673**
Net profit (adjusted)	**4,336**	**7,172**	**48**	**49**	**1,596**	**7,558**	**22,325**	**39,673**
EPS (reported)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
EPS (adjusted)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
EPS (adjusted fully-diluted)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
DPS (TWD)	**4.500**	**6.476**	**1.000**	**1.000**	**1.518**	**3.500**	**4.500**	**4.500**
EBIT	**5,456**	**9,851**	**1,042**	**1,095**	**2,670**	**9,142**	**24,773**	**43,198**
EBITDA	**9,833**	**14,958**	**6,394**	**7,228**	**9,914**	**17,313**	**33,761**	**53,085**

Cash flow (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Profit before tax	5,641	10,287	1,426	1,603	3,087	9,499	25,127	43,557
Depreciation and amortisation	4,377	5,107	5,351	6,133	7,244	8,171	8,988	9,887
Tax paid	(672)	(2,158)	(256)	(272)	(370)	(571)	(1,420)	(2,462)
Change in working capital	(1,347)	(176)	1,528	(906)	(1,959)	(3,452)	(2,982)	(3,707)
Other operational CF items	2,575	2,901	(1,793)	1,147	91	(634)	0	0
Cash flow from operations	**10,574**	**15,961**	**6,256**	**7,705**	**8,093**	**13,013**	**29,713**	**47,274**
Capex	(13,384)	(17,094)	(10,136)	(10,408)	(6,117)	(7,832)	(5,483)	(3,838)
Net (acquisitions)/disposals	0	0	0	0	0	0	0	0
Other investing CF items	489	559	(2,351)	1,301	(1,752)	(2,374)	(44)	(85)
Cash flow from investing	**(12,895)**	**(16,535)**	**(12,486)**	**(9,107)**	**(7,869)**	**(10,206)**	**(5,526)**	**(3,922)**
Change in debt	6,544	1,903	6,043	1,624	(1,684)	(8,891)	0	(45)
Net share issues/(repurchases)	0	0	0	0	0	0	0	0
Dividends paid	(451)	(2,443)	(3,438)	(454)	(457)	(774)	(1,785)	(2,294)
Other financing CF items	(94)	2,411	2,730	(173)	(1,130)	8,625	0	0
Cash flow from financing	**6,000**	**1,870**	**5,335**	**996**	**(3,270)**	**(1,041)**	**(1,785)**	**(2,340)**
Forex effect/others	(12)	56	(88)	71	(39)	51	39	35
Change in cash	**3,667**	**1,352**	**(983)**	**(335)**	**(3,084)**	**1,817**	**22,441**	**41,047**
Free cash flow	(2,809)	(1,133)	(3,880)	(2,703)	1,976	5,181	24,230	43,436

Source: FactSet, Daiwa forecasts

Financial summary continued …

Balance sheet (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & short-term investment	15,968	17,923	21,013	16,676	16,336	21,188	43,628	84,676
Inventory	3,466	3,481	2,612	2,981	4,828	6,237	8,747	10,924
Accounts receivable	5,679	5,050	4,299	5,284	6,804	10,300	16,405	22,824
Other current assets	1,312	1,647	1,445	4,735	970	1,080	1,080	1,080
Total current assets	**26,424**	**28,100**	**29,369**	**29,677**	**28,938**	**38,804**	**69,860**	**119,503**
Fixed assets	31,586	44,101	48,209	49,658	50,897	51,643	48,138	42,089
Goodwill & intangibles	0	0	0	0	0	0	0	0
Other non-current assets	376	432	489	100	138	434	478	562
Total assets	**58,386**	**72,633**	**78,067**	**79,435**	**79,973**	**90,882**	**118,476**	**162,154**
Short-term debt	1,100	377	1,409	2,452	3,053	915	915	869
Accounts payable	10,150	10,798	8,457	7,521	10,030	14,004	19,638	24,526
Other current liabilities	2,733	4,199	4,393	7,018	7,023	5,571	5,571	5,571
Total current liabilities	**13,983**	**15,374**	**14,259**	**16,990**	**20,106**	**20,490**	**26,124**	**30,966**
Long-term debt	9,387	10,770	15,280	13,779	11,197	6,219	6,219	6,219
Other non-current liabilities	2,027	7,674	9,369	8,788	7,297	6,318	6,318	6,318
Total liabilities	**25,397**	**33,818**	**38,908**	**39,557**	**38,600**	**33,027**	**38,661**	**43,503**
Share capital	4,508	4,528	4,544	4,566	4,568	5,269	5,269	5,269
Reserves/R.E./others	28,481	34,288	34,615	35,312	36,805	52,585	74,546	113,382
Shareholders' equity	**32,989**	**38,816**	**39,159**	**39,878**	**41,373**	**57,854**	**79,815**	**118,651**
Minority interests	0	0	0	0	0	0	0	0
Total equity & liabilities	**58,386**	**72,633**	**78,067**	**79,435**	**79,973**	**90,882**	**118,476**	**162,154**
EV	540,075	538,779	541,231	545,110	543,470	531,501	509,061	467,968
Net debt/(cash)	**(5,481)**	**(6,776)**	**(4,325)**	**(445)**	**(2,085)**	**(14,054)**	**(36,495)**	**(77,588)**
BVPS (TWD)	**64.478**	**75.804**	**69.748**	**69.314**	**71.656**	**92.357**	**131.414**	**202.420**

Key ratios (%)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Sales (YoY)	23.0	25.0	(35.6)	13.8	28.9	37.4	63.6	42.1
EBITDA (YoY)	70.8	52.1	(57.3)	13.1	37.1	74.6	95.0	57.2
Operating profit (YoY)	307.0	80.6	(89.4)	5.1	143.7	242.4	171.0	74.4
Net profit (YoY)	700.2	65.4	(99.3)	2.9	3,164.4	373.6	195.4	77.7
Core EPS (fully-diluted) (YoY)	698.1	65.3	(99.3)	2.5	3,153.4	322.9	195.4	77.7
Gross-profit margin	32.3	38.1	25.2	28.4	21.1	29.7	40.5	46.4
EBITDA margin	29.5	35.9	23.8	23.7	25.2	32.0	38.2	42.2
Operating-profit margin	16.4	23.6	3.9	3.6	6.8	16.9	28.0	34.4
Net profit margin	13.0	17.2	0.2	0.2	4.1	14.0	25.2	31.6
ROAE	13.9	20.0	0.1	0.1	3.9	15.2	32.4	40.0
ROAA	8.6	10.9	0.1	0.1	2.0	8.8	21.3	28.3
ROCE	14.0	21.1	2.0	2.0	4.8	15.2	32.6	40.6
ROIC	19.9	26.1	2.6	2.4	6.0	20.7	53.7	96.6
Net debt to equity	net cash	net cash	2.5	4.6	4.8	net cash	net cash	net cash
Effective tax rate	11.9	21.0	17.9	17.0	12.0	6.0	5.7	5.7
Accounts receivable (days)	55.2	47.0	63.6	57.3	56.1	57.7	55.1	57.0
Current ratio (x)	1.9	1.8	2.1	1.7	1.4	1.9	2.7	3.9
Net interest cover (x)	84.0	61.0	5.3	5.1	8.9	29.3	71.6	125.5
Net dividend payout	52.6	41.9	909.2	909.2	43.2	23.5	10.3	5.8
Free cash flow yield	n.a.	n.a.	n.a.	n.a.	0.4	0.9	4.4	8.0

Source: FactSet, Daiwa forecasts

Company profile

Kinsus was founded in 2000 as a subsidiary of Asustek. The company mainly focus on niche ball grid array (BGA) substrates to avoid price competition among mainstream products like graphics card and chipsets, and it has become top-3 flip chip manufacturer in global. Besides, its manufacturing sites are based in Taiwan and mainland China.

ESG analysis

ESG risks

Risks		Management	Analyst comments
G	Executive/board quality	2	Kinsus's leadership team comprises experienced professionals dedicated to steering the company towards sustainable growth. This diverse board composition ensures a comprehensive range of expertise and perspectives, contributing to Kinsus's strategic decision-making and corporate governamce.
	Capital management	2	Kinsus maintains a robust financial framework, ensuring the availability of resources necessary for operational efficiency and strategic investments. The company's financial health is regularly disclosed through annual reports, providing transparency to stakeholders.
	Related party & transaction	1	The company adheres to stringent corporate governance policies to manage related party transactions, ensuring transparency and compliance with regulatory standards. The company discloses such transactions in its financial reports, maintaining accountability and stakeholder trust. For instance, the company's annual general meeting documents provide details on related party transactions, ensuring transparency and accountability.
S	Supply chain management	1	Kinsus is committed to sustainable supply chain practices, emphasizing environmental, social, and governance considerations. The company has established comprehensive ESG policies, including commitments to labor standards, environmental health and safety, ethics, and conflict-free minerals. These policies guide the company's interactions with suppliers and partners, promoting sustainability throughout the supply chain.
E	Data security	3	The company recognizes the importance of information security in its operations. As part of its corporate governance framework, Kinsus is likely to implement standard data protection practices to safeguard sensitive information and ensure compliance with relevant regulations. Stakeholders are encouraged to consult the company's official communications or contact the company directly for detailed information on data security policies.

Note: Management score represents a company's ability to manage/benefit from certain ESG topics. The scores range from 1 to 3, with 1 being the strongest.

Update Date: 31 Jan 2026

Source: Daiwa, Company

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Rating	Percentage of total
Buy*	73.96%
Hold**	19.58%
Sell***	6.46%

Source: Daiwa

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Daiwa Capital Markets Hong Kong Limited

data

6 October 2026

MEMO

WinWay Technology (6515 TT)

Share price (6 Oct): **TWD6,130.00** 12-mth rating: **Buy (1)** **Information Technology: Taiwan**

Limited near-term catalysts before 2027 capacity unlock

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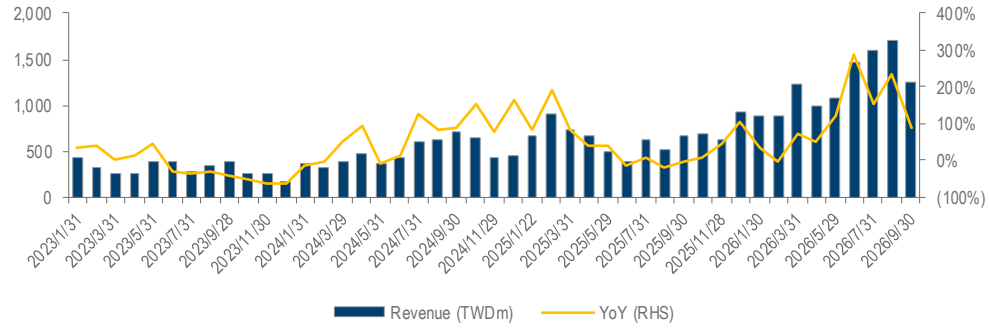
Summary: Winway reported September revenue of TWD1.25bn (-26.7% MoM; +87.7% YoY), brining 3Q26 revenue to a record high of TWD4.56bn (+29.3% QoQ; +152.6% YoY). The 3Q26 top-line result beat market consensus by 6.7% but missed Daiwa's estimate by 9.1%. Following August's exceptionally strong top line (TWD1.71bn), we had factored in a higher shipment run-rate for September. However, the miss stems primarily from the lumpiness of probe-card deliveries (high concentration to a single customer) and customer qualification timelines, rather than order cancellations or share loss.

We forecast 4Q26 revenue to track flattish to slightly down QoQ, reflecting lower probe-card shipment volumes compared to 3Q26, alongside typical year-end pull-in deceleration from customers. On profitability, we maintain our view that 2Q26 marked the trough for gross margin this year. As the revenue contribution from lower-margin MEMS probe cards declines, a more favourable product mix should drive a steady QoQ gross margin recovery, which may drive gross margin to reach 40% earlier than market expectations.

Although we remain positive on Winway's structural growth profile, near-term catalysts appear muted. This reflects the revenue lumpiness in probe cards, while additional socket capacity will not come online until March 2027—pushing the corresponding revenue recognition into 2Q27. We will revisit our financial model upon the company's 3Q26 results release. We currently have a Buy (1) rating on Winway with a 12-month TP of TWD10,000.

Winway monthly revenue trend



Source: Company

In the interests of timeliness, this document has not been edited.

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Rating	Percentage of total
Buy*	73.96%
Hold**	19.58%
Sell***	6.46%

Source: Daiwa

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King Slide

Maintain Buy: Sep sales -31% MoM from high bases; Aug earnings remain solid

Maintain Rating: BUY | PO: 14,888 TWD | Price: 13,100 TWD

Sep sales -31% MoM from high bases; 3Q26 sales in-line

King Slide's Sep sales of NT$4.46bn declined 31% MoM (+196% YoY), after the strong sales in the past few months (up 17-46% MoM in Apr-Jul and up 1% MoM in Aug). 3Q26 sales of NT$17.34bn (+60% QoQ, +297% YoY) were 0.5% above BofAe of NT$17.26bn and 0.9% below consensus of NT$17.49bn.

We are not surprised with the MoM sales drop in Sep as (1) the company's MoM sales trend has been volatile (Exhibit 1 and Exhibit 2), (2) sales bases in Apr-Aug were at record levels, (3) contribution from Nvidia Rubin GPU projects should be more meaningful in 4Q26 after hardware components ramp-up, and (4) partial general server orders/demand likely were pushed out due to key component shortage (e.g. memory, CPU).

Solid Aug EPS NT$40.07; pre-tax/net margin 74.6%/59.1%

In addition to Sep sales, on the same day (6 Oct), King Slide also announced its Aug unaudited key financial data as required by local regulations (Exhibit 3). Aug sales were NT$6,466mn (+344% YoY), while pre-tax profit was NT$4,826mn (+216% YoY) and net income was NT$3,819mn (+212% YoY). EPS came in at NT$40.07 (+212% YoY). Pre-tax margin and net margin were 74.6% and 59.1%, respectively.

Compared to July unaudited figures announced on 14 Aug (see our 17 Aug report), including sales of NT$6,407mn, pre-tax margin of 93.4%, net margin of 74.7%, and EPS of NT$50.24, we believe the margin and EPS differences mainly reflected FX impact under similar sales scale.

Combining Jul and Aug unaudited results, sales totaled NT$12.87bn and EPS reached NT$90.31. Pre-tax margin and net margin were 84.0% and 66.9%, respectively. Jul + Aug sales achieved 75%/74% of BofAe/consensus 3Q26 sales. Pre-tax profit, net income, and EPS all achieved 79% of BofAe (79.4% pre-tax margin, 62.9% net margin, and NT$113.87 EPS) and 77-78% of consensus (80.3% pre-tax margin, 63.1% net margin, and NT$117.62 EPS). We also noted that Jul + Aug EPS of NT$90.31 already surpassed EPS of NT$74.38 in 2Q26 with similar pre-tax/net margin (84.0%/66.9% in Jul + Aug, vs. 84.2%/65.4% in 2Q26).

Maintain Buy; expect a neutral-to-positive market reaction

We expect a neutral-to-positive market reaction in the near term in view of significant MoM sales decline in Sep but also still solid Aug unaudited data. We maintain Buy on King Slide as the server rail market leader with broad and diversified client/project base. We expect the company to post 3Q26 results after market close on 12 Nov (Thu).

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06 October 2026

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Stock Data

Price	13,100 TWD
Price Objective	14,888 TWD
Date Established	7-Aug-2026
Investment Opinion	C-1-7
52-Week Range	2,875 TWD-15,085 TWD
Mrkt Val / Shares Out (mn)	39,309 USD / 95.3
Market Value (mn)	1,248,392 TWD
Average Daily Value (mn)	201.14 USD
Free Float	67.5%
BofA Ticker / Exchange	KGSLF / TAI
Bloomberg / Reuters	2059 TT / 2059.TW
ROE (2026E)	77.4%
Net Dbt to Eqty (Dec-2025A)	-84.2%

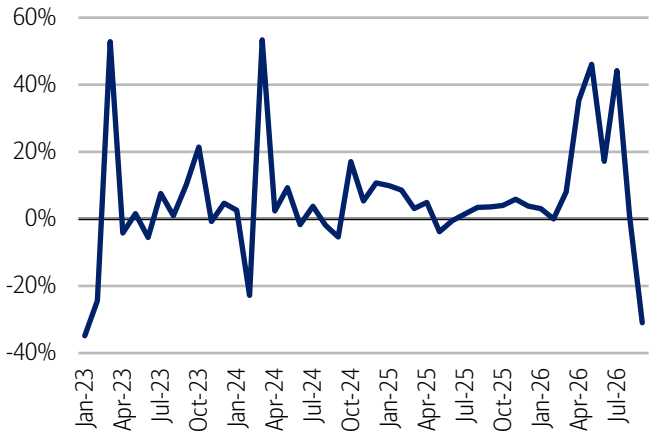
Acronyms

CPU: Central Processing Unit

FX: Foreign Exchange

GPU: Graphic Processing Unit

Exhibit 1: King Slide's sales MoM trend has been volatile in AI era
King Slide monthly sales MoM trend, Jan'23-Sep'26



Source: BofA Global Research, company data. AI: artificial intelligence

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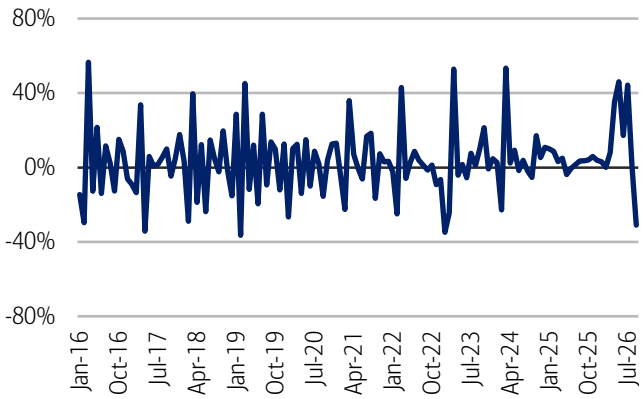
Exhibit 2: King Slide's sales MoM trend has been volatile in the past decade
King Slide monthly sales MoM trend, Jan'16-Sep'26



Source: BofA Global Research, company data

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Exhibit 3: Jul + Aug sales achieved 75%/74% of BofA/consensus 3Q26 estimate and EPS achieved 79%/77% of BofA/consensus 3Q26 estimate
King Slide Jul'26 and Aug'26 unaudited key financial data vs. 3Q26 BofAe/consensus

NT$mn	Jul'26	YoY %	Aug'26	YoY %	Jul + Aug Subtotal	3Q26 BofAe	Jul + Aug Run rate	3Q26 Consensus	Jul + Aug Run rate
Sales	6,407	336%	6,466	344%	12,873	17,260	75%	17,494	74%
Pretax profit	5,985	321%	4,826	216%	10,811	13,702	79%	14,049	77%
Net income	4,787	321%	3,819	212%	8,606	10,852	79%	11,038	78%
EPS (NT$)	50.24	321%	40.07	212%	90.31	113.87	79%	117.62	77%
Margin %									
Pre-tax margin	93.4%		74.6%		84.0%	79.4%		80.3%	
Net margin	74.7%		59.1%		66.9%	62.9%		63.1%	

Source: BofA Global Research estimates, Bloomberg, company data

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Price objective basis & risk

King Slide (KGSLF)

Our PO of NT$14,888 is based on 32x 2027E EPS. We use 32x P/E multiple (2.5SD above P/E average since listing) in view of strong AI server market demand, leading position in server rail, solid/high margin profiles, and double-digit sales/EPS CAGR in 2026-28E. Its early step into server market since 2001 with proven track record, much larger scale than peer, and 3,600+ patents as strong moats for high/above-peer margins. Its above-30x P/E is a similar case to other component sub-sector leaders in Taiwan.

Upside risks to our PO are: (1) even stronger AI server demand, (2) margin upside thanks to larger scale and/or better efficiency, (3) better kitchen slides market demand, (4) faster capacity expansion, (5) favorable FX movement (mainly USD/NTD) for GM and/or non-operating items.

Downside risks to our PO are: (1) softer AI server demand due to GPU/components shortage, (2) GM pressure due to higher raw material cost (e.g. steel, aluminum), (3) softer kitchen slides market demand under macro uncertainty/weakness, (4) delayed product certification, (5) slower capacity expansion, (6) unfavorable FX movement (mainly USD/NTD) to weigh on GM and/or non-operating items.

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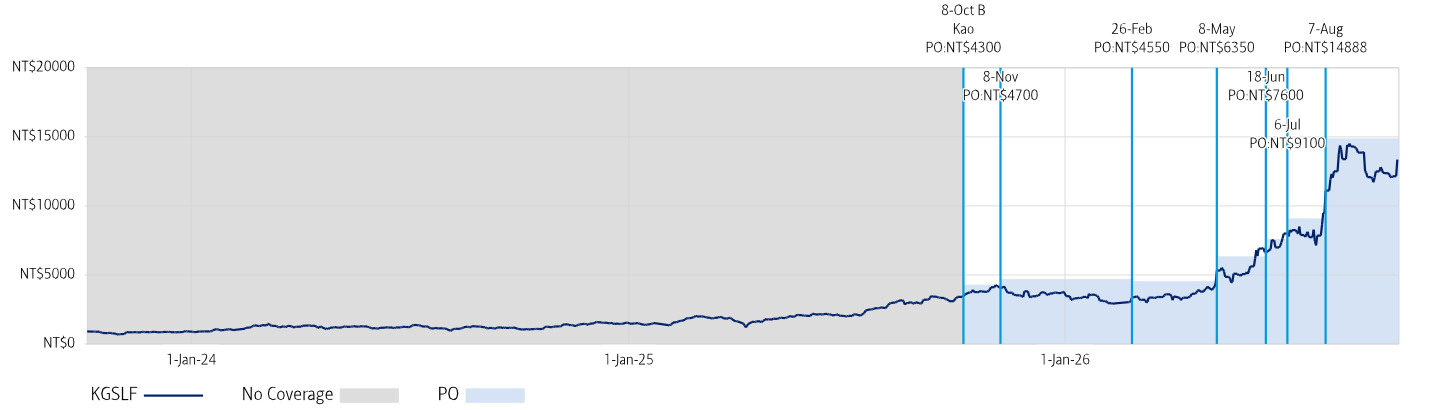
King Slide (KGSLF) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

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Equity Investment Rating Distribution: Technology Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	255	62.04%	Buy	157	61.57%
Hold	82	19.95%	Hold	46	56.10%
Sell	74	18.00%	Sell	31	41.89%

Equity Investment Rating Distribution: Global Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	2057	57.03%	Buy	1386	67.38%
Hold	806	22.35%	Hold	563	69.85%
Sell	744	20.63%	Sell	446	59.95%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2]Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

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October 6, 2026 11:43 AM GMT

Accton Technology Corporation | Asia Pacific

September Sales -1% MoM/ +66% YoY

In this report, we focus on Accton's monthly sales, which we believe could be a catalyst for share price movement.

Details:

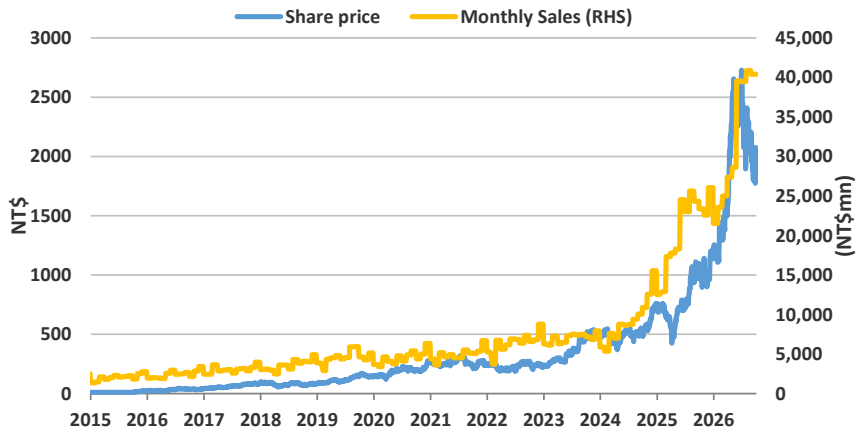
- September sales were NT$40,399mn (-1% MoM/+66% YoY).
- 3Q26 revenue was NT$120,744mn (+26% QoQ/+66% YoY), in line with MSe of NT$120,883mn (+27% QoQ/+66% YoY) and consensus of NT$117.390mn (+23% QoQ/+61% YoY) by 3%.

Our view:

- We believe the consignment arrangement of DRAM, PCB, etc., from August might have capped the upside on revenue growth, but it should be a favorable factor for gross margin, emerging from August/September and likely extending into 4Q26.
- Driven by the network switch business and its major customer's AI accelerator module ramp, we expect revenue to continue growing sequentially in 4Q26, with the momentum in 2027 strengthened by potential new customers and new products.

- Stay OW with a PT of NT$3,500, based on 30x 2027e P/E.

Exhibit 1: Accton's monthly sales vs. its share price since 2015



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Accton Technology Corporation (2345.TW, 2345 TT)
Top Pick

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$3,500.00
Up/downside to price target (%)	71
Shr price, close (Oct 6, 2026)	NT$2,050.00
52-Week Range	NT$2,825.00-884.00
Sh out, dil, curr (mn)	559
Mkt cap, curr (mn)	NT$1,145,761
EV, curr (mn)	NT$1,085,110
Avg daily trading value (mn)	NT$8,214

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	47.13	83.77	115.77	145.77
Prior EPS (NT$)**	-	-	-	-
EPS (NT$)§	45.42	86.65	118.19	165.38
Revenue, net (NT$ mn)	248,320	413,479	541,272	667,428
EBITDA (NT$ mn)	33,894	62,436	85,653	106,950
ModelWare net inc (NT$ mn)	26,342	46,817	64,707	81,472
P/E	25.1	24.5	17.7	14.1
P/BV	11.5	12.3	8.0	5.6
RNOA (%)	622.3	(1,479.8)	4,110.1	1,435.9
ROE (%)	72.6	81.2	69.2	56.9
EV/EBITDA	17.8	16.9	11.8	8.9
Div yld (%)	0.9	1.0	1.3	1.7
FCF yld ratio (%)**	4.5	3.7	5.3	6.8
Leverage (EOP) (%)	(101.7)	(96.2)	(94.8)	(94.4)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
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Valuation Methodology and Risks

Accton Technology Corporation (2345.TW)

Base case, P/E, which we believe better captures the company's earnings growth momentum. We assign a target P/E of 30x to our 2027 EPS estimate, which we view as justified by an operating profit CAGR of 48% for 2025-28, based on our estimates. This implies a PEG of around 0.6, vs. its average of 0.7 over 2019-25, when the stock was trading at a P/E of 22x for an operating profit CAGR of 32%.

Risks to Upside

- Stronger data center capex
- Faster network switch spec migration
- InfiniBand losing share in AI networks
- Stronger custom AI accelerator growth
- Stronger ODM direct business model adoption

Risks to Downside

- Weaker data center capex
- Slower network switch spec migration
- InfiniBand taking more share in AI networks
- Softer custom AI accelerator growth
- Softer ODM direct business model adoption

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(as of September 30, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
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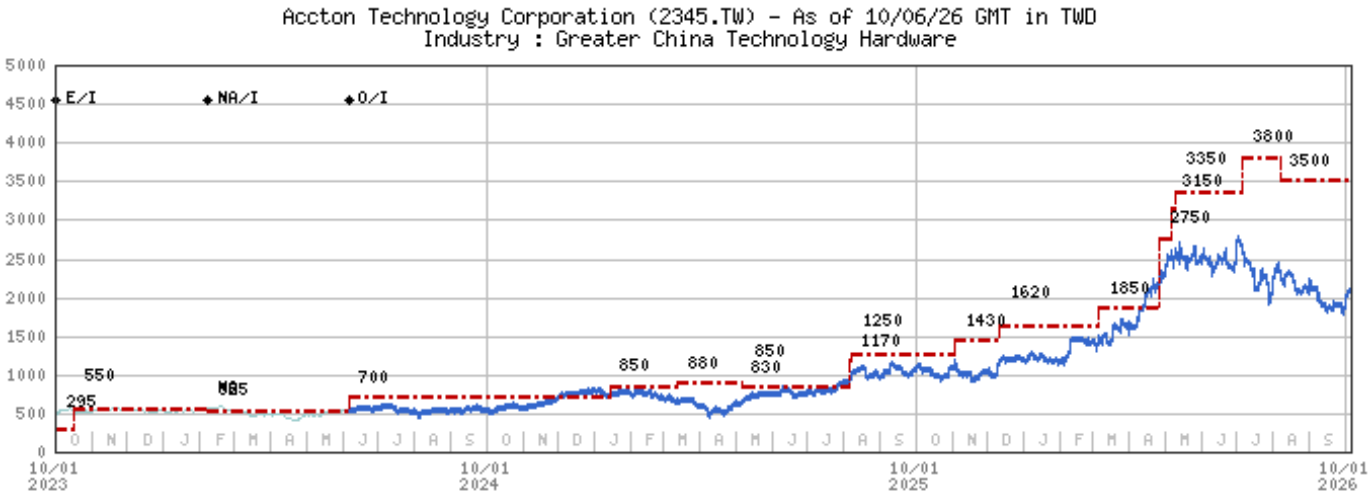
Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : O/I; 11/12/21 : E/I; 2/7/24 : NA/I; 6/6/24 : O/I

Price Target History: 7/28/21 : 336; 11/12/21 : 265; 1/10/22 : 280; 3/7/22 : 270; 5/13/22 : 235; 6/7/22 : 248; 8/12/22 : 275; 5/22/23 : 285; 8/10/23 : 295; 10/17/23 : 550; 2/7/24 : NA; 2/8/24 : 525; 6/6/24 : 700; 1/14/25 : 850; 3/13/25 : 880; 5/6/25 : 830; 5/9/25 : 850; 8/6/25 : 1170; 8/8/25 : 1250; 11/4/25 : 1430; 12/12/25 : 1620; 3/6/26 : 1850; 4/27/26 : 2750; 5/7/26 : 3150; 5/11/26 : 3350; 7/6/26 : 3800; 8/7/26 : 3500

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
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Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
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COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/06/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$39.36
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$23.90
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,045.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,855.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.43
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$52.30
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$5.00
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$38.14
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$64.10
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77

Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.26
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$180.00
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,079.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.80
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,050.00
Advantech (2395.TW)	O (01/20/2021)	NT$738.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,195.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,685.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$37.45
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,530.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,325.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$885.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,350.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,050.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,415.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$339.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.10
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,100.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$299.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.05
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$1,000.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$206.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.90
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,995.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$5.02
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$380.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,260.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$256.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.10
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,920.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.40
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,865.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,505.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$90.00
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$333.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,310.00

Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$567.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,290.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$625.00
Zhen Ding (4958.TW)	++	NT$574.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$551.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,755.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

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Morgan Stanley | RESEARCH

UPDATE

October 6, 2026 12:55 AM GMT

Bizlink | Asia Pacific

September Sales +5% MoM/ +39% YoY

In this report, we focus on Bizlink's monthly sales, which we believe could be a catalyst for its share price.

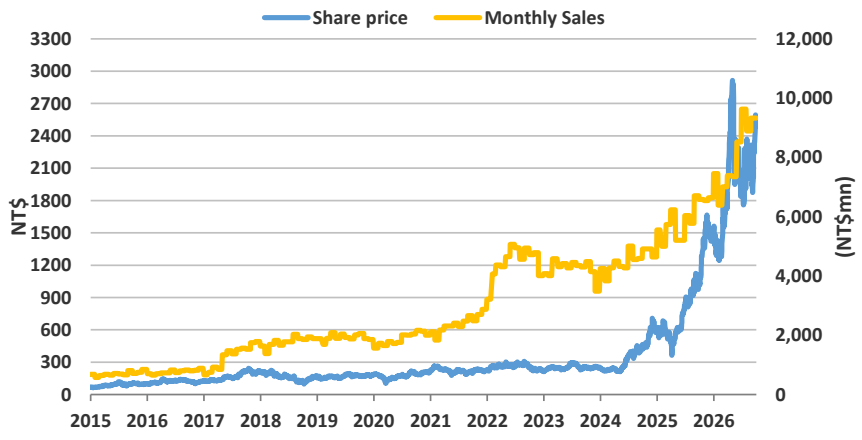
Details:

- September sales came in at NT$9,332mn (+5% MoM/+39% YoY)
- 3Q26 revenue reached NT$27,854mn (+20% QoQ/+50% YoY), slightly short of MSe of NT$28,521mn (+23% QoQ/+55% YoY) by 2% while exceeding consensus of NT$27,423mn (+18% QoQ/+49% YoY) by 2%.

Our view:

- We think 3Q26 revenue came in largely in-line with market expectations.
- Looking into 2H26, we expect HPC to remain the main growth driver, as power interconnects should continue to see momentum from AI/general servers. Meanwhile, data interconnects should grow along with new ASIC model ramp-up and GPU server demand. Moreover, semi cap equipment revenue growth should also sustain.
- We believe Bizlink has allocated increasing resources toward the optical interconnect space, including the collaboration on ALC with Credo, mini VCSEL optical module with ams-OSRAM and shuffle box with Corning. We think that these opportunities could help offset a potential slowdown in its AEC business as AEC solutions reach their scalability limits at higher data rates, likely from 3.2T.
- Stay OW.

Exhibit 1: Bizlink's monthly sales vs. its share price since 2015



Note: Past performance is no guarantee of future results. Results shown do not include transaction costs. Source: Company data, TEJ, Morgan Stanley Research.

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Bizlink (3665.TW, 3665 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$3,665.00
Up/downside to price target (%)	47
Shr price, close (Oct 5, 2026)	NT$2,485.00
52-Week Range	NT$3,010.00-1,000.00
Sh out, dil, curr (mn)	193
Mkt cap, curr (mn)	NT$480,515
EV, curr (mn)	NT$486,091
Avg daily trading value (mn)	NT$6,712

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	46.57	69.77	120.77	143.77
Prior EPS (NT$)**	-	-	-	-
EPS (NT$)§	47.71	68.93	119.70	154.48
Revenue, net (NT$ mn)	71,247	101,953	151,516	174,996
EBITDA (NT$ mn)	15,154	21,153	34,384	41,510
ModelWare net inc (NT$ mn)	9,005	13,491	23,353	27,800
P/E	32.6	35.6	20.6	17.3
P/BV	6.4	8.6	6.4	5.1
RNOA (%)	24.2	26.2	39.8	41.7
ROE (%)	24.9	28.9	40.8	36.5
EV/EBITDA	20.1	22.8	13.7	11.0
Div yld (%)	0.8	0.6	0.9	1.6
FCF yld ratio (%)**	(0.8)	1.5	2.9	5.1
Leverage (EOP) (%)	(0.9)	(8.3)	(19.1)	(32.5)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

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AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
Bizlink 3665.TW			
September 2026 monthly sales	06 Oct 2026	Unchanged	In-line

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Bizlink (3665.TW)

Base case, derived from 30x 2027e P/E. We think our target P/E multiple is justified by an earnings CAGR of 46% seen over 2025-28e, which implies a PEG of around 0.7. This is in line with the average of 0.7 in 2019-24, when the stock was trading at an average 13x P/E for a 19% earnings CAGR.

Risks to Upside

- Faster and wider AEC adoption
- Data center and semi equipment capex increases that drive stronger HPC and semi-cap business
- Stronger recovery for automotive business

Risks to Downside

- Lower-than-expected AEC adoption
- Intensifying competition
- Softer-than-expected AI capex
- Slower-than-expected growth in the semi capital equipment market
- Continued weakness in the auto industry

Disclosure Section

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(as of September 30, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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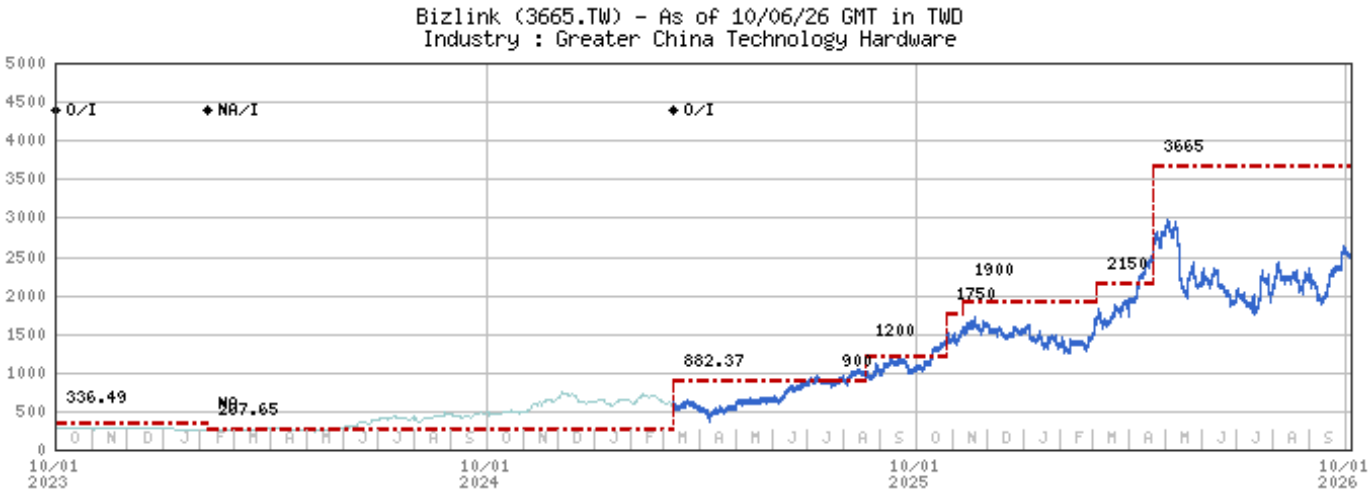
Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : NA/I; 8/18/22 : E/I; 2/20/23 : O/I; 2/7/24 : NA/I; 3/10/25 : O/I

Price Target History: 8/18/22 : 336.49; 11/11/22 : 293.23; 2/20/23 : 336.49; 2/7/24 : NA; 2/8/24 : 267.65; 3/10/25 : 882.37; 7/21/25 : 900; 8/19/25 : 1200; 10/27/25 : 1750; 11/10/25 : 1900; 3/4/26 : 2150; 4/22/26 : 3665

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -·- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) ▬
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/05/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$38.64
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.76
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,055.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,730.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.46
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$53.35
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.91
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$37.16
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$61.95
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$23.88

Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$179.10
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,111.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.62
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,075.00
Advantech (2395.TW)	O (01/20/2021)	NT$740.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,180.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,585.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$39.05
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,550.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,485.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$890.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,325.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,005.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$146.50
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$68.20
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,450.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$345.50
Innolux (3481.TW)	E (04/07/2025)	NT$52.30
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,355.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$298.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.40
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$995.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$207.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.80
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,700.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.86
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$388.50
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,215.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$254.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.30
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$3,000.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.92
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,875.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,495.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$88.70
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$331.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,325.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$606.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50

Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,150.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$626.00
Zhen Ding (4958.TW)	++	NT$576.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$570.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,780.00

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October 6, 2026 11:48 AM GMT

Chroma Ate Inc. | Asia Pacific

September Sales +22% MoM/ +206% YoY

In this report, we focus on Chroma's monthly sales, which we believe could be a catalyst for share price movement.

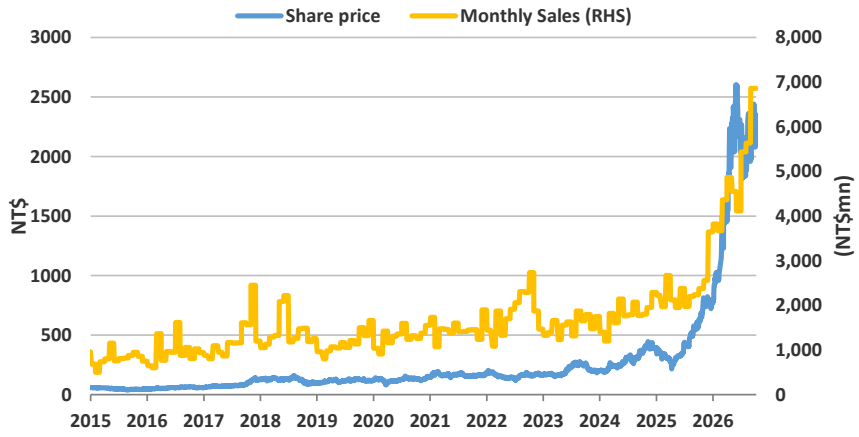
Details:

- September revenue was NT$6,860mn (22% MoM/+206% YoY), a record high.
- 3Q26 revenue came in at NT$17,926mn (+32% QoQ/+180% YoY), significantly higher than our estimate at NT$14,772mn (+9% QoQ/+130% YoY) and consensus at NT$14,555mn (+8% QoQ/+127% YoY).

Our view:

- We believe that the strength in 3Q26 was mainly driven by the semiconductor and photonics business, especially SLT, metrology, transceiver/CPO testing, etc.
- We think that Chroma might start to ship FT handlers and ovens in small volume from late 2026/early 2027, which represent new business opportunities for the coming years.
- Stay OW.

Exhibit 1: Chroma's monthly sales vs. its share price since 2015



Source: Company data, TEJ, Morgan Stanley Research. Note: Past performance is no guarantee of future results. Results shown do not include transaction costs.

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Chroma Ate Inc. (2360.TW, 2360 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$2,800.00
Up/downside to price target (%)	19
Shr price, close (Oct 6, 2026)	NT$2,350.00
52-Week Range	NT$2,795.00-599.00
Sh out, dil, curr (mn)	425
Mkt cap, curr (mn)	US$31,512
EV, curr (mn)	US$31,169
Avg daily trading value (mn)	US$182

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	27.49	43.97	62.77	74.77
EPS (NT$)§	26.88	44.69	66.84	86.80
Prior EPS (NT$)**	-	-	-	-
Revenue, net (NT$ mn)	28,311	55,154	76,117	89,224
EBITDA (NT$ mn)	10,038	22,684	32,742	38,930
ModelWare net inc (NT $ mn)	11,692	18,697	26,691	31,797
P/E	28.2	53.4	37.4	31.4
P/BV	10.1	22.4	16.9	13.7
RNOA (%)	49.4	80.3	95.7	98.3
ROE (%)	45.9	57.4	59.9	53.9
EV/EBITDA	31.8	43.3	29.7	24.7
Div yld (%)	1.2	0.8	1.3	1.9
FCF yld ratio (%)**	2.4	1.5	2.3	3.0
Leverage (EOP) (%)	(33.4)	(39.0)	(46.3)	(52.8)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

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AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
Chroma Ate Inc. 2360.TW			
September 2026 sales	06 Oct 2026	Unchanged	↑ Modest upside surprise

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Chroma Ate Inc. (2360.TW)

Base case, 45x 2027e P/E. We use P/E as our valuation methodology because we think it better captures Chroma's earnings growth momentum. The stock traded at an average valuation of ~19x P/E during 2014-25 with an operating profit CAGR of 20%, implying a PEG of 1.0. We think our target multiple of 45x is justified by the 61% operating profit CAGR we project for 2025-28.

Risks to Upside

- Better-than-expected macro conditions
- Stronger semi and photonics and power testing demand
- Margin improvement via better product mix
- Better opex control

Risks to Downside

- Macro slowdown
- Softer-than-expected AI capex
- Slower progress in the advanced packaging capacity expansion
- Weaker consumer electronics and automobile related demand
- Margin contraction arising from lack of revenue scale and increasing opex

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(as of September 30, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
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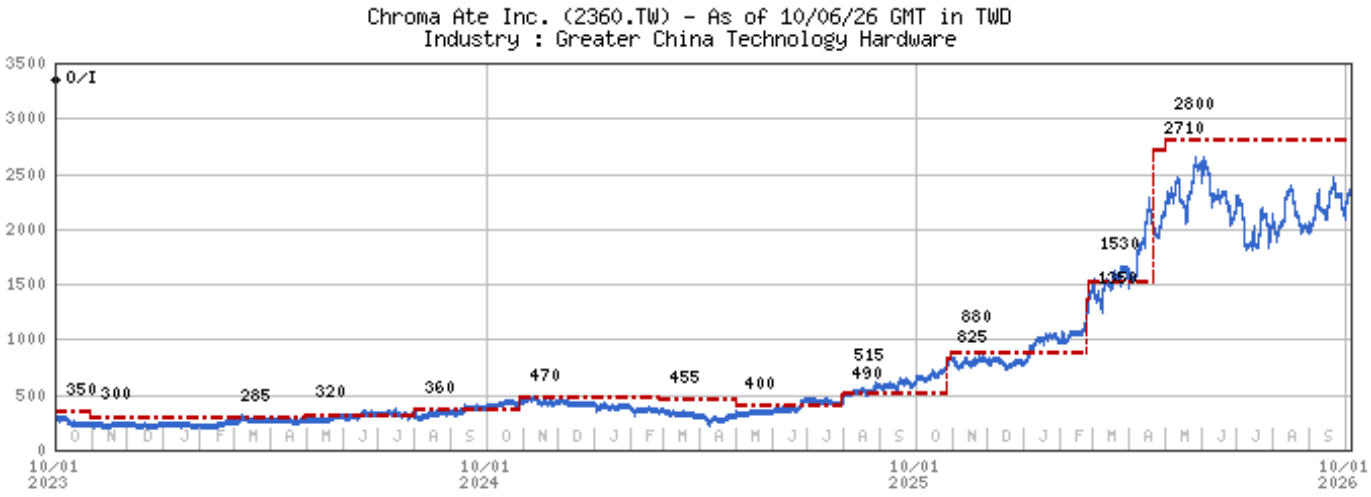
Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : O/I; 10/5/21 : O/I

Price Target History: 7/31/20 : 200; 10/5/21 : 220; 2/23/22 : 230; 2/24/23 : 215; 4/27/23 : 225; 7/7/23 : 320; 8/1/23 : 350; 10/31/23 : 300; 2/26/24 : 285; 4/30/24 : 320; 8/1/24 : 360; 10/30/24 : 470; 2/25/25 : 455; 5/1/25 : 400; 7/30/25 : 490; 8/1/25 : 515; 10/28/25 : 825; 10/31/25 : 880; 2/23/26 : 1350; 2/26/26 : 1530; 4/21/26 : 2710; 5/1/26 : 2800

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)

Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

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COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/06/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$39.36
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$23.90
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,045.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,855.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.43
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$52.30
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$5.00
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$38.14
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$64.10
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43

Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.26
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$180.00
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,079.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.80
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,050.00
Advantech (2395.TW)	O (01/20/2021)	NT$738.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,195.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,685.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$37.45
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,530.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,325.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$885.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,350.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,050.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,415.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$339.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.10
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,100.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$299.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.05
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$1,000.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$206.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.90
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,995.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$5.02
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$380.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,260.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$256.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.10
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,920.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.40
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,865.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,505.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$90.00
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$333.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,310.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$567.00

Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,290.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$625.00
Zhen Ding (4958.TW)	++	NT$574.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$551.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,755.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

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UPDATE

October 6, 2026 11:47 AM GMT

Hiwin Technologies Corp. | Asia Pacific

September Sales Flat MoM/ +30% YoY

In this report, we focus on Hiwin's monthly sales, which we believe could be a catalyst for its share price.

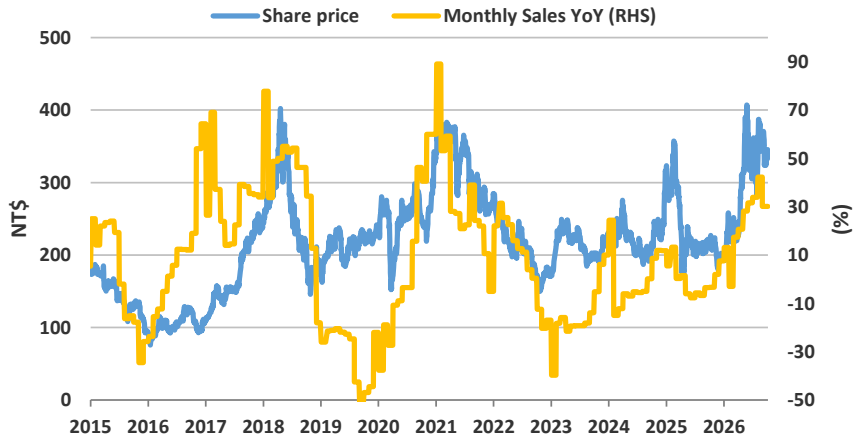
Details:

- September sales were NT$2,720mn (flat MoM/+30% YoY).
- 3Q26 revenue came in at NT$8,110mn (+8% QoQ/+35% YoY), largely in line with our estimate at NT$7,996mn (+6% QoQ/+33% YoY) and the consensus of NT$7,964mn (+6% QoQ/+33% YoY).

Our view:

- On top of the sequential growth on the top line, we believe that Hiwin should also further expand its gross margin on a QoQ basis in 3Q26, owing to the higher utilization and the tailwinds of price hikes.
- We think the momentum should remain favorable in the near term.
- We stay constructive on the stock, with current valuation at 29x 2027e P/E vs. a peak cycle level of 35-40x.

Exhibit 1: Hiwin's monthly sales YoY vs. its share price since 2015



Source: Past performance is no guarantee of future results. Results shown do not include transaction costs. Source: Company data, TEJ, Morgan Stanley Research.

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Hiwin Technologies Corp. (2049.TW, 2049 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$450.00
Up/downside to price target (%)	33
Shr price, close (Oct 6, 2026)	NT$339.50
52-Week Range	NT$426.00-180.50
Sh out, dil, curr (mn)	354
Mkt cap, curr (mn)	US$3,787
EV, curr (mn)	US$3,776
Avg daily trading value (mn)	US$45

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	4.31	9.77	11.77	13.77
Prior EPS (NT$)**	-	-	-	-
Revenue, net (NT$ mn)	24,263	29,992	34,688	39,123
EBITDA (NT$ mn)	4,001	6,485	7,488	8,431
ModelWare net inc (NT $ mn)	1,526	3,457	4,165	4,873
P/E	45.0	34.7	28.8	24.6
P/BV	1.4	2.2	2.0	1.9
RNOA (%)	3.6	8.7	11.1	12.7
ROE (%)	4.1	9.2	10.4	11.2
EV/EBITDA	12.6	17.9	15.3	13.3
Div yld (%)	1.2	0.6	0.9	1.1
FCF yld ratio (%)**	2.0	3.7	2.6	2.9

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
e = Morgan Stanley Research estimates

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AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
Hiwin Technologies Corp. 2049.TW			
September 2026 Sales	06 Oct 2026	Unchanged	In-line

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Hiwin Technologies Corp. (2049.TW)

Base case, P/E. We think this valuation methodology better captures Hiwin's earnings growth momentum, as its valuation tends to move with the business cycle. We apply a target multiple of 38x to our 2027 EPS estimate. We view this multiple as justified by the 47% operating profit CAGR we project for 2025-28.

Risks to Upside

- Stronger industrial automation demand pickup.
- Market share gains.
- Earlier-than-expected revenue contribution from humanoids.

Risks to Downside

- Weaker industrial automation demand.
- More pricing pressure.
- Market share loss.
- More broad-based impact on the global economy from geopolitical risks.

Disclosure Section

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The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Andy Meng, CFA; Derrick Yang.

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(as of September 30, 2026)
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For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
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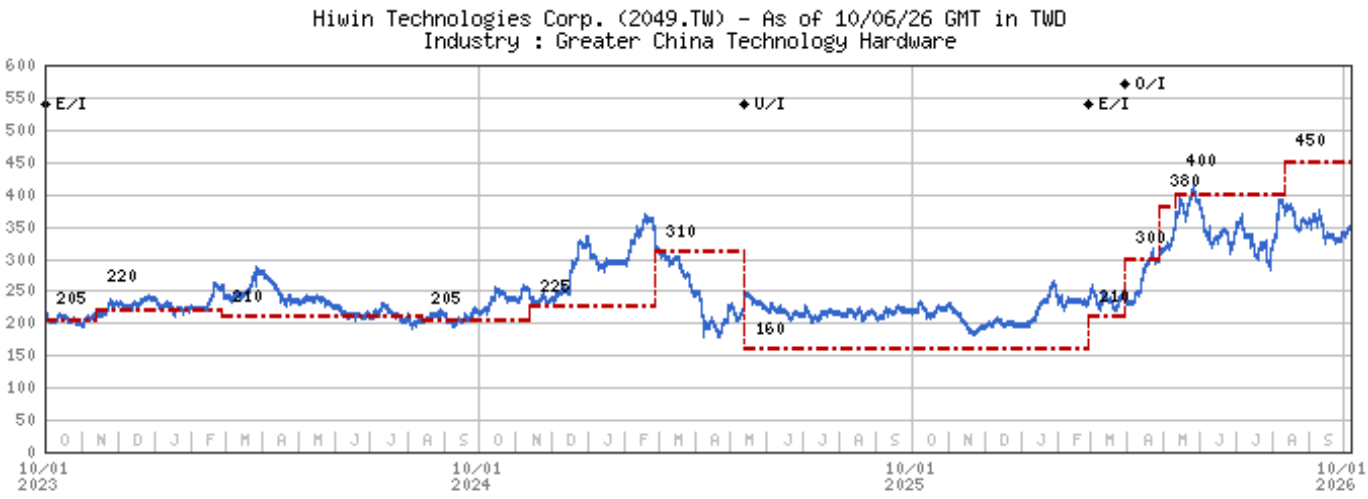
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : O/I; 8/11/22 : E/I; 2/21/23 : O/I; 8/11/23 : E/I; 5/12/25 : U/I; 2/27/26 : E/I; 3/30/26 : O/I

Price Target History: 8/5/21 : 480.04; 10/5/21 : 479.6; 2/28/22 : 341.16; 3/24/22 : 301.61; 7/20/22 : 295; 8/11/22 : 230; 11/21/22 : 200; 2/21/23 : 295; 2/27/23 : 283; 5/10/23 : 280; 8/11/23 : 205; 11/13/23 : 220; 2/27/24 : 210; 8/13/24 : 205; 11/12/24 : 225; 2/27/25 : 310; 5/12/25 : 160; 2/27/26 : 210; 3/30/26 : 300; 4/28/26 : 380; 5/11/26 : 400; 8/12/26 : 450

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)

Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/06/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$39.36
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$23.90
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,045.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,855.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.43
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$52.30
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$5.00
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$38.14
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$64.10
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43

Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.26
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$180.00
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,079.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.80
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,050.00
Advantech (2395.TW)	O (01/20/2021)	NT$738.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,195.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,685.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$37.45
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,530.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,325.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$885.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,350.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,050.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,415.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$339.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.10
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,100.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$299.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.05
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$1,000.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$206.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.90
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,995.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$5.02
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$380.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,260.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$256.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.10
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,920.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.40
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,865.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,505.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$90.00
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$333.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,310.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$567.00

Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,290.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$625.00
Zhen Ding (4958.TW)	++	NT$574.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$551.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,755.00

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ASIA PACIFIC INSIGHT

October 6, 2026 12:44 AM GMT

Greater China Technology Hardware

Integrated Power Solutions Driving Secular Demand

Increasing power consumption, along with the deployment of BBU/CBU/800/±400VDC/SST in AI clusters, could push the technology barrier higher and present growth opportunities for players with superior capabilities. Fuel cells could also emerge as a new opportunity amid power constraints.

Secular demand for power amid AI capex: AI-related power consumption will continue to increase, and we believe this will lead to secular demand for power-related components, such as power supply units (PSUs), battery backup units (BBUs), capacitor bank units (CBUs), coolant distribution units (CDUs), and so on. Moreover, the requirement for efficiency will inevitably necessitate the adoption of 800/±400 voltage direct current (VDC) in the power rack/center/block configuration (initially from late 2026 and broadly in 2028, we estimate), eventually extending to the deployment of solid state transformers (SST) in the longer term – all pointing to a more centralized power delivery architecture in the future. This trend of aggregation could raise the technology barriers amid the fusion of white/grey space, favoring players with better capabilities and execution. Overall, we forecast the datacenter PSU market size to grow at a 75% three-year CAGR to reach US$33bn in 2028.

Fuel cells to improve flexibility and resilience of AI power delivery: Despite the strong AI capex outlook (MSe global cloud capex +97%/20% YoY in 2026/2027), power has become a bottleneck, indicated by MS's forecast power shortfall of 5GW in 2026, scaling to a total 72GW through 2029 (see report). Fuel cells, among other alternative energy sources, could emerge as a good solution to pick up some slack and enhance efficiency through active micro grid management. Those who are able to offer comprehensive solutions from power sources to power delivery could be increasingly more competitive in the market.

Assuming coverage of Delta and Lite-On:

- Delta (OW, PT NT$2,500, based on 30x 2027/2028 P/E) should maintain its dominant position in the industry with its integrated power/thermal related solutions while fuel cells could add incremental growth opportunities in the longer term. We think a 30x P/E target multiple is justified by our estimated 63% earnings CAGR in 2025-28, driven by a 44% revenue CAGR and operating margin expansion to 22.1% in 2028 vs 15.1% in 2025,
- Lite-On (EW, PT NT$315, based on 17x 2027/2028 P/E), the second-largest player in AI power-related products, should also ride the tailwinds despite its slower progress in HVDC power racks and liquid cooling solutions. We think a 17x P/E target multiple is justified by our estimated 47% earnings CAGR in 2025-28 (29% revenue CAGR and 6.8ppt OPM expansion to 16.8% in 2028).

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GREATER CHINA TECHNOLOGY HARDWARE

Asia Pacific Industry View	**In-Line**

WHAT'S CHANGED

Delta Electronics Inc. (2308.TW)	**From**	**To**
Price Target	NT$2,700.00	**NT$2,500.00**
Lite-On Technology (2301.TW)	**From**	**To**
Price Target	NT$176.00	**NT$315.00**

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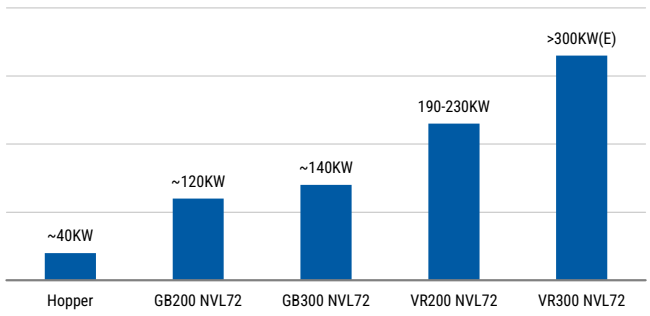
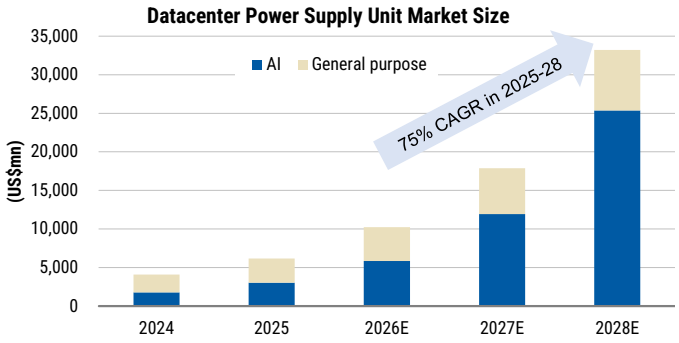
The Story in Charts

Exhibit 1: AI servers continuing to see higher TDP...



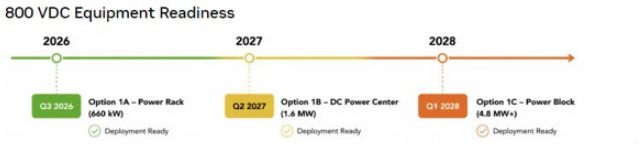
Source: Company data, Morgan Stanley Research estimates

Exhibit 2: ...which drives data center power supply demand



Source: Company data, Morgan Stanley Research estimates

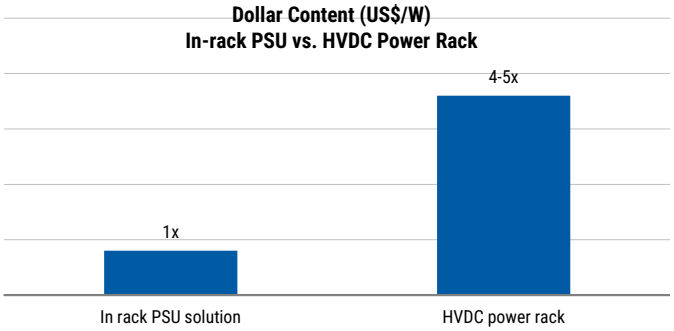
Exhibit 3: HVDC to see early adoption from late 2026...



Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

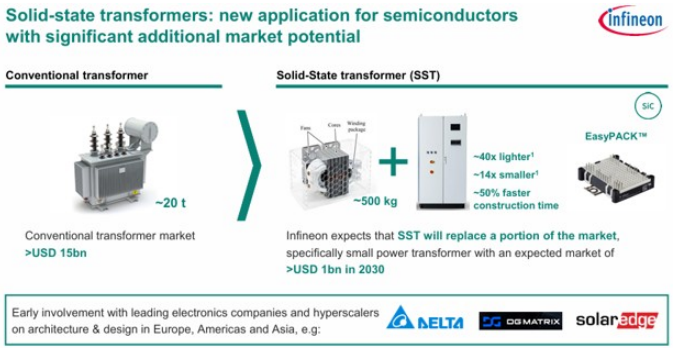
Source: Nvidia

Exhibit 4: ...with a much higher dollar content



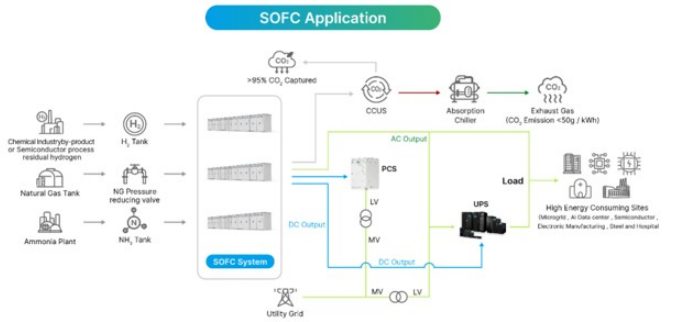
Source: Morgan Stanley Research

Exhibit 5: SST is also in the pipeline for more centralized power delivery



Source: Infineon

Exhibit 6: Fuel cells are a good option for better resilience in power architecture



Source: Delta

Executive Summary

AI cluster performance evolving to fulfill demand

The rapid adoption of artificial intelligence is driving a significant increase in global data center compute demand. Advances in foundation models, reasoning capabilities, agentic AI, and enterprise AI deployment are increasing the computational resources required for both training and inference. To support this growth, hyperscalers and model developers continue to deploy larger and more powerful AI clusters, driving demand for greater compute, memory, and networking capacity.

As AI infrastructure scales, cluster performance is increasingly determined by the efficiency of the entire system rather than the capabilities of individual processors alone. Compute, networking, cooling, and power infrastructure must all evolve together to support higher rack densities and larger deployments. While semiconductor innovation remains critical, supporting infrastructure is becoming an equally important factor in enabling future AI growth.

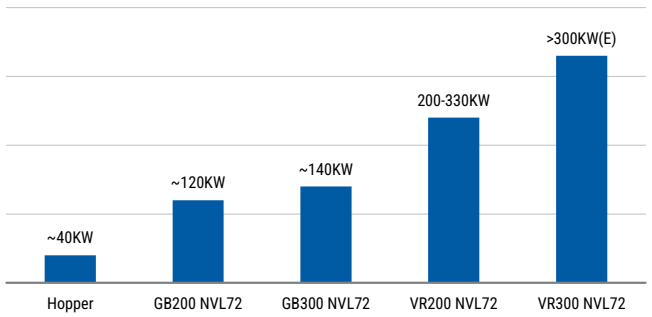
AI cluster performance is fundamentally a power problem

As AI clusters continue to scale in size and performance, power consumption is becoming one of the most important constraints on future infrastructure deployment. The power challenge is no longer limited to GPUs alone, as modern AI clusters comprise tightly integrated compute, networking, memory, storage, and cooling systems, all of which require increasing amounts of power as performance requirements rise. Consequently, the next phase of AI infrastructure investment should be viewed not only as a semiconductor upgrade cycle but also as a broad-based power infrastructure upgrade cycle.

Server power consumption has increased significantly across successive AI hardware generations. Nvidia's architecture roadmap illustrates the industry's migration from relatively modest server-level power budgets under the Ampere and Hopper generations toward rack-scale systems under Blackwell and future Rubin architectures. Estimated rack-level power consumption has increased from roughly 40kW for Hopper systems to approximately 120kW for GB200 NVL72 and 140kW for GB300 NVL72, with future Vera Rubin systems expected to approach 200-300kW per rack with current Vera Rubin systems reaching up to 330kW per rack while future Vera Rubin generations are expected to surpass these levels. As compute density continues to increase, power delivery infrastructure is becoming a critical enabler of system performance.

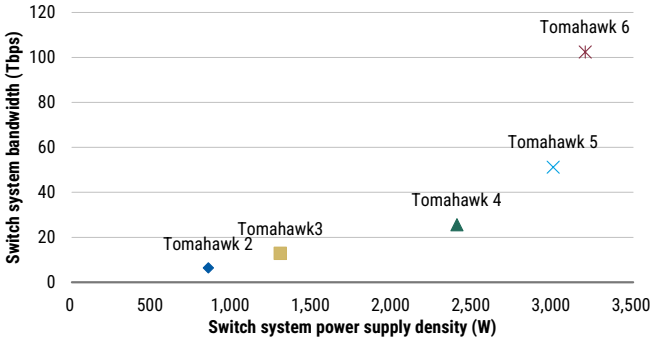
The increase in power consumption is not limited to compute hardware. Networking systems are also becoming increasingly power intensive as switch bandwidth scales from 100G architectures toward 800G and 1.6T generations. Despite substantial process-node improvements, switch ASIC power consumption has continued to rise due to increasing complexity and higher SerDes speeds. As a result, networking equipment now requires significantly larger power budgets than previous generations, creating additional demand for higher-capacity power supplies throughout the data center.

Exhibit 7: TDP roadmap for Nvidia racks



Source: Morgan Stanley Research

Exhibit 8: Power supply density for Tomahawks



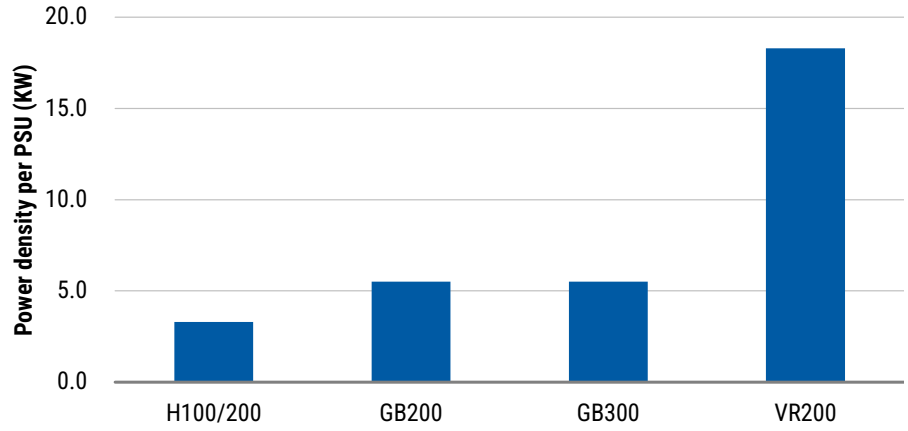
Source: Morgan Stanley Research

The growth in power consumption is driving a corresponding upgrade cycle for power-supply units (PSUs). Earlier AI server generations typically relied on multiple 3-3.3KW power supplies installed directly within individual server chassis. As the industry transitions toward rack-scale architectures, power conversion is increasingly centralized into dedicated power shelves capable of supporting substantially larger power loads. In Blackwell systems, power shelves utilize 5.5KW power supplies, while future Rubin-generation platforms are expected to increase PSU ratings to approximately 18KW, representing a significant increase in both PSU content and power density.

Delivering higher power within a similar footprint creates significant engineering challenges. Increasing PSU power density requires advances across multiple areas, including higher-performance power semiconductors such as silicon carbide (SiC) and gallium nitride (GaN), more efficient power-conversion topologies, smaller and more capable magnetic components, improved thermal-management solutions, and greater system integration. As output power increases, thermal constraints become increasingly important, making heat dissipation, airflow optimization, and mechanical design critical determinants of PSU performance.

Consequently, on top of the volume driven growth, the power-supply industry is benefiting from the technology driven dynamics, where efficiency, power density, and thermal performance are becoming key competitive differentiators. As AI rack power requirements continue to increase, PSU innovation will become an increasingly important component of the broader AI infrastructure investment cycle.

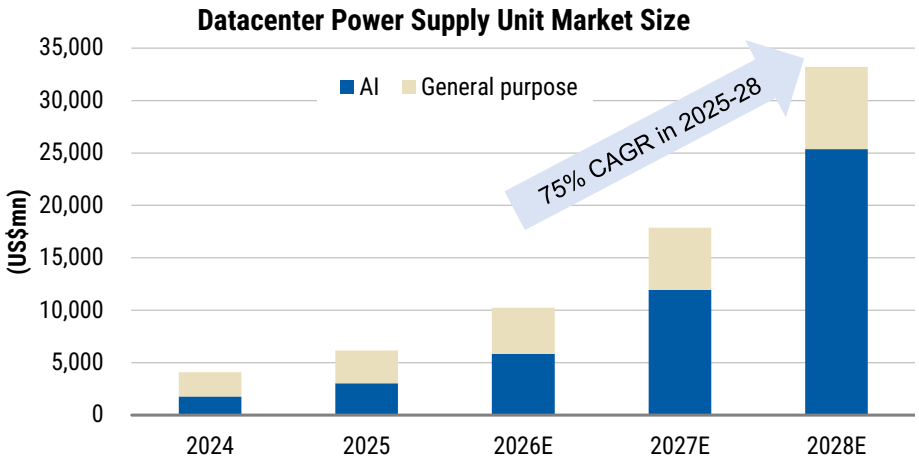
Exhibit 9: Power density per PSU trend in Nvidia server systems



Source: Nvidia, Morgan Stanley Research

Overall, we estimate the global datacenter power supply market will grow at a three-year CAGR of 75% to reach US$33bn in 2028 from US$6bn in 2025, driven mainly by server shipment growth, increasing power consumption per rack, and the adoption of HVDC power racks from 2027.

Exhibit 10: Global datacenter power supply market forecast



Source: Company data, Morgan Stanley Research estimates

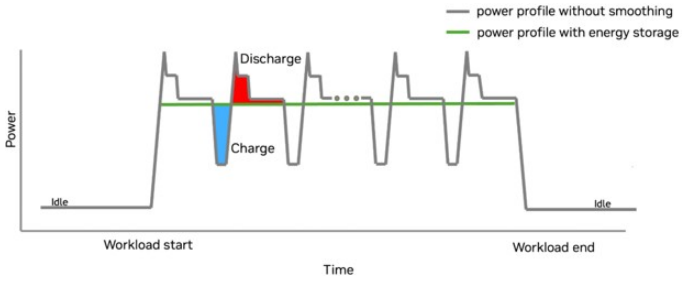
Complementary hardware for more resilient power delivery

As AI clusters become more power intensive, maintaining stable and resilient power delivery is emerging as a critical design priority. Traditional centralized power architectures are increasingly being complemented by localized energy-storage and power-conditioning technologies that can respond rapidly to dynamic workload demands.

Battery Backup Units (BBUs) represent one of the most important additions to modern AI power architectures. Positioned close to the computing workload, BBUs provide localized energy storage capable of responding significantly faster than facility-level backup systems. Their role extends beyond conventional backup power functionality. BBUs can

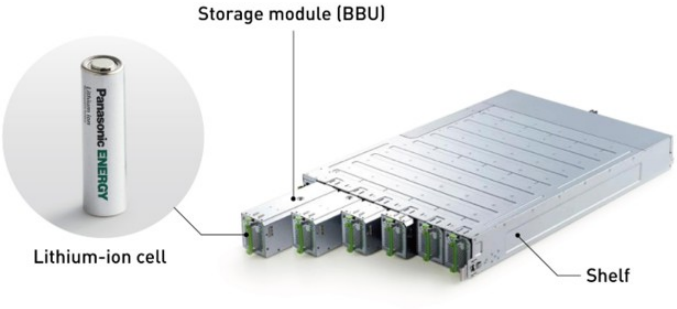
absorb transient fluctuations, smooth power ramps, and improve overall rack-level power stability. By providing local buffering capacity, BBUs help reduce the burden on upstream power distribution infrastructure while improving overall system resilience.

Exhibit 11: Concept of peak shaving for AI server power loads



Source: Nvidia

Exhibit 12: BBU



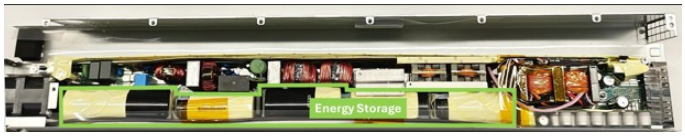
Source: Panasonic

Capacitor Backup Units (CBUs) address an even shorter-duration segment of the power stability challenge. Utilizing supercapacitors and other high-power-density storage technologies, CBUs can deliver and absorb power within milliseconds, making them particularly effective for managing rapid GPU power spikes. These systems act as shock absorbers within the electrical architecture, preventing sudden load changes from propagating throughout the broader power network. As AI rack power densities continue increasing, capacitor-based energy storage is becoming an attractive solution for improving power quality without requiring substantial oversizing of facility infrastructure.

Exhibit 13: CBU

Source: Panasonic

Exhibit 14: GB300 power shelf with integrated capacitors



Source: Nvidia

Together, BBUs and CBUs provide complementary benefits. BBUs are designed to handle longer-duration events, while CBUs are optimized for extremely fast transient response. The integration of these technologies enables operators to improve power quality, reduce stranded infrastructure capacity, enhance system reliability, and increase effective utilization of expensive electrical assets. As rack densities continue to rise, localized energy-storage solutions are likely to become a standard component of AI data center design rather than an optional enhancement.

The adoption of BBU and CBU architectures may also enable future increases in rack density. As operators attempt to push toward several hundred kilowatts per rack, localized power-buffering solutions can help mitigate transient peaks and reduce stress on grid-facing infrastructure. This allows operators to deploy higher-density computing resources without proportionately increasing upstream electrical capacity.

Exhibit 15: Comparison of CBU and BBU

Dimension	CBU	BBU
Primary storage medium	EDLC supercapacitors, LICs or high-capacitance	Typically lithium-ion battery cells managed by a BMS
Primary strength	High power, fast charge/discharge and frequent	Higher stored energy and longer sustained
Main function	Transient smoothing, peak shaving, voltage	Outage backup, source-transfer bridging and orderly workload
Response profile	Well suited to millisecond-class load support and	Fast switchover, but generally optimized for sustained support rather
Typical duration	Milliseconds to seconds; selected systems can extend toward tens of seconds or more	Commonly tens of seconds to several minutes, depending on battery and rack design
Power versus energy density	Higher power capability but lower energy storage per unit volume than a battery	Higher energy density, making it more economical for longer events
Cycle life	Eaton specifies more than one million cycles and up to 20 years for certain supercapacitor modules	Battery life is more sensitive to depth of discharge, temperature and repeated cycling; BMS supervision is therefore essential.
State management	Requires cell balancing, voltage/temperature monitoring and state-of-health tracking, but does not use a conventional battery fuel	Requires BMS functions including charge control, state of charge, state of health and cell protection
Best-fit use case	Repetitive AI power spikes, inrush events, PSU load leveling and bus stabilization	Power continuity through outages, brownouts or generator/source-transfer intervals

Source: Morgan Stanley Research

HVDC emerging as a necessity, with SST in the pipeline

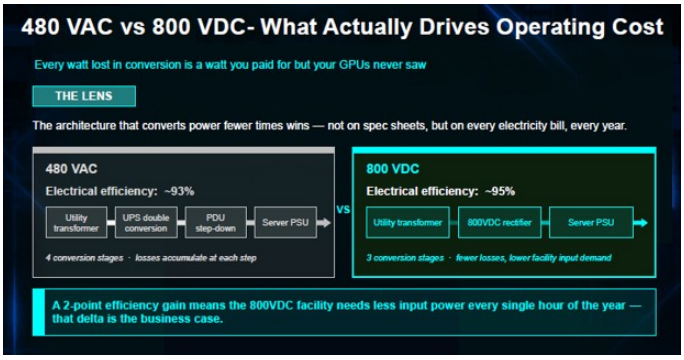
800VDC and ±400VDC progressing simultaneously

The rapid increase in AI rack power density is creating growing challenges for conventional power-distribution architectures. As rack-level power consumption rises from tens of kilowatts to several hundreds of KWs, electrical losses, cable size, thermal management requirements, and overall system complexity increase significantly. Under traditional low-voltage power architectures, delivering larger amounts of power requires proportionally higher current, which increases resistive losses and places greater stress on cables, connectors, busbars, and power-conversion equipment.

At the same time, every power-conversion stage introduces efficiency losses and generates additional heat. As AI deployments scale into multi-megawatt and eventually gigawatt-level campuses, even small efficiency improvements can translate into meaningful reductions in operating costs, cooling requirements, and infrastructure investment. Consequently, the industry is increasingly focused on power architectures that can reduce conversion stages, lower current levels, and improve overall power-delivery efficiency.

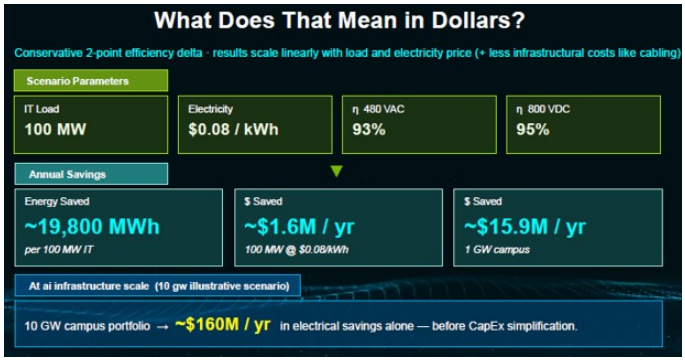
Against this backdrop, high-voltage direct current (HVDC) is emerging as one of the most important architectural upgrades for next-generation AI infrastructure. By increasing distribution voltage, HVDC allows the same amount of power to be delivered at a lower current, reducing conduction losses and improving system efficiency. Higher-voltage architectures also improve scalability for ultra-high-density rack deployments by alleviating some of the physical and thermal constraints associated with conventional power-distribution systems.

Exhibit 16: Better power efficiency for 800VDC architecture



Source: Delta

Exhibit 17: Better efficiency leads to cost savings



Source: Delta

Nvidia's roadmap illustrates the industry's gradual migration toward higher-voltage power architectures. Earlier AI server generations primarily utilized conventional power-delivery schemes centered around rack-level 48V distribution. As rack power requirements increased with Blackwell-generation systems, industry participants began evaluating higher-voltage architectures capable of supporting substantially greater power density. Looking ahead, the industry is increasingly converging around 800VDC as a potential next-generation standard for large-scale AI deployments, particularly as rack-level power consumption approaches several hundred kilowatts. The transition reflects the need to improve power-distribution efficiency while supporting the continued growth of AI computing infrastructure.

We believe that the adoption of the 800VDC power rack could start from late 4Q26, though it should be small scope deployment through 2027, since the in-rack power shelf is the reference design in the Vera Rubin NVL72 rack. Broader adoption should happen more likely from late 2027 and 2028, as the Vera Rubin Ultra rack further pushes up the power consumption.

Exhibit 18: HVDC deployment roadmap

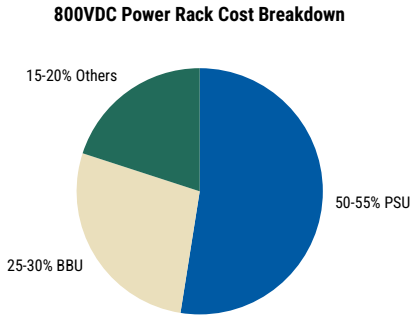
Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

Source: Nvidia

The 800VDC power rack BOM is primarily driven by the PSU, which accounts for 50–55% of total cost, followed by the BBU at 25–30%, with other components comprising the remaining 15–20%. This cost structure reflects the broader functionality integrated into an HVDC power rack, combining power conversion, backup power and supporting components within a centralized architecture.

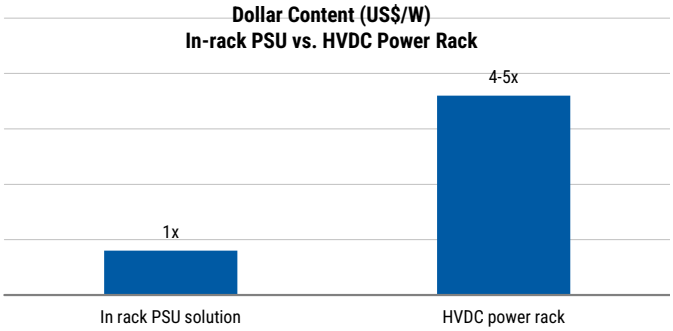
For power supply players, one of the most direct implications is that an HVDC power rack carries around 4–5x higher dollar content per watt compared with a conventional in-rack PSU solution, since the former integrates more components into a robust system and power supply players usually ship the entire rack instead of just the components.

Exhibit 19: BOM cost breakdown for an 800VDC power rack



Source: Morgan Stanley Research

Exhibit 20: Higher dollar content for HVDC power racks



Source: Morgan Stanley Research

However, the industry has not yet fully converged on a single architecture. While Nvidia's roadmap increasingly points toward 800VDC, several hyperscalers and members of the Open Compute Project (OCP) ecosystem have expressed interest in ±400VDC architectures. Both approaches deliver similar total voltage levels and pursue many of the same objectives, including lower current, higher efficiency, and improved scalability. The debate largely centers on implementation trade-offs rather than end goals.

Supporters of 800VDC generally highlight the advantages of a simpler single-bus architecture, potentially lower system complexity, and closer alignment with existing industrial high-voltage DC equipment. The architecture avoids the need to maintain balanced positive and negative voltage rails and may simplify certain aspects of power conversion and system integration. These characteristics make 800VDC particularly attractive in environments where operational simplicity and interoperability with existing electrical infrastructure are key priorities.

Advocates of ±400VDC, meanwhile, emphasize the advantages of a bipolar architecture. By creating positive and negative voltage rails around a neutral reference point, ±400VDC can potentially reduce insulation requirements, simplify certain fault-management scenarios, and provide improved flexibility for load balancing and future system expansion. Many hyperscalers also view the bipolar approach as being more compatible with long-term visions for highly distributed power systems incorporating energy storage, renewable energy, and microgrid functionality.

Importantly, the ultimate beneficiaries may be largely independent of which standard gains wider adoption. With the exception of certain power-semiconductor and protection-system requirements, many component suppliers across the power ecosystem remain

broadly agnostic between 800VDC and ±400VDC implementations. Higher-voltage architectures in either form require similar upgrades in power conversion, power management, cooling, connectors, magnetics, and infrastructure equipment. As a result, the industry's migration toward HVDC matters significantly more than the eventual selection of any single voltage standard.

Exhibit 21: Comparison of 800VDC and ±400VDC

Metric	±400VDC	800VDC
Voltage to ground	400V relative to ground	800V relative to ground
Voltage differential	800V	800V
Power distribution efficiency	High	Slightly higher
Scalability	Good	Excellent
Insulation requirements	Lower	Higher
Maintenance safety	Better	More challenging
System architecture simplicity	Dual-rail system	Single-bus system
Grounding Complexity	More complicated	Simpler
Electrical footprint	Small	Smaller
Ability to reduce conversion stages	Moderate	Highest
Long-distance distribution capability	Excellent	Excellent
Maturity of ecosystem	Better today	Early but rapidly developing
Industry players	OCP, hyperscalers	Nvidia
Long term industry direction	Transitional architecture	Likely end-state architecture

Source: Morgan Stanley Research

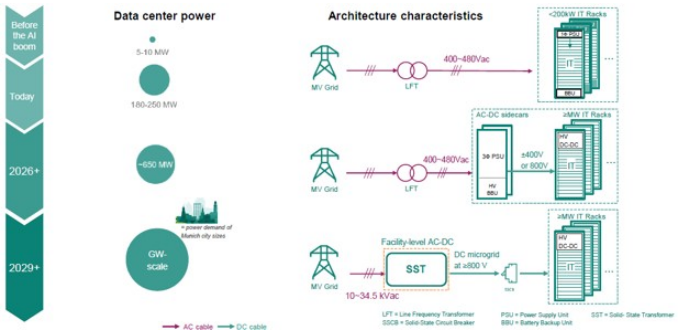
Solid-state transformers to follow after 800VDC/±400VDC matures

Looking further ahead, solid-state transformers (SSTs) represent a potential next-generation evolution of power infrastructure. Unlike traditional transformers that operate primarily as passive devices, SSTs leverage power electronics to provide voltage conversion, power conditioning, and bidirectional energy management within a single platform. This enables greater controllability and flexibility while supporting increasingly digitalized power networks.

Although SSTs remain at an early stage of commercialization and face challenges related to economics, reliability, and large-scale deployment, the technology aligns well with long-term data center trends. As AI campuses evolve into utility-scale power consumers incorporating energy storage, distributed generation, and advanced power management, SSTs could eventually become a key enabling technology.

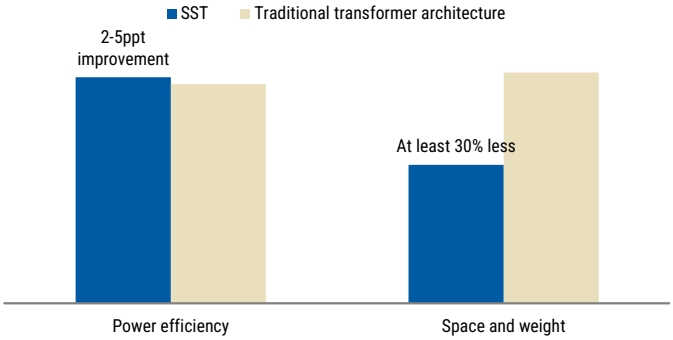
Therefore, while HVDC is expected to drive the next phase of power architecture upgrades, SSTs represent a longer-term technology roadmap that could further transform how future data centers distribute and manage electricity.

Exhibit 22: SST enabling more efficient power delivery architecture



Source: Infineon

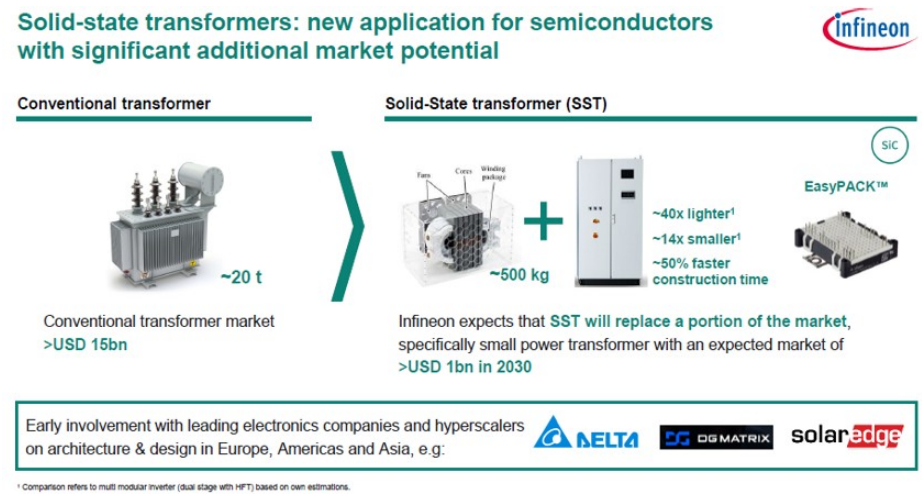
Exhibit 23: SST offers better power efficiency with at least 30% less space/weight



Source: Morgan Stanley Research

According to Infineon, SST could represent an emerging >US$1bn market opportunity by 2030, as the technology begins to replace a portion of the >US$15bn conventional transformer market, particularly in smaller-power applications. While SST penetration is expected to remain relatively modest versus the overall transformer market, the opportunity is significant for relevant players in the longer term.

Exhibit 24: SST to see initial adoption in 2030

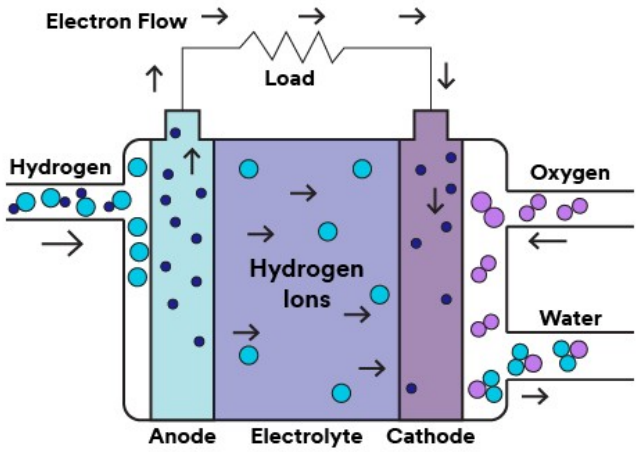


Source: Infineon

Fuel cells for better flexibility and resilience

Fuel cells generate electricity through an electrochemical reaction rather than combustion. Hydrogen, natural gas, biogas, or other fuels are supplied to the fuel cell stack, where chemical energy is converted directly into electricity, heat, and in some cases water. Unlike batteries, which store energy, fuel cells continuously generate power as long as fuel is supplied.

Exhibit 25: Fuel cell diagram



Source: 3M

Fuel cells can offer several benefits, including:

- **High energy efficiency:** Direct electrochemical conversion enables higher efficiency than conventional combustion-based power generation, reducing fuel consumption and operating costs.
- **Low carbon emissions**: Fuel cells produce minimal direct emissions and can support long-term decarbonization goals when powered by low-carbon fuels such as hydrogen.
- **Grid independence and energy security:** Provides reliable onsite power generation, reducing dependence on grid availability and improving energy resilience.
- **Long-duration energy storage enabler:** Hydrogen-based fuel-cell systems can complement renewable energy by storing energy over extended periods, supporting long-duration and seasonal energy storage.
- **Fuel flexibility:** Certain fuel-cell technologies can operate on hydrogen, natural gas, biogas, ammonia-derived hydrogen, and other hydrogen-rich fuels, allowing deployment using existing fuel infrastructure.
- **Low maintenance and quiet operation:** Limited moving parts reduce maintenance requirements while providing quieter operation than conventional combustion-based generators.
- **Scalable deployment:** Can be deployed across a wide range of power levels, from distributed backup systems to utility-scale installations, making fuel cells suitable for large AI data center campuses

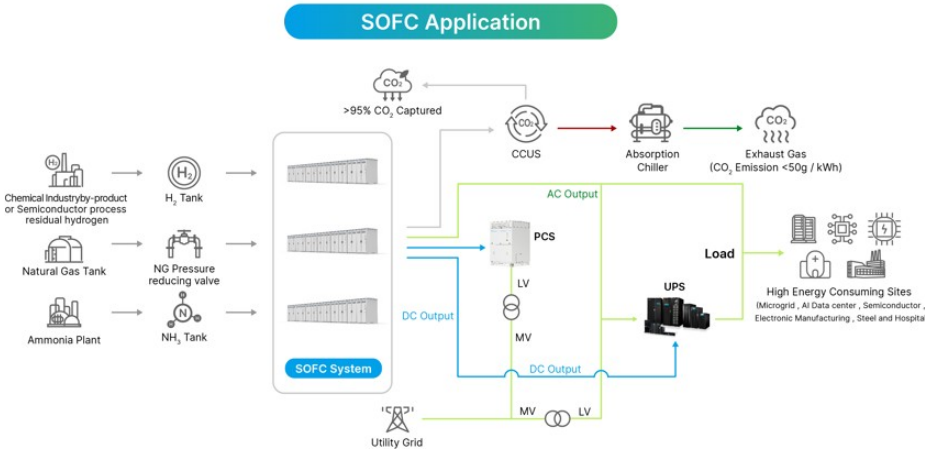
However, the following challenging are hurdles for wider deployment:

- **High system cost:** Fuel cells remain more expensive than conventional power-generation technologies, limiting large-scale adoption.
- **Limited hydrogen infrastructure:** Hydrogen production, transportation, storage, and refueling networks remain underdeveloped in most regions.
- **Durability requirements:** Long-term reliability and stack lifetime remain key challenges, particularly for large-scale and mission-critical deployments.
- **Regulatory and safety considerations:** Hydrogen handling requires specialized infrastructure and compliance with stringent safety regulations, increasing deployment complexity.
- **Lack of standardization:** Multiple fuel-cell technologies compete across different applications, creating uncertainty around long-term technology standards and adoption paths.

Fuel cells have the potential to fundamentally reshape future data center power architectures. Traditionally, data centers rely on grid electricity as the primary power source and diesel generators for backup power. However, as AI-driven workloads significantly increase power demand and power availability becomes a key deployment constraint, hyperscalers are increasingly evaluating fuel cells not only as a backup-power solution but also as a potential primary power source. This could accelerate the transition from a grid-centric architecture toward one incorporating distributed onsite generation, enabling data center deployment in regions where utility power capacity is limited.

Over the longer term, fuel cells could become a core component of AI data center microgrids that integrate renewable generation, battery storage, hydrogen storage, and intelligent energy-management systems. Such an architecture would improve scalability, resilience, and energy independence while reducing reliance on centralized utility infrastructure. In addition, because fuel cells inherently generate DC power, they align well with the industry's migration toward HVDC-based power distribution architectures. This could simplify power delivery by reducing conversion stages between generation and IT equipment, improving overall system efficiency and supporting next-generation high-density AI deployment

Exhibit 26: Fuel cell in datacenter power delivery architecture



Source: Delta

Stock picks

Delta (2308.TW, OW, PT NT$2,500)

Dominant supplier in the AI power supply offerings: Delta has been one of the most competitive players in power related components and systems, as its offerings spanning from the core power supply units, the complementary BBU/CBU for AI servers, the highly anticipated HVDC power rack, grey space power facilities to air/cooling solutions. We estimate that it has ~60% market share in the datacenter power supply market and is among the leading suppliers of the 800VDC/±400VDC power rack to Nvidia/hyperscalers in 2H26 (wider adoption likely from late 2027). Longer term, when the HVDC becomes more mature, the centralized power delivery architecture could migrate into SSTs, where Delta already has small volume shipment now. With the solid technology, close customer relationships and strong execution, we believe Delta should stay competitive even amid the fusion of the white space and grey space.

Extending from power delivery to power source: In addition to the offerings in the power delivery, Delta signed a licensing agreement with Ceres Power (CRW.GB) for its solid oxide fuel cell (SOFC) technology in early 2024. This could extend its business scope into the power source area. which we believe could add more value to its customers as it could help mitigate the power constraints for datacenter build and enhance the operational efficiency through active micro grid management.

Solid near term momentum: Despite the slower revenue growth owing to the component supply tightness in 3Q26, we expects its 4Q26 revenue to grow by double digit % QoQ on alleviated component supply and solid demand. 3Q26 GM should be flat to slightly up QoQ and 4Q26 might see further expansion on the tailwinds price hikes.

OW with PT of NT$2,500: We see power/thermal as one of the major secular AI related themes in the downstream tech space and Delta is well positioned to benefit, potentially driving a 63% earnings CAGR in 2025-28, in our estimate. We thus assume the coverage of Delta with an OW rating and the PT of NT$2,500, based on 30x 2027/2028 P/E. Our target multiple implies a PEG of 0.5 vs. the average PEG of 1.2 in 2020-25, when the stock was trading at an average 12M forward P/E of 22x with the earnings CAGR of 19%.

Lite-On (2301.TW, EW, PT NT$315)

Rising tide lifting all boats: Lite-On is the second largest power supply vendor globally and has been engaging with hyperscalers regarding the ASIC and Nvidia GPU servers. Looking ahead, it expects to expand its share in a couple of hyperscalers from late 2026. It should have over 50% of the total revenue generated from the cloud segment for PSUs, BBUs, and to a lesser extent, other mechanical components like liquid cooling, chassis/rack, etc. Moreover, it also has the HVDC power rack solutions ready and is qualifying with potential customers. The cloud shall remain as the major growth driver in the coming years.

Aspiration in the optical communication space: Lite-On has always had an opto business (10-20% of total revenue in 2026) in both visible and invisible light applications. Post the strategic investment at Dense Light (21% stake), an InP epi wafer and chip process manufacturer, it plans to leverage its expertise in packaging and provides the optical

modules and engine for the data interconnects. While the end applications looks promising, we think it might take some time to see more meaningful revenue contribution in Lite-On.

Cloud momentum extending into the coming quarters: We expect its 4Q26 revenue to grow by double digit QoQ driven mainly by the could segment and GM should normalize at ~25% in 2H26 vs. 27.2% in 2Q26 on some inventory write-off reversal and high margin product shipment delayed from the previous quarter.

EW with PT of NT$315: We think Lite-On will grow along with the AI capex cycle, driven by its power-related offerings (47% earnings CAGR in 2025-28, in our estimate), though execution and progress with new customers is worth monitoring. We assume coverage of Lite-On with an EW rating and a PT of NT$315, based on 17x 2027/2028 P/E. Our target multiple implies a PEG of 0.4 vs. the average PEG of 1.6 in 2020-25, when the stock was trading at an average 12M forward P/E of 14x on an earnings CAGR of 9%.

Panasonic Holdings (EW, PT ¥4,700, covered by Kazuo Yoshikawa)

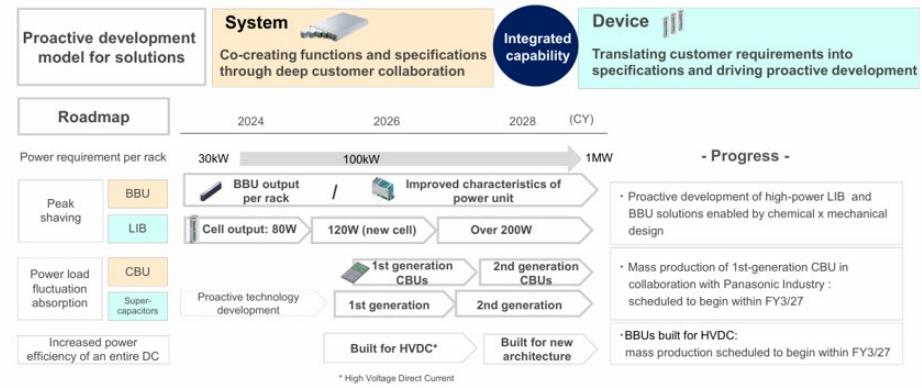
Panasonic has an approximately 80% share of the BBU market and is well positioned to benefit from rising AI data center power demand: The company expects data center energy storage revenue to grow from ¥322bn in FY3/26 to approximately ¥550bn in FY3/27 and around ¥1trn in FY3/29. Importantly, all of the business underpinning its FY3/29 revenue target has already been awarded, meaning agreements have been reached with customers on product development and orders.

BBU: Panasonic has begun limited mass production of new high-capacity BBU cells above 200W, with products adopted by a major GPU vendor and ramping from June-July. Discussions for next-generation server platforms are in the final stages, and the company expects to maintain its approximately 80% market share. It also sees scope for higher pricing as rack-level power consumption rises.

The company is also close to securing awards for BBUs used in dedicated power racks. As its medium-term plan includes only already-awarded business, additional design wins could provide upside.

CBU: CBUs are emerging as a new growth opportunity as GPU and ASIC power consumption increases. Panasonic is working with power-supply vendors to propose integrated BBU+CBU solutions and is receiving inquiries from US and Chinese hyperscalers, including customers that were not previously BBU users.

Exhibit 27: Panasonic Holdings: Strengthen proposal and development capabilities for the next generation



Source: Company data

Risks to our positive view

Risks to our positive view on power-related demand include:

- Component supply tightness leading to lower shipment of products
- Shortfall in hardware shipments, yield issues, electricity availability, etc. slowing down datacenter builds
- Lower-than-expected penetration of new power delivery architectures like 800VDC/±400VDC, SST, etc.
- More intense competition leading to margin erosion among most players
- Hyperscalers prove unable to finance aggressive capex through internal and external funding.

AI Cluster Performance Evolving to Fulfill Demand

AI adoption driving exponential growth in compute demand

AI is becoming a mainstream computing workload

Over the past three years, generative AI has evolved from a research application into a core technology platform. Consumer applications such as ChatGPT, Claude, Gemini, and Copilot now serve hundreds of millions of users globally, while enterprises are increasingly embedding AI into software development, customer service, document analysis, enterprise search, healthcare, financial research, and industrial automation.

Historically, software demand scaled roughly with the number of users. AI demand is fundamentally different because every user interaction consumes computational resources in real time. Unlike traditional cloud applications, where the bulk of computation occurs during software development, AI applications continuously consume compute whenever a model generates a response. Consequently, rising AI adoption directly translates into higher demand for data center compute resources.

Model complexity continues to increase

The AI industry has historically improved model performance through scaling laws: larger models trained on larger datasets using more compute generally produce better outcomes. While efficiency improvements continue, leading AI developers remain committed to scaling model capabilities.

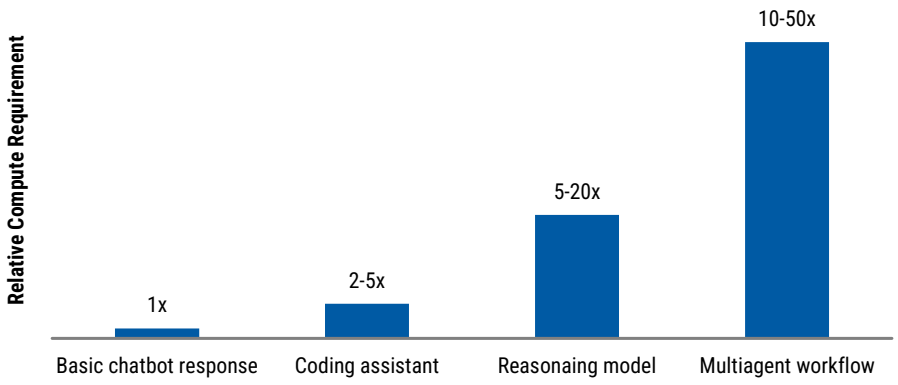
More importantly, the industry is moving beyond conventional language models toward significantly more compute-intensive architectures.

Reasoning models

Traditional LLMs generate answers directly from a prompt. New reasoning models perform multiple internal computational steps before producing a response. Instead of generating a few hundred output tokens, advanced reasoning systems may generate thousands or even tens of thousands of internal reasoning tokens while evaluating different solution paths. This dramatically increases inference workloads.

The industry is increasingly shifting from "parameter scaling" toward "test-time compute scaling," where additional compute is consumed during inference to improve answer quality. From an infrastructure perspective, this is highly significant because inference workloads occur continuously rather than once during model training.

Exhibit 28: Increasing demand for compute power from reasoning models



Source: Morgan Stanley Research estimates

Agentic AI

The next wave of AI applications involves autonomous agents capable of planning, executing, verifying, and refining tasks.

Instead of generating a single answer, an AI agent may search multiple information sources, run code, validate results, generate follow-up tasks, and produce a final response. Each step requires additional inference cycles.

A workflow that previously required one model invocation may now require dozens of model calls, substantially increasing compute consumption per user.

Multimodal AI

AI models are increasingly processing images, video, audio, sensor data, and text simultaneously.

Video understanding is particularly compute intensive because each second of video contains vastly more information than a text prompt.

As multimodal AI adoption grows, compute demand per query rises significantly, creating another structural driver of accelerator demand.

Training demand remains robust

Although inference is becoming the largest source of AI compute demand, training workloads continue to expand. Frontier model development increasingly requires clusters containing tens of thousands of accelerators operating simultaneously.

Training runs are becoming larger because developers are pursuing larger parameter counts, more training data, longer training durations, multimodal datasets and reinforcement learning and reasoning optimization

Consequently, each new generation of foundation models typically consumes substantially more compute than the previous generation.

Importantly, training infrastructure sets the baseline for future inference deployment. Every major model developer must maintain access to large-scale training clusters to remain competitive, supporting continued demand for high-end AI accelerators.

More compute requests to be fulfilled by better chip performance

Rising compute demand cannot be addressed simply by adding more servers. Modern AI workloads increasingly encounter bottlenecks related to memory bandwidth, interconnect performance, power consumption, and cooling. As a result, each new generation of AI hardware delivers improvements across multiple dimensions.

Compute performance

The latest AI accelerators provide several-fold performance improvements versus prior generations. For example, NVIDIA's Blackwell architecture was specifically designed to support both large-scale training and high-volume inference, with significant gains in AI throughput compared with previous Hopper-based systems. Performance improvements are necessary because reasoning and agentic workloads require substantially more operations per query.

Memory capacity and bandwidth

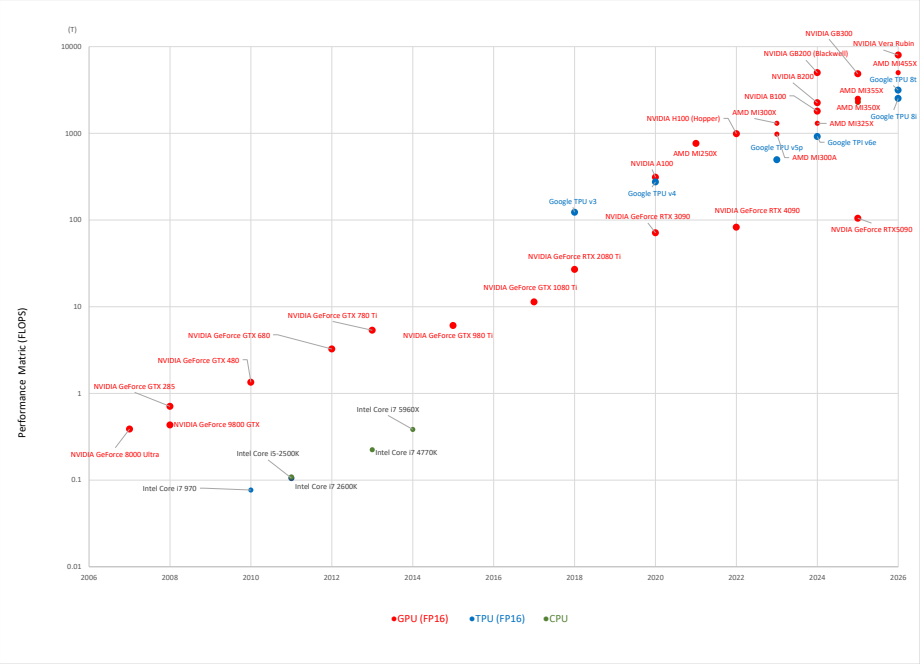
AI models are increasingly constrained by memory rather than raw compute, because longer models, longer context windows, reasoning models and agentic workflows all require more memory. As model complexity increases, modern high-end accelerators usually incorporate hundreds of gigabytes of High Bandwidth Memory (HBM), providing memory bandwidth measured in multiple terabytes per second.

Interconnect performance

As AI clusters scale from hundreds to thousands of accelerators, communication between processors becomes increasingly important. A large AI training cluster effectively behaves like a single massive computer. Consequently, networking technologies such as NVLink, InfiniBand, Ethernet, etc., have become critical enablers of AI scaling.

Many AI workloads can no longer be accelerated simply by increasing GPU performance; cluster interconnect bandwidth must also improve.

Exhibit 29: Compute power for HPCs continuing to evolve



Source: Company data, Morgan Stanley Research

AI Cluster Performance Is Fundamentally a Power Problem

As AI clusters become larger and more powerful, every component within the cluster must deliver higher throughput, lower latency, and greater bandwidth. Achieving these improvements inevitably requires more electrical power.

Therefore, the power challenge is no longer limited to GPUs. Modern AI clusters are highly integrated systems comprising compute servers, networking fabrics, memory subsystems, storage infrastructure, and cooling equipment. As cluster performance increases, power consumption rises across the entire infrastructure stack.

In our view, the next phase of AI infrastructure investment should be viewed not simply as a semiconductor upgrade cycle, but as a broad-based power consumption cycle driven by the need for ever-higher AI cluster performance.

Server systems seeing increasing TDP

Nvidia's AI-server power roadmap is best understood as a transition from chip-defined, air-cooled nodes to power-, cooling- and fabric-constrained rack-scale computers: the Ampere-era A100 SXM carried a disclosed 400W standard TDP—up to 500W in custom thermal designs—with 40/80GB HBM2/HBM2e and 600GB/s NVLink, while the eight-GPU, dual-AMD-EPYC DGX A100 was a conventional 6U server with a 6.5KW maximum input rating, meaning GPU silicon represented only roughly half of the node envelope once CPUs, NVSwitches, memory, NICs, drives, fans and conversion losses were included.

Hopper stepped the H100 SXM to a disclosed 700W TDP, 80GB HBM3, roughly 3TB/s memory bandwidth and 900GB/s fourth-generation NVLink, while H200 retained essentially the same Hopper compute architecture and power profile but substituted 141GB of faster HBM3e at 4.8TB/s—making H200 primarily a memory-capacity/bandwidth upgrade rather than a compute-TDP step-up—and Nvidia rates the eight-GPU, dual-Xeon DGX H100/H200 at 10.2KW maximum input, versus 6.5KW for DGX A100;

The architectural discontinuity arrives with GB200. Rather than treating an eight-GPU server as the scale-up unit and using InfiniBand/Ethernet primarily between nodes, Grace Blackwell NVL72 makes the entire rack the scale-up domain, integrating 18 one-U compute trays—each with two 72-core Grace Arm CPUs and four Blackwell GPUs—plus nine NVLink switch trays, management switches, , a 50–51V DC busbar, copper NVLink cartridges and liquid manifolds, so that 36 CPUs and 72 GPUs operate as one low-latency NVLink-connected system. Based on the same compute die, Nvidia upgraded the HBM content to further improve the performance of GB300 chip. Overall, the rack level power consumption is estimated at ~120KW for the GB200 NVL72 and ~140KW for the GB300 NVL72.

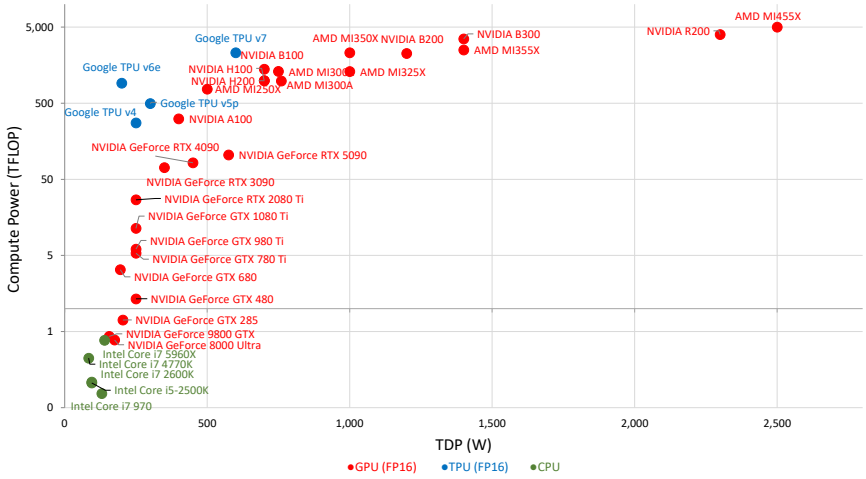
Vera Rubin—often called VR200/R200 in industry roadmaps—pushes the same rack-level philosophy further. Nvidia's disclosed architecture combines 72 dual-die Rubin GPUs with 36 new Vera CPUs, replacing Grace's 72 Arm Neoverse cores with 88 custom Olympus cores, raising CPU memory capacity and bandwidth to as much as 1.5TB LPDDR5X and 1.2TB/s, and doubling coherent CPU–GPU NVLink-C2C bandwidth to 1.8TB/s; Rubin itself moves from HBM3e to 288GB HBM4 at up to 22TB/s, introduces a third-

generation Transformer Engine and raises sixth-generation NVLink scale-up bandwidth to 3.6TB/s per GPU, while ConnectX-9 provides 1.6Tbps/GPU scale-out connectivity and BlueField-4 increasingly offloads storage, security and infrastructure processing.

Nvidia has not published a simple Rubin chip TDP in its primary datasheet, but our supply chain checks indicate roughly 1.8kW at an efficiency-oriented operating point and up to approximately 2.3kW at the performance mode, which is consistently corroborated by system vendors, with ASUS specifying 187kW Max-Q and 227kW Max-P for its fully liquid-cooled Vera Rubin NVL72.

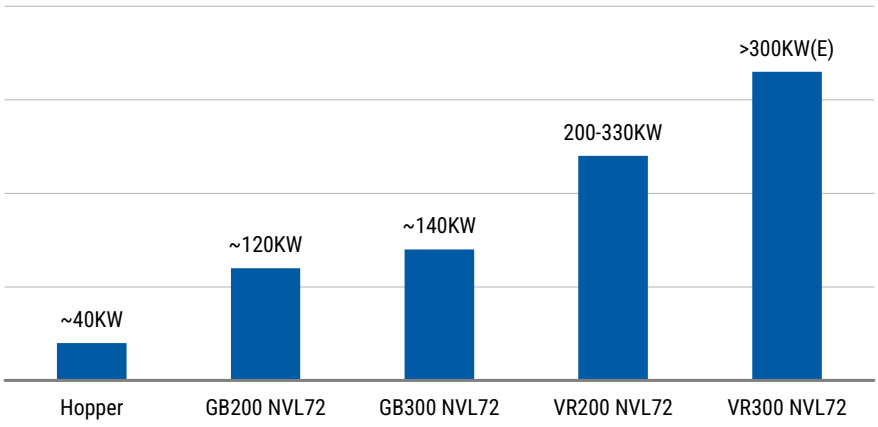
Looking ahead, we expect this trend to extend into the future along with the technology advancement and more critical workloads.

Exhibit 30: Compute power vs. TDP for HPCs



Source: Company data, Morgan Stanley Research

Exhibit 31: TDP roadmap for Nvidia racks



Source: Company data, Morgan Stanley Research

Networking consuming more power as well

The network switch system is another key devices within the datacenter and has been evolving over the past years, driven primarily by advances in the switch ASIC. From Tomahawk 1 (2014) to Tomahawk 6 (2025), the process node shrank from 28nm to 3nm, while aggregate switching bandwidth increased from 3.2 Tbps to 102.4 Tbps, representing a 32× increase. Over the same period, SerDes speed improved from 25 Gbps NRZ to 200 Gbps PAM4, enabling maximum port speeds to scale from 100G to 1.6T.

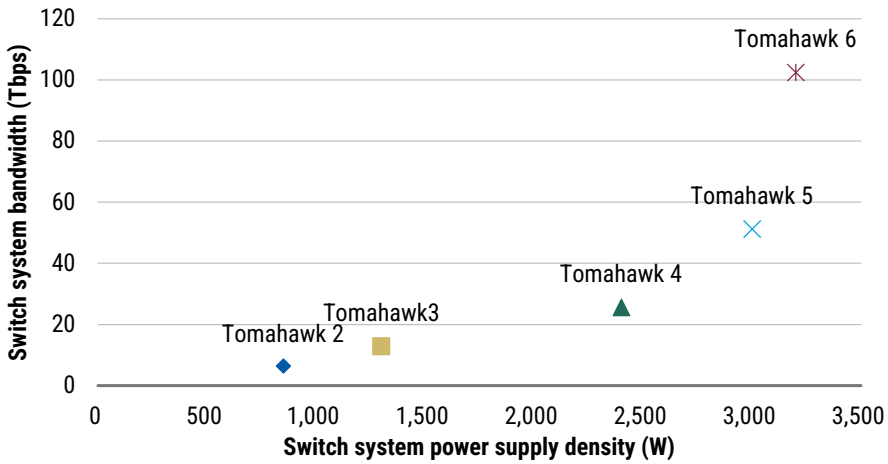
While process scaling significantly improved switching efficiency, it did not prevent ASIC power from rising. ASIC TDP increased from 150 W to 800 W, reflecting the growing complexity of higher radix switches and faster SerDes interfaces.

The impact of ASIC scaling is evident at the system level. As switch bandwidth increased from 100G-class systems to 800G and 1.6T platforms, power delivery requirements grew dramatically. Representative PSU ratings increased from 850W in Tomahawk 2 systems to 3,200W in Tomahawk 6 systems, nearly a 4× increase.

The rise in PSU power is driven not only by the higher ASIC TDP, but also by the growing power consumption of high-speed optics, increased cooling requirements, and larger overall system power budgets. As systems evolved from 100G to 1.6T ports, switch vendors were forced to add more fan redundancy, larger heatsinks, and in some cases larger chassis footprints. For example, Tomahawk 6 systems have expanded to 4U form factors with 7+1 fan redundancy to accommodate the thermal load associated with 102.4 Tbps switching capacity.

Alongside the increasing power consumption of the server racks, network switches are also witnessing higher power budgets as well.

Exhibit 32: Power supply density for different network switch generations



Source: Company data, Morgan Stanley Research

Power supply growing on both volume and specs

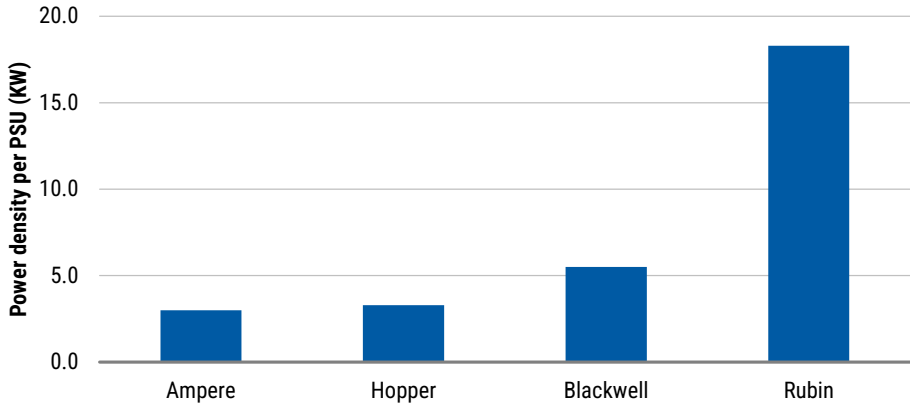
PSU spec upgrade trend

In response to the increasing power consumption of the AI servers, the power supply architecture has been migrating as well. in the A100 and H100/200 single system servers, 6 power supply units are placed directly inside the chassis, with each capable to deliver 3.0-3.3KW of power.

When Nvidia make a broad based move to the rack level server solution from the Blackwell generation, the power supply units were no longer placed directly within the chassis housing the HPCs that consume power; instead, power supply units are separated and grouped into power shelves. In the GB200/300 systems, there are 8 power shelves sitting on the same rack as the compute trays and switch trays, each enclosing six 5.5KW power supply units.

For the upcoming Vera Rubin rack, the power supply unit will be upgraded to 18.3KW of output to comprise a power shelf of 110KW, and each power shelf will take 3U of space on the rack vs. 1U in the Blackwell generation.

Exhibit 33: Power density for PSUs in Nvidia server systems



Source: Company data, Morgan Stanley Research

Challenges increasing the power density of PSUs

To come up with power supply unit with higher power density, manufacturers might need to solve the following potential hurdles:

Better power semiconductors (SiC and GaN)

The biggest improvement in PSU power density comes from replacing traditional silicon power chips with newer materials such as silicon carbide (SiC) and gallium nitride (GaN). These devices waste less energy during power conversion and can operate at much higher speeds. As a result, many supporting components inside the PSU—such as transformers and inductors—can be made significantly smaller. This allows PSU manufacturers to fit more output power into the same physical size while also improving efficiency. In many ways, SiC and GaN have become the foundation for the latest generation of high-density AI server power supplies.

More efficient power conversion designs

Modern PSUs use more sophisticated circuit designs, i.e., bridgeless PFC vs. conventional one, that reduce energy losses during power conversion. A conventional boost PFC includes a diode bridge and therefore incurs multiple semiconductor conduction drops. A bridgeless totem-pole PFC reduces the number of devices in the active current path and is now a preferred architecture for dense server PSUs. Instead of simply converting electricity from one voltage to another, newer architectures are designed to minimize wasted energy throughout the process. Higher efficiency means less heat is generated, which directly enables higher power density. This is particularly important in AI servers, where even small efficiency improvements can translate into meaningful reductions in heat generation and cooling requirements.

Smaller and smarter magnetic components

Transformers and inductors are often among the largest components inside a PSU. A major area of innovation is reducing their size without sacrificing performance. Manufacturers are increasingly using planar magnetics, where copper traces on circuit boards replace traditional wire windings. These designs are thinner, easier to cool, and more space-efficient. Some suppliers are also combining multiple magnetic functions into a single component, further reducing overall volume. As PSU power levels increase from 3 KW to 5.5 KW and beyond, shrinking magnetic components becomes one of the most important contributors to power-density improvement.

Improved thermal management

As power output increases, heat becomes the primary obstacle to higher density. Even highly efficient PSUs still generate a significant amount of heat. Therefore, manufacturers are investing heavily in better cooling solutions, including optimized airflow paths, advanced heatsinks, vapor chambers, and liquid-cooling technologies. The goal is to remove heat more effectively so that components can operate safely even when packed more tightly together. In many modern PSU designs, thermal engineering is just as important as electrical engineering in determining achievable power density.

Higher component integration

Another way to increase power density is to reduce the amount of unused space inside the PSU. Manufacturers are integrating more functions into fewer components and using three-dimensional layouts that make better use of available volume. Components are stacked, arranged vertically, or combined into integrated modules to maximize space utilization. This approach is similar to how smartphones have become more powerful without becoming larger—the same principle is increasingly being applied to server power supplies.

Higher efficiency creates more thermal headroom

Efficiency improvements provide a double benefit. First, less electricity is wasted. Second, less heat is generated. For example, a 5.5 kW PSU operating at 97.5% efficiency generates roughly 30% less heat than one operating at 96.5% efficiency. While a one-percentage-point improvement may appear small, it can remove tens of watts of heat from the system. This allows engineers to either increase output power within the same form factor or reduce cooling requirements while maintaining existing power levels.

Reducing the size of energy storage components

Server PSUs contain capacitors that store energy and maintain power delivery during short interruptions. Historically, these capacitors occupied a significant amount of space. Newer designs use smarter control techniques that allow manufacturers to achieve the same performance with smaller capacitor banks. Reducing capacitor volume frees up valuable internal space that can be used for higher power output or additional functionality.

For example, in Infineon's 3.3 KW example, the calculated capacitance for a conventional 10ms hold-up implementation was approximately 2.5 mF; allowing deeper discharge through a baby-boost stage reduced the requirement to about 900 µF. The 12 KW design similarly uses a bidirectional buck-boost buffer to achieve 20ms hold-up at full load without the capacitor volume that a passive solution would require. The penalty is additional semiconductors, an inductor, control software and failure modes, but the net volumetric benefit can be substantial.

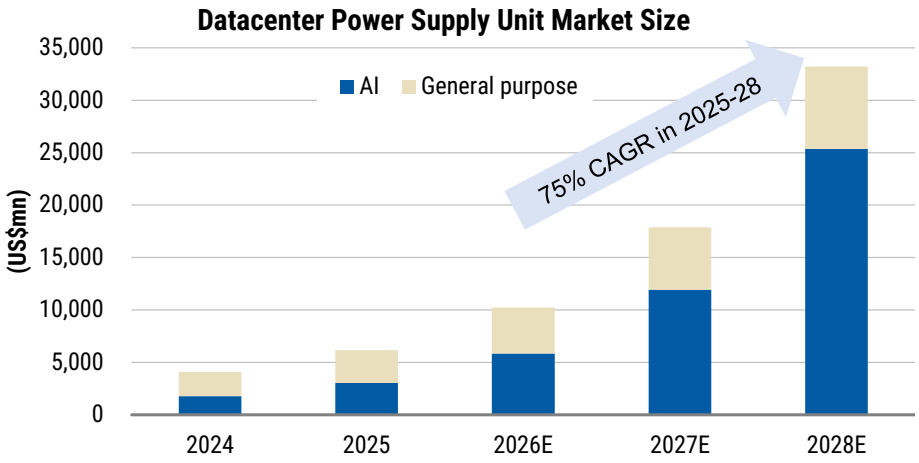
Advanced mechanical design

Power density is ultimately measured in watts per unit volume, making physical packaging extremely important. Manufacturers continuously optimize component placement, airflow channels, connector design, and enclosure structures to eliminate unused space. The best PSU designs are often not defined by a single breakthrough component but rather by hundreds of small mechanical optimizations that collectively improve volumetric efficiency.

Datacenter PSU market size to grow at a 75% CAGR in 2025-28

Overall, we estimate the global datacenter power supply market will grow at a three-year CAGR of 75% to reach US$33bn in 2028 from US$6bn in 2025, driven mainly by server shipment growth, increasing power consumption per rack, and the adoption of HVDC power racks from 2027.

Exhibit 34: Global datacenter power supply market forecast



Source: Company data, Morgan Stanley Research estimates

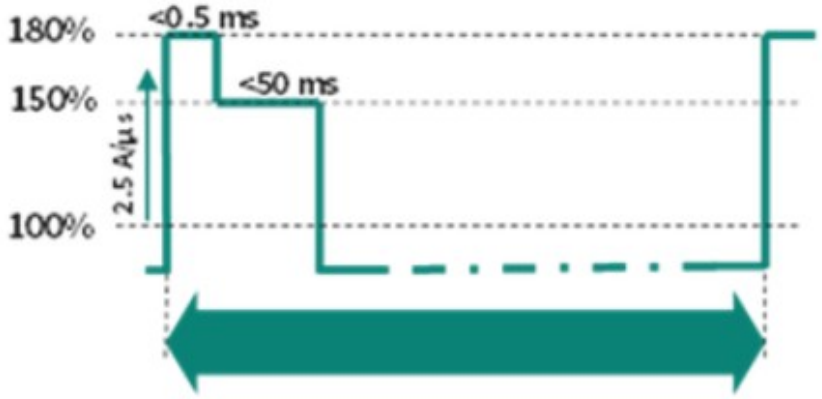
Complementary Hardware for More Resilient Power Delivery

Transient power spikes at AI servers

AI servers have a much more dynamic power profile than conventional enterprise servers because GPU utilization changes sharply throughout a workload. During large-model training, thousands of GPUs typically operate in a bulk-synchronous manner, and power consumption approaches the accelerator's design limit during tensor-compute phases, then falls during collective communications, check pointing, data movement, or periods in which GPUs wait for slower devices to synchronize. Inference can create a similar pattern as request arrivals, batching, context length and token-generation activity change rapidly.

Because the GPUs perform these transitions together, the power variation of one accelerator is multiplied across the server, rack and cluster rather than being averaged out by unrelated workloads, so the magnitude of the challenge is also rising with system density. Power spikes are therefore not simply the result of higher average consumption; they reflect the combination of higher accelerator power, more GPUs per system and increasingly synchronized workload behavior.

Exhibit 35: Transient power spikes at AI servers



Source: Infineon

System performance and reliability potentially compromised

The key issue is that power delivery systems are designed to supply a relatively stable amount of power, while AI workloads can change power demand almost instantaneously. When the load changes faster than the power-delivery system can react, a temporary mismatch occurs between power demanded by the GPUs and power supplied by the PSU, creating electrical instability throughout the server.

Sudden increases in GPU demand lead to voltage drops

When GPUs transition from a low-utilization state (communication, synchronization, checkpointing, etc.) to a full-compute state, their current demand can increase within microseconds. However, the PSU cannot instantly increase its output because its control loop requires time to detect the load change and adjust power conversion.

During this response period, the additional energy must come from capacitors located throughout the power chain. As these capacitors discharge, the voltage on the DC bus begins to fall, creating voltage drop.

If the drop becomes sufficiently large, several operational issues can occur:

- GPU throttling: The GPU reduces clock speed or performance to remain within safe operating limits.
- Computation errors: Voltage instability can affect timing margins in high-speed processors and memory.
- Unexpected resets or shutdowns: Protection circuits may trigger if voltage falls below specified thresholds.
- Training interruption: Large AI training jobs running across thousands of GPUs may fail if even a small number of nodes experience instability.

In a large AI cluster, a single server failure can force the entire training job to restart from the latest checkpoint, resulting in significant productivity loss.

Sudden decreases in GPU demand results in voltage spike

The opposite problem occurs when GPUs rapidly reduce power consumption.

For example, after a compute-intensive phase, GPUs may enter a communication phase where power consumption drops sharply. The PSU, however, is still delivering power based on the previous load condition. The excess energy has nowhere to go immediately and accumulates in the DC bus, causing a temporary voltage spike.

This can create:

- Overvoltage stress on components
- Protection-circuit activation
- Accelerated degradation of capacitors and voltage regulators
- Higher risk of component failure over time

Although individual spikes may be small, AI servers can experience these transitions continuously during training, causing repeated stress throughout the power-delivery chain.

Excessive current surges throughout the system

Power spikes are not only a voltage problem—they also create large current excursions.

A 1KW increase in power demand at 12V requires roughly 83A of additional current. In modern AI servers, where multiple GPUs can change utilization simultaneously, current swings can reach hundreds of amperes within a very short period.

These current surges lead to:

- Higher conductor losses
- Increased connector heating
- Greater stress on voltage-regulator modules (VRMs)
- Larger thermal fluctuations in power components
- As GPU power ratings rise to over 1KW, the magnitude of these transient currents becomes increasingly difficult to manage.

Thermal cycling and reliability degradation

Perhaps the most overlooked issue is the impact on long-term reliability.

Every power excursion creates a corresponding temperature excursion. When power rises, semiconductor junction temperatures increase; when power falls, temperatures decrease. This repeated heating and cooling cycle occurs thousands or even millions of times over a system's lifetime.

The result is accelerated wear on:

- Electrolytic capacitors
- Power MOSFETs
- Solder joints
- PCB interconnects
- Connectors and busbars

Consequently, even if average power remains within specification, transient power spikes can shorten component lifetime and increase failure rates.

Infrastructure overdesign and lower efficiency

To prevent instability, operators often provision the power infrastructure for the worst-case peak rather than average consumption.

For example:

- A rack averaging 80KW may need infrastructure sized for 120-140KW transient peaks.
- PSUs must be oversized to accommodate short-duration excursions.
- UPS systems must reserve capacity for sudden load increases.
- Cooling systems must handle peak thermal output.

This creates a significant amount of stranded capacity, where expensive electrical and cooling infrastructure remains underutilized most of the time. Furthermore, PSUs and UPS systems typically operate most efficiently near their optimal loading point; oversized systems therefore tend to operate at lower efficiency, increasing operating costs.

Conventional power delivery systems not designed to deal with power volatility

In the conventional power delivery architecture, the UPS protects the data hall against utility outages and power-quality disturbances, while the PSU converts upstream power into the tightly regulated DC output required by the server. Both remain essential in the modern AI server clusters, but neither was originally optimized to absorb large, repetitive and synchronized AI load steps economically.

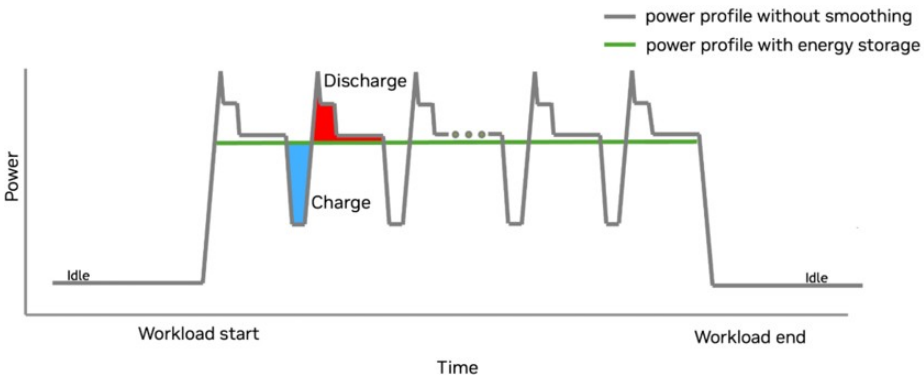
The PSU can use its internal capacitors to cover the initial transient, but capacitor volume is limited; once that stored energy is depleted, the PSU control loop must raise input current, potentially producing current overshoot and transferring the disturbance upstream.

The UPS is located farther from the accelerator and sees the aggregated load of many PSUs. A transient originating near the GPU can therefore occur on microsecond-to-millisecond timescales—including permitted GPU electrical-design-power excursions around 50 ms—before the complete upstream conversion chain can rebalance power.

This is a response time and location mismatch, rather than evidence that UPSs or PSUs are fundamentally incapable of supporting AI. New AI-tolerant UPS platforms are being designed for faster and larger load steps, but many legacy systems may still experience voltage excursions, nuisance alarms, lower partial-load efficiency or unwanted bypass transfers under steep synchronized loads.

To complement the conventional power delivery architecture, more AI server clusters are adopting BBUs or CBUs to shave peak loads and mitigate the above-mentioned issues.

Exhibit 36: Concept of peak shaving for AI server power loads



Source: Nvidia

Battery backup units seeing increasing importance

What's a BBU?

A BBU is an integrated battery, power-conversion and control system that supplies DC power to server loads when the normal power source is interrupted or temporarily insufficient. In an OCP-style rack, the BBU modules are housed in a dedicated shelf located near the PSU power shelf and connect directly to the common DC busbar. Under normal conditions, the PSUs power the servers and maintain the BBU's charge; when the bus

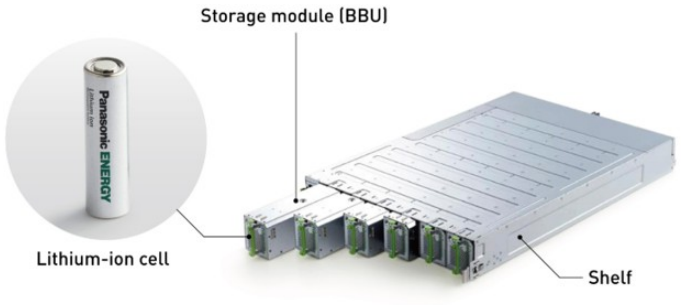
voltage falls below a programmed threshold, the BBU's bidirectional converter changes from charging or standby into discharge mode and supports the rack. Analog Devices' ORV3 reference design detects a bus voltage below 48.5V for more than 2ms and is required to ramp to full support within another 2ms, while maintaining the bus above the specified minimum voltage.

Key components within a BBU

- **Battery-cell pack:** The cells store the energy used during discharge. Lithium-ion cylindrical cells are commonly arranged in series and parallel to obtain the required voltage, power and runtime; the ORV3 reference module uses an interchangeable 11-series, 6-parallel battery pack. The cell must provide not only sufficient energy but also high discharge power, low internal resistance and predictable performance over its qualified life.
- **Bidirectional DC/DC converter:** The converter controls energy flow between the battery pack and the rack DC bus. It steps voltage in the appropriate direction during charging and discharging and regulates the output so that the battery's changing voltage does not destabilize the server bus. Multiphase designs divide the large current across several parallel power stages; TI's 12 KW design uses six 2 KW phases and reports peak efficiency above 99%, including above 99.5% when battery and bus voltages are close.
- **Battery management system:** The BMS measures individual cell voltages, current and temperature, balances cells and estimates state of charge and state of health. It also protects against overcharge, excessive discharge, overheating and short circuit. Accurate state-of-health information is important because the operator must know whether an apparently charged BBU can still deliver its required rack power and runtime after years of standby operation and periodic discharge events.
- **Power-path and protection circuitry:** Fuses, current sensors, disconnect switches, protection MOSFETs and hot-swap circuits isolate faults and prevent uncontrolled current flow. Typical protections include overvoltage, undervoltage, overcurrent and overtemperature shutdown. Hot-swap capability allows a failed or end-of-life module to be replaced without intentionally shutting down the entire rack, while active current sharing ensures that parallel BBU modules carry an appropriate portion of the load.
- **Control, monitoring and communications:** A module controller manages operating modes, converter control, fault handling, diagnostics and communication with the BMS and rack controller. Interfaces such as PMBus, Modbus, CAN, RS-485 or Ethernet can expose battery voltage, current, temperature, state of charge, state of health, fan speed and fault status to data-center management systems. Analog Devices' reference platform uses separate controllers for battery monitoring and overall BBU logic, communications and housekeeping.
- **Thermal-management system:** Fans, temperature sensors, airflow paths and mechanical spacing around cells remove heat during high-rate charging and discharging. Thermal management is both a performance and a safety requirement: excessive temperature accelerates battery ageing and can restrict available discharge power. Analog Devices' ORV3 design therefore incorporates fan control, airflow simulation and cell-to-cell spacing, while Murata's commercial 3 KW BBU includes internal forced-air cooling and overtemperature protection.

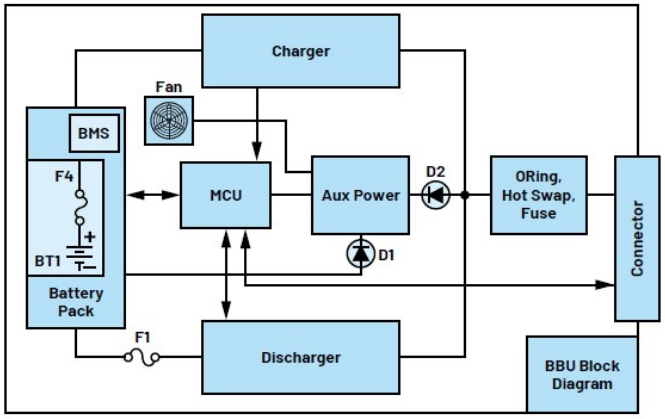
- **Shelf, backplane and mechanical integration:** The shelf distributes DC power and communications across several modules and provides retention, service access and cooling. Modularity allows power and runtime to be scaled by adding units and enables redundancy such as the ORV3 5+1 configuration. Murata specifies a 2U, hot-swappable 3 KW module, while Eaton's 19-inch BBU shelf accepts six 2.75 kW batteries and scales runtime according to the application requirement

Exhibit 37: BBU



Source: Panasonic

Exhibit 38: BBU diagram



Source: ADI

Key functions of BBUs

Backup power and outage ride-through

The BBU's fundamental function is to maintain rack operation during a brief loss of upstream power. The ride-through interval gives the data center time to restore the primary source, transfer to another source, preserve critical data, migrate workloads or perform an orderly shutdown. Rather than attempting long-duration operation, rack BBUs are generally optimized for high power over a relatively short period; the ORV3 example specifies 3KW per module for four minutes, and Murata guarantees a 240-second hold-up period at full-rated load under stated age and temperature conditions.

For AI systems, this protection is particularly valuable because a rack interruption can affect a synchronized workload extending well beyond the failed server. Local ride-through prevents a momentary upstream disturbance from immediately removing GPU, CPU, networking and storage power from the rack. Infineon consequently describes the server BBU as an integrated backup supply intended to prevent data loss and maintain AI workloads during outages or grid failures.

Peak shaving and load smoothing

A BBU can be operated as an active energy buffer. as it charges during lower-demand periods and discharges when server consumption temporarily exceeds a defined rack or PSU limit. This allows a relatively stable upstream power source to support a more volatile downstream AI workload. The economic benefit is that PSU, UPS and electrical capacity can be designed closer to sustained demand rather than every short theoretical peak, although the achievable benefit depends on the frequency, duration and predictability of those peaks.

The important limitation is battery cycling. A BBU designed primarily for emergency standby may not be optimized for continuous, repetitive peak shaving, so frequent cycling affects heat generation, state of charge, usable life and replacement schedules. Very short and highly repetitive transients may therefore be better supported by a CBU or hybrid battery-capacitor system, leaving the battery to cover longer events. Renesas correspondingly distinguishes distributed BBUs from hybrid systems that combine batteries with capacitors for fast transient response and longer-duration backup.

Power stabilization, redundancy and fault isolation

A rack BBU adds another independent source to the local power path. If a power shelf, upstream feed or AC source is interrupted, the BBU can sustain the rack while the fault is isolated or another source assumes the load. A modular shelf further improves resilience because one BBU failure need not remove the entire backup function; the remaining modules can continue operating within the available redundancy and power limits.

Distributed deployment also narrows the fault domain. A defective rack BBU affects a much smaller set of compute equipment than a failure within a centralized backup system serving a large room or data hall. The tradeoff is a much larger population of batteries, converters, controllers and fans that must be monitored, tested and eventually replaced. Digital state-of-health monitoring, fault reporting and hot-swappable construction are therefore central to the operating model rather than optional features.

BBUs vs. UPSs

While some of the functions that BBUs offer look similar to what traditional UPSs can do, we think BBUs and UPSs should be viewed as complementary rather than substitutive technologies, as they address different challenges within the datacenter power architecture.

A UPS is designed to provide facility- or row-level protection against utility outages, voltage sags, surges, and other power-quality disturbances, ensuring continuous AC power delivery to a broad range of IT equipment. In contrast, a BBU operates much closer to the AI workload, supplying DC power directly at the rack level and responding to localized power events that occur faster than the upstream power infrastructure can efficiently manage.

As AI racks become denser and more power intensive, the UPS continues to serve as the first line of defense against external power disruptions, while the BBU acts as a second layer of protection by stabilizing the rack DC bus, supporting transient load spikes, and providing short-duration ride-through capability. Together, they create a hierarchical power protection architecture in which the UPS ensures overall system availability and power quality, while the BBU enhances rack-level resilience and power-delivery performance. Rather than replacing the UPS, BBUs help reduce the burden on centralized backup systems, improve power utilization efficiency, and enable the reliable operation of next generation AI infrastructure with increasingly dynamic power profiles

Exhibit 39: Comparison of BBUs and UPSs

Dimension	BBU	UPS
Location	Inside the server rack, in a BBU shelf, or in an adjacent rack/sidecar close to the compute	Typically deployed at equipment room, row or facility level and protects multiple racks or other critical loads
Power path	Usually DC-coupled to the rack bus through a bidirectional DC/DC converter	Takes AC input and delivers AC output; the server PSU subsequently converts AC again for the server
Primary objective	Short rack-local ride-through, DC-bus support, modular redundancy and—when designed for it—transient or peak-load assistance	Broad power conditioning and uninterrupted AC supply during utility sags, surges, frequency disturbances and outages
Response	Monitors the local bus and can inject power within a few milliseconds; the cited ORV3 reference requires full support within 2ms after a qualifying 2ms event	An online double-conversion UPS already powers the load through its inverter and therefore has zero transfer time between utility and battery operation
Runtime	Generally optimized for short-duration, high-power support; OCP's reference example is four minutes at full load	Runtime varies widely with battery-bank sizing and system design; its broader function is to maintain protected AC loads until another source or operating response becomes available
Scalability	Added rack-by-rack or module-by-module; power and runtime can be matched to each compute deployment	Scaled centrally in larger KW blocks and requires capacity planning for the aggregate protected load
Fault domain	A module or shelf fault is normally localized to one rack or a limited group of racks	A centralized system protects a much broader load domain, so its redundancy and maintenance design are correspondingly critical

Source: Morgan Stanley Research

Capacitor backup unit (CBU) another complementary power device

What's a CBU?

A CBU is an integrated energy storage subsystem positioned close to the AI load, typically in a dedicated rack shelf, capacitor tray or adjacent power sidecar, and connected to the server rack's DC power path. Its principal storage medium may be an electric double-layer capacitor (EDLC or supercapacitor), lithium-ion capacitor (LIC), or a bank of high-capacitance electrolytic capacitors, depending on the required response time, energy duration, footprint and cost. Dedicated CBU products increasingly use EDLCs or LICs because their low internal resistance supports rapid, repetitive charge and discharge.

The CBU sits electrically alongside the PSU. Under steady-state conditions, the PSU supplies the AI rack and maintains the CBU at its target state of charge. When GPU demand temporarily exceeds the desired PSU output, the CBU discharges through a controlled power converter into the DC bus; when GPU demand falls, the converter reverses direction and recharges the capacitors at a controlled rate. This prevents the full GPU load step from passing through the PSU and upstream equipment. Infineon notes that a separate CBU must be compatible with, and potentially communicate with, the PSU shelves to coordinate rack power.

An alternative to deploying a standalone CBU shelf is to integrate the energy-storage capacitors directly into the power shelf, effectively combining power conversion and transient-energy buffering within the same subsystem. NVIDIA's GB300 NVL72 architecture is a good example: its enhanced power shelves incorporate a substantial capacitor bank alongside the PSU modules, with NVIDIA allocating approximately 65 joules of stored energy per GPU.

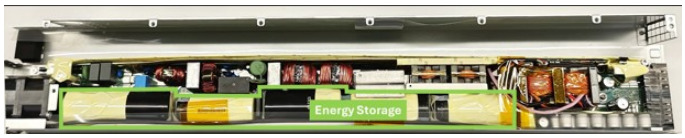
NVIDIA reported that this approach reduced peak upstream power demand by roughly 30% in a measured Megatron LLM workload. Compared with a separate CBU, integrating capacitors into the power shelf can save valuable rack space, shorten the electrical path between the energy buffer and the load, and simplify coordination between the PSU and

energy-storage system, since charging, discharging, and PSU output can be managed as one integrated power architecture. The trade-off is that capacitor capacity becomes more tightly coupled to the power-shelf design, potentially reducing flexibility in independently scaling or servicing the energy-storage system. Nevertheless, as AI racks become increasingly power dense, integrating transient-energy storage directly into the power shelf could become an attractive alternative to standalone CBUs, particularly where rack space and fast transient response are key design priorities.

Exhibit 40: CBU

Source: Panasonic

Exhibit 41: GB300 power shelf with integrated capacitors



Source: Nvidia

Key components within a CBU

- **Capacitor bank:** The storage section contains many capacitor cells arranged in series to reach the required voltage and in parallel to provide the required power and energy. Individual EDLC cells commonly operate at only a few volts, so high-voltage systems require long series strings; Abracon cites approximately 2.7 V as a typical supercapacitor cell rating, while TI's NVIDIA-oriented reference architecture demonstrates an 800 V CBU using EDLC cells. LICs can provide more stored energy than conventional EDLCs, while EDLCs generally emphasize lower resistance, high peak power and very high cycling endurance. Delta, for example, offers a power-capacitance shelf using fast-charge/discharge LICs, rated to provide 15 seconds of hold-up at 20 KW.
- **Cell-balancing system:** Because cell capacitance, leakage current and internal resistance vary across a series string, the voltage does not divide perfectly equally. A balancing circuit prevents an individual cell from exceeding its rated voltage during charging or being reverse-biased during deep discharge. Passive balancing uses resistors to bleed excess charge, while active approaches use MOSFETs, amplifiers or switched converters to redistribute energy more efficiently; the design tradeoff is between cost, balancing speed, component count and standby loss.
- **Bidirectional DC/DC converter:** This is the active interface between the capacitor bank and the rack DC bus. During a GPU power spike, it converts the capacitor's changing voltage into a regulated bus contribution; during lower demand, it reverses power flow to recharge the bank. The converter also allows more of the capacitors' stored energy to be used than would be possible with a direct connection, because capacitor voltage declines materially during discharge.

Infineon specifically identifies a capacitor bank plus bidirectional converter as the main CBU structure, while Skeleton Technologies notes that converter-based integration provides greater energy utilization and more precise power-flow control than a direct bus connection.

- **Controller and management software:** The controller determines when the CBU should charge, remain idle or discharge, based on bus voltage, PSU output, rack power limits and capacitor state. It may also calculate state of health, manage balancing, regulate recharge current and coordinate multiple CBU modules.
- **Protection and switching:** A high-power capacitor bank can release very large fault current, so fuses, contactors, pre-charge circuits and overvoltage, overcurrent and overtemperature protection remain essential. Pre-charge limits inrush current when the CBU is connected to the DC bus, while contactors or solid-state switches isolate the bank during a fault or service event. Skeleton's reference system integrates switchgear, fuse-fault detection, pre-charge/discharge contactors and protection against high-current and overheating events; industry discussions also highlight the need for faster DC fault isolation as CBU voltage rises.
- **Monitoring and communications:** Voltage, current, temperature, equivalent series resistance and state-of-health measurements allow operators to identify degradation before the CBU can no longer meet its specified peak-power requirement. Panasonic emphasizes voltage, temperature and ESR monitoring for predictive maintenance, while Skeleton provides single-cell voltage sensing, multiple temperature sensors and communication bus to its master controller.
- **Thermal and mechanical system:** Although supercapacitors are efficient, high repetitive current produces resistive heating, particularly in cells, busbars, converters and connectors. The shelf therefore requires designed airflow, temperature sensing and, at higher power levels, potentially liquid cooling. Skeleton's cabinet uses controlled fan cooling and also offers liquid-cooled configurations, while the Open Compute Project presentation from Panasonic identifies lower-resistance capacitors and liquid cooling as development priorities for future higher-output CBUs

Main functions of CBU

Transient power stabilization

One of the most fundamental roles of a CBU is to stabilize the rack DC power bus when GPU power consumption changes very rapidly. AI workloads frequently alternate between compute-intensive phases and lower-power communication, synchronization, or idle phases. Because many GPUs operate in parallel, these transitions can create very large changes in rack power within milliseconds.

When GPU demand suddenly increases, the PSU cannot instantaneously increase its output. The CBU therefore discharges stored energy into the DC bus during this short response gap, helping prevent voltage drop and excessive current excursions. Conversely, when GPU demand suddenly falls, the CBU can absorb excess energy while the PSU reduces its output, helping limit voltage overshoot.

The key objective here is power quality and system stability, rather than reducing the overall power capacity required by the rack. In other words, the CBU acts as a fast

electrical shock absorber between the highly dynamic GPU load and the slower PSU. This becomes increasingly important as AI racks move above 100 KW and synchronized GPU workloads create larger absolute power swings.

Peak shaving

Another major function is peak shaving, whose objective is to limit the maximum power that the PSU and upstream infrastructure need to supply.

The benefit is potentially significant because the PSU, connectors, protection devices, and portions of the upstream power infrastructure no longer need to be oversized solely for short-duration GPU peaks. It can also allow PSUs to operate closer to their optimal loading range, improving utilization and potentially efficiency.

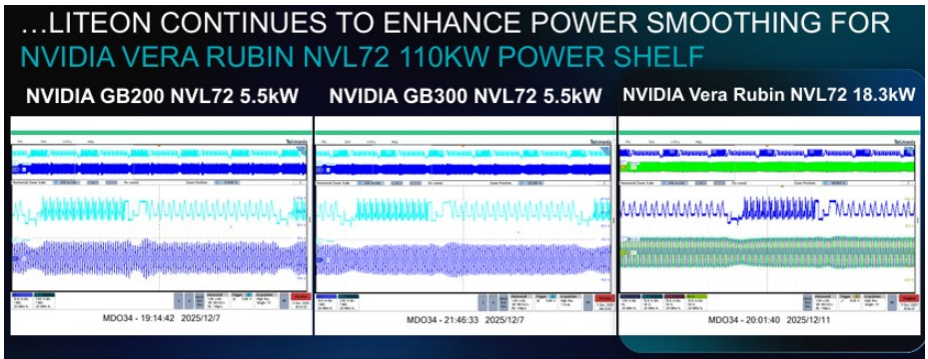
Short duration ride-through

The third function is ride-through, where the CBU temporarily powers the rack when the normal power source is interrupted altogether. This is fundamentally different from the first two functions because the issue is no longer a mismatch between GPU demand and PSU response—the upstream power source itself has temporarily disappeared.

A sufficiently sized CBU can maintain the DC bus during events such as a brief AC interruption, PSU switchover, source-transfer event, or temporary power-shelf failure. Depending on the amount and type of capacitance installed, this support can range from milliseconds to several seconds. For example, some LIC-based capacitance shelves are designed to provide multi-second hold-up at high rack power.

However, ride-through is generally a secondary function for a CBU. Capacitors have high power density but relatively low energy density, meaning they can deliver a large amount of power very quickly but cannot economically sustain that power for long periods. As the required backup duration increases from seconds toward minutes, a battery-based BBU generally becomes more suitable.

Exhibit 42: Capacitors help smooth power delivery



	Ipk-max	Ipk-min	Max/Min Ratio
NVIDIA GB200 NVL72	15.9A	8.3A	91.57%
NVIDIA GB300 NVL72	13.3A	8.6A	54.65%
NVIDIA Vera Rubin NVL72	**12.8A**	**10.3A**	**24.27%**

Source: Lite On

CBUs vs. BBUs

The two technologies should consequently be regarded as complementary rather than directly substitutive. A CBU handles the high-power, short-duration and high-cycle portion of the load profile, while a BBU supplies the larger amount of energy required when the upstream source remains unavailable. Some describe BBUs and CBUs as coexisting in the same power sidecar, and hybrid architectures can allocate fast transients to capacitors while preserving battery charge and cycle life for longer outages.

The CBU's major advantages are rapid bidirectional response, long cycle life, high efficiency and the ability to absorb repeated GPU peaks without the same electrochemical wear mechanism as a conventional battery. It can also recharge quickly between workload cycles and may require less routine replacement. However, the CBU has materially lower energy density, its voltage falls continuously during discharge, and large banks require balancing, high-current protection, cooling and valuable rack or sidecar space.

Exhibit 43: Comparison of CBUs and BBUs

Dimension	CBU	BBU
Primary storage medium	EDLC supercapacitors, LICs or high-capacitance	Typically lithium-ion battery cells managed by a BMS
Primary strength	High power, fast charge/discharge and frequent	Higher stored energy and longer sustained
Main function	Transient smoothing, peak shaving, voltage	Outage backup, source-transfer bridging and orderly workload
Response profile	Well suited to millisecond-class load support and	Fast switchover, but generally optimized for sustained support rather
Typical duration	Milliseconds to seconds; selected systems can extend toward tens of seconds or more	Commonly tens of seconds to several minutes, depending on battery and rack design
Power versus energy density	Higher power capability but lower energy storage per unit volume than a battery	Higher energy density, making it more economical for longer events
Cycle life	Eaton specifies more than one million cycles and up to 20 years for certain supercapacitor modules	Battery life is more sensitive to depth of discharge, temperature and repeated cycling; BMS supervision is therefore essential.
State management	Requires cell balancing, voltage/temperature monitoring and state-of-health tracking, but does not use a conventional battery fuel	Requires BMS functions including charge control, state of charge, state of health and cell protection
Best-fit use case	Repetitive AI power spikes, inrush events, PSU load leveling and bus stabilization	Power continuity through outages, brownouts or generator/source-transfer intervals

Source: Morgan Stanley Research

HVDC Emerging, with SST in the Pipeline

HVDC coming along with the increasing power consumption of AI clusters

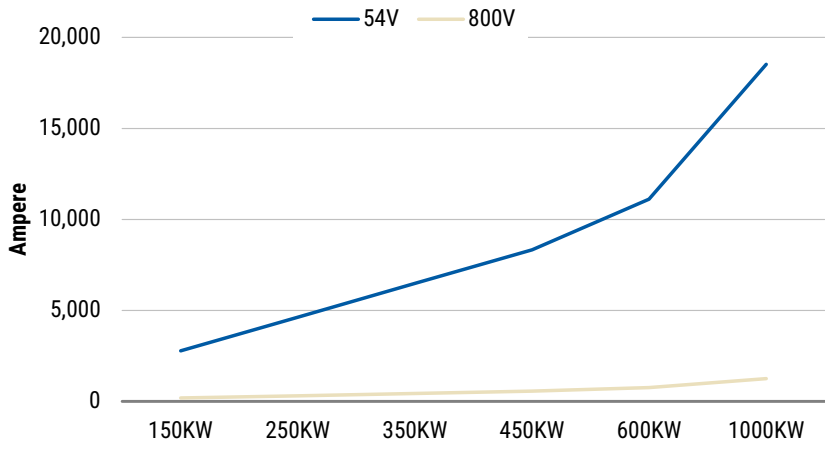
Why do we need HVDC?

Low voltage turns higher rack power into an extreme-current problem

The fundamental relationship is that at a given rack load, increasing the bulk-distribution voltage from 54 V to 800 V reduces current by approximately 14.8 times. A balanced ±400 V architecture also presents 800 V between its +400 V and –400 V conductors, so its bulk-transfer current is broadly equivalent to a 0–800 V architecture.

This current differential has an outsized mechanical consequence. For the same conductor resistance, I^2R loss at 54V would theoretically be approximately 219 times that at 800 V, though real systems avoid this level of loss by adding copper, parallel conductors and cooling. The result is not a 219-fold difference in actual operating loss, but a trade-off in which low-voltage designs consume increasing amounts of busbar cross-section, connector area, rack volume and cooling capacity to remain within voltage-drop and temperature limits.

Exhibit 44: Transmission current for different power scales: 54V vs 800V



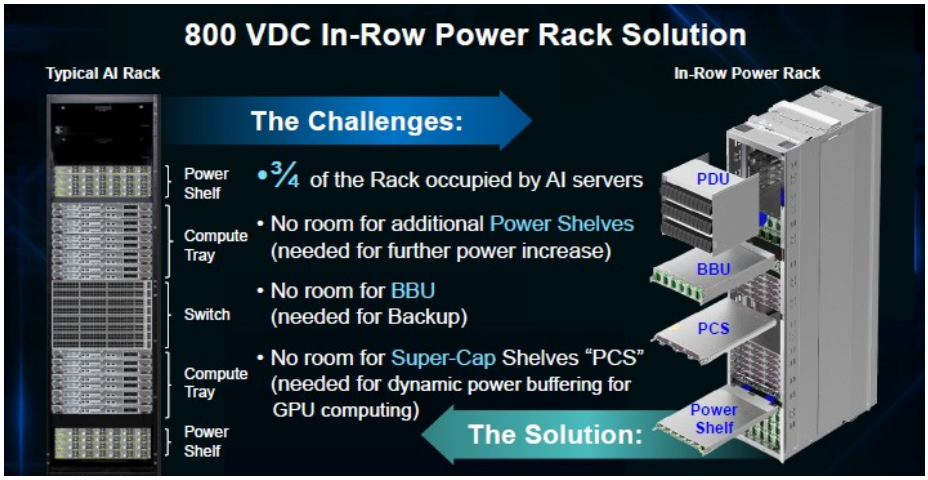
Source: Delta

Copper, connectors and thermal management become system constraints

At several thousand amperes, distribution cannot rely on a single ordinary conductor or connector. Designers must use broad laminated busbars, multiple parallel contacts, tightly controlled joint resistance and temperature monitoring. A small increase in contact resistance produces disproportionately high heat at these currents, raising the risk of hot spots, connector ageing and nuisance protection events. OCP's 100KW HVDC-to-LVDC shelf illustrates where the current problem moves: only two 62.5 A HVDC inputs are required on the ±400V side, while the 49–50V output approaches 2,000A and therefore uses a substantial bolted output connection with dedicated thermistors.

The same issue appears in power-shelf packaging. Delta notes that a typical AI rack may already allocate roughly three-quarters of its volume to servers, leaving insufficient room for additional power shelves, BBUs or supercapacitor shelves as power rises. Moving AC/DC conversion, protection and storage into an adjacent power rack therefore releases compute space and separates the thermal design of power electronics from that of GPUs. Microsoft's Mt Diablo rationale similarly states that disaggregated power can enable up to 35% more accelerators in the IT rack and allows power shelf count to be sized independently for each workload configuration

Exhibit 45: Not enough room for power devices on the traditional AI server rack



Source: Delta

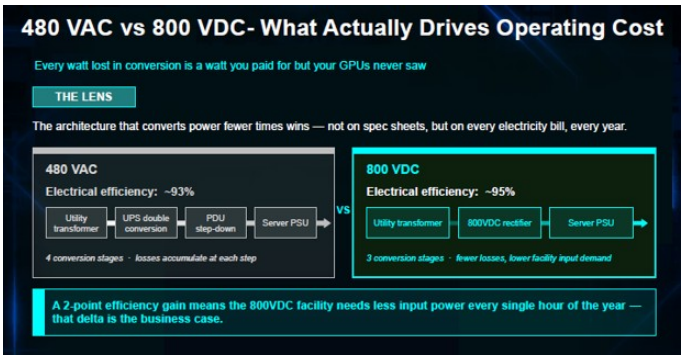
HVDC enables denser, fewer and more efficient power stages

As AI rack power scales from today's GB200/GB300 systems to Vera Rubin and future multi-hundred-kilowatt racks, power distribution efficiency becomes increasingly important. Our checks indicate that a conventional 480VAC architecture delivers an end-to-end electrical efficiency of roughly 93%, while an 800VDC architecture improves efficiency to about 95% by eliminating one power conversion stage and simplifying the power path from grid to rack.

The higher-voltage 800VDC distribution also reduces current levels for the same power load, lowering resistive (I^2R) losses in cables and distribution equipment and enabling more efficient delivery to increasingly power-dense AI racks.

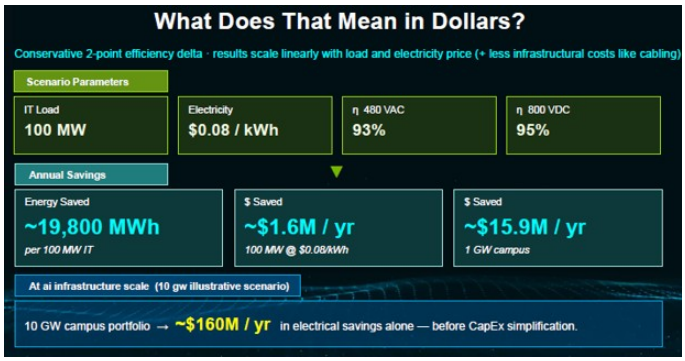
While a 2ppts of efficiency improvement may appear modest, the savings become substantial at hyperscale AI data center deployments. Delta's example shows that for a 100MW IT load, improving facility efficiency from 93% to 95% reduces annual energy consumption by approximately 19,800MWh, translating into roughly US$1.6 million of annual electricity cost savings assuming a power price of US$0.08/kWh. At a 1GW AI campus, the savings scale to nearly US$16 million per year, and at emerging multi-gigawatt AI factory deployments, the annual benefit can exceed US$100 million. Consequently, the transition from conventional AC distribution and 54VDC rack power architectures toward 800VDC distribution is not only a technical necessity for supporting future 500kW-to-1MW AI racks, but also a compelling economic proposition that lowers operating expenses and improves total cost of ownership through reduced energy losses and simplified electrical infrastructure.

Exhibit 46: Better power efficiency for 800VDC architecture



Source: Delta

Exhibit 47: Better efficiency leads to cost savings



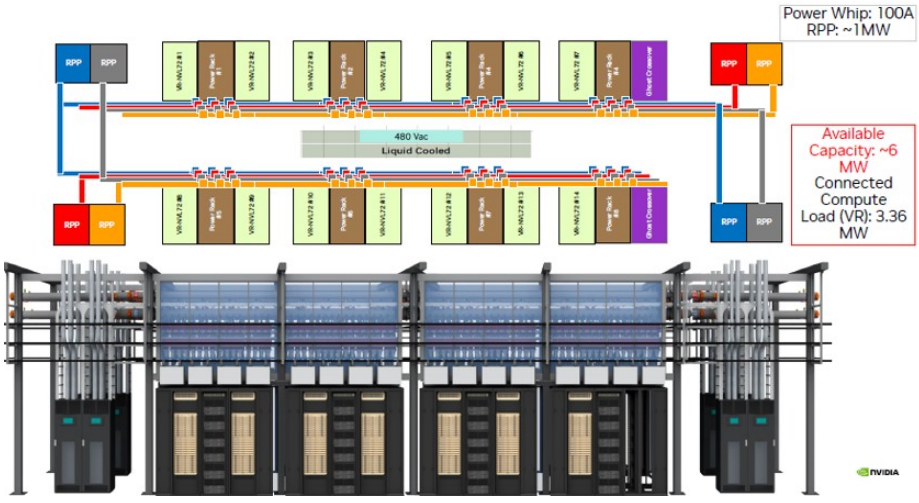
Source: Delta

What's the deployment roadmap for HVDC?

The deployment roadmap for 800VDC is best understood as a multi-stage evolution rather than a single-step transition. Both Nvidia and major hyperscalers recognize that existing 480VAC infrastructure cannot efficiently support the rapid increase in rack power. As rack densities increase, conventional power whips, busbars, connectors, and PSU architectures increasingly become bottlenecks, driving the need for higher voltage distribution

The first phase of deployment focuses on introducing 800VDC power racks within existing AC-based data centers. In this architecture, the facility remains largely unchanged, but a dedicated power rack converts 415-480VAC into 800VDC before distributing power locally to AI racks. This approach minimizes disruption to existing infrastructure while allowing operators to deploy Vera Rubin-class systems that exceed the practical limits of traditional power shelves. Nvidia refers to this as the row-level power rack architecture, which serves as the industry's bridge solution between legacy AC facilities and future native DC environments.

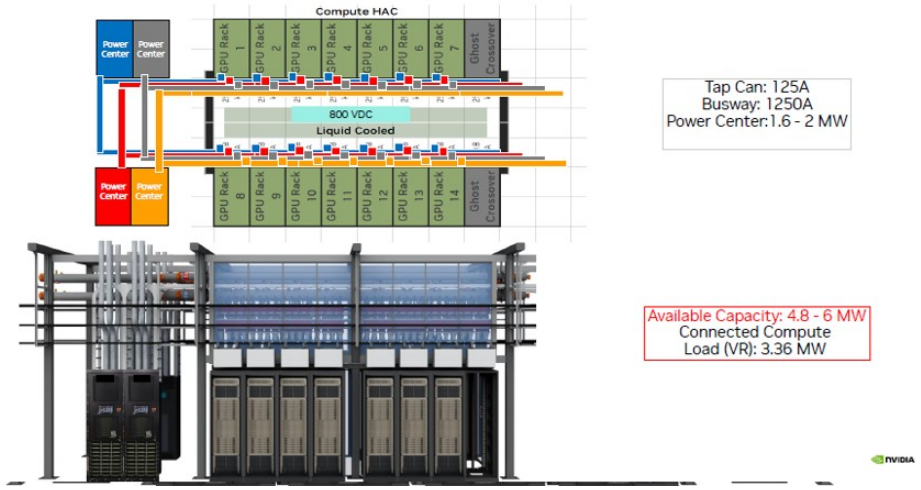
Exhibit 48: Phase I - 800VDC power rack



Source: Nvidia

The second phase expands 800VDC beyond individual rows into data hall-level power centers. Instead of deploying a dedicated power rack for every few compute cabinets, centralized rectifier or power-center systems distribute 800VDC through DC busways across an entire hall. Nvidia's roadmap identifies 1.6-2MW power centers and 1250A DC busway architectures as the next step, improving power density while reducing equipment duplication. This stage also aligns well with the deployment strategies being discussed by hyperscalers and the OCP community because it preserves existing medium-voltage and low-voltage infrastructure while moving a significant portion of distribution into DC

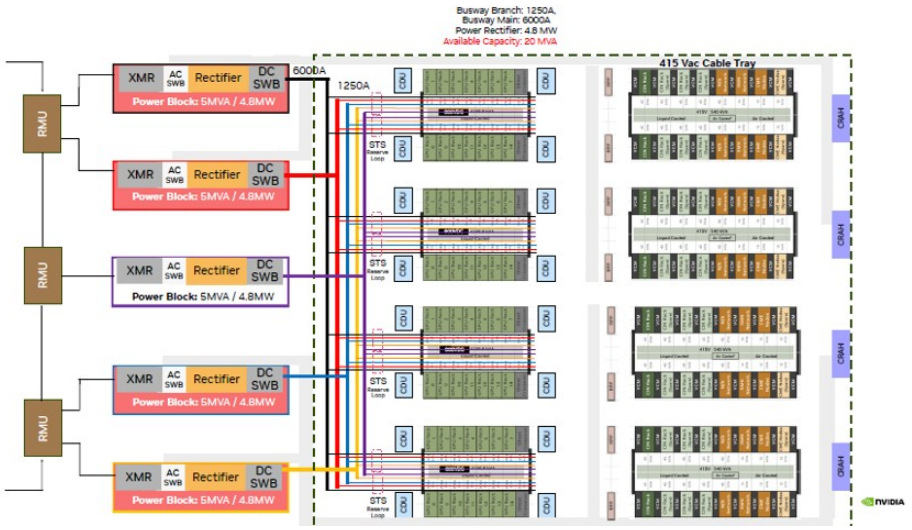
Exhibit 49: Phase II – 800VDC power center



Source: Nvidia

The third phase is the transition toward facility-wide 800VDC distribution, where AC is converted to DC near the utility entrance or medium-voltage substation and distributed throughout the data center via high-capacity DC switchboards and busways. Nvidia describes this as the DC power block architecture, utilizing approximately 4.8MW power blocks, 6000A DC busways, and integrated redundancy schemes. At this stage, the architecture begins to resemble an "AI factory," where power distribution is simplified, energy storage can be integrated directly onto the DC bus, and multiple megawatts of compute capacity can be deployed as modular building blocks

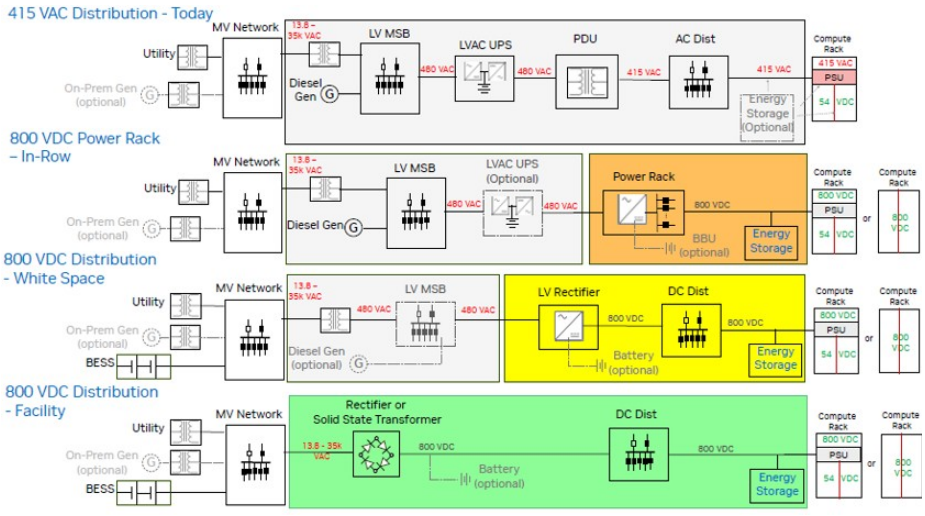
Exhibit 50: Phase III – power block



Source: Nvidia

The final stage is the emergence of native 800VDC compute racks, which Nvidia identifies as the long-term destination. Today, even in 800VDC deployments, power is generally stepped down to 54V before entering the servers. Future systems, however, are expected to accept 800VDC directly within the rack and potentially at the server level, eliminating another conversion stage and further reducing power losses. Nvidia's public roadmap shows a progression from current 54V architectures, to 800VDC-fed PSUs, and ultimately to 1MW-class native 800VDC racks capable of supporting next-generation AI accelerators

Exhibit 51: 800VDC native server racks



Source: Nvidia

In sum, deployment is likely to occur gradually because the ecosystem remains under development. NVIDIA notes that several enabling technologies—including DC protection devices, busways, connectors, grounding standards, arc-flash modeling, and solid-state transformers— are still undergoing certification and standardization. The roadmap presented by NVIDIA suggests a phased commercialization timeline, with 800VDC power racks reaching deployment readiness in 2026, hall-level power centers in 2027, and facility-scale DC power blocks becoming deployment-ready around 2028. Industry

collaboration among NVIDIA, Microsoft, Google, and OCP participants is focused on ensuring interoperability and regulatory acceptance before broader hyperscale adoption.

Exhibit 52: HVDC deployment roadmap

Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

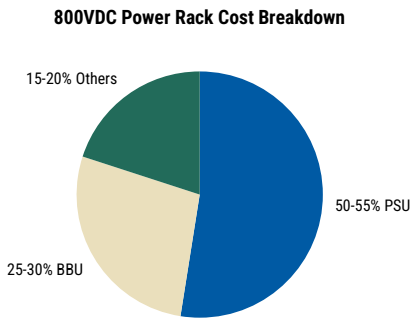
Source: Nvidia

800VDC represents 4-5x dollar content vs. in-rack power supply architecture

The 800VDC power rack BOM is primarily driven by the PSU, which accounts for 50–55% of the total cost, followed by the BBU at 25–30%, with other components comprising the remaining 15–20%. This cost structure reflects the broader functionality integrated into an HVDC power rack, combining power conversion, backup power and supporting components within a centralized architecture.

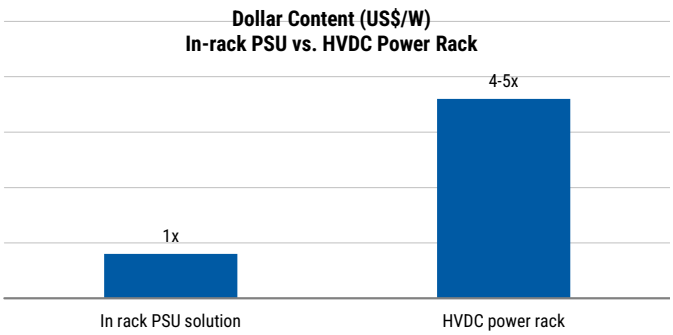
For power supply players, one of the most implications is that an HVDC power rack carries around 4–5x higher dollar content per watt compared with a conventional in-rack PSU solution, since the former integrates more components into a robust system and power supply players usually ship the entire rack instead of just the components.

Exhibit 53: BOM cost breakdown for an 800VDC power rack



Source: Morgan Stanley Research

Exhibit 54: Higher dollar content for HVDC power racks



Source: Morgan Stanley Research

800VDC vs. ±400VDC?

The debate between ±400VDC and 800VDC centers on balancing near-term practicality against long-term scalability. While both architectures deliver an identical 800V differential voltage and therefore achieve similar reductions in current, copper loss, and cable size versus traditional 54V rack power systems, they differ significantly in implementation.

Under a ±400VDC architecture, each conductor operates at only 400V relative to ground, resulting in less stringent insulation, creepage, clearance, and protection requirements. This makes the architecture easier to deploy, maintain, and certify, which explains why many hyperscalers and OCP members view ±400VDC as an attractive migration path toward high-density AI racks.

By contrast, NVIDIA and several power infrastructure vendors are increasingly advocating for a single-bus 800VDC architecture because it simplifies overall power distribution, reduces conversion stages, and is better suited for future 500kW to 1MW rack deployments.

We believe that ±400VDC appears well positioned as a near-term bridge architecture due to its lower deployment risk and more mature ecosystem, whereas 800VDC is increasingly viewed as the long-term end-state architecture for AI factories as rack power scales toward the megawatt era.

Exhibit 55: Comparison of 800VDC and ±400VDC

Metric	±400VDC	800VDC
Voltage to ground	400V relative to ground	800V relative to ground
Voltage differential	800V	800V
Power distribution efficiency	High	Slightly higher
Scalability	Good	Excellent
Insulation requirements	Lower	Higher
Maintenance safety	Better	More challenging
System architecture simplicity	Dual-rail system	Single-bus system
Grounding Complexity	More complicated	Simpler
Electrical footprint	Small	Smaller
Ability to reduce conversion stages	Moderate	Highest
Long-distance distribution capability	Excellent	Excellent
Maturity of ecosystem	Better today	Early but rapidly developing
Industry players	OCP, hyperscalers	Nvidia
Long term industry direction	Transitional architecture	Likely end-state architecture

Source: Morgan Stanley Research

From a component supplier perspective, the transition toward HVDC is far more important than which architecture ultimately wins, as most hardware content increases under either scenario. PSU vendors, BBU suppliers, busbar manufacturers, connector providers, power-whip suppliers, magnetics vendors, and rack-level DC-DC converter manufacturers are largely architecture-agnostic, since both ±400VDC and 800VDC require a similar shift toward high-voltage power distribution, centralized power shelves, and rack-level conversion.

Our understanding is that the primary exception is the power semiconductor industry, where the choice of architecture could materially influence device content and technology adoption. A ±400VDC system can often be supported using existing 650V-900V MOSFET and SiC platforms, benefiting suppliers with exposure to lower-voltage power devices. By contrast, 800VDC architectures increasingly require 1200V-class SiC MOSFETs, higher-

voltage gate drivers, hot-swap controllers, isolation components, and advanced protection circuitry, driving greater semiconductor content per watt and raising technology barriers.

Solid state transformer potentially being the holy grail in AI datacenter power delivery

What is a solid-state transformer ?

A datacenter solid state transformer (SST) integrates power electronics to replace or augment traditional line-frequency transformers, and it's typically built around three major functional stages. The first is the medium-voltage AC-to-DC rectifier stage, which uses high-voltage semiconductor switches—often silicon carbide (SiC) or gallium nitride (GaN) devices due to their superior switching speed and thermal performance—to convert incoming grid-level medium voltage AC into a regulated DC bus. The second is the isolated DC-DC conversion stage, usually implemented as a dual active bridge (DAB) or LLC resonant converter, which provides galvanic isolation through a high-frequency transformer (operating in the kHz to tens-of-kHz range rather than 50/60 Hz), dramatically shrinking the magnetic core and transformer size compared to conventional units while stepping voltage down toward datacenter distribution levels. The third is the low-voltage DC-AC or DC-DC output stage, which conditions power into the form needed by the facility—either regulated DC for direct DC-bus distribution to IT racks or inverted AC for legacy PDU/UPS infrastructure. Supporting these power stages are the control and sensing subsystems (digital controllers, gate drivers, voltage/current sensors) that manage switching, fault protection, and grid synchronization; thermal management components (heat sinks, cold plates, or liquid cooling loops) to handle the concentrated losses in the semiconductor devices; and increasingly, integrated energy storage interfaces or DC microgrid ports that allow the SST to natively support battery buffering, renewable integration, and finer-grained power quality control than a passive transformer ever could

A Solid-State Transformer (SST) is a power-electronics-based transformer that replaces the traditional 50/60Hz iron-core transformer with a combination of power semiconductors, control electronics, and a medium-frequency transformer. Unlike conventional transformers that perform only voltage conversion, SSTs actively process power through multiple stages, enabling both AC and DC inputs/outputs, bidirectional power flow, voltage regulation, reactive power compensation, and intelligent grid control

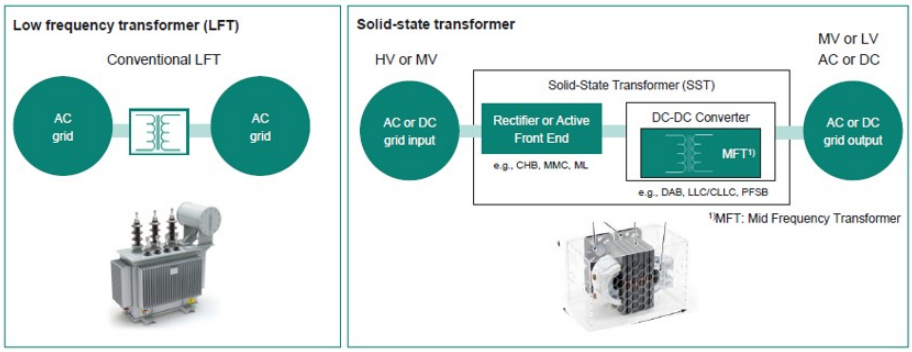
A SST is typically built around three major functional stages:

- The first is the medium-voltage AC-to-DC rectifier stage, which uses high-voltage semiconductor switches—often silicon carbide (SiC) or gallium nitride (GaN) devices due to their superior switching speed and thermal performance—to convert incoming grid-level medium voltage AC into a regulated DC bus.
- The second is the isolated DC-DC conversion stage, usually implemented as a dual active bridge (DAB) or LLC resonant converter, which provides galvanic isolation through a high-frequency transformer (operating in the KHz to tens-of-LHz range rather than 50/60 Hz), dramatically shrinking the magnetic core and transformer size compared to conventional units while stepping voltage down toward datacenter distribution levels.
- The third is the low-voltage DC-AC or DC-DC output stage, which conditions

power into the form needed by the facility—either regulated DC for direct DC-bus distribution to IT racks or inverted AC for legacy PDU/UPS infrastructure.

Supporting these power stages are the control and sensing subsystems (digital controllers, gate drivers, voltage/current sensors) that manage switching, fault protection, and grid synchronization; thermal management components (heat sinks, cold plates, or liquid cooling loops) to handle the concentrated losses in the semiconductor devices; and increasingly, integrated energy storage interfaces or DC microgrid ports that allow the SST to natively support battery buffering, renewable integration, and finer-grained power quality control than a passive transformer ever could.

Exhibit 56: Solid state vs. traditional transformers



Source: Infineon

Why do we need SSTs?

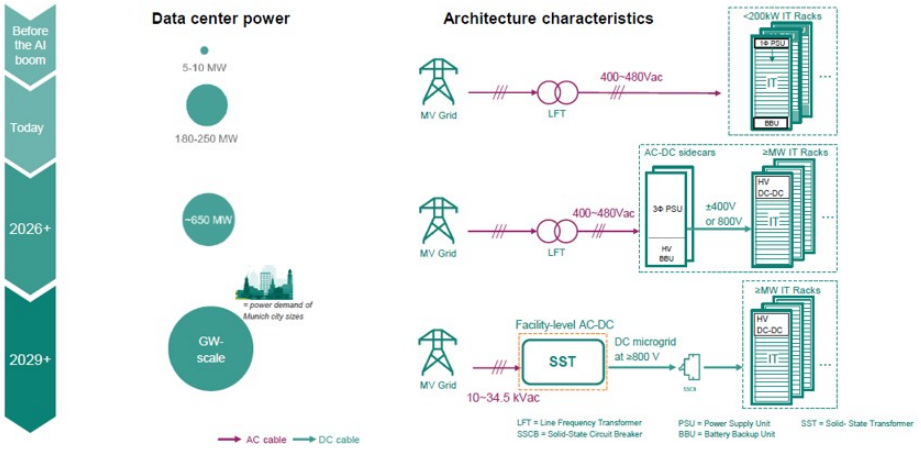
Improving efficiency in an era of gigawatt data centers

One of the strongest drivers for SST adoption is the rapid increase in AI data center power demand, which has progressed from 5-10MW facilities in the pre-AI era to campuses exceeding 650MW today, with gigawatt-scale facilities expected beyond 2029.

Traditional architectures use medium-voltage AC distribution, followed by low-frequency transformers and multiple power conversion stages before power reaches IT racks. SSTs allow facilities to convert 10-34.5KV medium-voltage power directly into ±400V or 800V DC distribution networks. This reduces conversion losses and improves overall grid-to-chip efficiency.

The efficiency advantage could be at the order of about 2-5ppt vs, the traditional transformer architecture, but at a 1GW AI campus, this magnitude of improvement could translate into roughly 20-50MW of continuous power savings. Over the lifetime of a facility, this can result in substantial reductions in electricity cost and carbon emissions. SSTs are therefore becoming increasingly attractive for hyperscaler operators seeking lower total cost of ownership (TCO).

Exhibit 57: SST enabling more efficient power delivery architecture



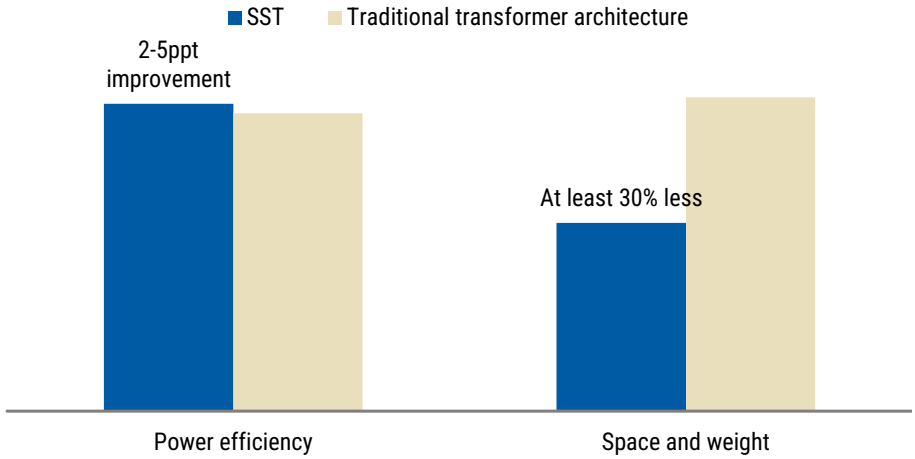
Source: Infineon

Higher power density and smaller footprint

Because SSTs use medium-frequency transformers instead of bulky 50/60Hz transformers, they can achieve significantly higher power density. Infineon indicates that SST systems can reduce both footprint and weight by at least 30% compared with conventional transformer-based installations. They also accelerate site energization and facilitate modular deployment

This becomes particularly valuable in urban data centers, utility substations, EV charging hubs, and energy storage sites where space constraints directly affect project economics.

Exhibit 58: SST offers better power efficiency with at least 30% less space/weight



Source: Infineon, Delta Electronics, Morgan Stanley Research

Supporting distributed energy resources

As renewable energy penetration increases, future grids will require seamless integration of solar generation, battery storage, fuel cells, gas turbines, and other distributed energy resources.

SSTs provide intelligent power routing, bidirectional operation, and dynamic voltage control. This allows them to function as energy hubs that connect renewable generation,

storage, and loads through a common DC infrastructure. Such capabilities are difficult or impossible to achieve using passive transformers alone.

Improved grid intelligence

Unlike conventional transformers, SSTs can actively regulate voltage and frequency, provide reactive power compensation, support grid-forming functionality, and improve overall power quality. These capabilities make SSTs an important enabler for future smart grids, microgrids, and virtual power plants.

What's the deployment status now?

Despite decades of research, SSTs remain at an early commercial deployment stage. While we started to see the SST models being unveiled by relevant suppliers and there start to be some pilot adoption, we believe that the 800VDC power rack/center/block architecture shall be deployed first, and SSTs will take more time, as more fundamental design change in the datacenter and all hardware inside will involved. We believe wider adoption probably won't happen before 2030.

Exhibit 59: SST from Delta

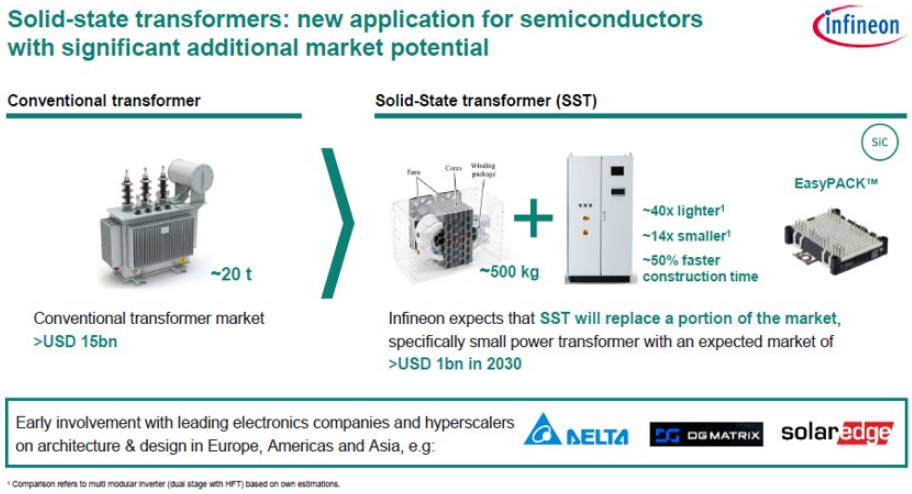
Source: Delta

Exhibit 60: SST from Eaton

Source: Eaton

According to Infineon, SST could represent a >US$1bn market opportunity by 2030, as the technology begins to replace a portion of the >US$15bn conventional transformer market, particularly in smaller-power applications. While SST penetration is expected to remain relatively modest versus the overall transformer market, the opportunity is significant for relevant players in the longer term.

Exhibit 61: SST to see initial adoption in 2030



Source: Infineon

Key hurdles for the wider adoption of SSTs

- **High system cost remains the primary barrier:** Despite their technical advantages, SSTs remain significantly more expensive than conventional transformers. Traditional transformers consist primarily of steel cores, copper windings, insulation materials, and cooling systems, all of which benefit from mature manufacturing processes and economies of scale. In contrast, SSTs require large numbers of high-voltage power semiconductors, gate drivers, digital controllers, sensors, protection systems, and sophisticated thermal management solutions. The extensive use of silicon carbide (SiC) devices, which remain substantially more expensive than conventional silicon-based components, further increases the overall system cost. Although SSTs can deliver efficiency gains, improved controllability, and reduced footprint, the economic benefit is often insufficient to offset the substantial upfront capital expenditure in many utility applications. Consequently, SST deployment is currently limited to applications where efficiency, power density, or system flexibility carry exceptionally high economic value, such as AI data centers, megawatt EV charging stations, and grid-scale battery storage systems.
- **Long term reliability has yet to be proven:** Electric utilities and large infrastructure operators place an extremely high value on reliability because transformers are expected to operate continuously for several decades with minimal maintenance. Conventional transformers routinely achieve service lifetimes exceeding 30 to 40 years and have accumulated more than a century of operating history. SSTs, however, are fundamentally different devices that incorporate thousands of semiconductor junctions, complex control electronics, communication networks, cooling systems, and software algorithms. While each individual technology may be mature, the long-term reliability of the complete SST system remains largely unproven at scale. Operators therefore remain cautious about replacing a passive asset with an active power-electronic system whose failure modes are more complex and less understood. For critical infrastructure such as substations, industrial facilities, or hyperscale data centers, reliability concerns often outweigh potential efficiency improvements.

- **DC protection technology ss still evolving:** One of the most significant technical challenges associated with SST deployment is fault protection, particularly in medium-voltage and high-voltage DC networks. Unlike AC systems, where current naturally crosses zero multiple times each cycle and facilitates fault interruption, DC currents continue flowing continuously during fault conditions. As a result, fault currents can rise extremely quickly, making them more difficult to detect and interrupt safely. Future SST-based power architectures therefore require advanced protection devices such as solid-state circuit breakers (SSCBs), which can react far faster than conventional mechanical breakers. However, DC protection technologies are still evolving, and there is currently no universally accepted approach equivalent to the mature protection schemes available in AC grids. The absence of standardized protection architectures increases deployment risk and remains one of the key obstacles to the wider adoption of SST-based DC distribution systems.
- **System complexity creates engineering, qualification and maintenance challenges:** Compared with traditional transformers, SSTs are considerably more complex systems that require the integration of multiple high-performance subsystems. A complete SST includes medium-voltage power converters, isolated DC/DC stages, high-frequency transformers, semiconductor modules, digital controllers, communication interfaces, and advanced protection mechanisms. As voltage and power levels increase, the number of converter modules and semiconductor devices required can rise significantly. Although the introduction of higher-voltage SiC devices can reduce subsystem count and simplify design, SST architectures remain substantially more complicated than conventional transformers. This complexity increases development costs, certification requirements, maintenance needs, and operational risks, all of which slow adoption among traditionally conservative utility customers.
- **Industry standards and ecosystems are still developing:** The transformer industry benefits from decades of established standards, testing procedures, maintenance practices, and supply chain infrastructure. SSTs, by comparison, represent an emerging technology with limited deployment history and evolving standards. Many utilities remain uncertain regarding qualification procedures, maintenance requirements, spare-parts strategies, and long-term asset management. Furthermore, widespread SST adoption will require parallel development of supporting technologies such as DC protection systems, DC switchgear, energy management software, and medium-voltage DC distribution networks. Until the broader ecosystem matures, many customers will continue to view SSTs as a promising but relatively high-risk alternative to proven transformer technologies.

Fuel Cells for Better Flexibility and Resilience

What are fuel cells?

Fuel cells are electrochemical energy-conversion devices that generate electricity by combining hydrogen and oxygen without combustion. Unlike conventional power generation technologies, which convert fuel into heat and then into electricity through multiple intermediate steps, fuel cells produce electricity directly through an electrochemical reaction, resulting in higher efficiency and significantly lower emissions.

When powered by hydrogen, the primary byproducts are water and heat, making fuel cells one of the cleanest forms of power generation available today.

Fuel cells can be deployed across a wide range of applications, including passenger vehicles, heavy-duty trucks, buses, trains, ships, stationary power generation, backup power systems, and increasingly data centers. As governments and industries pursue decarbonization while seeking reliable and dispatchable power sources.

Major fuel-cell technologies include:

- SOFC (Solid Oxide Fuel Cell): Preferred for stationary power and data centers.
- PEMFC (Proton Exchange Membrane Fuel Cell): Dominant in transportation and backup power.
- MCFC (Molten Carbonate Fuel Cell): Large-scale distributed generation.
- PAFC (Phosphoric Acid Fuel Cell): Commercial combined heat and power.
- DMFC (Direct Methanol Fuel Cell): Portable applications

Exhibit 62: Fuel cell offering - Delta

Source: Delta

Exhibit 63: Fuel cell offering - Bloom Energy

Source: Bloom Energy

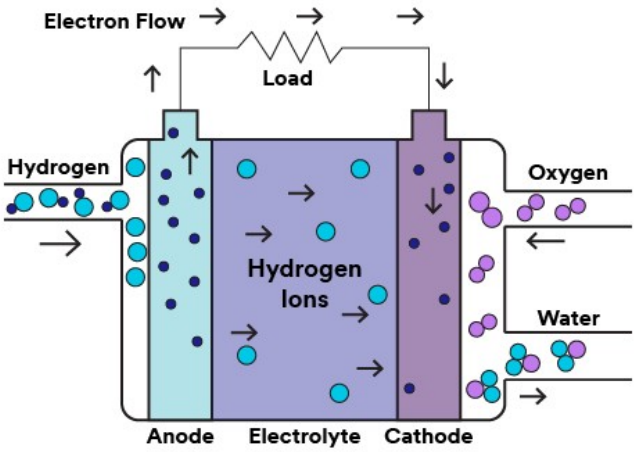
How do fuel cells work?

While there are several fuel-cell architectures, the fundamental operating principle is largely the same: fuel is supplied to the anode, oxygen is supplied to the cathode, and an electrolyte in between allows charged ions to move while forcing electrons to flow through an external circuit. This flow of electrons generates usable electricity, while the

reaction also produces heat and water as byproducts. Unlike conventional generators, which rely on burning fuel to drive a turbine or engine, fuel cells produce electricity directly through electrochemical reactions, allowing them to achieve significantly higher efficiency and lower emissions.

Using solid oxide fuel cell (SOFC) technology as an example, air is supplied to the cathode side of the cell, where oxygen molecules gain electrons and form oxygen ions (O^{2-}). These oxygen ions migrate through a ceramic electrolyte toward the anode side of the cell. At the same time, hydrogen or another hydrogen-rich fuel is supplied to the anode. When the oxygen ions reach the anode, they react with the fuel to produce water, heat, and electrons. The generated electrons cannot pass through the electrolyte and therefore travel through an external electrical circuit before returning to the cathode, creating a continuous flow of electricity. As long as fuel and air continue to be supplied, the fuel cell can generate power continuously.

Exhibit 64: Fuel cell diagram



Source: 3M

What benefits can fuel cells offer?

- **High energy efficiency:** Fuel cells convert the chemical energy of hydrogen directly into electricity through an electrochemical reaction, avoiding the multiple energy-conversion steps associated with combustion-based power generation. This allows fuel cells to achieve electrical efficiencies of over 60%, with some systems exceeding 80% total efficiency when waste heat is recovered through combined heat-and-power configurations, compared with roughly 35-45% for conventional gas-fired power generation. The efficiency advantage translates directly into lower fuel consumption, reduced operating costs, and improved energy utilization over the lifetime of the system.
- **Low carbon emissions:** When powered by hydrogen, fuel cells produce only water and heat as byproducts, generating no direct carbon emissions at the point of use. Even when operating on hydrogen derived from natural gas, fuel cells typically produce significantly lower greenhouse gas emissions than conventional combustion-based power systems. As green hydrogen production scales, fuel cells offer a potential pathway toward near-zero-carbon power generation for transportation, industrial facilities, and distributed energy systems.

- **Grid independence and energy security:** Fuel cells can provide reliable on-site power generation independent of the electrical grid, improving energy security and resilience. This makes them attractive for mission-critical applications such as data centers, hospitals, telecommunications infrastructure, defense installations, and industrial facilities that require uninterrupted power. In regions facing grid congestion or reliability concerns, fuel cells can also serve as a distributed power source located directly at the point of consumption.
- **Long duration energy storage enabler:** Fuel cells play an important role in the emerging hydrogen economy by enabling excess renewable energy from solar and wind farms to be stored as hydrogen and later converted back into electricity. Unlike batteries, which become increasingly expensive for longer storage durations, hydrogen storage can economically provide energy over days, weeks, or even seasonal periods. This capability could become increasingly valuable as renewable energy penetration rises worldwide.
- **Fuel flexibility:** Certain fuel-cell technologies, particularly Solid Oxide Fuel Cells (SOFCs), can operate on a variety of fuels including hydrogen, natural gas, biogas, ammonia-derived hydrogen, and other hydrogen-rich fuels. This flexibility allows users to deploy fuel-cell systems today using existing fuel infrastructure while maintaining a pathway toward pure hydrogen operation as hydrogen supply chains mature.
- **Quiet operation and reduced maintenance:** Because fuel cells contain relatively few moving parts compared with combustion engines and turbine-based generators, they operate quietly and often require less routine maintenance. The absence of combustion also reduces mechanical wear and vibration, which can improve system reliability and lower lifecycle operating costs, particularly in stationary power applications.
- **Scalability across multiple applications:** Fuel cells can be deployed across a wide range of power levels, from portable power systems measured in watts to utility-scale installations generating multiple megawatts. This scalability allows the technology to serve residential buildings, commercial facilities, industrial plants, transportation platforms, microgrids, and large-scale data centers, making fuel cells one of the most versatile clean-energy technologies available today.

Limitations and constraints of fuel cells

- **High system cost:** Despite significant technological progress, fuel cells remain more expensive than many conventional power-generation technologies. The cost challenge stems from complex stack manufacturing, specialized materials, balance-of-plant components, and relatively low production volumes compared with mature technologies such as internal combustion engines, gas turbines, and batteries. While costs have declined over the past decade, achieving widespread commercial adoption will likely require further manufacturing scale, supply-chain development, and technological improvements.
- **Limited hydrogen infrastructure:** One of the biggest challenges facing fuel-cell deployment is the lack of hydrogen production, transportation, storage, and refueling infrastructure. Unlike electricity, which is already widely distributed through established grids, hydrogen supply networks remain in the early stages of

development in most regions. This infrastructure gap creates a chicken-and-egg challenge, where end users are reluctant to adopt fuel cell technologies without reliable hydrogen availability, while infrastructure developers are hesitant to invest until demand reaches sufficient scale.

- **Durability and lifetime requirements:** Although fuel-cell durability has improved substantially, long-term reliability remains a key focus area, particularly for commercial transportation and utility-scale installations. Continuous exposure to heat, humidity, contaminants, and repeated operating cycles can gradually degrade fuel cell performance over time. Customers in sectors such as transportation, power generation, and industrial infrastructure often require operating lifetimes measured in tens of thousands of hours, creating significant engineering challenges for system developers.
- **Regulatory and safety considerations:** Hydrogen is a highly flammable gas and requires specialized handling, storage, and transportation systems. Although hydrogen technologies have well-established safety standards, new fueling stations, pipelines, and storage facilities must comply with stringent regulations. Developing these standards and obtaining approvals can extend deployment timelines and increase project costs.
- **Technology fragmentation and lack of standardization:** The fuel cell industry encompasses multiple technology platforms, including PEM, SOFC, MCFC, and other architectures, each optimized for different applications. While this diversity encourages innovation, it can also slow industry standardization and create uncertainty for customers evaluating long-term technology choices. Unlike the battery industry, which is increasingly converging around a few dominant chemistries, the fuel cell sector remains relatively fragmented.

Implications to datacenter power architecture

Transition from backup power to primary power source

Traditionally, data centers rely on grid electricity for primary power and diesel generators for backup power during outages. Fuel cells are increasingly being evaluated not only as a replacement for diesel backup generators but also as a potential primary power source. As AI-driven workloads substantially increase power density and overall electricity demand, hyperscalers are exploring fuel cells as a means of deploying computing capacity in locations where grid power is constrained or unavailable. This could fundamentally shift data-center power architecture from a grid-centric design to one that incorporates distributed on-site generation.

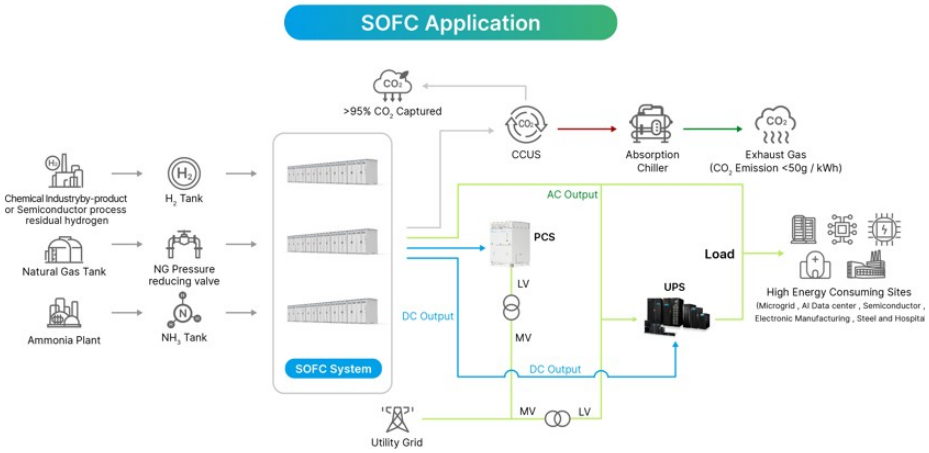
Potential long term evolution toward microgrid-based datacenters

In the longer term, fuel cells could become a core component of data-center microgrids combining renewable generation, battery storage, hydrogen storage, and intelligent energy management systems. Rather than functioning as passive consumers of grid electricity, future AI-focused data centers may increasingly operate as self-contained energy ecosystems capable of managing their own generation, storage, and load balancing. Such an architecture could provide greater scalability, resilience, and energy independence than traditional grid-connected designs

Potential simplification of power distribution architecture

Fuel cells inherently generate DC power, which aligns more closely with the power requirements of modern servers and IT equipment. Current data centers typically involve multiple AC/DC and DC/DC conversion stages between the utility grid and the processor, each introducing efficiency losses. Over time, fuel cell adoption could encourage a greater migration toward direct DC power distribution architectures, reducing conversion losses and improving overall facility efficiency. Moreover, its DC output fits well with the shift into HVDC power delivery architecture for datacenters.

Exhibit 65: Fuel cells in datacenter power delivery architecture



Source: Delta

Risk Reward – Delta Electronics Inc. (2308.TW)

Strong combination: Power + cooling for AI infrastructure

PRICE TARGET NT$2,500.00

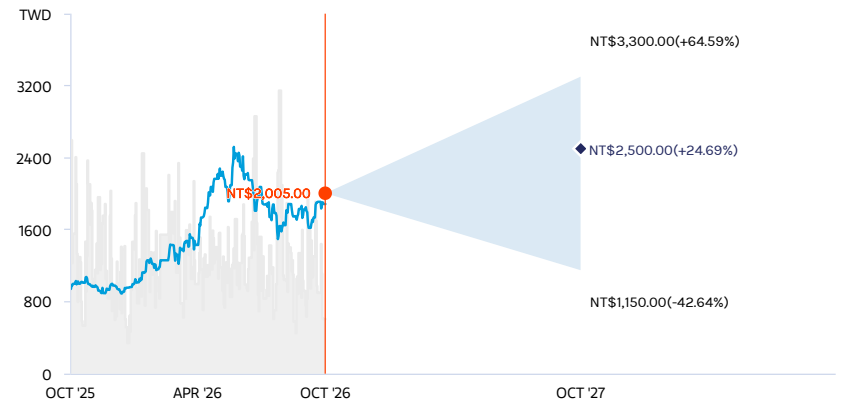
Base case: P/E multiple of 30x based on average 2027-28 EPS. We believe our target multiple is justified by the 63% earnings CAGR we estimate for 2025-28, implying a PEG of 0.5, versus the stock's ~23x P/E and 23% earnings CAGR during 2022-25, which implied a PEG of 1.0.

Consensus Price Target Distribution NT$2,200.00 NT$2,621.29 MS PT NT$4,120.00

Source: Refinitiv, Morgan Stanley Research

◆ Mean ◆ Morgan Stanley Estimates

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We view AI server power supplies and power management/cooling solutions for data centers as key growth drivers for the coming decade.
- Our expectation for a re-rating reflects expanding gross/operating margins and increasing exposure to AI-related businesses.
- Our 2026-28 earnings estimates reflect strength in AI server power supplies and liquid cooling.
- Our price target implies a P/E of 30x based on average 2027-28 EPS.

Consensus Rating Distribution

100%	Overweight
0%	Equal-weight
0%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth:	*Positive*
Self-help:	*Positive*
Technology Diffusion:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$3,300.00

40x average 2027-28e P/E

Strong sales growth with significant margin expansion: We assume a greater-than-expected contribution from higher-margin businesses such as power electronics and liquid cooling as well as greater synergies from acquired companies.

BASE CASE NT$2,500.00

30x average 2027-28e P/E

Strong YoY revenue growth in 2026-28, coupled with operating leverage: We expect steady growth in AI server power supply and data center infrastructure to be the key drivers. We see favorable mix and scale merits driving profit to grow more than double in 2027 vs. 2025.

BEAR CASE NT$1,150.00

14x average 2027-28e P/E

Moderate sales growth with slight margin improvement: We assume further profit drag from legacy products, smaller profit contribution from higher-margin businesses, higher opex and/or less meaningful contribution from acquired companies.

Risk Reward – Delta Electronics Inc. (2308.TW)

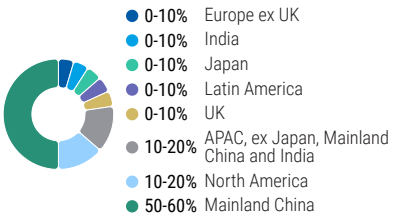
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Power Electronics YoY (%)	25.0	48.9	52.6	51.8
Automation YoY (%)	4.8	10.9	5.2	5.2
Infrastructure YoY (%)	81.6	44.6	51.0	48.2

INVESTMENT DRIVERS

- Transition to EV-related business and data infrastructure
- Operating margin recovery
- FINI holdings

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster-than-expected global economic growth
- Falling raw material prices
- Margin improvement

RISKS TO DOWNSIDE

- Global economic slowdown
- Ongoing component supply tightness
- Rising raw material prices
- Margin contraction as a result of higher operating expenses or faster ASP erosion

OWNERSHIP POSITIONING

Inst. Owners, % Active 64.7%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	746,914	775,166 / 774,709	807,877
EBITDA (NT$, mn)	163,606	169,213 / 175,311	191,720
Net income (NT$, mn)	96,767	103,307 / 105,731	111,870
EPS (NT$)	37.25	39.77 / 40.50	43.07

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Financial Summary – Delta

Exhibit 66: Delta: Financial Summary

Consolidated Income Statements

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Net sales	**421,148**	**554,885**	**775,166**	**1,131,877**	**1,659,520**
COGS	(284,567)	(364,729)	(494,374)	(707,257)	(1,028,632)
Gross profit	136,580	190,157	280,792	424,620	630,887
Operating expenses	(88,928)	(106,224)	(142,491)	(191,135)	(263,905)
Operating income	**47,652**	**83,932**	**138,300**	**233,484**	**366,983**
Non-operating income	**3,664**	**3,934**	**10,365**	**7,100**	**7,300**
Interest income	1,859	1,501	1,687	2,100	2,300
Investment income	(162)	(200)	372	-	-
Exchange gain	1,041	644	2,091	-	-
Other	926	1,989	6,215	5,000	5,000
Pre-tax income	**51,316**	**87,866**	**148,666**	**240,584**	**374,283**
Income tax	(10,925)	(19,930)	(32,925)	(52,189)	(81,172)
Minority interests	(5,163)	(7,828)	(12,433)	(20,158)	(31,363)
Net income	**35,229**	**60,108**	**103,307**	**168,237**	**261,747**
Modelware EPS (NT$)	**13.56**	**23.14**	**39.77**	**64.77**	**100.77**

Consolidated Cash Flow Statements

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Operating CF	**72,895**	**98,474**	**116,226**	**177,240**	**263,652**
Net Profits	35,229	60,108	103,307	168,237	261,747
Depreciation & Amort.	25,016	27,839	30,912	33,985	37,058
Change in WC	(2,486)	(24,190)	(55,485)	(86,295)	(125,847)
Other adjustments	15,137	34,717	37,491	61,313	90,694
Investing CF	**(40,360)**	**(53,214)**	**(70,000)**	**(70,000)**	**(60,000)**
Change of FA (Capex)	(33,435)	(45,950)	(70,000)	(70,000)	(60,000)
Change of L/T invest.	(4,739)	-	-	-	-
Other adjustments	(1,644)	(3,517)	-	-	-
Financing CF	**(11,118)**	**(9,401)**	**36,107**	**(12,414)**	**(54,050)**
Increase in L/T debt	8,552	(11,308)	12,530	2,978	3,829
Increase in S/T debt	(1,417)	446	29	212	272
Issuance of stock	-	-	-	-	-
Cash dividends	(18,696)	(20,313)	(30,131)	(51,787)	(84,335)
Other adjustments	444	21,774	53,680	36,183	26,183
FX adjust.	8,586	(2,146)	-	-	-
Net change in cash	**30,003**	**33,713**	**82,333**	**94,826**	**149,602**

Consolidated Balance Sheets

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Cash	**117,459**	**151,172**	**233,505**	**328,332**	**477,934**
Mkt securities	4,333	3,941	3,941	3,941	3,941
A/R	89,921	121,717	170,036	248,282	364,023
Inventory	83,856	101,478	142,217	205,395	298,726
Others	15,355	17,473	24,410	35,642	52,257
Current Assets	**310,925**	**395,781**	**574,109**	**821,592**	**1,196,881**
Long-term investments	4,740	4,876	4,876	4,876	4,876
Fixed assets	115,710	142,040	127,820	127,652	124,410
Other assets	100,523	96,922	96,922	96,922	96,922
Total Assets	**531,898**	**639,619**	**803,727**	**1,051,042**	**1,423,089**
S/T borrowings	1,918	2,364	2,393	2,605	2,877
AP/NP	69,223	94,452	128,025	183,154	266,379
Other ST liabilities	81,142	112,849	157,649	230,195	337,504
Total current liabilities	**152,283**	**209,665**	**288,067**	**415,954**	**606,759**
L/T debt	32,209	21,143	33,673	36,650	40,480
Other LT liabilities	67,840	83,422	83,422	83,422	83,422
Total Liabilities	**252,332**	**314,230**	**405,162**	**536,026**	**730,661**
Common shares	25,975	25,975	25,975	25,975	25,975
Retained Earnings	102,429	140,925	214,101	330,552	507,964
Other SH equity	151,161	158,489	158,489	158,489	158,489
Shareholders' equity	**279,565**	**325,389**	**398,565**	**515,016**	**692,428**
Total Liab./SE	**531,898**	**639,619**	**803,727**	**1,051,042**	**1,423,089**

Consolidated Financial Ratios

	2024	2025	2026E	2027E	2028E
Margins (%)					
Gross margin	32.4	34.3	36.2	37.5	38.0
Operating margin	11.3	15.1	17.8	20.6	22.1
Pretax margin	12.2	15.8	19.2	21.3	22.6
Net margin	8.4	10.8	13.3	14.9	15.8
YoY growth (%)					
Sales	5	32	40	46	47
Operating profits	16	76	65	69	57
Pretax profits	5	71	69	62	56
Net profits	5	71	72	63	56
Others					
Cash div payout (%)	52%	50%	50%	50%	50%
Adjusted cash div (NT$)	7.0	11.6	19.9	32.5	50.5
Yield (%)	0%	1%	1%	2%	3%
Net Debt/Equity (%)	-31%	-40%	-51%	-57%	-63%
Liabilities/Equity (%)	90%	97%	102%	104%	106%
Liabilities/Assets (%)	47%	49%	50%	51%	51%
ROAE (%)	13%	20%	29%	37%	43%
ROAA (%)	7%	10%	14%	18%	21%
AR/NR days	73	70	69	67	67
Inventory days	103	93	90	90	89
AP/NP days	79	82	82	80	80
Cash conversion	97	80	77	77	77

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 67: Delta: Quarterly Financials

Delta (2308.TW) (NT$ mn)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E
Net sales	**159,353**	**183,256**	**201,379**	**231,177**	**231,791**	**265,526**	**302,930**	**331,630**	**554,885**	**775,166**	**1,131,877**
COGS	(100,392)	(117,948)	(129,404)	(146,630)	(146,821)	(166,230)	(188,191)	(206,015)	(364,729)	(494,374)	(707,257)
Gross profit	**58,961**	**65,308**	**71,976**	**84,547**	**84,970**	**99,295**	**114,739**	**125,615**	**190,157**	**280,792**	**424,620**
Operating expenses	(30,544)	(34,738)	(36,983)	(40,226)	(40,422)	(45,405)	(50,589)	(54,719)	(106,224)	(142,491)	(191,135)
SG&A	(16,886)	(18,956)	(20,208)	(21,963)	(22,067)	(24,959)	(27,870)	(30,178)	(57,482)	(78,014)	(105,074)
R&D exp.	(13,657)	(15,782)	(16,775)	(18,263)	(18,356)	(20,445)	(22,720)	(24,541)	(48,743)	(64,477)	(86,061)
Operating income	**28,417**	**30,570**	**34,992**	**44,321**	**44,548**	**53,891**	**64,149**	**70,896**	**83,932**	**138,300**	**233,484**
Non-operating income	**2,676**	**4,430**	**1,630**	**1,630**	**1,775**	**1,775**	**1,775**	**1,775**	**3,934**	**10,365**	**7,100**
Interest income	428	398	430	430	525	525	525	525	1,501	1,687	2,100
Investment income	487	(115)	-	-	-	-	-	-	(200)	372	-
Disposal of investment	-	-	-	-	-	-	-	-	-	-	-
Exchange gain	1,688	403	-	-	-	-	-	-	644	2,091	-
Other	72	3,743	1,200	1,200	1,250	1,250	1,250	1,250	1,989	6,215	5,000
Pre-tax income	**31,093**	**35,000**	**36,622**	**45,951**	**46,323**	**55,666**	**65,924**	**72,671**	**87,866**	**148,666**	**240,584**
Income tax	(7,258)	(7,831)	(7,910)	(9,925)	(10,423)	(11,968)	(14,174)	(15,624)	(19,930)	(32,925)	(52,189)
Minority interest	(3,279)	(2,033)	(3,158)	(3,963)	(3,841)	(4,676)	(5,537)	(6,104)	(7,828)	(12,433)	(20,158)
Net income	**20,556**	**25,136**	**25,554**	**32,063**	**32,059**	**39,022**	**46,213**	**50,943**	**60,108**	**103,307**	**168,237**
EPS (NT$)	**7.91**	**9.68**	**9.84**	**12.34**	**12.34**	**15.02**	**17.79**	**19.61**	**23.14**	**39.77**	**64.77**
Margins (%)											
Gross Margin	37.0	35.6	35.7	36.6	36.7	37.4	37.9	37.9	34.3	36.2	37.5
Operating Margin	17.8	16.7	17.4	19.2	19.2	20.3	21.2	21.4	15.1	17.8	20.6
Pretax Margin	19.5	19.1	18.2	19.9	20.0	21.0	21.8	21.9	15.8	19.2	21.3
Net Margin	12.9	13.7	12.7	13.9	13.8	14.7	15.3	15.4	10.8	13.3	14.9
QoQ Growth (%)											
Sales	(1)	15	10	15	0	15	14	9			
Gross Profit	5	11	10	17	1	17	16	9			
Operating Profit	8	8	14	27	1	21	19	11			
Pretax Profit	21	13	5	25	1	20	18	10			
Net Profit	19	22	2	25	(0)	22	18	10			
YoY Growth (%)											
Sales	34	48	34	43	45	45	50	43	32	40	46
Gross Profit	56	48	37	51	44	52	59	49	39	48	51
Operating Profit	102	64	41	68	57	76	83	60	76	65	69
Pretax Profit	99	79	36	79	49	59	80	58	71	69	62
Net Profit	101	80	37	85	56	55	81	59	71	72	63

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

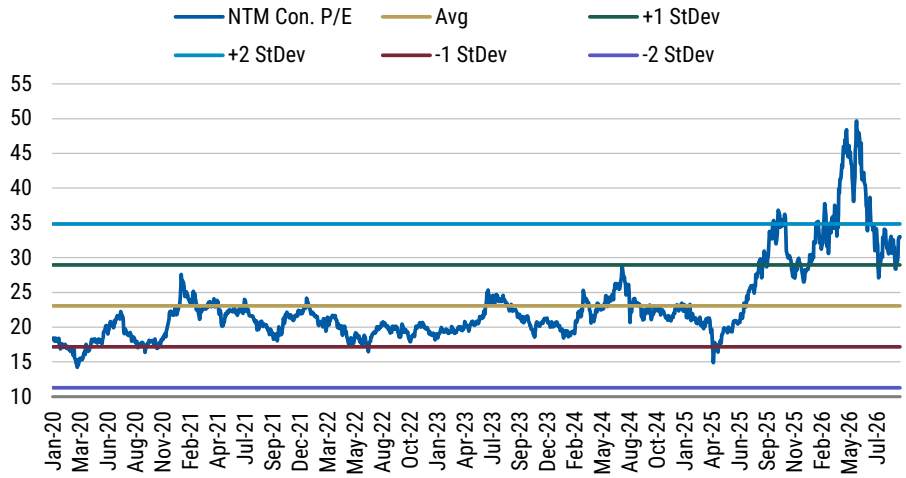
Valuation Methodology – Delta

Our price target (base case scenario value) is NT$2,500, derived from 30x 2027-28e P/E: We use price/earnings as our valuation methodology because we believe it appropriately captures Delta's earnings growth potential. We believe our target P/E multiple of 30x is justified by an earnings CAGR of 63% over 2025-28, which implies a PEG of 0.5. This is lower than the average of 1.0 in 2022-25, when the stock was trading at an average P/E of 23x for a 23% earnings CAGR.

Bull case scenario value – NT$3,300: We apply a 40x P/E multiple to our average base case estimate for 2027-2028. This scenario assumes a greater-than-expected contribution from higher-margin businesses such as power electronics and liquid cooling as well as greater synergies from acquired companies.

Bear case scenario value – NT$1,150: We apply a 14x P/E multiple to our average base case estimate for 2027-2028. This scenario assumes further profit drag from legacy products, smaller profit contribution from higher-margin businesses, higher opex and/or less meaningful contribution from acquired companies.

Exhibit 68: Delta: NTM P/E chart



Source: FactSet consensus estimates, Morgan Stanley Research

Why Does Delta Trade at a Significant Valuation Discount to Delta Thailand?

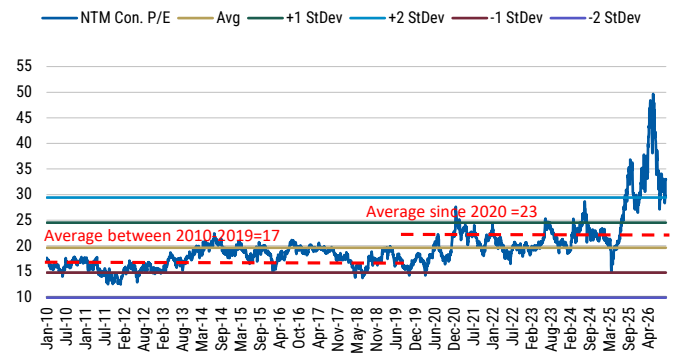
Discussion of the valuation gap between Delta Electronics (2308.TW, currently 33x 12M forward consensus P/E) and Delta Electronics Thailand (DELTA.TH, 71x currently) has been ongoing for several years. If we look at their valuations before 2020, Delta was actually trading at a premium, with an average 17x 12M forward P/E in 2010-2020 vs. Delta Thailand's 13x. This made sense, as Delta is much more competitive in the global arena and Delta Thailand was set up as a subsidiary focused on manufacturing.

The situation flipped after 2020. Though Delta has been re-rated to trade at an average 23x 12M forward P/E since 2020, Delta Thailand has seen its valuation surge to an average of 60x over the same period. This cannot be explained by moves in their respective markets, either, as TWSE went from 14x forward P/E in 2010-20 to 16x after 2020, while the SET went from 13x to 16x.

We think this could be attributable to a few factors:

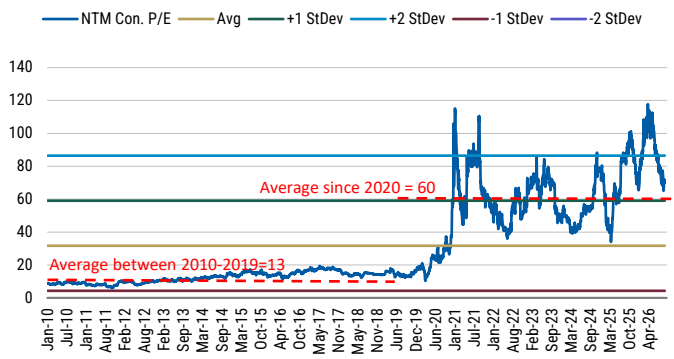
- **Different positioning within the Delta group:** Delta group has increased its stake in Delta Thailand to over 60% from about 20% in 2019 through a tender offer. Since then, there has been much more collaboration between Delta group and Delta Thailand. As Thailand emerged as one of the major manufacturing hubs within the Delta group, its earnings growth (28% in 2020-25) outpaced that of Delta Taiwan (19%).
- **Scarcity in the Thai stock market:** Delta Thailand is one of the relatively few large-cap Thai-listed companies that offer direct exposure to AI/data-center power and EV-related themes, whereas there are many such alternatives in Taiwan.
- **Limited free float:** As the Delta group owns a ~62% stake in Delta Thailand and there have been a few quite stable long-term shareholders, we think the effective free float is probably 20% or less.

Exhibit 69: Delta NTM P/E



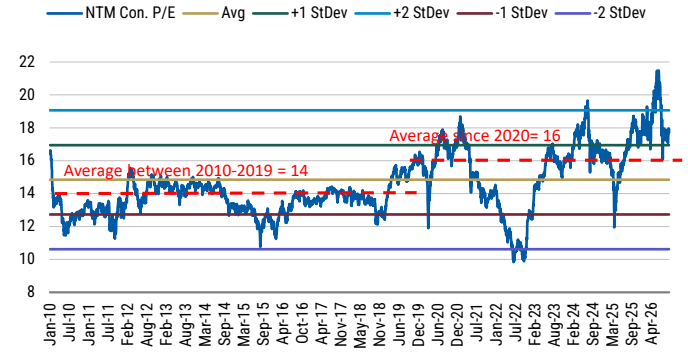
Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 70: DET NTM P/E



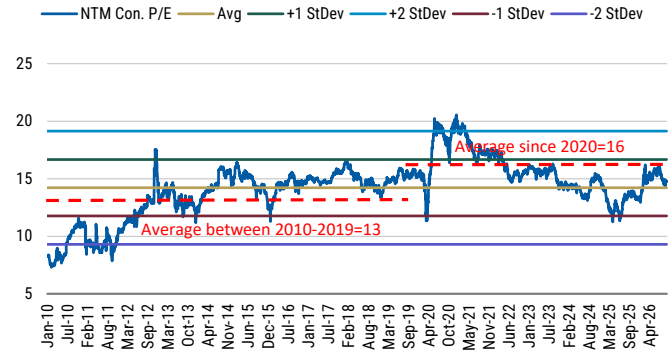
Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 71: TWSE NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

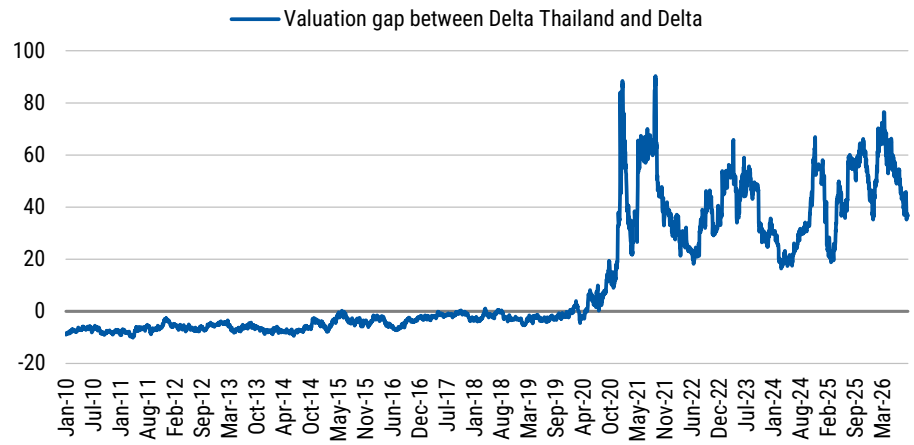
Exhibit 72: SET Index NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 73:

Valuation gap between Delta Thailand and Delta based on 12M forward P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Global Footprint – Delta

Comprehensive Global Operations

Aggressive capacity expansion plans

The company has been continuously diversifying its production base away from China owing to the trade turmoil back in 2018-2019, and then by supply chain disruptions during Covid. Management explicitly set a target of lowering China production to 60% of total capacity (from >70% previously) for the first time back in 2020. New capacity was added in Thailand, India, and Taiwan during 2020-2021. It then saw demand surge from various sectors, including EVs, automation and telecom, during 2022-23, followed by AI-related power suppliers and cooling solutions emerging in 2024. In order to grasp the emerging opportunities, relevant plant build-outs were planned accordingly, on top of the ongoing geographical diversification. It expanded its presence—production sites, R&D, and sales teams—in Europe and the US to better cope with operational disruptions and capture domestic demand, in addition to its existing setups in Taiwan, Thailand, and India. There was then a substantial increase in capex of NT$12.2bn in 2025 (vs. NT$5.6-6bn in 2023-24) as the company further accelerated its expansion plan, mainly on AI-related power/cooling demand and increasing capacity tightness, specifically in Thailand and the US. Delta continued to expand in Thailand by adding three new plants, with another three under construction for power and liquid cooling system offerings. In view of the capacity tightness reserved for AI-related business, it also started to prioritize margin-accretive business—i.e., de-emphasize business with net margins below 10%—which further optimized product mix.

Looking ahead into 2026 and beyond, expansion is likely to remain aggressive in order to fulfill substantial power and liquid cooling turnkey solution demand for large-scale data centers, while management also indicated that the new capacity added at the end of 2025 has already been fully taken. With capacity staying tight for AI power and cooling offerings, the company will continue to expand actively in China, Thailand, the US, and Tainan, Taiwan. It also plans to add mass production lines in Taoyuan, scheduled to be ready by 2028, in addition to the pilot run lines in Tainan. Against this backdrop, the capex budget has been revised up several times and is now roughly NT$50bn for 2026.

Exhibit 74: Delta's global footprint

Source: Company data, Morgan Stanley Research. Note: Data as of 2024.

Financial Analysis – Delta

44% Revenue CAGR in 2025-28e

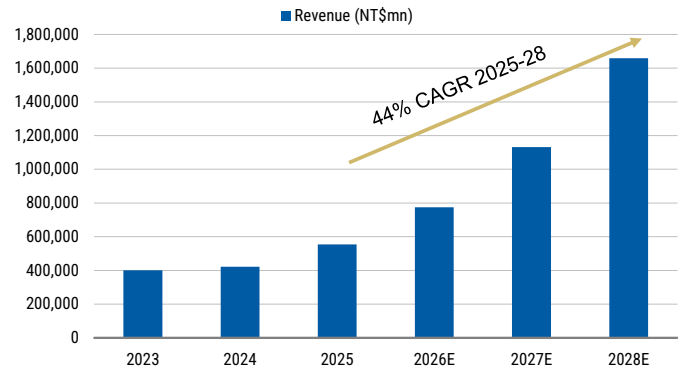
Delta's revenue growth was moderate in 2020 at merely 5%, making it the only year of the Covid disruption (2019-22) that did not see double-digit growth. Growth picked up to 11% in 2021, underpinned by surging demand for cloud, networking, and work-from-home infrastructure amid the pandemic, while the next uplift to 22% growth in 2022 was mainly fueled by increasing demand for EV- and data center-related components (server power supplies, networking switches, energy storage systems, etc). Revenue momentum subsequently moderated in 2023, growing just 4%, as the EV component business was impacted by the UAW strike in the US and automation demand in China stayed subdued, despite sustained demand for AI server power supplies and modules. Growth in 2024 was at a similar level, as strength in data center offerings were not enough to offset weakness in its legacy businesses.

Revenue surged 32% in 2025 to a record NT$554,885mn, driven by AI server rack power and cooling solutions, along with further spec upgrades and value growth.

We expect revenue to expand at a 44% CAGR over 2025-28, supported by ongoing AI server shipments, spec upgrades, 800 VDC power solutions, and turnkey liquid cooling offerings. As the upgrade to 800V HVDC power architecture is likely to take years given the complex design and high safety requirements, we expect AI server power solution upgrades to be separated into a few stages – from power shelves in AI server racks to standalone side-car power racks, from 50V DC to 800V DC output, from 72kW per power shelf to >250kW, etc, as well as power wattage increases for next-generation GPU rack designs.

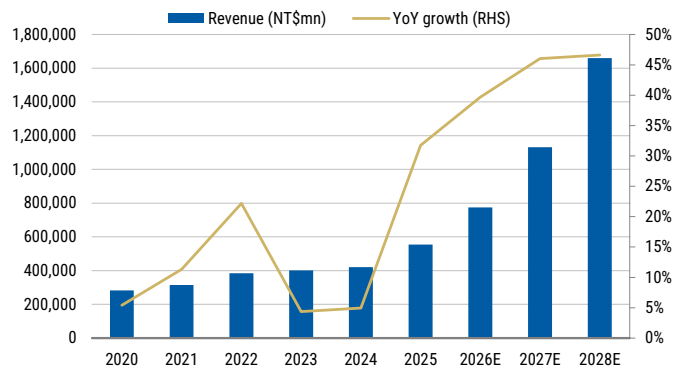
By segment, Power Electronics has been the primary revenue driver, contributing at least 50% over the past years despite the company separating the Mobility segment from Power Electronics since 2024 so as to further distinguish revenue generated from EVs/ autos. We expect the segment to remain the key revenue contributor, supported by strong demand from AI server-related power solutions, and estimate it will reach 58% of revenue by 2028 vs. 50% in 2025. Infrastructure, the second-largest revenue contributor historically, is expected to grow along with the Power Electronics segment, fueled by demand for data center power-related solutions (i.e. UPSs, CDUs, sidecars, etc) as well as energy infrastructure-related offerings (i.e. SSTs (solid-state transformers), fuel cells, modular power solutions, micro grid management, etc.). We model the segment's revenue contribution to reach 35% by 2028 from 24% in 2024 and 33% in 2025. For Automation and Mobility, we expect their revenue contributions to decline sequentially throughout the years, with Automation dropping from 10% in 2025 to 4% in 2028 and Mobility falling from 7% in 2025 to 2% in 2028.

Exhibit 75: Delta: Revenue CAGR, 2025-28e



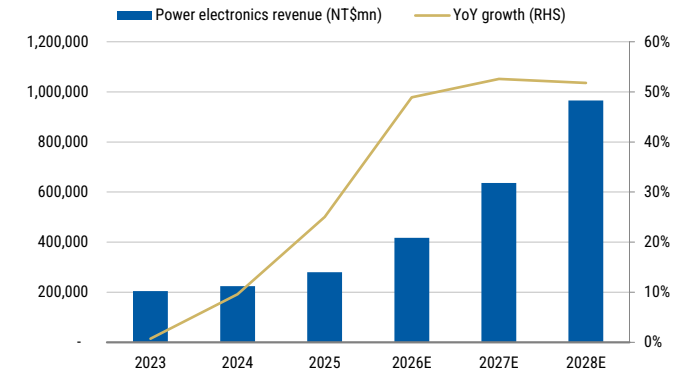
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 76: Delta: Annual revenue growth



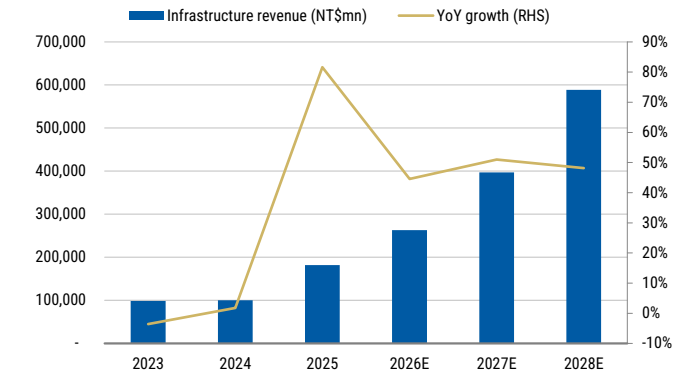
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 77: Delta: Power electronics revenue



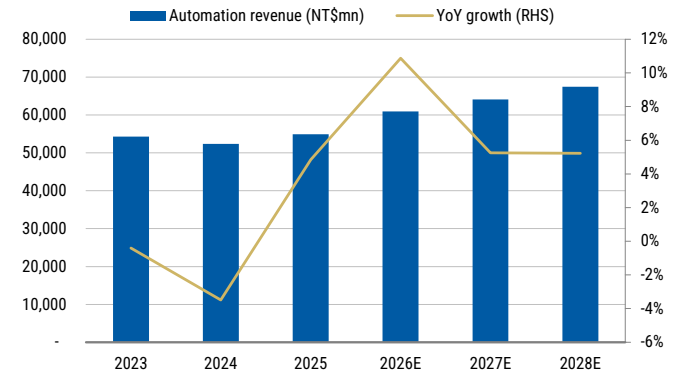
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 78: Delta: Infrastructure revenue



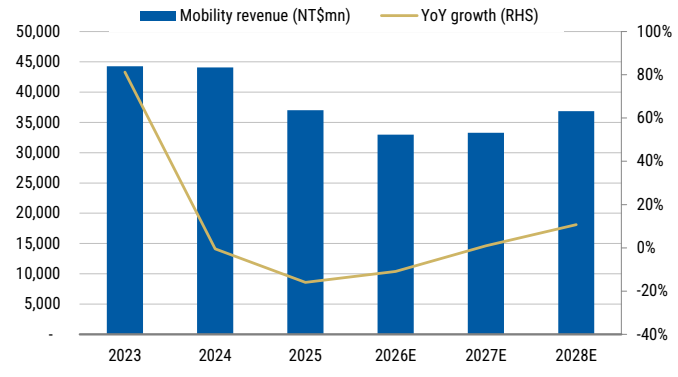
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 79: Delta: Automation revenue



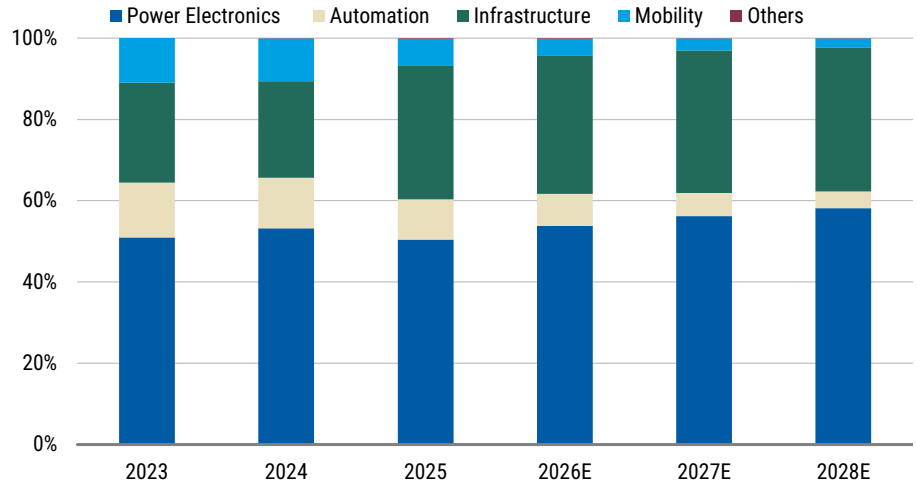
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 80: Delta: Mobility revenue



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 81:
Delta: Revenue breakdown by segment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

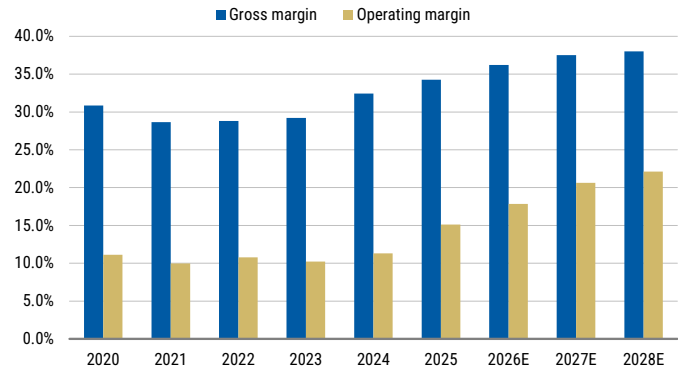
Continuous margin expansion on favorable product mix

Delta's gross margin stayed largely between 28-29% during 2021-23, as revenue was mainly driven by relatively lower-margin products, including EV- and consumer/IT-related products, while server-related offerings had yet to scale up. 2024 served as an inflection point, with the product mix changing structurally, supported by robust demand for AI data center-related offerings, while its legacy businesses stayed relatively weak. Gross margin thus grew by 3.2ppt YoY during the year, from 29.2% in 2023 to 32.4% in 2024. Increasing AI computing needs further increased power consumption and cooling requirements per rack, boosting demand for high-density AI power and cooling products along with related system-level solutions in 2025, which lifted gross margin by another 1.8ppt YoY to 34.3% for the year.

The opex ratio was largely stable, fluctuating around 18-19% between 2020 and 2023. It then increased to 19% in 2023 and 21.1% in 2024, mainly due to operating deleverage, higher R&D costs, and ongoing capacity expansion. The ratio declined back to 19.1% in 2025 on the back of an enlarged revenue base, despite R&D-related investments remaining elevated.

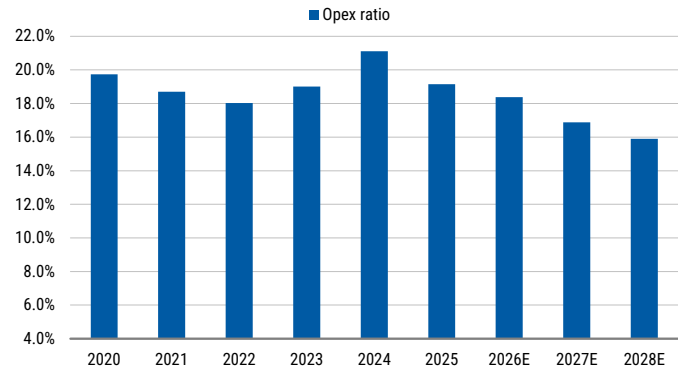
Looking ahead, we expect gross margin to continue expanding on the back of increasing revenue contribution from AI-related power supplies. We thus pencil in a 2.0ppt gross margin expansion to 36.2% in 2026, followed by further increases of 1.3ppt and 0.5ppt YoY in 2027 and 2028, respectively, reaching 38.0% by 2028. On the opex ratio, we expect Delta to continue benefiting from operating leverage in the coming years, and thus estimate operating margin to improve more substantially than gross margin, expanding by 2.7ppt YoY in 2026 to 17.8%, followed by further expansions of 2.8ppt and 1.5ppt in 2027 and 2028, reaching 20.6% and 22.1%, respectively.

Exhibit 82: Delta: Margin trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 83: Delta: Opex ratio



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Net profit to grow at a 63% CAGR in 2025-28

Delta's operating profit grew sequentially throughout 2020-22, supported by gross margin expansion and a lower opex ratio, from NT$31,426mn in 2020 to NT$41,439mn in 2022. It then saw a slight 1% YoY decline in 2023 due to higher operating expenses as the company further stepped up its R&D investments, despite a 0.4ppt gross margin expansion in 2023. Operating margin also edged down accordingly by 0.6ppt to 10.2% during the year.

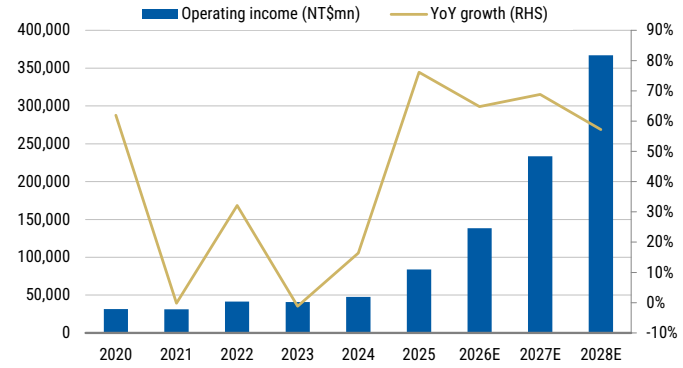
Operating profit returned to a growth trajectory from 2024 onward, growing 16% YoY in 2024 and 76% YoY in 2025, as AI-related power offerings started to scale up. Operating margin expanded faster than gross margin, supported by operating leverage and disciplined cost controls, increasing by 1.1ppt in 2024 to 11.3% and another 3.8ppt to 15.1% in 2025.

With the tax rate staying largely stable at ~20% during 2020-23 before starting to edge up in 2024 and reaching 23% in 2025, Delta's net profit growth was largely aligned with its operating profit growth trajectory. 2023 was an exception, as net profit still managed to grow 2% YoY despite a 1% YoY decline in operating profit, mainly due to higher FX gains.

The net margin trend has been largely aligned with that of operating margin in recent years. After dropping from 9.0% in 2020 to 8.5% in 2021, as lower gross margin was offset by a lower opex ratio, net margin hovered around the 8.3-8.5% level during 2021-24. Net margin was largely flat YoY in 2024 despite a 1.1ppt expansion in operating margin, mainly due to lower non-operating income and a slightly higher tax rate. We then saw a surge in 2025, with net margin expanding 2.5ppt YoY to 10.8%, benefiting from a larger contribution from higher-margin AI power offerings.

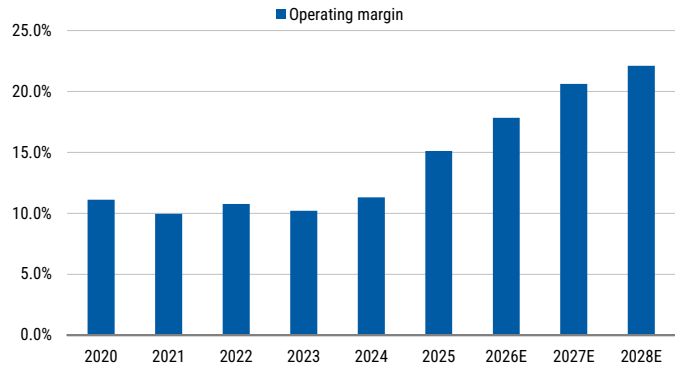
Looking ahead, we forecast net profit to grow 72% YoY in 2026e, followed by 63% YoY growth in 2027e and 56% YoY growth in 2028e, implying a 63% CAGR in 2025-28e, assuming a stable tax rate of 22%. Net margin is likely to rise accordingly, from 10.8% in 2025 to 13.3% in 2026, followed by further expansion to 14.9% in 2027 and 15.8% in 2028.

Exhibit 84: Delta: Operating income trend



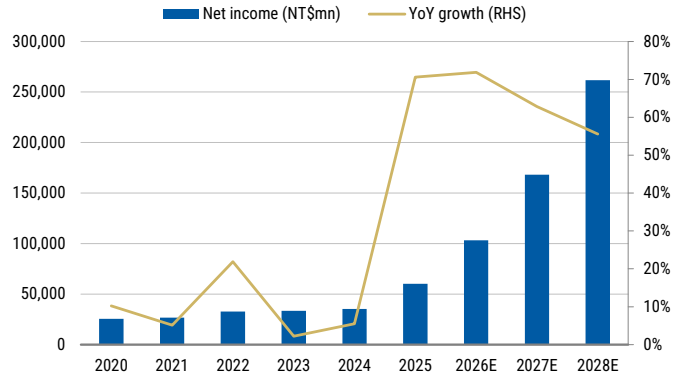
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 85: Delta: Operating margin



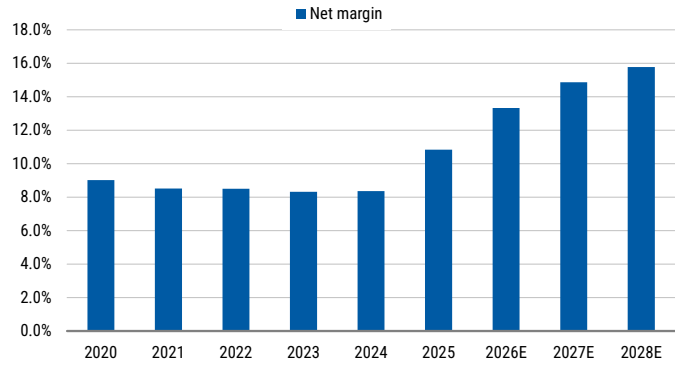
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 86: Delta: Net profit trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 87: Delta: Net margin



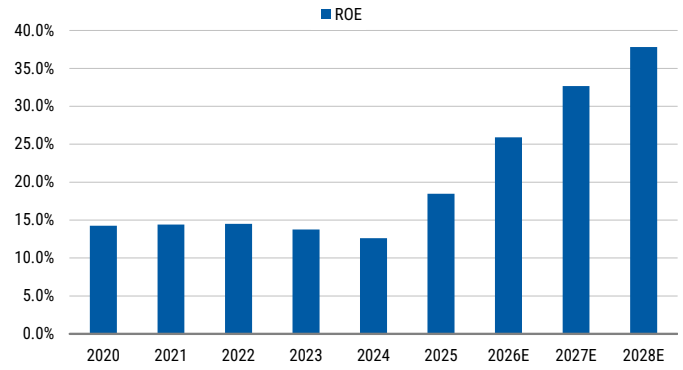
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

ROE to rise alongside margin expansion

ROE rose gradually but steadily during 2020-22, from 14.3% to 14.5%, mainly driven by improving asset turnover, offset by lower net margin. ROE edged down to 13.8% in 2023 and fell further to 12.6% in 2024 on the back of lower asset turnover, while net margin and financial leverage remained at similar levels. It then rebounded to 18.5% in 2025, supported by both improving asset turnover and net margin expansion, which was fueled by an increasingly favorable product mix and operating leverage.

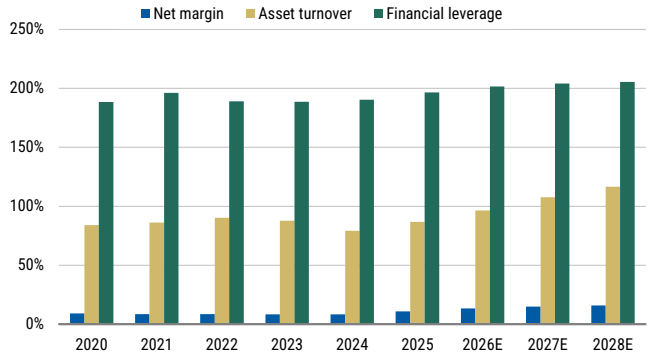
Looking ahead, we expect ROE to continue rising, reaching 25.9% in 2026, 32.7% in 2027, and 37.8% in 2028, primarily driven by continued margin expansion and improving asset turnover. Net margin is projected to expand from 10.8% in 2025 to 15.8% in 2028, supported by gross margin improvement—from 34.3% to 38.0%—and a declining operating expense ratio from 19.1% in 2025 to 15.9% in 2028.

Exhibit 88: Delta: ROE trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 89: Delta: DuPont analysis



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

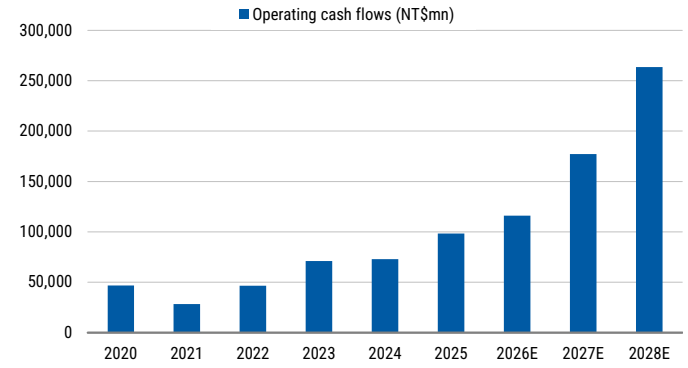
Improving balance sheet and continuous free cash flow generation

Delta has generated positive operating cash flow in recent years. Operating cash flow declined in 2021, mainly due to higher inventory buildup amid supply chain disruptions during the year. Free cash flow generation has also remained positive, while fluctuating with capital expenditures over the years. Free cash flow was exceptionally low at NT $5,292mn in 2021, as the company started to step up capex mainly for geographic capacity diversification and greater production flexibility—adding new capacity in Thailand, India, Tainan/Zhongli in Taiwan, and Wuhu, Wujiang, and Chongqing in Mainland China. Both absolute capex and the capex-to-sales ratio have trended upward since 2022, as Delta continued to add capacity for EV-related offerings and power and cooling solutions during 2023-24, with further acceleration in 2025 due to capacity tightness for AI power and cooling offerings, especially in Thailand and the US.

Looking ahead, the company aims to continue adding capacity worldwide to fulfill substantial demand for power and liquid-cooling turnkey solutions for large-scale data centers operated by major hyperscalers beyond 2027. Capacity remains tight for AI power and cooling offerings, so it continues to expand actively in China, Thailand, the US, and Tainan. Management also plans to add mass-production lines in Taoyuan, Taiwan, scheduled to be ready by 2028, in addition to the pilot-run lines in Tainan. Accordingly, we now estimate capex to rise to NT$70bn in 2026 from ~NT$46bn in 2025 and stay at a similar level in 2027 before edging down slightly to NT$60bn in 2028.

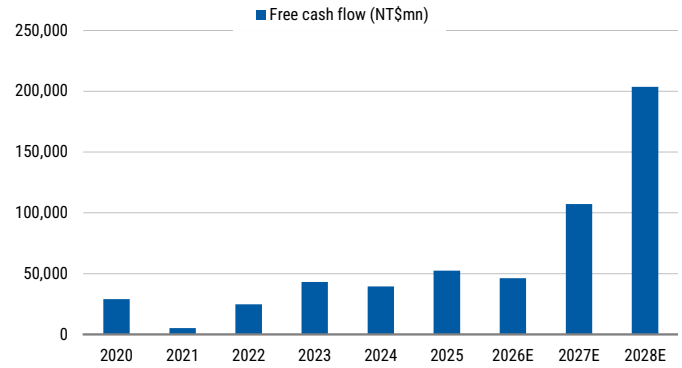
Delta's net cash position has been rising since 2021, with its net cash-to-equity ratio improving from just 2% in 2021 to 40% in 2025. We expect this ratio to continue rising, reaching 51% in 2026, 57% in 2027, and 63% in 2028, on strong free cash flow generation despite a higher total debt level amid ongoing capacity expansion.

Exhibit 90: Delta: Positive operating cash flow



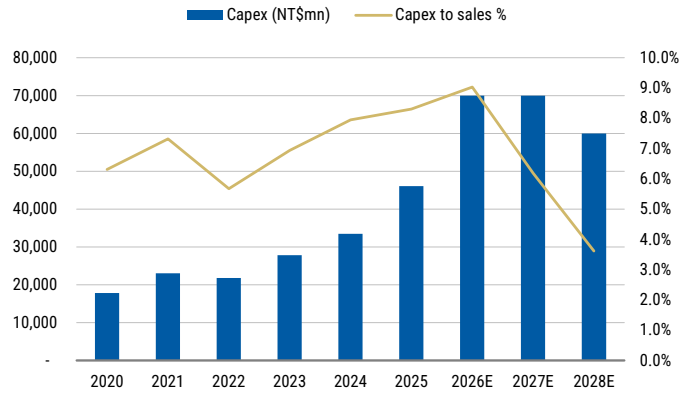
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 91: Delta: Improving free cash flow generation



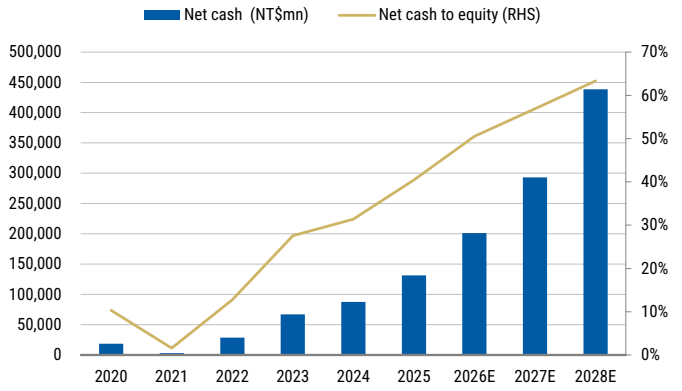
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 92: Delta: Capex and capex-to-sales ratio



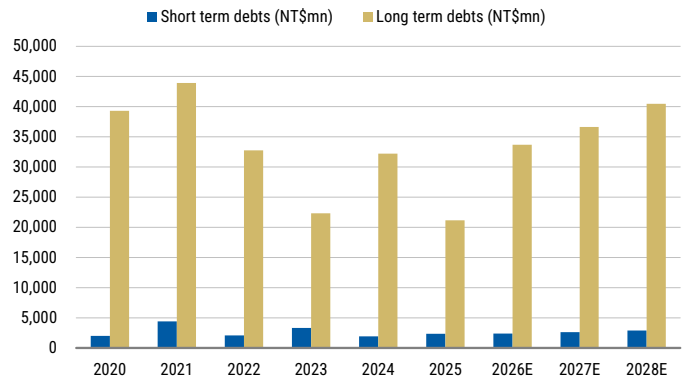
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 93: Delta: Net cash position



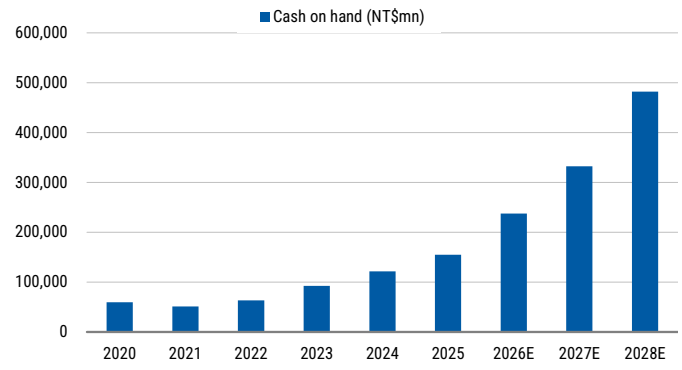
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 94: Delta: Debt position



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 95: Delta: Cash on hand



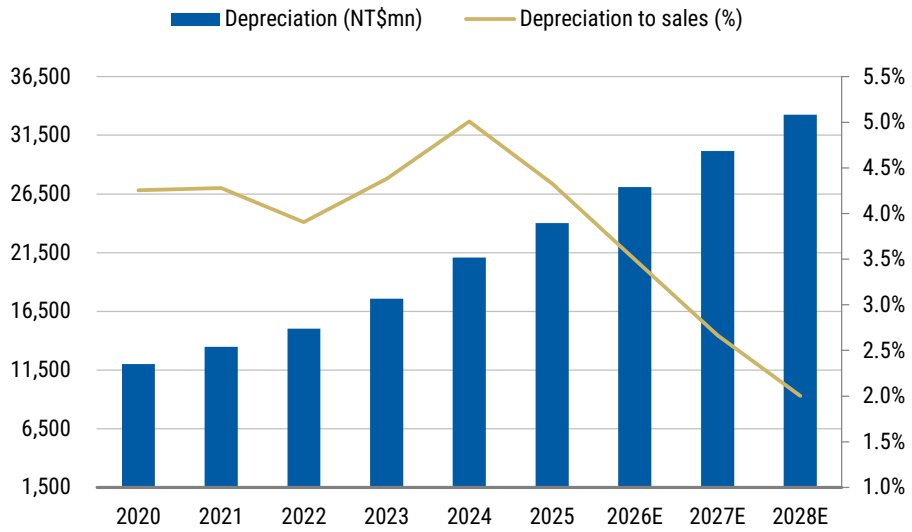
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

After edging down slightly in 2022 from a relatively high 7.3% in 2021, Delta's capex-to-sales ratio has been rising alongside its aggressive capacity expansion plans since 2022, despite an expanding revenue base. Looking ahead, as we expect capex to increase further in 2026, we project a capex-to-sales ratio of 9.0% for the year on robust revenue growth, before declining to 6.2% in 2027 and 3.6% in 2028.

The depreciation-to-sales ratio hovered around the 4-5% level during 2020-25. We

estimate the ratio will decline sequentially as the revenue base grows faster than capex and depreciation, from 4.3% in 2025 to 3.5% in 2026, 2.7% in 2027, and 2.0% in 2028.

Exhibit 96: Delta: Depreciation and depreciation to sales

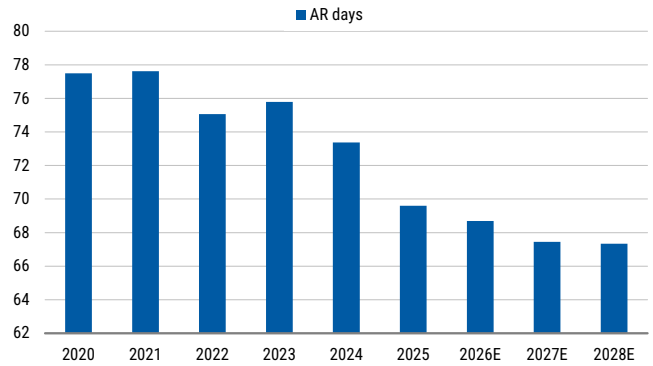


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Delta's cash conversion cycle (CCC) rose from 77 days in 2020 to a peak of 101 days in 2023 before improving to 80 days in 2025. By 2025, inventory days had increased from 80 days in 2020 to 93 days, while AP days had risen to 82 days after falling from 81 days in 2020 to 75 days in 2023. This was partly offset by improving AR days, which declined from 77 days in 2020 to 70 days in 2025.

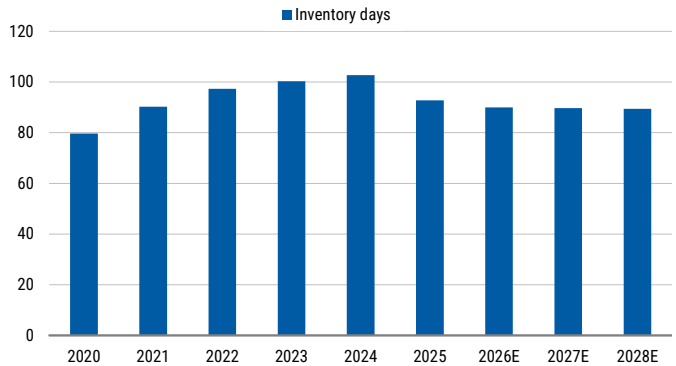
Looking ahead, we expect the overall cash conversion cycle to improve further and remain at 77 days throughout 2026-28, mainly supported by improving AR days and a modest decline in inventory days.

Exhibit 97: Delta: AR days



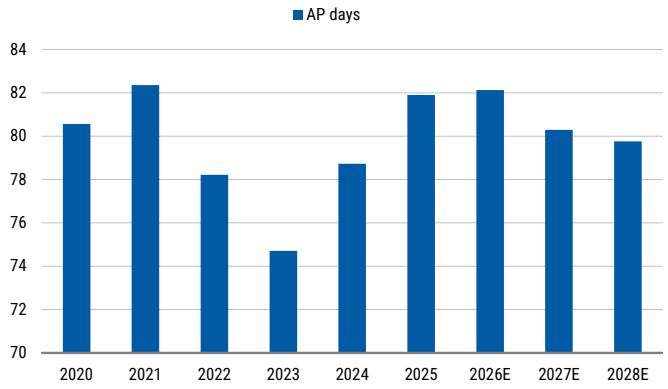
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 98: Delta: Inventory days



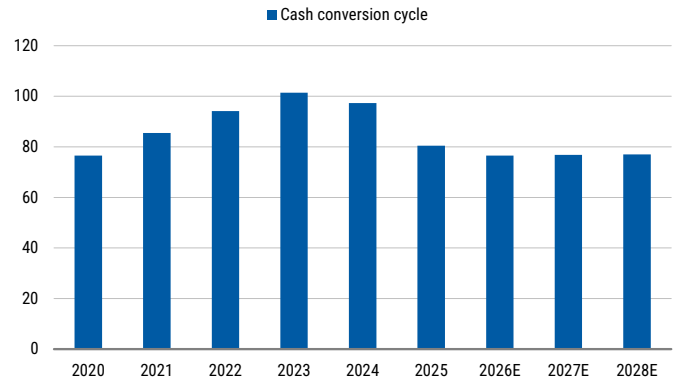
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 99: Delta: AP days



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 100: Delta: Cash conversion cycle



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Company Background – Delta

Company snapshot

Delta Electronics is a global provider of power and thermal management technologies, with a focus on energy-efficient products and integrated solutions. Founded in 1971, the company serves a broad range of applications spanning components, power and system products, thermal management, mobility, industrial and building automation, data centers, telecom, and energy infrastructure, across industries including automotive, healthcare, data centers, and power grids. By combining its capabilities in power electronics, digital technologies, and system integration, Delta aims to support customers in improving operational efficiency and advancing smart manufacturing, smart cities, and more sustainable infrastructure.

The company is headquartered in Taipei, Taiwan. Its global network consists of 293 operations, including 165 sales offices, 55 manufacturing sites, and 73 R&D centers, supporting customers worldwide.

Exhibit 101: Delta: Key milestones

Year	Key Milestones
1971	Delta Electronics, Inc. was established
1976	Achieved >US$1mn of revenue
1977	Moved plant to Taoyuan Guishan Industrial Park
1983	- Mass production of switching power supplies
1988	- Listed on the Taiwan Stock Exchange (code 2308) - Mass production of brushless DC fans - Invested in the establishment of Delta Electronics (Thailand) PCL, with an original stake of 6%
1989	- Established first Japan office in Tokyo - Established Delta Power Electronics laboratory in Raleigh, US
1992	- Inauguration of the first plant in China (Dongguan) - Established Delta Greentech (China) Co., Ltd. in Shanghai
1995	Delta Electronics (Thailand) PCL was listed on Thailand Stock Exchange
1999	- Inaugurated its new Taipei HQ - Established the Delta Power Electronics Center (DPEC) in Shanghai
2000	Established the Delta Power Electronics Science and Education Development program
2001	Inaugurated its Wuhu plant in China
2003	Acquired ASCOM Energy Systems and its plant in Slovakia
2006	Inaugurated Delta's Tainan Plant I
2008	Inaugurated its Rudrapur plant in India and Wuhu plant in China
2010	Inaugurated Chenzhou plant in China
2011	Inaugurated Shanghai R&D and Operations Center and Taoyuan R&D Center in Taiwan
2012	Mr. Yancey Hai was elected Chairman and Mr. Ping Cheng was appointed as CEO
2015	- Acquired Eltek ASA, a leading provider of telecom power, industrial power, and data center power solutions - Inaugurated Delta Americas' headquarters green building and Taoyuan Plant 5
2017	- Announced organizational adjustment to focus on Power Electronics, Automation, and Infrastructure as main business categories - Acquired a controlling stake in VIVOTEK
2019	- Acquired a controlling stake in Delta Electronics (Thailand) Public Company Limited - Acquired lighting-solutions brand Amerlux
2020	Accquired Trihedral, a Canadian SCADA and IIoT software company, to strengthen industrial automation and system-integration businesses
2021	Acquired March Networks, a global video-surveillance provider, and Universal Instruments, a leader in precision automation solutions for electronics manufacturing
2022	- Inaugurated a new facility at the Automotive Campus in Helmond, the Netherlands - Purchased land and a factory in Plano, Texas, United States
2023	Acquired TB&C, a leading provider of automotive high-voltage hybrid components
2024	- Mr. Ping Cheng selected as Chairman while concurrently serves as CEO - Adjusted business categories into Power Electronics, Mobility, Automation, and Infrastructure
2025	- Inaugurated new North East Asia HQ in Tokyo - Acquired ACT Genomics and Noda RF Technologies

Source: Company data, Morgan Stanley Research

Product offerings

Delta's key offerings can be categorized into four key areas: Power Electronics, Infrastructure, Mobility, and Automation.

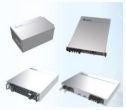
1. Power Electronics

Delta operates in switching power supplies, thermal management, and passive components, supplying products for ICT, cloud computing, EVs, and industrial and commercial applications. The company frames its business around energy efficiency and

sustainability and now also offers grid-to-chip power and thermal management solutions for data centers.

- Components: Includes essential passive and magnetic components used in portable devices, AI cloud computing, automotive electronics, and IoT.
- Power and System: Contains power products and systems for AI, cloud computing, IoT, client devices, and industrial, medical, lighting, e-mobility, and electrical-appliance applications.
- Fans and Thermal Management: This mainly covers air and liquid cooling solutions for AI cloud and edge computing, EVs, as well as industrial, commercial, and residential sectors. Key products includes DC and EC fans, thermal modules, liquid cooling cold plates, and coolant distribution units, etc.

Exhibit 102: Delta: Key offerings for the power electronics segment

POWER ELECTRONICS							
Components							
Inductors	RF Inductors	Standard Power Module (DC/DC Converter & AC/DC Module)	Transformers	Networking	EMI Filters	Solenoids	Current Sensing Resistor
				Wire-Wound Common Mode Choke			
Power and System							
Embedded Power	Adapters	Industrial Power	Medical power	Industrial Battery Charging	USB Socket Outlets	High Voltage Power	Hydrogen Power
				Modular Charging System			
Fans and Thermal Management							
Chip Level Cooling Solution	DC Brushless Fans and Blowers	Thermal Module	Cold Plate Loop	In-rack CDU	Rack Manifold	Electical Pump	In-row CDU
Communication Equipment Cooling	Energy Infrastructure and Industrial Automation Cooling	Portable Device and Equipment Cooling	Infrastructure Ventilation	Automotive Thermal			
DC Brushless Fans and Blowers	DC Brushless Fans and Blowers	Laptop cooling	DC Brushless Fans and Blowers				

Source: Company data, Morgan Stanley Research

2. Infrastructure

Delta supplies ICT and energy infrastructure to support demand related to AI, smart cities, and IoT. Its ICT infrastructure solutions target data centers and telecom networks, while its energy infrastructure offerings integrate AIoT, power conversion, and microgrid control across energy generation, storage, and management.

- ICT Infrastructure: It mainly provides infrastructure for AI computing and 5G

communications under this category, ranging from telecom cell sites to data centers. The portfolio includes power, air and liquid cooling, networking equipment, and management software.

- Energy Infrastructure: The company offers renewable energy, AIoT-based energy management, and power conversion technologies for low-carbon transformation. Solutions include solar inverters, wind power converters, energy storage, EV charging, and management platforms for smart grids. It also supplies semiconductor high-voltage power supplies, automated testing instruments, and variable-frequency drives for industrial processes.
- Display and Visualization: Delta's display technologies serve control rooms, public information systems, projection mapping, and exhibition or event spaces. Offerings range from large-scale and 8K projection systems to integrated video-wall solutions and can be customized for mission-critical operations.

Exhibit 103: Delta: Key offerings for the infrastructure segment

INFRASTRUCTURE							
ICT Infrastructure							
Uninterruptible Power Supplies (UPS)	Battery Backup System (BBU)	800 VDC In-Row Power	Power Distribution Unit (PDU)	Rack-Mount Remote Power Panel (rRPP)	Liquid and Air Cooling	Modular Rack & Rack Mount Accessories	Lithium-ion UPS Battery System
Ethernet Switches	Indoor/Outdoor Telecom Power Systems	Renewable Hybrid Power Systems	Small Cell Power Systems	Li-ion Battery Backup Systems	Campus/Enterprise Networks		
Energy Infrastructure							
EV Charger	Power Conversion System (PCS)	All-in-One ESS solution	Distributed ESS Solutions	Solar Inverter	Wind Power Converter	Solid State Transformer Platforms	Medium Voltage Drives
High Voltage Power	DC EV Charger ATS	Power Supply ATS	Solid Oxide Fuel Cell (SOFC)	Hydrogen Energy Solution	Active Power Filter (APF) / Static Var Generator (SVG)		
Display and Visualization							
Business and Education Multimedia Projector	High Performance Projector	DLP, LED and LCD Video Wall	Wireless Collaboration and Signage	Vivitek			
				vivitek			

Source: Company data, Morgan Stanley Research

3. Mobility

Delta applies its power electronics and motor technologies to smart mobility, supplying

power systems, drive technologies, and integrated powertrain solutions. It works with automakers to develop electric vehicles, positioning the business as part of the low-carbon transition and sustainable urban mobility.

- EV Powertrain System: The company's powertrain portfolio for EV makers includes On-board Charging Modules (OBCM), DC/DC converters, power electronics combos, and traction inverters, along with X-in-One integrated solutions.

Exhibit 104: Delta: Key offerings for the mobility segment

MOBILITY						
EV Powertrain System						
EV Power Electronics	Traction Products	x-EV Power Management	In-Vehicle Network and AI Driving Computing	Automotive Thermal	Low Voltage Powertrain	EV Charging

Source: Company data, Morgan Stanley Research

4. Automation

Delta provides automation solutions for industrial and building applications through integrated software and hardware. Its industrial automation offerings support smart production and operational efficiency, while its building automation solutions target energy-efficient, secure, and sustainable spaces.

- Industrial Automation: This segment provides automation solutions across industries by utilizing AI, big data, IoT, and digitized manufacturing models. Relevant offerings support diverse production needs, including customized, small-volume and high-variety production, as well as mass production.
- Building Automation: Delta's Building Automation Solutions use an IoT architecture and integrate technologies from five brands: Amerlux lighting, Delta Controls and LOYTEC IP-based building control, and March Networks and VIVOTEK surveillance. The management platform enables operators to manage energy use, occupant comfort, and sustainability goals through a single interface.

Exhibit 105: Delta: Key offerings for the automation segment

AUTOMATION							
Industrial Automation							
Inverters - AC Motor Drives	Drive System for Motion Control Solutions	Motors	Servo Systems - AC Servo Motors and Drives	CNC Solution	PC-Based Motion Control Solution	Linear Motion	Embedded Motion Controller
Industrial Ethernet Solution	Temperature Controllers	High Voltage Power	SCARA Robot	Electric Servo Press			
Building Automation							
Delta Controls	LOYTEC	Indoor Air Quality Monitor	LED Lighting - Industrial/ Road/Fishing/Commercial	Smart Surveillance - VIVOTEK	Healthy Lighting		

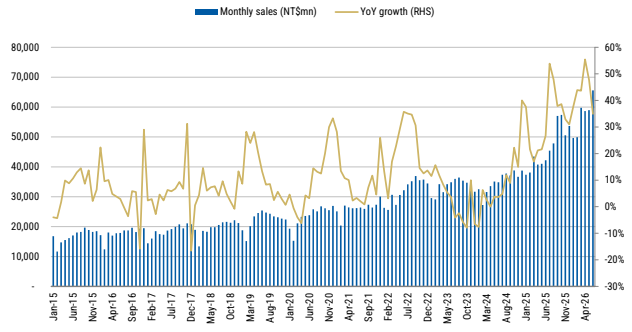
Source: Company data, Morgan Stanley Research

Monthly sales vs. share price

Delta's monthly sales show that seasonality has been an important factor over the years. Following a substantial increase in monthly sales since 2024, traditional seasonal patterns have become less pronounced as revenue from AI-related offerings has started to outweigh them, while liquid-cooling-related offerings have added a new layer of monthly sales volatility. The correlation between the company's monthly sales and share price movements has remained high, at ~87% since 2023.

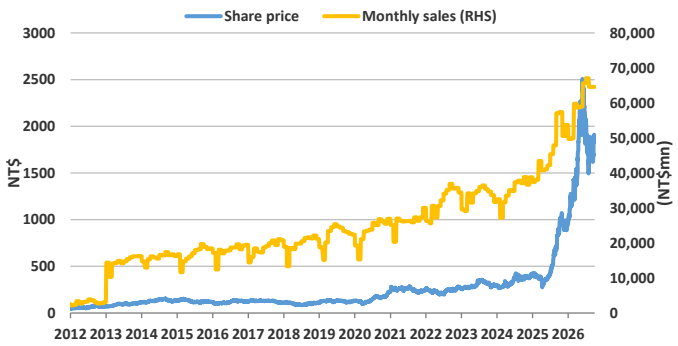
The company reported August monthly sales of NT$64,586mn (-4% MoM/+35% YoY). We saw the Power Electronics segment decline sequentially from a high base, offset by growth in Automation and slight growth in the Infrastructure segment.

Exhibit 106: Delta: Monthly sales vs. YoY sales growth



Source: TEJ, Morgan Stanley Research

Exhibit 107: Delta: Monthly sales vs. share price



Source: TEJ, Morgan Stanley Research

Key management & shareholding structure

Delta's board currently consists of eight directors and five independent non-executive directors. Its founder, Chung-Hua Cheng (Bruce Cheng), and his sons, Ping Cheng and An Cheng, together owned a 7% stake as of March 2026. Excluding the founder and his family, the management team held a stake of ~0.1% as of March 2026.

The FINI holding percentage as of July 14, 2026, was 63%, versus a previous peak of 78% in February 2011 and a historical average (since 2010) of 69%.

Exhibit 108: Delta: Shareholding structure

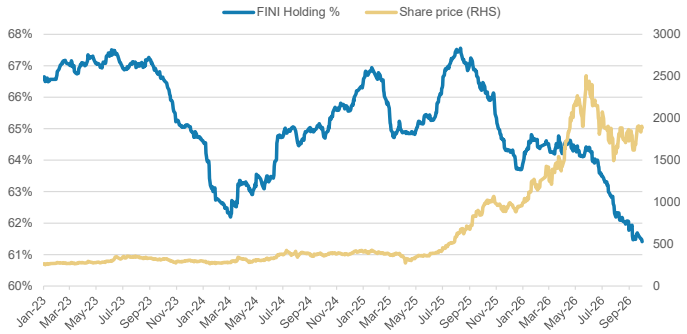
Shareholders	%
Dejico International LTD	10.30%
Deltron Holding LTD	8.40%
New Labor Retirement Pension Fund	3.75%
Chung-Hua Cheng (Bruce CH Cheng)*	3.15%
Government of Singapore-GOS-EFM C	2.28%
Ping Cheng*	2.14%
An Cheng (Victor Cheng)*	1.94%
Yuanta/P-shares Taiwan Top 50 ETF	1.43%
Labor Retirement Reserve Fund (The Old Fund)	1.42%
Labor Insurance Fund	1.32%
Total	**36.13%**

Source: Company data, Morgan Stanley Research. Note: Data as of March 2026. (*) are shown as insiders.

Exhibit 109: Delta: FINI holdings vs. share price since 2010

Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 110: Delta: FINI holdings vs. share price since 2023



Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 111:

Delta: Management profiles

Name	Title	Education	Work Experience
Bruce C.H. Cheng	Founder and Honorary Chairman Chairman, Delta Electronics Foundation	Bachelor of Electrical Engineering — National Cheng Kung University	- 2012–present — Honorary Chairman, Delta Electronics - 1990–present — Founded and chairs the Delta Electronics Foundation - 1971–2012 — Founder and Chairman, Delta Electronics - Before 1971 — Worked at TRW and Air Asia Co.
Ping Cheng	Chairman and CEO	Bachelor of Business Administration — California State University, East Bay	- Joined Delta in 1988 - 2024–present — Chairman - 2012–present — CEO of Delta Electronics - 2010–2012 — Corporate Senior Vice President and President of China Region - 2008–2010 — Senior Vice President & General Manager of Power Supply Business Group - 1998–2008 — Deputy General Manager of Component Business Group (Delta China); Executive Vice President of China Region
Mark Ko	Vice Chairman	Bachelor of Control Engineering — National Chiao Tung University	- Joined Delta in 1988 - 2012–present — Vice Chairman of Delta Electronics - 2004–2012 — President and Chief Operating Officer of Delta Group - 1994–2004 — General Manager of Power System Business Group
Simon Chang	Board Member President and COO	EMBA — Chung Yuan Christian University	- Joined Delta in 1981 - 2017–present — Board Member; President and Chief Operating Officer, Delta Electronics - Before 2017 — Senior Vice President and General Manager of Industrial Automation Business Group - From 1995 — Director of Industrial Automation Division
Victor Cheng	Board Member CEO of Delta Electronics (Thailand) PCL.	-Master of Electrical Engineering — Santa Clara University -Bachelor of Electrical Engineering — Santa Clara University	- Joined Delta in 1993 - 2024–present — Board Member of Delta; CEO of Delta Electronics (Thailand) PCL. - 2023–2024 — Executive Vice President of Infrastructure Business - 2017–2023 — General Manager of Information and Communications Technology Business Group - 2014–2017 — General Manager of Power System Business Group - 2002–2019 — President, then Chairman of Delta Networks - 1999–2002 — Head of Video Display Business Unit - 1993–1999 — International Sales Director to Assistant General Manager of Video Display Business Unit
Shan-Shan Guo	Board Member Chief Brand Officer Vice Chairman, Delta Electronics Foundation	- Master of Marketing and Communications — University of Michigan - Bachelor of Foreign Languages and Literatures — National Taiwan University	- Joined Delta in 2010 - 2011–present — Chief Brand Officer of Delta
Johnson Lee	Executive Vice President Infrastructure Business (IFB)	- EMBA — National Taiwan University - Bachelor of Industrial Engineering and Systems Management — Feng Chia University	- Joined Delta in 1985 - 2024–present — Executive Vice President of Infrastructure Business - 2012–2017 — Chief Operating Officer of Delta Electronics - 2004–2012 — General Manager of Power Supplies Business Group - Earlier — Procurement manager; General Manager of materials management
Ted Shyy	Executive Vice President Power Electronics Business (PEB)	Master in Materials Science and Engineering — National Cheng Kung University	- Joined Delta in 1988 - 2024–present — Executive Vice President of Power Electronics Business - 2014–present — Chief Executive Officer of Cyntec (concurrent role) - 2011–2014 — General Manager of Delta Components Business Group - 2003–2014 — General Manager of Cyntec - 1991–2003 — Rotation in Research and Development, production and marketing, under Cyntec, a joint venture between Delta and Japan's SUSUMU
James Tang	Executive Vice President Mobility Business (EVB)	- EMBA — National Taiwan University - Bachelor of Management Science — National Cheng Kung University	- Joined Delta in 1988 - 2024–present — Executive Vice President of Mobility Business; continues as General Manager of Electric Vehicle Solutions Business Group - 2017–present — General Manager of Electric Vehicle Solutions Business Group - 2008–2016 — General Manager across Power Supplies, Consumer and Lighting Solutions, and Embedded Power Supplies business groups - 1988–2008 — Procurement; Materials Manager; Director of Personal Computer Business Group
Jimmy Yiin	Executive Vice President Global Business Operations (GBO)	Master of Business Administration — National Taiwan University and Fudan University	- Joined Delta in 1992 - 2023–present — Executive Vice President of Global Business Operations - 2019–2022 — General Manager of Power System Business Group - 2014–2019 — Vice Chairman of Delta Greentech (China); General Manager of Delta Electronics (Shanghai) - 2008–2013 — General Manager of Delta Electronics (Jiangsu) - 1999–2007 — Purchasing manager/materials section chief of Delta Dongguan and Delta Networks
Rock Huang	Executive Vice President Global Manufacturing	Master of Industrial Engineering and Management — National Taipei University of Technology	- Joined Delta in 1987; ~40 years of experience in the manufacturing field - 2026–present — Executive Vice President of Global Manufacturing - 2020–2026 — President of Global Manufacturing - 2019–2020 — President of Delta China - 2018–2019 — Executive Vice President of Delta China - 2007–2018 — General Manager of Southern China, leading Dongguan team

Source: Company data, Morgan Stanley Research

Risk Reward – Lite-On Technology (2301.TW)

Stronger ASIC server power supply but fair risk-reward

PRICE TARGET NT$315.00

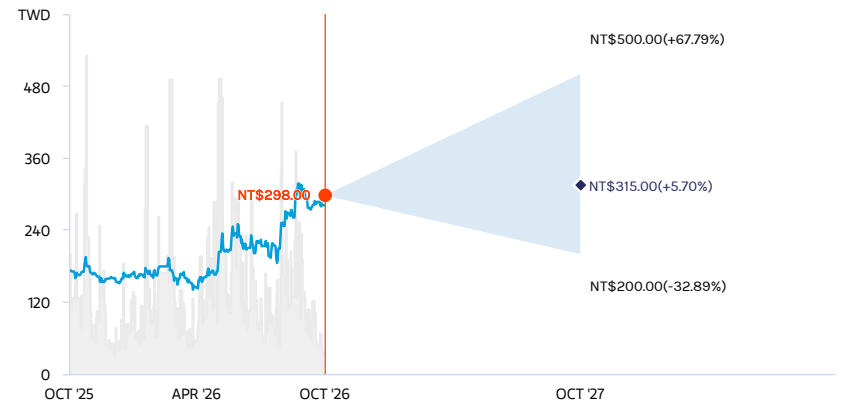
Base case: P/E multiple of 17x based on average 2027-28 EPS. We believe our target multiple is justified by the 47% earnings CAGR we estimate for 2025-28, implying a PEG of 0.4, versus the stock's 16x P/E and 2% earnings CAGR during 2022-25, which implied a PEG of 7.2.

Consensus Price Target Distribution NT$176.00 NT$300.50 MS PT NT$400.00

Source: Refinitiv, Morgan Stanley Research

◆ Mean ◆ Morgan Stanley Estimates

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- We expect Lite-On Tech to benefit from strong AI server power demand in 2026-27, with the company gaining more traction on ASIC platforms.
- We are mindful of weakness in consumer/IT demand amid rising input costs, which could weigh on Lite-On Tech's profit in 2026-27.
- Liquid-cooling sidecar CDU system shipments may take more time to ramp up.

Consensus Rating Distribution

77% Overweight
23% Equal-weight
0% Underweight
● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian:	*Negative*
New Data Era:	*Positive*
Technology Diffusion:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$500.00

27x average 2027-28e P/E

Better-than-expected margin expansion: Operating efficiency and accelerating production automation drive margin expansion, while new-business revenue comes in earlier than expected. Growth in cloud computing and automotive electronics is stronger than expected.

BASE CASE NT$315.00

17x average 2027-28e P/E

AI server projects progress, but some business uncertainties remain: We expect margin improvement from a better product mix as AI server power supply projects ramp up. However, we are mindful of weakness in consumer/IT demand amid rising input costs.

BEAR CASE NT$200.00

11x average 2027-28e P/E

Operating margin contraction: Operating efficiency improves less than expected, while pricing competition intensifies amid macro volatility; automotive electronics take longer to ramp up.

Risk Reward – Lite-On Technology (2301.TW)

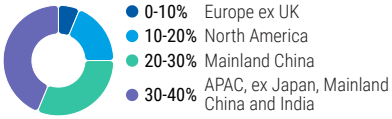
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Opto-electronics OPM (%)	8.3	7.9	7.5	7.5
Information Technologies OPM (%)	9.2	7.7	7.8	8.6
Opto-electronics revenue YoY (%)	0.2	8.0	1.7	2.3
Information Technologies revenue YoY (%)	7.7	(5.2)	0.5	(0.2)

INVESTMENT DRIVERS

- LED adoption (outdoor lighting/auto)
- Margin expansion
- Raw material prices, particularly for steel and copper

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger optical business growth
- Stronger data center/server demand
- PC/NB demand growth
- Downtrend in raw material prices

RISKS TO DOWNSIDE

- Macro demand weakness
- Weaker data center/server demand
- PC/NB demand slowdown
- Uptrend in raw material prices

OWNERSHIP POSITIONING

Inst. Owners, % Active 65.2%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	210,118	224,829 / 219,960	229,691
EBITDA (NT$, mn)	31,349	33,549 / 34,893	38,115
Net income (NT$, mn)	23,541	25,187 / 24,910	26,835
EPS (NT$)	10.30	11.07 / 10.88	11.79

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Financial Summary – Lite-On

Exhibit 112: Financial Summary – Lite-On

Consolidated Income Statement

NT$ mn	2024	2025	2026E	2027E	2028E
Net sales	**137,134**	**166,085**	**224,829**	**297,964**	**357,670**
COGS	-107,504	-128,046	-168,703	-219,712	-259,785
Gross profit	**29,630**	**38,039**	**56,126**	**78,252**	**97,885**
Operating expenses	-16,648	-21,301	-26,044	-33,189	-37,645
- Promotion	-5,184	-7,629	-6,902	-8,782	-10,247
- ADM	-3,990	-4,965	-7,384	-9,340	-10,162
- R&D	-7,476	-8,721	-11,787	-15,037	-17,200
Operating income	**12,981**	**16,738**	**30,082**	**45,063**	**60,240**
Non-operating income	2,653	2,975	2,876	2,500	2,500
Interest income	1,491	1,119	799	800	800
Investment income	2,808	-956	1,436	1,200	1,200
Disposal of investment	75	-537	-77	0	0
Exchange gain	2,808	1,091	1,390	0	0
Other	-4,528	2,258	-672	500	500
Pre-tax income	**15,635**	**19,713**	**32,958**	**47,563**	**62,740**
Income tax	-3,674	-4,601	-7,767	-11,227	-14,793
Minority	19	-4	3	0	0
Reported net income	**11,942**	**15,116**	**25,187**	**36,337**	**47,947**
EPS (NT$)	**5.21**	**6.64**	**11.07**	**15.97**	**21.07**

Consolidated Cash Flow Statement

NT$ mn	2024	2025	2026E	2027E	2028E
Cash flow from Operations	**12,411**	**14,533**	**34,885**	**43,140**	**51,224**
Net profits	11,942	15,116	25,187	36,337	47,947
Depreciation + Amortization	4,047	3,668	3,994	4,065	4,137
Inv. Gain/Loss (Equity)	-15	-86	0	0	0
Disposal of Investment	9	568	0	0	0
Working Capital Change	-3,886	-2,099	-12,395	-12,092	-10,109
Other adjustments	313	-2,635	18,098	14,830	9,248
Cash flow from Investing	**-4,843**	**-7,395**	**-13,287**	**-8,993**	**-11,919**
Capex	-3,283	-6,795	-13,287	-8,993	-11,919
Change of LT investment	-2,234	-43	0	0	0
Change of ST investment	1,453	-417	0	0	0
Other adjustments	-779	-140	0	0	0
Cash flow from financing	**-3,363**	**-23,099**	**-19,866**	**-28,659**	**-37,818**
Increase in L/T debt	0	0	0	0	0
Increase in S/T debt	7,584	-10,116	0	0	0
Cash Dividend Paid	-10,372	-10,297	-19,866	-28,660	-37,818
Issuance of stock	0	0	0	0	0
Other adjustments	-574	-2,686	0	0	0
Exchange rate adjustment	3,736	-3,748	0	0	0
Net change in cash	**7,942**	**-19,709**	**1,732**	**5,487**	**1,487**

Consolidated Balance Sheet

NT$ mn	2024	2025	2026E	2027E	2028E
Cash	100,684	80,974	82,706	88,193	89,680
Mkt Securities	933	1,009	1,009	1,009	1,009
AR/NR	37,182	42,318	60,960	80,790	96,978
Inventory	26,817	34,262	42,083	54,807	64,804
Others	3,501	5,458	5,458	5,458	5,458
Current Assets	**169,117**	**164,021**	**192,216**	**230,257**	**257,928**
Long-term investments	8,195	7,748	7,748	7,748	7,748
Fixed assets	18,776	24,752	34,495	39,873	48,104
Deferred Asset	1,628	2,246	2,246	2,246	2,246
Intangible Assets	3,156	3,060	2,610	2,160	1,710
Other assets	3,811	3,860	3,860	3,860	3,860
Total Assets	**204,683**	**205,687**	**243,175**	**286,144**	**321,597**
S/T borrowings	30,187	20,029	20,029	20,029	20,029
AP/NP	43,126	53,608	67,676	88,138	104,213
Other ST liabilities	34,556	35,960	54,059	68,888	78,137
Total Current Liabilities	**107,869**	**109,597**	**141,764**	**177,056**	**202,379**
LT debt(include CB)	3,000	3,000	3,000	3,000	3,000
Other LT liabilities	2,903	3,111	3,111	3,111	3,111
Total Liabilities	**113,772**	**115,709**	**147,875**	**183,167**	**208,491**
Common shares	23,473	23,168	23,168	23,168	23,168
Retained Earnings	46,277	42,393	47,714	55,390	65,519
Other shareholders' equity	21,161	24,418	24,418	24,418	24,418
Total Equity	**90,911**	**89,979**	**95,300**	**102,976**	**113,105**
Total Liab. & Shldr's Equity	**204,683**	**205,687**	**243,175**	**286,143**	**321,596**

Consolidated Financial Ratios

	2024	2025	2026E	2027E	2028E
Margins					
Gross margin	21.6%	22.9%	25.0%	26.3%	27.4%
Operating margin	9.5%	10.1%	13.4%	15.1%	16.8%
Pretax margin	11.4%	11.9%	14.7%	16.0%	17.5%
Net margin	8.7%	9.1%	11.2%	12.2%	13.4%
YoY growth					
Sales	-7.6%	21.1%	35.4%	32.5%	20.0%
Operating profits	-10.6%	28.9%	79.7%	49.8%	33.7%
Pretax profits	-14.9%	26.1%	67.2%	44.3%	31.9%
Net profits	-18.0%	26.6%	66.6%	44.3%	32.0%
EPS	-18.1%	27.5%	66.6%	44.3%	32.0%
Net Debt/Equity	-75.3%	-65.5%	-63.7%	-64.3%	-59.8%
Liabilities/Equity	125.1%	128.6%	155.2%	177.9%	184.3%
Liabilities/Assets	55.6%	56.3%	60.8%	64.0%	64.8%
ROAE	13.6%	16.7%	27.2%	36.7%	44.4%
ROAA	6.1%	7.4%	11.2%	13.7%	15.8%
AR/NR Turnover (days)	92.2	87.4	83.8	86.8	90.7
AP/NP Turnover (days)	142.7	137.9	131.2	129.4	135.1
Inventory Turnover (days)	89.3	87.1	82.6	80.5	84.0
Cash conversion cycle (days)	38.9	36.5	35.2	37.9	39.6

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 113: Lite-on: Quarterly Financials

NT$ mn	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	2025	2026E	2027E
Net sales	**36,417**	**40,421**	**44,891**	**44,356**	**43,407**	**52,703**	**59,526**	**69,193**	**166,085**	**224,829**	**297,964**
COGS	-28,195	-31,479	-33,648	-34,724	-34,004	-38,355	-44,867	-51,477	-128,046	-168,703	-219,712
Gross profit	**8,222**	**8,942**	**11,243**	**9,633**	**9,403**	**14,348**	**14,659**	**17,716**	**38,039**	**56,126**	**78,252**
Operating expenses	-4,546	-5,215	-6,560	-4,980	-5,304	-6,146	-6,851	-7,743	-21,301	-26,044	-33,189
- Promotion	-1,532	-2,051	-2,535	-1,512	-1,500	-1,623	-1,786	-1,993	-7,629	-6,902	-8,782
- ADM	-998	-1,048	-1,792	-1,127	-1,457	-1,783	-1,964	-2,180	-4,965	-7,384	-9,340
- R&D	-2,023	-2,110	-2,244	-2,345	-2,341	-2,787	-3,095	-3,563	-8,721	-11,787	-15,037
Operating income	**3,675**	**3,727**	**4,683**	**4,653**	**4,099**	**8,203**	**7,807**	**9,973**	**16,738**	**30,082**	**45,063**
Non-operating income	994	536	886	560	871	1,166	445	395	2,975	2,876	2,500
Net interest income	424	323	184	188	214	195	195	195	1,119	799	800
Net investment income	596	-3,235	1,389	293	99	1,237	50	50	-956	1,436	1,200
Net gain of investment dispos	137	-702	23	5	1	-78	0	0	-537	-77	0
Exchange gain	379	2,540	-751	-122	284	1,106	0	0	1,091	1,390	0
Others	-542	1,609	41	195	272	-1,295	200	150	2,258	-672	500
Pre-tax income	4,669	4,262	5,569	5,213	4,970	9,368	8,252	10,368	19,713	32,958	47,563
Income tax	-1,214	-1,108	-924	-1,355	-1,195	-2,248	-1,939	-2,385	-4,601	-7,767	-11,227
Net income	**3,456**	**3,155**	**4,645**	**3,860**	**3,775**	**7,116**	**6,313**	**7,983**	**15,116**	**25,187**	**36,337**
EPS (NT$)	**1.51**	**1.39**	**2.05**	**1.70**	**1.66**	**3.14**	**2.77**	**3.51**	**6.64**	**11.07**	**15.97**
Margins											
Gross margin	22.6%	22.1%	25.0%	21.7%	21.7%	27.2%	24.6%	25.6%	22.9%	25.0%	26.3%
Operating margin	10.1%	9.2%	10.4%	10.5%	9.4%	15.6%	13.1%	14.4%	10.1%	13.4%	15.1%
Pre-tax margin	12.8%	10.5%	12.4%	11.8%	11.4%	17.8%	13.9%	15.0%	11.9%	14.7%	16.0%
Net margin	9.5%	7.8%	10.3%	8.7%	8.7%	13.5%	10.6%	11.5%	9.1%	11.2%	12.2%
QoQ Growth											
Sales	-4.9%	11.0%	11.1%	-1.2%	-2.1%	21.4%	12.9%	16.2%			
Gross profit	1.0%	8.8%	25.7%	-14.3%	-2.4%	52.6%	2.2%	20.9%			
Operating profit	5.1%	1.4%	25.7%	-0.6%	-11.9%	100.1%	-4.8%	27.7%			
Pre-tax profit	17.1%	-8.7%	30.7%	-6.4%	-4.7%	88.5%	-11.9%	25.6%			
Net profit	13.3%	-8.7%	47.2%	-16.9%	-2.2%	88.5%	-11.3%	26.5%			
YoY Growth											
Sales	26.6%	21.4%	22.1%	15.8%	19.2%	30.4%	32.6%	56.0%	21.1%	35.4%	32.5%
Gross profit	40.6%	21.0%	36.3%	18.3%	14.4%	60.5%	30.4%	83.9%	28.4%	47.5%	39.4%
Operating profit	64.8%	13.6%	19.3%	33.1%	11.5%	120.1%	66.7%	114.3%	28.9%	79.7%	49.8%
Pre-tax profit	49.2%	4.4%	25.6%	30.7%	6.4%	119.8%	48.2%	98.9%	26.1%	67.2%	44.3%
Net profit	44.7%	1.3%	37.0%	26.6%	9.3%	125.5%	35.9%	106.8%	26.6%	66.6%	44.3%

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

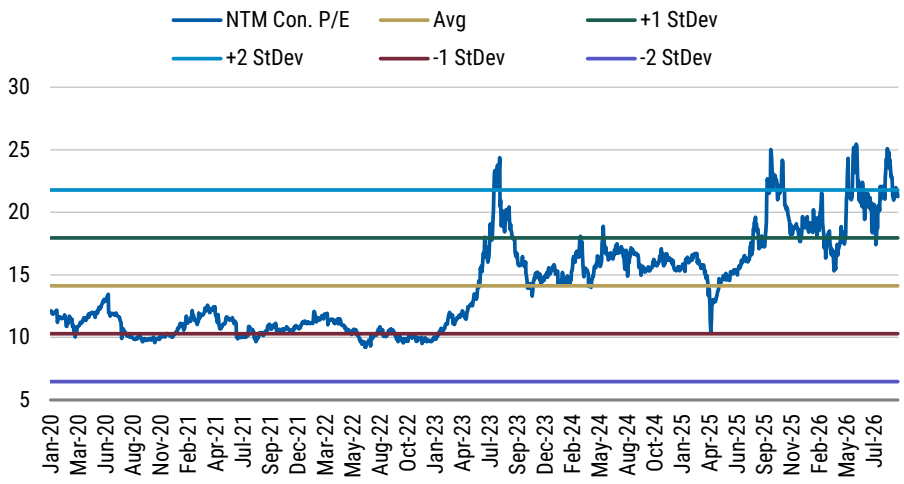
Valuation Methodology – Lite-On

Our price target (base case scenario value) is NT$315, derived from 17x 2027-28e P/E: We use price/earnings as our valuation methodology because we believe it appropriately captures Lite-On's earnings growth potential. We believe our target P/E multiple of 17x is justified by an earnings CAGR of 47% over 2025-28, implying a PEG of 0.4. This is lower than the average PEG of 7.2 in 2022-25, when the stock was trading at an average P/E of 16x for a 2% earnings CAGR.

Bull case scenario value – NT$500: We apply a 27x P/E multiple to our average base case estimate for 2027-28. This scenario assumes better-than-expected margin expansion, driven by improved operating efficiency and accelerating production automation, while new business revenue comes in earlier than expected. Growth in cloud computing and automotive electronics is stronger than expected.

Bear case scenario value – NT$200: We apply an 11x P/E multiple to our average base case estimate for 2027-28. This scenario assumes weaker-than-expected improvement in operating efficiency, while pricing competition intensifies amid macro volatility; automotive electronics take longer to ramp up.

Exhibit 114: Lite-on: NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Global Footprint – Lite-On

More capacity expansion to come

Lite-On's production footprint spans the globe, with 15 sites in total, although it remains heavily weighted toward China. The company has been expanding aggressively, both to meet strong demand for AI power supplies and to support its strategy of geographic risk diversification and proximity to customers.

Looking at its expansion plans in recent years, the company has continued to demonstrate its commitment to diversifying production away from China and increasing proximity to customers. It completed its Thailand plant expansion and announced the second-phase expansion of its Vietnam facility in 2021. It then established automotive electronics and cloud power production lines in Dallas, Texas, in 2H22, along with the official launch of the second phase of its Hai Phong plant in Vietnam, which focuses on networking and consumer products. In 2023, it began expanding its second Kaohsiung plant and announced additional capacity expansion in Vietnam. In 2024, it committed a further US $120mn to establish a wholly owned subsidiary in Vietnam and began progressively shifting production of high-end power products to that facility. In 2025, the company aimed to double capacity as demand for AI server power supplies and battery backup units (BBUs) surged, focusing primarily on capacity expansion in the US, Kaohsiung, and Vietnam.

For 2026, the company expects capacity allocation of ~40% in China, 27% in Vietnam, 20% in Taiwan, and 10% in the US, with the remaining non-China footprint (i.e., Mexico, Brazil, and Thailand) accounting for ~3% of total capacity. It has also announced investments of US$919mn and US$149mn in its US and Vietnam subsidiaries, respectively, to continue expanding power-related capacity, mainly for PSUs. The capex budget for 2026 has also been raised to ~NT$13bn, versus ~NT$7bn in 2025.

Exhibit 115: Lite-on's global footprint

Source: Lite-on.

Financial Analysis – Lite-On

29% revenue CAGR in 2025-28

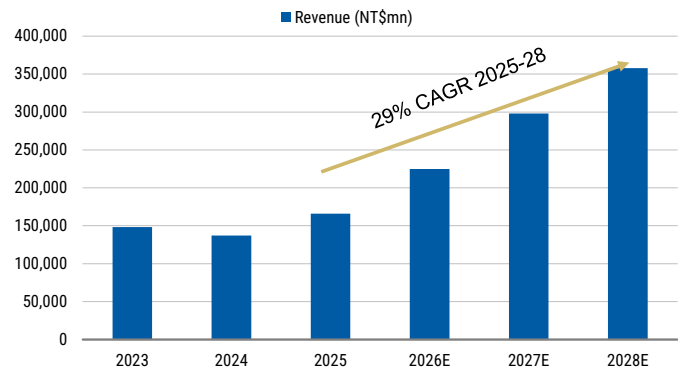
Lite-On's revenue growth was relatively moderate during 2020-22, with revenue expanding at a CAGR of 5% over the period. The company benefited from surging cloud computing demand, which drove increasing demand for power supplies, along with rising demand from EV/automotive electronics, although growth was offset by elevated component costs and supply-chain disruptions during COVID-19. Revenue then fell by 14% in 2023 and 8% in 2024 due to a post-pandemic demand pullback across the consumer electronics market and PC/NB model transitions, coupled with softening demand from EV-related businesses. The IT segment declined 20%/11% YoY, while the opto-electronics segment declined 11%/9% YoY in 2023/2024, respectively.

Revenue rebounded in 2025, growing 21% YoY, as the company began to benefit from surging demand for AI/cloud server power supplies. Contributions from ramping high-voltage/AI power infrastructure products began to increase, coupled with a partial recovery in the IT segment, mainly fueled by power adapter shipments for game consoles and gaming PCs/NBs, as well as related power offerings for low-Earth-orbit applications.

Looking ahead, we estimate revenue to expand at a 29% CAGR in 2025-28e, driven by the AI server power demand boom, while we remain mindful of weakness in consumer/IT demand amid elevated component costs.

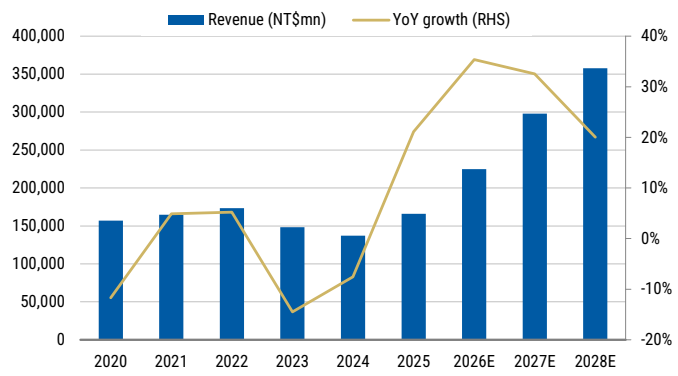
By product segment, IT-related products were the primary revenue driver in past years, while 2025 marked a crossover point, with the contribution from the Cloud & AIoT segment surpassing that of the IT segment, mainly supported by AI power-related offerings. IT revenue contribution declined from 72% in 2020 to 38% in 2025, and we estimate it will reach 17% by 2028. For the Cloud & AIoT segment, revenue contribution grew substantially from 27% in 2021 (0% in 2020) to 45% in 2025; we estimate it will account for 74% of total revenue by 2028, on the back of surging AI power-related demand. For opto-electronics, we expect revenue contribution to decline from 17% in 2025 to 9% in 2028.

Exhibit 116: Lite-on: Revenue CAGR, 2025-28e



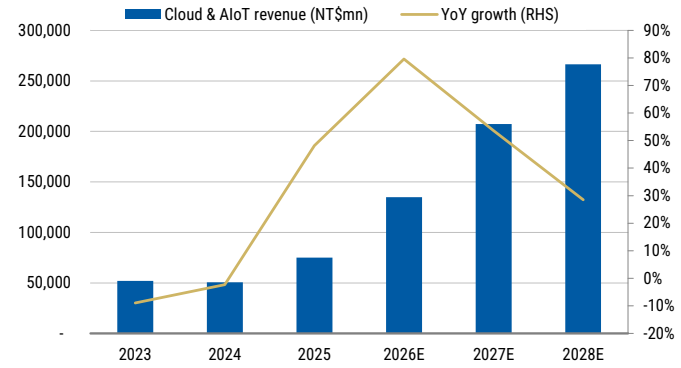
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 117: Lite-on: Annual revenue growth



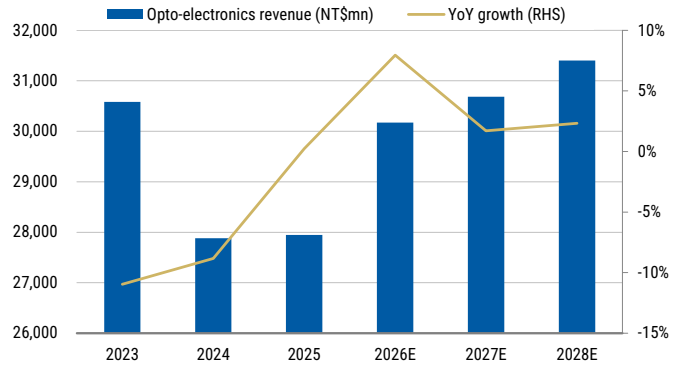
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 118: Lite-on: Cloud & AIoT revenue



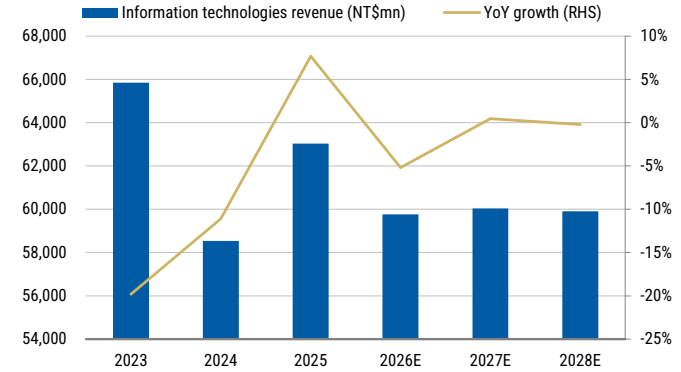
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 119: Lite-on: Opto-electronics revenue



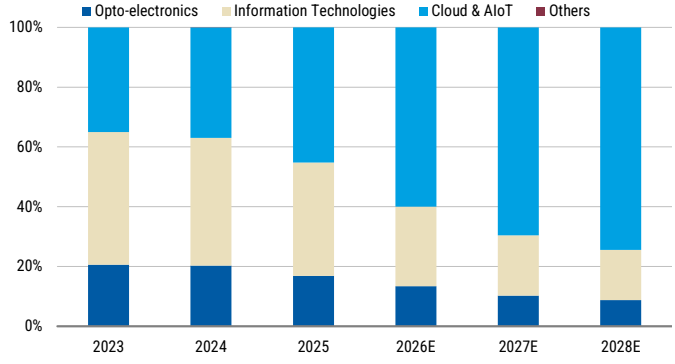
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 120: Lite-on: IT revenue



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 121: Lite-on: Revenue breakdown by segment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

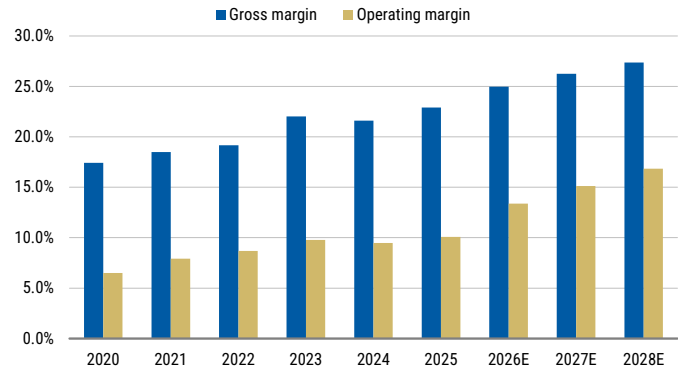
Continuing margin expansion

Lite-On's gross margin rose steadily over the years, from 17.4% in 2020 to 22.0% in 2023, thanks to a continually improving product mix. Gross margin dipped to 21.6% in 2024, mainly due to a smaller revenue base, while contributions from AI/cloud had yet to scale and margins were weighed down by learning-curve costs at the new Vietnam site. It then rose to 22.9% in 2025 as the AI server power business began to scale alongside rising BBU (battery backup unit) shipments, resulting in a more favorable product mix.

The opex ratio was largely stable at 10.5-11.0% during 2020-22. It then increased to 12.2% in 2023 and 12.1% in 2024, mainly due to operating deleverage and higher R&D costs. The opex ratio edged up further to 12.8% in 2025, mainly due to ongoing investment in R&D to support new business development in AI server power supplies, BBUs, liquid cooling, etc., along with a one-off tariff prepayment.

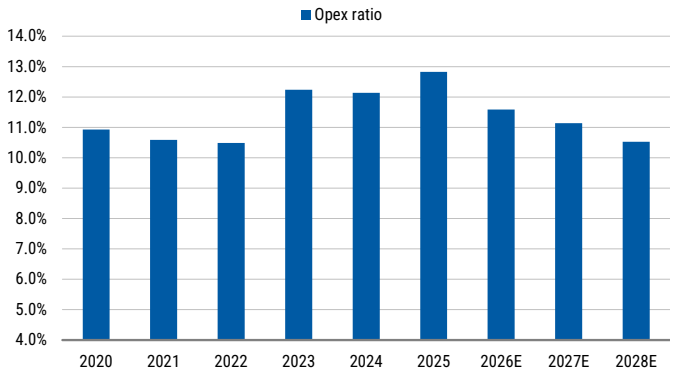
Looking ahead, we expect margins to continue expanding on a favorable product mix. We now pencil in a 2.1ppt gross margin improvement in 2026 to 25.0%, followed by expansions of 1.3ppt and 1.1ppt in 2027 and 2028, to 26.3% and 27.4%, respectively. For the opex ratio, we expect it to decline sequentially as the revenue base expands. Accordingly, we estimate operating margin to improve by 3.3ppt to 13.4% in 2026, followed by expansions of 1.7ppt in both 2027 and 2028, reaching 15.1% and 16.8%.

Exhibit 122: Lite-on: Margin trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 123: Lite-on: Opex ratio



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Net profit to grow at a 47% CAGR in 2025-28

Operating profit grew steadily alongside gross margin expansion during 2020-22, from NT $10,206mn to NT$15,073mn. It then fell 4% YoY in 2023 and a further 11% YoY in 2024 due to higher operating expenses, mainly from R&D costs, despite gross margin expansion in 2023. Operating margin increased by 1.1ppt YoY to 9.8% in 2023 amid a smaller revenue base, before declining by 0.3ppt YoY to 9.5% in 2024 on lower gross margin and an elevated opex ratio.

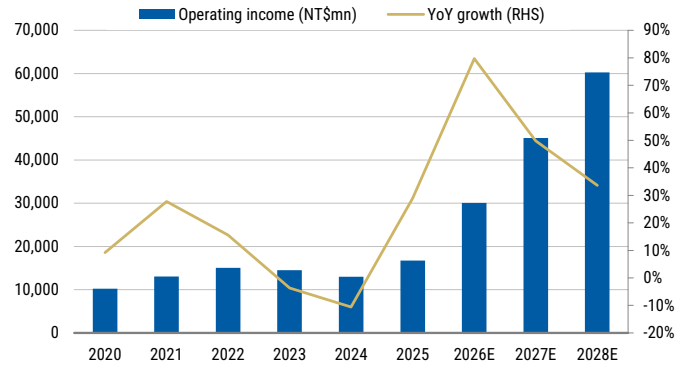
Operating profit rebounded by 29% YoY to NT$16,738mn in 2025, supported by both an expanding revenue base and strengthening gross margin. Operating margin also improved, increasing by 0.6ppt YoY to 10.1%, as the recovery in gross margin was partly offset by higher opex.

With the tax rate remaining relatively stable at 20-24%, Lite-On's net profit growth has generally tracked its operating profit growth. An exception occurred in 2023, when net profit grew 3% YoY despite a 4% decline in operating profit, mainly due to higher FX gains during the year.

Net margin has broadly tracked operating margin in recent years. It reached a record high of 9.8% in 2023 on improving gross margin, despite a higher opex ratio. It then declined to 8.7% in 2024 on gross margin erosion before rebounding to 9.1% in 2025, benefiting from increasing demand for AI power supplies.

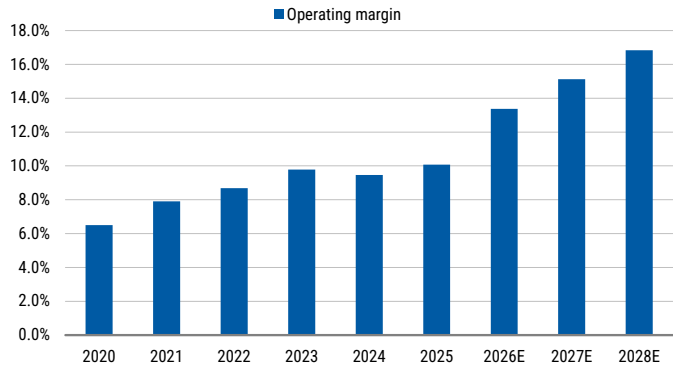
Looking ahead, we forecast net profit to grow 67% YoY in 2026e, followed by 44% YoY growth in 2027e and 32% YoY growth in 2028e, implying a 47% CAGR in 2025-28e. Net margin is likely to rise to 11.2% in 2026, 12.2% in 2027, and 13.4% in 2028.

Exhibit 124: Lite-on: 35% CAGR for operating profit in 2024-27



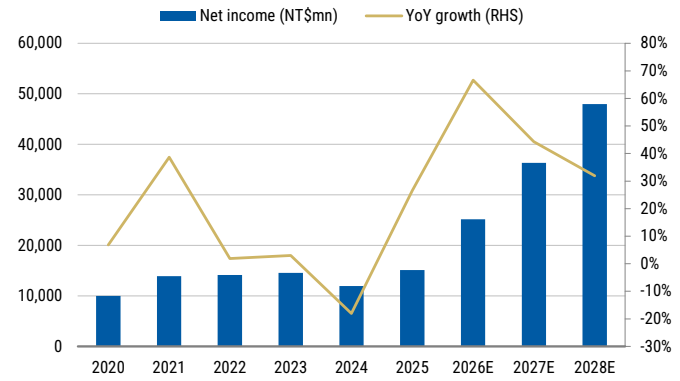
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 125: Lite-on: Operating margin



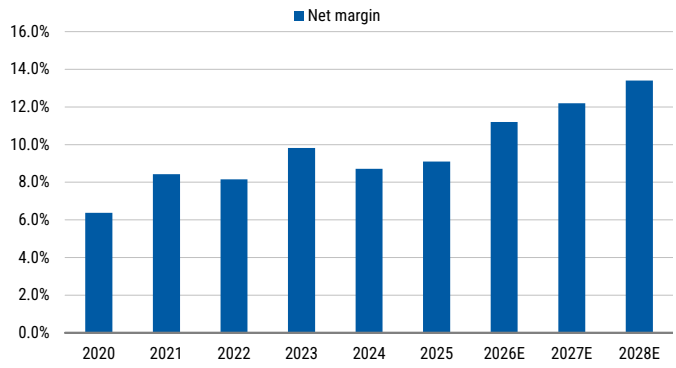
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 126: Lite-on: Net profit trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 127: Lite-on: Net margin



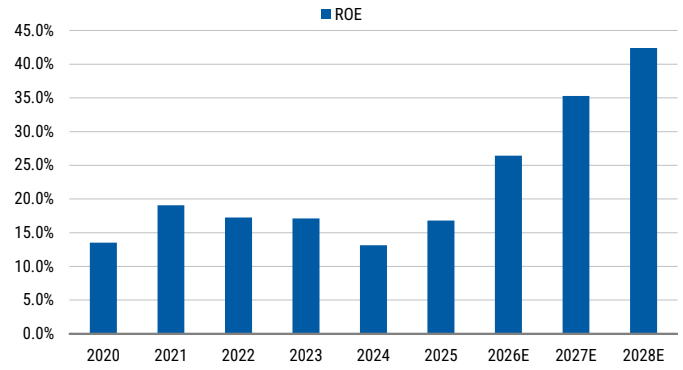
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

ROE to rise

ROE movements were highly correlated with changes in net margin during 2020-22, as asset turnover and financial leverage were largely stable over the period. ROE edged down slightly in 2023 despite higher net margin, which increased from 8.2% to 9.8%, due to lower asset turnover and financial leverage. ROE fell further to a trough of 13.1% in 2024 as net margin and asset turnover declined further. It then rebounded to 16.8% in 2025, primarily supported by net margin expansion, fueled by a favorable product mix, along with an improvement in asset turnover back to the 80% level.

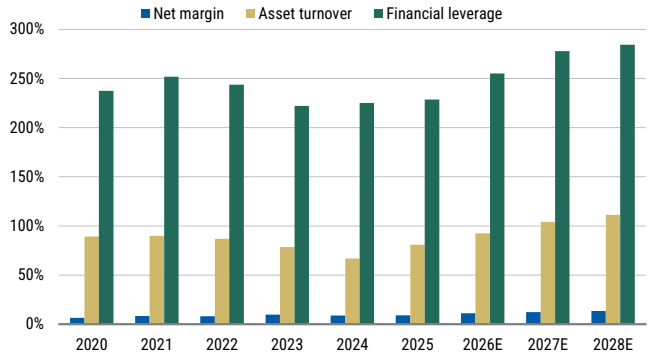
Looking ahead, we expect ROE to continue rising, reaching 26.4% in 2026, 35.3% in 2027, and 42.4% in 2028, primarily driven by continued margin expansion and improving asset turnover. Net margin is projected to increase from 9.1% in 2025 to 13.4% by 2028, supported by gross margin improvement from 22.9% to 27.4% and a decline in the operating expense ratio from 12.8% in 2025 to 10.5% in 2028.

Exhibit 128: ROE trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 129: DuPont analysis



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Improving balance sheet and continuous free cash flow generation

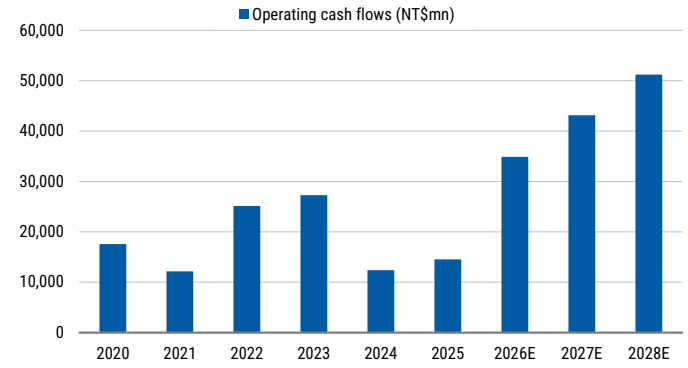
Lite-On has consistently generated positive operating cash flow in recent years, although cash flow dropped notably in 2021, 2024, and 2025, mainly due to higher working-capital-related cash outflows. The company has also consistently generated positive free cash flow, with capital expenditures generally at the NT$3-4bn level over the years. An exception occurred in 2025, when capital expenditure rose to NT$6.7bn, mainly for power-related capacity expansion in Vietnam, Taiwan, and the US.

As the company has announced investments of US$919mn and US$149mn in its US and Vietnam subsidiaries, respectively, alongside continued expansion of power-related capacity, mainly for PSUs, we now estimate capex to rise to ~NT$13bn in 2026, versus ~NT $7bn in 2025. Capex should then decline to NT$9bn in 2027 as investment levels normalize.

Lite-On's net cash position has remained relatively healthy, with its net cash-to-equity ratio improving from 65% in 2020 to 80% in 2023. It then declined to 75% in 2024 and 66% in 2025, as total debt increased by NT$7.7bn in 2024, while cash on hand declined in 2025, likely due to increasing capital investment.

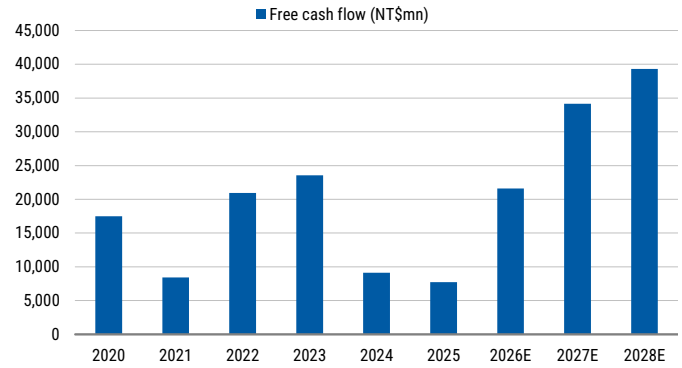
Looking ahead, we assume the total debt position will remain stable despite ongoing capex investment. The net cash position should continue to grow sequentially on the back of positive free cash flow generation.

Exhibit 130: Lite-on: Positive operating cash flow



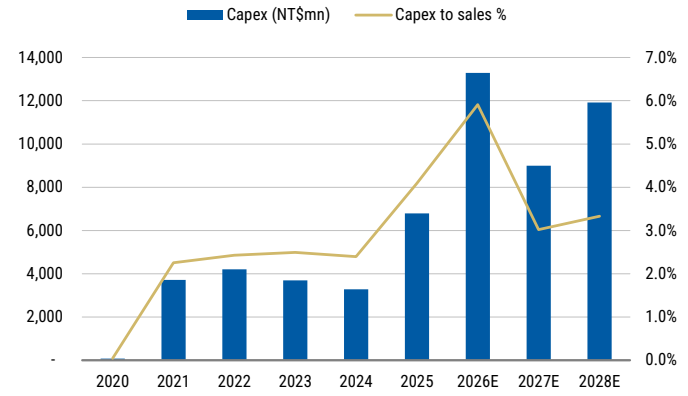
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 131: Lite-on: Improving free cash flow generation



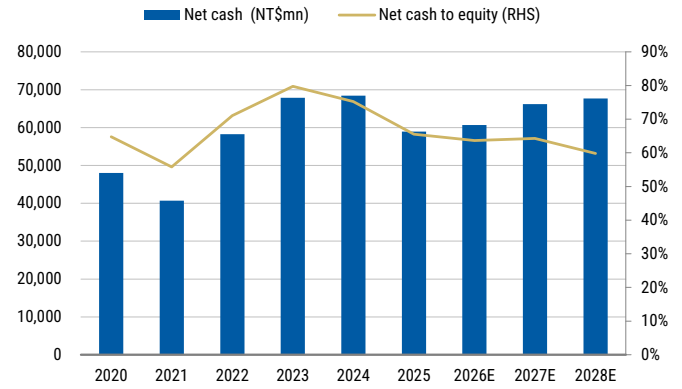
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 132: Lite-on: Capex and capex-to-sales ratio



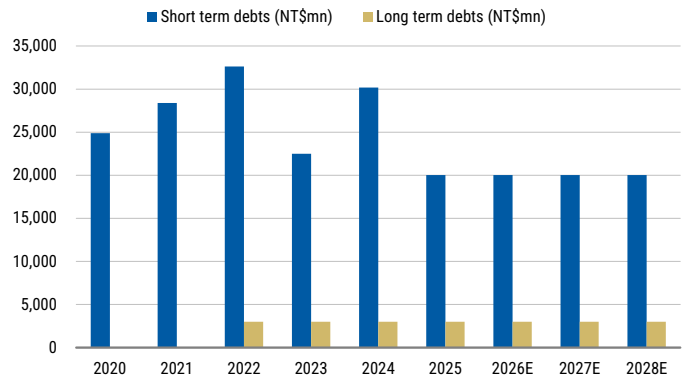
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 133: Lite-on: Net cash position



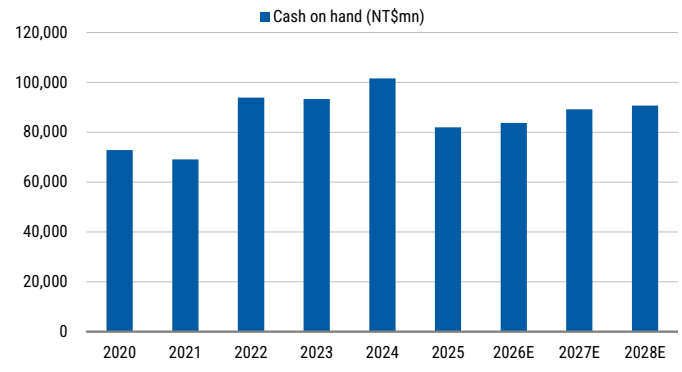
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 134: Lite-on: Debt position



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 135: Lite-on: Cash on hand

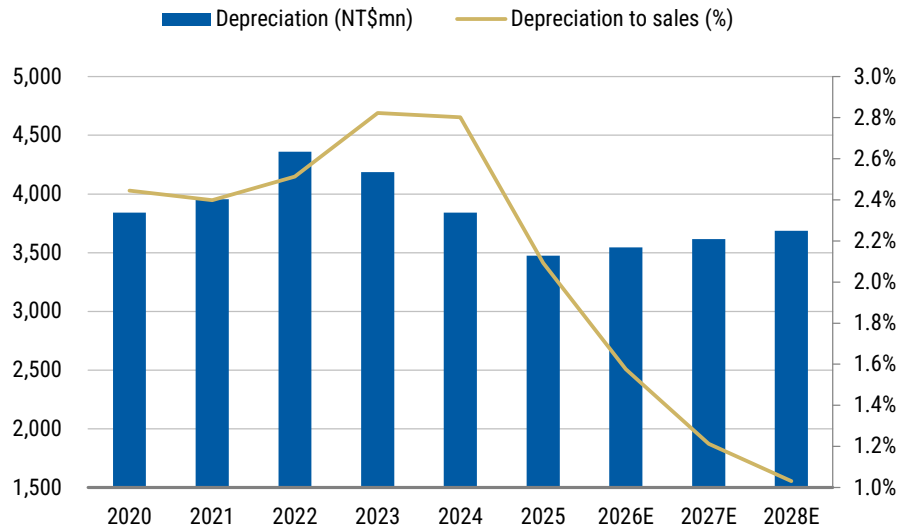


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Lite-On's average capex-to-sales ratio was around 2.3-2.4% during 2021-24, then rose to 4.1% in 2025 as the company began aggressively expanding global capacity, despite a larger revenue base during the year. Looking ahead, we project the capex-to-sales ratio to rise to 5.9% in 2026 amid continued production-site development, before declining to 3.0-3.3% in 2027-28 as investments normalize.

The depreciation-to-sales ratio remained at 2.4-2.5% during 2020-22 before rising to 2.8% in 2023, in line with higher capex during the year. The ratio then declined to 2.1% in 2025 on a larger revenue base. Looking ahead, we expect the ratio to decline further as the revenue base grows faster than capex and depreciation, falling to 1.6% in 2026 and 1.0% in 2028.

Exhibit 136: Lite-on: Depreciation and depreciation to sales

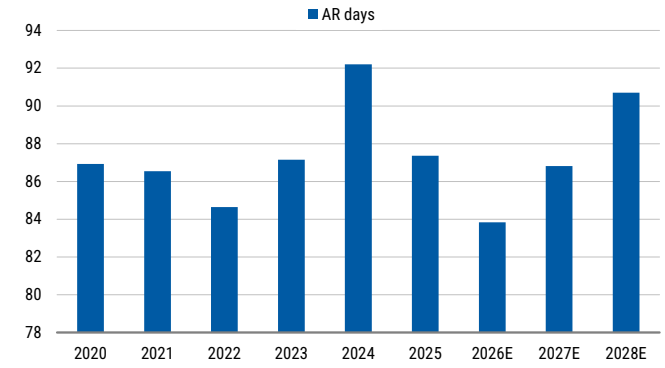


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Lite-On's cash conversion cycle (CCC) fluctuated within a relatively narrow band of roughly 35-44 days between 2020 and 2025, reflecting generally stable but modestly shifting working-capital efficiency. CCC was at a relatively low level of 35 days in 2020, supported by lower inventory days and higher AP days, before rising steadily to 44 days in 2022 as both inventory days (78) and AR days (85) increased, while AP days were slightly lower (119). From 2023 onward, the cycle improved meaningfully, falling to 36 days as AP days extended sharply from 119 in 2022 to 135. This improvement partially reversed in 2024, with CCC ticking up to 39 days as inventory and AR days both increased to 89 and 92 days, respectively, even as AP days extended further to 143. CCC then eased again to 37 days in 2025 as AP days moderated slightly to 138, alongside stable inventory and receivables levels.

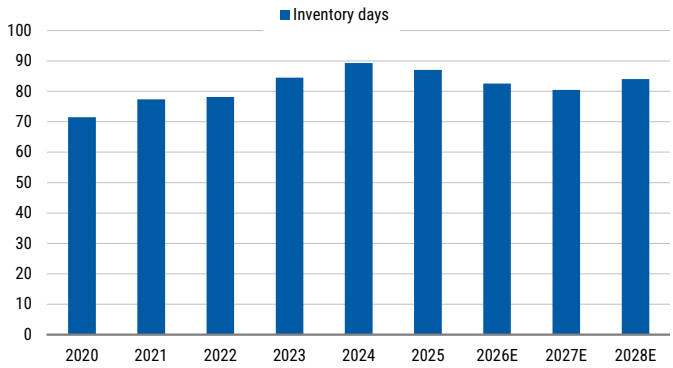
Looking ahead, we expect the overall cash conversion cycle to edge up slightly from 37 days in 2025 to 40 days in 2028, as increasing AR days and lower AP days are partially offset by improving inventory days.

Exhibit 137: Lite-on: AR days



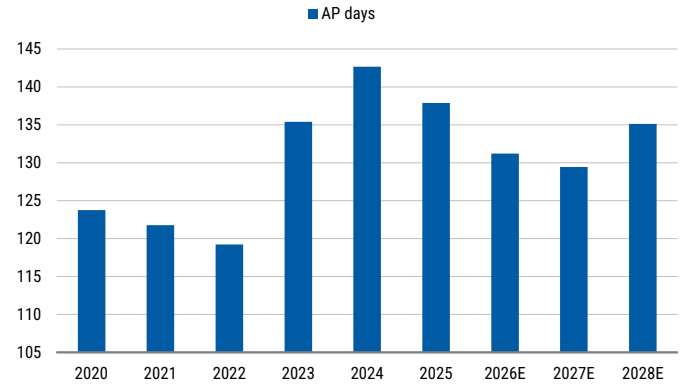
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 138: Lite-on: Inventory days



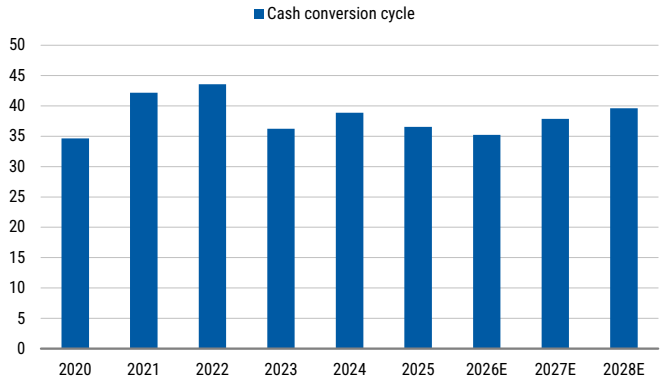
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 139: Lite-on: AP days



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 140: Lite-on: Cash conversion cycle



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Company Background – Lite-On

Company snapshot

Founded in 1975, Lite-On Technology is a Taiwan-based manufacturer of optoelectronic components, power supplies, and information technology products. The company produced Taiwan's first LEDs and became the first technology company listed on the Taiwan Stock Exchange in 1983. It mainly operates through four business segments: Information Products (power supplies, servers, networking equipment, keyboards, and chassis), Optoelectronics (LED packaging, LED lighting, automotive electronic components, and camera modules), Storage (storage components), and other supporting businesses. Additional product lines include energy management solutions, such as data center power supply systems and cloud server chassis; networking and surveillance products; and automotive electronics, including ADAS and EV charging solutions.

Lite-On is headquartered in Taipei, Taiwan. The company operates manufacturing sites, R&D centers, and sales offices across Asia (Taiwan, China, Japan, and Singapore), the US, and Europe (the UK, the Netherlands, and Germany).

Exhibit 141: Lite-on: Key milestones

Year	Key Milestones
1975	Established LITEON's first LED factory in Zhonghe, Taiwan
1983	Became the first electronics company listed on the Taiwan Stock Exchange (Ticker: 2301)
1985	Began manufacturing power supply units.
1989	Opened first overseas plant in Bangkok, Thailand, for optoelectronic products.
1990	• Set up a switching power supply plant in Penang, Malaysia. • Silitek acquired Maxi Switch, the No. 3 U.S. keyboard maker.
1991	Acquired NPE's switching power supply plant in the UK.
1994	Established a subsidiary in California, USA.
1999	Spun off the Optoelectronics business unit as LITEON IT Corp.
2002	LITEON Electronics, Silitek, LITEON Technology, and GVC merged into LITEON Technology Corporation (Ticker: 2301)
2003	Moved global headquarters to Neihu Science Park.
2006	• Acquired BenQ's Optical Disk Drive business, becoming a top-3 global ODD maker. • Partnered with Philips to enter the automotive disc drive market. • Acquired 100% of LITEON Automotive Corp., expanding into automotive electronics.
2007	• Acquired 100% of Li Shin International via stock swap, strengthening power supply resources for LCD TVs and other products. • Completed tender offer for Perlos, the world's largest handset casing maker.
2009	Opened East China Regional HQ in Wujin, Changzhou.
2010	• Invested in Epyon Power, a supplier of fast-charging technology. • Completed the world's first Optoelectronic Semiconductor Carbon Footprint and Type III Environmental Product Declarations.
2011	Completed the world's first Product Category Rules for Circuit Boards.
2013	Acquired Power Innovations (US), strengthening cloud and power management capabilities.
2014	Selected for the Carbon Disclosure Project's Climate Disclosure Leadership Index.
2016	Opened a new operations center in Kaohsiung's Nanzih Export Processing Zone.
2018	Divested the mobile camera module and mechanical business to focus resources.
2019	Acquired 100% of LOJ via tender offer.
2020	• Launched succession plan: Tom Soong named as Chairman, Anson Chiu named as President. • Completed sale of the SSD business to KIOXIA Holdings.
2021	• Launched an Employee Stock Ownership Trust (ESOT). • Joined the O-RAN Alliance and launched 5G small cells.
2022	• Began reconstructing the Zhonghe office into the "LITEON Zhonghe Digitech Center." • Inaugurated Phase II of the LITEON Vietnam plant in Haiphong.
2023	Opened its first corporate store, 'LITEON Collection,' in Neihu Science Park.
2024	• Took a 19.99% strategic stake in COSEL. • Broke ground on a new low-carbon building in Kaohsiung.
2025	• Returned to COMPUTEX with next-gen AI solutions, winning the Sustainable Design Award (Bronze). • Reopened its founding Zhonghe headquarters after a three-year reconstruction.

Source: Company data, Morgan Stanley Research

Exhibit 142: Lite-on: Global operation locations

Source: Company data

Exhibit 143: Lite-on: Global footprint

Headquarter	
Taipei City, Taiwan	
Sales Offices	
Milpitas, California, U.S.A.	
Plano, Texas, U.S.A.	
Tokyo, Japan	
Shanghai, China	
Singapore	
Hoofddorp, NL	
Cramlington, Northumberland, UK	
Wetzlar, DE	
Subsidiary	**Entity name**
American Fork, Utah, U.S.A.	Power Innovations International

Source: Company data, Morgan Stanley Research

Product offerings

Lite-On's key offerings can be categorized into six key areas: Optoelectronic Semiconductors, Energy Management, Consumer Electronics, Networking and Surveillance, Automotive Electronics, and Roadway Intellimation System.

1. Optoelectronic semiconductors

Its product portfolio spans a broad technological range, encompassing illumination and sensing applications, low- to high-power devices, standard and custom-made solutions, and components ranging from LEDs to lasers and from visible light to infrared and ultraviolet spectra.

Exhibit 144: Lite-on: Offerings for optoelectronic semiconductors

Optoelectronic Semiconductors						
Visible, UV & Automotive LEDs				Couplers	Optical Sensors	IR Emitters & Detectors
Illumination / UV-C / Automotive				Isolation & Signal Transfer	Ambient / Proximity / Gesture	Infrared Sensing
Automotive LED	SMD LED	Flash LED	UV LED	Photocouplers	Optical Sensors	IR Emitters & Detectors

Source: Company data, Morgan Stanley Research

2. Energy management

The product range serves the communications, industrial automation, smart home, medical, office, and automotive electronics industries. It offers a comprehensive suite of platinum- to titanium-efficiency-grade power supplies (CRPS, GPU, and SLIM series) suitable for both standard and AI servers, supporting AC or DC input with high power density, high reliability, and modular scalability. Power-shelf and battery-backup-unit (BBU) solutions, along with rack-cooling and liquid-cooling technologies for data centers, cloud infrastructure, and AI/machine-learning platforms, are also included in this segment.

Exhibit 145: Lite-on: Offerings for energy management

Energy Management			
Data Center Power Supply System	**Cloud Server Chassis**	**Power Supply Module**	**Magnetic Components**
CRPS / GPU / SLIM, Platinum-Titanium	Open Rack Standard V3	PoE / Industrial / LED Driver	Transformers & Inductors
			Includes components apply for cloud infrastructure, industrial power supply, white appliances, electric vehicle and consumer electric, etc
Server Power Networking Power Open Compute Project	Server Rack Server Chassis	POE Injector Industrial Power Adapter LED Driver	Magnetic Components

Source: Company data, Morgan Stanley Research

3. Consumer electronics

This segment is committed to advancing home entertainment and office solutions through feature-rich product design intended to enhance the overall user experience. Offerings include office/gaming solutions, laptop keyboard modules and touchpads, laptop power adapters, USB Type-C/PD chargers, etc.

Exhibit 146: Lite-on: Offerings for consumer electronics

Consumer Electronics		
3C Accessories	**Power Supply Module**	
Office / Gaming / Laptop peripherals	PC & Networking Power	
Office Solutions Gaming Solutions Laptop Keyboard Module Touchpad Laptop Power Adapters USB Type C/PD Charger	PC Power Supply	Networking Adapter

Source: Company data, Morgan Stanley Research

4. Networking and surveillance

The product portfolio covers networking solutions, video surveillance, AIoT solutions, and server-related offerings. The solutions are designed to be user-friendly and adaptable to emerging trends in digital intelligence.

Exhibit 147: Lite-on: Offerings for networking and surveillance

Networking and Surveillance		
Video Surveillance	**Access Control & Alarms**	**Networking Solutions**
Home / Building / Module	Smart Hub / Access / Alarm	5G / WLAN / Switch / Module
Home Security Camera Building Surveillance Camera Camera Module	Smart Hub Access Control Alarm Kits	5G Total Solution Wireless LAN Wired Switch Wireless Module

Source: Company data, Morgan Stanley Research

5. Automotive electronics

Lite-On is able to leverage its core competencies in optoelectronics and power management, along with more than four decades of expertise in automotive electronics. This segment is dedicated to advancing vehicle-charging technology, ADAS (Advanced Driver Assistance Systems) solutions, and automotive lighting modules through high-end engineering. AC charging solutions are offered across multiple current ratings (32/40A, 48A, 80A, and an EU-specific variant), while DC charging solutions include a 30kW fast-charging offering.

Exhibit 148: Lite-on: Offerings for automotive electronics

Automotive Electronics		
ADAS Solutions	**AC Charging Solutions**	**DC Charging Solutions**
Vision & Lighting	32/40A - 48A - 80A - EU	Fast Charging
Automotive Vision Solutions Vehicle Lighting Solutions	IC 32/40A IC 48A IC 80A IC EU	DC 30kW

Source: Company data, Morgan Stanley Research

6. Roadway Intellimation System

This segment is mainly operated under the LEOTEK brand, a subsidiary of the LITEON Technology Group with dual headquarters in Silicon Valley and Taipei. It has served road infrastructure across more than 30 countries—including North America, Europe, the Middle East, and the Asia-Pacific region—since 1992. The segment comprises intelligent roadway lighting, traffic signal systems, and AI-driven predictive maintenance solutions, all developed to support smarter and more efficient road infrastructure.

Exhibit 149: Lite-on: Offerings for Roadway Intellimation System

Roadway Intellimation System		
Roadway Lighting Intellimation	**Intelligent Traffic Signal System**	**AI Predictive Maintenance Solution**
Smart LED Streetlights	AI-driven Signal Control	Condition Monitoring
		Includes AI warning and maintenance management, traffic flow data analysis and computation, traffic-adaptive signal control, and priority passage for emergency vehicles, etc
Roadway Lighting Intellimation	Intelligent Traffic Signal System	AI Predictive Maintenance Solution

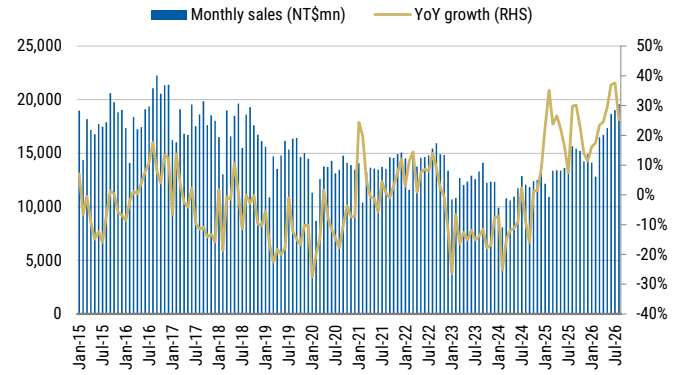
Source: Company data, Morgan Stanley Research

Monthly sales vs. share price

Lite-On's monthly sales have hovered around the NT$10-20bn level since 2013. Notably, we see a high correlation of ~76% between the company's monthly sales and its share price movements since 2023.

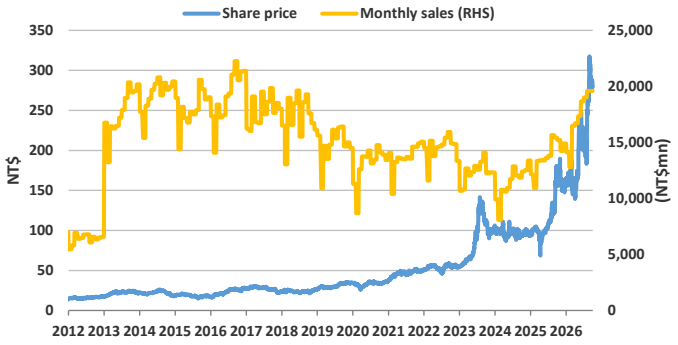
The company reported August monthly sales of NT$19,592mn (+3% MoM/+25% YoY).

Exhibit 150: Lite-on: Monthly sales vs. sales YoY growth



Source: TEJ, Morgan Stanley Research

Exhibit 151: Lite-on: Monthly sales vs. share price



Source: TEJ, Morgan Stanley Research

Key management & shareholding structure

Lite-On's board currently consists of five directors and four independent non-executive directors.

Its founder, Raymond Soong, and current Chairman, Tom Soong, together owned a stake of ~5% as of June 2026. Excluding the founder and Chairman, the management team owned a stake of ~0.1% as of June 2026.

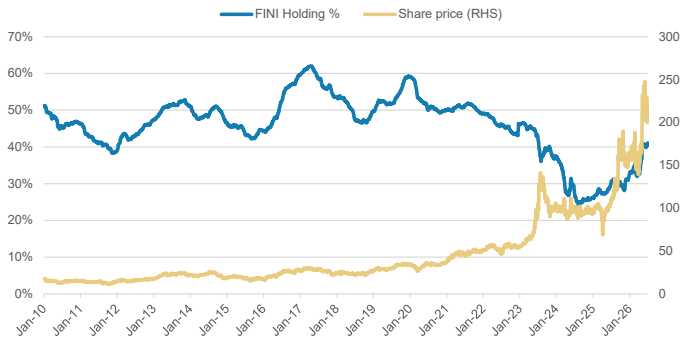
The FINI holding percentage as of July 7, 2026, was 41%, versus a previous peak of 62% in April 2017 and a historical average (since 2010) of 46%.

Exhibit 152: Lite-on: Shareholding structure

Shareholders	Ownership
Cathay MSCI Taiwan ESG Sustainability High Dividend Yield ETF	5.6%
Yuanta/P-shares Taiwan Dividend Plus ETF	5.5%
Ta-Rong Investment Co., Ltd	3.7%
Raymond Soong*	3.4%
Fuh Hwa Taiwan Technology Dividend Highlight ETF	3.3%
Labor Pension fund	2.6%
Ming-Hsing Investment Co., Ltd	2.6%
Ta-Sung Investment Co., Ltd*	2.1%
Chunghwa Post Co., Ltd	2.0%
Lite-On Technology Corporation (Treasury shares)*	1.7%
Total	**32.6%**

Source: Company data, Morgan Stanley Research. Note: Data as of August 2025. (*) are shown as insiders.

Exhibit 153: Lite-on: FINI holdings vs. share price since 2010



Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 154: Lite-on: FINI holdings vs. share price since 2023

Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 155:

Lite-on: Management profiles

Name	Title	Education	Work Experience
Tom Soong	Chairman and Chief Sustainability Officer, LITEON Technology	Electrical Engineering, University of Southern California, USA; International Business Program Attendance, NTU-FUDAN EMBA	- 20+ years at LITEON - Special Assistant to Chairman & Vice Chairman, LITEON - CEO of LITEON Smart Life and Applications Business Group - General Manager, LITEON Shanghai Operational Center
Anson Chiu	President and Director, LITEON Technology	Bachelor of Industrial Management, Lunghwa University of Science and Technology	- Joined LITEON Technology in 1987 - 35+ years in global supply chain and business development' - CEO, LITEON Power Conversion Business Group - General Manager, LITEON Power Conversion Business Group - Director, NB Business Unit, Power Conversion Business Group
Raymond Soong	Founder and Director, LITEON Technology and LITEON Cultural Foundation	Fellow, Industrial Technology Research Institute; Honorary Doctor of Management, National Chiao Tung University; Honorary Doctor of Management, Taipei University of Technology	- Chairman, LITEON Group - Chairman & Founder, LITEON Cultural Foundation - Chief Engineer, Texas Instruments Taiwan Ltd. - Board of Councilors member, Doctorate College of Technology, USC
Harvey Chang	Director Representative, Lite-On Technology Corp.	MBA, The Wharton School, University of Pennsylvania	- Independent Director, LITEON Technology - Chairman, IC Broadcasting - Chairman, TVBS and Catchplay Media Holdings Ltd. - President & CEO, Taiwan Mobile - Senior VP & CFO, TSMC
Sarina Lin	Director Representative, Lite-On; Advisor to Chairman's Office, Lite-On	Doctor of Laws, National Taiwan University	- Chief Legal Officer, Lite-On - Special Assistant to Chairman and Attorney, MediaTek Inc. - Director, MediaTek Foundation - SVP, CHRO & Chief Legal Officer, MStar Semiconductor - Director & CEO, MStar Semiconductor Charity Foundation - Director and Supervisor, MStar Semiconductor
Albert Hsueh	Independent Director, LITEON Technology;	MBA, Bloomsburg University, Pennsylvania, U.S.A.	- Chairman, PricewaterhouseCoopers Taiwan - Director, Corporate Governance Association in Taiwan - Professor, National Taiwan University and Tsinghua University, College of Technology Management
Mike Yang	Independent Director, Lite-On Technology Corp.	Master of Science, Arizona State University	- Associate Vice President, ADI Corp. Procurement Dept. - Director, LIONS Taiwan Technology Inc. - Director, Quanta Culture & Education Foundation - Cloud computing and global technology leadership
MK Lu	Independent Director, Lite-On Technology Corp.	Honorary Doctor, Tatung University; Honorary Doctor of Engineering, National Chiao Tung University; Department of Electrical Engineering, Tatung University	- Chairman, Tatung Company - Chairman, Sino-American Silicon Products Inc. - Chairman, Actron Technology Corporation - President, Lite-On Semiconductor Corp. - President, Vishay Lite-On Power Semiconductor Corp. - Vice President, Silitek Corp.
Jesse Ding	Independent Director, Lite-On Technology Corp.	PhD program, The Wharton School, University of Pennsylvania; MBA, University of Detroit, USA	- Chairman, EnTie Commercial Bank - General Manager, EnTie Commercial Bank - General Manager, TAIPEI FUBON COMMERCIAL BANK CO., LTD - Financial accounting and leadership decision-making

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Panasonic Holdings (6752.T)

Our target price is based on P/E 19.5x our FY3/28 EPS forecast.

We apply: 36x to EPS generated by AI infrastructure-related businesses, 5x to EPS attributable to IRA subsidies in the automotive battery business, and 13.5x to EPS from all other businesses.

Risks to Upside

- Acceleration of restructuring and exit/scale-down of unprofitable businesses
- Stronger than expected growth of datacenter BBU, automotive battery business

Risks to Downside

- Deteriorating visibility for energy business
- Expectations for business portfolio revisions sink

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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(as of September 30, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
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Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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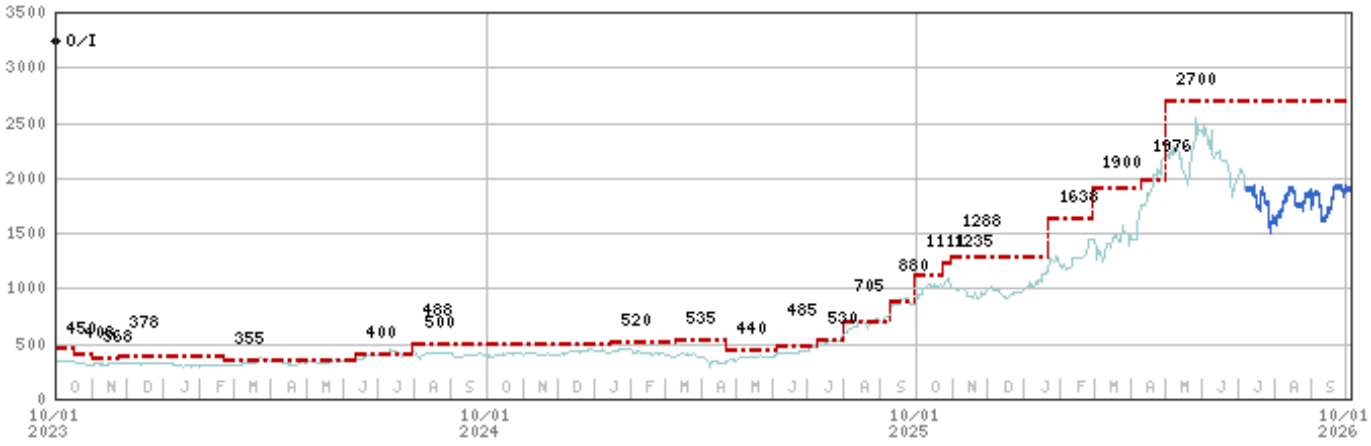
Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Delta Electronics Inc. (2308.TW) - As of 10/05/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : O/I

Price Target History: 9/9/21 : 345; 10/29/21 : 325; 2/25/22 : 303; 7/29/22 : 312; 9/21/22 : 326; 10/6/22 : 350; 11/23/22 : 368; 8/2/23 : 450; 10/17/23 : 408; 11/1/23 : 368; 11/23/23 : 378; 2/21/24 : 355; 6/12/24 : 400; 7/30/24 : 488; 8/1/24 : 500; 1/15/25 : 520; 3/11/25 : 535; 4/23/25 : 440; 6/4/25 : 485; 7/9/25 : 530; 8/1/25 : 705; 9/8/25 : 880; 9/30/25 : 1111; 10/24/25 : 1235; 10/30/25 : 1288; 1/21/26 : 1638; 2/27/26 : 1900; 4/10/26 : 1976; 5/1/26 : 2700

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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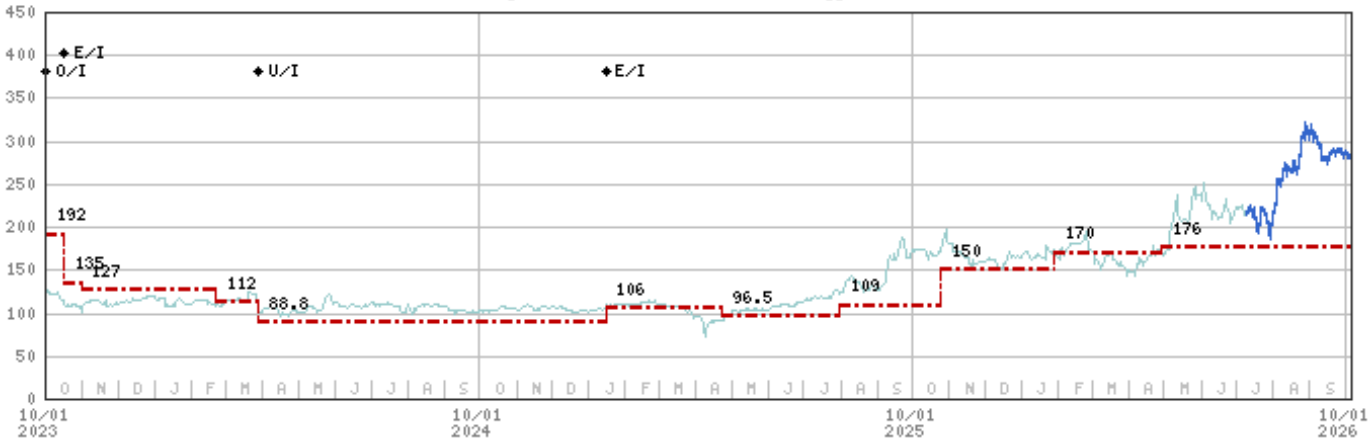
Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Lite-On Technology (2301.TW) - As of 10/05/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : E/I; 7/31/23 : O/I; 10/17/23 : E/I; 3/28/24 : U/I; 1/15/25 : E/I

Price Target History: 4/30/21 : 62; 7/19/22 : 65; 2/21/23 : 61.2; 7/31/23 : 192; 10/17/23 : 135; 10/31/23 : 127; 2/21/24 : 112; 3/28/24 : 88.8; 1/15/25 : 106; 4/23/25 : 96.5; 7/31/25 : 109; 10/24/25 : 150; 1/28/26 : 170; 4/29/26 : 176

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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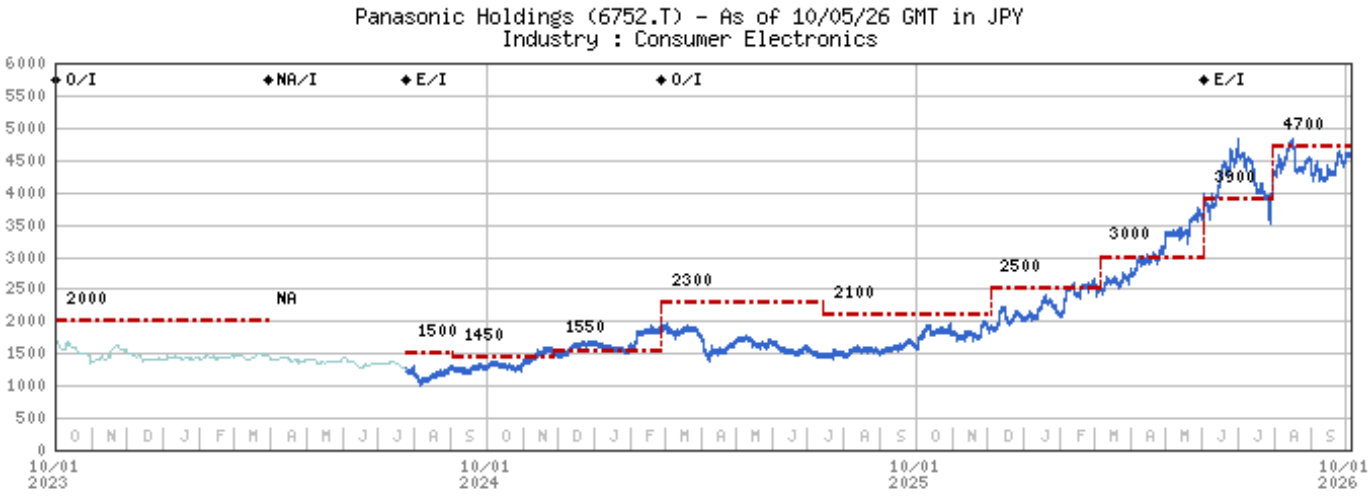

Stock Rating History: 10/1/21 : O/I; 3/29/24 : NA/I; 7/25/24 : E/I; 2/26/25 : O/I; 6/3/26 : E/I

Price Target History: 9/22/21 : 1900; 3/1/22 : 1700; 6/13/22 : 1500; 12/1/22 : 1600; 2/20/23 : 1500; 6/7/23 : 2000; 3/29/24 : NA; 7/25/24 : 1500; 9/3/24 : 1450; 11/28/24 : 1550; 2/26/25 : 2300; 7/14/25 : 2100; 12/3/25 : 2500; 3/6/26 : 3000; 6/3/26 : 3900; 7/31/26 : 4700

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/05/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$38.64
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.76
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,055.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,730.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.46
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$53.35
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.91
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$37.16
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$61.95
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43

Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$23.88
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$179.10
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,111.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.62
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,075.00
Advantech (2395.TW)	O (01/20/2021)	NT$740.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,180.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,585.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$39.05
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,550.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,485.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$890.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,325.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,005.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$146.50
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$68.20
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,450.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$345.50
Innolux (3481.TW)	E (04/07/2025)	NT$52.30
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,355.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$298.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.40
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$995.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$207.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.80
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,700.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.86
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$388.50
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,215.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$254.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.30
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$3,000.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.92
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,875.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,495.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$88.70
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$331.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,325.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$606.00

Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,150.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$626.00
Zhen Ding (4958.TW)	++	NT$576.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$570.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,780.00

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ASIA PACIFIC INSIGHT

October 6, 2026 12:44 AM GMT

Greater China Technology Hardware

Integrated Power Solutions Driving Secular Demand

Increasing power consumption, along with the deployment of BBU/CBU/800/±400VDC/SST in AI clusters, could push the technology barrier higher and present growth opportunities for players with superior capabilities. Fuel cells could also emerge as a new opportunity amid power constraints.

Secular demand for power amid AI capex: AI-related power consumption will continue to increase, and we believe this will lead to secular demand for power-related components, such as power supply units (PSUs), battery backup units (BBUs), capacitor bank units (CBUs), coolant distribution units (CDUs), and so on. Moreover, the requirement for efficiency will inevitably necessitate the adoption of 800/±400 voltage direct current (VDC) in the power rack/center/block configuration (initially from late 2026 and broadly in 2028, we estimate), eventually extending to the deployment of solid state transformers (SST) in the longer term – all pointing to a more centralized power delivery architecture in the future. This trend of aggregation could raise the technology barriers amid the fusion of white/grey space, favoring players with better capabilities and execution. Overall, we forecast the datacenter PSU market size to grow at a 75% three-year CAGR to reach US$33bn in 2028.

Fuel cells to improve flexibility and resilience of AI power delivery: Despite the strong AI capex outlook (MSe global cloud capex +97%/20% YoY in 2026/2027), power has become a bottleneck, indicated by MS's forecast power shortfall of 5GW in 2026, scaling to a total 72GW through 2029 (see report). Fuel cells, among other alternative energy sources, could emerge as a good solution to pick up some slack and enhance efficiency through active micro grid management. Those who are able to offer comprehensive solutions from power sources to power delivery could be increasingly more competitive in the market.

Assuming coverage of Delta and Lite-On:

- Delta (OW, PT NT$2,500, based on 30x 2027/2028 P/E) should maintain its dominant position in the industry with its integrated power/thermal related solutions while fuel cells could add incremental growth opportunities in the longer term. We think a 30x P/E target multiple is justified by our estimated 63% earnings CAGR in 2025-28, driven by a 44% revenue CAGR and operating margin expansion to 22.1% in 2028 vs 15.1% in 2025,
- Lite-On (EW, PT NT$315, based on 17x 2027/2028 P/E), the second-largest player in AI power-related products, should also ride the tailwinds despite its slower progress in HVDC power racks and liquid cooling solutions. We think a 17x P/E target multiple is justified by our estimated 47% earnings CAGR in 2025-28 (29% revenue CAGR and 6.8ppt OPM expansion to 16.8% in 2028).

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GREATER CHINA TECHNOLOGY HARDWARE

Asia Pacific Industry View	**In-Line**

WHAT'S CHANGED

Delta Electronics Inc. (2308.TW)	**From**	**To**
Price Target	NT$2,700.00	**NT$2,500.00**
Lite-On Technology (2301.TW)	**From**	**To**
Price Target	NT$176.00	**NT$315.00**

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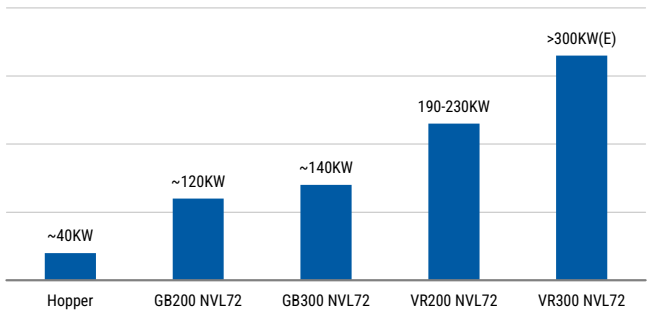
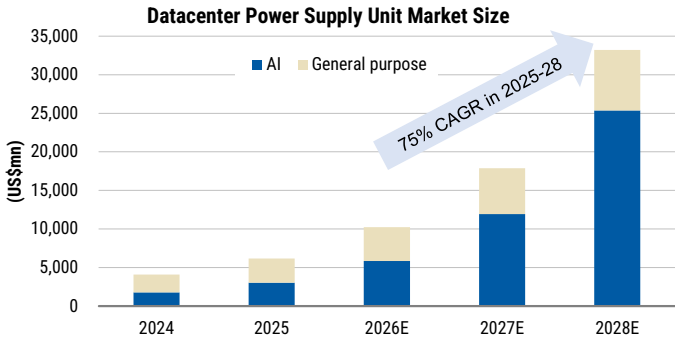
The Story in Charts

Exhibit 1: AI servers continuing to see higher TDP...



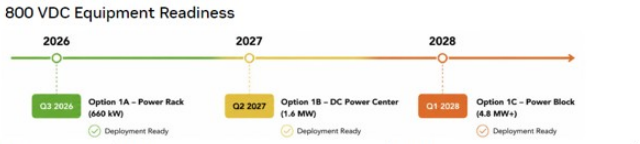
Source: Company data, Morgan Stanley Research estimates

Exhibit 2: ...which drives data center power supply demand



Source: Company data, Morgan Stanley Research estimates

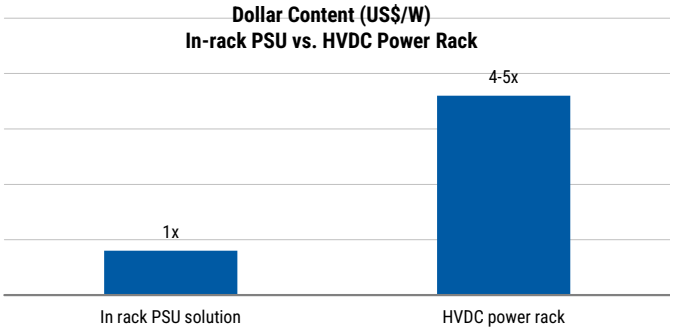
Exhibit 3: HVDC to see early adoption from late 2026...



Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

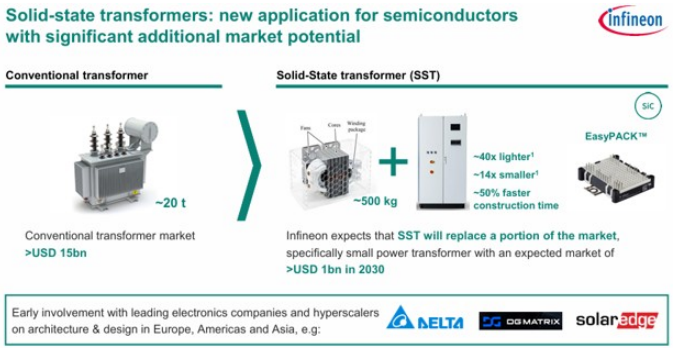
Source: Nvidia

Exhibit 4: ...with a much higher dollar content



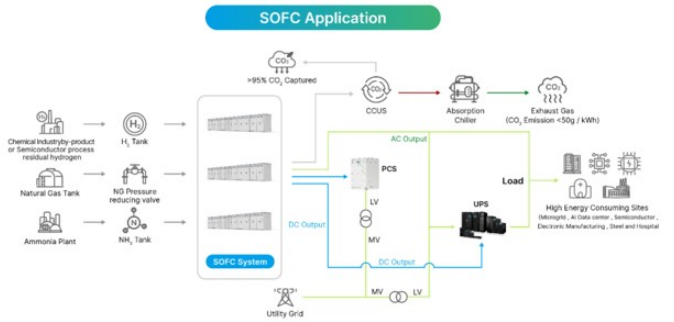
Source: Morgan Stanley Research

Exhibit 5: SST is also in the pipeline for more centralized power delivery



Source: Infineon

Exhibit 6: Fuel cells are a good option for better resilience in power architecture



Source: Delta

Executive Summary

AI cluster performance evolving to fulfill demand

The rapid adoption of artificial intelligence is driving a significant increase in global data center compute demand. Advances in foundation models, reasoning capabilities, agentic AI, and enterprise AI deployment are increasing the computational resources required for both training and inference. To support this growth, hyperscalers and model developers continue to deploy larger and more powerful AI clusters, driving demand for greater compute, memory, and networking capacity.

As AI infrastructure scales, cluster performance is increasingly determined by the efficiency of the entire system rather than the capabilities of individual processors alone. Compute, networking, cooling, and power infrastructure must all evolve together to support higher rack densities and larger deployments. While semiconductor innovation remains critical, supporting infrastructure is becoming an equally important factor in enabling future AI growth.

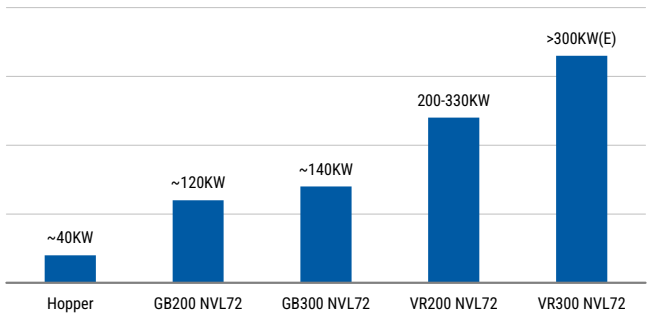
AI cluster performance is fundamentally a power problem

As AI clusters continue to scale in size and performance, power consumption is becoming one of the most important constraints on future infrastructure deployment. The power challenge is no longer limited to GPUs alone, as modern AI clusters comprise tightly integrated compute, networking, memory, storage, and cooling systems, all of which require increasing amounts of power as performance requirements rise. Consequently, the next phase of AI infrastructure investment should be viewed not only as a semiconductor upgrade cycle but also as a broad-based power infrastructure upgrade cycle.

Server power consumption has increased significantly across successive AI hardware generations. Nvidia's architecture roadmap illustrates the industry's migration from relatively modest server-level power budgets under the Ampere and Hopper generations toward rack-scale systems under Blackwell and future Rubin architectures. Estimated rack-level power consumption has increased from roughly 40kW for Hopper systems to approximately 120kW for GB200 NVL72 and 140kW for GB300 NVL72, with future Vera Rubin systems expected to approach 200-300kW per rack with current Vera Rubin systems reaching up to 330kW per rack while future Vera Rubin generations are expected to surpass these levels. As compute density continues to increase, power delivery infrastructure is becoming a critical enabler of system performance.

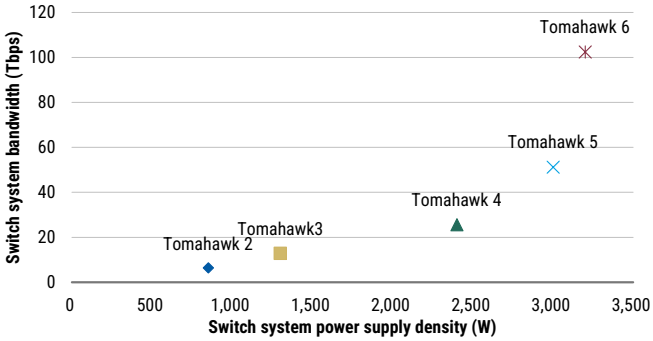
The increase in power consumption is not limited to compute hardware. Networking systems are also becoming increasingly power intensive as switch bandwidth scales from 100G architectures toward 800G and 1.6T generations. Despite substantial process-node improvements, switch ASIC power consumption has continued to rise due to increasing complexity and higher SerDes speeds. As a result, networking equipment now requires significantly larger power budgets than previous generations, creating additional demand for higher-capacity power supplies throughout the data center.

Exhibit 7: TDP roadmap for Nvidia racks



Source: Morgan Stanley Research

Exhibit 8: Power supply density for Tomahawks



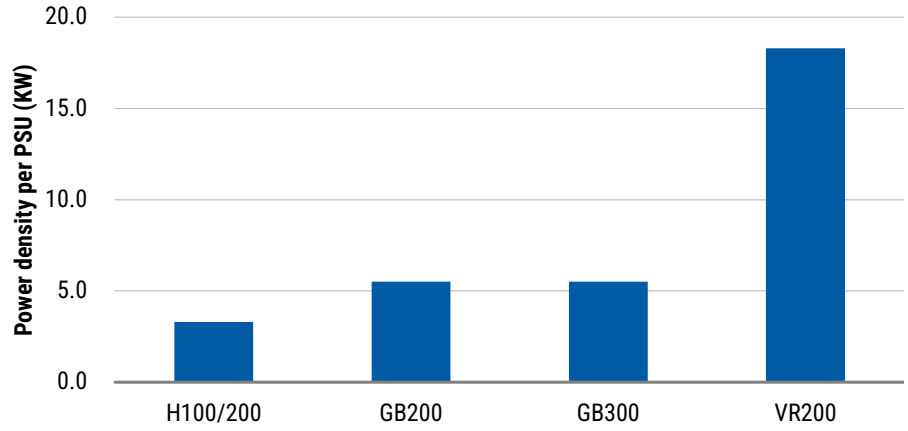
Source: Morgan Stanley Research

The growth in power consumption is driving a corresponding upgrade cycle for power-supply units (PSUs). Earlier AI server generations typically relied on multiple 3-3.3KW power supplies installed directly within individual server chassis. As the industry transitions toward rack-scale architectures, power conversion is increasingly centralized into dedicated power shelves capable of supporting substantially larger power loads. In Blackwell systems, power shelves utilize 5.5KW power supplies, while future Rubin-generation platforms are expected to increase PSU ratings to approximately 18KW, representing a significant increase in both PSU content and power density.

Delivering higher power within a similar footprint creates significant engineering challenges. Increasing PSU power density requires advances across multiple areas, including higher-performance power semiconductors such as silicon carbide (SiC) and gallium nitride (GaN), more efficient power-conversion topologies, smaller and more capable magnetic components, improved thermal-management solutions, and greater system integration. As output power increases, thermal constraints become increasingly important, making heat dissipation, airflow optimization, and mechanical design critical determinants of PSU performance.

Consequently, on top of the volume driven growth, the power-supply industry is benefiting from the technology driven dynamics, where efficiency, power density, and thermal performance are becoming key competitive differentiators. As AI rack power requirements continue to increase, PSU innovation will become an increasingly important component of the broader AI infrastructure investment cycle.

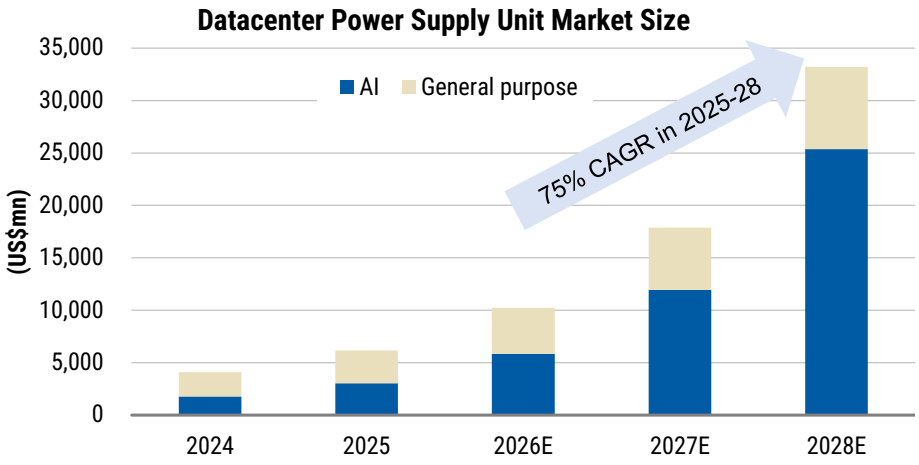
Exhibit 9: Power density per PSU trend in Nvidia server systems



Source: Nvidia, Morgan Stanley Research

Overall, we estimate the global datacenter power supply market will grow at a three-year CAGR of 75% to reach US$33bn in 2028 from US$6bn in 2025, driven mainly by server shipment growth, increasing power consumption per rack, and the adoption of HVDC power racks from 2027.

Exhibit 10: Global datacenter power supply market forecast



Source: Company data, Morgan Stanley Research estimates

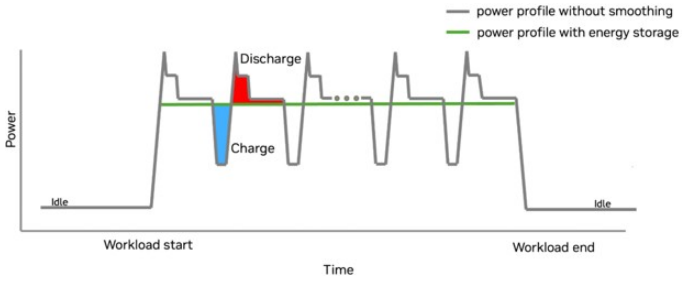
Complementary hardware for more resilient power delivery

As AI clusters become more power intensive, maintaining stable and resilient power delivery is emerging as a critical design priority. Traditional centralized power architectures are increasingly being complemented by localized energy-storage and power-conditioning technologies that can respond rapidly to dynamic workload demands.

Battery Backup Units (BBUs) represent one of the most important additions to modern AI power architectures. Positioned close to the computing workload, BBUs provide localized energy storage capable of responding significantly faster than facility-level backup systems. Their role extends beyond conventional backup power functionality. BBUs can

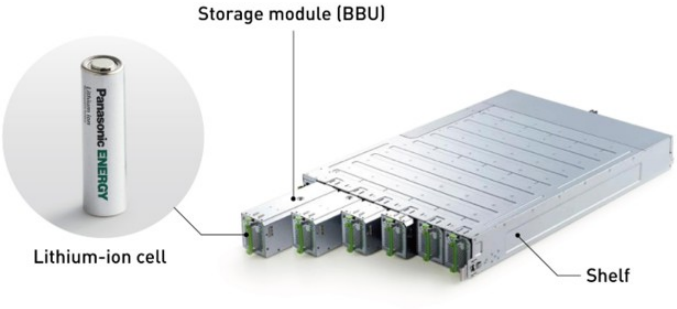
absorb transient fluctuations, smooth power ramps, and improve overall rack-level power stability. By providing local buffering capacity, BBUs help reduce the burden on upstream power distribution infrastructure while improving overall system resilience.

Exhibit 11: Concept of peak shaving for AI server power loads



Source: Nvidia

Exhibit 12: BBU



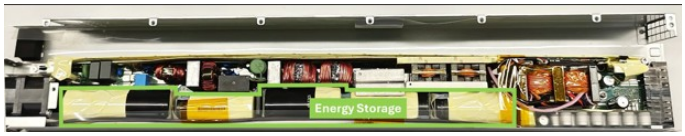
Source: Panasonic

Capacitor Backup Units (CBUs) address an even shorter-duration segment of the power stability challenge. Utilizing supercapacitors and other high-power-density storage technologies, CBUs can deliver and absorb power within milliseconds, making them particularly effective for managing rapid GPU power spikes. These systems act as shock absorbers within the electrical architecture, preventing sudden load changes from propagating throughout the broader power network. As AI rack power densities continue increasing, capacitor-based energy storage is becoming an attractive solution for improving power quality without requiring substantial oversizing of facility infrastructure.

Exhibit 13: CBU

Source: Panasonic

Exhibit 14: GB300 power shelf with integrated capacitors



Source: Nvidia

Together, BBUs and CBUs provide complementary benefits. BBUs are designed to handle longer-duration events, while CBUs are optimized for extremely fast transient response. The integration of these technologies enables operators to improve power quality, reduce stranded infrastructure capacity, enhance system reliability, and increase effective utilization of expensive electrical assets. As rack densities continue to rise, localized energy-storage solutions are likely to become a standard component of AI data center design rather than an optional enhancement.

The adoption of BBU and CBU architectures may also enable future increases in rack density. As operators attempt to push toward several hundred kilowatts per rack, localized power-buffering solutions can help mitigate transient peaks and reduce stress on grid-facing infrastructure. This allows operators to deploy higher-density computing resources without proportionately increasing upstream electrical capacity.

Exhibit 15: Comparison of CBU and BBU

Dimension	CBU	BBU
Primary storage medium	EDLC supercapacitors, LICs or high-capacitance	Typically lithium-ion battery cells managed by a BMS
Primary strength	High power, fast charge/discharge and frequent	Higher stored energy and longer sustained
Main function	Transient smoothing, peak shaving, voltage	Outage backup, source-transfer bridging and orderly workload
Response profile	Well suited to millisecond-class load support and	Fast switchover, but generally optimized for sustained support rather
Typical duration	Milliseconds to seconds; selected systems can extend toward tens of seconds or more	Commonly tens of seconds to several minutes, depending on battery and rack design
Power versus energy density	Higher power capability but lower energy storage per unit volume than a battery	Higher energy density, making it more economical for longer events
Cycle life	Eaton specifies more than one million cycles and up to 20 years for certain supercapacitor modules	Battery life is more sensitive to depth of discharge, temperature and repeated cycling; BMS supervision is therefore essential.
State management	Requires cell balancing, voltage/temperature monitoring and state-of-health tracking, but does not use a conventional battery fuel	Requires BMS functions including charge control, state of charge, state of health and cell protection
Best-fit use case	Repetitive AI power spikes, inrush events, PSU load leveling and bus stabilization	Power continuity through outages, brownouts or generator/source-transfer intervals

Source: Morgan Stanley Research

HVDC emerging as a necessity, with SST in the pipeline

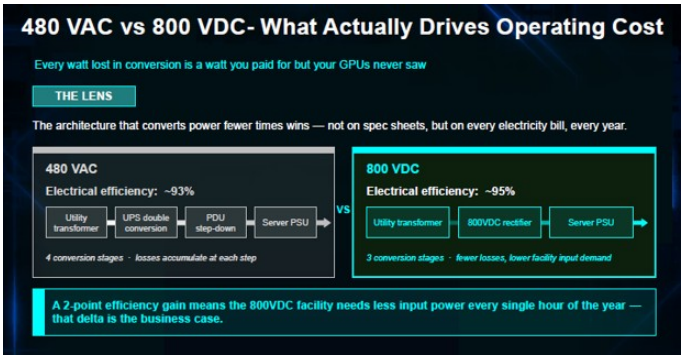
800VDC and ±400VDC progressing simultaneously

The rapid increase in AI rack power density is creating growing challenges for conventional power-distribution architectures. As rack-level power consumption rises from tens of kilowatts to several hundreds of KWs, electrical losses, cable size, thermal management requirements, and overall system complexity increase significantly. Under traditional low-voltage power architectures, delivering larger amounts of power requires proportionally higher current, which increases resistive losses and places greater stress on cables, connectors, busbars, and power-conversion equipment.

At the same time, every power-conversion stage introduces efficiency losses and generates additional heat. As AI deployments scale into multi-megawatt and eventually gigawatt-level campuses, even small efficiency improvements can translate into meaningful reductions in operating costs, cooling requirements, and infrastructure investment. Consequently, the industry is increasingly focused on power architectures that can reduce conversion stages, lower current levels, and improve overall power-delivery efficiency.

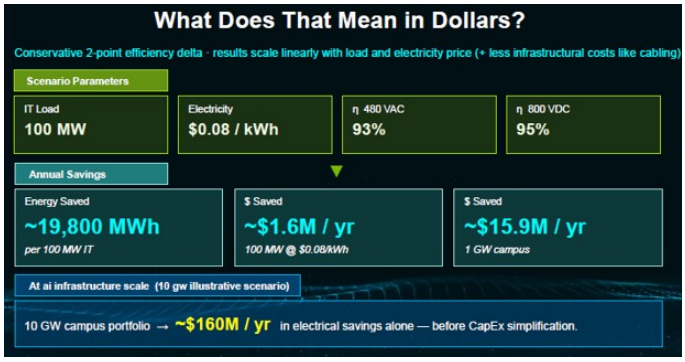
Against this backdrop, high-voltage direct current (HVDC) is emerging as one of the most important architectural upgrades for next-generation AI infrastructure. By increasing distribution voltage, HVDC allows the same amount of power to be delivered at a lower current, reducing conduction losses and improving system efficiency. Higher-voltage architectures also improve scalability for ultra-high-density rack deployments by alleviating some of the physical and thermal constraints associated with conventional power-distribution systems.

Exhibit 16: Better power efficiency for 800VDC architecture



Source: Delta

Exhibit 17: Better efficiency leads to cost savings



Source: Delta

Nvidia's roadmap illustrates the industry's gradual migration toward higher-voltage power architectures. Earlier AI server generations primarily utilized conventional power-delivery schemes centered around rack-level 48V distribution. As rack power requirements increased with Blackwell-generation systems, industry participants began evaluating higher-voltage architectures capable of supporting substantially greater power density. Looking ahead, the industry is increasingly converging around 800VDC as a potential next-generation standard for large-scale AI deployments, particularly as rack-level power consumption approaches several hundred kilowatts. The transition reflects the need to improve power-distribution efficiency while supporting the continued growth of AI computing infrastructure.

We believe that the adoption of the 800VDC power rack could start from late 4Q26, though it should be small scope deployment through 2027, since the in-rack power shelf is the reference design in the Vera Rubin NVL72 rack. Broader adoption should happen more likely from late 2027 and 2028, as the Vera Rubin Ultra rack further pushes up the power consumption.

Exhibit 18: HVDC deployment roadmap

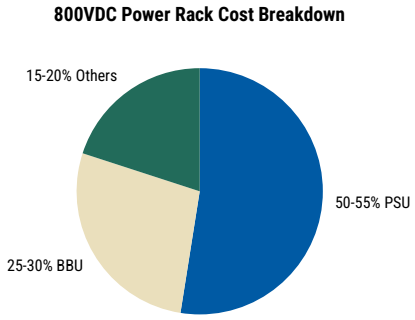
Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

Source: Nvidia

The 800VDC power rack BOM is primarily driven by the PSU, which accounts for 50–55% of total cost, followed by the BBU at 25–30%, with other components comprising the remaining 15–20%. This cost structure reflects the broader functionality integrated into an HVDC power rack, combining power conversion, backup power and supporting components within a centralized architecture.

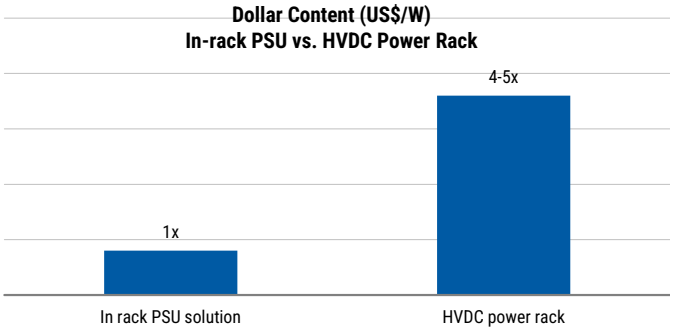
For power supply players, one of the most direct implications is that an HVDC power rack carries around 4–5x higher dollar content per watt compared with a conventional in-rack PSU solution, since the former integrates more components into a robust system and power supply players usually ship the entire rack instead of just the components.

Exhibit 19: BOM cost breakdown for an 800VDC power rack



Source: Morgan Stanley Research

Exhibit 20: Higher dollar content for HVDC power racks



Source: Morgan Stanley Research

However, the industry has not yet fully converged on a single architecture. While Nvidia's roadmap increasingly points toward 800VDC, several hyperscalers and members of the Open Compute Project (OCP) ecosystem have expressed interest in ±400VDC architectures. Both approaches deliver similar total voltage levels and pursue many of the same objectives, including lower current, higher efficiency, and improved scalability. The debate largely centers on implementation trade-offs rather than end goals.

Supporters of 800VDC generally highlight the advantages of a simpler single-bus architecture, potentially lower system complexity, and closer alignment with existing industrial high-voltage DC equipment. The architecture avoids the need to maintain balanced positive and negative voltage rails and may simplify certain aspects of power conversion and system integration. These characteristics make 800VDC particularly attractive in environments where operational simplicity and interoperability with existing electrical infrastructure are key priorities.

Advocates of ±400VDC, meanwhile, emphasize the advantages of a bipolar architecture. By creating positive and negative voltage rails around a neutral reference point, ±400VDC can potentially reduce insulation requirements, simplify certain fault-management scenarios, and provide improved flexibility for load balancing and future system expansion. Many hyperscalers also view the bipolar approach as being more compatible with long-term visions for highly distributed power systems incorporating energy storage, renewable energy, and microgrid functionality.

Importantly, the ultimate beneficiaries may be largely independent of which standard gains wider adoption. With the exception of certain power-semiconductor and protection-system requirements, many component suppliers across the power ecosystem remain

broadly agnostic between 800VDC and ±400VDC implementations. Higher-voltage architectures in either form require similar upgrades in power conversion, power management, cooling, connectors, magnetics, and infrastructure equipment. As a result, the industry's migration toward HVDC matters significantly more than the eventual selection of any single voltage standard.

Exhibit 21: Comparison of 800VDC and ±400VDC

Metric	±400VDC	800VDC
Voltage to ground	400V relative to ground	800V relative to ground
Voltage differential	800V	800V
Power distribution efficiency	High	Slightly higher
Scalability	Good	Excellent
Insulation requirements	Lower	Higher
Maintenance safety	Better	More challenging
System architecture simplicity	Dual-rail system	Single-bus system
Grounding Complexity	More complicated	Simpler
Electrical footprint	Small	Smaller
Ability to reduce conversion stages	Moderate	Highest
Long-distance distribution capability	Excellent	Excellent
Maturity of ecosystem	Better today	Early but rapidly developing
Industry players	OCP, hyperscalers	Nvidia
Long term industry direction	Transitional architecture	Likely end-state architecture

Source: Morgan Stanley Research

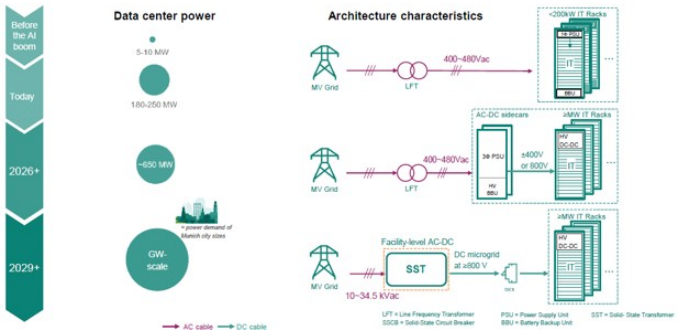
Solid-state transformers to follow after 800VDC/±400VDC matures

Looking further ahead, solid-state transformers (SSTs) represent a potential next-generation evolution of power infrastructure. Unlike traditional transformers that operate primarily as passive devices, SSTs leverage power electronics to provide voltage conversion, power conditioning, and bidirectional energy management within a single platform. This enables greater controllability and flexibility while supporting increasingly digitalized power networks.

Although SSTs remain at an early stage of commercialization and face challenges related to economics, reliability, and large-scale deployment, the technology aligns well with long-term data center trends. As AI campuses evolve into utility-scale power consumers incorporating energy storage, distributed generation, and advanced power management, SSTs could eventually become a key enabling technology.

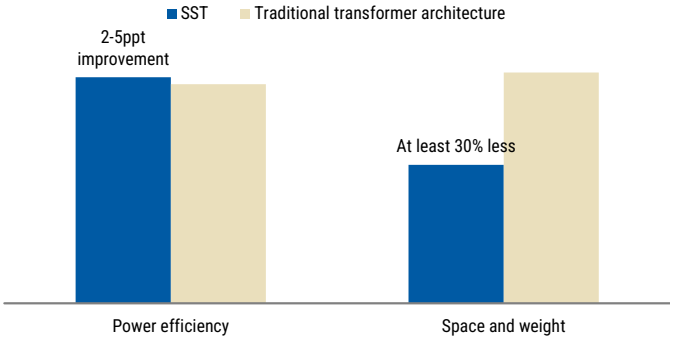
Therefore, while HVDC is expected to drive the next phase of power architecture upgrades, SSTs represent a longer-term technology roadmap that could further transform how future data centers distribute and manage electricity.

Exhibit 22: SST enabling more efficient power delivery architecture



Source: Infineon

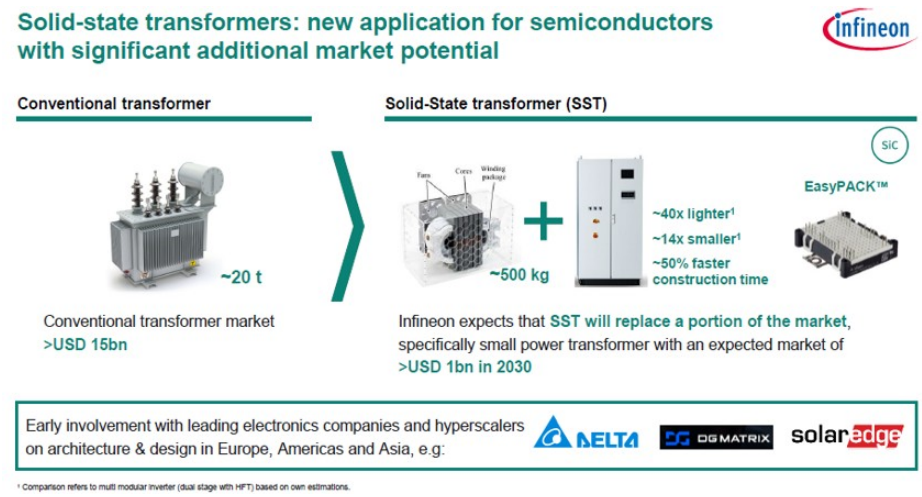
Exhibit 23: SST offers better power efficiency with at least 30% less space/weight



Source: Morgan Stanley Research

According to Infineon, SST could represent an emerging >US$1bn market opportunity by 2030, as the technology begins to replace a portion of the >US$15bn conventional transformer market, particularly in smaller-power applications. While SST penetration is expected to remain relatively modest versus the overall transformer market, the opportunity is significant for relevant players in the longer term.

Exhibit 24: SST to see initial adoption in 2030

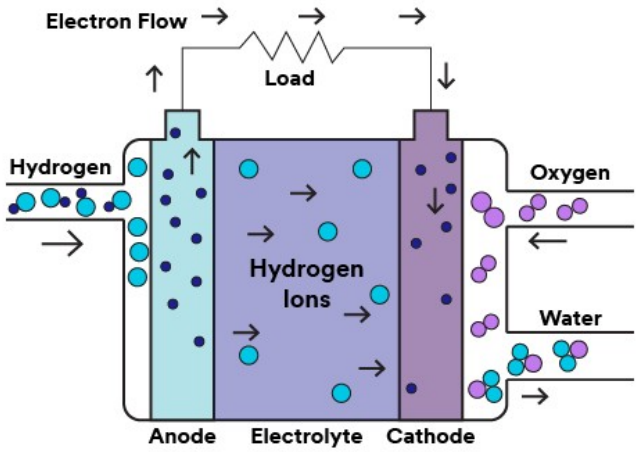


Source: Infineon

Fuel cells for better flexibility and resilience

Fuel cells generate electricity through an electrochemical reaction rather than combustion. Hydrogen, natural gas, biogas, or other fuels are supplied to the fuel cell stack, where chemical energy is converted directly into electricity, heat, and in some cases water. Unlike batteries, which store energy, fuel cells continuously generate power as long as fuel is supplied.

Exhibit 25: Fuel cell diagram



Source: 3M

Fuel cells can offer several benefits, including:

- **High energy efficiency:** Direct electrochemical conversion enables higher efficiency than conventional combustion-based power generation, reducing fuel consumption and operating costs.
- **Low carbon emissions**: Fuel cells produce minimal direct emissions and can support long-term decarbonization goals when powered by low-carbon fuels such as hydrogen.
- **Grid independence and energy security:** Provides reliable onsite power generation, reducing dependence on grid availability and improving energy resilience.
- **Long-duration energy storage enabler:** Hydrogen-based fuel-cell systems can complement renewable energy by storing energy over extended periods, supporting long-duration and seasonal energy storage.
- **Fuel flexibility:** Certain fuel-cell technologies can operate on hydrogen, natural gas, biogas, ammonia-derived hydrogen, and other hydrogen-rich fuels, allowing deployment using existing fuel infrastructure.
- **Low maintenance and quiet operation:** Limited moving parts reduce maintenance requirements while providing quieter operation than conventional combustion-based generators.
- **Scalable deployment:** Can be deployed across a wide range of power levels, from distributed backup systems to utility-scale installations, making fuel cells suitable for large AI data center campuses

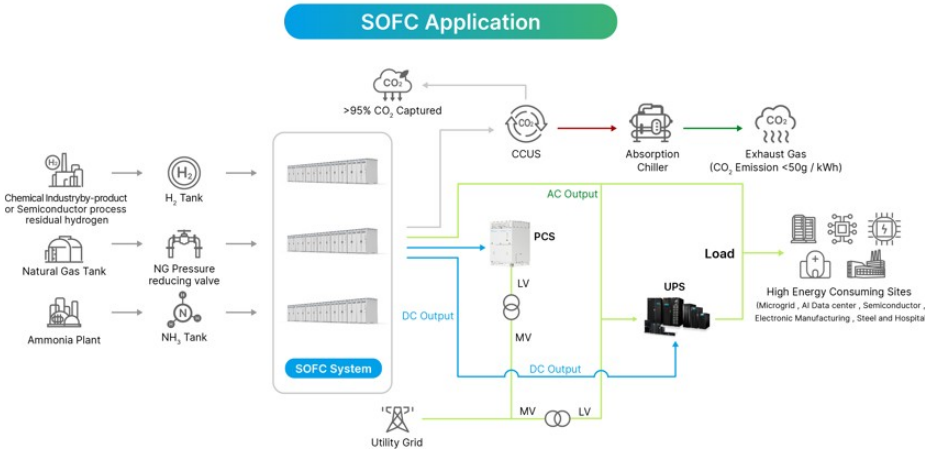
However, the following challenging are hurdles for wider deployment:

- **High system cost:** Fuel cells remain more expensive than conventional power-generation technologies, limiting large-scale adoption.
- **Limited hydrogen infrastructure:** Hydrogen production, transportation, storage, and refueling networks remain underdeveloped in most regions.
- **Durability requirements:** Long-term reliability and stack lifetime remain key challenges, particularly for large-scale and mission-critical deployments.
- **Regulatory and safety considerations:** Hydrogen handling requires specialized infrastructure and compliance with stringent safety regulations, increasing deployment complexity.
- **Lack of standardization:** Multiple fuel-cell technologies compete across different applications, creating uncertainty around long-term technology standards and adoption paths.

Fuel cells have the potential to fundamentally reshape future data center power architectures. Traditionally, data centers rely on grid electricity as the primary power source and diesel generators for backup power. However, as AI-driven workloads significantly increase power demand and power availability becomes a key deployment constraint, hyperscalers are increasingly evaluating fuel cells not only as a backup-power solution but also as a potential primary power source. This could accelerate the transition from a grid-centric architecture toward one incorporating distributed onsite generation, enabling data center deployment in regions where utility power capacity is limited.

Over the longer term, fuel cells could become a core component of AI data center microgrids that integrate renewable generation, battery storage, hydrogen storage, and intelligent energy-management systems. Such an architecture would improve scalability, resilience, and energy independence while reducing reliance on centralized utility infrastructure. In addition, because fuel cells inherently generate DC power, they align well with the industry's migration toward HVDC-based power distribution architectures. This could simplify power delivery by reducing conversion stages between generation and IT equipment, improving overall system efficiency and supporting next-generation high-density AI deployment

Exhibit 26: Fuel cell in datacenter power delivery architecture



Source: Delta

Stock picks

Delta (2308.TW, OW, PT NT$2,500)

Dominant supplier in the AI power supply offerings: Delta has been one of the most competitive players in power related components and systems, as its offerings spanning from the core power supply units, the complementary BBU/CBU for AI servers, the highly anticipated HVDC power rack, grey space power facilities to air/cooling solutions. We estimate that it has ~60% market share in the datacenter power supply market and is among the leading suppliers of the 800VDC/±400VDC power rack to Nvidia/hyperscalers in 2H26 (wider adoption likely from late 2027). Longer term, when the HVDC becomes more mature, the centralized power delivery architecture could migrate into SSTs, where Delta already has small volume shipment now. With the solid technology, close customer relationships and strong execution, we believe Delta should stay competitive even amid the fusion of the white space and grey space.

Extending from power delivery to power source: In addition to the offerings in the power delivery, Delta signed a licensing agreement with Ceres Power (CRW.GB) for its solid oxide fuel cell (SOFC) technology in early 2024. This could extend its business scope into the power source area. which we believe could add more value to its customers as it could help mitigate the power constraints for datacenter build and enhance the operational efficiency through active micro grid management.

Solid near term momentum: Despite the slower revenue growth owing to the component supply tightness in 3Q26, we expects its 4Q26 revenue to grow by double digit % QoQ on alleviated component supply and solid demand. 3Q26 GM should be flat to slightly up QoQ and 4Q26 might see further expansion on the tailwinds price hikes.

OW with PT of NT$2,500: We see power/thermal as one of the major secular AI related themes in the downstream tech space and Delta is well positioned to benefit, potentially driving a 63% earnings CAGR in 2025-28, in our estimate. We thus assume the coverage of Delta with an OW rating and the PT of NT$2,500, based on 30x 2027/2028 P/E. Our target multiple implies a PEG of 0.5 vs. the average PEG of 1.2 in 2020-25, when the stock was trading at an average 12M forward P/E of 22x with the earnings CAGR of 19%.

Lite-On (2301.TW, EW, PT NT$315)

Rising tide lifting all boats: Lite-On is the second largest power supply vendor globally and has been engaging with hyperscalers regarding the ASIC and Nvidia GPU servers. Looking ahead, it expects to expand its share in a couple of hyperscalers from late 2026. It should have over 50% of the total revenue generated from the cloud segment for PSUs, BBUs, and to a lesser extent, other mechanical components like liquid cooling, chassis/rack, etc. Moreover, it also has the HVDC power rack solutions ready and is qualifying with potential customers. The cloud shall remain as the major growth driver in the coming years.

Aspiration in the optical communication space: Lite-On has always had an opto business (10-20% of total revenue in 2026) in both visible and invisible light applications. Post the strategic investment at Dense Light (21% stake), an InP epi wafer and chip process manufacturer, it plans to leverage its expertise in packaging and provides the optical

modules and engine for the data interconnects. While the end applications looks promising, we think it might take some time to see more meaningful revenue contribution in Lite-On.

Cloud momentum extending into the coming quarters: We expect its 4Q26 revenue to grow by double digit QoQ driven mainly by the could segment and GM should normalize at ~25% in 2H26 vs. 27.2% in 2Q26 on some inventory write-off reversal and high margin product shipment delayed from the previous quarter.

EW with PT of NT$315: We think Lite-On will grow along with the AI capex cycle, driven by its power-related offerings (47% earnings CAGR in 2025-28, in our estimate), though execution and progress with new customers is worth monitoring. We assume coverage of Lite-On with an EW rating and a PT of NT$315, based on 17x 2027/2028 P/E. Our target multiple implies a PEG of 0.4 vs. the average PEG of 1.6 in 2020-25, when the stock was trading at an average 12M forward P/E of 14x on an earnings CAGR of 9%.

Panasonic Holdings (EW, PT ¥4,700, covered by Kazuo Yoshikawa)

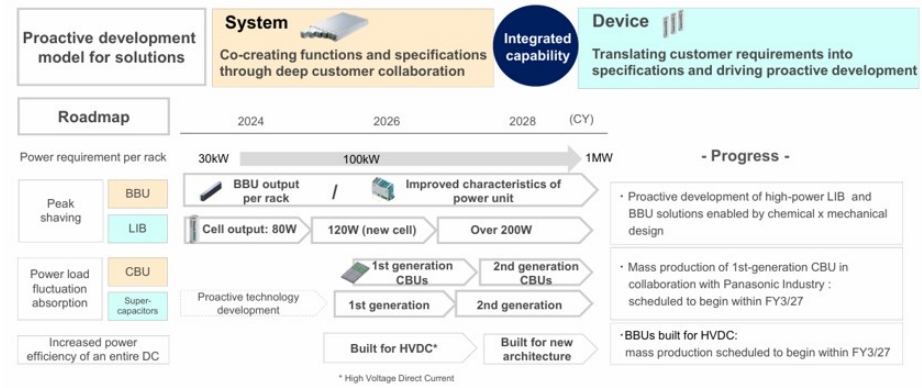
Panasonic has an approximately 80% share of the BBU market and is well positioned to benefit from rising AI data center power demand: The company expects data center energy storage revenue to grow from ¥322bn in FY3/26 to approximately ¥550bn in FY3/27 and around ¥1trn in FY3/29. Importantly, all of the business underpinning its FY3/29 revenue target has already been awarded, meaning agreements have been reached with customers on product development and orders.

BBU: Panasonic has begun limited mass production of new high-capacity BBU cells above 200W, with products adopted by a major GPU vendor and ramping from June-July. Discussions for next-generation server platforms are in the final stages, and the company expects to maintain its approximately 80% market share. It also sees scope for higher pricing as rack-level power consumption rises.

The company is also close to securing awards for BBUs used in dedicated power racks. As its medium-term plan includes only already-awarded business, additional design wins could provide upside.

CBU: CBUs are emerging as a new growth opportunity as GPU and ASIC power consumption increases. Panasonic is working with power-supply vendors to propose integrated BBU+CBU solutions and is receiving inquiries from US and Chinese hyperscalers, including customers that were not previously BBU users.

Exhibit 27: Panasonic Holdings: Strengthen proposal and development capabilities for the next generation



Source: Company data

Risks to our positive view

Risks to our positive view on power-related demand include:

- Component supply tightness leading to lower shipment of products
- Shortfall in hardware shipments, yield issues, electricity availability, etc. slowing down datacenter builds
- Lower-than-expected penetration of new power delivery architectures like 800VDC/±400VDC, SST, etc.
- More intense competition leading to margin erosion among most players
- Hyperscalers prove unable to finance aggressive capex through internal and external funding.

AI Cluster Performance Evolving to Fulfill Demand

AI adoption driving exponential growth in compute demand

AI is becoming a mainstream computing workload

Over the past three years, generative AI has evolved from a research application into a core technology platform. Consumer applications such as ChatGPT, Claude, Gemini, and Copilot now serve hundreds of millions of users globally, while enterprises are increasingly embedding AI into software development, customer service, document analysis, enterprise search, healthcare, financial research, and industrial automation.

Historically, software demand scaled roughly with the number of users. AI demand is fundamentally different because every user interaction consumes computational resources in real time. Unlike traditional cloud applications, where the bulk of computation occurs during software development, AI applications continuously consume compute whenever a model generates a response. Consequently, rising AI adoption directly translates into higher demand for data center compute resources.

Model complexity continues to increase

The AI industry has historically improved model performance through scaling laws: larger models trained on larger datasets using more compute generally produce better outcomes. While efficiency improvements continue, leading AI developers remain committed to scaling model capabilities.

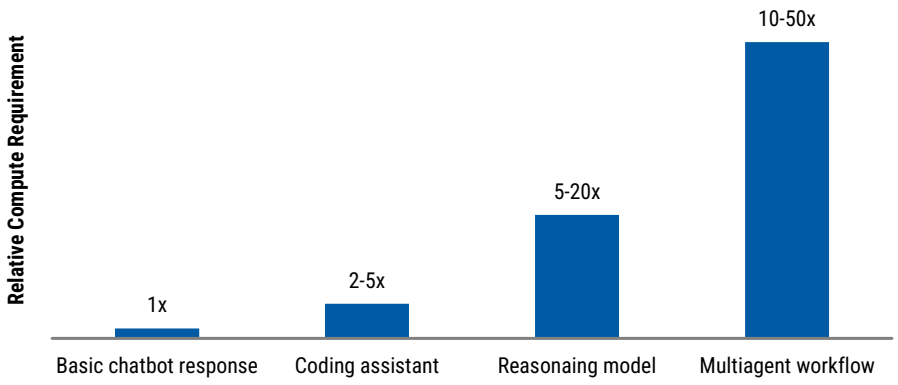
More importantly, the industry is moving beyond conventional language models toward significantly more compute-intensive architectures.

Reasoning models

Traditional LLMs generate answers directly from a prompt. New reasoning models perform multiple internal computational steps before producing a response. Instead of generating a few hundred output tokens, advanced reasoning systems may generate thousands or even tens of thousands of internal reasoning tokens while evaluating different solution paths. This dramatically increases inference workloads.

The industry is increasingly shifting from "parameter scaling" toward "test-time compute scaling," where additional compute is consumed during inference to improve answer quality. From an infrastructure perspective, this is highly significant because inference workloads occur continuously rather than once during model training.

Exhibit 28: Increasing demand for compute power from reasoning models



Source: Morgan Stanley Research estimates

Agentic AI

The next wave of AI applications involves autonomous agents capable of planning, executing, verifying, and refining tasks.

Instead of generating a single answer, an AI agent may search multiple information sources, run code, validate results, generate follow-up tasks, and produce a final response. Each step requires additional inference cycles.

A workflow that previously required one model invocation may now require dozens of model calls, substantially increasing compute consumption per user.

Multimodal AI

AI models are increasingly processing images, video, audio, sensor data, and text simultaneously.

Video understanding is particularly compute intensive because each second of video contains vastly more information than a text prompt.

As multimodal AI adoption grows, compute demand per query rises significantly, creating another structural driver of accelerator demand.

Training demand remains robust

Although inference is becoming the largest source of AI compute demand, training workloads continue to expand. Frontier model development increasingly requires clusters containing tens of thousands of accelerators operating simultaneously.

Training runs are becoming larger because developers are pursuing larger parameter counts, more training data, longer training durations, multimodal datasets and reinforcement learning and reasoning optimization

Consequently, each new generation of foundation models typically consumes substantially more compute than the previous generation.

Importantly, training infrastructure sets the baseline for future inference deployment. Every major model developer must maintain access to large-scale training clusters to remain competitive, supporting continued demand for high-end AI accelerators.

More compute requests to be fulfilled by better chip performance

Rising compute demand cannot be addressed simply by adding more servers. Modern AI workloads increasingly encounter bottlenecks related to memory bandwidth, interconnect performance, power consumption, and cooling. As a result, each new generation of AI hardware delivers improvements across multiple dimensions.

Compute performance

The latest AI accelerators provide several-fold performance improvements versus prior generations. For example, NVIDIA's Blackwell architecture was specifically designed to support both large-scale training and high-volume inference, with significant gains in AI throughput compared with previous Hopper-based systems. Performance improvements are necessary because reasoning and agentic workloads require substantially more operations per query.

Memory capacity and bandwidth

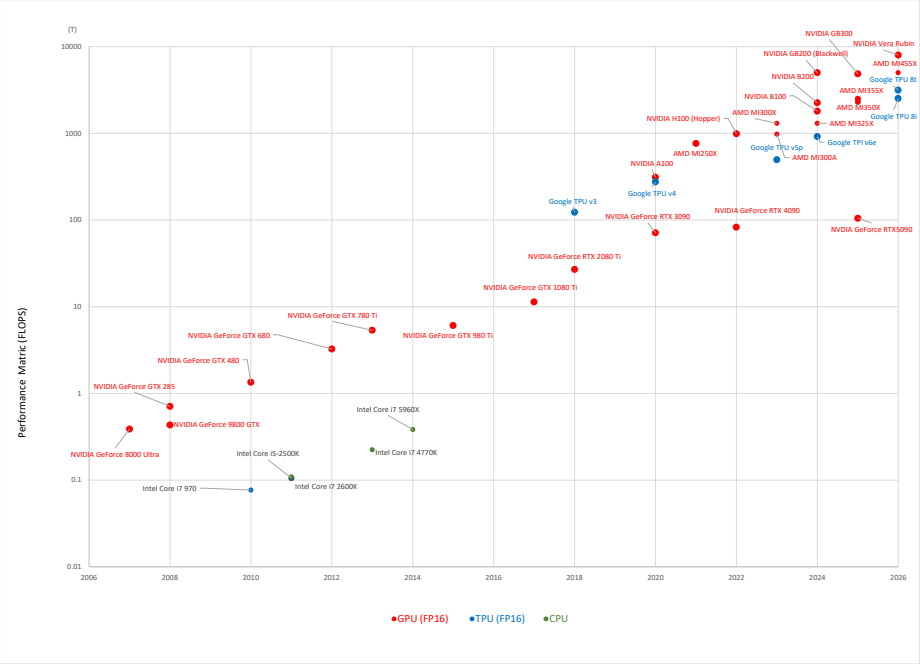
AI models are increasingly constrained by memory rather than raw compute, because longer models, longer context windows, reasoning models and agentic workflows all require more memory. As model complexity increases, modern high-end accelerators usually incorporate hundreds of gigabytes of High Bandwidth Memory (HBM), providing memory bandwidth measured in multiple terabytes per second.

Interconnect performance

As AI clusters scale from hundreds to thousands of accelerators, communication between processors becomes increasingly important. A large AI training cluster effectively behaves like a single massive computer. Consequently, networking technologies such as NVLink, InfiniBand, Ethernet, etc., have become critical enablers of AI scaling.

Many AI workloads can no longer be accelerated simply by increasing GPU performance; cluster interconnect bandwidth must also improve.

Exhibit 29: Compute power for HPCs continuing to evolve



Source: Company data, Morgan Stanley Research

AI Cluster Performance Is Fundamentally a Power Problem

As AI clusters become larger and more powerful, every component within the cluster must deliver higher throughput, lower latency, and greater bandwidth. Achieving these improvements inevitably requires more electrical power.

Therefore, the power challenge is no longer limited to GPUs. Modern AI clusters are highly integrated systems comprising compute servers, networking fabrics, memory subsystems, storage infrastructure, and cooling equipment. As cluster performance increases, power consumption rises across the entire infrastructure stack.

In our view, the next phase of AI infrastructure investment should be viewed not simply as a semiconductor upgrade cycle, but as a broad-based power consumption cycle driven by the need for ever-higher AI cluster performance.

Server systems seeing increasing TDP

Nvidia's AI-server power roadmap is best understood as a transition from chip-defined, air-cooled nodes to power-, cooling- and fabric-constrained rack-scale computers: the Ampere-era A100 SXM carried a disclosed 400W standard TDP—up to 500W in custom thermal designs—with 40/80GB HBM2/HBM2e and 600GB/s NVLink, while the eight-GPU, dual-AMD-EPYC DGX A100 was a conventional 6U server with a 6.5KW maximum input rating, meaning GPU silicon represented only roughly half of the node envelope once CPUs, NVSwitches, memory, NICs, drives, fans and conversion losses were included.

Hopper stepped the H100 SXM to a disclosed 700W TDP, 80GB HBM3, roughly 3TB/s memory bandwidth and 900GB/s fourth-generation NVLink, while H200 retained essentially the same Hopper compute architecture and power profile but substituted 141GB of faster HBM3e at 4.8TB/s—making H200 primarily a memory-capacity/bandwidth upgrade rather than a compute-TDP step-up—and Nvidia rates the eight-GPU, dual-Xeon DGX H100/H200 at 10.2KW maximum input, versus 6.5KW for DGX A100;

The architectural discontinuity arrives with GB200. Rather than treating an eight-GPU server as the scale-up unit and using InfiniBand/Ethernet primarily between nodes, Grace Blackwell NVL72 makes the entire rack the scale-up domain, integrating 18 one-U compute trays—each with two 72-core Grace Arm CPUs and four Blackwell GPUs—plus nine NVLink switch trays, management switches, , a 50–51V DC busbar, copper NVLink cartridges and liquid manifolds, so that 36 CPUs and 72 GPUs operate as one low-latency NVLink-connected system. Based on the same compute die, Nvidia upgraded the HBM content to further improve the performance of GB300 chip. Overall, the rack level power consumption is estimated at ~120KW for the GB200 NVL72 and ~140KW for the GB300 NVL72.

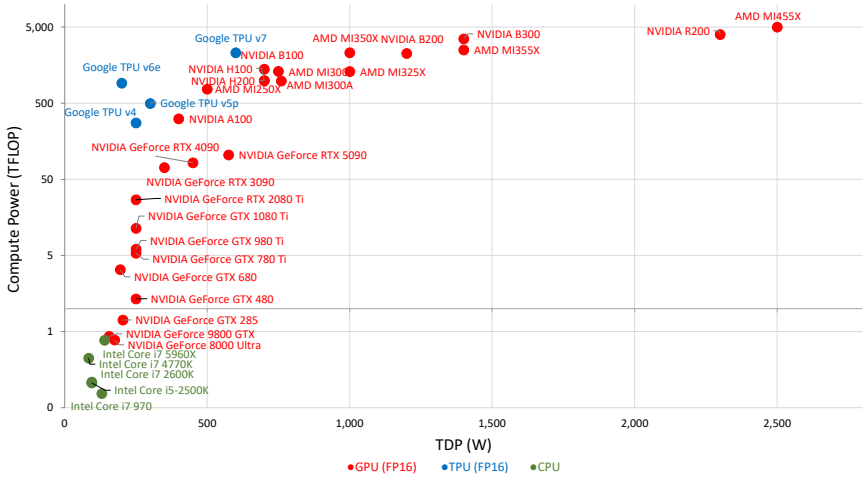
Vera Rubin—often called VR200/R200 in industry roadmaps—pushes the same rack-level philosophy further. Nvidia's disclosed architecture combines 72 dual-die Rubin GPUs with 36 new Vera CPUs, replacing Grace's 72 Arm Neoverse cores with 88 custom Olympus cores, raising CPU memory capacity and bandwidth to as much as 1.5TB LPDDR5X and 1.2TB/s, and doubling coherent CPU–GPU NVLink-C2C bandwidth to 1.8TB/s; Rubin itself moves from HBM3e to 288GB HBM4 at up to 22TB/s, introduces a third-

generation Transformer Engine and raises sixth-generation NVLink scale-up bandwidth to 3.6TB/s per GPU, while ConnectX-9 provides 1.6Tbps/GPU scale-out connectivity and BlueField-4 increasingly offloads storage, security and infrastructure processing.

Nvidia has not published a simple Rubin chip TDP in its primary datasheet, but our supply chain checks indicate roughly 1.8kW at an efficiency-oriented operating point and up to approximately 2.3kW at the performance mode, which is consistently corroborated by system vendors, with ASUS specifying 187kW Max-Q and 227kW Max-P for its fully liquid-cooled Vera Rubin NVL72.

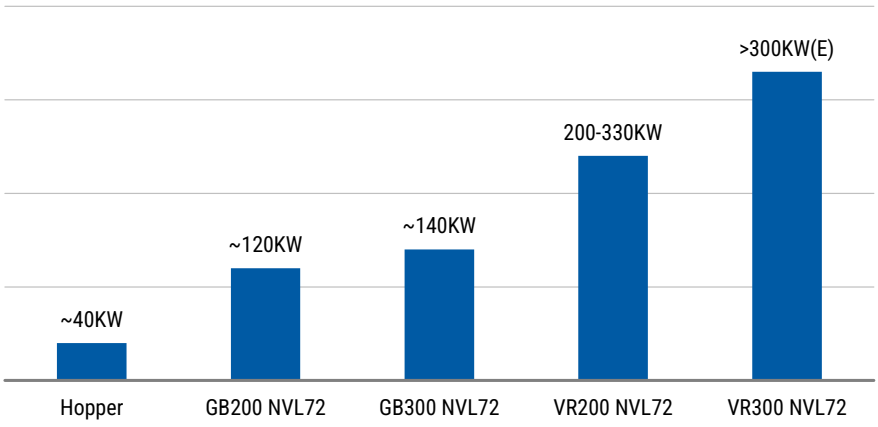
Looking ahead, we expect this trend to extend into the future along with the technology advancement and more critical workloads.

Exhibit 30: Compute power vs. TDP for HPCs



Source: Company data, Morgan Stanley Research

Exhibit 31: TDP roadmap for Nvidia racks



Source: Company data, Morgan Stanley Research

Networking consuming more power as well

The network switch system is another key devices within the datacenter and has been evolving over the past years, driven primarily by advances in the switch ASIC. From Tomahawk 1 (2014) to Tomahawk 6 (2025), the process node shrank from 28nm to 3nm, while aggregate switching bandwidth increased from 3.2 Tbps to 102.4 Tbps, representing a 32× increase. Over the same period, SerDes speed improved from 25 Gbps NRZ to 200 Gbps PAM4, enabling maximum port speeds to scale from 100G to 1.6T.

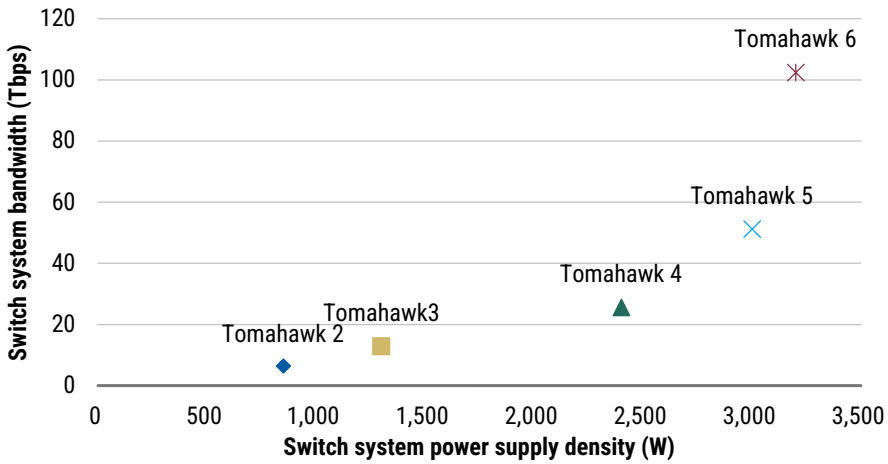
While process scaling significantly improved switching efficiency, it did not prevent ASIC power from rising. ASIC TDP increased from 150 W to 800 W, reflecting the growing complexity of higher radix switches and faster SerDes interfaces.

The impact of ASIC scaling is evident at the system level. As switch bandwidth increased from 100G-class systems to 800G and 1.6T platforms, power delivery requirements grew dramatically. Representative PSU ratings increased from 850W in Tomahawk 2 systems to 3,200W in Tomahawk 6 systems, nearly a 4× increase.

The rise in PSU power is driven not only by the higher ASIC TDP, but also by the growing power consumption of high-speed optics, increased cooling requirements, and larger overall system power budgets. As systems evolved from 100G to 1.6T ports, switch vendors were forced to add more fan redundancy, larger heatsinks, and in some cases larger chassis footprints. For example, Tomahawk 6 systems have expanded to 4U form factors with 7+1 fan redundancy to accommodate the thermal load associated with 102.4 Tbps switching capacity.

Alongside the increasing power consumption of the server racks, network switches are also witnessing higher power budgets as well.

Exhibit 32: Power supply density for different network switch generations



Source: Company data, Morgan Stanley Research

Power supply growing on both volume and specs

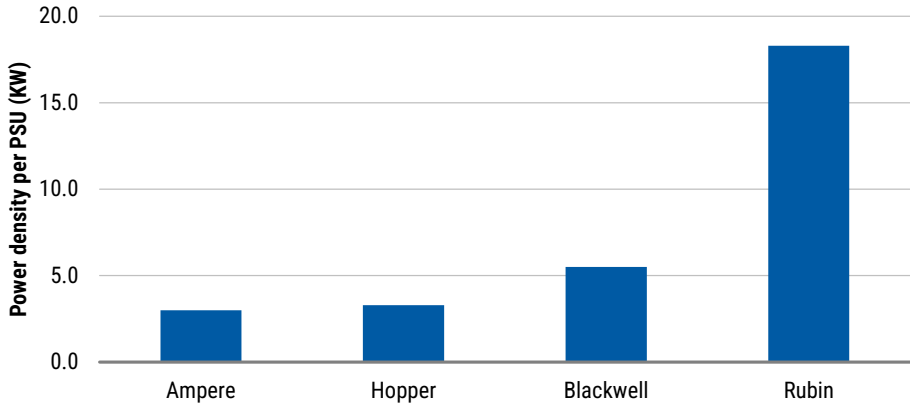
PSU spec upgrade trend

In response to the increasing power consumption of the AI servers, the power supply architecture has been migrating as well. in the A100 and H100/200 single system servers, 6 power supply units are placed directly inside the chassis, with each capable to deliver 3.0-3.3KW of power.

When Nvidia make a broad based move to the rack level server solution from the Blackwell generation, the power supply units were no longer placed directly within the chassis housing the HPCs that consume power; instead, power supply units are separated and grouped into power shelves. In the GB200/300 systems, there are 8 power shelves sitting on the same rack as the compute trays and switch trays, each enclosing six 5.5KW power supply units.

For the upcoming Vera Rubin rack, the power supply unit will be upgraded to 18.3KW of output to comprise a power shelf of 110KW, and each power shelf will take 3U of space on the rack vs. 1U in the Blackwell generation.

Exhibit 33: Power density for PSUs in Nvidia server systems



Source: Company data, Morgan Stanley Research

Challenges increasing the power density of PSUs

To come up with power supply unit with higher power density, manufacturers might need to solve the following potential hurdles:

Better power semiconductors (SiC and GaN)

The biggest improvement in PSU power density comes from replacing traditional silicon power chips with newer materials such as silicon carbide (SiC) and gallium nitride (GaN). These devices waste less energy during power conversion and can operate at much higher speeds. As a result, many supporting components inside the PSU—such as transformers and inductors—can be made significantly smaller. This allows PSU manufacturers to fit more output power into the same physical size while also improving efficiency. In many ways, SiC and GaN have become the foundation for the latest generation of high-density AI server power supplies.

More efficient power conversion designs

Modern PSUs use more sophisticated circuit designs, i.e., bridgeless PFC vs. conventional one, that reduce energy losses during power conversion. A conventional boost PFC includes a diode bridge and therefore incurs multiple semiconductor conduction drops. A bridgeless totem-pole PFC reduces the number of devices in the active current path and is now a preferred architecture for dense server PSUs. Instead of simply converting electricity from one voltage to another, newer architectures are designed to minimize wasted energy throughout the process. Higher efficiency means less heat is generated, which directly enables higher power density. This is particularly important in AI servers, where even small efficiency improvements can translate into meaningful reductions in heat generation and cooling requirements.

Smaller and smarter magnetic components

Transformers and inductors are often among the largest components inside a PSU. A major area of innovation is reducing their size without sacrificing performance. Manufacturers are increasingly using planar magnetics, where copper traces on circuit boards replace traditional wire windings. These designs are thinner, easier to cool, and more space-efficient. Some suppliers are also combining multiple magnetic functions into a single component, further reducing overall volume. As PSU power levels increase from 3 KW to 5.5 KW and beyond, shrinking magnetic components becomes one of the most important contributors to power-density improvement.

Improved thermal management

As power output increases, heat becomes the primary obstacle to higher density. Even highly efficient PSUs still generate a significant amount of heat. Therefore, manufacturers are investing heavily in better cooling solutions, including optimized airflow paths, advanced heatsinks, vapor chambers, and liquid-cooling technologies. The goal is to remove heat more effectively so that components can operate safely even when packed more tightly together. In many modern PSU designs, thermal engineering is just as important as electrical engineering in determining achievable power density.

Higher component integration

Another way to increase power density is to reduce the amount of unused space inside the PSU. Manufacturers are integrating more functions into fewer components and using three-dimensional layouts that make better use of available volume. Components are stacked, arranged vertically, or combined into integrated modules to maximize space utilization. This approach is similar to how smartphones have become more powerful without becoming larger—the same principle is increasingly being applied to server power supplies.

Higher efficiency creates more thermal headroom

Efficiency improvements provide a double benefit. First, less electricity is wasted. Second, less heat is generated. For example, a 5.5 kW PSU operating at 97.5% efficiency generates roughly 30% less heat than one operating at 96.5% efficiency. While a one-percentage-point improvement may appear small, it can remove tens of watts of heat from the system. This allows engineers to either increase output power within the same form factor or reduce cooling requirements while maintaining existing power levels.

Reducing the size of energy storage components

Server PSUs contain capacitors that store energy and maintain power delivery during short interruptions. Historically, these capacitors occupied a significant amount of space. Newer designs use smarter control techniques that allow manufacturers to achieve the same performance with smaller capacitor banks. Reducing capacitor volume frees up valuable internal space that can be used for higher power output or additional functionality.

For example, in Infineon's 3.3 KW example, the calculated capacitance for a conventional 10ms hold-up implementation was approximately 2.5 mF; allowing deeper discharge through a baby-boost stage reduced the requirement to about 900 µF. The 12 KW design similarly uses a bidirectional buck-boost buffer to achieve 20ms hold-up at full load without the capacitor volume that a passive solution would require. The penalty is additional semiconductors, an inductor, control software and failure modes, but the net volumetric benefit can be substantial.

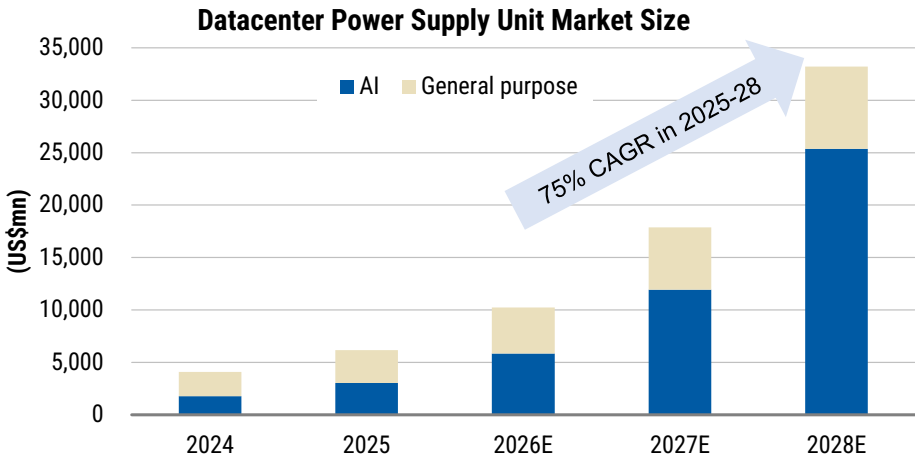
Advanced mechanical design

Power density is ultimately measured in watts per unit volume, making physical packaging extremely important. Manufacturers continuously optimize component placement, airflow channels, connector design, and enclosure structures to eliminate unused space. The best PSU designs are often not defined by a single breakthrough component but rather by hundreds of small mechanical optimizations that collectively improve volumetric efficiency.

Datacenter PSU market size to grow at a 75% CAGR in 2025-28

Overall, we estimate the global datacenter power supply market will grow at a three-year CAGR of 75% to reach US$33bn in 2028 from US$6bn in 2025, driven mainly by server shipment growth, increasing power consumption per rack, and the adoption of HVDC power racks from 2027.

Exhibit 34: Global datacenter power supply market forecast



Source: Company data, Morgan Stanley Research estimates

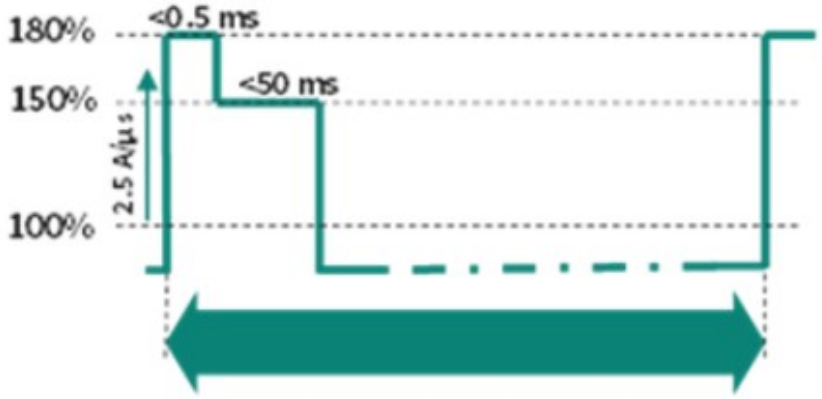
Complementary Hardware for More Resilient Power Delivery

Transient power spikes at AI servers

AI servers have a much more dynamic power profile than conventional enterprise servers because GPU utilization changes sharply throughout a workload. During large-model training, thousands of GPUs typically operate in a bulk-synchronous manner, and power consumption approaches the accelerator's design limit during tensor-compute phases, then falls during collective communications, check pointing, data movement, or periods in which GPUs wait for slower devices to synchronize. Inference can create a similar pattern as request arrivals, batching, context length and token-generation activity change rapidly.

Because the GPUs perform these transitions together, the power variation of one accelerator is multiplied across the server, rack and cluster rather than being averaged out by unrelated workloads, so the magnitude of the challenge is also rising with system density. Power spikes are therefore not simply the result of higher average consumption; they reflect the combination of higher accelerator power, more GPUs per system and increasingly synchronized workload behavior.

Exhibit 35: Transient power spikes at AI servers



Source: Infineon

System performance and reliability potentially compromised

The key issue is that power delivery systems are designed to supply a relatively stable amount of power, while AI workloads can change power demand almost instantaneously. When the load changes faster than the power-delivery system can react, a temporary mismatch occurs between power demanded by the GPUs and power supplied by the PSU, creating electrical instability throughout the server.

Sudden increases in GPU demand lead to voltage drops

When GPUs transition from a low-utilization state (communication, synchronization, checkpointing, etc.) to a full-compute state, their current demand can increase within microseconds. However, the PSU cannot instantly increase its output because its control loop requires time to detect the load change and adjust power conversion.

During this response period, the additional energy must come from capacitors located throughout the power chain. As these capacitors discharge, the voltage on the DC bus begins to fall, creating voltage drop.

If the drop becomes sufficiently large, several operational issues can occur:

- GPU throttling: The GPU reduces clock speed or performance to remain within safe operating limits.
- Computation errors: Voltage instability can affect timing margins in high-speed processors and memory.
- Unexpected resets or shutdowns: Protection circuits may trigger if voltage falls below specified thresholds.
- Training interruption: Large AI training jobs running across thousands of GPUs may fail if even a small number of nodes experience instability.

In a large AI cluster, a single server failure can force the entire training job to restart from the latest checkpoint, resulting in significant productivity loss.

Sudden decreases in GPU demand results in voltage spike

The opposite problem occurs when GPUs rapidly reduce power consumption.

For example, after a compute-intensive phase, GPUs may enter a communication phase where power consumption drops sharply. The PSU, however, is still delivering power based on the previous load condition. The excess energy has nowhere to go immediately and accumulates in the DC bus, causing a temporary voltage spike.

This can create:

- Overvoltage stress on components
- Protection-circuit activation
- Accelerated degradation of capacitors and voltage regulators
- Higher risk of component failure over time

Although individual spikes may be small, AI servers can experience these transitions continuously during training, causing repeated stress throughout the power-delivery chain.

Excessive current surges throughout the system

Power spikes are not only a voltage problem—they also create large current excursions.

A 1KW increase in power demand at 12V requires roughly 83A of additional current. In modern AI servers, where multiple GPUs can change utilization simultaneously, current swings can reach hundreds of amperes within a very short period.

These current surges lead to:

- Higher conductor losses
- Increased connector heating
- Greater stress on voltage-regulator modules (VRMs)
- Larger thermal fluctuations in power components
- As GPU power ratings rise to over 1KW, the magnitude of these transient currents becomes increasingly difficult to manage.

Thermal cycling and reliability degradation

Perhaps the most overlooked issue is the impact on long-term reliability.

Every power excursion creates a corresponding temperature excursion. When power rises, semiconductor junction temperatures increase; when power falls, temperatures decrease. This repeated heating and cooling cycle occurs thousands or even millions of times over a system's lifetime.

The result is accelerated wear on:

- Electrolytic capacitors
- Power MOSFETs
- Solder joints
- PCB interconnects
- Connectors and busbars

Consequently, even if average power remains within specification, transient power spikes can shorten component lifetime and increase failure rates.

Infrastructure overdesign and lower efficiency

To prevent instability, operators often provision the power infrastructure for the worst-case peak rather than average consumption.

For example:

- A rack averaging 80KW may need infrastructure sized for 120-140KW transient peaks.
- PSUs must be oversized to accommodate short-duration excursions.
- UPS systems must reserve capacity for sudden load increases.
- Cooling systems must handle peak thermal output.

This creates a significant amount of stranded capacity, where expensive electrical and cooling infrastructure remains underutilized most of the time. Furthermore, PSUs and UPS systems typically operate most efficiently near their optimal loading point; oversized systems therefore tend to operate at lower efficiency, increasing operating costs.

Conventional power delivery systems not designed to deal with power volatility

In the conventional power delivery architecture, the UPS protects the data hall against utility outages and power-quality disturbances, while the PSU converts upstream power into the tightly regulated DC output required by the server. Both remain essential in the modern AI server clusters, but neither was originally optimized to absorb large, repetitive and synchronized AI load steps economically.

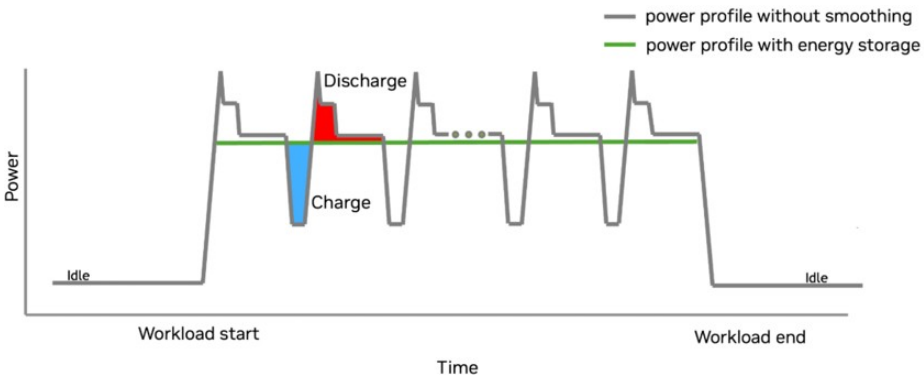
The PSU can use its internal capacitors to cover the initial transient, but capacitor volume is limited; once that stored energy is depleted, the PSU control loop must raise input current, potentially producing current overshoot and transferring the disturbance upstream.

The UPS is located farther from the accelerator and sees the aggregated load of many PSUs. A transient originating near the GPU can therefore occur on microsecond-to-millisecond timescales—including permitted GPU electrical-design-power excursions around 50 ms—before the complete upstream conversion chain can rebalance power.

This is a response time and location mismatch, rather than evidence that UPSs or PSUs are fundamentally incapable of supporting AI. New AI-tolerant UPS platforms are being designed for faster and larger load steps, but many legacy systems may still experience voltage excursions, nuisance alarms, lower partial-load efficiency or unwanted bypass transfers under steep synchronized loads.

To complement the conventional power delivery architecture, more AI server clusters are adopting BBUs or CBUs to shave peak loads and mitigate the above-mentioned issues.

Exhibit 36: Concept of peak shaving for AI server power loads



Source: Nvidia

Battery backup units seeing increasing importance

What's a BBU?

A BBU is an integrated battery, power-conversion and control system that supplies DC power to server loads when the normal power source is interrupted or temporarily insufficient. In an OCP-style rack, the BBU modules are housed in a dedicated shelf located near the PSU power shelf and connect directly to the common DC busbar. Under normal conditions, the PSUs power the servers and maintain the BBU's charge; when the bus

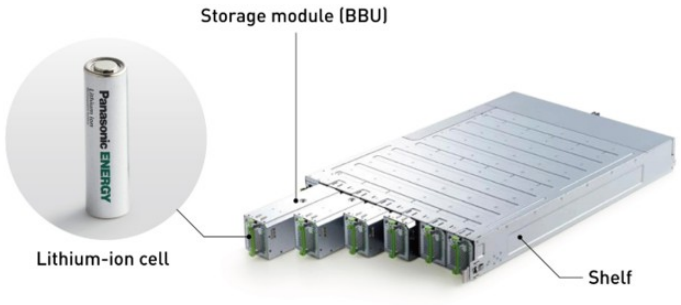
voltage falls below a programmed threshold, the BBU's bidirectional converter changes from charging or standby into discharge mode and supports the rack. Analog Devices' ORV3 reference design detects a bus voltage below 48.5V for more than 2ms and is required to ramp to full support within another 2ms, while maintaining the bus above the specified minimum voltage.

Key components within a BBU

- **Battery-cell pack:** The cells store the energy used during discharge. Lithium-ion cylindrical cells are commonly arranged in series and parallel to obtain the required voltage, power and runtime; the ORV3 reference module uses an interchangeable 11-series, 6-parallel battery pack. The cell must provide not only sufficient energy but also high discharge power, low internal resistance and predictable performance over its qualified life.
- **Bidirectional DC/DC converter:** The converter controls energy flow between the battery pack and the rack DC bus. It steps voltage in the appropriate direction during charging and discharging and regulates the output so that the battery's changing voltage does not destabilize the server bus. Multiphase designs divide the large current across several parallel power stages; TI's 12 KW design uses six 2 KW phases and reports peak efficiency above 99%, including above 99.5% when battery and bus voltages are close.
- **Battery management system:** The BMS measures individual cell voltages, current and temperature, balances cells and estimates state of charge and state of health. It also protects against overcharge, excessive discharge, overheating and short circuit. Accurate state-of-health information is important because the operator must know whether an apparently charged BBU can still deliver its required rack power and runtime after years of standby operation and periodic discharge events.
- **Power-path and protection circuitry:** Fuses, current sensors, disconnect switches, protection MOSFETs and hot-swap circuits isolate faults and prevent uncontrolled current flow. Typical protections include overvoltage, undervoltage, overcurrent and overtemperature shutdown. Hot-swap capability allows a failed or end-of-life module to be replaced without intentionally shutting down the entire rack, while active current sharing ensures that parallel BBU modules carry an appropriate portion of the load.
- **Control, monitoring and communications:** A module controller manages operating modes, converter control, fault handling, diagnostics and communication with the BMS and rack controller. Interfaces such as PMBus, Modbus, CAN, RS-485 or Ethernet can expose battery voltage, current, temperature, state of charge, state of health, fan speed and fault status to data-center management systems. Analog Devices' reference platform uses separate controllers for battery monitoring and overall BBU logic, communications and housekeeping.
- **Thermal-management system:** Fans, temperature sensors, airflow paths and mechanical spacing around cells remove heat during high-rate charging and discharging. Thermal management is both a performance and a safety requirement: excessive temperature accelerates battery ageing and can restrict available discharge power. Analog Devices' ORV3 design therefore incorporates fan control, airflow simulation and cell-to-cell spacing, while Murata's commercial 3 KW BBU includes internal forced-air cooling and overtemperature protection.

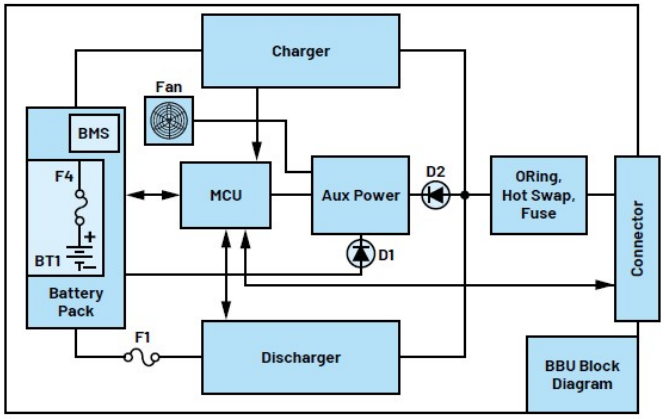
- **Shelf, backplane and mechanical integration:** The shelf distributes DC power and communications across several modules and provides retention, service access and cooling. Modularity allows power and runtime to be scaled by adding units and enables redundancy such as the ORV3 5+1 configuration. Murata specifies a 2U, hot-swappable 3 KW module, while Eaton's 19-inch BBU shelf accepts six 2.75 kW batteries and scales runtime according to the application requirement

Exhibit 37: BBU



Source: Panasonic

Exhibit 38: BBU diagram



Source: ADI

Key functions of BBUs

Backup power and outage ride-through

The BBU's fundamental function is to maintain rack operation during a brief loss of upstream power. The ride-through interval gives the data center time to restore the primary source, transfer to another source, preserve critical data, migrate workloads or perform an orderly shutdown. Rather than attempting long-duration operation, rack BBUs are generally optimized for high power over a relatively short period; the ORV3 example specifies 3KW per module for four minutes, and Murata guarantees a 240-second hold-up period at full-rated load under stated age and temperature conditions.

For AI systems, this protection is particularly valuable because a rack interruption can affect a synchronized workload extending well beyond the failed server. Local ride-through prevents a momentary upstream disturbance from immediately removing GPU, CPU, networking and storage power from the rack. Infineon consequently describes the server BBU as an integrated backup supply intended to prevent data loss and maintain AI workloads during outages or grid failures.

Peak shaving and load smoothing

A BBU can be operated as an active energy buffer. as it charges during lower-demand periods and discharges when server consumption temporarily exceeds a defined rack or PSU limit. This allows a relatively stable upstream power source to support a more volatile downstream AI workload. The economic benefit is that PSU, UPS and electrical capacity can be designed closer to sustained demand rather than every short theoretical peak, although the achievable benefit depends on the frequency, duration and predictability of those peaks.

The important limitation is battery cycling. A BBU designed primarily for emergency standby may not be optimized for continuous, repetitive peak shaving, so frequent cycling affects heat generation, state of charge, usable life and replacement schedules. Very short and highly repetitive transients may therefore be better supported by a CBU or hybrid battery-capacitor system, leaving the battery to cover longer events. Renesas correspondingly distinguishes distributed BBUs from hybrid systems that combine batteries with capacitors for fast transient response and longer-duration backup.

Power stabilization, redundancy and fault isolation

A rack BBU adds another independent source to the local power path. If a power shelf, upstream feed or AC source is interrupted, the BBU can sustain the rack while the fault is isolated or another source assumes the load. A modular shelf further improves resilience because one BBU failure need not remove the entire backup function; the remaining modules can continue operating within the available redundancy and power limits.

Distributed deployment also narrows the fault domain. A defective rack BBU affects a much smaller set of compute equipment than a failure within a centralized backup system serving a large room or data hall. The tradeoff is a much larger population of batteries, converters, controllers and fans that must be monitored, tested and eventually replaced. Digital state-of-health monitoring, fault reporting and hot-swappable construction are therefore central to the operating model rather than optional features.

BBUs vs. UPSs

While some of the functions that BBUs offer look similar to what traditional UPSs can do, we think BBUs and UPSs should be viewed as complementary rather than substitutive technologies, as they address different challenges within the datacenter power architecture.

A UPS is designed to provide facility- or row-level protection against utility outages, voltage sags, surges, and other power-quality disturbances, ensuring continuous AC power delivery to a broad range of IT equipment. In contrast, a BBU operates much closer to the AI workload, supplying DC power directly at the rack level and responding to localized power events that occur faster than the upstream power infrastructure can efficiently manage.

As AI racks become denser and more power intensive, the UPS continues to serve as the first line of defense against external power disruptions, while the BBU acts as a second layer of protection by stabilizing the rack DC bus, supporting transient load spikes, and providing short-duration ride-through capability. Together, they create a hierarchical power protection architecture in which the UPS ensures overall system availability and power quality, while the BBU enhances rack-level resilience and power-delivery performance. Rather than replacing the UPS, BBUs help reduce the burden on centralized backup systems, improve power utilization efficiency, and enable the reliable operation of next generation AI infrastructure with increasingly dynamic power profiles

Exhibit 39: Comparison of BBUs and UPSs

Dimension	BBU	UPS
Location	Inside the server rack, in a BBU shelf, or in an adjacent rack/sidecar close to the compute	Typically deployed at equipment room, row or facility level and protects multiple racks or other critical loads
Power path	Usually DC-coupled to the rack bus through a bidirectional DC/DC converter	Takes AC input and delivers AC output; the server PSU subsequently converts AC again for the server
Primary objective	Short rack-local ride-through, DC-bus support, modular redundancy and—when designed for it—transient or peak-load assistance	Broad power conditioning and uninterrupted AC supply during utility sags, surges, frequency disturbances and outages
Response	Monitors the local bus and can inject power within a few milliseconds; the cited ORV3 reference requires full support within 2ms after a qualifying 2ms event	An online double-conversion UPS already powers the load through its inverter and therefore has zero transfer time between utility and battery operation
Runtime	Generally optimized for short-duration, high-power support; OCP's reference example is four minutes at full load	Runtime varies widely with battery-bank sizing and system design; its broader function is to maintain protected AC loads until another source or operating response becomes available
Scalability	Added rack-by-rack or module-by-module; power and runtime can be matched to each compute deployment	Scaled centrally in larger KW blocks and requires capacity planning for the aggregate protected load
Fault domain	A module or shelf fault is normally localized to one rack or a limited group of racks	A centralized system protects a much broader load domain, so its redundancy and maintenance design are correspondingly critical

Source: Morgan Stanley Research

Capacitor backup unit (CBU) another complementary power device

What's a CBU?

A CBU is an integrated energy storage subsystem positioned close to the AI load, typically in a dedicated rack shelf, capacitor tray or adjacent power sidecar, and connected to the server rack's DC power path. Its principal storage medium may be an electric double-layer capacitor (EDLC or supercapacitor), lithium-ion capacitor (LIC), or a bank of high-capacitance electrolytic capacitors, depending on the required response time, energy duration, footprint and cost. Dedicated CBU products increasingly use EDLCs or LICs because their low internal resistance supports rapid, repetitive charge and discharge.

The CBU sits electrically alongside the PSU. Under steady-state conditions, the PSU supplies the AI rack and maintains the CBU at its target state of charge. When GPU demand temporarily exceeds the desired PSU output, the CBU discharges through a controlled power converter into the DC bus; when GPU demand falls, the converter reverses direction and recharges the capacitors at a controlled rate. This prevents the full GPU load step from passing through the PSU and upstream equipment. Infineon notes that a separate CBU must be compatible with, and potentially communicate with, the PSU shelves to coordinate rack power.

An alternative to deploying a standalone CBU shelf is to integrate the energy-storage capacitors directly into the power shelf, effectively combining power conversion and transient-energy buffering within the same subsystem. NVIDIA's GB300 NVL72 architecture is a good example: its enhanced power shelves incorporate a substantial capacitor bank alongside the PSU modules, with NVIDIA allocating approximately 65 joules of stored energy per GPU.

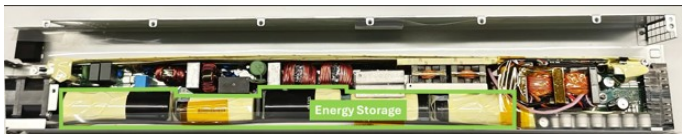
NVIDIA reported that this approach reduced peak upstream power demand by roughly 30% in a measured Megatron LLM workload. Compared with a separate CBU, integrating capacitors into the power shelf can save valuable rack space, shorten the electrical path between the energy buffer and the load, and simplify coordination between the PSU and

energy-storage system, since charging, discharging, and PSU output can be managed as one integrated power architecture. The trade-off is that capacitor capacity becomes more tightly coupled to the power-shelf design, potentially reducing flexibility in independently scaling or servicing the energy-storage system. Nevertheless, as AI racks become increasingly power dense, integrating transient-energy storage directly into the power shelf could become an attractive alternative to standalone CBUs, particularly where rack space and fast transient response are key design priorities.

Exhibit 40: CBU

Source: Panasonic

Exhibit 41: GB300 power shelf with integrated capacitors



Source: Nvidia

Key components within a CBU

- **Capacitor bank:** The storage section contains many capacitor cells arranged in series to reach the required voltage and in parallel to provide the required power and energy. Individual EDLC cells commonly operate at only a few volts, so high-voltage systems require long series strings; Abracon cites approximately 2.7 V as a typical supercapacitor cell rating, while TI's NVIDIA-oriented reference architecture demonstrates an 800 V CBU using EDLC cells. LICs can provide more stored energy than conventional EDLCs, while EDLCs generally emphasize lower resistance, high peak power and very high cycling endurance. Delta, for example, offers a power-capacitance shelf using fast-charge/discharge LICs, rated to provide 15 seconds of hold-up at 20 KW.
- **Cell-balancing system:** Because cell capacitance, leakage current and internal resistance vary across a series string, the voltage does not divide perfectly equally. A balancing circuit prevents an individual cell from exceeding its rated voltage during charging or being reverse-biased during deep discharge. Passive balancing uses resistors to bleed excess charge, while active approaches use MOSFETs, amplifiers or switched converters to redistribute energy more efficiently; the design tradeoff is between cost, balancing speed, component count and standby loss.
- **Bidirectional DC/DC converter:** This is the active interface between the capacitor bank and the rack DC bus. During a GPU power spike, it converts the capacitor's changing voltage into a regulated bus contribution; during lower demand, it reverses power flow to recharge the bank. The converter also allows more of the capacitors' stored energy to be used than would be possible with a direct connection, because capacitor voltage declines materially during discharge.

Infineon specifically identifies a capacitor bank plus bidirectional converter as the main CBU structure, while Skeleton Technologies notes that converter-based integration provides greater energy utilization and more precise power-flow control than a direct bus connection.

- **Controller and management software:** The controller determines when the CBU should charge, remain idle or discharge, based on bus voltage, PSU output, rack power limits and capacitor state. It may also calculate state of health, manage balancing, regulate recharge current and coordinate multiple CBU modules.
- **Protection and switching:** A high-power capacitor bank can release very large fault current, so fuses, contactors, pre-charge circuits and overvoltage, overcurrent and overtemperature protection remain essential. Pre-charge limits inrush current when the CBU is connected to the DC bus, while contactors or solid-state switches isolate the bank during a fault or service event. Skeleton's reference system integrates switchgear, fuse-fault detection, pre-charge/discharge contactors and protection against high-current and overheating events; industry discussions also highlight the need for faster DC fault isolation as CBU voltage rises.
- **Monitoring and communications:** Voltage, current, temperature, equivalent series resistance and state-of-health measurements allow operators to identify degradation before the CBU can no longer meet its specified peak-power requirement. Panasonic emphasizes voltage, temperature and ESR monitoring for predictive maintenance, while Skeleton provides single-cell voltage sensing, multiple temperature sensors and communication bus to its master controller.
- **Thermal and mechanical system:** Although supercapacitors are efficient, high repetitive current produces resistive heating, particularly in cells, busbars, converters and connectors. The shelf therefore requires designed airflow, temperature sensing and, at higher power levels, potentially liquid cooling. Skeleton's cabinet uses controlled fan cooling and also offers liquid-cooled configurations, while the Open Compute Project presentation from Panasonic identifies lower-resistance capacitors and liquid cooling as development priorities for future higher-output CBUs

Main functions of CBU

Transient power stabilization

One of the most fundamental roles of a CBU is to stabilize the rack DC power bus when GPU power consumption changes very rapidly. AI workloads frequently alternate between compute-intensive phases and lower-power communication, synchronization, or idle phases. Because many GPUs operate in parallel, these transitions can create very large changes in rack power within milliseconds.

When GPU demand suddenly increases, the PSU cannot instantaneously increase its output. The CBU therefore discharges stored energy into the DC bus during this short response gap, helping prevent voltage drop and excessive current excursions. Conversely, when GPU demand suddenly falls, the CBU can absorb excess energy while the PSU reduces its output, helping limit voltage overshoot.

The key objective here is power quality and system stability, rather than reducing the overall power capacity required by the rack. In other words, the CBU acts as a fast

electrical shock absorber between the highly dynamic GPU load and the slower PSU. This becomes increasingly important as AI racks move above 100 KW and synchronized GPU workloads create larger absolute power swings.

Peak shaving

Another major function is peak shaving, whose objective is to limit the maximum power that the PSU and upstream infrastructure need to supply.

The benefit is potentially significant because the PSU, connectors, protection devices, and portions of the upstream power infrastructure no longer need to be oversized solely for short-duration GPU peaks. It can also allow PSUs to operate closer to their optimal loading range, improving utilization and potentially efficiency.

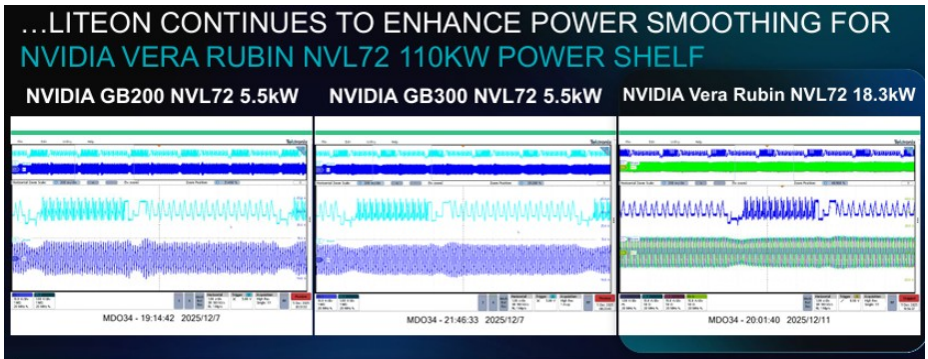
Short duration ride-through

The third function is ride-through, where the CBU temporarily powers the rack when the normal power source is interrupted altogether. This is fundamentally different from the first two functions because the issue is no longer a mismatch between GPU demand and PSU response—the upstream power source itself has temporarily disappeared.

A sufficiently sized CBU can maintain the DC bus during events such as a brief AC interruption, PSU switchover, source-transfer event, or temporary power-shelf failure. Depending on the amount and type of capacitance installed, this support can range from milliseconds to several seconds. For example, some LIC-based capacitance shelves are designed to provide multi-second hold-up at high rack power.

However, ride-through is generally a secondary function for a CBU. Capacitors have high power density but relatively low energy density, meaning they can deliver a large amount of power very quickly but cannot economically sustain that power for long periods. As the required backup duration increases from seconds toward minutes, a battery-based BBU generally becomes more suitable.

Exhibit 42: Capacitors help smooth power delivery



	Ipk-max	Ipk-min	Max/Min Ratio
NVIDIA GB200 NVL72	15.9A	8.3A	91.57%
NVIDIA GB300 NVL72	13.3A	8.6A	54.65%
NVIDIA Vera Rubin NVL72	**12.8A**	**10.3A**	**24.27%**

Source: Lite On

CBUs vs. BBUs

The two technologies should consequently be regarded as complementary rather than directly substitutive. A CBU handles the high-power, short-duration and high-cycle portion of the load profile, while a BBU supplies the larger amount of energy required when the upstream source remains unavailable. Some describe BBUs and CBUs as coexisting in the same power sidecar, and hybrid architectures can allocate fast transients to capacitors while preserving battery charge and cycle life for longer outages.

The CBU's major advantages are rapid bidirectional response, long cycle life, high efficiency and the ability to absorb repeated GPU peaks without the same electrochemical wear mechanism as a conventional battery. It can also recharge quickly between workload cycles and may require less routine replacement. However, the CBU has materially lower energy density, its voltage falls continuously during discharge, and large banks require balancing, high-current protection, cooling and valuable rack or sidecar space.

Exhibit 43: Comparison of CBUs and BBUs

Dimension	CBU	BBU
Primary storage medium	EDLC supercapacitors, LICs or high-capacitance	Typically lithium-ion battery cells managed by a BMS
Primary strength	High power, fast charge/discharge and frequent	Higher stored energy and longer sustained
Main function	Transient smoothing, peak shaving, voltage	Outage backup, source-transfer bridging and orderly workload
Response profile	Well suited to millisecond-class load support and	Fast switchover, but generally optimized for sustained support rather
Typical duration	Milliseconds to seconds; selected systems can extend toward tens of seconds or more	Commonly tens of seconds to several minutes, depending on battery and rack design
Power versus energy density	Higher power capability but lower energy storage per unit volume than a battery	Higher energy density, making it more economical for longer events
Cycle life	Eaton specifies more than one million cycles and up to 20 years for certain supercapacitor modules	Battery life is more sensitive to depth of discharge, temperature and repeated cycling; BMS supervision is therefore essential.
State management	Requires cell balancing, voltage/temperature monitoring and state-of-health tracking, but does not use a conventional battery fuel	Requires BMS functions including charge control, state of charge, state of health and cell protection
Best-fit use case	Repetitive AI power spikes, inrush events, PSU load leveling and bus stabilization	Power continuity through outages, brownouts or generator/source-transfer intervals

Source: Morgan Stanley Research

HVDC Emerging, with SST in the Pipeline

HVDC coming along with the increasing power consumption of AI clusters

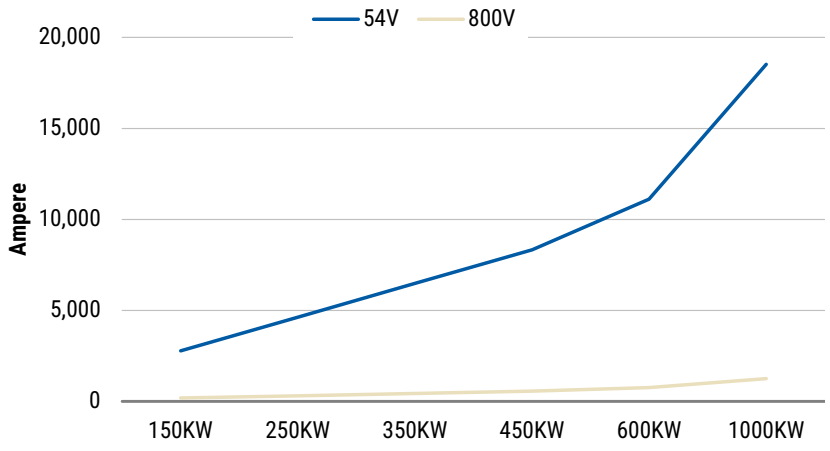
Why do we need HVDC?

Low voltage turns higher rack power into an extreme-current problem

The fundamental relationship is that at a given rack load, increasing the bulk-distribution voltage from 54 V to 800 V reduces current by approximately 14.8 times. A balanced ±400 V architecture also presents 800 V between its +400 V and –400 V conductors, so its bulk-transfer current is broadly equivalent to a 0–800 V architecture.

This current differential has an outsized mechanical consequence. For the same conductor resistance, I^2R loss at 54V would theoretically be approximately 219 times that at 800 V, though real systems avoid this level of loss by adding copper, parallel conductors and cooling. The result is not a 219-fold difference in actual operating loss, but a trade-off in which low-voltage designs consume increasing amounts of busbar cross-section, connector area, rack volume and cooling capacity to remain within voltage-drop and temperature limits.

Exhibit 44: Transmission current for different power scales: 54V vs 800V



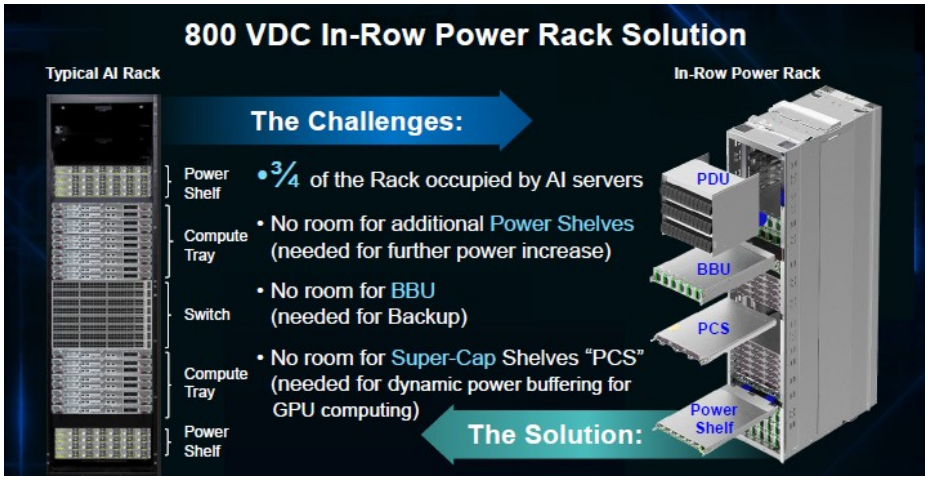
Source: Delta

Copper, connectors and thermal management become system constraints

At several thousand amperes, distribution cannot rely on a single ordinary conductor or connector. Designers must use broad laminated busbars, multiple parallel contacts, tightly controlled joint resistance and temperature monitoring. A small increase in contact resistance produces disproportionately high heat at these currents, raising the risk of hot spots, connector ageing and nuisance protection events. OCP's 100KW HVDC-to-LVDC shelf illustrates where the current problem moves: only two 62.5 A HVDC inputs are required on the ±400V side, while the 49–50V output approaches 2,000A and therefore uses a substantial bolted output connection with dedicated thermistors.

The same issue appears in power-shelf packaging. Delta notes that a typical AI rack may already allocate roughly three-quarters of its volume to servers, leaving insufficient room for additional power shelves, BBUs or supercapacitor shelves as power rises. Moving AC/DC conversion, protection and storage into an adjacent power rack therefore releases compute space and separates the thermal design of power electronics from that of GPUs. Microsoft's Mt Diablo rationale similarly states that disaggregated power can enable up to 35% more accelerators in the IT rack and allows power shelf count to be sized independently for each workload configuration

Exhibit 45: Not enough room for power devices on the traditional AI server rack



Source: Delta

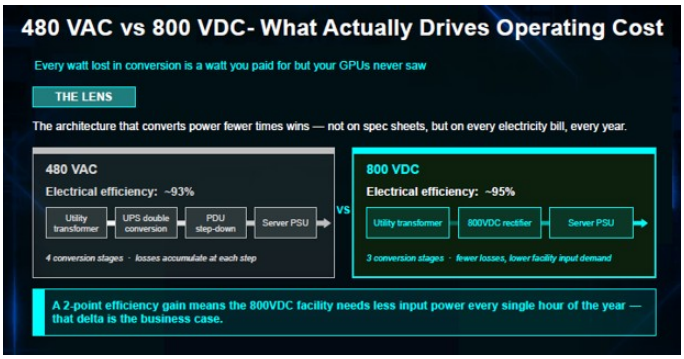
HVDC enables denser, fewer and more efficient power stages

As AI rack power scales from today's GB200/GB300 systems to Vera Rubin and future multi-hundred-kilowatt racks, power distribution efficiency becomes increasingly important. Our checks indicate that a conventional 480VAC architecture delivers an end-to-end electrical efficiency of roughly 93%, while an 800VDC architecture improves efficiency to about 95% by eliminating one power conversion stage and simplifying the power path from grid to rack.

The higher-voltage 800VDC distribution also reduces current levels for the same power load, lowering resistive (I^2R) losses in cables and distribution equipment and enabling more efficient delivery to increasingly power-dense AI racks.

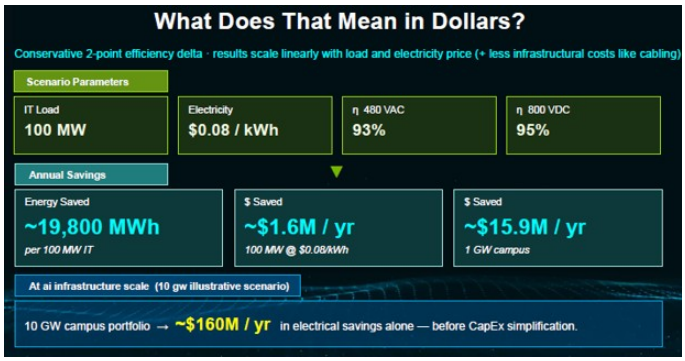
While a 2ppts of efficiency improvement may appear modest, the savings become substantial at hyperscale AI data center deployments. Delta's example shows that for a 100MW IT load, improving facility efficiency from 93% to 95% reduces annual energy consumption by approximately 19,800MWh, translating into roughly US$1.6 million of annual electricity cost savings assuming a power price of US$0.08/kWh. At a 1GW AI campus, the savings scale to nearly US$16 million per year, and at emerging multi-gigawatt AI factory deployments, the annual benefit can exceed US$100 million. Consequently, the transition from conventional AC distribution and 54VDC rack power architectures toward 800VDC distribution is not only a technical necessity for supporting future 500kW-to-1MW AI racks, but also a compelling economic proposition that lowers operating expenses and improves total cost of ownership through reduced energy losses and simplified electrical infrastructure.

Exhibit 46: Better power efficiency for 800VDC architecture



Source: Delta

Exhibit 47: Better efficiency leads to cost savings



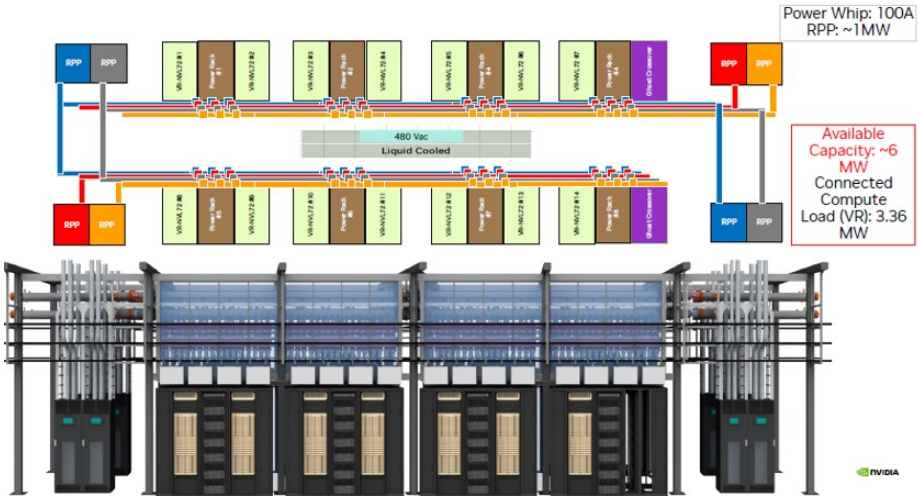
Source: Delta

What's the deployment roadmap for HVDC?

The deployment roadmap for 800VDC is best understood as a multi-stage evolution rather than a single-step transition. Both Nvidia and major hyperscalers recognize that existing 480VAC infrastructure cannot efficiently support the rapid increase in rack power. As rack densities increase, conventional power whips, busbars, connectors, and PSU architectures increasingly become bottlenecks, driving the need for higher voltage distribution

The first phase of deployment focuses on introducing 800VDC power racks within existing AC-based data centers. In this architecture, the facility remains largely unchanged, but a dedicated power rack converts 415-480VAC into 800VDC before distributing power locally to AI racks. This approach minimizes disruption to existing infrastructure while allowing operators to deploy Vera Rubin-class systems that exceed the practical limits of traditional power shelves. Nvidia refers to this as the row-level power rack architecture, which serves as the industry's bridge solution between legacy AC facilities and future native DC environments.

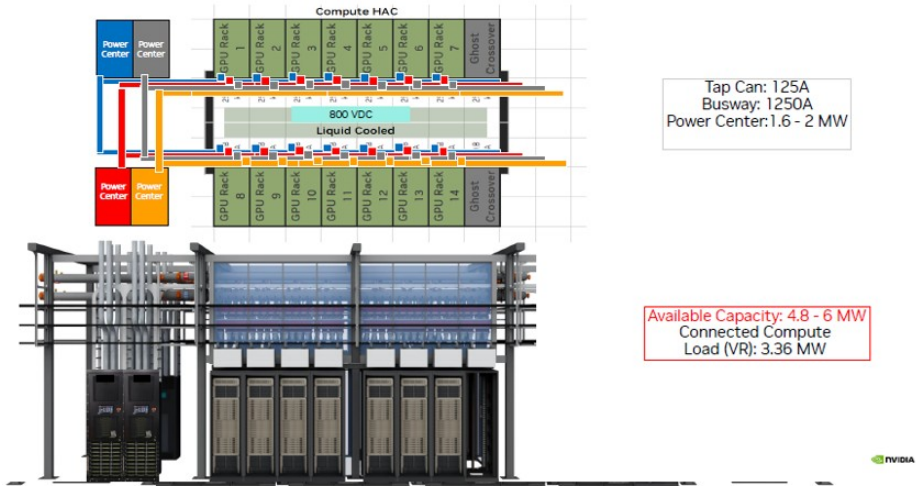
Exhibit 48: Phase I - 800VDC power rack



Source: Nvidia

The second phase expands 800VDC beyond individual rows into data hall-level power centers. Instead of deploying a dedicated power rack for every few compute cabinets, centralized rectifier or power-center systems distribute 800VDC through DC busways across an entire hall. Nvidia's roadmap identifies 1.6-2MW power centers and 1250A DC busway architectures as the next step, improving power density while reducing equipment duplication. This stage also aligns well with the deployment strategies being discussed by hyperscalers and the OCP community because it preserves existing medium-voltage and low-voltage infrastructure while moving a significant portion of distribution into DC

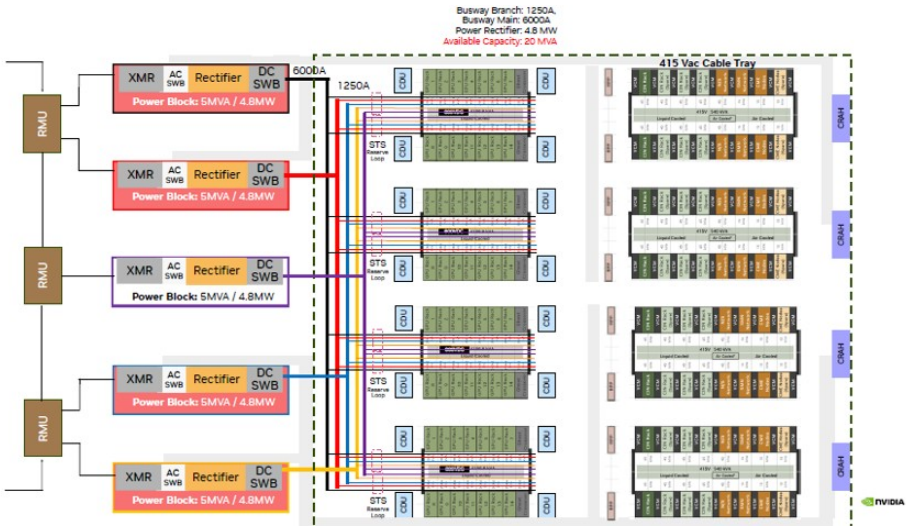
Exhibit 49: Phase II – 800VDC power center



Source: Nvidia

The third phase is the transition toward facility-wide 800VDC distribution, where AC is converted to DC near the utility entrance or medium-voltage substation and distributed throughout the data center via high-capacity DC switchboards and busways. Nvidia describes this as the DC power block architecture, utilizing approximately 4.8MW power blocks, 6000A DC busways, and integrated redundancy schemes. At this stage, the architecture begins to resemble an "AI factory," where power distribution is simplified, energy storage can be integrated directly onto the DC bus, and multiple megawatts of compute capacity can be deployed as modular building blocks

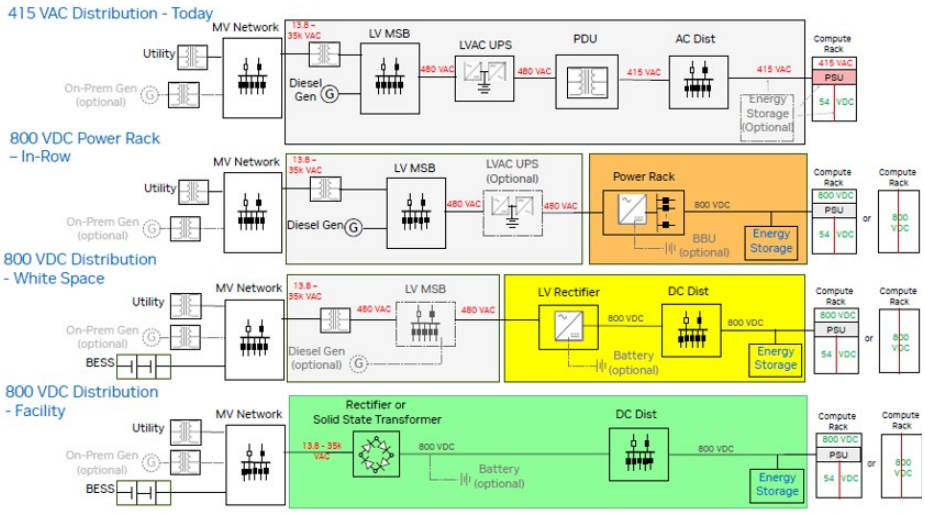
Exhibit 50: Phase III – power block



Source: Nvidia

The final stage is the emergence of native 800VDC compute racks, which Nvidia identifies as the long-term destination. Today, even in 800VDC deployments, power is generally stepped down to 54V before entering the servers. Future systems, however, are expected to accept 800VDC directly within the rack and potentially at the server level, eliminating another conversion stage and further reducing power losses. Nvidia's public roadmap shows a progression from current 54V architectures, to 800VDC-fed PSUs, and ultimately to 1MW-class native 800VDC racks capable of supporting next-generation AI accelerators

Exhibit 51: 800VDC native server racks



Source: Nvidia

In sum, deployment is likely to occur gradually because the ecosystem remains under development. NVIDIA notes that several enabling technologies—including DC protection devices, busways, connectors, grounding standards, arc-flash modeling, and solid-state transformers— are still undergoing certification and standardization. The roadmap presented by NVIDIA suggests a phased commercialization timeline, with 800VDC power racks reaching deployment readiness in 2026, hall-level power centers in 2027, and facility-scale DC power blocks becoming deployment-ready around 2028. Industry

collaboration among NVIDIA, Microsoft, Google, and OCP participants is focused on ensuring interoperability and regulatory acceptance before broader hyperscale adoption.

Exhibit 52: HVDC deployment roadmap

Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

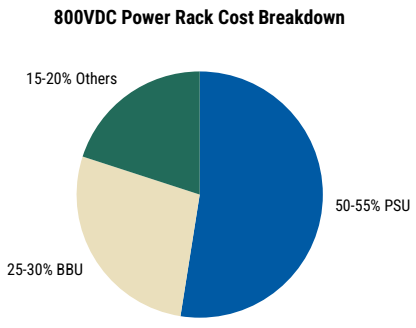
Source: Nvidia

800VDC represents 4-5x dollar content vs. in-rack power supply architecture

The 800VDC power rack BOM is primarily driven by the PSU, which accounts for 50–55% of the total cost, followed by the BBU at 25–30%, with other components comprising the remaining 15–20%. This cost structure reflects the broader functionality integrated into an HVDC power rack, combining power conversion, backup power and supporting components within a centralized architecture.

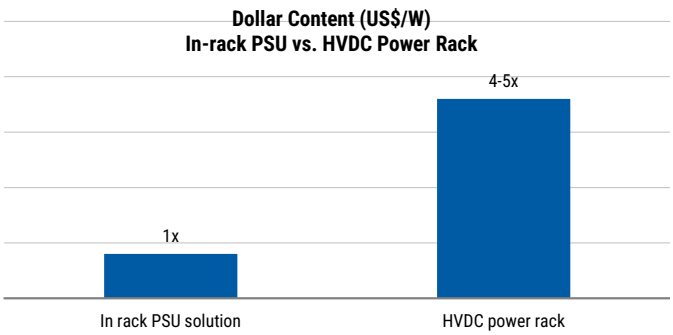
For power supply players, one of the most implications is that an HVDC power rack carries around 4–5x higher dollar content per watt compared with a conventional in-rack PSU solution, since the former integrates more components into a robust system and power supply players usually ship the entire rack instead of just the components.

Exhibit 53: BOM cost breakdown for an 800VDC power rack



Source: Morgan Stanley Research

Exhibit 54: Higher dollar content for HVDC power racks



Source: Morgan Stanley Research

800VDC vs. ±400VDC?

The debate between ±400VDC and 800VDC centers on balancing near-term practicality against long-term scalability. While both architectures deliver an identical 800V differential voltage and therefore achieve similar reductions in current, copper loss, and cable size versus traditional 54V rack power systems, they differ significantly in implementation.

Under a ±400VDC architecture, each conductor operates at only 400V relative to ground, resulting in less stringent insulation, creepage, clearance, and protection requirements. This makes the architecture easier to deploy, maintain, and certify, which explains why many hyperscalers and OCP members view ±400VDC as an attractive migration path toward high-density AI racks.

By contrast, NVIDIA and several power infrastructure vendors are increasingly advocating for a single-bus 800VDC architecture because it simplifies overall power distribution, reduces conversion stages, and is better suited for future 500kW to 1MW rack deployments.

We believe that ±400VDC appears well positioned as a near-term bridge architecture due to its lower deployment risk and more mature ecosystem, whereas 800VDC is increasingly viewed as the long-term end-state architecture for AI factories as rack power scales toward the megawatt era.

Exhibit 55: Comparison of 800VDC and ±400VDC

Metric	±400VDC	800VDC
Voltage to ground	400V relative to ground	800V relative to ground
Voltage differential	800V	800V
Power distribution efficiency	High	Slightly higher
Scalability	Good	Excellent
Insulation requirements	Lower	Higher
Maintenance safety	Better	More challenging
System architecture simplicity	Dual-rail system	Single-bus system
Grounding Complexity	More complicated	Simpler
Electrical footprint	Small	Smaller
Ability to reduce conversion stages	Moderate	Highest
Long-distance distribution capability	Excellent	Excellent
Maturity of ecosystem	Better today	Early but rapidly developing
Industry players	OCP, hyperscalers	Nvidia
Long term industry direction	Transitional architecture	Likely end-state architecture

Source: Morgan Stanley Research

From a component supplier perspective, the transition toward HVDC is far more important than which architecture ultimately wins, as most hardware content increases under either scenario. PSU vendors, BBU suppliers, busbar manufacturers, connector providers, power-whip suppliers, magnetics vendors, and rack-level DC-DC converter manufacturers are largely architecture-agnostic, since both ±400VDC and 800VDC require a similar shift toward high-voltage power distribution, centralized power shelves, and rack-level conversion.

Our understanding is that the primary exception is the power semiconductor industry, where the choice of architecture could materially influence device content and technology adoption. A ±400VDC system can often be supported using existing 650V-900V MOSFET and SiC platforms, benefiting suppliers with exposure to lower-voltage power devices. By contrast, 800VDC architectures increasingly require 1200V-class SiC MOSFETs, higher-

voltage gate drivers, hot-swap controllers, isolation components, and advanced protection circuitry, driving greater semiconductor content per watt and raising technology barriers.

Solid state transformer potentially being the holy grail in AI datacenter power delivery

What is a solid-state transformer ?

A datacenter solid state transformer (SST) integrates power electronics to replace or augment traditional line-frequency transformers, and it's typically built around three major functional stages. The first is the medium-voltage AC-to-DC rectifier stage, which uses high-voltage semiconductor switches—often silicon carbide (SiC) or gallium nitride (GaN) devices due to their superior switching speed and thermal performance—to convert incoming grid-level medium voltage AC into a regulated DC bus. The second is the isolated DC-DC conversion stage, usually implemented as a dual active bridge (DAB) or LLC resonant converter, which provides galvanic isolation through a high-frequency transformer (operating in the kHz to tens-of-kHz range rather than 50/60 Hz), dramatically shrinking the magnetic core and transformer size compared to conventional units while stepping voltage down toward datacenter distribution levels. The third is the low-voltage DC-AC or DC-DC output stage, which conditions power into the form needed by the facility—either regulated DC for direct DC-bus distribution to IT racks or inverted AC for legacy PDU/UPS infrastructure. Supporting these power stages are the control and sensing subsystems (digital controllers, gate drivers, voltage/current sensors) that manage switching, fault protection, and grid synchronization; thermal management components (heat sinks, cold plates, or liquid cooling loops) to handle the concentrated losses in the semiconductor devices; and increasingly, integrated energy storage interfaces or DC microgrid ports that allow the SST to natively support battery buffering, renewable integration, and finer-grained power quality control than a passive transformer ever could

A Solid-State Transformer (SST) is a power-electronics-based transformer that replaces the traditional 50/60Hz iron-core transformer with a combination of power semiconductors, control electronics, and a medium-frequency transformer. Unlike conventional transformers that perform only voltage conversion, SSTs actively process power through multiple stages, enabling both AC and DC inputs/outputs, bidirectional power flow, voltage regulation, reactive power compensation, and intelligent grid control

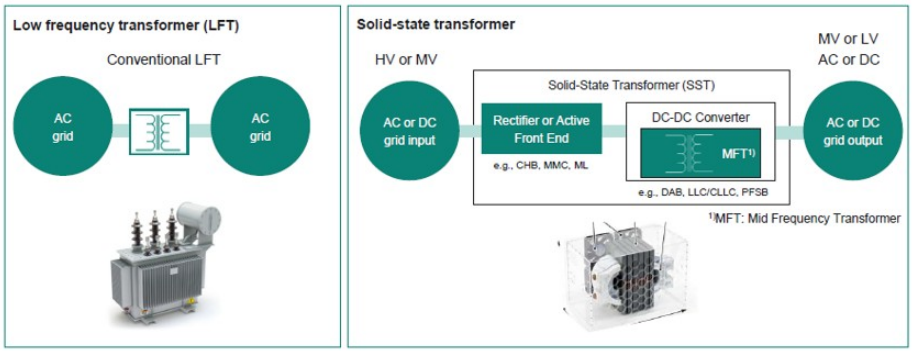
A SST is typically built around three major functional stages:

- The first is the medium-voltage AC-to-DC rectifier stage, which uses high-voltage semiconductor switches—often silicon carbide (SiC) or gallium nitride (GaN) devices due to their superior switching speed and thermal performance—to convert incoming grid-level medium voltage AC into a regulated DC bus.
- The second is the isolated DC-DC conversion stage, usually implemented as a dual active bridge (DAB) or LLC resonant converter, which provides galvanic isolation through a high-frequency transformer (operating in the KHz to tens-of-LHz range rather than 50/60 Hz), dramatically shrinking the magnetic core and transformer size compared to conventional units while stepping voltage down toward datacenter distribution levels.
- The third is the low-voltage DC-AC or DC-DC output stage, which conditions

power into the form needed by the facility—either regulated DC for direct DC-bus distribution to IT racks or inverted AC for legacy PDU/UPS infrastructure.

Supporting these power stages are the control and sensing subsystems (digital controllers, gate drivers, voltage/current sensors) that manage switching, fault protection, and grid synchronization; thermal management components (heat sinks, cold plates, or liquid cooling loops) to handle the concentrated losses in the semiconductor devices; and increasingly, integrated energy storage interfaces or DC microgrid ports that allow the SST to natively support battery buffering, renewable integration, and finer-grained power quality control than a passive transformer ever could.

Exhibit 56: Solid state vs. traditional transformers



Source: Infineon

Why do we need SSTs?

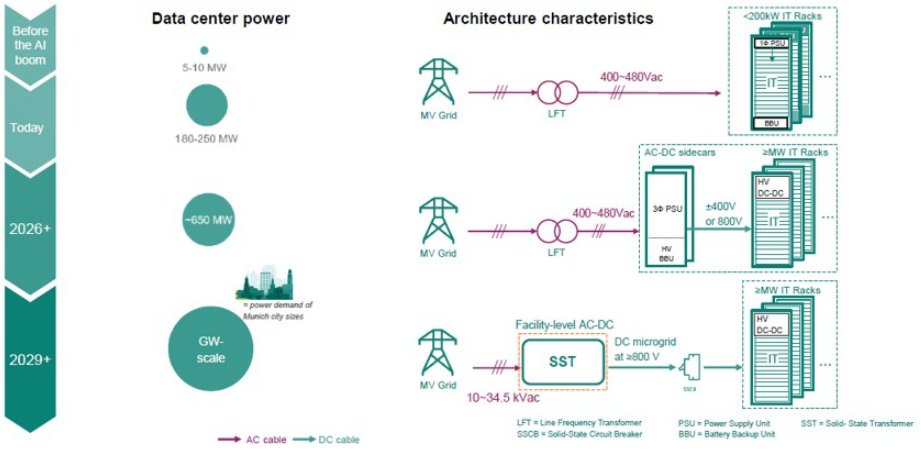
Improving efficiency in an era of gigawatt data centers

One of the strongest drivers for SST adoption is the rapid increase in AI data center power demand, which has progressed from 5-10MW facilities in the pre-AI era to campuses exceeding 650MW today, with gigawatt-scale facilities expected beyond 2029.

Traditional architectures use medium-voltage AC distribution, followed by low-frequency transformers and multiple power conversion stages before power reaches IT racks. SSTs allow facilities to convert 10-34.5KV medium-voltage power directly into ±400V or 800V DC distribution networks. This reduces conversion losses and improves overall grid-to-chip efficiency.

The efficiency advantage could be at the order of about 2-5ppt vs, the traditional transformer architecture, but at a 1GW AI campus, this magnitude of improvement could translate into roughly 20-50MW of continuous power savings. Over the lifetime of a facility, this can result in substantial reductions in electricity cost and carbon emissions. SSTs are therefore becoming increasingly attractive for hyperscaler operators seeking lower total cost of ownership (TCO).

Exhibit 57: SST enabling more efficient power delivery architecture



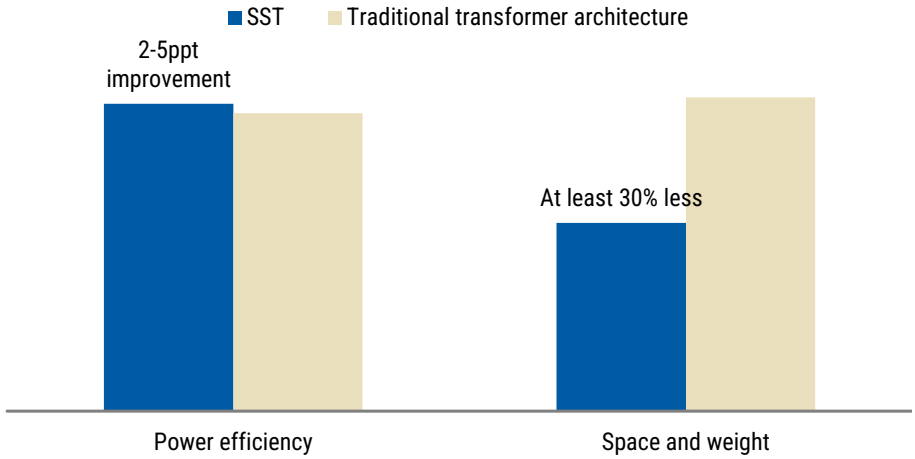
Source: Infineon

Higher power density and smaller footprint

Because SSTs use medium-frequency transformers instead of bulky 50/60Hz transformers, they can achieve significantly higher power density. Infineon indicates that SST systems can reduce both footprint and weight by at least 30% compared with conventional transformer-based installations. They also accelerate site energization and facilitate modular deployment

This becomes particularly valuable in urban data centers, utility substations, EV charging hubs, and energy storage sites where space constraints directly affect project economics.

Exhibit 58: SST offers better power efficiency with at least 30% less space/weight



Source: Infineon, Delta Electronics, Morgan Stanley Research

Supporting distributed energy resources

As renewable energy penetration increases, future grids will require seamless integration of solar generation, battery storage, fuel cells, gas turbines, and other distributed energy resources.

SSTs provide intelligent power routing, bidirectional operation, and dynamic voltage control. This allows them to function as energy hubs that connect renewable generation,

storage, and loads through a common DC infrastructure. Such capabilities are difficult or impossible to achieve using passive transformers alone.

Improved grid intelligence

Unlike conventional transformers, SSTs can actively regulate voltage and frequency, provide reactive power compensation, support grid-forming functionality, and improve overall power quality. These capabilities make SSTs an important enabler for future smart grids, microgrids, and virtual power plants.

What's the deployment status now?

Despite decades of research, SSTs remain at an early commercial deployment stage. While we started to see the SST models being unveiled by relevant suppliers and there start to be some pilot adoption, we believe that the 800VDC power rack/center/block architecture shall be deployed first, and SSTs will take more time, as more fundamental design change in the datacenter and all hardware inside will involved. We believe wider adoption probably won't happen before 2030.

Exhibit 59: SST from Delta

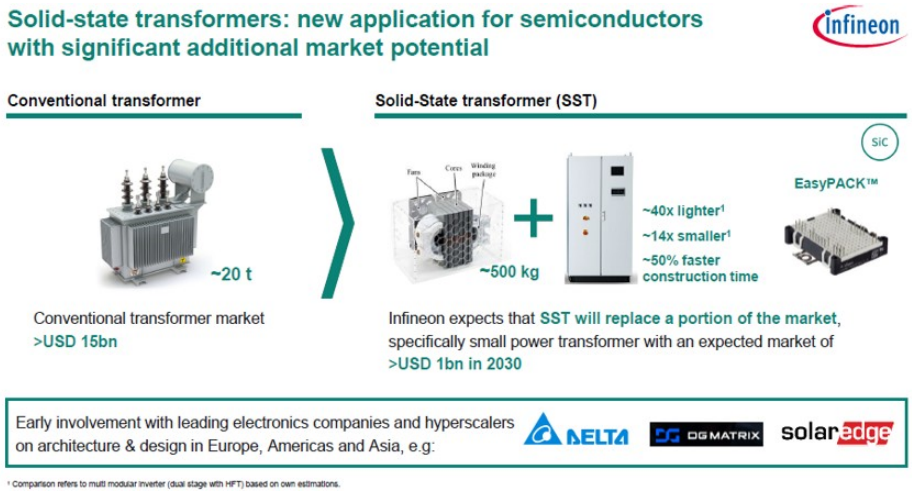
Source: Delta

Exhibit 60: SST from Eaton

Source: Eaton

According to Infineon, SST could represent a >US$1bn market opportunity by 2030, as the technology begins to replace a portion of the >US$15bn conventional transformer market, particularly in smaller-power applications. While SST penetration is expected to remain relatively modest versus the overall transformer market, the opportunity is significant for relevant players in the longer term.

Exhibit 61: SST to see initial adoption in 2030



Source: Infineon

Key hurdles for the wider adoption of SSTs

- **High system cost remains the primary barrier:** Despite their technical advantages, SSTs remain significantly more expensive than conventional transformers. Traditional transformers consist primarily of steel cores, copper windings, insulation materials, and cooling systems, all of which benefit from mature manufacturing processes and economies of scale. In contrast, SSTs require large numbers of high-voltage power semiconductors, gate drivers, digital controllers, sensors, protection systems, and sophisticated thermal management solutions. The extensive use of silicon carbide (SiC) devices, which remain substantially more expensive than conventional silicon-based components, further increases the overall system cost. Although SSTs can deliver efficiency gains, improved controllability, and reduced footprint, the economic benefit is often insufficient to offset the substantial upfront capital expenditure in many utility applications. Consequently, SST deployment is currently limited to applications where efficiency, power density, or system flexibility carry exceptionally high economic value, such as AI data centers, megawatt EV charging stations, and grid-scale battery storage systems.
- **Long term reliability has yet to be proven:** Electric utilities and large infrastructure operators place an extremely high value on reliability because transformers are expected to operate continuously for several decades with minimal maintenance. Conventional transformers routinely achieve service lifetimes exceeding 30 to 40 years and have accumulated more than a century of operating history. SSTs, however, are fundamentally different devices that incorporate thousands of semiconductor junctions, complex control electronics, communication networks, cooling systems, and software algorithms. While each individual technology may be mature, the long-term reliability of the complete SST system remains largely unproven at scale. Operators therefore remain cautious about replacing a passive asset with an active power-electronic system whose failure modes are more complex and less understood. For critical infrastructure such as substations, industrial facilities, or hyperscale data centers, reliability concerns often outweigh potential efficiency improvements.

- **DC protection technology ss still evolving:** One of the most significant technical challenges associated with SST deployment is fault protection, particularly in medium-voltage and high-voltage DC networks. Unlike AC systems, where current naturally crosses zero multiple times each cycle and facilitates fault interruption, DC currents continue flowing continuously during fault conditions. As a result, fault currents can rise extremely quickly, making them more difficult to detect and interrupt safely. Future SST-based power architectures therefore require advanced protection devices such as solid-state circuit breakers (SSCBs), which can react far faster than conventional mechanical breakers. However, DC protection technologies are still evolving, and there is currently no universally accepted approach equivalent to the mature protection schemes available in AC grids. The absence of standardized protection architectures increases deployment risk and remains one of the key obstacles to the wider adoption of SST-based DC distribution systems.
- **System complexity creates engineering, qualification and maintenance challenges:** Compared with traditional transformers, SSTs are considerably more complex systems that require the integration of multiple high-performance subsystems. A complete SST includes medium-voltage power converters, isolated DC/DC stages, high-frequency transformers, semiconductor modules, digital controllers, communication interfaces, and advanced protection mechanisms. As voltage and power levels increase, the number of converter modules and semiconductor devices required can rise significantly. Although the introduction of higher-voltage SiC devices can reduce subsystem count and simplify design, SST architectures remain substantially more complicated than conventional transformers. This complexity increases development costs, certification requirements, maintenance needs, and operational risks, all of which slow adoption among traditionally conservative utility customers.
- **Industry standards and ecosystems are still developing:** The transformer industry benefits from decades of established standards, testing procedures, maintenance practices, and supply chain infrastructure. SSTs, by comparison, represent an emerging technology with limited deployment history and evolving standards. Many utilities remain uncertain regarding qualification procedures, maintenance requirements, spare-parts strategies, and long-term asset management. Furthermore, widespread SST adoption will require parallel development of supporting technologies such as DC protection systems, DC switchgear, energy management software, and medium-voltage DC distribution networks. Until the broader ecosystem matures, many customers will continue to view SSTs as a promising but relatively high-risk alternative to proven transformer technologies.

Fuel Cells for Better Flexibility and Resilience

What are fuel cells?

Fuel cells are electrochemical energy-conversion devices that generate electricity by combining hydrogen and oxygen without combustion. Unlike conventional power generation technologies, which convert fuel into heat and then into electricity through multiple intermediate steps, fuel cells produce electricity directly through an electrochemical reaction, resulting in higher efficiency and significantly lower emissions.

When powered by hydrogen, the primary byproducts are water and heat, making fuel cells one of the cleanest forms of power generation available today.

Fuel cells can be deployed across a wide range of applications, including passenger vehicles, heavy-duty trucks, buses, trains, ships, stationary power generation, backup power systems, and increasingly data centers. As governments and industries pursue decarbonization while seeking reliable and dispatchable power sources.

Major fuel-cell technologies include:

- SOFC (Solid Oxide Fuel Cell): Preferred for stationary power and data centers.
- PEMFC (Proton Exchange Membrane Fuel Cell): Dominant in transportation and backup power.
- MCFC (Molten Carbonate Fuel Cell): Large-scale distributed generation.
- PAFC (Phosphoric Acid Fuel Cell): Commercial combined heat and power.
- DMFC (Direct Methanol Fuel Cell): Portable applications

Exhibit 62: Fuel cell offering - Delta

Source: Delta

Exhibit 63: Fuel cell offering - Bloom Energy

Source: Bloom Energy

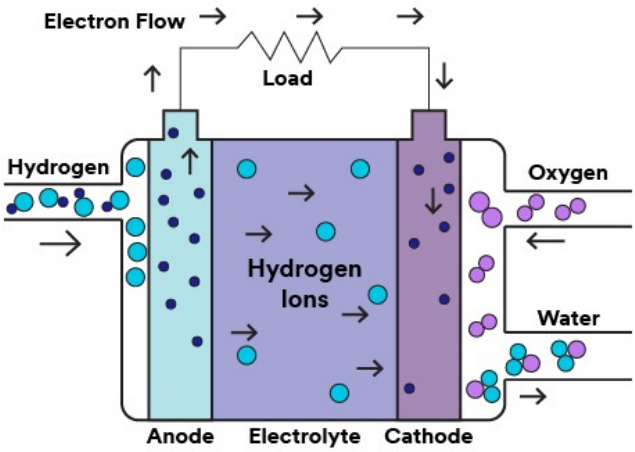
How do fuel cells work?

While there are several fuel-cell architectures, the fundamental operating principle is largely the same: fuel is supplied to the anode, oxygen is supplied to the cathode, and an electrolyte in between allows charged ions to move while forcing electrons to flow through an external circuit. This flow of electrons generates usable electricity, while the

reaction also produces heat and water as byproducts. Unlike conventional generators, which rely on burning fuel to drive a turbine or engine, fuel cells produce electricity directly through electrochemical reactions, allowing them to achieve significantly higher efficiency and lower emissions.

Using solid oxide fuel cell (SOFC) technology as an example, air is supplied to the cathode side of the cell, where oxygen molecules gain electrons and form oxygen ions (O^{2-}). These oxygen ions migrate through a ceramic electrolyte toward the anode side of the cell. At the same time, hydrogen or another hydrogen-rich fuel is supplied to the anode. When the oxygen ions reach the anode, they react with the fuel to produce water, heat, and electrons. The generated electrons cannot pass through the electrolyte and therefore travel through an external electrical circuit before returning to the cathode, creating a continuous flow of electricity. As long as fuel and air continue to be supplied, the fuel cell can generate power continuously.

Exhibit 64: Fuel cell diagram



Source: 3M

What benefits can fuel cells offer?

- **High energy efficiency:** Fuel cells convert the chemical energy of hydrogen directly into electricity through an electrochemical reaction, avoiding the multiple energy-conversion steps associated with combustion-based power generation. This allows fuel cells to achieve electrical efficiencies of over 60%, with some systems exceeding 80% total efficiency when waste heat is recovered through combined heat-and-power configurations, compared with roughly 35-45% for conventional gas-fired power generation. The efficiency advantage translates directly into lower fuel consumption, reduced operating costs, and improved energy utilization over the lifetime of the system.
- **Low carbon emissions:** When powered by hydrogen, fuel cells produce only water and heat as byproducts, generating no direct carbon emissions at the point of use. Even when operating on hydrogen derived from natural gas, fuel cells typically produce significantly lower greenhouse gas emissions than conventional combustion-based power systems. As green hydrogen production scales, fuel cells offer a potential pathway toward near-zero-carbon power generation for transportation, industrial facilities, and distributed energy systems.

- **Grid independence and energy security:** Fuel cells can provide reliable on-site power generation independent of the electrical grid, improving energy security and resilience. This makes them attractive for mission-critical applications such as data centers, hospitals, telecommunications infrastructure, defense installations, and industrial facilities that require uninterrupted power. In regions facing grid congestion or reliability concerns, fuel cells can also serve as a distributed power source located directly at the point of consumption.
- **Long duration energy storage enabler:** Fuel cells play an important role in the emerging hydrogen economy by enabling excess renewable energy from solar and wind farms to be stored as hydrogen and later converted back into electricity. Unlike batteries, which become increasingly expensive for longer storage durations, hydrogen storage can economically provide energy over days, weeks, or even seasonal periods. This capability could become increasingly valuable as renewable energy penetration rises worldwide.
- **Fuel flexibility:** Certain fuel-cell technologies, particularly Solid Oxide Fuel Cells (SOFCs), can operate on a variety of fuels including hydrogen, natural gas, biogas, ammonia-derived hydrogen, and other hydrogen-rich fuels. This flexibility allows users to deploy fuel-cell systems today using existing fuel infrastructure while maintaining a pathway toward pure hydrogen operation as hydrogen supply chains mature.
- **Quiet operation and reduced maintenance:** Because fuel cells contain relatively few moving parts compared with combustion engines and turbine-based generators, they operate quietly and often require less routine maintenance. The absence of combustion also reduces mechanical wear and vibration, which can improve system reliability and lower lifecycle operating costs, particularly in stationary power applications.
- **Scalability across multiple applications:** Fuel cells can be deployed across a wide range of power levels, from portable power systems measured in watts to utility-scale installations generating multiple megawatts. This scalability allows the technology to serve residential buildings, commercial facilities, industrial plants, transportation platforms, microgrids, and large-scale data centers, making fuel cells one of the most versatile clean-energy technologies available today.

Limitations and constraints of fuel cells

- **High system cost:** Despite significant technological progress, fuel cells remain more expensive than many conventional power-generation technologies. The cost challenge stems from complex stack manufacturing, specialized materials, balance-of-plant components, and relatively low production volumes compared with mature technologies such as internal combustion engines, gas turbines, and batteries. While costs have declined over the past decade, achieving widespread commercial adoption will likely require further manufacturing scale, supply-chain development, and technological improvements.
- **Limited hydrogen infrastructure:** One of the biggest challenges facing fuel-cell deployment is the lack of hydrogen production, transportation, storage, and refueling infrastructure. Unlike electricity, which is already widely distributed through established grids, hydrogen supply networks remain in the early stages of

development in most regions. This infrastructure gap creates a chicken-and-egg challenge, where end users are reluctant to adopt fuel cell technologies without reliable hydrogen availability, while infrastructure developers are hesitant to invest until demand reaches sufficient scale.

- **Durability and lifetime requirements:** Although fuel-cell durability has improved substantially, long-term reliability remains a key focus area, particularly for commercial transportation and utility-scale installations. Continuous exposure to heat, humidity, contaminants, and repeated operating cycles can gradually degrade fuel cell performance over time. Customers in sectors such as transportation, power generation, and industrial infrastructure often require operating lifetimes measured in tens of thousands of hours, creating significant engineering challenges for system developers.
- **Regulatory and safety considerations:** Hydrogen is a highly flammable gas and requires specialized handling, storage, and transportation systems. Although hydrogen technologies have well-established safety standards, new fueling stations, pipelines, and storage facilities must comply with stringent regulations. Developing these standards and obtaining approvals can extend deployment timelines and increase project costs.
- **Technology fragmentation and lack of standardization:** The fuel cell industry encompasses multiple technology platforms, including PEM, SOFC, MCFC, and other architectures, each optimized for different applications. While this diversity encourages innovation, it can also slow industry standardization and create uncertainty for customers evaluating long-term technology choices. Unlike the battery industry, which is increasingly converging around a few dominant chemistries, the fuel cell sector remains relatively fragmented.

Implications to datacenter power architecture

Transition from backup power to primary power source

Traditionally, data centers rely on grid electricity for primary power and diesel generators for backup power during outages. Fuel cells are increasingly being evaluated not only as a replacement for diesel backup generators but also as a potential primary power source. As AI-driven workloads substantially increase power density and overall electricity demand, hyperscalers are exploring fuel cells as a means of deploying computing capacity in locations where grid power is constrained or unavailable. This could fundamentally shift data-center power architecture from a grid-centric design to one that incorporates distributed on-site generation.

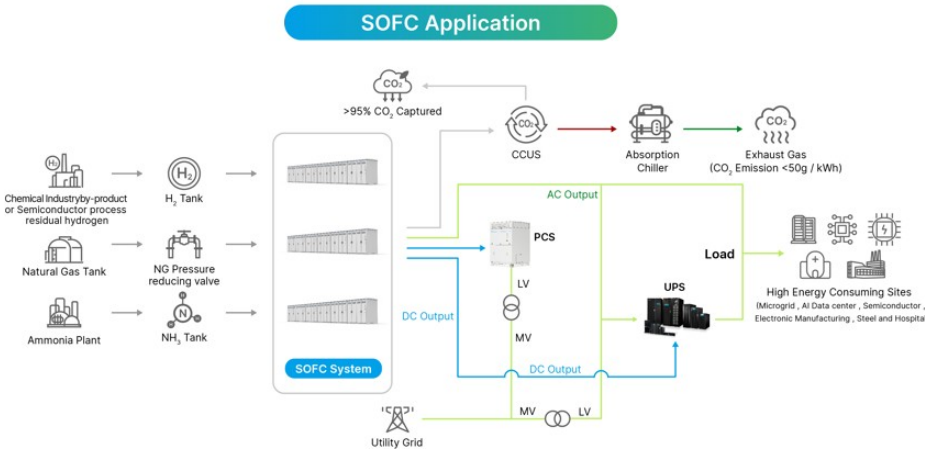
Potential long term evolution toward microgrid-based datacenters

In the longer term, fuel cells could become a core component of data-center microgrids combining renewable generation, battery storage, hydrogen storage, and intelligent energy management systems. Rather than functioning as passive consumers of grid electricity, future AI-focused data centers may increasingly operate as self-contained energy ecosystems capable of managing their own generation, storage, and load balancing. Such an architecture could provide greater scalability, resilience, and energy independence than traditional grid-connected designs

Potential simplification of power distribution architecture

Fuel cells inherently generate DC power, which aligns more closely with the power requirements of modern servers and IT equipment. Current data centers typically involve multiple AC/DC and DC/DC conversion stages between the utility grid and the processor, each introducing efficiency losses. Over time, fuel cell adoption could encourage a greater migration toward direct DC power distribution architectures, reducing conversion losses and improving overall facility efficiency. Moreover, its DC output fits well with the shift into HVDC power delivery architecture for datacenters.

Exhibit 65: Fuel cells in datacenter power delivery architecture



Source: Delta

Risk Reward – Delta Electronics Inc. (2308.TW)

Strong combination: Power + cooling for AI infrastructure

PRICE TARGET NT$2,500.00

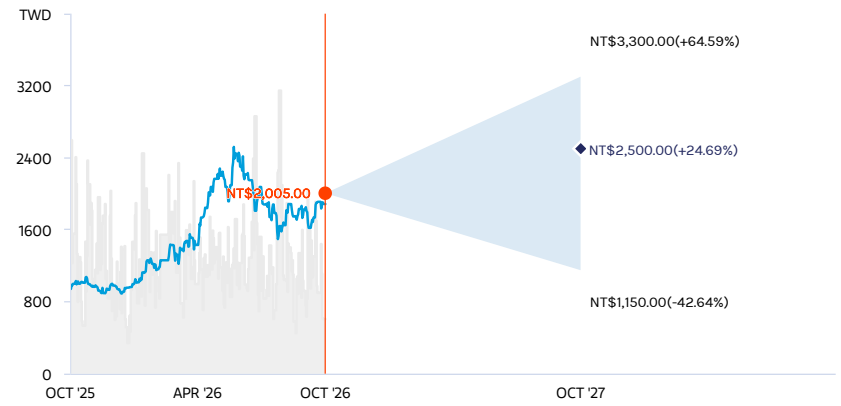
Base case: P/E multiple of 30x based on average 2027-28 EPS. We believe our target multiple is justified by the 63% earnings CAGR we estimate for 2025-28, implying a PEG of 0.5, versus the stock's ~23x P/E and 23% earnings CAGR during 2022-25, which implied a PEG of 1.0.

Consensus Price Target Distribution NT$2,200.00 NT$2,621.29 MS PT NT$4,120.00

Source: Refinitiv, Morgan Stanley Research

◆ Mean ◆ Morgan Stanley Estimates

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We view AI server power supplies and power management/cooling solutions for data centers as key growth drivers for the coming decade.
- Our expectation for a re-rating reflects expanding gross/operating margins and increasing exposure to AI-related businesses.
- Our 2026-28 earnings estimates reflect strength in AI server power supplies and liquid cooling.
- Our price target implies a P/E of 30x based on average 2027-28 EPS.

Consensus Rating Distribution

100%	Overweight
0%	Equal-weight
0%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth:	*Positive*
Self-help:	*Positive*
Technology Diffusion:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$3,300.00

40x average 2027-28e P/E

Strong sales growth with significant margin expansion: We assume a greater-than-expected contribution from higher-margin businesses such as power electronics and liquid cooling as well as greater synergies from acquired companies.

BASE CASE NT$2,500.00

30x average 2027-28e P/E

Strong YoY revenue growth in 2026-28, coupled with operating leverage: We expect steady growth in AI server power supply and data center infrastructure to be the key drivers. We see favorable mix and scale merits driving profit to grow more than double in 2027 vs. 2025.

BEAR CASE NT$1,150.00

14x average 2027-28e P/E

Moderate sales growth with slight margin improvement: We assume further profit drag from legacy products, smaller profit contribution from higher-margin businesses, higher opex and/or less meaningful contribution from acquired companies.

Risk Reward – Delta Electronics Inc. (2308.TW)

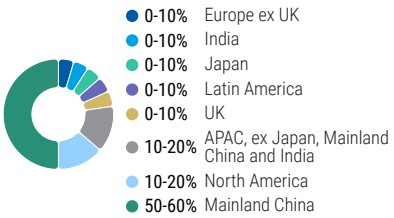
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Power Electronics YoY (%)	25.0	48.9	52.6	51.8
Automation YoY (%)	4.8	10.9	5.2	5.2
Infrastructure YoY (%)	81.6	44.6	51.0	48.2

INVESTMENT DRIVERS

- Transition to EV-related business and data infrastructure
- Operating margin recovery
- FINI holdings

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster-than-expected global economic growth
- Falling raw material prices
- Margin improvement

RISKS TO DOWNSIDE

- Global economic slowdown
- Ongoing component supply tightness
- Rising raw material prices
- Margin contraction as a result of higher operating expenses or faster ASP erosion

OWNERSHIP POSITIONING

Inst. Owners, % Active 64.7%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	746,914	775,166 / 774,709	807,877
EBITDA (NT$, mn)	163,606	169,213 / 175,311	191,720
Net income (NT$, mn)	96,767	103,307 / 105,731	111,870
EPS (NT$)	37.25	39.77 / 40.50	43.07

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Financial Summary – Delta

Exhibit 66: Delta: Financial Summary

Consolidated Income Statements

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Net sales	**421,148**	**554,885**	**775,166**	**1,131,877**	**1,659,520**
COGS	(284,567)	(364,729)	(494,374)	(707,257)	(1,028,632)
Gross profit	136,580	190,157	280,792	424,620	630,887
Operating expenses	(88,928)	(106,224)	(142,491)	(191,135)	(263,905)
Operating income	**47,652**	**83,932**	**138,300**	**233,484**	**366,983**
Non-operating income	**3,664**	**3,934**	**10,365**	**7,100**	**7,300**
Interest income	1,859	1,501	1,687	2,100	2,300
Investment income	(162)	(200)	372	-	-
Exchange gain	1,041	644	2,091	-	-
Other	926	1,989	6,215	5,000	5,000
Pre-tax income	**51,316**	**87,866**	**148,666**	**240,584**	**374,283**
Income tax	(10,925)	(19,930)	(32,925)	(52,189)	(81,172)
Minority interests	(5,163)	(7,828)	(12,433)	(20,158)	(31,363)
Net income	**35,229**	**60,108**	**103,307**	**168,237**	**261,747**
Modelware EPS (NT$)	**13.56**	**23.14**	**39.77**	**64.77**	**100.77**

Consolidated Cash Flow Statements

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Operating CF	**72,895**	**98,474**	**116,226**	**177,240**	**263,652**
Net Profits	35,229	60,108	103,307	168,237	261,747
Depreciation & Amort.	25,016	27,839	30,912	33,985	37,058
Change in WC	(2,486)	(24,190)	(55,485)	(86,295)	(125,847)
Other adjustments	15,137	34,717	37,491	61,313	90,694
Investing CF	**(40,360)**	**(53,214)**	**(70,000)**	**(70,000)**	**(60,000)**
Change of FA (Capex)	(33,435)	(45,950)	(70,000)	(70,000)	(60,000)
Change of L/T invest.	(4,739)	-	-	-	-
Other adjustments	(1,644)	(3,517)	-	-	-
Financing CF	**(11,118)**	**(9,401)**	**36,107**	**(12,414)**	**(54,050)**
Increase in L/T debt	8,552	(11,308)	12,530	2,978	3,829
Increase in S/T debt	(1,417)	446	29	212	272
Issuance of stock	-	-	-	-	-
Cash dividends	(18,696)	(20,313)	(30,131)	(51,787)	(84,335)
Other adjustments	444	21,774	53,680	36,183	26,183
FX adjust.	8,586	(2,146)	-	-	-
Net change in cash	**30,003**	**33,713**	**82,333**	**94,826**	**149,602**

Consolidated Balance Sheets

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Cash	**117,459**	**151,172**	**233,505**	**328,332**	**477,934**
Mkt securities	4,333	3,941	3,941	3,941	3,941
A/R	89,921	121,717	170,036	248,282	364,023
Inventory	83,856	101,478	142,217	205,395	298,726
Others	15,355	17,473	24,410	35,642	52,257
Current Assets	**310,925**	**395,781**	**574,109**	**821,592**	**1,196,881**
Long-term investments	4,740	4,876	4,876	4,876	4,876
Fixed assets	115,710	142,040	127,820	127,652	124,410
Other assets	100,523	96,922	96,922	96,922	96,922
Total Assets	**531,898**	**639,619**	**803,727**	**1,051,042**	**1,423,089**
S/T borrowings	1,918	2,364	2,393	2,605	2,877
AP/NP	69,223	94,452	128,025	183,154	266,379
Other ST liabilities	81,142	112,849	157,649	230,195	337,504
Total current liabilities	**152,283**	**209,665**	**288,067**	**415,954**	**606,759**
L/T debt	32,209	21,143	33,673	36,650	40,480
Other LT liabilities	67,840	83,422	83,422	83,422	83,422
Total Liabilities	**252,332**	**314,230**	**405,162**	**536,026**	**730,661**
Common shares	25,975	25,975	25,975	25,975	25,975
Retained Earnings	102,429	140,925	214,101	330,552	507,964
Other SH equity	151,161	158,489	158,489	158,489	158,489
Shareholders' equity	**279,565**	**325,389**	**398,565**	**515,016**	**692,428**
Total Liab./SE	**531,898**	**639,619**	**803,727**	**1,051,042**	**1,423,089**

Consolidated Financial Ratios

	2024	2025	2026E	2027E	2028E
Margins (%)					
Gross margin	32.4	34.3	36.2	37.5	38.0
Operating margin	11.3	15.1	17.8	20.6	22.1
Pretax margin	12.2	15.8	19.2	21.3	22.6
Net margin	8.4	10.8	13.3	14.9	15.8
YoY growth (%)					
Sales	5	32	40	46	47
Operating profits	16	76	65	69	57
Pretax profits	5	71	69	62	56
Net profits	5	71	72	63	56
Others					
Cash div payout (%)	52%	50%	50%	50%	50%
Adjusted cash div (NT$)	7.0	11.6	19.9	32.5	50.5
Yield (%)	0%	1%	1%	2%	3%
Net Debt/Equity (%)	-31%	-40%	-51%	-57%	-63%
Liabilities/Equity (%)	90%	97%	102%	104%	106%
Liabilities/Assets (%)	47%	49%	50%	51%	51%
ROAE (%)	13%	20%	29%	37%	43%
ROAA (%)	7%	10%	14%	18%	21%
AR/NR days	73	70	69	67	67
Inventory days	103	93	90	90	89
AP/NP days	79	82	82	80	80
Cash conversion	97	80	77	77	77

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 67: Delta: Quarterly Financials

Delta (2308.TW) (NT$ mn)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E
Net sales	**159,353**	**183,256**	**201,379**	**231,177**	**231,791**	**265,526**	**302,930**	**331,630**	**554,885**	**775,166**	**1,131,877**
COGS	(100,392)	(117,948)	(129,404)	(146,630)	(146,821)	(166,230)	(188,191)	(206,015)	(364,729)	(494,374)	(707,257)
Gross profit	**58,961**	**65,308**	**71,976**	**84,547**	**84,970**	**99,295**	**114,739**	**125,615**	**190,157**	**280,792**	**424,620**
Operating expenses	(30,544)	(34,738)	(36,983)	(40,226)	(40,422)	(45,405)	(50,589)	(54,719)	(106,224)	(142,491)	(191,135)
SG&A	(16,886)	(18,956)	(20,208)	(21,963)	(22,067)	(24,959)	(27,870)	(30,178)	(57,482)	(78,014)	(105,074)
R&D exp.	(13,657)	(15,782)	(16,775)	(18,263)	(18,356)	(20,445)	(22,720)	(24,541)	(48,743)	(64,477)	(86,061)
Operating income	**28,417**	**30,570**	**34,992**	**44,321**	**44,548**	**53,891**	**64,149**	**70,896**	**83,932**	**138,300**	**233,484**
Non-operating income	**2,676**	**4,430**	**1,630**	**1,630**	**1,775**	**1,775**	**1,775**	**1,775**	**3,934**	**10,365**	**7,100**
Interest income	428	398	430	430	525	525	525	525	1,501	1,687	2,100
Investment income	487	(115)	-	-	-	-	-	-	(200)	372	-
Disposal of investment	-	-	-	-	-	-	-	-	-	-	-
Exchange gain	1,688	403	-	-	-	-	-	-	644	2,091	-
Other	72	3,743	1,200	1,200	1,250	1,250	1,250	1,250	1,989	6,215	5,000
Pre-tax income	**31,093**	**35,000**	**36,622**	**45,951**	**46,323**	**55,666**	**65,924**	**72,671**	**87,866**	**148,666**	**240,584**
Income tax	(7,258)	(7,831)	(7,910)	(9,925)	(10,423)	(11,968)	(14,174)	(15,624)	(19,930)	(32,925)	(52,189)
Minority interest	(3,279)	(2,033)	(3,158)	(3,963)	(3,841)	(4,676)	(5,537)	(6,104)	(7,828)	(12,433)	(20,158)
Net income	**20,556**	**25,136**	**25,554**	**32,063**	**32,059**	**39,022**	**46,213**	**50,943**	**60,108**	**103,307**	**168,237**
EPS (NT$)	**7.91**	**9.68**	**9.84**	**12.34**	**12.34**	**15.02**	**17.79**	**19.61**	**23.14**	**39.77**	**64.77**
Margins (%)											
Gross Margin	37.0	35.6	35.7	36.6	36.7	37.4	37.9	37.9	34.3	36.2	37.5
Operating Margin	17.8	16.7	17.4	19.2	19.2	20.3	21.2	21.4	15.1	17.8	20.6
Pretax Margin	19.5	19.1	18.2	19.9	20.0	21.0	21.8	21.9	15.8	19.2	21.3
Net Margin	12.9	13.7	12.7	13.9	13.8	14.7	15.3	15.4	10.8	13.3	14.9
QoQ Growth (%)											
Sales	(1)	15	10	15	0	15	14	9			
Gross Profit	5	11	10	17	1	17	16	9			
Operating Profit	8	8	14	27	1	21	19	11			
Pretax Profit	21	13	5	25	1	20	18	10			
Net Profit	19	22	2	25	(0)	22	18	10			
YoY Growth (%)											
Sales	34	48	34	43	45	45	50	43	32	40	46
Gross Profit	56	48	37	51	44	52	59	49	39	48	51
Operating Profit	102	64	41	68	57	76	83	60	76	65	69
Pretax Profit	99	79	36	79	49	59	80	58	71	69	62
Net Profit	101	80	37	85	56	55	81	59	71	72	63

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

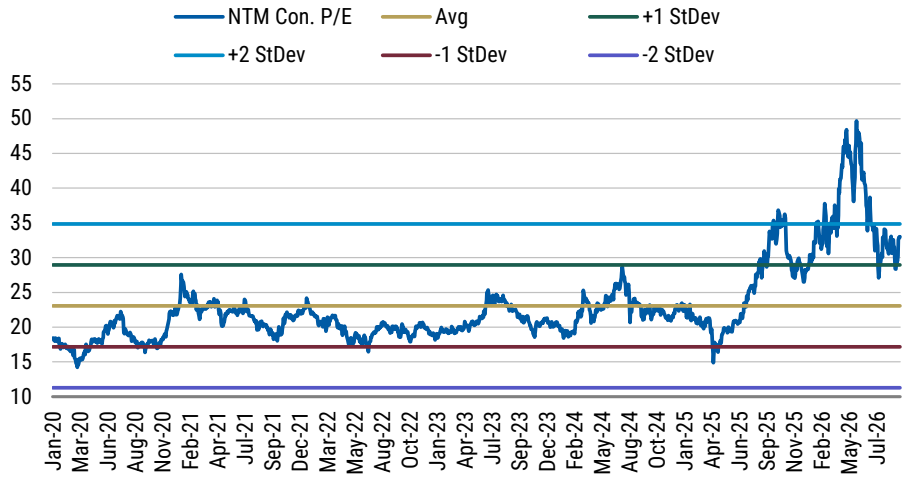
Valuation Methodology – Delta

Our price target (base case scenario value) is NT$2,500, derived from 30x 2027-28e P/E: We use price/earnings as our valuation methodology because we believe it appropriately captures Delta's earnings growth potential. We believe our target P/E multiple of 30x is justified by an earnings CAGR of 63% over 2025-28, which implies a PEG of 0.5. This is lower than the average of 1.0 in 2022-25, when the stock was trading at an average P/E of 23x for a 23% earnings CAGR.

Bull case scenario value – NT$3,300: We apply a 40x P/E multiple to our average base case estimate for 2027-2028. This scenario assumes a greater-than-expected contribution from higher-margin businesses such as power electronics and liquid cooling as well as greater synergies from acquired companies.

Bear case scenario value – NT$1,150: We apply a 14x P/E multiple to our average base case estimate for 2027-2028. This scenario assumes further profit drag from legacy products, smaller profit contribution from higher-margin businesses, higher opex and/or less meaningful contribution from acquired companies.

Exhibit 68: Delta: NTM P/E chart



Source: FactSet consensus estimates, Morgan Stanley Research

Why Does Delta Trade at a Significant Valuation Discount to Delta Thailand?

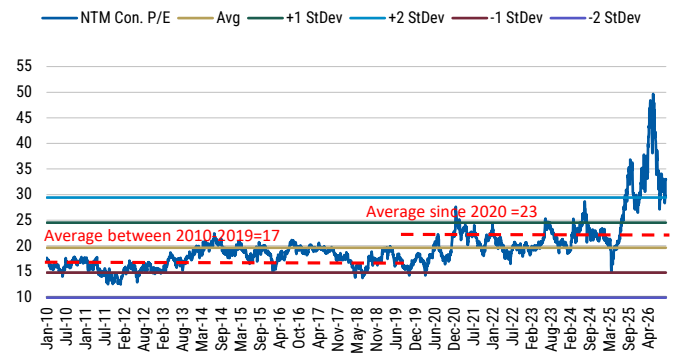
Discussion of the valuation gap between Delta Electronics (2308.TW, currently 33x 12M forward consensus P/E) and Delta Electronics Thailand (DELTA.TH, 71x currently) has been ongoing for several years. If we look at their valuations before 2020, Delta was actually trading at a premium, with an average 17x 12M forward P/E in 2010-2020 vs. Delta Thailand's 13x. This made sense, as Delta is much more competitive in the global arena and Delta Thailand was set up as a subsidiary focused on manufacturing.

The situation flipped after 2020. Though Delta has been re-rated to trade at an average 23x 12M forward P/E since 2020, Delta Thailand has seen its valuation surge to an average of 60x over the same period. This cannot be explained by moves in their respective markets, either, as TWSE went from 14x forward P/E in 2010-20 to 16x after 2020, while the SET went from 13x to 16x.

We think this could be attributable to a few factors:

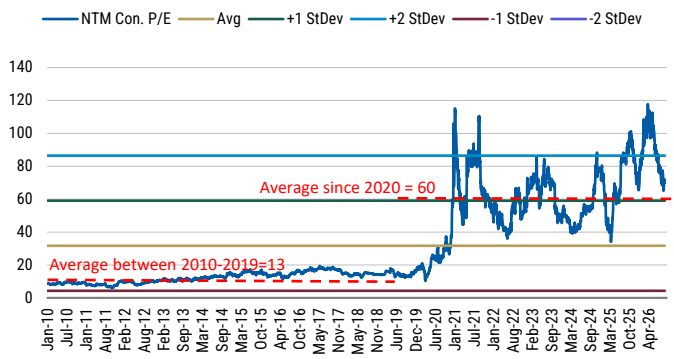
- **Different positioning within the Delta group:** Delta group has increased its stake in Delta Thailand to over 60% from about 20% in 2019 through a tender offer. Since then, there has been much more collaboration between Delta group and Delta Thailand. As Thailand emerged as one of the major manufacturing hubs within the Delta group, its earnings growth (28% in 2020-25) outpaced that of Delta Taiwan (19%).
- **Scarcity in the Thai stock market:** Delta Thailand is one of the relatively few large-cap Thai-listed companies that offer direct exposure to AI/data-center power and EV-related themes, whereas there are many such alternatives in Taiwan.
- **Limited free float:** As the Delta group owns a ~62% stake in Delta Thailand and there have been a few quite stable long-term shareholders, we think the effective free float is probably 20% or less.

Exhibit 69: Delta NTM P/E



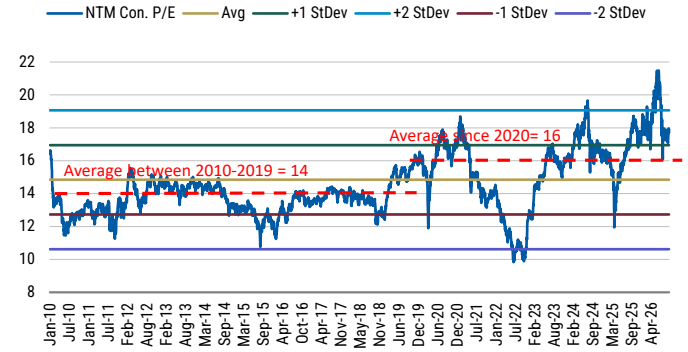
Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 70: DET NTM P/E



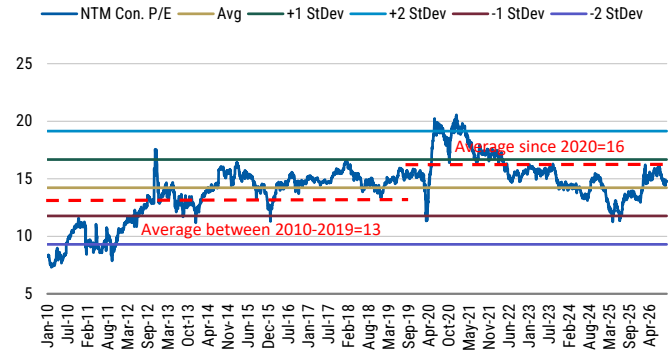
Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 71: TWSE NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

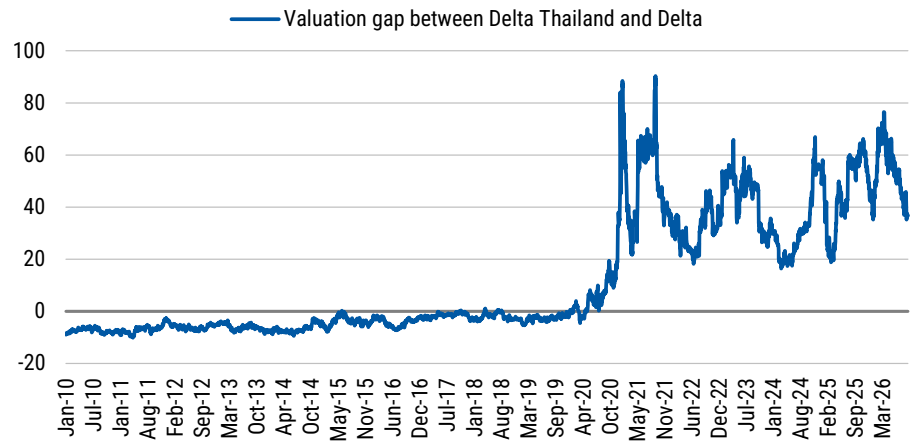
Exhibit 72: SET Index NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 73:

Valuation gap between Delta Thailand and Delta based on 12M forward P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Global Footprint – Delta

Comprehensive Global Operations

Aggressive capacity expansion plans

The company has been continuously diversifying its production base away from China owing to the trade turmoil back in 2018-2019, and then by supply chain disruptions during Covid. Management explicitly set a target of lowering China production to 60% of total capacity (from >70% previously) for the first time back in 2020. New capacity was added in Thailand, India, and Taiwan during 2020-2021. It then saw demand surge from various sectors, including EVs, automation and telecom, during 2022-23, followed by AI-related power suppliers and cooling solutions emerging in 2024. In order to grasp the emerging opportunities, relevant plant build-outs were planned accordingly, on top of the ongoing geographical diversification. It expanded its presence—production sites, R&D, and sales teams—in Europe and the US to better cope with operational disruptions and capture domestic demand, in addition to its existing setups in Taiwan, Thailand, and India. There was then a substantial increase in capex of NT$12.2bn in 2025 (vs. NT$5.6-6bn in 2023-24) as the company further accelerated its expansion plan, mainly on AI-related power/cooling demand and increasing capacity tightness, specifically in Thailand and the US. Delta continued to expand in Thailand by adding three new plants, with another three under construction for power and liquid cooling system offerings. In view of the capacity tightness reserved for AI-related business, it also started to prioritize margin-accretive business—i.e., de-emphasize business with net margins below 10%—which further optimized product mix.

Looking ahead into 2026 and beyond, expansion is likely to remain aggressive in order to fulfill substantial power and liquid cooling turnkey solution demand for large-scale data centers, while management also indicated that the new capacity added at the end of 2025 has already been fully taken. With capacity staying tight for AI power and cooling offerings, the company will continue to expand actively in China, Thailand, the US, and Tainan, Taiwan. It also plans to add mass production lines in Taoyuan, scheduled to be ready by 2028, in addition to the pilot run lines in Tainan. Against this backdrop, the capex budget has been revised up several times and is now roughly NT$50bn for 2026.

Exhibit 74: Delta's global footprint

Source: Company data, Morgan Stanley Research. Note: Data as of 2024.

Financial Analysis – Delta

44% Revenue CAGR in 2025-28e

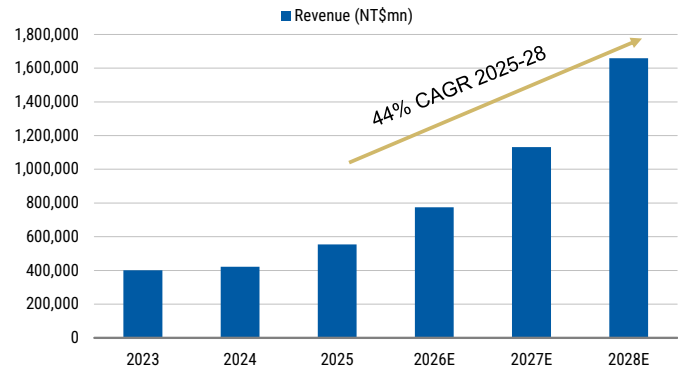
Delta's revenue growth was moderate in 2020 at merely 5%, making it the only year of the Covid disruption (2019-22) that did not see double-digit growth. Growth picked up to 11% in 2021, underpinned by surging demand for cloud, networking, and work-from-home infrastructure amid the pandemic, while the next uplift to 22% growth in 2022 was mainly fueled by increasing demand for EV- and data center-related components (server power supplies, networking switches, energy storage systems, etc). Revenue momentum subsequently moderated in 2023, growing just 4%, as the EV component business was impacted by the UAW strike in the US and automation demand in China stayed subdued, despite sustained demand for AI server power supplies and modules. Growth in 2024 was at a similar level, as strength in data center offerings were not enough to offset weakness in its legacy businesses.

Revenue surged 32% in 2025 to a record NT$554,885mn, driven by AI server rack power and cooling solutions, along with further spec upgrades and value growth.

We expect revenue to expand at a 44% CAGR over 2025-28, supported by ongoing AI server shipments, spec upgrades, 800 VDC power solutions, and turnkey liquid cooling offerings. As the upgrade to 800V HVDC power architecture is likely to take years given the complex design and high safety requirements, we expect AI server power solution upgrades to be separated into a few stages – from power shelves in AI server racks to standalone side-car power racks, from 50V DC to 800V DC output, from 72kW per power shelf to >250kW, etc, as well as power wattage increases for next-generation GPU rack designs.

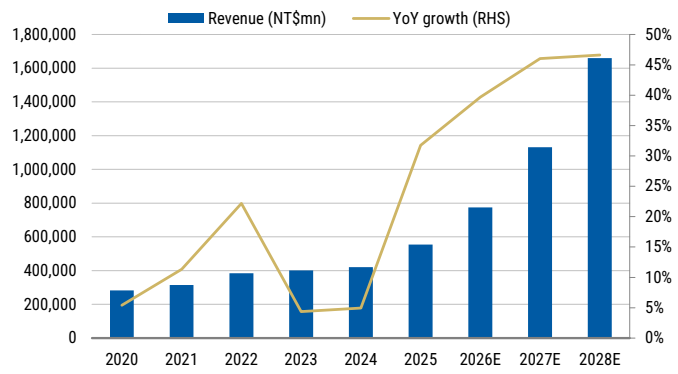
By segment, Power Electronics has been the primary revenue driver, contributing at least 50% over the past years despite the company separating the Mobility segment from Power Electronics since 2024 so as to further distinguish revenue generated from EVs/ autos. We expect the segment to remain the key revenue contributor, supported by strong demand from AI server-related power solutions, and estimate it will reach 58% of revenue by 2028 vs. 50% in 2025. Infrastructure, the second-largest revenue contributor historically, is expected to grow along with the Power Electronics segment, fueled by demand for data center power-related solutions (i.e. UPSs, CDUs, sidecars, etc) as well as energy infrastructure-related offerings (i.e. SSTs (solid-state transformers), fuel cells, modular power solutions, micro grid management, etc.). We model the segment's revenue contribution to reach 35% by 2028 from 24% in 2024 and 33% in 2025. For Automation and Mobility, we expect their revenue contributions to decline sequentially throughout the years, with Automation dropping from 10% in 2025 to 4% in 2028 and Mobility falling from 7% in 2025 to 2% in 2028.

Exhibit 75: Delta: Revenue CAGR, 2025-28e



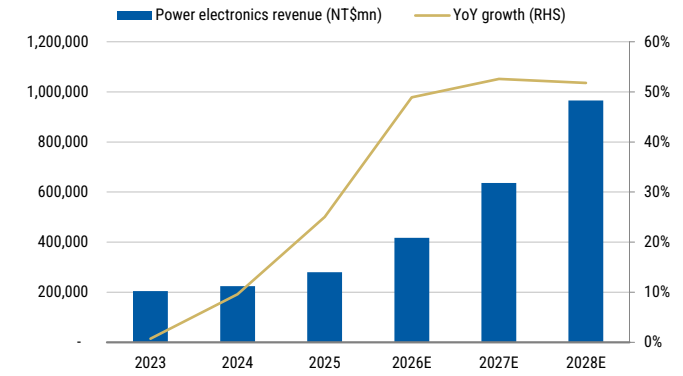
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 76: Delta: Annual revenue growth



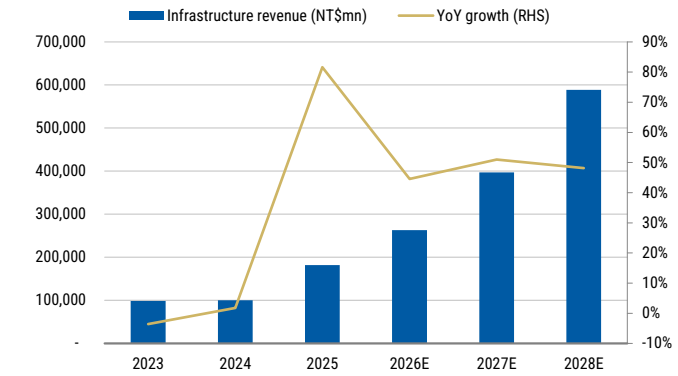
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 77: Delta: Power electronics revenue



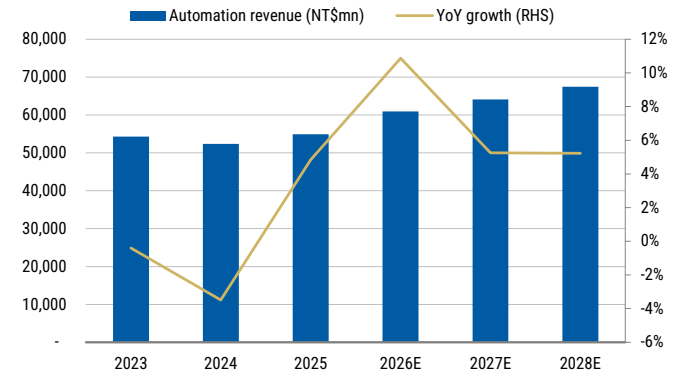
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 78: Delta: Infrastructure revenue



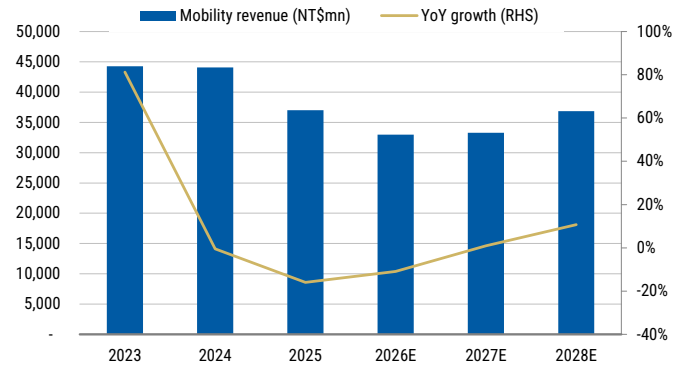
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 79: Delta: Automation revenue



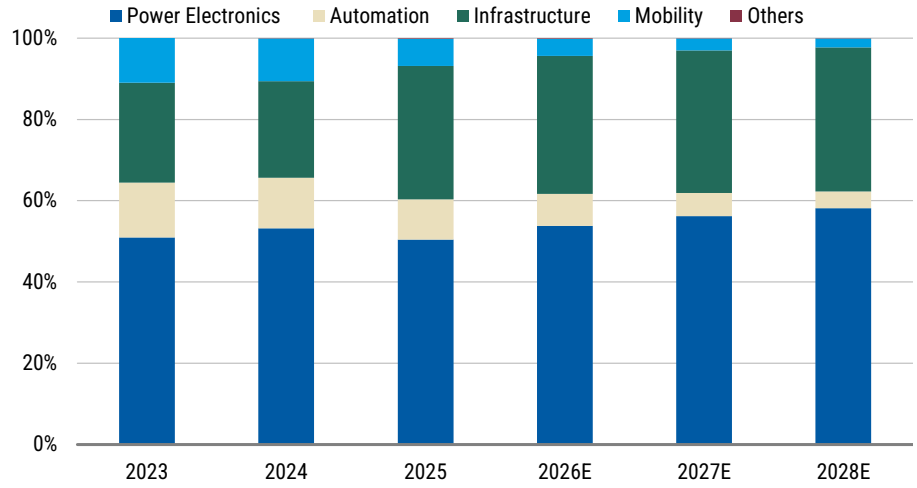
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 80: Delta: Mobility revenue



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 81:
Delta: Revenue breakdown by segment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

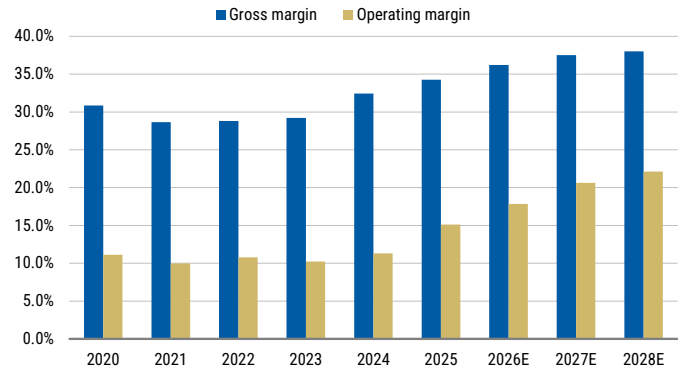
Continuous margin expansion on favorable product mix

Delta's gross margin stayed largely between 28-29% during 2021-23, as revenue was mainly driven by relatively lower-margin products, including EV- and consumer/IT-related products, while server-related offerings had yet to scale up. 2024 served as an inflection point, with the product mix changing structurally, supported by robust demand for AI data center-related offerings, while its legacy businesses stayed relatively weak. Gross margin thus grew by 3.2ppt YoY during the year, from 29.2% in 2023 to 32.4% in 2024. Increasing AI computing needs further increased power consumption and cooling requirements per rack, boosting demand for high-density AI power and cooling products along with related system-level solutions in 2025, which lifted gross margin by another 1.8ppt YoY to 34.3% for the year.

The opex ratio was largely stable, fluctuating around 18-19% between 2020 and 2023. It then increased to 19% in 2023 and 21.1% in 2024, mainly due to operating deleverage, higher R&D costs, and ongoing capacity expansion. The ratio declined back to 19.1% in 2025 on the back of an enlarged revenue base, despite R&D-related investments remaining elevated.

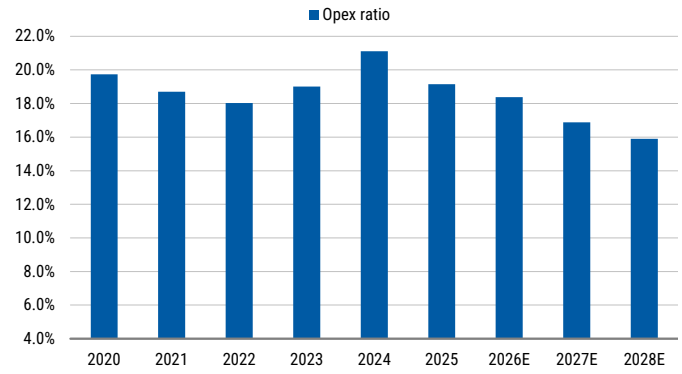
Looking ahead, we expect gross margin to continue expanding on the back of increasing revenue contribution from AI-related power supplies. We thus pencil in a 2.0ppt gross margin expansion to 36.2% in 2026, followed by further increases of 1.3ppt and 0.5ppt YoY in 2027 and 2028, respectively, reaching 38.0% by 2028. On the opex ratio, we expect Delta to continue benefiting from operating leverage in the coming years, and thus estimate operating margin to improve more substantially than gross margin, expanding by 2.7ppt YoY in 2026 to 17.8%, followed by further expansions of 2.8ppt and 1.5ppt in 2027 and 2028, reaching 20.6% and 22.1%, respectively.

Exhibit 82: Delta: Margin trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 83: Delta: Opex ratio



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Net profit to grow at a 63% CAGR in 2025-28

Delta's operating profit grew sequentially throughout 2020-22, supported by gross margin expansion and a lower opex ratio, from NT$31,426mn in 2020 to NT$41,439mn in 2022. It then saw a slight 1% YoY decline in 2023 due to higher operating expenses as the company further stepped up its R&D investments, despite a 0.4ppt gross margin expansion in 2023. Operating margin also edged down accordingly by 0.6ppt to 10.2% during the year.

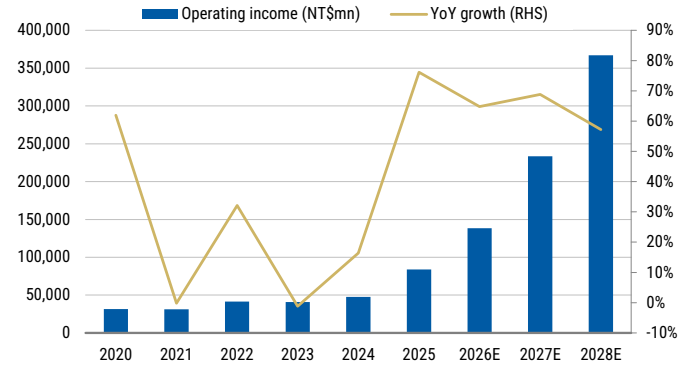
Operating profit returned to a growth trajectory from 2024 onward, growing 16% YoY in 2024 and 76% YoY in 2025, as AI-related power offerings started to scale up. Operating margin expanded faster than gross margin, supported by operating leverage and disciplined cost controls, increasing by 1.1ppt in 2024 to 11.3% and another 3.8ppt to 15.1% in 2025.

With the tax rate staying largely stable at ~20% during 2020-23 before starting to edge up in 2024 and reaching 23% in 2025, Delta's net profit growth was largely aligned with its operating profit growth trajectory. 2023 was an exception, as net profit still managed to grow 2% YoY despite a 1% YoY decline in operating profit, mainly due to higher FX gains.

The net margin trend has been largely aligned with that of operating margin in recent years. After dropping from 9.0% in 2020 to 8.5% in 2021, as lower gross margin was offset by a lower opex ratio, net margin hovered around the 8.3-8.5% level during 2021-24. Net margin was largely flat YoY in 2024 despite a 1.1ppt expansion in operating margin, mainly due to lower non-operating income and a slightly higher tax rate. We then saw a surge in 2025, with net margin expanding 2.5ppt YoY to 10.8%, benefiting from a larger contribution from higher-margin AI power offerings.

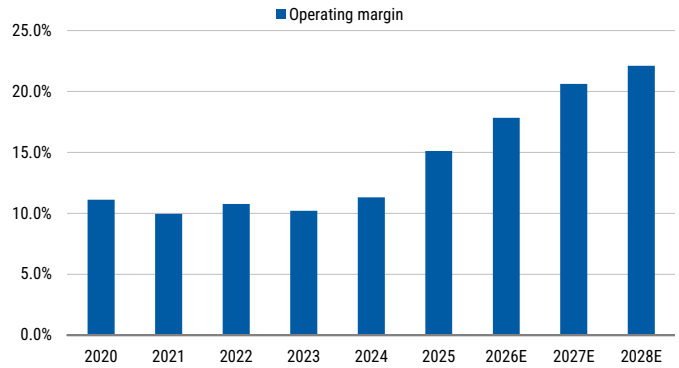
Looking ahead, we forecast net profit to grow 72% YoY in 2026e, followed by 63% YoY growth in 2027e and 56% YoY growth in 2028e, implying a 63% CAGR in 2025-28e, assuming a stable tax rate of 22%. Net margin is likely to rise accordingly, from 10.8% in 2025 to 13.3% in 2026, followed by further expansion to 14.9% in 2027 and 15.8% in 2028.

Exhibit 84: Delta: Operating income trend



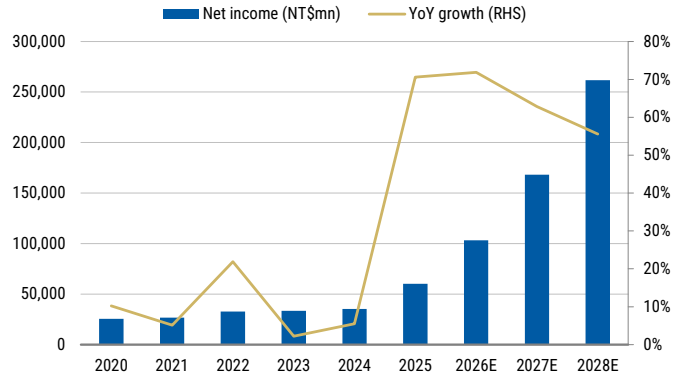
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 85: Delta: Operating margin



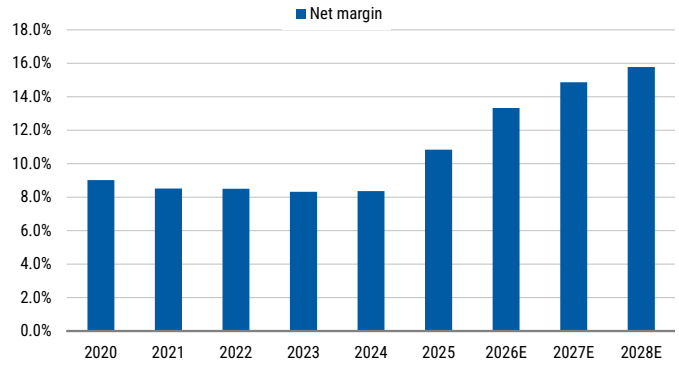
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 86: Delta: Net profit trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 87: Delta: Net margin



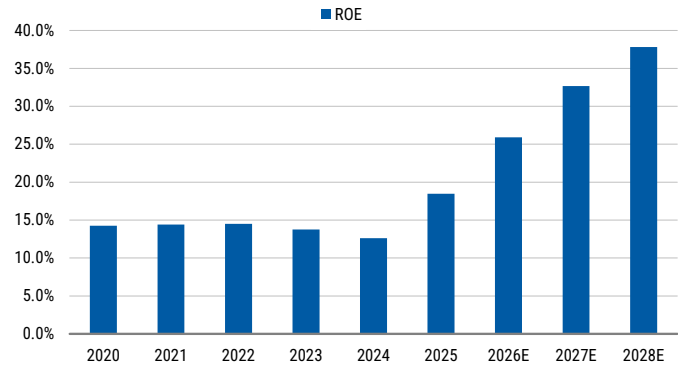
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

ROE to rise alongside margin expansion

ROE rose gradually but steadily during 2020-22, from 14.3% to 14.5%, mainly driven by improving asset turnover, offset by lower net margin. ROE edged down to 13.8% in 2023 and fell further to 12.6% in 2024 on the back of lower asset turnover, while net margin and financial leverage remained at similar levels. It then rebounded to 18.5% in 2025, supported by both improving asset turnover and net margin expansion, which was fueled by an increasingly favorable product mix and operating leverage.

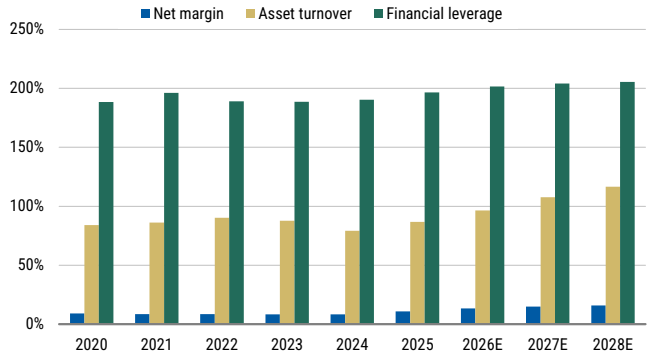
Looking ahead, we expect ROE to continue rising, reaching 25.9% in 2026, 32.7% in 2027, and 37.8% in 2028, primarily driven by continued margin expansion and improving asset turnover. Net margin is projected to expand from 10.8% in 2025 to 15.8% in 2028, supported by gross margin improvement—from 34.3% to 38.0%—and a declining operating expense ratio from 19.1% in 2025 to 15.9% in 2028.

Exhibit 88: Delta: ROE trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 89: Delta: DuPont analysis



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

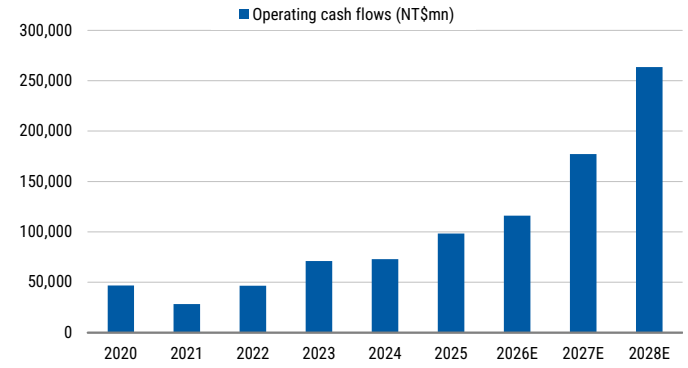
Improving balance sheet and continuous free cash flow generation

Delta has generated positive operating cash flow in recent years. Operating cash flow declined in 2021, mainly due to higher inventory buildup amid supply chain disruptions during the year. Free cash flow generation has also remained positive, while fluctuating with capital expenditures over the years. Free cash flow was exceptionally low at NT $5,292mn in 2021, as the company started to step up capex mainly for geographic capacity diversification and greater production flexibility—adding new capacity in Thailand, India, Tainan/Zhongli in Taiwan, and Wuhu, Wujiang, and Chongqing in Mainland China. Both absolute capex and the capex-to-sales ratio have trended upward since 2022, as Delta continued to add capacity for EV-related offerings and power and cooling solutions during 2023-24, with further acceleration in 2025 due to capacity tightness for AI power and cooling offerings, especially in Thailand and the US.

Looking ahead, the company aims to continue adding capacity worldwide to fulfill substantial demand for power and liquid-cooling turnkey solutions for large-scale data centers operated by major hyperscalers beyond 2027. Capacity remains tight for AI power and cooling offerings, so it continues to expand actively in China, Thailand, the US, and Tainan. Management also plans to add mass-production lines in Taoyuan, Taiwan, scheduled to be ready by 2028, in addition to the pilot-run lines in Tainan. Accordingly, we now estimate capex to rise to NT$70bn in 2026 from ~NT$46bn in 2025 and stay at a similar level in 2027 before edging down slightly to NT$60bn in 2028.

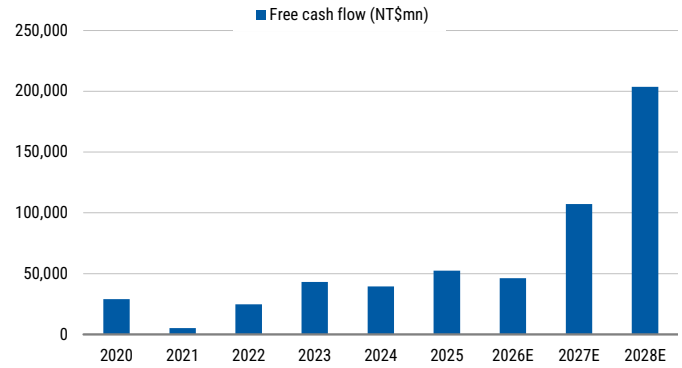
Delta's net cash position has been rising since 2021, with its net cash-to-equity ratio improving from just 2% in 2021 to 40% in 2025. We expect this ratio to continue rising, reaching 51% in 2026, 57% in 2027, and 63% in 2028, on strong free cash flow generation despite a higher total debt level amid ongoing capacity expansion.

Exhibit 90: Delta: Positive operating cash flow



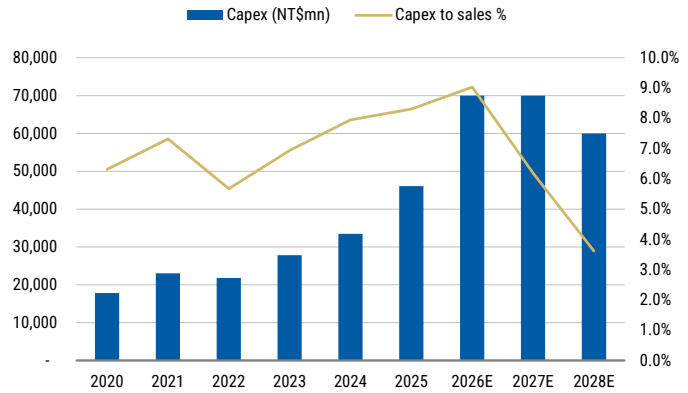
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 91: Delta: Improving free cash flow generation



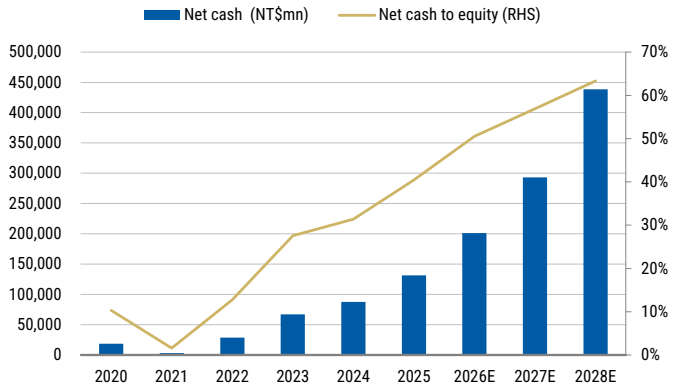
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 92: Delta: Capex and capex-to-sales ratio



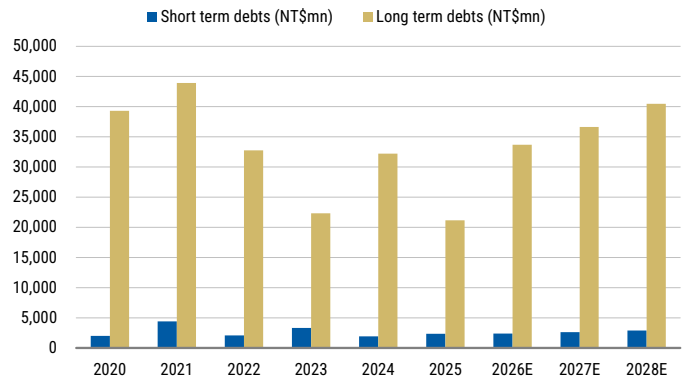
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 93: Delta: Net cash position



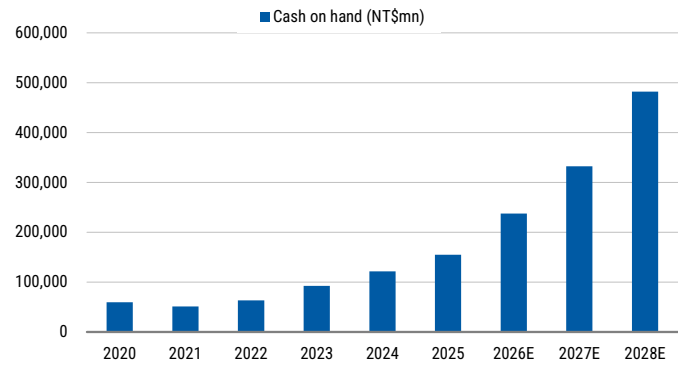
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 94: Delta: Debt position



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 95: Delta: Cash on hand



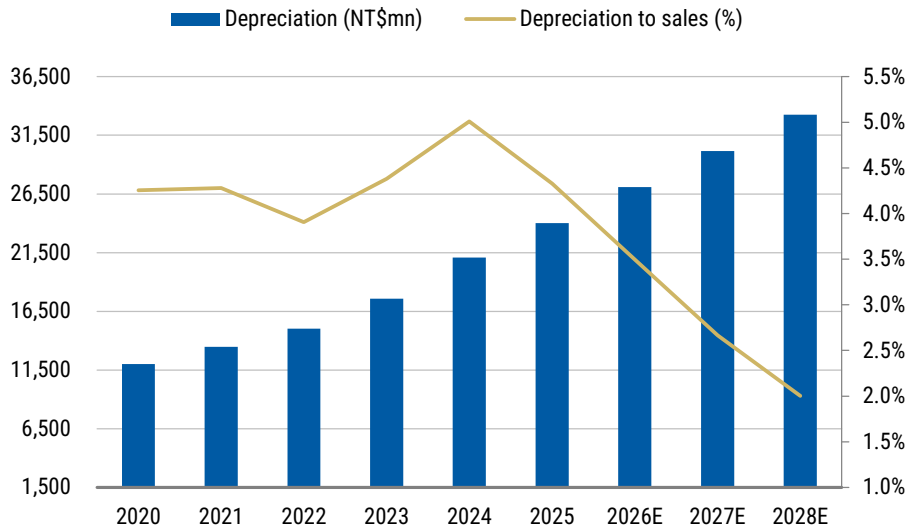
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

After edging down slightly in 2022 from a relatively high 7.3% in 2021, Delta's capex-to-sales ratio has been rising alongside its aggressive capacity expansion plans since 2022, despite an expanding revenue base. Looking ahead, as we expect capex to increase further in 2026, we project a capex-to-sales ratio of 9.0% for the year on robust revenue growth, before declining to 6.2% in 2027 and 3.6% in 2028.

The depreciation-to-sales ratio hovered around the 4-5% level during 2020-25. We

estimate the ratio will decline sequentially as the revenue base grows faster than capex and depreciation, from 4.3% in 2025 to 3.5% in 2026, 2.7% in 2027, and 2.0% in 2028.

Exhibit 96: Delta: Depreciation and depreciation to sales

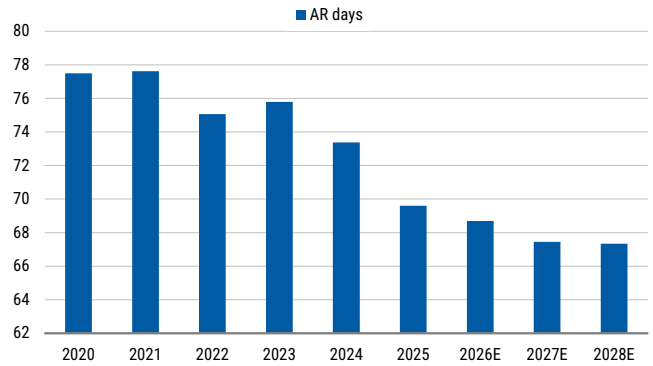


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Delta's cash conversion cycle (CCC) rose from 77 days in 2020 to a peak of 101 days in 2023 before improving to 80 days in 2025. By 2025, inventory days had increased from 80 days in 2020 to 93 days, while AP days had risen to 82 days after falling from 81 days in 2020 to 75 days in 2023. This was partly offset by improving AR days, which declined from 77 days in 2020 to 70 days in 2025.

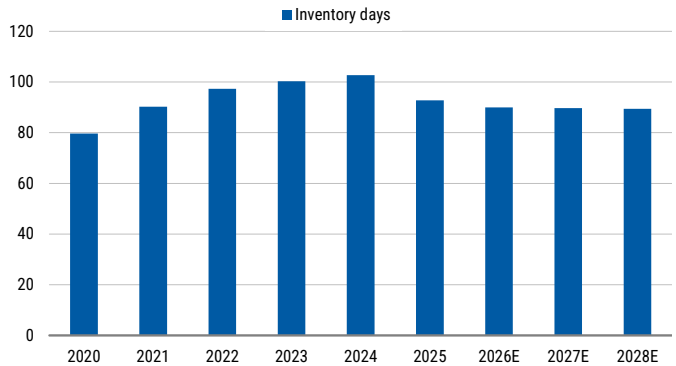
Looking ahead, we expect the overall cash conversion cycle to improve further and remain at 77 days throughout 2026-28, mainly supported by improving AR days and a modest decline in inventory days.

Exhibit 97: Delta: AR days



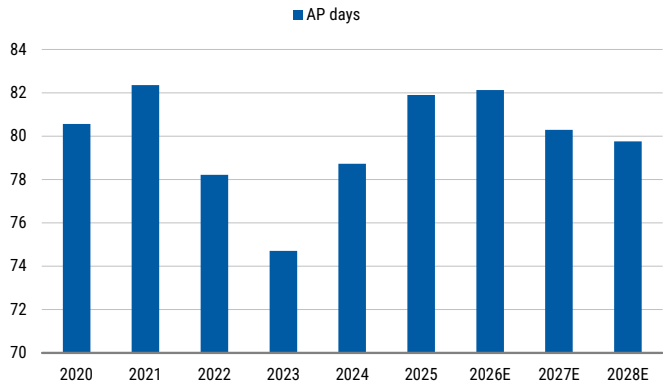
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 98: Delta: Inventory days



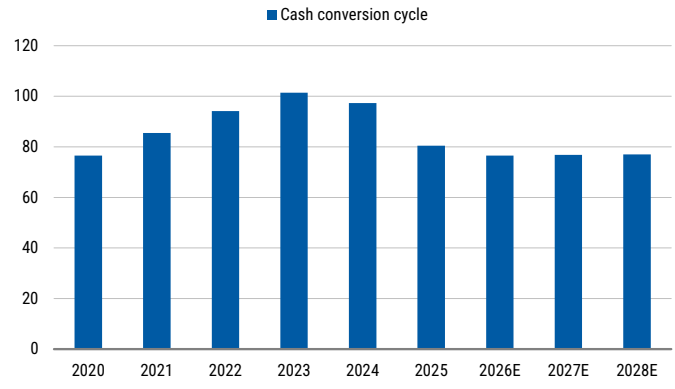
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 99: Delta: AP days



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 100: Delta: Cash conversion cycle



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Company Background – Delta

Company snapshot

Delta Electronics is a global provider of power and thermal management technologies, with a focus on energy-efficient products and integrated solutions. Founded in 1971, the company serves a broad range of applications spanning components, power and system products, thermal management, mobility, industrial and building automation, data centers, telecom, and energy infrastructure, across industries including automotive, healthcare, data centers, and power grids. By combining its capabilities in power electronics, digital technologies, and system integration, Delta aims to support customers in improving operational efficiency and advancing smart manufacturing, smart cities, and more sustainable infrastructure.

The company is headquartered in Taipei, Taiwan. Its global network consists of 293 operations, including 165 sales offices, 55 manufacturing sites, and 73 R&D centers, supporting customers worldwide.

Exhibit 101: Delta: Key milestones

Year	Key Milestones
1971	Delta Electronics, Inc. was established
1976	Achieved >US$1mn of revenue
1977	Moved plant to Taoyuan Guishan Industrial Park
1983	- Mass production of switching power supplies
1988	- Listed on the Taiwan Stock Exchange (code 2308) - Mass production of brushless DC fans - Invested in the establishment of Delta Electronics (Thailand) PCL, with an original stake of 6%
1989	- Established first Japan office in Tokyo - Established Delta Power Electronics laboratory in Raleigh, US
1992	- Inauguration of the first plant in China (Dongguan) - Established Delta Greentech (China) Co., Ltd. in Shanghai
1995	Delta Electronics (Thailand) PCL was listed on Thailand Stock Exchange
1999	- Inaugurated its new Taipei HQ - Established the Delta Power Electronics Center (DPEC) in Shanghai
2000	Established the Delta Power Electronics Science and Education Development program
2001	Inaugurated its Wuhu plant in China
2003	Acquired ASCOM Energy Systems and its plant in Slovakia
2006	Inaugurated Delta's Tainan Plant I
2008	Inaugurated its Rudrapur plant in India and Wuhu plant in China
2010	Inaugurated Chenzhou plant in China
2011	Inaugurated Shanghai R&D and Operations Center and Taoyuan R&D Center in Taiwan
2012	Mr. Yancey Hai was elected Chairman and Mr. Ping Cheng was appointed as CEO
2015	- Acquired Eltek ASA, a leading provider of telecom power, industrial power, and data center power solutions - Inaugurated Delta Americas' headquarters green building and Taoyuan Plant 5
2017	- Announced organizational adjustment to focus on Power Electronics, Automation, and Infrastructure as main business categories - Acquired a controlling stake in VIVOTEK
2019	- Acquired a controlling stake in Delta Electronics (Thailand) Public Company Limited - Acquired lighting-solutions brand Amerlux
2020	Accquired Trihedral, a Canadian SCADA and IIoT software company, to strengthen industrial automation and system-integration businesses
2021	Acquired March Networks, a global video-surveillance provider, and Universal Instruments, a leader in precision automation solutions for electronics manufacturing
2022	- Inaugurated a new facility at the Automotive Campus in Helmond, the Netherlands - Purchased land and a factory in Plano, Texas, United States
2023	Acquired TB&C, a leading provider of automotive high-voltage hybrid components
2024	- Mr. Ping Cheng selected as Chairman while concurrently serves as CEO - Adjusted business categories into Power Electronics, Mobility, Automation, and Infrastructure
2025	- Inaugurated new North East Asia HQ in Tokyo - Acquired ACT Genomics and Noda RF Technologies

Source: Company data, Morgan Stanley Research

Product offerings

Delta's key offerings can be categorized into four key areas: Power Electronics, Infrastructure, Mobility, and Automation.

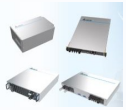
1. Power Electronics

Delta operates in switching power supplies, thermal management, and passive components, supplying products for ICT, cloud computing, EVs, and industrial and commercial applications. The company frames its business around energy efficiency and

sustainability and now also offers grid-to-chip power and thermal management solutions for data centers.

- Components: Includes essential passive and magnetic components used in portable devices, AI cloud computing, automotive electronics, and IoT.
- Power and System: Contains power products and systems for AI, cloud computing, IoT, client devices, and industrial, medical, lighting, e-mobility, and electrical-appliance applications.
- Fans and Thermal Management: This mainly covers air and liquid cooling solutions for AI cloud and edge computing, EVs, as well as industrial, commercial, and residential sectors. Key products includes DC and EC fans, thermal modules, liquid cooling cold plates, and coolant distribution units, etc.

Exhibit 102: Delta: Key offerings for the power electronics segment

POWER ELECTRONICS							
Components							
Inductors	RF Inductors	Standard Power Module (DC/DC Converter & AC/DC Module)	Transformers	Networking	EMI Filters	Solenoids	Current Sensing Resistor
				Wire-Wound Common Mode Choke			
Power and System							
Embedded Power	Adapters	Industrial Power	Medical power	Industrial Battery Charging	USB Socket Outlets	High Voltage Power	Hydrogen Power
				Modular Charging System			
Fans and Thermal Management							
Chip Level Cooling Solution	DC Brushless Fans and Blowers	Thermal Module	Cold Plate Loop	In-rack CDU	Rack Manifold	Electical Pump	In-row CDU
Communication Equipment Cooling	Energy Infrastructure and Industrial Automation Cooling	Portable Device and Equipment Cooling	Infrastructure Ventilation	Automotive Thermal			
DC Brushless Fans and Blowers	DC Brushless Fans and Blowers	Laptop cooling	DC Brushless Fans and Blowers				

Source: Company data, Morgan Stanley Research

2. Infrastructure

Delta supplies ICT and energy infrastructure to support demand related to AI, smart cities, and IoT. Its ICT infrastructure solutions target data centers and telecom networks, while its energy infrastructure offerings integrate AIoT, power conversion, and microgrid control across energy generation, storage, and management.

- ICT Infrastructure: It mainly provides infrastructure for AI computing and 5G

communications under this category, ranging from telecom cell sites to data centers. The portfolio includes power, air and liquid cooling, networking equipment, and management software.

- Energy Infrastructure: The company offers renewable energy, AIoT-based energy management, and power conversion technologies for low-carbon transformation. Solutions include solar inverters, wind power converters, energy storage, EV charging, and management platforms for smart grids. It also supplies semiconductor high-voltage power supplies, automated testing instruments, and variable-frequency drives for industrial processes.
- Display and Visualization: Delta's display technologies serve control rooms, public information systems, projection mapping, and exhibition or event spaces. Offerings range from large-scale and 8K projection systems to integrated video-wall solutions and can be customized for mission-critical operations.

Exhibit 103: Delta: Key offerings for the infrastructure segment

INFRASTRUCTURE							
ICT Infrastructure							
Uninterruptible Power Supplies (UPS)	Battery Backup System (BBU)	800 VDC In-Row Power	Power Distribution Unit (PDU)	Rack-Mount Remote Power Panel (rRPP)	Liquid and Air Cooling	Modular Rack & Rack Mount Accessories	Lithium-ion UPS Battery System
Ethernet Switches	Indoor/Outdoor Telecom Power Systems	Renewable Hybrid Power Systems	Small Cell Power Systems	Li-ion Battery Backup Systems	Campus/Enterprise Networks		
Energy Infrastructure							
EV Charger	Power Conversion System (PCS)	All-in-One ESS solution	Distributed ESS Solutions	Solar Inverter	Wind Power Converter	Solid State Transformer Platforms	Medium Voltage Drives
High Voltage Power	DC EV Charger ATS	Power Supply ATS	Solid Oxide Fuel Cell (SOFC)	Hydrogen Energy Solution	Active Power Filter (APF) / Static Var Generator (SVG)		
Display and Visualization							
Business and Education Multimedia Projector	High Performance Projector	DLP, LED and LCD Video Wall	Wireless Collaboration and Signage	Vivitek			
				vivitek			

Source: Company data, Morgan Stanley Research

3. Mobility

Delta applies its power electronics and motor technologies to smart mobility, supplying

power systems, drive technologies, and integrated powertrain solutions. It works with automakers to develop electric vehicles, positioning the business as part of the low-carbon transition and sustainable urban mobility.

- EV Powertrain System: The company's powertrain portfolio for EV makers includes On-board Charging Modules (OBCM), DC/DC converters, power electronics combos, and traction inverters, along with X-in-One integrated solutions.

Exhibit 104: Delta: Key offerings for the mobility segment

MOBILITY						
EV Powertrain System						
EV Power Electronics	Traction Products	x-EV Power Management	In-Vehicle Network and AI Driving Computing	Automotive Thermal	Low Voltage Powertrain	EV Charging

Source: Company data, Morgan Stanley Research

4. Automation

Delta provides automation solutions for industrial and building applications through integrated software and hardware. Its industrial automation offerings support smart production and operational efficiency, while its building automation solutions target energy-efficient, secure, and sustainable spaces.

- Industrial Automation: This segment provides automation solutions across industries by utilizing AI, big data, IoT, and digitized manufacturing models. Relevant offerings support diverse production needs, including customized, small-volume and high-variety production, as well as mass production.
- Building Automation: Delta's Building Automation Solutions use an IoT architecture and integrate technologies from five brands: Amerlux lighting, Delta Controls and LOYTEC IP-based building control, and March Networks and VIVOTEK surveillance. The management platform enables operators to manage energy use, occupant comfort, and sustainability goals through a single interface.

Exhibit 105: Delta: Key offerings for the automation segment

AUTOMATION							
Industrial Automation							
Inverters - AC Motor Drives	Drive System for Motion Control Solutions	Motors	Servo Systems - AC Servo Motors and Drives	CNC Solution	PC-Based Motion Control Solution	Linear Motion	Embedded Motion Controller
Industrial Ethernet Solution	Temperature Controllers	High Voltage Power	SCARA Robot	Electric Servo Press			
Building Automation							
Delta Controls	LOYTEC	Indoor Air Quality Monitor	LED Lighting - Industrial/ Road/Fishing/Commercial	Smart Surveillance - VIVOTEK	Healthy Lighting		

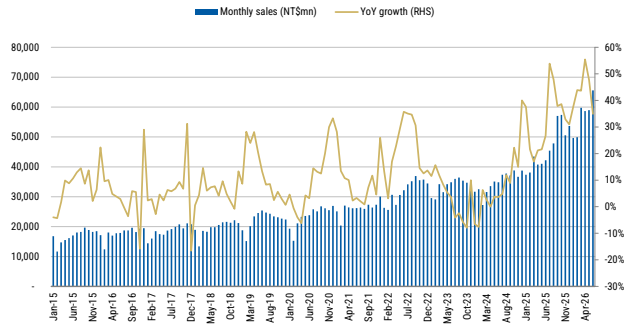
Source: Company data, Morgan Stanley Research

Monthly sales vs. share price

Delta's monthly sales show that seasonality has been an important factor over the years. Following a substantial increase in monthly sales since 2024, traditional seasonal patterns have become less pronounced as revenue from AI-related offerings has started to outweigh them, while liquid-cooling-related offerings have added a new layer of monthly sales volatility. The correlation between the company's monthly sales and share price movements has remained high, at ~87% since 2023.

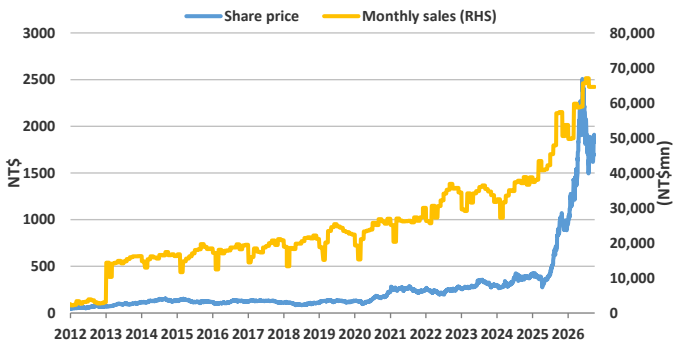
The company reported August monthly sales of NT$64,586mn (-4% MoM/+35% YoY). We saw the Power Electronics segment decline sequentially from a high base, offset by growth in Automation and slight growth in the Infrastructure segment.

Exhibit 106: Delta: Monthly sales vs. YoY sales growth



Source: TEJ, Morgan Stanley Research

Exhibit 107: Delta: Monthly sales vs. share price



Source: TEJ, Morgan Stanley Research

Key management & shareholding structure

Delta's board currently consists of eight directors and five independent non-executive directors. Its founder, Chung-Hua Cheng (Bruce Cheng), and his sons, Ping Cheng and An Cheng, together owned a 7% stake as of March 2026. Excluding the founder and his family, the management team held a stake of ~0.1% as of March 2026.

The FINI holding percentage as of July 14, 2026, was 63%, versus a previous peak of 78% in February 2011 and a historical average (since 2010) of 69%.

Exhibit 108: Delta: Shareholding structure

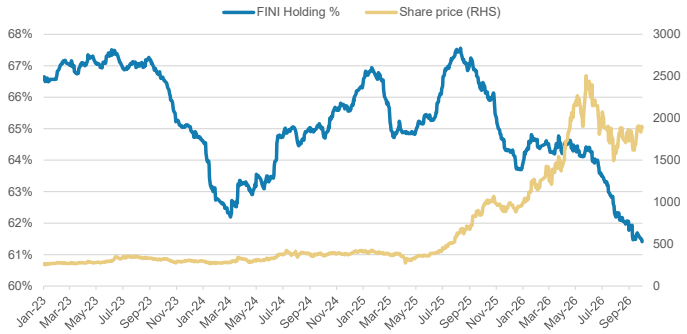
Shareholders	%
Dejico International LTD	10.30%
Deltron Holding LTD	8.40%
New Labor Retirement Pension Fund	3.75%
Chung-Hua Cheng (Bruce CH Cheng)*	3.15%
Government of Singapore-GOS-EFM C	2.28%
Ping Cheng*	2.14%
An Cheng (Victor Cheng)*	1.94%
Yuanta/P-shares Taiwan Top 50 ETF	1.43%
Labor Retirement Reserve Fund (The Old Fund)	1.42%
Labor Insurance Fund	1.32%
Total	**36.13%**

Source: Company data, Morgan Stanley Research. Note: Data as of March 2026. (*) are shown as insiders.

Exhibit 109: Delta: FINI holdings vs. share price since 2010

Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 110: Delta: FINI holdings vs. share price since 2023



Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 111:

Delta: Management profiles

Name	Title	Education	Work Experience
Bruce C.H. Cheng	Founder and Honorary Chairman Chairman, Delta Electronics Foundation	Bachelor of Electrical Engineering — National Cheng Kung University	- 2012–present — Honorary Chairman, Delta Electronics - 1990–present — Founded and chairs the Delta Electronics Foundation - 1971–2012 — Founder and Chairman, Delta Electronics - Before 1971 — Worked at TRW and Air Asia Co.
Ping Cheng	Chairman and CEO	Bachelor of Business Administration — California State University, East Bay	- Joined Delta in 1988 - 2024–present — Chairman - 2012–present — CEO of Delta Electronics - 2010–2012 — Corporate Senior Vice President and President of China Region - 2008–2010 — Senior Vice President & General Manager of Power Supply Business Group - 1998–2008 — Deputy General Manager of Component Business Group (Delta China); Executive Vice President of China Region
Mark Ko	Vice Chairman	Bachelor of Control Engineering — National Chiao Tung University	- Joined Delta in 1988 - 2012–present — Vice Chairman of Delta Electronics - 2004–2012 — President and Chief Operating Officer of Delta Group - 1994–2004 — General Manager of Power System Business Group
Simon Chang	Board Member President and COO	EMBA — Chung Yuan Christian University	- Joined Delta in 1981 - 2017–present — Board Member; President and Chief Operating Officer, Delta Electronics - Before 2017 — Senior Vice President and General Manager of Industrial Automation Business Group - From 1995 — Director of Industrial Automation Division
Victor Cheng	Board Member CEO of Delta Electronics (Thailand) PCL.	-Master of Electrical Engineering — Santa Clara University -Bachelor of Electrical Engineering — Santa Clara University	- Joined Delta in 1993 - 2024–present — Board Member of Delta; CEO of Delta Electronics (Thailand) PCL. - 2023–2024 — Executive Vice President of Infrastructure Business - 2017–2023 — General Manager of Information and Communications Technology Business Group - 2014–2017 — General Manager of Power System Business Group - 2002–2019 — President, then Chairman of Delta Networks - 1999–2002 — Head of Video Display Business Unit - 1993–1999 — International Sales Director to Assistant General Manager of Video Display Business Unit
Shan-Shan Guo	Board Member Chief Brand Officer Vice Chairman, Delta Electronics Foundation	- Master of Marketing and Communications — University of Michigan - Bachelor of Foreign Languages and Literatures — National Taiwan University	- Joined Delta in 2010 - 2011–present — Chief Brand Officer of Delta
Johnson Lee	Executive Vice President Infrastructure Business (IFB)	- EMBA — National Taiwan University - Bachelor of Industrial Engineering and Systems Management — Feng Chia University	- Joined Delta in 1985 - 2024–present — Executive Vice President of Infrastructure Business - 2012–2017 — Chief Operating Officer of Delta Electronics - 2004–2012 — General Manager of Power Supplies Business Group - Earlier — Procurement manager; General Manager of materials management
Ted Shyy	Executive Vice President Power Electronics Business (PEB)	Master in Materials Science and Engineering — National Cheng Kung University	- Joined Delta in 1988 - 2024–present — Executive Vice President of Power Electronics Business - 2014–present — Chief Executive Officer of Cyntec (concurrent role) - 2011–2014 — General Manager of Delta Components Business Group - 2003–2014 — General Manager of Cyntec - 1991–2003 — Rotation in Research and Development, production and marketing, under Cyntec, a joint venture between Delta and Japan's SUSUMU
James Tang	Executive Vice President Mobility Business (EVB)	- EMBA — National Taiwan University - Bachelor of Management Science — National Cheng Kung University	- Joined Delta in 1988 - 2024–present — Executive Vice President of Mobility Business; continues as General Manager of Electric Vehicle Solutions Business Group - 2017–present — General Manager of Electric Vehicle Solutions Business Group - 2008–2016 — General Manager across Power Supplies, Consumer and Lighting Solutions, and Embedded Power Supplies business groups - 1988–2008 — Procurement; Materials Manager; Director of Personal Computer Business Group
Jimmy Yiin	Executive Vice President Global Business Operations (GBO)	Master of Business Administration — National Taiwan University and Fudan University	- Joined Delta in 1992 - 2023–present — Executive Vice President of Global Business Operations - 2019–2022 — General Manager of Power System Business Group - 2014–2019 — Vice Chairman of Delta Greentech (China); General Manager of Delta Electronics (Shanghai) - 2008–2013 — General Manager of Delta Electronics (Jiangsu) - 1999–2007 — Purchasing manager/materials section chief of Delta Dongguan and Delta Networks
Rock Huang	Executive Vice President Global Manufacturing	Master of Industrial Engineering and Management — National Taipei University of Technology	- Joined Delta in 1987; ~40 years of experience in the manufacturing field - 2026–present — Executive Vice President of Global Manufacturing - 2020–2026 — President of Global Manufacturing - 2019–2020 — President of Delta China - 2018–2019 — Executive Vice President of Delta China - 2007–2018 — General Manager of Southern China, leading Dongguan team

Source: Company data, Morgan Stanley Research

Risk Reward – Lite-On Technology (2301.TW)

Stronger ASIC server power supply but fair risk-reward

PRICE TARGET NT$315.00

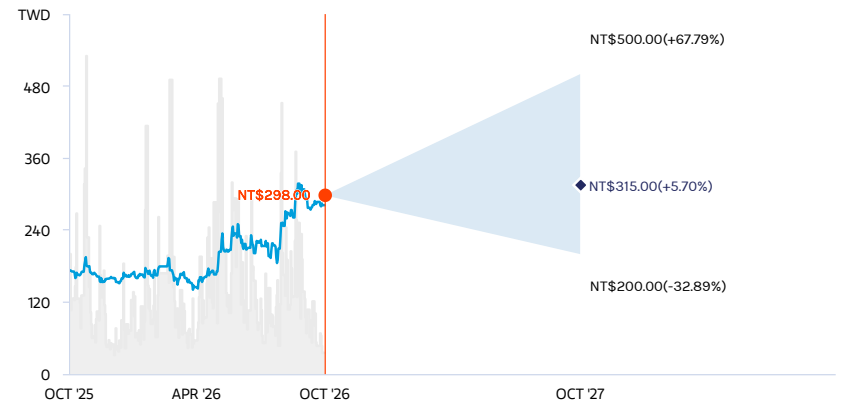
Base case: P/E multiple of 17x based on average 2027-28 EPS. We believe our target multiple is justified by the 47% earnings CAGR we estimate for 2025-28, implying a PEG of 0.4, versus the stock's 16x P/E and 2% earnings CAGR during 2022-25, which implied a PEG of 7.2.

Consensus Price Target Distribution NT$176.00 NT$300.50 MS PT NT$400.00

Source: Refinitiv, Morgan Stanley Research

◆ Mean ◆ Morgan Stanley Estimates

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- We expect Lite-On Tech to benefit from strong AI server power demand in 2026-27, with the company gaining more traction on ASIC platforms.
- We are mindful of weakness in consumer/IT demand amid rising input costs, which could weigh on Lite-On Tech's profit in 2026-27.
- Liquid-cooling sidecar CDU system shipments may take more time to ramp up.

Consensus Rating Distribution

77% Overweight
23% Equal-weight
0% Underweight
● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian:	*Negative*
New Data Era:	*Positive*
Technology Diffusion:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$500.00

27x average 2027-28e P/E

Better-than-expected margin expansion: Operating efficiency and accelerating production automation drive margin expansion, while new-business revenue comes in earlier than expected. Growth in cloud computing and automotive electronics is stronger than expected.

BASE CASE NT$315.00

17x average 2027-28e P/E

AI server projects progress, but some business uncertainties remain: We expect margin improvement from a better product mix as AI server power supply projects ramp up. However, we are mindful of weakness in consumer/IT demand amid rising input costs.

BEAR CASE NT$200.00

11x average 2027-28e P/E

Operating margin contraction: Operating efficiency improves less than expected, while pricing competition intensifies amid macro volatility; automotive electronics take longer to ramp up.

Risk Reward – Lite-On Technology (2301.TW)

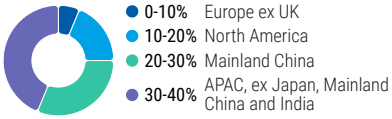
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Opto-electronics OPM (%)	8.3	7.9	7.5	7.5
Information Technologies OPM (%)	9.2	7.7	7.8	8.6
Opto-electronics revenue YoY (%)	0.2	8.0	1.7	2.3
Information Technologies revenue YoY (%)	7.7	(5.2)	0.5	(0.2)

INVESTMENT DRIVERS

- LED adoption (outdoor lighting/auto)
- Margin expansion
- Raw material prices, particularly for steel and copper

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger optical business growth
- Stronger data center/server demand
- PC/NB demand growth
- Downtrend in raw material prices

RISKS TO DOWNSIDE

- Macro demand weakness
- Weaker data center/server demand
- PC/NB demand slowdown
- Uptrend in raw material prices

OWNERSHIP POSITIONING

Inst. Owners, % Active 65.2%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	210,118	224,829 / 219,960	229,691
EBITDA (NT$, mn)	31,349	33,549 / 34,893	38,115
Net income (NT$, mn)	23,541	25,187 / 24,910	26,835
EPS (NT$)	10.30	11.07 / 10.88	11.79

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Financial Summary – Lite-On

Exhibit 112: Financial Summary – Lite-On

Consolidated Income Statement

NT$ mn	2024	2025	2026E	2027E	2028E
Net sales	**137,134**	**166,085**	**224,829**	**297,964**	**357,670**
COGS	-107,504	-128,046	-168,703	-219,712	-259,785
Gross profit	**29,630**	**38,039**	**56,126**	**78,252**	**97,885**
Operating expenses	-16,648	-21,301	-26,044	-33,189	-37,645
- Promotion	-5,184	-7,629	-6,902	-8,782	-10,247
- ADM	-3,990	-4,965	-7,384	-9,340	-10,162
- R&D	-7,476	-8,721	-11,787	-15,037	-17,200
Operating income	**12,981**	**16,738**	**30,082**	**45,063**	**60,240**
Non-operating income	2,653	2,975	2,876	2,500	2,500
Interest income	1,491	1,119	799	800	800
Investment income	2,808	-956	1,436	1,200	1,200
Disposal of investment	75	-537	-77	0	0
Exchange gain	2,808	1,091	1,390	0	0
Other	-4,528	2,258	-672	500	500
Pre-tax income	**15,635**	**19,713**	**32,958**	**47,563**	**62,740**
Income tax	-3,674	-4,601	-7,767	-11,227	-14,793
Minority	19	-4	3	0	0
Reported net income	**11,942**	**15,116**	**25,187**	**36,337**	**47,947**
EPS (NT$)	**5.21**	**6.64**	**11.07**	**15.97**	**21.07**

Consolidated Cash Flow Statement

NT$ mn	2024	2025	2026E	2027E	2028E
Cash flow from Operations	**12,411**	**14,533**	**34,885**	**43,140**	**51,224**
Net profits	11,942	15,116	25,187	36,337	47,947
Depreciation + Amortization	4,047	3,668	3,994	4,065	4,137
Inv. Gain/Loss (Equity)	-15	-86	0	0	0
Disposal of Investment	9	568	0	0	0
Working Capital Change	-3,886	-2,099	-12,395	-12,092	-10,109
Other adjustments	313	-2,635	18,098	14,830	9,248
Cash flow from Investing	**-4,843**	**-7,395**	**-13,287**	**-8,993**	**-11,919**
Capex	-3,283	-6,795	-13,287	-8,993	-11,919
Change of LT investment	-2,234	-43	0	0	0
Change of ST investment	1,453	-417	0	0	0
Other adjustments	-779	-140	0	0	0
Cash flow from financing	**-3,363**	**-23,099**	**-19,866**	**-28,659**	**-37,818**
Increase in L/T debt	0	0	0	0	0
Increase in S/T debt	7,584	-10,116	0	0	0
Cash Dividend Paid	-10,372	-10,297	-19,866	-28,660	-37,818
Issuance of stock	0	0	0	0	0
Other adjustments	-574	-2,686	0	0	0
Exchange rate adjustment	3,736	-3,748	0	0	0
Net change in cash	**7,942**	**-19,709**	**1,732**	**5,487**	**1,487**

Consolidated Balance Sheet

NT$ mn	2024	2025	2026E	2027E	2028E
Cash	100,684	80,974	82,706	88,193	89,680
Mkt Securities	933	1,009	1,009	1,009	1,009
AR/NR	37,182	42,318	60,960	80,790	96,978
Inventory	26,817	34,262	42,083	54,807	64,804
Others	3,501	5,458	5,458	5,458	5,458
Current Assets	**169,117**	**164,021**	**192,216**	**230,257**	**257,928**
Long-term investments	8,195	7,748	7,748	7,748	7,748
Fixed assets	18,776	24,752	34,495	39,873	48,104
Deferred Asset	1,628	2,246	2,246	2,246	2,246
Intangible Assets	3,156	3,060	2,610	2,160	1,710
Other assets	3,811	3,860	3,860	3,860	3,860
Total Assets	**204,683**	**205,687**	**243,175**	**286,144**	**321,597**
S/T borrowings	30,187	20,029	20,029	20,029	20,029
AP/NP	43,126	53,608	67,676	88,138	104,213
Other ST liabilities	34,556	35,960	54,059	68,888	78,137
Total Current Liabilities	**107,869**	**109,597**	**141,764**	**177,056**	**202,379**
LT debt(include CB)	3,000	3,000	3,000	3,000	3,000
Other LT liabilities	2,903	3,111	3,111	3,111	3,111
Total Liabilities	**113,772**	**115,709**	**147,875**	**183,167**	**208,491**
Common shares	23,473	23,168	23,168	23,168	23,168
Retained Earnings	46,277	42,393	47,714	55,390	65,519
Other shareholders' equity	21,161	24,418	24,418	24,418	24,418
Total Equity	**90,911**	**89,979**	**95,300**	**102,976**	**113,105**
Total Liab. & Shldr's Equity	**204,683**	**205,687**	**243,175**	**286,143**	**321,596**

Consolidated Financial Ratios

	2024	2025	2026E	2027E	2028E
Margins					
Gross margin	21.6%	22.9%	25.0%	26.3%	27.4%
Operating margin	9.5%	10.1%	13.4%	15.1%	16.8%
Pretax margin	11.4%	11.9%	14.7%	16.0%	17.5%
Net margin	8.7%	9.1%	11.2%	12.2%	13.4%
YoY growth					
Sales	-7.6%	21.1%	35.4%	32.5%	20.0%
Operating profits	-10.6%	28.9%	79.7%	49.8%	33.7%
Pretax profits	-14.9%	26.1%	67.2%	44.3%	31.9%
Net profits	-18.0%	26.6%	66.6%	44.3%	32.0%
EPS	-18.1%	27.5%	66.6%	44.3%	32.0%
Net Debt/Equity	-75.3%	-65.5%	-63.7%	-64.3%	-59.8%
Liabilities/Equity	125.1%	128.6%	155.2%	177.9%	184.3%
Liabilities/Assets	55.6%	56.3%	60.8%	64.0%	64.8%
ROAE	13.6%	16.7%	27.2%	36.7%	44.4%
ROAA	6.1%	7.4%	11.2%	13.7%	15.8%
AR/NR Turnover (days)	92.2	87.4	83.8	86.8	90.7
AP/NP Turnover (days)	142.7	137.9	131.2	129.4	135.1
Inventory Turnover (days)	89.3	87.1	82.6	80.5	84.0
Cash conversion cycle (days)	38.9	36.5	35.2	37.9	39.6

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 113: Lite-on: Quarterly Financials

NT$ mn	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	2025	2026E	2027E
Net sales	**36,417**	**40,421**	**44,891**	**44,356**	**43,407**	**52,703**	**59,526**	**69,193**	**166,085**	**224,829**	**297,964**
COGS	-28,195	-31,479	-33,648	-34,724	-34,004	-38,355	-44,867	-51,477	-128,046	-168,703	-219,712
Gross profit	**8,222**	**8,942**	**11,243**	**9,633**	**9,403**	**14,348**	**14,659**	**17,716**	**38,039**	**56,126**	**78,252**
Operating expenses	-4,546	-5,215	-6,560	-4,980	-5,304	-6,146	-6,851	-7,743	-21,301	-26,044	-33,189
- Promotion	-1,532	-2,051	-2,535	-1,512	-1,500	-1,623	-1,786	-1,993	-7,629	-6,902	-8,782
- ADM	-998	-1,048	-1,792	-1,127	-1,457	-1,783	-1,964	-2,180	-4,965	-7,384	-9,340
- R&D	-2,023	-2,110	-2,244	-2,345	-2,341	-2,787	-3,095	-3,563	-8,721	-11,787	-15,037
Operating income	**3,675**	**3,727**	**4,683**	**4,653**	**4,099**	**8,203**	**7,807**	**9,973**	**16,738**	**30,082**	**45,063**
Non-operating income	994	536	886	560	871	1,166	445	395	2,975	2,876	2,500
Net interest income	424	323	184	188	214	195	195	195	1,119	799	800
Net investment income	596	-3,235	1,389	293	99	1,237	50	50	-956	1,436	1,200
Net gain of investment dispos	137	-702	23	5	1	-78	0	0	-537	-77	0
Exchange gain	379	2,540	-751	-122	284	1,106	0	0	1,091	1,390	0
Others	-542	1,609	41	195	272	-1,295	200	150	2,258	-672	500
Pre-tax income	4,669	4,262	5,569	5,213	4,970	9,368	8,252	10,368	19,713	32,958	47,563
Income tax	-1,214	-1,108	-924	-1,355	-1,195	-2,248	-1,939	-2,385	-4,601	-7,767	-11,227
Net income	**3,456**	**3,155**	**4,645**	**3,860**	**3,775**	**7,116**	**6,313**	**7,983**	**15,116**	**25,187**	**36,337**
EPS (NT$)	**1.51**	**1.39**	**2.05**	**1.70**	**1.66**	**3.14**	**2.77**	**3.51**	**6.64**	**11.07**	**15.97**
Margins											
Gross margin	22.6%	22.1%	25.0%	21.7%	21.7%	27.2%	24.6%	25.6%	22.9%	25.0%	26.3%
Operating margin	10.1%	9.2%	10.4%	10.5%	9.4%	15.6%	13.1%	14.4%	10.1%	13.4%	15.1%
Pre-tax margin	12.8%	10.5%	12.4%	11.8%	11.4%	17.8%	13.9%	15.0%	11.9%	14.7%	16.0%
Net margin	9.5%	7.8%	10.3%	8.7%	8.7%	13.5%	10.6%	11.5%	9.1%	11.2%	12.2%
QoQ Growth											
Sales	-4.9%	11.0%	11.1%	-1.2%	-2.1%	21.4%	12.9%	16.2%			
Gross profit	1.0%	8.8%	25.7%	-14.3%	-2.4%	52.6%	2.2%	20.9%			
Operating profit	5.1%	1.4%	25.7%	-0.6%	-11.9%	100.1%	-4.8%	27.7%			
Pre-tax profit	17.1%	-8.7%	30.7%	-6.4%	-4.7%	88.5%	-11.9%	25.6%			
Net profit	13.3%	-8.7%	47.2%	-16.9%	-2.2%	88.5%	-11.3%	26.5%			
YoY Growth											
Sales	26.6%	21.4%	22.1%	15.8%	19.2%	30.4%	32.6%	56.0%	21.1%	35.4%	32.5%
Gross profit	40.6%	21.0%	36.3%	18.3%	14.4%	60.5%	30.4%	83.9%	28.4%	47.5%	39.4%
Operating profit	64.8%	13.6%	19.3%	33.1%	11.5%	120.1%	66.7%	114.3%	28.9%	79.7%	49.8%
Pre-tax profit	49.2%	4.4%	25.6%	30.7%	6.4%	119.8%	48.2%	98.9%	26.1%	67.2%	44.3%
Net profit	44.7%	1.3%	37.0%	26.6%	9.3%	125.5%	35.9%	106.8%	26.6%	66.6%	44.3%

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

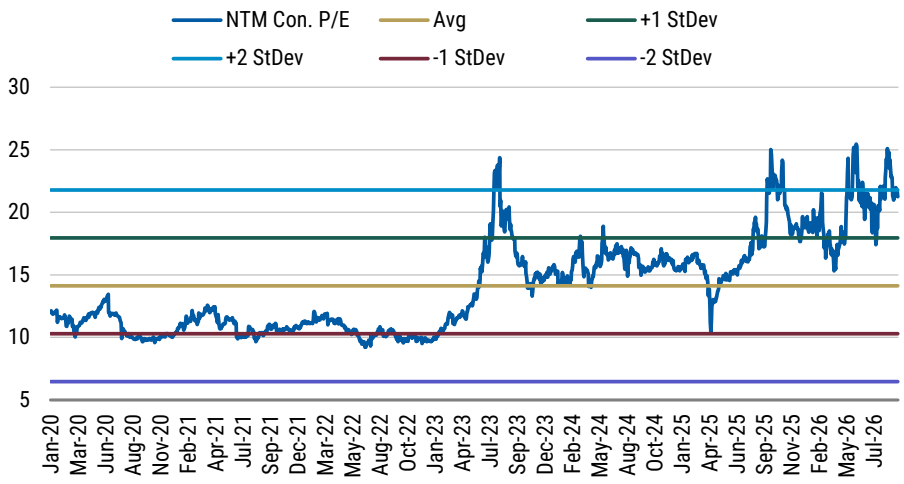
Valuation Methodology – Lite-On

Our price target (base case scenario value) is NT$315, derived from 17x 2027-28e P/E: We use price/earnings as our valuation methodology because we believe it appropriately captures Lite-On's earnings growth potential. We believe our target P/E multiple of 17x is justified by an earnings CAGR of 47% over 2025-28, implying a PEG of 0.4. This is lower than the average PEG of 7.2 in 2022-25, when the stock was trading at an average P/E of 16x for a 2% earnings CAGR.

Bull case scenario value – NT$500: We apply a 27x P/E multiple to our average base case estimate for 2027-28. This scenario assumes better-than-expected margin expansion, driven by improved operating efficiency and accelerating production automation, while new business revenue comes in earlier than expected. Growth in cloud computing and automotive electronics is stronger than expected.

Bear case scenario value – NT$200: We apply an 11x P/E multiple to our average base case estimate for 2027-28. This scenario assumes weaker-than-expected improvement in operating efficiency, while pricing competition intensifies amid macro volatility; automotive electronics take longer to ramp up.

Exhibit 114: Lite-on: NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Global Footprint – Lite-On

More capacity expansion to come

Lite-On's production footprint spans the globe, with 15 sites in total, although it remains heavily weighted toward China. The company has been expanding aggressively, both to meet strong demand for AI power supplies and to support its strategy of geographic risk diversification and proximity to customers.

Looking at its expansion plans in recent years, the company has continued to demonstrate its commitment to diversifying production away from China and increasing proximity to customers. It completed its Thailand plant expansion and announced the second-phase expansion of its Vietnam facility in 2021. It then established automotive electronics and cloud power production lines in Dallas, Texas, in 2H22, along with the official launch of the second phase of its Hai Phong plant in Vietnam, which focuses on networking and consumer products. In 2023, it began expanding its second Kaohsiung plant and announced additional capacity expansion in Vietnam. In 2024, it committed a further US $120mn to establish a wholly owned subsidiary in Vietnam and began progressively shifting production of high-end power products to that facility. In 2025, the company aimed to double capacity as demand for AI server power supplies and battery backup units (BBUs) surged, focusing primarily on capacity expansion in the US, Kaohsiung, and Vietnam.

For 2026, the company expects capacity allocation of ~40% in China, 27% in Vietnam, 20% in Taiwan, and 10% in the US, with the remaining non-China footprint (i.e., Mexico, Brazil, and Thailand) accounting for ~3% of total capacity. It has also announced investments of US$919mn and US$149mn in its US and Vietnam subsidiaries, respectively, to continue expanding power-related capacity, mainly for PSUs. The capex budget for 2026 has also been raised to ~NT$13bn, versus ~NT$7bn in 2025.

Exhibit 115: Lite-on's global footprint

Source: Lite-on.

Financial Analysis – Lite-On

29% revenue CAGR in 2025-28

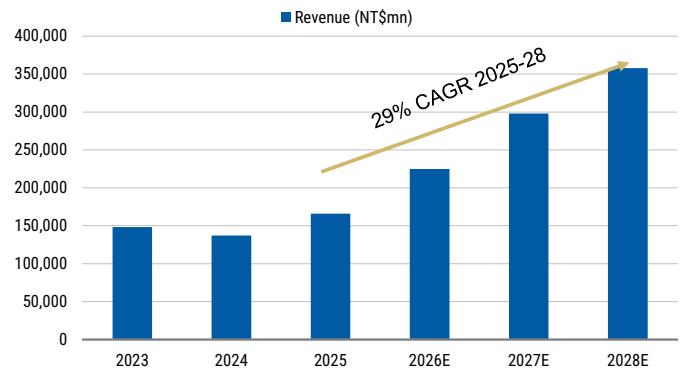
Lite-On's revenue growth was relatively moderate during 2020-22, with revenue expanding at a CAGR of 5% over the period. The company benefited from surging cloud computing demand, which drove increasing demand for power supplies, along with rising demand from EV/automotive electronics, although growth was offset by elevated component costs and supply-chain disruptions during COVID-19. Revenue then fell by 14% in 2023 and 8% in 2024 due to a post-pandemic demand pullback across the consumer electronics market and PC/NB model transitions, coupled with softening demand from EV-related businesses. The IT segment declined 20%/11% YoY, while the opto-electronics segment declined 11%/9% YoY in 2023/2024, respectively.

Revenue rebounded in 2025, growing 21% YoY, as the company began to benefit from surging demand for AI/cloud server power supplies. Contributions from ramping high-voltage/AI power infrastructure products began to increase, coupled with a partial recovery in the IT segment, mainly fueled by power adapter shipments for game consoles and gaming PCs/NBs, as well as related power offerings for low-Earth-orbit applications.

Looking ahead, we estimate revenue to expand at a 29% CAGR in 2025-28e, driven by the AI server power demand boom, while we remain mindful of weakness in consumer/IT demand amid elevated component costs.

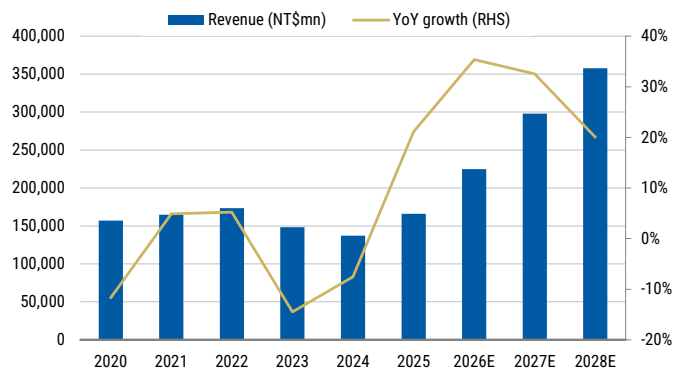
By product segment, IT-related products were the primary revenue driver in past years, while 2025 marked a crossover point, with the contribution from the Cloud & AIoT segment surpassing that of the IT segment, mainly supported by AI power-related offerings. IT revenue contribution declined from 72% in 2020 to 38% in 2025, and we estimate it will reach 17% by 2028. For the Cloud & AIoT segment, revenue contribution grew substantially from 27% in 2021 (0% in 2020) to 45% in 2025; we estimate it will account for 74% of total revenue by 2028, on the back of surging AI power-related demand. For opto-electronics, we expect revenue contribution to decline from 17% in 2025 to 9% in 2028.

Exhibit 116: Lite-on: Revenue CAGR, 2025-28e



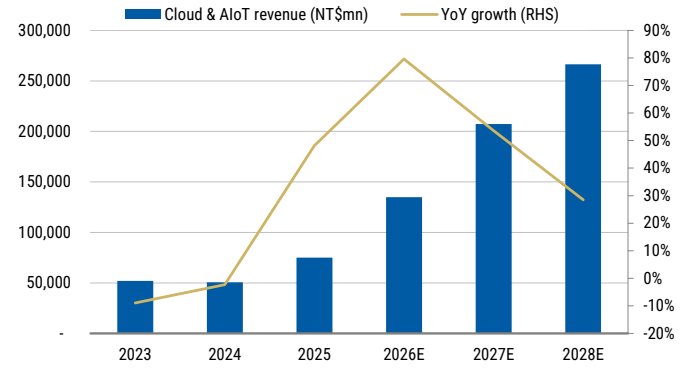
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 117: Lite-on: Annual revenue growth



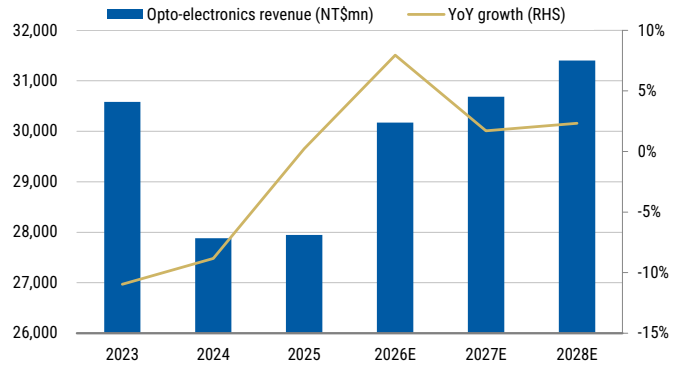
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 118: Lite-on: Cloud & AIoT revenue



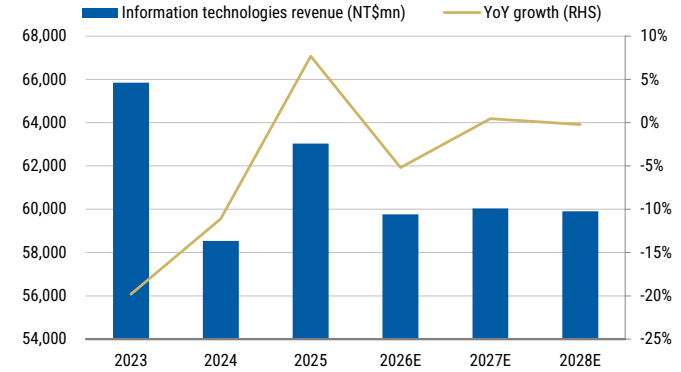
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 119: Lite-on: Opto-electronics revenue



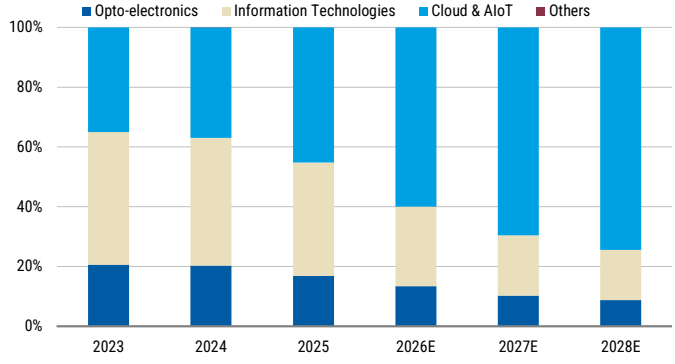
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 120: Lite-on: IT revenue



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 121: Lite-on: Revenue breakdown by segment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

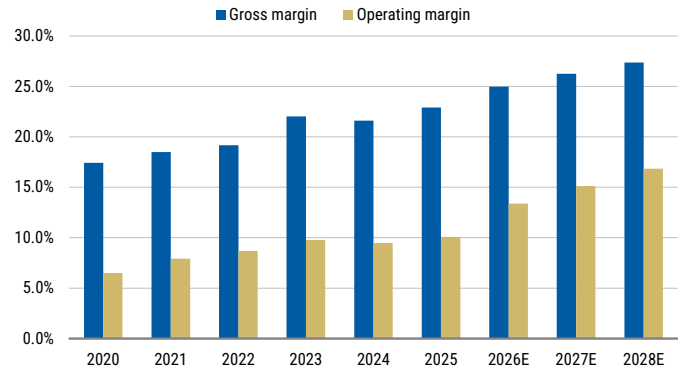
Continuing margin expansion

Lite-On's gross margin rose steadily over the years, from 17.4% in 2020 to 22.0% in 2023, thanks to a continually improving product mix. Gross margin dipped to 21.6% in 2024, mainly due to a smaller revenue base, while contributions from AI/cloud had yet to scale and margins were weighed down by learning-curve costs at the new Vietnam site. It then rose to 22.9% in 2025 as the AI server power business began to scale alongside rising BBU (battery backup unit) shipments, resulting in a more favorable product mix.

The opex ratio was largely stable at 10.5-11.0% during 2020-22. It then increased to 12.2% in 2023 and 12.1% in 2024, mainly due to operating deleverage and higher R&D costs. The opex ratio edged up further to 12.8% in 2025, mainly due to ongoing investment in R&D to support new business development in AI server power supplies, BBUs, liquid cooling, etc., along with a one-off tariff prepayment.

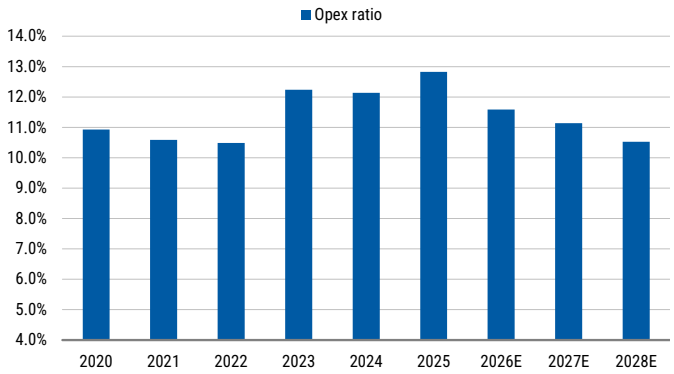
Looking ahead, we expect margins to continue expanding on a favorable product mix. We now pencil in a 2.1ppt gross margin improvement in 2026 to 25.0%, followed by expansions of 1.3ppt and 1.1ppt in 2027 and 2028, to 26.3% and 27.4%, respectively. For the opex ratio, we expect it to decline sequentially as the revenue base expands. Accordingly, we estimate operating margin to improve by 3.3ppt to 13.4% in 2026, followed by expansions of 1.7ppt in both 2027 and 2028, reaching 15.1% and 16.8%.

Exhibit 122: Lite-on: Margin trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 123: Lite-on: Opex ratio



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Net profit to grow at a 47% CAGR in 2025-28

Operating profit grew steadily alongside gross margin expansion during 2020-22, from NT $10,206mn to NT$15,073mn. It then fell 4% YoY in 2023 and a further 11% YoY in 2024 due to higher operating expenses, mainly from R&D costs, despite gross margin expansion in 2023. Operating margin increased by 1.1ppt YoY to 9.8% in 2023 amid a smaller revenue base, before declining by 0.3ppt YoY to 9.5% in 2024 on lower gross margin and an elevated opex ratio.

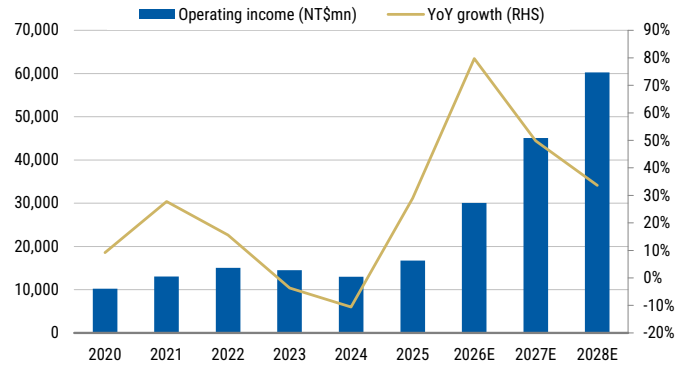
Operating profit rebounded by 29% YoY to NT$16,738mn in 2025, supported by both an expanding revenue base and strengthening gross margin. Operating margin also improved, increasing by 0.6ppt YoY to 10.1%, as the recovery in gross margin was partly offset by higher opex.

With the tax rate remaining relatively stable at 20-24%, Lite-On's net profit growth has generally tracked its operating profit growth. An exception occurred in 2023, when net profit grew 3% YoY despite a 4% decline in operating profit, mainly due to higher FX gains during the year.

Net margin has broadly tracked operating margin in recent years. It reached a record high of 9.8% in 2023 on improving gross margin, despite a higher opex ratio. It then declined to 8.7% in 2024 on gross margin erosion before rebounding to 9.1% in 2025, benefiting from increasing demand for AI power supplies.

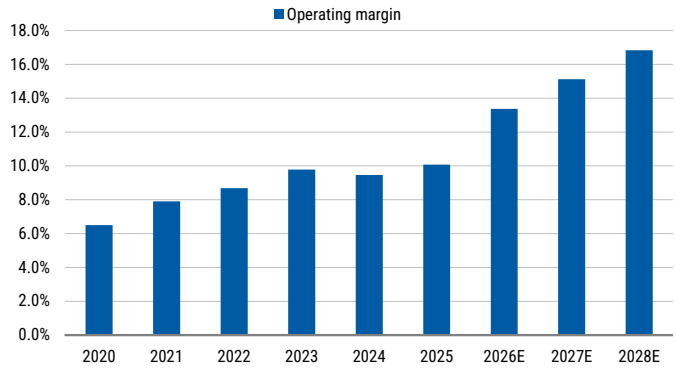
Looking ahead, we forecast net profit to grow 67% YoY in 2026e, followed by 44% YoY growth in 2027e and 32% YoY growth in 2028e, implying a 47% CAGR in 2025-28e. Net margin is likely to rise to 11.2% in 2026, 12.2% in 2027, and 13.4% in 2028.

Exhibit 124: Lite-on: 35% CAGR for operating profit in 2024-27



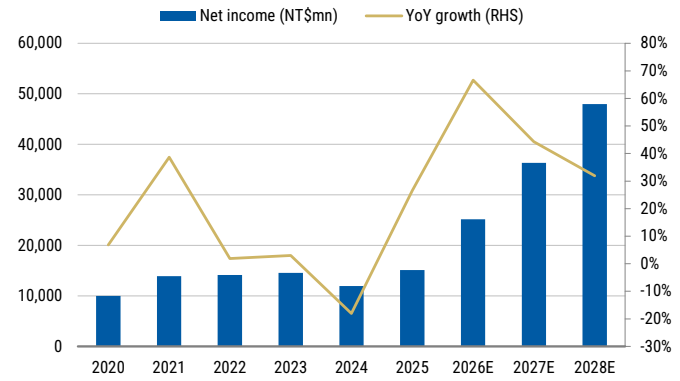
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 125: Lite-on: Operating margin



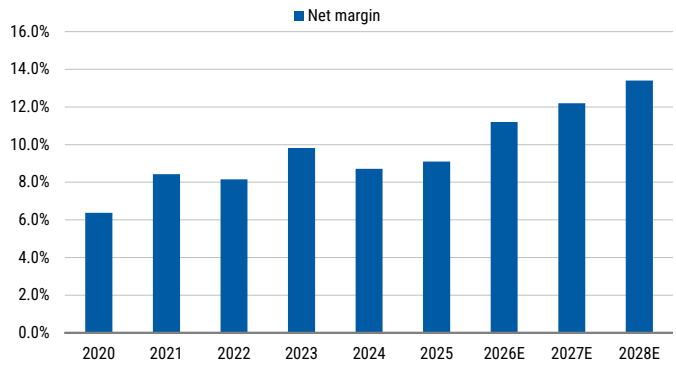
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 126: Lite-on: Net profit trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 127: Lite-on: Net margin



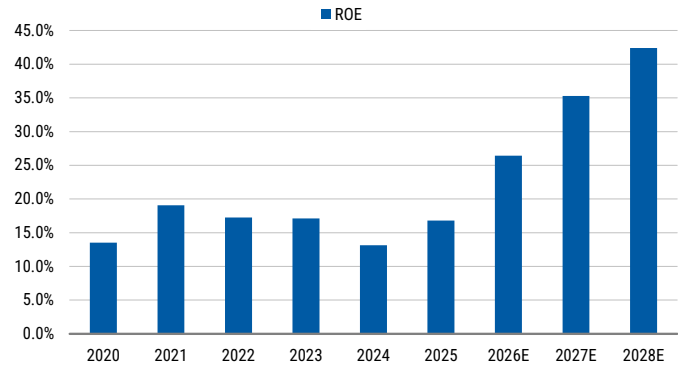
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

ROE to rise

ROE movements were highly correlated with changes in net margin during 2020-22, as asset turnover and financial leverage were largely stable over the period. ROE edged down slightly in 2023 despite higher net margin, which increased from 8.2% to 9.8%, due to lower asset turnover and financial leverage. ROE fell further to a trough of 13.1% in 2024 as net margin and asset turnover declined further. It then rebounded to 16.8% in 2025, primarily supported by net margin expansion, fueled by a favorable product mix, along with an improvement in asset turnover back to the 80% level.

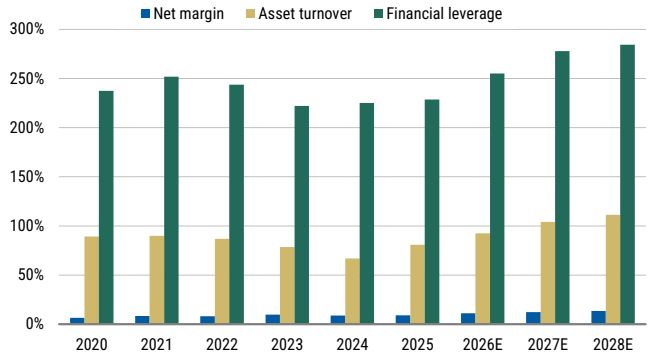
Looking ahead, we expect ROE to continue rising, reaching 26.4% in 2026, 35.3% in 2027, and 42.4% in 2028, primarily driven by continued margin expansion and improving asset turnover. Net margin is projected to increase from 9.1% in 2025 to 13.4% by 2028, supported by gross margin improvement from 22.9% to 27.4% and a decline in the operating expense ratio from 12.8% in 2025 to 10.5% in 2028.

Exhibit 128: ROE trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 129: DuPont analysis



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Improving balance sheet and continuous free cash flow generation

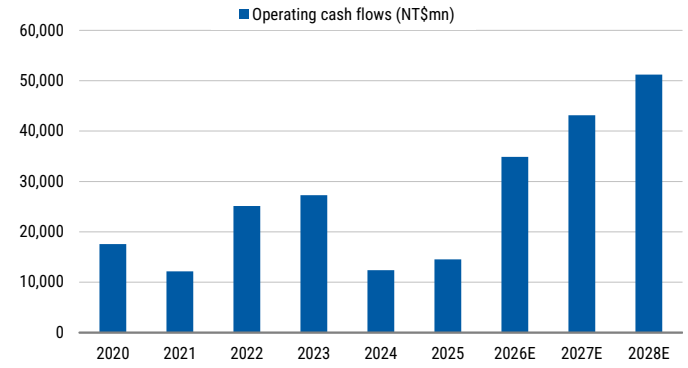
Lite-On has consistently generated positive operating cash flow in recent years, although cash flow dropped notably in 2021, 2024, and 2025, mainly due to higher working-capital-related cash outflows. The company has also consistently generated positive free cash flow, with capital expenditures generally at the NT$3-4bn level over the years. An exception occurred in 2025, when capital expenditure rose to NT$6.7bn, mainly for power-related capacity expansion in Vietnam, Taiwan, and the US.

As the company has announced investments of US$919mn and US$149mn in its US and Vietnam subsidiaries, respectively, alongside continued expansion of power-related capacity, mainly for PSUs, we now estimate capex to rise to ~NT$13bn in 2026, versus ~NT $7bn in 2025. Capex should then decline to NT$9bn in 2027 as investment levels normalize.

Lite-On's net cash position has remained relatively healthy, with its net cash-to-equity ratio improving from 65% in 2020 to 80% in 2023. It then declined to 75% in 2024 and 66% in 2025, as total debt increased by NT$7.7bn in 2024, while cash on hand declined in 2025, likely due to increasing capital investment.

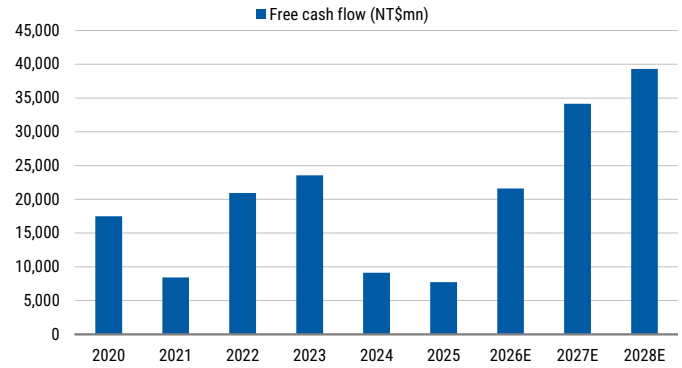
Looking ahead, we assume the total debt position will remain stable despite ongoing capex investment. The net cash position should continue to grow sequentially on the back of positive free cash flow generation.

Exhibit 130: Lite-on: Positive operating cash flow



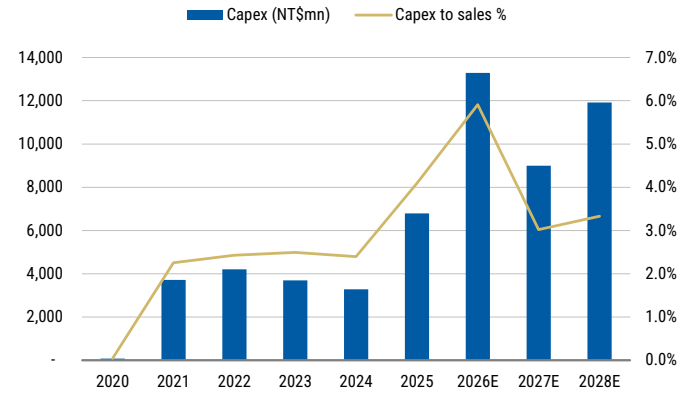
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 131: Lite-on: Improving free cash flow generation



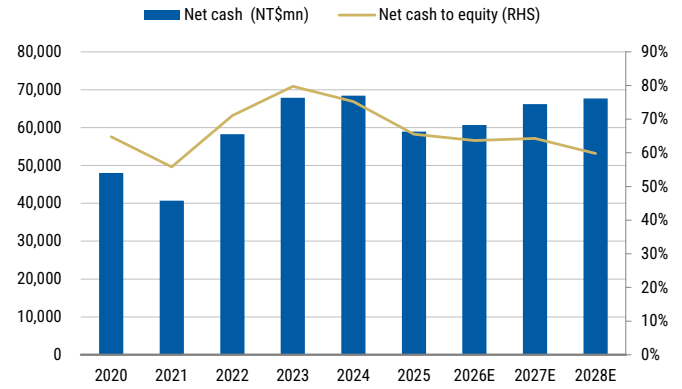
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 132: Lite-on: Capex and capex-to-sales ratio



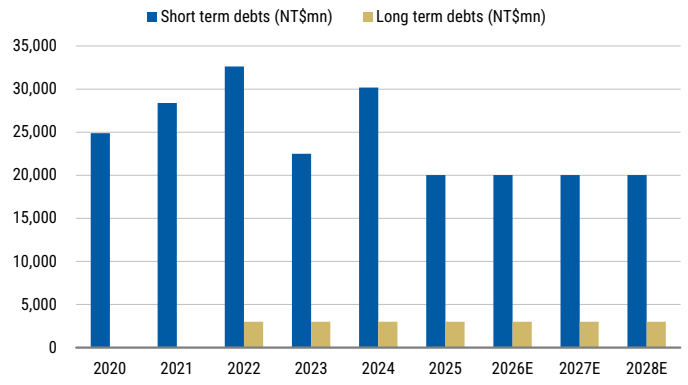
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 133: Lite-on: Net cash position



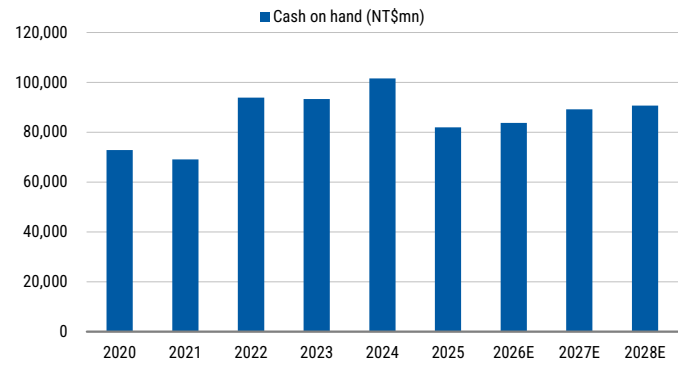
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 134: Lite-on: Debt position



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 135: Lite-on: Cash on hand

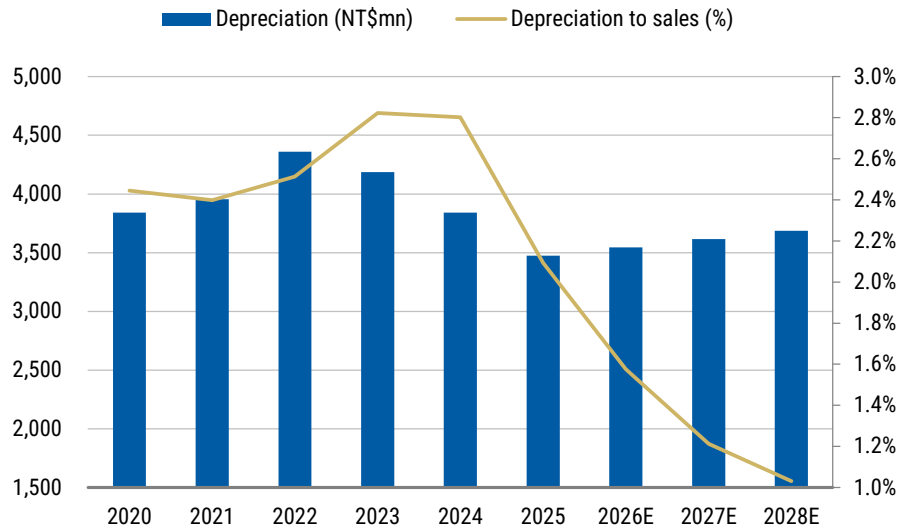


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Lite-On's average capex-to-sales ratio was around 2.3-2.4% during 2021-24, then rose to 4.1% in 2025 as the company began aggressively expanding global capacity, despite a larger revenue base during the year. Looking ahead, we project the capex-to-sales ratio to rise to 5.9% in 2026 amid continued production-site development, before declining to 3.0-3.3% in 2027-28 as investments normalize.

The depreciation-to-sales ratio remained at 2.4-2.5% during 2020-22 before rising to 2.8% in 2023, in line with higher capex during the year. The ratio then declined to 2.1% in 2025 on a larger revenue base. Looking ahead, we expect the ratio to decline further as the revenue base grows faster than capex and depreciation, falling to 1.6% in 2026 and 1.0% in 2028.

Exhibit 136: Lite-on: Depreciation and depreciation to sales

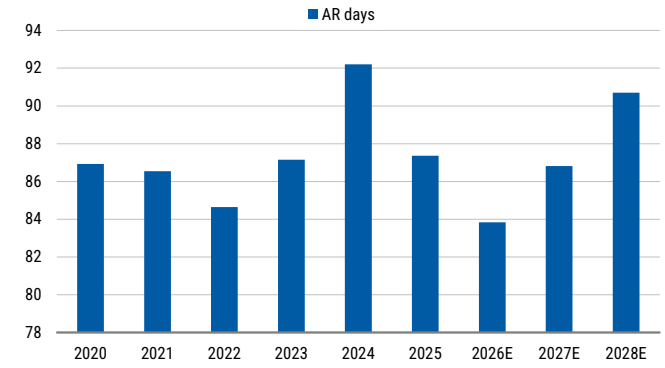


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Lite-On's cash conversion cycle (CCC) fluctuated within a relatively narrow band of roughly 35-44 days between 2020 and 2025, reflecting generally stable but modestly shifting working-capital efficiency. CCC was at a relatively low level of 35 days in 2020, supported by lower inventory days and higher AP days, before rising steadily to 44 days in 2022 as both inventory days (78) and AR days (85) increased, while AP days were slightly lower (119). From 2023 onward, the cycle improved meaningfully, falling to 36 days as AP days extended sharply from 119 in 2022 to 135. This improvement partially reversed in 2024, with CCC ticking up to 39 days as inventory and AR days both increased to 89 and 92 days, respectively, even as AP days extended further to 143. CCC then eased again to 37 days in 2025 as AP days moderated slightly to 138, alongside stable inventory and receivables levels.

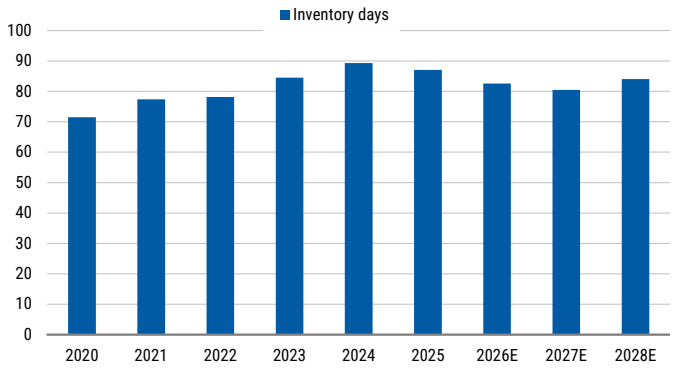
Looking ahead, we expect the overall cash conversion cycle to edge up slightly from 37 days in 2025 to 40 days in 2028, as increasing AR days and lower AP days are partially offset by improving inventory days.

Exhibit 137: Lite-on: AR days



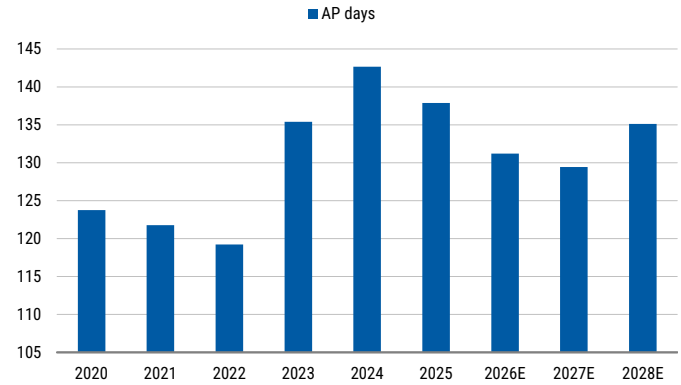
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 138: Lite-on: Inventory days



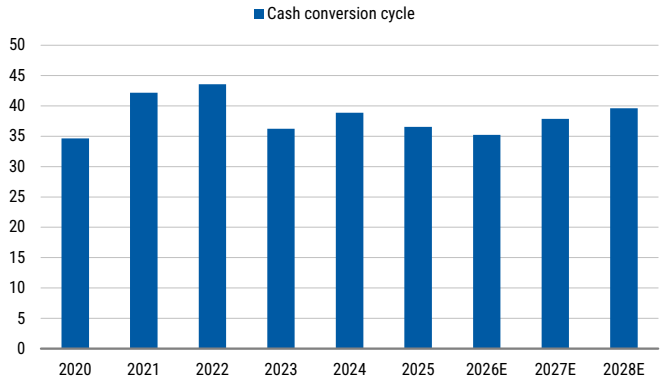
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 139: Lite-on: AP days



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 140: Lite-on: Cash conversion cycle



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Company Background – Lite-On

Company snapshot

Founded in 1975, Lite-On Technology is a Taiwan-based manufacturer of optoelectronic components, power supplies, and information technology products. The company produced Taiwan's first LEDs and became the first technology company listed on the Taiwan Stock Exchange in 1983. It mainly operates through four business segments: Information Products (power supplies, servers, networking equipment, keyboards, and chassis), Optoelectronics (LED packaging, LED lighting, automotive electronic components, and camera modules), Storage (storage components), and other supporting businesses. Additional product lines include energy management solutions, such as data center power supply systems and cloud server chassis; networking and surveillance products; and automotive electronics, including ADAS and EV charging solutions.

Lite-On is headquartered in Taipei, Taiwan. The company operates manufacturing sites, R&D centers, and sales offices across Asia (Taiwan, China, Japan, and Singapore), the US, and Europe (the UK, the Netherlands, and Germany).

Exhibit 141: Lite-on: Key milestones

Year	Key Milestones
1975	Established LITEON's first LED factory in Zhonghe, Taiwan
1983	Became the first electronics company listed on the Taiwan Stock Exchange (Ticker: 2301)
1985	Began manufacturing power supply units.
1989	Opened first overseas plant in Bangkok, Thailand, for optoelectronic products.
1990	• Set up a switching power supply plant in Penang, Malaysia. • Silitek acquired Maxi Switch, the No. 3 U.S. keyboard maker.
1991	Acquired NPE's switching power supply plant in the UK.
1994	Established a subsidiary in California, USA.
1999	Spun off the Optoelectronics business unit as LITEON IT Corp.
2002	LITEON Electronics, Silitek, LITEON Technology, and GVC merged into LITEON Technology Corporation (Ticker: 2301)
2003	Moved global headquarters to Neihu Science Park.
2006	• Acquired BenQ's Optical Disk Drive business, becoming a top-3 global ODD maker. • Partnered with Philips to enter the automotive disc drive market. • Acquired 100% of LITEON Automotive Corp., expanding into automotive electronics.
2007	• Acquired 100% of Li Shin International via stock swap, strengthening power supply resources for LCD TVs and other products. • Completed tender offer for Perlos, the world's largest handset casing maker.
2009	Opened East China Regional HQ in Wujin, Changzhou.
2010	• Invested in Epyon Power, a supplier of fast-charging technology. • Completed the world's first Optoelectronic Semiconductor Carbon Footprint and Type III Environmental Product Declarations.
2011	Completed the world's first Product Category Rules for Circuit Boards.
2013	Acquired Power Innovations (US), strengthening cloud and power management capabilities.
2014	Selected for the Carbon Disclosure Project's Climate Disclosure Leadership Index.
2016	Opened a new operations center in Kaohsiung's Nanzih Export Processing Zone.
2018	Divested the mobile camera module and mechanical business to focus resources.
2019	Acquired 100% of LOJ via tender offer.
2020	• Launched succession plan: Tom Soong named as Chairman, Anson Chiu named as President. • Completed sale of the SSD business to KIOXIA Holdings.
2021	• Launched an Employee Stock Ownership Trust (ESOT). • Joined the O-RAN Alliance and launched 5G small cells.
2022	• Began reconstructing the Zhonghe office into the "LITEON Zhonghe Digitech Center." • Inaugurated Phase II of the LITEON Vietnam plant in Haiphong.
2023	Opened its first corporate store, 'LITEON Collection,' in Neihu Science Park.
2024	• Took a 19.99% strategic stake in COSEL. • Broke ground on a new low-carbon building in Kaohsiung.
2025	• Returned to COMPUTEX with next-gen AI solutions, winning the Sustainable Design Award (Bronze). • Reopened its founding Zhonghe headquarters after a three-year reconstruction.

Source: Company data, Morgan Stanley Research

Exhibit 142: Lite-on: Global operation locations

Source: Company data

Exhibit 143: Lite-on: Global footprint

Headquarter	
Taipei City, Taiwan	
Sales Offices	
Milpitas, California, U.S.A.	
Plano, Texas, U.S.A.	
Tokyo, Japan	
Shanghai, China	
Singapore	
Hoofddorp, NL	
Cramlington, Northumberland, UK	
Wetzlar, DE	
Subsidiary	**Entity name**
American Fork, Utah, U.S.A.	Power Innovations International

Source: Company data, Morgan Stanley Research

Product offerings

Lite-On's key offerings can be categorized into six key areas: Optoelectronic Semiconductors, Energy Management, Consumer Electronics, Networking and Surveillance, Automotive Electronics, and Roadway Intellimation System.

1. Optoelectronic semiconductors

Its product portfolio spans a broad technological range, encompassing illumination and sensing applications, low- to high-power devices, standard and custom-made solutions, and components ranging from LEDs to lasers and from visible light to infrared and ultraviolet spectra.

Exhibit 144: Lite-on: Offerings for optoelectronic semiconductors

Optoelectronic Semiconductors						
Visible, UV & Automotive LEDs				Couplers	Optical Sensors	IR Emitters & Detectors
Illumination / UV-C / Automotive				Isolation & Signal Transfer	Ambient / Proximity / Gesture	Infrared Sensing
Automotive LED	SMD LED	Flash LED	UV LED	Photocouplers	Optical Sensors	IR Emitters & Detectors

Source: Company data, Morgan Stanley Research

2. Energy management

The product range serves the communications, industrial automation, smart home, medical, office, and automotive electronics industries. It offers a comprehensive suite of platinum- to titanium-efficiency-grade power supplies (CRPS, GPU, and SLIM series) suitable for both standard and AI servers, supporting AC or DC input with high power density, high reliability, and modular scalability. Power-shelf and battery-backup-unit (BBU) solutions, along with rack-cooling and liquid-cooling technologies for data centers, cloud infrastructure, and AI/machine-learning platforms, are also included in this segment.

Exhibit 145: Lite-on: Offerings for energy management

Energy Management			
Data Center Power Supply System	**Cloud Server Chassis**	**Power Supply Module**	**Magnetic Components**
CRPS / GPU / SLIM, Platinum-Titanium	Open Rack Standard V3	PoE / Industrial / LED Driver	Transformers & Inductors
			Includes components apply for cloud infrastructure, industrial power supply, white appliances, electric vehicle and consumer electric, etc
Server Power Networking Power Open Compute Project	Server Rack Server Chassis	POE Injector Industrial Power Adapter LED Driver	Magnetic Components

Source: Company data, Morgan Stanley Research

3. Consumer electronics

This segment is committed to advancing home entertainment and office solutions through feature-rich product design intended to enhance the overall user experience. Offerings include office/gaming solutions, laptop keyboard modules and touchpads, laptop power adapters, USB Type-C/PD chargers, etc.

Exhibit 146: Lite-on: Offerings for consumer electronics

Consumer Electronics		
3C Accessories	**Power Supply Module**	
Office / Gaming / Laptop peripherals	PC & Networking Power	
Office Solutions Gaming Solutions Laptop Keyboard Module Touchpad Laptop Power Adapters USB Type C/PD Charger	PC Power Supply	Networking Adapter

Source: Company data, Morgan Stanley Research

4. Networking and surveillance

The product portfolio covers networking solutions, video surveillance, AIoT solutions, and server-related offerings. The solutions are designed to be user-friendly and adaptable to emerging trends in digital intelligence.

Exhibit 147: Lite-on: Offerings for networking and surveillance

Networking and Surveillance		
Video Surveillance	**Access Control & Alarms**	**Networking Solutions**
Home / Building / Module	Smart Hub / Access / Alarm	5G / WLAN / Switch / Module
Home Security Camera Building Surveillance Camera Camera Module	Smart Hub Access Control Alarm Kits	5G Total Solution Wireless LAN Wired Switch Wireless Module

Source: Company data, Morgan Stanley Research

5. Automotive electronics

Lite-On is able to leverage its core competencies in optoelectronics and power management, along with more than four decades of expertise in automotive electronics. This segment is dedicated to advancing vehicle-charging technology, ADAS (Advanced Driver Assistance Systems) solutions, and automotive lighting modules through high-end engineering. AC charging solutions are offered across multiple current ratings (32/40A, 48A, 80A, and an EU-specific variant), while DC charging solutions include a 30kW fast-charging offering.

Exhibit 148: Lite-on: Offerings for automotive electronics

Automotive Electronics		
ADAS Solutions	**AC Charging Solutions**	**DC Charging Solutions**
Vision & Lighting	32/40A - 48A - 80A - EU	Fast Charging
Automotive Vision Solutions Vehicle Lighting Solutions	IC 32/40A IC 48A IC 80A IC EU	DC 30kW

Source: Company data, Morgan Stanley Research

6. Roadway Intellimation System

This segment is mainly operated under the LEOTEK brand, a subsidiary of the LITEON Technology Group with dual headquarters in Silicon Valley and Taipei. It has served road infrastructure across more than 30 countries—including North America, Europe, the Middle East, and the Asia-Pacific region—since 1992. The segment comprises intelligent roadway lighting, traffic signal systems, and AI-driven predictive maintenance solutions, all developed to support smarter and more efficient road infrastructure.

Exhibit 149: Lite-on: Offerings for Roadway Intellimation System

Roadway Intellimation System		
Roadway Lighting Intellimation	**Intelligent Traffic Signal System**	**AI Predictive Maintenance Solution**
Smart LED Streetlights	AI-driven Signal Control	Condition Monitoring
		Includes AI warning and maintenance management, traffic flow data analysis and computation, traffic-adaptive signal control, and priority passage for emergency vehicles, etc
Roadway Lighting Intellimation	Intelligent Traffic Signal System	AI Predictive Maintenance Solution

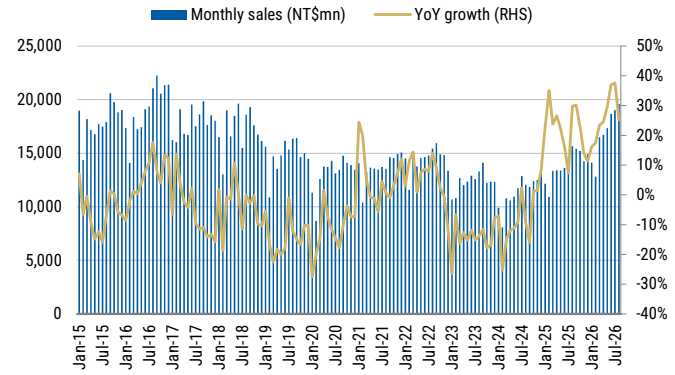
Source: Company data, Morgan Stanley Research

Monthly sales vs. share price

Lite-On's monthly sales have hovered around the NT$10-20bn level since 2013. Notably, we see a high correlation of ~76% between the company's monthly sales and its share price movements since 2023.

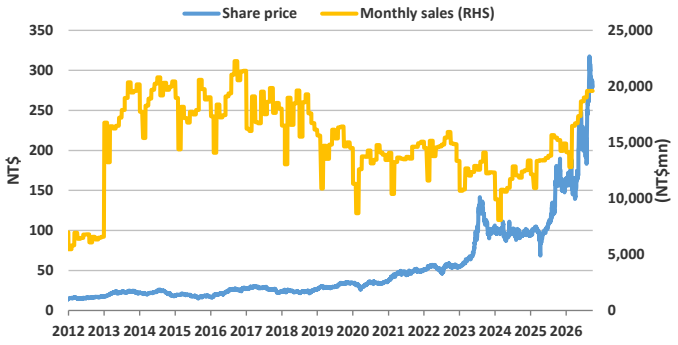
The company reported August monthly sales of NT$19,592mn (+3% MoM/+25% YoY).

Exhibit 150: Lite-on: Monthly sales vs. sales YoY growth



Source: TEJ, Morgan Stanley Research

Exhibit 151: Lite-on: Monthly sales vs. share price



Source: TEJ, Morgan Stanley Research

Key management & shareholding structure

Lite-On's board currently consists of five directors and four independent non-executive directors.

Its founder, Raymond Soong, and current Chairman, Tom Soong, together owned a stake of ~5% as of June 2026. Excluding the founder and Chairman, the management team owned a stake of ~0.1% as of June 2026.

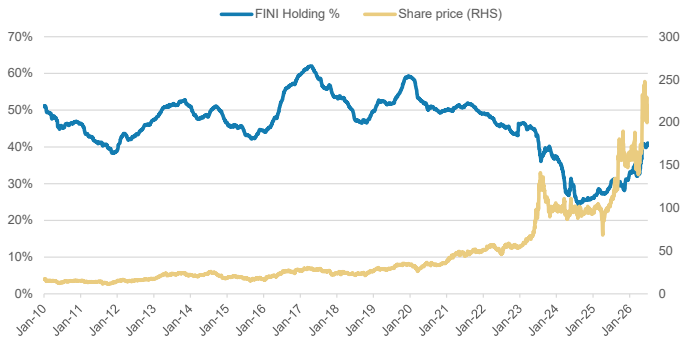
The FINI holding percentage as of July 7, 2026, was 41%, versus a previous peak of 62% in April 2017 and a historical average (since 2010) of 46%.

Exhibit 152: Lite-on: Shareholding structure

Shareholders	Ownership
Cathay MSCI Taiwan ESG Sustainability High Dividend Yield ETF	5.6%
Yuanta/P-shares Taiwan Dividend Plus ETF	5.5%
Ta-Rong Investment Co., Ltd	3.7%
Raymond Soong*	3.4%
Fuh Hwa Taiwan Technology Dividend Highlight ETF	3.3%
Labor Pension fund	2.6%
Ming-Hsing Investment Co., Ltd	2.6%
Ta-Sung Investment Co., Ltd*	2.1%
Chunghwa Post Co., Ltd	2.0%
Lite-On Technology Corporation (Treasury shares)*	1.7%
Total	**32.6%**

Source: Company data, Morgan Stanley Research. Note: Data as of August 2025. (*) are shown as insiders.

Exhibit 153: Lite-on: FINI holdings vs. share price since 2010



Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 154: Lite-on: FINI holdings vs. share price since 2023

Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 155:

Lite-on: Management profiles

Name	Title	Education	Work Experience
Tom Soong	Chairman and Chief Sustainability Officer, LITEON Technology	Electrical Engineering, University of Southern California, USA; International Business Program Attendance, NTU-FUDAN EMBA	- 20+ years at LITEON - Special Assistant to Chairman & Vice Chairman, LITEON - CEO of LITEON Smart Life and Applications Business Group - General Manager, LITEON Shanghai Operational Center
Anson Chiu	President and Director, LITEON Technology	Bachelor of Industrial Management, Lunghwa University of Science and Technology	- Joined LITEON Technology in 1987 - 35+ years in global supply chain and business development' - CEO, LITEON Power Conversion Business Group - General Manager, LITEON Power Conversion Business Group - Director, NB Business Unit, Power Conversion Business Group
Raymond Soong	Founder and Director, LITEON Technology and LITEON Cultural Foundation	Fellow, Industrial Technology Research Institute; Honorary Doctor of Management, National Chiao Tung University; Honorary Doctor of Management, Taipei University of Technology	- Chairman, LITEON Group - Chairman & Founder, LITEON Cultural Foundation - Chief Engineer, Texas Instruments Taiwan Ltd. - Board of Councilors member, Doctorate College of Technology, USC
Harvey Chang	Director Representative, Lite-On Technology Corp.	MBA, The Wharton School, University of Pennsylvania	- Independent Director, LITEON Technology - Chairman, IC Broadcasting - Chairman, TVBS and Catchplay Media Holdings Ltd. - President & CEO, Taiwan Mobile - Senior VP & CFO, TSMC
Sarina Lin	Director Representative, Lite-On; Advisor to Chairman's Office, Lite-On	Doctor of Laws, National Taiwan University	- Chief Legal Officer, Lite-On - Special Assistant to Chairman and Attorney, MediaTek Inc. - Director, MediaTek Foundation - SVP, CHRO & Chief Legal Officer, MStar Semiconductor - Director & CEO, MStar Semiconductor Charity Foundation - Director and Supervisor, MStar Semiconductor
Albert Hsueh	Independent Director, LITEON Technology;	MBA, Bloomsburg University, Pennsylvania, U.S.A.	- Chairman, PricewaterhouseCoopers Taiwan - Director, Corporate Governance Association in Taiwan - Professor, National Taiwan University and Tsinghua University, College of Technology Management
Mike Yang	Independent Director, Lite-On Technology Corp.	Master of Science, Arizona State University	- Associate Vice President, ADI Corp. Procurement Dept. - Director, LIONS Taiwan Technology Inc. - Director, Quanta Culture & Education Foundation - Cloud computing and global technology leadership
MK Lu	Independent Director, Lite-On Technology Corp.	Honorary Doctor, Tatung University; Honorary Doctor of Engineering, National Chiao Tung University; Department of Electrical Engineering, Tatung University	- Chairman, Tatung Company - Chairman, Sino-American Silicon Products Inc. - Chairman, Actron Technology Corporation - President, Lite-On Semiconductor Corp. - President, Vishay Lite-On Power Semiconductor Corp. - Vice President, Silitek Corp.
Jesse Ding	Independent Director, Lite-On Technology Corp.	PhD program, The Wharton School, University of Pennsylvania; MBA, University of Detroit, USA	- Chairman, EnTie Commercial Bank - General Manager, EnTie Commercial Bank - General Manager, TAIPEI FUBON COMMERCIAL BANK CO., LTD - Financial accounting and leadership decision-making

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Panasonic Holdings (6752.T)

Our target price is based on P/E 19.5x our FY3/28 EPS forecast.

We apply: 36x to EPS generated by AI infrastructure-related businesses, 5x to EPS attributable to IRA subsidies in the automotive battery business, and 13.5x to EPS from all other businesses.

Risks to Upside

- Acceleration of restructuring and exit/scale-down of unprofitable businesses
- Stronger than expected growth of datacenter BBU, automotive battery business

Risks to Downside

- Deteriorating visibility for energy business
- Expectations for business portfolio revisions sink

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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(as of September 30, 2026)
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For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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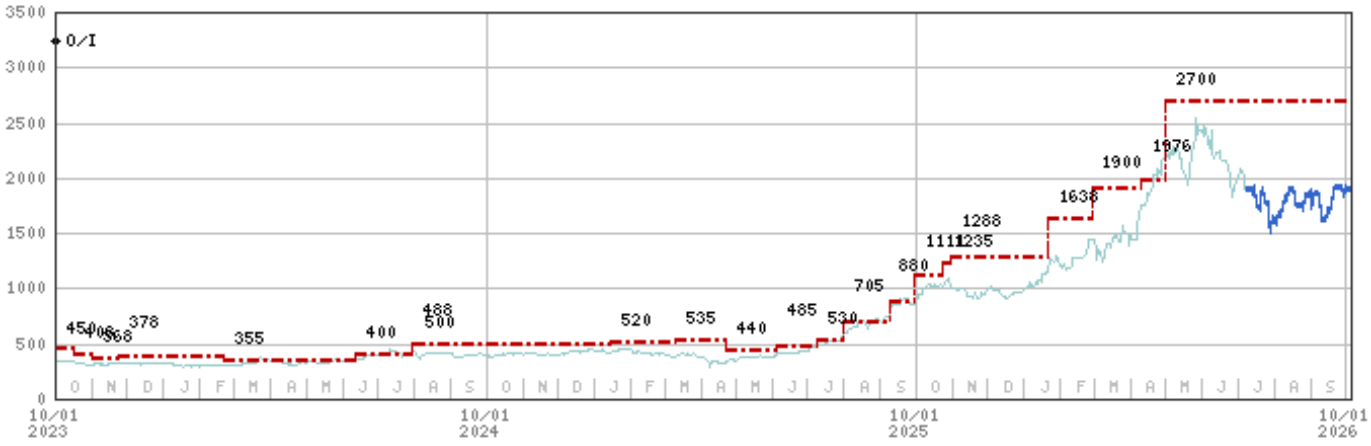
Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Delta Electronics Inc. (2308.TW) - As of 10/05/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : O/I

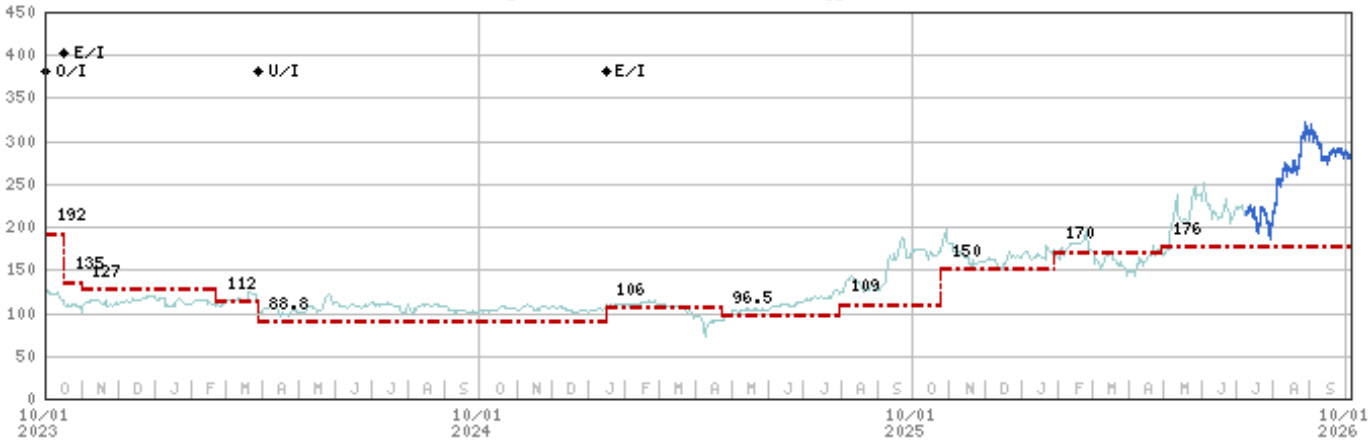
Price Target History: 9/9/21 : 345; 10/29/21 : 325; 2/25/22 : 303; 7/29/22 : 312; 9/21/22 : 326; 10/6/22 : 350; 11/23/22 : 368; 8/2/23 : 450; 10/17/23 : 408; 11/1/23 : 368; 11/23/23 : 378; 2/21/24 : 355; 6/12/24 : 400; 7/30/24 : 488; 8/1/24 : 500; 1/15/25 : 520; 3/11/25 : 535; 4/23/25 : 440; 6/4/25 : 485; 7/9/25 : 530; 8/1/25 : 705; 9/8/25 : 880; 9/30/25 : 1111; 10/24/25 : 1235; 10/30/25 : 1288; 1/21/26 : 1638; 2/27/26 : 1900; 4/10/26 : 1976; 5/1/26 : 2700

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Lite-On Technology (2301.TW) - As of 10/05/26 GMT in TWD
Industry : Greater China Technology Hardware



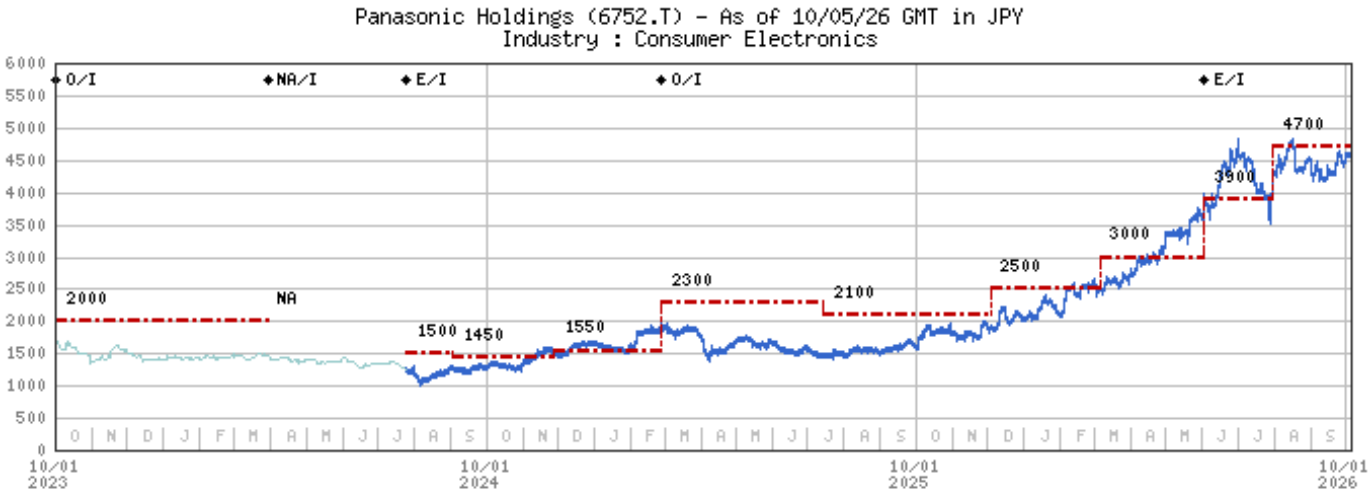
Stock Rating History: 10/1/21 : E/I; 7/31/23 : O/I; 10/17/23 : E/I; 3/28/24 : U/I; 1/15/25 : E/I

Price Target History: 4/30/21 : 62; 7/19/22 : 65; 2/21/23 : 61.2; 7/31/23 : 192; 10/17/23 : 135; 10/31/23 : 127; 2/21/24 : 112; 3/28/24 : 88.8; 1/15/25 : 106; 4/23/25 : 96.5; 7/31/25 : 109; 10/24/25 : 150; 1/28/26 : 170; 4/29/26 : 176

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.



Stock Rating History: 10/1/21 : O/I; 3/29/24 : NA/I; 7/25/24 : E/I; 2/26/25 : O/I; 6/3/26 : E/I

Price Target History: 9/22/21 : 1900; 3/1/22 : 1700; 6/13/22 : 1500; 12/1/22 : 1600; 2/20/23 : 1500; 6/7/23 : 2000; 3/29/24 : NA; 7/25/24 : 1500; 9/3/24 : 1450; 11/28/24 : 1550; 2/26/25 : 2300; 7/14/25 : 2100; 12/3/25 : 2500; 3/6/26 : 3000; 6/3/26 : 3900; 7/31/26 : 4700

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/05/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$38.64
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.76
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,055.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,730.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.46
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$53.35
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.91
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$37.16
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$61.95
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43

Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$23.88
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$179.10
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,111.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.62
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,075.00
Advantech (2395.TW)	O (01/20/2021)	NT$740.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,180.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,585.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$39.05
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,550.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,485.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$890.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,325.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,005.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$146.50
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$68.20
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,450.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$345.50
Innolux (3481.TW)	E (04/07/2025)	NT$52.30
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,355.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$298.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.40
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$995.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$207.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.80
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,700.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.86
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$388.50
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,215.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$254.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.30
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$3,000.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.92
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,875.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,495.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$88.70
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$331.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,325.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$606.00

Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,150.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$626.00
Zhen Ding (4958.TW)	++	NT$576.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$570.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,780.00

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October 6, 2026 11:56 AM GMT

King Slide Works Co. Ltd. | Asia Pacific

Preliminary August Profit and September Monthly Sales (-31% MoM/+196% YoY)

Sentiment to stay soft in near term but 2027 outlook is promising.

King Slide announced its preliminary profit for August, per TWSE regulatory requirements:

- Revenue was NT$6,466mn (+1% MoM/+344% YoY), 33% of our estimate for 3Q26 of NT$19,555mn (+81% QoQ/+348% YoY) and 39% of consensus' NT $16,755mn (+55% QoQ/+284% YoY).
- Pretax income was NT$4,826mn (+216% YoY; pretax margin at 74.6%), 29% of our 3Q26 pretax income estimate of NT$16,377mn (margin at 83.7%) and 34% of consensus' NT$14,094mn (margin at 84.1%).
- Net income was NT$3,819mn (+212% YoY; EPS NT$40.07; NPM 59.1%), 30% of our 3Q26 net income estimate of NT$12,938mn (EPS NT$135.76; NPM 66.2%) and 34% consensus' NT$11,118mn (EPS NT$114.38; NPM 66.4%).

September monthly sales

- September sales of NT$4,464mn (-31% MoM/+196% YoY), driving 3Q26 revenue to NT$17,338mn (+60% QoQ/+297% YoY), were short of our estimate of NT$19,555mn (+81% QoQ/+348% YoY) and but ahead of the consensus forecast of NT$16,755mn (+55% QoQ/+284% YoY).

Our view:

- We think the sequential decline in September revenue was driven by slower momentum for some projects, which entered the later part of their product cycles. While sentiment should stay soft in the near term, we think the outlook into 2027 is still promising as customers start new product cycles.
- We maintain our view that King Slide's competitiveness in the comprehensive patent portfolio and strong execution are entrenched and accentuated amid fast generation migration in AI applications.
- Stay OW with PT of NT$15,000 (30x 2027e P/E).

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King Slide Works Co. Ltd. (2059.TW, 2059 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$15,000.00
Up/downside to price target (%)	15
Shr price, close (Oct 6, 2026)	NT$13,100.00
52-Week Range	NT$15,085.00-2,875.00
Sh out, dil, curr (mn)	95
Mkt cap, curr (mn)	NT$1,248,392
EV, curr (mn)	NT$1,224,690
Avg daily trading value (mn)	NT$4,468

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	103.23	382.77	502.77	559.77
Prior EPS (NT$)**	-	-	-	-
EPS (NT$)§	100.25	343.67	499.35	574.20
Revenue, net (NT$ mn)	17,501	55,414	74,250	82,820
EBITDA (NT$ mn)	12,449	45,194	60,021	67,047
ModelWare net inc (NT $ mn)	9,837	36,477	47,913	53,345
P/E	36.3	34.2	26.1	23.4
RNOA (%)	216.6	818.3	534.3	549.4
ROE (%)	46.2	130.2	90.9	62.3
EV/EBITDA	26.8	26.7	19.5	17.0
Div yld (%)	0.9	0.9	1.2	1.3
FCF yld ratio (%)**	2.9	2.6	3.8	4.2
Leverage (EOP) (%)	(84.6)	(83.2)	(88.8)	(91.5)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

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AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
King Slide Works Co. Ltd. 2059.TW			
September 2026 sales	06 Oct 2026	Unchanged	↓ Modest downside surprise
Preliminary August Profit Announcement	06 Oct 2026	Unchanged	↓ Modest downside surprise

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

King Slide Works Co. Ltd. (2059.TW)

Base case, 30x 2027e P/E. We use price/earnings as our valuation methodology because we believe it better captures King Slide's earnings growth potential. We believe our target multiple is justified by an operating profit CAGR of 77% seen for 2025-28, implying a PEG of 0.4 vs. the average of 1.2 over 2010-24, when the stock was trading at 17-18x P/E with an operating profit CAGR of 14%.

Risks to Upside

- Stronger-than-expected AI capex
- Stronger server demand
- Faster market share gains

Risks to Downside

- Softer AI capex
- Weaker server demand
- Sluggish kitchen furniture demand
- Competition leading to slower share gains.

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(as of September 30, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
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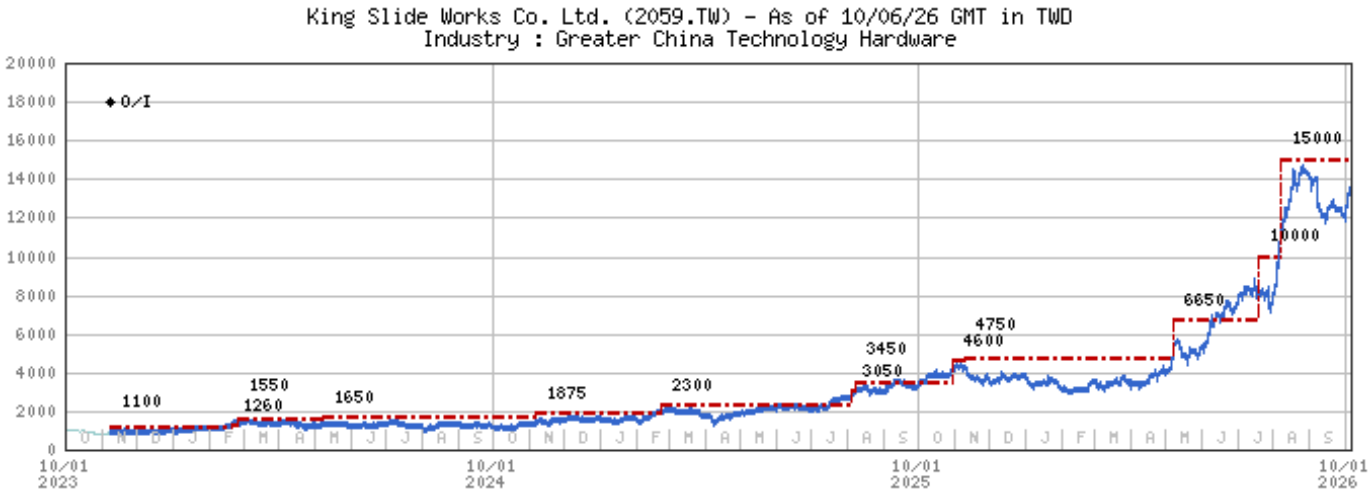
Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 11/8/23 : O/I

Price Target History: 11/8/23 : 1100; 2/20/24 : 1260; 2/26/24 : 1550; 5/8/24 : 1650; 11/7/24 : 1875; 2/22/25 : 2300; 8/4/25 : 3050; 8/8/25 : 3450; 10/30/25 : 4600; 11/10/25 : 4750; 5/8/26 : 6650; 7/20/26 : 10000; 8/7/26 : 15000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -·- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) ▬
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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/06/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$39.36
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$23.90
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,045.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,855.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.43
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$52.30
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$5.00
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$38.14
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$64.10
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.26

Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$180.00
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,079.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.80
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,050.00
Advantech (2395.TW)	O (01/20/2021)	NT$738.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,195.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,685.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$37.45
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,530.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,325.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$885.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,350.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,050.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,415.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$339.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.10
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,100.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$299.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.05
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$1,000.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$206.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.90
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,995.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$5.02
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$380.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,260.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$256.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.10
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,920.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.40
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,865.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,505.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$90.00
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$333.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,310.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$567.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50

Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,290.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$625.00
Zhen Ding (4958.TW)	++	NT$574.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$551.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,755.00

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* Historical prices are not split adjusted.

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Kaori

NT$1,600.00 - HIGH CONVICTION OUTPERFORM

Research provided by SinoPac

6 October 2026

Taiwan

Technology

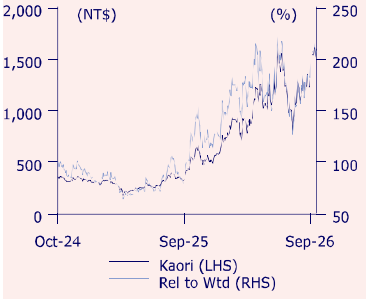
Reuters	8996.TW
Bloomberg	8996 TT
Priced on 6 October 2026	Taiwan Wtd @ 49,712.0
12M hi/lo	NT$1,655.00/401.00
12M price target	NT$2,710.00
±% potential	+69%
Shares in issue	92.5m
Free float (est.)	81.0%
Market cap	US$4.7bn
3M ADV	US$129.9m
Foreign s'holding	35.1%

Major shareholders
TU Shui-Cheng 3.6%
Tsai He-Ling 2.2%

SinoPac analysts:
Kevin SW Chen, SinoPac
Vanessa Huang, SinoPac

Stock performance (%)

	1M	3M	12M
Absolute	23.1	12.3	290.2
Relative	15.3	5.7	110.1
Abs (US$)	22.7	13.3	275.1



Source: Bloomberg

Growth in multiples

We forecast revenue to more than double next year

Bloom Energy's planned capacity expansion of up to 10GW strengthens Kaori's long-term SOFC growth outlook, while higher liquid cooling content and shipments should sustain liquid cooling momentum into 2027. CSP CDU heat exchangers and semiconductor equipment cooling could provide additional growth drivers from 4Q26. We raise our 2027/28 EPS forecasts by 14%/12% and increase our target price to NT$2,710 from NT$2,300. Reiterate HC O-PF rating.

Bloom Energy to double its capacity

Bloom Energy (BE) has announced plans to acquire a 158k sqft facility in Fremont, US, comparable in size to its existing Fremont manufacturing plant, which can produce up to 5GW. We believe the acquisition could support an eventual doubling of its fuel cell stack manufacturing capacity to 10GW. We expect its supply chain to support shipments of 2/4/6GW in 2026/27/28 to meet strong solid oxide fuel cell (SOFC). demand. To support this ramp, Kaori is expanding capacity to 6-7x its current level by end-2027. We now forecast Kaori's sales from BE to grow by 185%/66% YoY in 2027/28, versus our previous estimate of 101%/77%.

Strong liquid cooling momentum into 2027

With SpaceX planning an additional 6-8GW of datacentre capacity in 2027, we believe GPU server suppliers Supermicro and Dell will be key beneficiaries, supporting continued growth across their supply chains from 2H26 into 2027. Higher content value for Rubin manifolds, together with stronger CDU and rear-door heat exchanger (RDHx) shipments, should support Kaori's liquid cooling momentum. We forecast its liquid cooling revenue to grow by 60%/25% YoY in 2027/28.

Diversifying the growth engines

While SOFC and Supermicro-related liquid cooling remain Kaori's main growth drivers, we believe plate heat exchangers for CSP CDUs and liquid cooling applications in semiconductor equipment could emerge as additional growth drivers from 4Q26. These opportunities should broaden Kaori's customer and application exposure, supporting its longer-term growth outlook.

Earnings revisions and valuation

We raise our 2027/28SP EPS forecasts by 14%/12% to reflect higher expected demand and shipments for BE's fuel cells, alongside a stronger liquid cooling outlook. We increase our TP to NT$2,710 from NT$2,300, based on an unchanged 40x 2H27-1H28SP PE. Reiterate High Conviction Outperform rating.

Financials

Year to 31 December	**24A**	**25A**	**26SP**	**27SP**	**28SP**
Revenue (NT$m)	4,003	6,581	14,390	30,154	46,912
Rev forecast change (%)	-	-	*0.5*	*15.6*	*13.8*
Net profit (NT$m)	593	830	2,028	4,665	7,643
NP forecast change (%)	-	-	*(0.7)*	*15.7*	*13.4*
EPS (NT$)	6.56	9.07	21.47	48.35	79.21
SP/consensus (5) (EPS%)	-	-	97	114	107
EPS growth (% YoY)	1.7	38.2	136.8	125.2	63.8
PE (x)	244.0	176.5	74.5	33.1	20.2
Dividend yield (%)	0.2	0.2	0.5	0.9	1.5
ROE (%)	19.4	21.9	40.4	59.8	59.5
Net debt/equity (%)	(5.2)	(6.7)	22.1	10.9	3.2

Source: SinoPac

Change in target price

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Financials at a glance

Year to 31 December	2024A	2025A	2026SP	(% YoY)	2027SP	2028SP
Profit & Loss (NT$m)						
Revenue	**4,003**	**6,581**	**14,390**	***118.7***	**30,154**	**46,912**
Cogs (ex-D&A)	(2,650)	(4,587)	(10,010)		(20,537)	(31,642)
Gross Profit (ex-D&A)	**1,354**	**1,994**	**4,380**	***119.7***	**9,616**	**15,270**
SG&A and other expenses	(548)	(775)	(1,468)		(2,756)	(4,037)
Op Ebitda	**806**	**1,218**	**2,912**	***139***	**6,861**	**11,233**
Depreciation/amortisation	(159)	(195)	(367)		(997)	(1,631)
Op Ebit	**647**	**1,024**	**2,546**	***148.7***	**5,864**	**9,602**
Net interest inc/(exp)	(14)	(10)	(8)		(7)	(7)
Other non-Op items	117	39	49	*26.1*	(2)	(2)
Profit before tax	**750**	**1,052**	**2,586**	***145.9***	**5,855**	**9,593**
Taxation	(157)	(222)	(558)		(1,190)	(1,950)
Profit after tax	**593**	**830**	**2,028**	***144.5***	**4,665**	**7,643**
Minority interest	0	0	0		0	0
Net profit	**593**	**830**	**2,028**	***144.5***	**4,665**	**7,643**
Adjusted profit	**593**	**830**	**2,028**	***144.5***	**4,665**	**7,643**
Cashflow (NT$m)	**2024A**	**2025A**	**2026SP**	**(% YoY)**	**2027SP**	**2028SP**
Operating profit	**647**	**1,024**	**2,546**	***148.7***	**5,864**	**9,602**
Depreciation/amortisation	159	195	367	*88.4*	997	1,631
Working capital changes	331	(416)	(3)		(1,625)	(4,867)
Other items	(96)	(205)	(518)		(1,199)	(1,959)
Net operating cashflow	**1,041**	**598**	**2,392**	***300.1***	**4,037**	**4,406**
Capital expenditure	(190)	(313)	(3,617)		(3,000)	(2,460)
Free cashflow	**851**	**285**	**(1,225)**		**1,037**	**1,946**
M&A/Others	(4)	94	0		0	0
Net investing cashflow	**(195)**	**(219)**	**(3,617)**		**(3,000)**	**(2,460)**
Increase in loans	787	(81)	200		200	200
Dividends	(358)	(366)	(365)		(811)	(1,400)
Net equity raised/other	(4)	(132)	5		5	5
Net financing cashflow	**426**	**(579)**	**(160)**		**(606)**	**(1,195)**
Incr/(decr) in net cash	1,273	(200)	(1,385)		431	752
Exch rate movements	8	5	5	*0*	5	5
Balance sheet (NT$m)	**2024A**	**2025A**	**2026SP**	**(% YoY)**	**2027SP**	**2028SP**
Cash & equivalents	1,871	1,677	298	*(82.2)*	734	1,491
Accounts receivable	822	1,817	1,799	*(1)*	3,769	5,864
Other current assets	1,163	2,086	2,308	*10.6*	3,822	8,551
Fixed assets	1,966	2,075	5,332	*156.9*	7,335	8,164
Investments	145	252	298	*18*	298	298
Intangible assets	9	15	15	*0*	15	15
Other non-current assets	127	204	127	*(37.8)*	127	127
Total assets	**6,103**	**8,127**	**10,176**	***25.2***	**16,100**	**24,510**
Short-term debt	958	1,174	1,174	*0*	1,174	1,174
Accounts payable	362	1,529	1,729	*13.1*	3,589	5,545
Other current liabs	557	872	872	*0*	872	872
Long-term debt/CBs	735	224	424	*89.4*	624	824
Provisions/other LT liabs	53	152	97	*(36.1)*	107	118
Shareholder funds	3,395	4,171	5,874	*40.8*	9,728	15,971
Minorities/other equity	42	6	6	*0*	6	6
Total liabs & equity	**6,103**	**8,127**	**10,176**	***25.2***	**16,100**	**24,510**
Ratio analysis	**2024A**	**2025A**	**2026SP**	**(% YoY)**	**2027SP**	**2028SP**
Revenue growth (% YoY)	(7.4)	64.4	118.7		109.5	55.6
Ebitda margin (%)	20.1	18.5	20.2		22.8	23.9
Ebit margin (%)	16.2	15.6	17.7		19.4	20.5
Net profit growth (%)	2.9	39.9	144.5		130.0	63.8
Op cashflow growth (% YoY)	53.3	(42.6)	300.1		68.8	9.1
Capex/sales (%)	4.8	4.8	25.1		9.9	5.2
Net debt/equity (%)	(5.2)	(6.7)	22.1		10.9	3.2
Net debt/Ebitda (x)	-	-	0.4		0.2	0.0
ROE (%)	19.4	21.9	40.4		59.8	59.5
ROIC (%)	15.7	23.2	37.1		53.1	56.9

Source: SinoPac

Figure 1

Forecast revisions

	2026SP			2027SP			2028SP		
(NT$m)	old	new	% ch	old	new	% ch	old	new	% ch
Revenue	14,318	14,390	0%	26,078	30,154	16%	41,233	46,912	14%
Gross profit	4,027	4,014	0%	7,463	8,620	16%	12,024	13,639	13%
Opex	1,460	1,468	0%	2,387	2,756	15%	3,549	4,037	14%
Operating profit	2,566	2,546	0%	5,076	5,864	16%	8,475	9,602	13%
Pretax profit	2,605	2,586	0%	5,060	5,855	16%	8,460	9,593	13%
Net profit	2,043	2,028	0%	4,032	4,665	16%	6,740	7,643	13%
EPS (NT$)	21.40	21.02	(1%)	42.23	48.35	14%	70.60	79.21	12%
Margin									
Gross margin	28.1%	27.9%		28.6%	28.6%		29.2%	29.1%	
Operating margin	17.9%	17.7%		19.5%	19.4%		20.6%	20.5%	
Net margin	14.3%	14.1%		15.5%	15.5%		16.3%	16.3%	

Source: SinoPac

Figure 2

Quarterly P&L

(NT$m)	1Q26	2Q26	3Q26SP	4Q26SP	1Q27SP	2Q27SP	3Q27SP	4Q27SP	2025	2026SP	2027SP	2028SP
Sales	3,423	2,305	3,743	4,919	5,395	6,637	8,334	9,788	6,581	14,390	30,154	46,912
Gross profit	993	599	1,030	1,391	1,529	1,890	2,383	2,818	1,799	4,014	8,620	13,639
Opex	285	301	385	497	513	611	742	891	775	1,468	2,756	4,037
Operating profit	709	298	644	895	1,016	1,280	1,642	1,927	1,024	2,546	5,864	9,602
Non-OP	51	(6)	(3)	(2)	(2)	(2)	(2)	(2)	28	41	(9)	(9)
Pretax profit	760	293	641	893	1,014	1,278	1,639	1,925	1,052	2,586	5,855	9,593
Net profit	606	201	511	711	808	1,018	1,306	1,533	830	2,028	4,665	7,643
EPS (NT$)	6.54	2.16	5.40	7.53	8.37	10.55	13.54	15.89	8.97	21.02	48.35	79.21
Growth												
Sales QoQ	49%	(33%)	62%	34%	10%	23%	26%	17%				
Sales YoY	238%	55%	110%	115%	58%	188%	127%	99%	64%	119%	111%	56%
EPS QoQ	90%	(67%)	150%	35%	11%	26%	28%	17%				
EPS YoY	347%	62%	91%	119%	28%	388%	143%	111%	37%	134%	128%	64%
Margin												
Gross margin	29.0%	26.0%	27.5%	28.3%	28.3%	28.5%	28.6%	28.8%	27.3%	27.9%	28.6%	29.1%
Opex	8.3%	13.1%	10.3%	10.1%	9.5%	9.2%	8.9%	9.1%	11.8%	10.2%	9.1%	8.6%
OP margin	20.7%	12.9%	17.2%	18.2%	18.8%	19.3%	19.7%	19.7%	15.6%	17.7%	19.4%	20.5%
Non-OP	1.5%	(0.2%)	-0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.4%	0.3%	0.0%	0.0%
Net margin	17.7%	8.7%	13.6%	14.5%	15.0%	15.3%	15.7%	15.7%	12.6%	14.1%	15.5%	16.3%

Source: SinoPac

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Solid
Meh

Figure 3

Sales mix and GM assumptions

Sales (NT$m)	2025	2026SP	2027SP	2028SP
Heat Energy Product	4,878	12,073	25,439	37,548
Plate Heat Exchanger	1,702	2,317	4,715	9,364
Total	6,581	14,390	30,154	46,912
YoY (%)	2025	2026SP	2027SP	2028SP
Heat Energy Product	104%	147%	112%	48%
Plate Heat Exchanger	5%	36%	101%	99%
Total	64%	119%	111%	56%
Sales mix (%)	2025	2026SP	2027SP	2028SP
Heat Energy Product	74%	84%	84%	80%
Plate Heat Exchanger	26%	16%	16%	20%
Total	100%	100%	100%	100%
Gross margin (%)	2025	2026SP	2027SP	2028SP
Heat Energy Product	27%	28%	28%	28%
Plate Heat Exchanger	29%	29%	30%	32%
Total	27%	28%	29%	29%

Source: SinoPac

Figure 4

Forward PE

Source: SinoPac

Figure 5

Forward PB

Source: SinoPac

Figure 6

Target price change

	Target PE (x)	Valuation period	12-month rolling EPS (NT$)	Target Price (NT$)
Old	40	2H27-1H28	57.6	2,300
New	40	2H27-1H28	67.1	2,710

Source: SinoPac

Investment thesis

Kaori is well positioned to capitalise on rising demand for liquid cooling, and it expects related product shipments to ramp up. The fuel-cell business should be a huge growth driver in 2026-27 to meet AI power infrastructure demand in the US. In the longer term, we believe the company will also benefit from heat-pump demand as Europe is phasing out gas boilers, despite some noise about near-term demand. Kaori has been planning capacity expansion earlier than peers to fully capture demand, with rising exposure to large clients. We thus expect the company to enjoy robust earnings in coming years.

Catalysts

The major growth catalysts largely depend on fuel-cell projects for AI datacentres taking off, increasing content value in liquid cooling offerings, and a clearer outlook for hydrogen electrolyzer development.

Valuation details

We adopt a PE-based valuation methodology as we believe the market will mainly focus on the earnings growth outlook, which is better reflected with PE metrics. We base our valuation on 40x PE, which is around +1sd of its five-year average and supported by an 93% EPS Cagr in 2026-28SP, to account for the surging demand for AI server liquid cooling and power infrastructure. We are positive on the company's long-term outlook given its growing stationary fuel cell demand from AI datacentres, as well as its efforts to diversify customer exposure for liquid cooling business. We believe the company can maintain its competitive edge over other cooling players given its expertise in vacuum welding technology to manufacture high-quality products.

Investment risks

Downside risks: 1) increased competition for Bloom Energy's fuel cell business or new technology solutions replacing fuel cells; 2) slower kick-off of new fuel cell projects for AI datacentres; 3) weaker-than-expected liquid cooling adoption; 4) slower rollout of Europe/US government subsidies for heat pump installations.

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Detailed financials

Profit & Loss (NT$m)

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Revenue	**2,844**	**4,326**	**4,003**	**6,581**	**14,390**	**30,154**	**46,912**
Cogs (ex-D&A)	(1,944)	(2,973)	(2,650)	(4,587)	(10,010)	(20,537)	(31,642)
Gross Profit (ex-D&A)	**899**	**1,353**	**1,354**	**1,994**	**4,380**	**9,616**	**15,270**
Research & development costs	(81)	(84)	(113)	(218)	(364)	(551)	(807)
Selling & marketing expenses	(132)	(139)	(139)	(159)	(335)	(689)	(1,009)
Other SG&A	(227)	(289)	(296)	(398)	(769)	(1,516)	(2,221)
Other Op Expenses ex-D&A	-	-	-	-	-	-	-
Op Ebitda	**460**	**841**	**806**	**1,218**	**2,912**	**6,861**	**11,233**
Depreciation/amortisation	(114)	(129)	(159)	(195)	(367)	(997)	(1,631)
Op Ebit	**346**	**711**	**647**	**1,024**	**2,546**	**5,864**	**9,602**
Interest income	2	8	9	20	24	26	26
Interest expense	(13)	(33)	(22)	(31)	(32)	(33)	(32)
Net interest inc/(exp)	(12)	(25)	(14)	(10)	(8)	(7)	(7)
Associates/investments	(6)	(2)	12	3	0	0	0
Forex/other income	57	31	50	16	40	(2)	(2)
Asset sales/other cash items	-	-	-	-	-	-	-
Provisions/other non-cash items	5	14	55	20	9	0	0
Asset revaluation/Exceptional items	-	-	-	-	-	-	-
Profit before tax	**390**	**730**	**750**	**1,052**	**2,586**	**5,855**	**9,593**
Taxation	(89)	(153)	(157)	(222)	(558)	(1,190)	(1,950)
Profit after tax	**301**	**577**	**593**	**830**	**2,028**	**4,665**	**7,643**
Preference dividends	-	-	-	-	-	-	-
Profit for period	**301**	**577**	**593**	**830**	**2,028**	**4,665**	**7,643**
Minority interest	0	0	0	0	0	0	0
Net profit	**301**	**577**	**593**	**830**	**2,028**	**4,665**	**7,643**
Extraordinaries/others	0	0	0	0	0	0	0
Profit avail to ordinary shares	301	577	593	830	2,028	4,665	7,643
Dividends	(134)	(134)	(358)	(366)	(365)	(811)	(1,400)
Retained profit	167	442	236	464	1,663	3,854	6,243
Adjusted profit	**301**	**577**	**593**	**830**	**2,028**	**4,665**	**7,643**
EPS (NT$)	3.4	6.4	6.6	9.1	21.5	48.4	79.2
Adj EPS [pre excep] (NT$)	3.4	6.4	6.6	9.1	21.5	48.4	79.2
Core EPS (NT$)	3.4	6.4	6.6	9.1	21.5	48.4	79.2
DPS (NT$)	1.5	4.0	2.9	3.9	8.4	14.5	23.8

Profit & loss ratios

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Growth (%)							
Revenue growth (% YoY)	27.4	52.1	(7.4)	64.4	118.7	109.5	55.6
Ebitda growth (% YoY)	34.1	82.7	(4.2)	51.3	139.0	135.6	63.7
Ebit growth (% YoY)	50.3	105.5	(9.1)	58.3	148.7	130.3	63.7
Net profit growth (%)	101.8	91.5	2.9	39.9	144.5	130.0	63.8
EPS growth (% YoY)	101.8	91.5	1.7	38.2	136.8	125.2	63.8
Adj EPS growth (% YoY)	101.8	91.5	1.7	38.2	136.8	125.2	63.8
DPS growth (% YoY)	(0.3)	166.3	(27.5)	36.8	113.1	72.5	63.8
Core EPS growth (% YoY)	101.8	91.5	1.7	38.2	136.8	125.2	63.8
Margins (%)							
Gross margin (%)	31.6	31.3	33.8	30.3	30.4	31.9	32.6
Ebitda margin (%)	16.2	19.4	20.1	18.5	20.2	22.8	23.9
Ebit margin (%)	12.2	16.4	16.2	15.6	17.7	19.4	20.5
Net profit margin (%)	10.6	13.3	14.8	12.6	14.1	15.5	16.3
Core profit margin	10.6	13.3	14.8	12.6	14.1	15.5	16.3
Op cashflow margin	(10.1)	15.7	26.0	9.1	16.6	13.4	9.4
Returns (%)							
ROE (%)	15.4	24.3	19.4	21.9	40.4	59.8	59.5
ROA (%)	7.1	12.5	9.4	11.3	21.8	35.6	37.7
ROIC (%)	9.8	17.4	15.7	23.2	37.1	53.1	56.9
ROCE (%)	12.1	21.2	19.2	28.6	46.0	65.2	70.4
Other key ratios (%)							
Effective tax rate (%)	22.8	21.0	20.9	21.1	21.6	20.3	20.3
Ebitda/net int exp (x)	39.6	33.5	57.9	116.4	365.4	1,036.5	1,723.6
Exceptional or extraord. inc/PBT (%)	-	-	-	-	-	-	-
Dividend payout (%)	44.4	61.7	44.0	43.5	39.2	30.0	30.0

Source: SinoPac

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Balance sheet (NT$m)

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Cash & equivalents	249	591	1,871	1,677	298	734	1,491
Accounts receivable	580	798	822	1,817	1,799	3,769	5,864
Inventories	1,180	1,235	1,052	1,853	2,075	3,589	8,318
Other current assets	236	78	110	232	232	232	232
Current assets	**2,245**	**2,702**	**3,856**	**5,580**	**4,404**	**8,325**	**15,906**
Fixed assets	1,625	1,839	1,966	2,075	5,332	7,335	8,164
Investments	159	132	145	252	298	298	298
Goodwill	0	0	0	0	0	0	0
Other intangible assets	0	3	9	15	15	15	15
Other non-current assets	128	154	127	204	127	127	127
Total assets	**4,156**	**4,829**	**6,103**	**8,127**	**10,176**	**16,100**	**24,510**
Short term loans/OD	838	255	958	1,174	1,174	1,174	1,174
Accounts payable	255	206	362	1,529	1,729	3,589	5,545
Accrued expenses	-	-	-	-	-	-	-
Taxes payable	0	0	0	0	0	0	0
Other current liabs	383	535	557	872	872	872	872
Current liabilities	**1,476**	**996**	**1,877**	**3,575**	**3,775**	**5,635**	**7,591**
Long-term debt/leases/other	595	1,092	735	224	424	624	824
Convertible bonds	-	-	-	-	-	-	-
Provisions/other LT liabs	26	38	53	152	97	107	118
Total liabilities	**2,097**	**2,125**	**2,666**	**3,950**	**4,296**	**6,366**	**8,533**
Share capital	894	894	915	925	965	965	965
Retained earnings	368	778	956	1,451	2,911	6,298	11,777
Reserves/others	784	1,037	1,524	1,795	1,998	2,465	3,229
Shareholder funds	**2,045**	**2,709**	**3,395**	**4,171**	**5,874**	**9,728**	**15,971**
Minorities/other equity	14	(5)	42	6	6	6	6
Total equity	**2,059**	**2,704**	**3,437**	**4,177**	**5,880**	**9,734**	**15,977**
Total liabs & equity	**4,156**	**4,829**	**6,103**	**8,127**	**10,176**	**16,100**	**24,510**
Total debt	1,433	1,346	1,694	1,397	1,597	1,797	1,997
Net debt	1,184	755	(178)	(280)	1,300	1,063	506
Adjusted EV	144,017	143,628	146,063	147,447	155,381	155,145	154,587
BVPS (NT$)	22.9	30.3	37.1	45.1	60.9	100.8	165.5

Balance sheet ratios

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Key ratios							
Current ratio (x)	1.5	2.7	2.1	1.6	1.2	1.5	2.1
Growth in total assets (% YoY)	21.8	16.2	26.4	33.2	25.2	58.2	52.2
Growth in capital employed (% YoY)	30.8	6.7	(5.8)	19.6	84.2	50.4	52.7
Net debt to operating cashflow (x)	(4.1)	1.1	-	-	0.5	0.3	0.1
Gross debt to operating cashflow (x)	(5.0)	2.0	1.6	2.3	0.7	0.4	0.5
Gross debt to Ebitda (x)	3.1	1.6	2.1	1.1	0.5	0.3	0.2
Net debt/Ebitda (x)	2.6	0.9	-	-	0.4	0.2	0.0
Gearing							
Net debt/equity (%)	57.5	27.9	(5.2)	(6.7)	22.1	10.9	3.2
Gross debt/equity (%)	69.6	49.8	49.3	33.5	27.2	18.5	12.5
Interest cover (x)	26.1	21.6	29.2	33.8	79.9	181.2	296.5
Debt cover (x)	(0.2)	0.5	0.6	0.4	1.5	2.2	2.2
Net cash per share (NT$)	(13.2)	(8.4)	2.0	3.1	(13.8)	(11.0)	(5.2)
Working capital analysis							
Inventory days	155.7	142.1	148.6	110.9	69.1	48.0	65.3
Debtor days	60.4	58.1	73.9	73.2	45.9	33.7	37.5
Creditor days	39.6	27.1	36.9	72.2	57.3	45.1	50.1
Working capital/Sales (%)	47.7	31.7	26.6	22.8	10.5	10.4	17.0
Capital employed analysis							
Sales/Capital employed (%)	87.7	125.1	122.8	168.8	200.4	279.3	284.6
EV/Capital employed (%)	4,441.3	4,152.4	4,481.3	3,783.3	2,164.1	1,436.9	937.8
Working capital/Capital employed (%)	41.9	39.6	32.7	38.5	21.0	29.0	48.5
Fixed capital/Capital employed (%)	50.1	53.2	60.3	53.3	74.3	67.9	49.5
Other ratios (%)							
PB (x)	69.9	52.8	43.1	35.5	26.3	15.9	9.7
EV/Ebitda (x)	313.1	170.9	181.3	121.0	53.4	22.6	13.8
EV/OCF (x)	(500.3)	211.5	140.3	246.6	65.0	38.4	35.1
EV/FCF (x)	(372.1)	435.0	171.6	517.6	(126.9)	149.6	79.4
EV/Sales (x)	50.6	33.2	36.5	22.4	10.8	5.1	3.3
Capex/depreciation (%)	91.9	284.1	126.0	169.7	1,004.2	301.0	150.8

Source: SinoPac

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Cashflow (NT$m)

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Operating profit	346	711	647	1,024	2,546	5,864	9,602
Operating adjustments	(45)	(135)	(54)	(194)	(518)	(1,199)	(1,959)
Depreciation/amortisation	114	129	159	195	367	997	1,631
Working capital changes	(711)	(55)	331	(416)	(3)	(1,625)	(4,867)
Interest paid / other financial expenses	0	0	0	0	0	0	0
Tax paid	-	-	-	-	-	-	-
Other non-cash operating items	8	28	(42)	(11)	0	0	0
Net operating cashflow	(288)	679	1,041	598	2,392	4,037	4,406
Capital expenditure	(99)	(349)	(190)	(313)	(3,617)	(3,000)	(2,460)
Free cashflow	(387)	330	851	285	(1,225)	1,037	1,946
Acq/inv/disposals	26	-	-	92	-	-	-
Int, invt & associate div	(93)	3	(4)	2	-	-	-
Net investing cashflow	(167)	(346)	(195)	(219)	(3,617)	(3,000)	(2,460)
Increase in loans	341	139	787	(81)	200	200	200
Dividends	(134)	(134)	(358)	(366)	(365)	(811)	(1,400)
Net equity raised/others	(3)	(3)	(4)	(132)	5	5	5
Net financing cashflow	204	2	426	(579)	(160)	(606)	(1,195)
Incr/(decr) in net cash	(251)	335	1,273	(200)	(1,385)	431	752
Exch rate movements	2	(4)	8	5	5	5	5
Opening cash	499	259	591	1,871	1,677	298	734
Closing cash	249	591	1,871	1,677	298	734	1,491
OCF PS (NT$)	(3.2)	7.6	11.5	6.5	25.3	41.8	45.7
FCF PS (NT$)	(4.3)	3.7	9.4	3.1	(13.0)	10.7	20.2

Cashflow ratio analysis

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Growth (%)							
Op cashflow growth (% YoY)	(197.5)	nm	53.3	(42.6)	300.1	68.8	9.1
FCF growth (% YoY)	(298.8)	-	157.7	(66.5)	(529.9)	-	87.7
Capex growth (%)	(1.4)	251.5	(45.5)	64.5	1,055.3	(17.1)	(18.0)
Other key ratios (%)							
Capex/sales (%)	3.5	8.1	4.8	4.8	25.1	9.9	5.2
Capex/op cashflow (%)	(34.5)	51.4	18.3	52.4	151.2	74.3	55.8
Operating cashflow payout ratio (%)	-	52.4	25.1	60.4	33.2	34.7	52.0
Cashflow payout ratio (%)	-	19.7	34.3	61.2	15.3	20.1	31.8
Free cashflow payout ratio (%)	-	40.6	42.0	128.4	-	78.2	71.9

DuPont analysis

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Ebit margin (%)	12.2	16.4	16.2	15.6	17.7	19.4	20.5
Asset turnover (x)	0.8	1.0	0.7	0.9	1.6	2.3	2.3
Interest burden (x)	1.1	1.0	1.2	1.0	1.0	1.0	1.0
Tax burden (x)	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Return on assets (%)	7.1	12.5	9.4	11.3	21.8	35.6	37.7
Leverage (x)	1.9	1.9	1.8	1.9	1.8	1.7	1.6
ROE (%)	15.4	24.3	19.4	21.9	40.4	59.8	59.5

EVA® analysis

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Ebit adj for tax	267	562	511	807	1,996	4,672	7,650
Average invested capital	2,721	3,237	3,266	3,482	5,388	8,793	13,455
ROIC (%)	9.8	17.4	15.7	23.2	37.1	53.1	56.9
Cost of equity (%)	9.7	9.7	9.7	9.7	9.7	9.7	9.7
Cost of debt (adj for tax)	2.0	2.1	2.1	2.1	2.0	2.1	2.1
Weighted average cost of capital (%)	7.5	7.5	7.5	7.5	7.5	7.5	7.5
EVA/IC (%)	2.3	9.8	8.1	15.6	29.5	45.6	49.3
EVA (NT$m)	62	318	265	545	1,590	4,009	6,635

Source: SinoPac

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Companies mentioned

Kaori (8996 TT - NT$1,600.00 - HIGH CONVICTION OUTPERFORM)
Bloom Energy (N-R)
Dell (N-R)
Supermicro (N-R)

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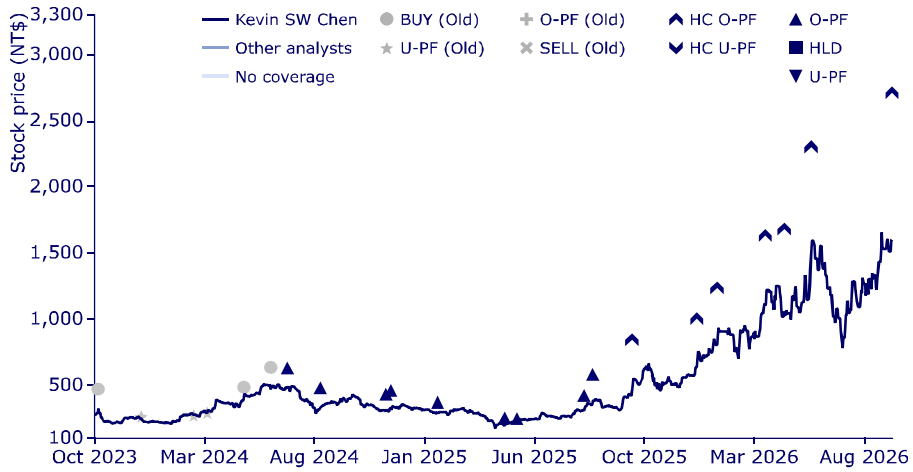
Recommendation history of Kaori Heat Treatment Co Ltd 8996 TT



Date	Rec	Target	Date	Rec	Target
LATEST	HC O-PF	2,710.00	20 Jan 2025	O-PF	370.00
17 Jun 2026	HC O-PF	2,300.00	17 Nov 2024	O-PF	460.00
11 May 2026	HC O-PF	1,680.00	10 Nov 2024	O-PF	430.00
15 Apr 2026	HC O-PF	1,630.00	12 Aug 2024	O-PF	480.00
08 Feb 2026	HC O-PF	1,234.00	28 Jun 2024	O-PF	630.00
11 Jan 2026	HC O-PF	1,000.00	05 Jun 2024	BUY	630.00
14 Oct 2025	HC O-PF	840.00	29 Apr 2024	BUY	480.00
21 Aug 2025	O-PF	580.00	10 Mar 2024	U-PF	290.00
09 Aug 2025	O-PF	420.00	20 Feb 2024	U-PF	271.00
09 May 2025	O-PF	245.00	10 Dec 2023	U-PF	260.00
22 Apr 2025	O-PF	250.00	12 Oct 2023	BUY	466.00

Note: At 6pm (HKT) on 28 June 2024, we changed our ratings to Outperform (O-PF), Hold (HLD) and Underperform (U-PF); and on 16 December 2024, we added the ratings: High-Conviction Outperform (HC O-PF) and High-Conviction Underperform (HC U-PF). Please see Key to CLSA/SinoPac investment rankings below for details. Source: CLSA, SinoPac

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